

Hartshorne Exercise Solutions

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Appendix A

A.6 Appendix A Exercises

Exercise A.6.1

Permitted context.

Work over an algebraically closed field k . Chow groups are cycles modulo the subgroup generated by pushforwards of principal divisors on normalizations of integral subvarieties. Proper pushforward, pullback between smooth varieties, the projection formula, localization, and intersections with Cartier divisors are available. A proper generically finite map pushes a principal divisor to the divisor of the field norm; if its image has smaller dimension, the corresponding pushforward is zero, using degree zero of a principal divisor on the generic proper curve in the dimension-one fiber case.

Problem. Show that rational equivalence is generated by relations between the fibers at 0, 1 of cycles W on $X \times \mathbb{A}^1$ meeting those fibers properly.

Solution. First take an integral component of W dominating $\mathbb{A}^1$. Close it in $X \times \mathbb{P}^1$ and normalize. The rational function $t/(t-1)$ has divisor the difference of its fibers at 0, 1, with their Cartier intersection multiplicities. It has neither zero nor pole over infinity. Projection to X is proper, so pushing this principal divisor shows that the two fiber cycles are rationally equivalent. Components over other fixed parameters meet neither specified fiber; components inside those fibers are excluded by proper intersection. Additivity proves this direction for arbitrary cycles.

Conversely a generating relation is $\nu_* \operatorname{div}(f)$, where $\nu : \tilde{V} \rightarrow V \subseteq X$ is normalization and $f \in k(V)^\times$. Constant f gives the zero relation. Otherwise take the closure of the graph of $t = f/(1+f)$ in $\tilde{V} \times \mathbb{P}^1$, and push its cycle to $X \times \mathbb{P}^1$. The graph is birational to $\tilde{V}$. At codimension-one points, normality gives a discrete valuation ring, and the rational map to $\mathbb{P}^1$ extends there. The fiber at 0 is the zero divisor of f , and the fiber at 1 is its pole divisor, with their orders. Any additional divisorial components over indeterminacy map into codimension at least two in V , so have zero pushforward in the required dimension. Restrict the resulting cycle to $X \times \mathbb{A}^1$; its fibers meet properly and their difference is exactly $\nu_* \operatorname{div}(f)$. Adding constant families and taking sums generates all the original rational-equivalence relations. $\square$

Exercise A.6.2

Permitted context.

Let $f : X \rightarrow X'$ be a proper dominant generically finite map of normal varieties over k . Normalization in a finite function-field extension is finite. A proper birational map to a normal variety is an isomorphism over codimension-one points. A finite torsion-free module over a discrete valuation ring is free. The norm is the determinant of multiplication on the finite-dimensional extension field.

Problem. Prove that f_* carries linearly equivalent Weil divisors to linearly equivalent Weil divisors.

Solution. It suffices to prove

$$f_* \operatorname{div}_X(g) = \operatorname{div}_{X'} N_{k(X)/k(X')}(g) \quad (g \in k(X)^\times).$$

Let Z be the normalization of X' in $k(X)$, and factor

$$X \xrightarrow{h} Z \xrightarrow{g_0} X'.$$

Here h is proper birational and g_0 is finite. Over a neighborhood of every codimension-one point of X' , h is an isomorphism at the relevant codimension-one points; exceptional divisors with smaller-dimensional image contribute zero. Removing closed subsets of codimension at least two therefore reduces the calculation to a finite map, flat at these points by torsion-freeness over a discrete valuation ring.

Fix a prime divisor D of X' . Over its local discrete valuation ring R , the finite normalization algebra is a free R -module. For an integral nonzero g , the order of its determinant is the length of the multiplication cokernel. Decomposing that finite-length module at the primes E above D gives

$$v_D(N(g)) = \sum_{E \mapsto D} [k(E) : k(D)] v_E(g).$$

Write a general g as a quotient to obtain the same equality. The right side is exactly the coefficient in divisor pushforward. This proof uses no separability assumption. Equality at every prime divisor proves the formula, and hence the claim. $\square$

Exercise A.6.3

Permitted context.

Work over an algebraically closed field k . Chow groups are cycles modulo the subgroup generated by pushforwards of principal divisors on normalizations of integral subvarieties. Proper pushforward, pullback between smooth varieties, the projection formula, localization, and intersections with Cartier divisors are available. A proper generically finite map pushes a principal divisor to the divisor of the field norm; if its image has smaller dimension, the corresponding pushforward is zero, using degree zero of a principal divisor on the generic proper curve in the dimension-one fiber case.

A projective r -fold admits a finite linear projection to $\mathbb{P}^r$ with center disjoint from it. Over the principal ideal domain $k[t]$, torsion-free modules are flat. The Hilbert polynomial is constant in a projective flat family. Fibers of a one-parameter cycle are rationally equivalent.

Problem. Use projection to prove directly that a degree- d integral subvariety $Y \subseteq \mathbb{P}^n$ is rationally equivalent to d times a linear subspace of its dimension.

Solution. Choose coordinates $[u_0 : \cdots : u_r : v_1 : \cdots : v_{n-r}]$ so that the u_i have no common zero on Y and define a finite surjection $Y \rightarrow L = \mathbb{P}^r$. The map

$$Y \times \mathbb{A}^1 \longrightarrow \mathbb{P}^n \times \mathbb{A}^1, \quad ([u : v], t) \longmapsto ([u : tv], t)$$

is everywhere defined and proper. Give its image W the reduced scheme structure. It is integral and dominates the parameter line, so its structure sheaf is torsion-free, hence flat over $k[t]$. For $t \neq 0$, its fiber is an automorphic image of Y . For $t = 0$, its support is precisely L , by finiteness and surjectivity of the projection. Its associated fiber cycle is consequently $m[L]$ for a positive integer m . Constancy of the Hilbert polynomial makes its degree equal to d ; as $\deg L = 1$, $m = d$. The relation between the fibers at 1 and 0 is $[Y] \sim d[L]$, as required. $\square$

Exercise A.6.4**Permitted context.**

Work over an algebraically closed field k . Chow groups are cycles modulo the subgroup generated by pushforwards of principal divisors on normalizations of integral subvarieties. Proper pushforward, pullback between smooth varieties, the projection formula, localization, and intersections with Cartier divisors are available. A proper generically finite map pushes a principal divisor to the divisor of the field norm; if its image has smaller dimension, the corresponding pushforward is zero, using degree zero of a principal divisor on the generic proper curve in the dimension-one fiber case.

Let $\pi : X = \mathbb{P}(\mathcal{E}) \rightarrow C$ be a ruled surface over a smooth projective curve. It has a section s : a rank-two bundle on a smooth curve has a line quotient after saturating a rational quotient. Every fiber is $\mathbb{P}^1$, on which any two points are rationally equivalent. On a smooth curve, zero-cycles modulo rational equivalence form its Picard group.

Problem. Prove $A^2(X) \cong \text{Pic } C$.

Solution. Proper maps give $\pi_* : A^2(X) \rightarrow A^1(C)$ and $s_* : A^1(C) \rightarrow A^2(X)$, with $\pi_* s_* = \text{id}$. For each closed point $x \in X$, the fiber over $\pi(x)$ contains both x and $s(\pi(x))$; a rational function on that projective line has divisor their difference. Thus $[x] = [s(\pi(x))]$ in $A^2(X)$. Point cycles generate, so $s_* \pi_* = \text{id}$ as well. The resulting inverse isomorphisms identify this group with $A^1(C) = \text{Pic } C$. $\square$

Exercise A.6.5**Permitted context.**

Work over an algebraically closed field k . Chow groups are cycles modulo the subgroup generated by pushforwards of principal divisors on normalizations of integral subvarieties. Proper pushforward, pullback between smooth varieties, the projection formula, localization, and intersections with Cartier divisors are available. A proper generically finite map pushes a principal divisor to the divisor of the field norm; if its image has smaller dimension, the corresponding pushforward is zero, using degree zero of a principal divisor on the generic proper curve in the dimension-one fiber case.

Let $\pi : \tilde{X} \rightarrow X$ blow up a point P of a smooth projective surface. Its exceptional curve $E \cong \mathbb{P}^1$ has normal bundle $\mathcal{O}_E(-1)$. The map is an isomorphism off E . Divisor classes satisfy $\text{Pic } \tilde{X} = \pi^* \text{Pic } X \oplus \mathbb{Z}[E]$. One may choose two very ample divisors meeting transversely at P and at their other intersection points.

Problem. Prove that $A(\tilde{X}) = \pi^* A(X) \oplus \mathbb{Z}[E]$ as graded groups, and describe multiplication, including $E^2 = -\pi^*[P]$.

Solution. Let q be the class of a point of E . The normal-bundle self-intersection calculation gives $E^2 = -q$. Also $E \cdot \pi^* D = 0$ for every divisor D on X , because its pulled-back line bundle restricts trivially to E .

To identify $\pi^*[P]$, choose the two transverse divisors in the context, with distinct tangent directions at P . Their total transforms are their strict transforms plus E . Over P the strict transforms are disjoint, each meeting E transversely in a point. Expanding their intersection therefore gives $q + q + E^2 = q$ above P . Away from P the map is an isomorphism, so subtracting the other intersection points proves $\pi^*[P] = q$. Hence $E^2 = -\pi^*[P]$.

The projection formula gives $\pi_*\pi^* = \text{id}$. For any zero-cycle class α on $\widetilde{X}$, the class $\alpha - \pi^*\pi_*\alpha$ restricts to zero off E . By localization it is a multiple nq . Pushing down gives $n[P] = 0$, and pulling back then gives $nq = 0$; thus π^* is an isomorphism on zero-cycles. It is an isomorphism in degree zero as well. The Picard splitting handles degree one, proving the graded group decomposition. The relations just computed and the fact that π^* is a ring map determine multiplication; all products of total codimension greater than two vanish. $\square$

Exercise A.6.6

Permitted context.

For a smooth projective variety, Chern classes multiply in an exact sequence of vector bundles. The self-intersection formula for a regular immersion $i : Y \rightarrow X$ of codimension r is $i^*i_*1 = c_r(N_{Y/X})$. Hirzebruch–Riemann–Roch says $\chi(\mathcal{F}) = \deg(\text{ch}(\mathcal{F}) \text{td}(T_X))_{\dim X}$. The degree-at-most-three Todd expansion is

$$\text{td}(T_X) = 1 + \frac{c_1}{2} + \frac{c_1^2 + c_2}{12} + \frac{c_1c_2}{24},$$

$$\text{and } \text{ch}(\mathcal{O}(D)) = \exp(D).$$

Problem. Identify the self-intersection of the diagonal of a smooth projective n -fold with $c_n(T_X)$.

Solution. For $i : X \cong \Delta \hookrightarrow X \times X$, restriction of the ambient tangent bundle is $T_X \oplus T_X$. The tangent map is $v \mapsto (v, v)$, whose quotient is canonically T_X by $(v, w) \mapsto w - v$. Thus $N_{\Delta/(X \times X)} \cong T_X$. Self-intersection gives $i^*i_*1 = c_n(T_X)$ in $A^n(X)$. Equivalently in the ambient Chow ring, $[\Delta]^2 = i_*c_n(T_X)$. The latter formula makes explicit which group contains the ordinary product of the two diagonal classes. $\square$

Exercise A.6.7

Permitted context.

For a smooth projective variety, Chern classes multiply in an exact sequence of vector bundles. The self-intersection formula for a regular immersion $i : Y \rightarrow X$ of codimension r is $i^*i_*1 = c_r(N_{Y/X})$. Hirzebruch–Riemann–Roch says $\chi(\mathcal{F}) = \deg(\text{ch}(\mathcal{F}) \text{td}(T_X))_{\dim X}$. The degree-at-most-three Todd expansion is

$$\text{td}(T_X) = 1 + \frac{c_1}{2} + \frac{c_1^2 + c_2}{12} + \frac{c_1c_2}{24},$$

$$\text{and } \text{ch}(\mathcal{O}(D)) = \exp(D).$$

Write K for the canonical divisor, so $c_1(T_X) = -K$. For a threefold, $\chi(\mathcal{O}_X) = 1 - p_a(X)$. Products of classes of total degree three below denote their degrees.

Problem. For a smooth projective threefold, prove

$$1 - p_a = \frac{c_1c_2}{24}, \quad \chi(\mathcal{O}(D)) = \frac{D(D-K)(2D-K)}{12} + \frac{Dc_2}{12} + 1 - p_a.$$

Solution. Apply Riemann–Roch to the structure sheaf: its character is one, so only the degree-three Todd term contributes, giving $\chi(\mathcal{O}_X) = c_1 c_2 / 24 = 1 - p_a$. For $\mathcal{O}(D)$, multiply the Todd expansion by $1 + D + D^2/2 + D^3/6$. The top-degree term is

$$\frac{D^3}{6} + \frac{D^2 c_1}{4} + \frac{D(c_1^2 + c_2)}{12} + \frac{c_1 c_2}{24}.$$

Substituting $c_1 = -K$ and the first identity gives the formula, since $2D^3 - 3D^2 K + DK^2 = D(D - K)(2D - K)$. $\square$

Exercise A.6.8

Permitted context.

For a smooth projective variety, Chern classes multiply in an exact sequence of vector bundles. The self-intersection formula for a regular immersion $i : Y \rightarrow X$ of codimension r is $i^* i_* 1 = c_r(N_{Y/X})$. Hirzebruch–Riemann–Roch says $\chi(\mathcal{F}) = \deg(\text{ch}(\mathcal{F}) \text{td}(T_X))_{\dim X}$. The degree-at-most-three Todd expansion is

$$\text{td}(T_X) = 1 + \frac{c_1}{2} + \frac{c_1^2 + c_2}{12} + \frac{c_1 c_2}{24},$$

and $\text{ch}(\mathcal{O}(D)) = \exp(D)$.

On $\mathbb{P}^3$, $A(\mathbb{P}^3) = \mathbb{Z}[h]/(h^4)$, $\deg h^3 = 1$, and the Euler sequence gives $\text{td}(T_{\mathbb{P}^3}) = 1 + 2h + 11h^2/6 + h^3$. For a rank-two bundle with $c_1 = ah$, $c_2 = bh^2$,

$$\text{ch}(\mathcal{E}) = 2 + ah + (a^2 - 2b)h^2/2 + (a^3 - 3ab)h^3/6.$$

Problem. Prove that the two integer Chern numbers of a rank-two bundle on $\mathbb{P}^3$ satisfy $ab \equiv 0 \pmod{2}$.

Solution. The coefficient of h^3 in the product gives

$$\chi(\mathcal{E}) = 2 + \frac{11a}{6} + a^2 - 2b + \frac{a^3 - 3ab}{6} = 1 + \binom{a+3}{3} - 2b - \frac{ab}{2}.$$

The Euler characteristic and the binomial expression are integers for every integer a , including negative ones. Therefore $ab/2$ is an integer, proving the parity assertion. $\square$

Exercise A.6.9**Permitted context.**

For a smooth projective variety, Chern classes multiply in an exact sequence of vector bundles. The self-intersection formula for a regular immersion $i : Y \rightarrow X$ of codimension r is $i^*i_*1 = c_r(N_{Y/X})$. Hirzebruch–Riemann–Roch says $\chi(\mathcal{F}) = \deg(\text{ch}(\mathcal{F}) \text{td}(T_X))_{\dim X}$. The degree-at-most-three Todd expansion is

$$\text{td}(T_X) = 1 + \frac{c_1}{2} + \frac{c_1^2 + c_2}{12} + \frac{c_1 c_2}{24},$$

and $\text{ch}(\mathcal{O}(D)) = \exp(D)$.

For a smooth surface in $\mathbb{P}^4$, with $d = H^2$, the normal sequence and self-intersection give

$$d^2 - 10d - 5HK - 2K^2 + 12\chi(\mathcal{O}_X) = 0.$$

On the rational ruled surface $\mathbb{F}_e$, $C_0^2 = -e$, $C_0F = 1$, $F^2 = 0$, $K = -2C_0 - (e + 2)F$, $K^2 = 8$, $\chi = 1$. A divisor $aC_0 + bF$ is very ample exactly when $a \geq 1$, $b \geq ae + 1$. The quadric realizes $\mathbb{F}_0$ in $\mathbb{P}^3$; $|C_0 + 2F|$ realizes $\mathbb{F}_1$ as the cubic scroll in $\mathbb{P}^4$. A K3 surface has $K = 0$, $\chi = 2$; an abelian surface has $K = 0$, $\chi = 0$.

Problem. (a) Check the surface formula on the rational cubic scroll.

(b) Determine the possible degrees of a K3 surface embedded in $\mathbb{P}^4$.

(c) Determine the degree of an abelian surface embedded in $\mathbb{P}^4$.

(d) Determine which $\mathbb{F}_e$, $e \geq 0$, embed in $\mathbb{P}^4$.

Solution. (a) On $\mathbb{F}_1$, take $H = C_0 + 2F$. Then $d = 3$, $HK = -5$, $K^2 = 8$, $\chi = 1$, so the left side is $9 - 30 + 25 - 16 + 12 = 0$.

(b) The formula becomes $d^2 - 10d + 24 = 0$, hence $d = 4$ or 6 . These possibilities are realized by a smooth quartic in a linear $\mathbb{P}^3$, and a smooth $(2, 3)$ complete intersection in $\mathbb{P}^4$, respectively.

(c) It becomes $d^2 - 10d = 0$. Since $d > 0$, it gives $d = 10$.

(d) Let the hyperplane divisor of an embedding be $H = aC_0 + bF$. Put $t = 2b - ae$. Very ampleness gives $a \geq 1$, $t \geq ae + 2 \geq 2$. Then $d = at$, $HK = -t - 2a$, and the formula is

$$a^2t^2 - 10at + 5t + 10a - 4 = 0.$$

For $a \geq 2$, this polynomial increases for $t \geq 2$, since its derivative there is at least $4a^2 - 10a + 5 > 0$. At $t = 2$ its value is $2(2a - 3)(a - 1) > 0$, a contradiction. Thus $a = 1$, and the equation is $(t - 2)(t - 3) = 0$. If $t = 2$, $2b - e = 2$ and $b \geq e + 1$ imply $e = 0$, $b = 1$. If $t = 3$, the same inequalities and parity imply $e = 1$, $b = 2$. Both occur by the stated embeddings, with the quadric included in a hyperplane of $\mathbb{P}^4$. Thus exactly $\mathbb{F}_0, \mathbb{F}_1$ admit such embeddings. $\square$

Exercise A.6.10**Permitted context.**

For a smooth projective variety, Chern classes multiply in an exact sequence of vector bundles. The self-intersection formula for a regular immersion $i : Y \rightarrow X$ of codimension r is $i^*i_*1 = c_r(N_{Y/X})$. Hirzebruch–Riemann–Roch says $\chi(\mathcal{F}) = \deg(\text{ch}(\mathcal{F}) \text{td}(T_X))_{\dim X}$. The degree-at-most-three Todd expansion is

$$\text{td}(T_X) = 1 + \frac{c_1}{2} + \frac{c_1^2 + c_2}{12} + \frac{c_1c_2}{24},$$

and $\text{ch}(\mathcal{O}(D)) = \exp(D)$.

The tangent bundle of an abelian variety is trivial. For a smooth r -fold in $\mathbb{P}^n$, its normal bundle has rank $n - r$. Chern classes above the rank vanish. The Euler sequence gives $c(T_{\mathbb{P}^n}) = (1 + h)^{n+1}$.

Problem. Prove that an abelian threefold cannot be embedded in $\mathbb{P}^5$.

Solution. If it were embedded, its normal bundle N would have rank two. The tangent-normal exact sequence and triviality of the tangent bundle of X would give

$$c(N) = c(T_{\mathbb{P}^5}|_X) = (1 + H)^6.$$

The degree-three term would be $c_3(N) = 20H^3$. But rank two forces $c_3(N) = 0$. Taking degrees yields $20 \deg X = 0$ in $\mathbb{Z}$, impossible since an embedded projective threefold has positive degree. This is an integer Chow-group argument, so it does not disappear in characteristics two or five. $\square$

Appendix B

B.6 Appendix B Exercises

Exercise B.6.1

Permitted context.

All algebraic schemes are of finite type over $\mathbb{C}$. Analytification preserves and detects reducedness, smoothness and connectedness; algebraic separatedness is equivalent to analytic Hausdorffness, and algebraic properness to analytic properness. The analytic local rings are faithfully flat over the corresponding algebraic local rings; analytification is faithful and exact on coherent sheaves. A smooth separated algebraic curve is its smooth projective completion minus finitely many points.

A bounded holomorphic function on a punctured disc extends across the puncture. Holomorphic functions on a compact connected Riemann surface are constant.

Problem. Show that the open unit disc is not the analytification of an algebraic scheme.

Solution. If X_h were the unit disc, the comparison facts would make X a smooth connected separated curve. Its smooth projective completion $\overline{X}$ adds finitely many points. The coordinate of the disc pulls back to a bounded nonconstant holomorphic function on X_h . Near every added point, it is a bounded function on a punctured coordinate disc, so extends holomorphically. It would then be a nonconstant holomorphic function on the compact connected Riemann surface $\overline{X}_h$, a contradiction. The same contradiction applies if there were no added points. $\square$

Exercise B.6.2

Permitted context.

All algebraic schemes are of finite type over $\mathbb{C}$. Analytification preserves and detects reducedness, smoothness and connectedness; algebraic separatedness is equivalent to analytic Hausdorffness, and algebraic properness to analytic properness. The analytic local rings are faithfully flat over the corresponding algebraic local rings; analytification is faithful and exact on coherent sheaves. A smooth separated algebraic curve is its smooth projective completion minus finitely many points.

Weierstrass's product theorem supplies an entire function with any prescribed discrete zero set and positive integral multiplicities. Every ideal in $\mathbb{C}[t]$ is principal.

Problem. Let $|z_n| \rightarrow \infty$, and let $\mathcal{J} \subseteq \mathcal{O}_{\mathbb{C}}$ be the sheaf of holomorphic functions vanishing at these points. Show that it is not the analytification of an algebraic ideal in $\mathcal{O}_{\mathbb{A}^1}$, but is abstractly isomorphic to the analytification of a coherent algebraic sheaf.

Solution. Use the set of distinct z_n ; it is infinite and discrete, regardless of any repetitions in the sequence. A nonzero algebraic ideal is generated by a nonzero polynomial, whose zero set is finite, so its analytic ideal cannot equal $\mathcal{J}$. The zero ideal cannot work either, since $\mathcal{J}$ is nonzero. Choose an entire function g having simple zeros exactly at the specified set. Locally the vanishing ideal is (g) : at a zero both are generated by a local parameter; elsewhere g is a unit. Multiplication by g therefore gives an isomorphism $\mathcal{O}_{\mathbb{C}} \cong \mathcal{J}$. Taking the algebraic sheaf $\mathcal{F} = \mathcal{O}_{\mathbb{A}^1}$ gives $\mathcal{F}_h \cong \mathcal{J}$ as sheaves. This isomorphism does not identify their specified inclusions as algebraic ideals. $\square$

Exercise B.6.3**Permitted context.**

All algebraic schemes are of finite type over $\mathbb{C}$. Analytification preserves and detects reducedness, smoothness and connectedness; algebraic separatedness is equivalent to analytic Hausdorffness, and algebraic properness to analytic properness. The analytic local rings are faithfully flat over the corresponding algebraic local rings; analytification is faithful and exact on coherent sheaves. A smooth separated algebraic curve is its smooth projective completion minus finitely many points.

The punctured algebraic plane $U = \mathbb{A}^2 \setminus \{0\}$ has $\text{Pic } U = 0$: removing codimension two preserves the divisor class group of the normal factorial affine plane, and on a smooth variety Picard and divisor class groups agree. A nowhere-zero holomorphic function on $\mathbb{C}^* \times \mathbb{C}$ has a holomorphic logarithm exactly when its winding number around the $\mathbb{C}^*$ factor is zero. Cauchy's integral theorem applies to each variable separately.

Problem. On $\mathbb{C}^2 \setminus \{0\}$, glue trivial analytic line bundles on $U_1 = \{z \neq 0\}$, $U_2 = \{w \neq 0\}$ by $\exp(-1/(zw))$ on the overlap. Prove that the resulting bundle is not algebraizable.

Solution. Any coherent algebraization of the analytic line bundle would itself be invertible: faithful flatness of analytic local rings descends local freeness of rank one. Such an algebraic line bundle would be trivial by $\text{Pic } U = 0$. Suppose the analytic bundle were trivial. Its transition would be a quotient a/b , with a a holomorphic unit on U_1 and b a holomorphic unit on U_2 . Its winding along each coordinate circle is zero, since it is the exponential of the single-valued function $-1/(zw)$. Along a z -circle, b has winding zero because it is a unit on a full z -disc; hence a has winding zero too. Similarly b has zero winding in its w -direction. The logarithm criterion gives $a = e^A, b = e^B$ with A, B holomorphic on their respective opens. Connectedness of the overlap then gives

$$-\frac{1}{zw} = A - B + 2\pi in$$

for a fixed integer n . Integrate against $dz dw$ over two coordinate circles. The left side has integral $-(2\pi i)^2$. The A -term integrates to zero first in w , since A is holomorphic across $w = 0$; the B -term integrates to zero first in z ; the constant term also gives zero. This contradiction proves nontriviality and hence nonalgebraizability. $\square$

Exercise B.6.4**Permitted context.**

All algebraic schemes are of finite type over $\mathbb{C}$. Analytification preserves and detects reducedness, smoothness and connectedness; algebraic separatedness is equivalent to analytic Hausdorffness, and algebraic properness to analytic properness. The analytic local rings are faithfully flat over the corresponding algebraic local rings; analytification is faithful and exact on coherent sheaves. A smooth separated algebraic curve is its smooth projective completion minus finitely many points.

Holomorphic functions on a reduced compact connected complex analytic space are constant (apply the maximum principle on normalizations of its irreducible components). A separated finite-type scheme has a proper compactification. In an integral variety, an affine neighborhood of a boundary point can be cut by hypersurfaces to obtain a curve through that point not contained in a prescribed proper closed subset. A nonproper separated integral algebraic curve is affine. Projective GAGA identifies global sections of coherent sheaves.

Problem. Prove that for reduced proper X , algebraic and analytic global regular functions agree. Prove the converse detection statement for separated X : if X is not proper, some coherent $\mathcal{F}$ has a nonsurjective comparison map on global sections. Explain why separatedness is necessary in the converse.

Solution. If X is reduced and proper, both algebraic and analytic functions are constant on each connected component. Algebraically, properness forces the image of a morphism to the affine line to be finite; reducedness identifies the ring with these componentwise constants. Analytically use compactness and the maximum principle. The connected components correspond under analytification and are finite in number. Both rings are thus the same finite product of copies of $\mathbb{C}$, with the comparison map its identity.

Now let X be separated but not proper. Choose a proper compactification $j : X \hookrightarrow \overline{X}$, and let D be a reduced irreducible component of the closure of X_{red} meeting $\overline{X} \setminus X$. Select a boundary point of D . Curve selection inside D , followed by closure in $\overline{X}$, gives a proper integral curve $\overline{C}$ through that point meeting X but not contained in it. Its intersection $C = \overline{C} \cap X$ is closed in X , nonempty, and a proper open of $\overline{C}$; it is a nonproper affine integral curve. Choose a nonconstant $h \in \Gamma(C, \mathcal{O}_C)$. The analytic function e^h is not algebraic regular: on the smooth projective model of C , the rational function h has a pole somewhere outside C , and e^h has an essential singularity there. An algebraic function would be meromorphic. For the closed inclusion $i : C \hookrightarrow X$, take $\mathcal{F} = i_* \mathcal{O}_C$, a coherent sheaf. Its analytic global sections contain e^h , absent from the image of its algebraic global sections, proving the claim.

Without separatedness the converse as stated is false. Glue two copies of $\mathbb{P}^1$ along their identical complements of one point, leaving that point doubled. The resulting finite-type scheme is not separated, hence not proper. For any coherent sheaf, sections on each of the two projective opens algebraize by GAGA. An analytic global section gives two such algebraic sections agreeing analytically on the overlap; faithfulness makes them agree algebraically, so they glue. Thus every coherent global-section comparison is an isomorphism on this nonproper scheme. $\square$

Exercise B.6.5**Permitted context.**

All algebraic schemes are of finite type over $\mathbb{C}$. Analytification preserves and detects reducedness, smoothness and connectedness; algebraic separatedness is equivalent to analytic Hausdorffness, and algebraic properness to analytic properness. The analytic local rings are faithfully flat over the corresponding algebraic local rings; analytification is faithful and exact on coherent sheaves. A smooth separated algebraic curve is its smooth projective completion minus finitely many points.

Chow's theorem makes a closed analytic subspace of complex projective space algebraic. A finite birational map onto a normal curve is an isomorphism. Proper maps of algebraic curves with finite fibers are finite. A holomorphic map to a coordinate disc extends across a puncture if its coordinate is bounded.

Problem. If two nonsingular affine algebraic curves have isomorphic analytic spaces, prove that the algebraic curves are isomorphic.

Solution. Complete the curves to smooth projective curves $\overline{X}, \overline{X}'$, adding finite nonempty sets S, S' . Let $f : X_h \rightarrow X'_h$ be the biholomorphism. At $p \in S$, its cluster set in compact $\overline{X}'_h$ is nonempty and connected: it is the intersection of the nested compact connected closures of images of punctured small discs. It contains no point of X'_h , since convergence to such a point and continuity of f^{-1} would force convergence inside X_h . It is therefore a connected subset of finite S' , hence a single point p' . Thus f extends continuously at p . In a coordinate disc at p' , the extended map is holomorphic by the removable-singularity theorem. Apply the same argument to f^{-1} . The extensions are inverse biholomorphisms of the compactifications and identify their missing sets.

The graph of this biholomorphism is a closed analytic subspace of the projective product. By Chow it is algebraic. Each projection of its reduced algebraic graph is a proper bijective map of smooth curves, generically of degree one in characteristic zero; thus it is finite birational, hence an isomorphism by normality. The resulting algebraic isomorphism restricts to $X \cong X'$, since it matches S, S' . $\square$

Exercise B.6.6**Permitted context.**

All algebraic schemes are of finite type over $\mathbb{C}$. Analytification preserves and detects reducedness, smoothness and connectedness; algebraic separatedness is equivalent to analytic Hausdorffness, and algebraic properness to analytic properness. The analytic local rings are faithfully flat over the corresponding algebraic local rings; analytification is faithful and exact on coherent sheaves. A smooth separated algebraic curve is its smooth projective completion minus finitely many points.

Projective GAGA gives an equivalence of coherent algebraic and analytic sheaves, compatible with global sections and morphisms. Faithful flatness detects invertibility and surjectivity of coherent maps. A line bundle with $n+1$ generating sections defines a morphism to $\mathbb{P}^n$, uniquely up to isomorphism of that data.

Problem. Prove that an analytic morphism $f : X_h \rightarrow Y_h$ between analytifications of projective schemes has a unique algebraic origin, allowing nonreduced schemes.

Solution. Embed Y in $\mathbb{P}^n$. The composite analytic map pulls back $\mathcal{O}(1)$ with its $n+1$ standard sections. GAGA algebraizes this coherent sheaf as $\mathcal{L}$ on X and the sections as $s_0, \dots, s_n$. Faithful flatness at closed points makes $\mathcal{L}$ invertible and the evaluation map $\mathcal{O}_X^{n+1} \rightarrow \mathcal{L}$ surjective: its analytic cokernel is zero, so the algebraic cokernel is zero. The resulting morphism $g : X \rightarrow \mathbb{P}^n$ has analytification equal to the composite with f .

Let $\mathcal{I}_Y$ be the homogeneous ideal sheaf of Y . The map $g^*\mathcal{I}_Y \rightarrow \mathcal{O}_X$ analytifies to zero because the analytic morphism factors through Y_h . Faithfulness forces the original map to be zero as well. Thus g factors through the closed subscheme Y , giving the desired algebraic morphism, including its nilpotent structure. For uniqueness, an analytic identification of the pulled-back line bundles and generating sections algebraizes uniquely by full faithfulness of GAGA. The associated maps to $\mathbb{P}^n$ coincide; their factorizations through the fixed closed immersion of Y coincide as well. $\square$

Appendix C

C.5 Appendix C Exercises

Exercise C.5.1

Permitted context.

For a scheme of finite type over $\mathbb{F}_q$, put $N_r(X) = \#X(\mathbb{F}_{q^r})$ and

$$Z(X, t) = \exp \left(\sum_{r \geq 1} N_r(X) t^r / r \right) \in 1 + t\mathbb{Q}[[t]].$$

Formal logarithm and exponential are inverse over $\mathbb{Q}$, and $-\log(1 - at) = \sum_{r \geq 1} a^r t^r / r$.

Problem. For a disjoint decomposition into locally closed $\mathbb{F}_q$ -subschemes X_i , prove $Z(X, t) = \prod_i Z(X_i, t)$.

Solution. Every $\mathbb{F}_{q^r}$ -point lies in exactly one stratum, so $N_r(X) = \sum_i N_r(X_i)$. For a finite decomposition, substitute this in the defining logarithm and exponentiate to obtain the product. The same proof allows an infinite decomposition: for each bounded range of r , only finitely many strata can contribute points, since each $X(\mathbb{F}_{q^r})$ is finite. Thus the sums and product are well defined coefficient by coefficient in the t -adic topology. $\square$

Exercise C.5.2

Permitted context.

For a scheme of finite type over $\mathbb{F}_q$, put $N_r(X) = \#X(\mathbb{F}_{q^r})$ and

$$Z(X, t) = \exp \left(\sum_{r \geq 1} N_r(X) t^r / r \right) \in 1 + t\mathbb{Q}[[t]].$$

Formal logarithm and exponential are inverse over $\mathbb{Q}$, and $-\log(1 - at) = \sum_{r \geq 1} a^r t^r / r$.

For smooth projective dimension n , the Weil properties ask for rationality, a functional equation $Z(1/(q^n t)) = \pm q^{nE/2} t^E Z(t)$, and factors P_i with reciprocal roots of absolute value $q^{i/2}$. Their degrees should equal the complex Betti numbers for a smooth lift, and E is the top tangent Chern number. The Euler sequence gives $c(T_{\mathbb{P}^n}) = (1 + h)^{n+1}$. Complex projective n -space has one cell in each even real dimension $0, 2, \dots, 2n$.

Problem. Compute $Z(\mathbb{P}^n, t)$ and verify all the Weil properties for projective space.

Solution. Counting nonzero vectors modulo nonzero scalar gives

$$N_r = 1 + q^r + \dots + q^{nr}.$$

The defining logarithm therefore yields

$$Z(\mathbb{P}^n, t) = \prod_{j=0}^n (1 - q^j t)^{-1},$$

a rational function with integer factors. The top tangent Chern number is $E = n + 1$. Direct substitution gives

$$Z(\mathbb{P}^n, 1/(q^n t)) = (-1)^{n+1} q^{n(n+1)/2} t^{n+1} Z(\mathbb{P}^n, t),$$

the required equation. Take $P_{2j} = 1 - q^j t$ and $P_{2j+1} = 1$. Each reciprocal root has absolute value $q^j = q^{(2j)/2}$; there are no odd-degree roots. The factor degrees give one Betti number in each even degree and zero in odd degrees, matching the cell decomposition of $\mathbb{P}^n(\mathbb{C})$. Their alternating sum is $n + 1 = E$. $\square$

Exercise C.5.3

Permitted context.

For a scheme of finite type over $\mathbb{F}_q$, put $N_r(X) = \#X(\mathbb{F}_{q^r})$ and

$$Z(X, t) = \exp \left(\sum_{r \geq 1} N_r(X) t^r / r \right) \in 1 + t\mathbb{Q}[[t]].$$

Formal logarithm and exponential are inverse over $\mathbb{Q}$, and $-\log(1 - at) = \sum_{r \geq 1} a^r t^r / r$.

Problem. Prove $Z(X \times \mathbb{A}^1, t) = Z(X, qt)$.

Solution. For every r , the affine line has q^r rational points, so $N_r(X \times \mathbb{A}^1) = q^r N_r(X)$. The logarithm of the left side is $\sum N_r(X)(qt)^r / r$, exactly that of the right side. Exponentiating proves the identity. $\square$

Exercise C.5.4

Permitted context.

For a scheme of finite type over $\mathbb{F}_q$, put $N_r(X) = \#X(\mathbb{F}_{q^r})$ and

$$Z(X, t) = \exp \left(\sum_{r \geq 1} N_r(X) t^r / r \right) \in 1 + t\mathbb{Q}[[t]].$$

Formal logarithm and exponential are inverse over $\mathbb{Q}$, and $-\log(1 - at) = \sum_{r \geq 1} a^r t^r / r$.

A closed point of degree d over $\mathbb{F}_q$ has residue field $\mathbb{F}_{q^d}$ and contributes d points over $\mathbb{F}_{q^r}$ if $d \mid r$, and none otherwise. The arithmetic Euler product is $\zeta_X(s) = \prod_{x \text{ closed}} (1 - \#k(x)^{-s})^{-1}$.

Problem. For X of finite type over $\mathbb{F}_q$, prove $\zeta_X(s) = Z(X, q^{-s})$, interpreting the equality first as a formal Euler product and then where it converges.

Solution. Let b_d be the number of closed points of degree d . It is finite, and $N_r = \sum_{d|r} db_d$. Consequently

$$\log \prod_{d \geq 1} (1 - t^d)^{-b_d} = \sum_{d, m \geq 1} b_d t^{dm} / m = \sum_{r \geq 1} N_r t^r / r.$$

Each coefficient uses finitely many divisors, so the formal rearrangement is valid. Exponentiating gives the closed-point Euler product for $Z(X, t)$. Substitution $t = q^{-s}$ gives $\#k(x)^{-s} = t^{\deg x}$ at each factor, hence the desired identity. It is also an analytic identity for sufficiently large $\operatorname{Re} s$: finite affine covers and embeddings bound N_r by Cq^{Mr} , ensuring absolute convergence when $|t| < q^{-M}$. $\square$

Exercise C.5.5

Permitted context.

For a scheme of finite type over $\mathbb{F}_q$, put $N_r(X) = \#X(\mathbb{F}_{q^r})$ and

$$Z(X, t) = \exp \left(\sum_{r \geq 1} N_r(X) t^r / r \right) \in 1 + t\mathbb{Q}[[t]].$$

Formal logarithm and exponential are inverse over $\mathbb{Q}$, and $-\log(1 - at) = \sum_{r \geq 1} a^r t^r / r$.

Let C be smooth, projective, geometrically connected of genus g . Assume its Weil properties:

$$Z(C, t) = \frac{P(t)}{(1-t)(1-qt)}, \quad P(t) = \prod_{i=1}^{2g} (1 - \alpha_i t), \quad P(t) = q^g t^{2g} P(1/(qt)).$$

In particular $P(0) = 1$.

Problem. Prove that $N_1, \dots, N_g$, together with q, g , determine every N_r .

Solution. Taking logarithms gives $s_r := \sum_i \alpha_i^r = 1 + q^r - N_r$. Write $P(t) = \sum_{j=0}^{2g} c_j t^j$, $c_0 = 1$. Logarithmic differentiation, or Newton's identities, gives

$$j c_j = - \sum_{i=1}^j s_i c_{j-i}.$$

Thus $s_1, \dots, s_g$ determine $c_1, \dots, c_g$ recursively over $\mathbb{Q}$. Reciprocity gives $c_{2g-j} = q^{g-j} c_j$ for $0 \leq j \leq g$, determining every remaining coefficient. The rational function $Z(C, t)$ is now determined, and its formal logarithm recovers every N_r . If $g = 0$, no initial counts are needed: $P = 1$, so $N_r = 1 + q^r$. $\square$

Exercise C.5.6**Permitted context.**

For a scheme of finite type over $\mathbb{F}_q$, put $N_r(X) = \#X(\mathbb{F}_{q^r})$ and

$$Z(X, t) = \exp \left(\sum_{r \geq 1} N_r(X) t^r / r \right) \in 1 + t\mathbb{Q}[[t]].$$

Formal logarithm and exponential are inverse over $\mathbb{Q}$, and $-\log(1 - at) = \sum_{r \geq 1} a^r t^r / r$.

For an elliptic curve $E/\mathbb{F}_q$, Frobenius F and its dual satisfy $F\bar{F} = [q]$ and $F + \bar{F} = [a]$ for an integer a , with $|a| \leq 2\sqrt{q}$. The degree formula is $\deg(1 - F^r) = 1 + q^r - (F^r + \bar{F}^r)$, where the last sum is identified with its integer scalar. The isogeny $1 - F^r$ is separable because its tangent map is the identity. Its degree equals its number of geometric kernel points.

Problem. Use these elliptic-curve results to compute the zeta function and verify the Weil properties for E . Show explicitly that the Hasse bound is equivalent to the required absolute values of the two reciprocal roots.

Solution. The kernel of $1 - F^r$ is $E(\mathbb{F}_{q^r})$, so its degree is N_r . Let α, β be the complex roots of $T^2 - aT + q$. The identities for Frobenius imply the same recurrence for $F^r + \bar{F}^r$ as for $\alpha^r + \beta^r$, with initial values $2, a$. Therefore

$$N_r = 1 + q^r - \alpha^r - \beta^r, \quad Z(E, t) = \frac{(1 - \alpha t)(1 - \beta t)}{(1 - t)(1 - qt)} = \frac{1 - at + qt^2}{(1 - t)(1 - qt)}.$$

This proves rationality and integrality. Substitution verifies $Z(E, 1/(qt)) = Z(E, t)$, the functional equation with Euler number zero. The inequality $|a| \leq 2\sqrt{q}$ says the real-coefficient quadratic has either conjugate nonreal roots or a repeated real root. Since their product is q , both have modulus $\sqrt{q}$. Conversely these two moduli imply $|a| = |\alpha + \beta| \leq 2\sqrt{q}$. Thus the Hasse bound gives exactly the Riemann-hypothesis condition. Finally the degrees of the factors are $1, 2, 1$, the Betti numbers of a complex genus-one curve; their alternating sum is zero. $\square$

Exercise C.5.7**Permitted context.**

For a scheme of finite type over $\mathbb{F}_q$, put $N_r(X) = \#X(\mathbb{F}_{q^r})$ and

$$Z(X, t) = \exp \left(\sum_{r \geq 1} N_r(X) t^r / r \right) \in 1 + t\mathbb{Q}[[t]].$$

Formal logarithm and exponential are inverse over $\mathbb{Q}$, and $-\log(1 - at) = \sum_{r \geq 1} a^r t^r / r$.

For a smooth projective geometrically connected genus- g curve, assume

$$Z(C, t) = \frac{\prod_{i=1}^{2g} (1 - \alpha_i t)}{(1 - t)(1 - qt)}, \quad N_r = 1 - a_r + q^r, \quad |a_r| \leq 2gq^{r/2}.$$

The functional equation makes the multiset of nonzero roots $\{\alpha_i\}$ invariant under $\alpha \mapsto q/\alpha$.

Problem. (a) Show $a_r = \sum_i \alpha_i^r$.

(b) Prove that the bounds $|a_r| \leq 2gq^{r/2}$ for all r are equivalent to $|\alpha_i| \leq \sqrt{q}$ for all i .

(c) Use the functional equation to obtain equality of every modulus.

Solution. (a) Expand the formal logarithms of the numerator and denominator and compare the coefficient of t^r/r . This gives $N_r = 1 + q^r - \sum_i \alpha_i^r$, as required.

(b) If all root moduli are at most $\sqrt{q}$, the triangle inequality gives the bound immediately. Conversely the assumed bounds make

$$A(t) = \sum_{r \geq 1} a_r t^{r-1}$$

holomorphic for $|t| < q^{-1/2}$. Near zero, part (a) identifies it with the rational function

$$\sum_{i=1}^{2g} \frac{\alpha_i}{1 - \alpha_i t}.$$

If some $|\alpha_i| > \sqrt{q}$, this function has a pole inside that disk. Terms with the same α_i add a positive integer multiple of the same nonzero pole; different values have different poles, so cancellation is impossible. This contradicts the holomorphic power series, by uniqueness of analytic continuation. All moduli are therefore at most $\sqrt{q}$.

(c) The paired root q/α_i also satisfies that upper bound. It gives $|\alpha_i| \geq \sqrt{q}$, hence equality. For genus zero the multiset is empty and the conclusion is vacuous. $\square$

Chapter I

I.1 Affine Varieties

Exercise I.1.1

Permitted context.

Let k be algebraically closed, of any characteristic. For an affine algebraic set W , its coordinate ring is $A(W) = k[x, y]/I(W)$. The Nullstellensatz identifies $I(V(f))$ with $\sqrt{(f)}$; an irreducible polynomial in $k[x, y]$ generates a prime ideal. Changes of affine coordinates induce isomorphisms of coordinate rings. A homogeneous binary quadratic over k is a product of two linear forms, possibly proportional.

Problem. (a) For $Y = V(y - x^2)$, show that $A(Y) \cong k[t]$.

(b) For $Z = V(xy - 1)$, prove that $A(Z)$ is not isomorphic as a k -algebra to $k[t]$.

(c) Classify $A(V(f))$ for every irreducible polynomial f of total degree two in $k[x, y]$: show it is one of the two rings in (a),(b), and give a criterion deciding which.

Solution. (a) Substitution $x \mapsto t$, $y \mapsto t^2$ is a surjective homomorphism with kernel $(y - x^2)$, as division by the monic linear polynomial in y shows. Its quotient is $k[t]$, a domain, so the ideal is prime and is the vanishing ideal of Y . Hence $A(Y) \cong k[t]$.

(b) There are inverse homomorphisms

$$k[x, y]/(xy - 1) \longleftrightarrow k[t, t^{-1}], \quad x \mapsto t, \quad y \mapsto t^{-1}.$$

Thus this quotient is a domain and equals $A(Z)$. It has the nonconstant unit x , with inverse y . In $k[t]$, degrees add on nonzero products, so every unit is a nonzero constant. A k -algebra isomorphism fixes k and preserves units, giving the contradiction.

(c) Write $f = f_2 + f_1 + f_0$ by total degree. If the two factors of f_2 are linearly independent, choose linear coordinates and scale f to obtain

$$f = xy + ax + by + c = (x + b)(y + a) + (c - ab).$$

Irreducibility forces $c - ab \neq 0$. Translating and then scaling one coordinate gives the equation $uv = 1$, so $A(V(f)) \cong A(Z)$.

If those factors are proportional, linear coordinates and a scalar put f in the form $x^2 + ax + by + c$. Here $b \neq 0$: otherwise this quadratic in x splits over k and is reducible. Solving for y identifies the quotient with $k[x]$, so $A(V(f)) \cong A(Y)$. This criterion uses only the factorization of the leading homogeneous part: a repeated factor gives $A(Y)$, and two distinct factors give $A(Z)$. The argument uses no division by two, and applies also in characteristic two. $\square$

Exercise I.1.2**Permitted context.**

Let k be an algebraically closed field. For an algebraic set $X \subseteq \mathbb{A}_k^n$, write $I(X)$ for its vanishing ideal and $A(X) = k[x_1, \dots, x_n]/I(X)$ for its affine coordinate ring. One may use that an affine algebraic set is irreducible exactly when its coordinate ring is a domain, and that $\dim k[t] = 1$.

Problem. Let

$$Y = \{(t, t^2, t^3) : t \in k\} \subseteq \mathbb{A}_k^3.$$

Prove that Y is a one-dimensional affine variety, determine generators for $I(Y)$, and identify $A(Y)$ with a polynomial ring in one variable.

Solution. Consider the homomorphism

$$\phi : k[x, y, z] \longrightarrow k[t], \quad x \longmapsto t, \quad y \longmapsto t^2, \quad z \longmapsto t^3.$$

It is surjective because its image contains t . The polynomials $y - x^2$ and $z - x^3$ lie in its kernel. Conversely,

$$k[x, y, z]/(y - x^2, z - x^3) \cong k[x],$$

and under this isomorphism the map induced by ϕ is $k[x] \rightarrow k[t]$, $x \mapsto t$, which is an isomorphism. Hence

$$\ker \phi = (y - x^2, z - x^3).$$

Since k is infinite, a polynomial $f \in k[x, y, z]$ vanishes at every point of Y if and only if the one-variable polynomial $\phi(f) = f(t, t^2, t^3)$ vanishes at every $t \in k$, which is equivalent to $\phi(f) = 0$. Hence $I(Y) = \ker \phi$.

The common zero set of these two generators consists precisely of the points (t, t^2, t^3) , so $Y = V(y - x^2, z - x^3)$ is closed. Its coordinate ring is

$$A(Y) \cong k[x, y, z]/(y - x^2, z - x^3) \cong k[t].$$

This ring is a domain, so Y is irreducible, and its Krull dimension is 1. Thus Y is a one-dimensional affine variety and the displayed two polynomials generate $I(Y)$. $\square$

Exercise I.1.3**Permitted context.**

Let k be algebraically closed. Closed irreducible subsets of affine space correspond to prime vanishing ideals. A finite union of irreducible closed subsets, none contained in another, is the irreducible-component decomposition.

Problem. In $\mathbb{A}_k^3$ with coordinates x, y, z , let

$$Y = V(x^2 - yz, xz - x).$$

Express Y as the union of its irreducible components and give the prime ideal of each component.

Solution. At a point of Y , the second equation is $x(z - 1) = 0$. If $x = 0$, the first equation becomes $yz = 0$. This gives the two lines

$$L_y = V(x, z), \quad L_z = V(x, y).$$

If $z = 1$, the first equation gives $y = x^2$, producing the curve

$$C = V(z - 1, y - x^2).$$

These cases exhaust all points, so

$$Y = L_y \cup L_z \cup C.$$

The corresponding coordinate rings are

$$k[x, y, z]/(x, z) \cong k[y], \quad k[x, y, z]/(x, y) \cong k[z],$$

and

$$k[x, y, z]/(z - 1, y - x^2) \cong k[x].$$

They are domains, so the three defining ideals are prime and the three closed sets are irreducible. The lines are distinct, and neither line is contained in C ; likewise C is not contained in either line. Therefore these are exactly the irreducible components of Y , with prime ideals

$$I(L_y) = (x, z), \quad I(L_z) = (x, y), \quad I(C) = (z - 1, y - x^2).$$

□

Exercise I.1.4

Permitted context.

Let k be algebraically closed. The affine line $\mathbb{A}_k^1$ is irreducible, so any two nonempty Zariski-open subsets of it intersect. The product topology on $\mathbb{A}_k^1 \times \mathbb{A}_k^1$ has a basis consisting of products $U \times V$ of open subsets.

Problem. Identify the set $\mathbb{A}_k^2$ with $\mathbb{A}_k^1 \times \mathbb{A}_k^1$. Prove that the Zariski topology on $\mathbb{A}_k^2$ is not the product of the Zariski topologies on the two affine-line factors.

Solution. The diagonal

$$\Delta = \{(a, a) : a \in k\} = V(x - y)$$

is closed in the Zariski topology of $\mathbb{A}_k^2$.

It is not closed in the product topology. Indeed, choose $a, b \in k$ with $a \neq b$. If the complement of Δ were product-open, there would be nonempty Zariski-open neighborhoods U of a and V of b such that

$$(U \times V) \cap \Delta = \emptyset.$$

This condition is equivalent to $U \cap V = \emptyset$, contradicting irreducibility of $\mathbb{A}_k^1$. Thus Δ is Zariski-closed in $\mathbb{A}_k^2$ but not closed in the product topology, so the two topologies differ. □

Exercise I.1.5**Permitted context.**

Let k be algebraically closed. The affine coordinate ring of $X \subseteq \mathbb{A}_k^n$ is

$$A(X) = k[x_1, \dots, x_n]/I(X).$$

The Hilbert Nullstellensatz gives $I(V(I)) = \sqrt{I}$ for every ideal $I \subseteq k[x_1, \dots, x_n]$.

The empty algebraic set and the zero ring are allowed in this correspondence.

Problem. Characterize the k -algebras that occur as affine coordinate rings of algebraic subsets of some affine space. Prove that these are exactly the finitely generated reduced k -algebras.

Solution. Suppose first that $B = A(X)$ for an algebraic set $X \subseteq \mathbb{A}_k^n$. Then B is generated over k by the residue classes of the coordinate functions. Moreover, $I(X)$ is radical: if f^m vanishes on X , then f vanishes on X . Therefore the quotient $k[x_1, \dots, x_n]/I(X)$ has no nonzero nilpotents, so B is reduced.

Conversely, let B be a finitely generated reduced k -algebra. Choose generators and a surjection

$$k[x_1, \dots, x_n] \longrightarrow B$$

with kernel I . Since B is reduced, I is radical. Set $X = V(I) \subseteq \mathbb{A}_k^n$. The Nullstellensatz yields

$$I(X) = I(V(I)) = \sqrt{I} = I.$$

Consequently

$$A(X) = k[x_1, \dots, x_n]/I(X) \cong k[x_1, \dots, x_n]/I \cong B.$$

Thus finite generation and reducedness are both necessary and sufficient. □

Exercise I.1.6**Permitted context.**

A nonempty topological space is irreducible if it is not the union of two proper closed subsets. Equivalently, every two nonempty open subsets intersect.

Problem. Prove both statements below.

1. Every nonempty open subset of an irreducible space is dense and is irreducible in its subspace topology.
2. If a subset is irreducible in its subspace topology, then its closure in the ambient space is irreducible.

Solution. Let X be irreducible and let $U \subseteq X$ be nonempty and open. Every nonempty open subset $V \subseteq X$ meets U , which is precisely the assertion that $\overline{U} = X$.

To prove that U is irreducible, let U_1, U_2 be nonempty open subsets of U . Write $U_i = U \cap V_i$ with V_i open in X . Each $U \cap V_i$ is a nonempty open subset of X . Irreducibility of X therefore gives

$$U_1 \cap U_2 = (U \cap V_1) \cap (U \cap V_2) \neq \emptyset.$$

Hence U is irreducible.

Now let $Y \subseteq X$ be irreducible and suppose

$$\overline{Y} = F_1 \cup F_2$$

with F_1, F_2 closed in $\overline{Y}$. Intersecting with Y gives

$$Y = (Y \cap F_1) \cup (Y \cap F_2).$$

Since Y is irreducible, it is contained in one of the F_i , say F_1 . The set F_1 is closed in $\overline{Y}$, so it contains the closure of Y inside $\overline{Y}$, which is all of $\overline{Y}$. Thus $F_1 = \overline{Y}$, proving that $\overline{Y}$ is irreducible. $\square$

Exercise I.1.7

Permitted context.

A topological space is noetherian if every descending chain of closed subsets stabilizes. A space is quasi-compact if every open cover has a finite subcover. One may use that a quasi-compact subset of a Hausdorff space is closed.

A finite union may have no members; that union is the empty set.

Problem. (a) Prove that the following conditions on a topological space X are equivalent: descending chains of closed subsets stabilize; every nonempty family of closed subsets has a minimal member; ascending chains of open subsets stabilize; every nonempty family of open subsets has a maximal member.

- (b) Prove that every noetherian space is quasi-compact.
- (c) Prove that every subspace of a noetherian space is noetherian.
- (d) Prove that a noetherian Hausdorff space is a finite discrete space.

Solution. (a) Assume descending chains of closed sets stabilize. If a nonempty family of closed sets had no minimal member, one could begin with one member and repeatedly choose a strictly smaller member, producing a nonstabilizing descending chain. Thus every such family has a minimal member. Conversely, a minimal member of the family formed by the terms of a descending chain forces the chain to stabilize.

Taking complements reverses inclusion and exchanges closed sets with open sets. It therefore carries the first condition to the ascending-chain condition for open sets and the minimal-member condition for closed sets to the maximal-member condition for open sets. All four conditions are equivalent.

(b) Let $\{U_i\}_{i \in I}$ cover X , and consider the family of all finite unions of the U_i . By part (a), this family has a maximal member V . If $V \neq X$, choose $x \in X \setminus V$ and then U_j containing x . The finite union $V \cup U_j$ strictly contains V , a contradiction. Hence $V = X$, giving a finite subcover.

(c) Let $Y \subseteq X$, and let

$$F_1 \supseteq F_2 \supseteq \cdots$$

be closed subsets of Y . For each i , because F_i is closed in Y ,

$$F_i = Y \cap \overline{F_i}^X.$$

The closures in X form a descending chain and hence stabilize. Intersecting with Y shows that the original chain stabilizes. Thus Y is noetherian.

(d) By parts (b) and (c), every subset of X is quasi-compact. Because X is Hausdorff, every quasi-compact subset is closed. Hence every subset is closed, so X is discrete. Part (b) applied to the cover of X by singleton subsets now shows that X is finite. $\square$

Exercise I.1.8

Permitted context.

Let k be algebraically closed and let Y be an affine variety with coordinate ring $A = A(Y)$. One may use that irreducible closed subsets $Z \subseteq Y$ correspond to prime ideals $I(Z)/I(Y)$ of $A(Y)$, that primes minimal over a nonzero principal ideal in a finitely generated integral k -algebra have height one, and that

$$\dim(A/\mathfrak{p}) + \text{ht}(\mathfrak{p}) = \dim A.$$

Problem. Let $Y \subseteq \mathbb{A}_k^n$ be an affine variety of dimension r , and let H be a hypersurface that does not contain Y . Prove that every irreducible component of $Y \cap H$ has dimension $r - 1$.

Solution. Write $A = A(Y)$, and let $f \in A$ be the restriction to Y of an equation defining H . Since $Y \not\subseteq H$, the element f is nonzero. If $Y \cap H$ is empty there is nothing to prove, so assume it is nonempty. Then f is not a unit.

An irreducible component Z of $Y \cap H = V_Y(f)$ corresponds, through the affine variety-coordinate ring correspondence, to a prime ideal $\mathfrak{p} \subseteq A$ minimal over (f) . The ring A is a domain, and the principal ideal theorem therefore gives

$$\text{ht}(\mathfrak{p}) = 1.$$

The dimension formula now yields

$$\dim Z = \dim(A/\mathfrak{p}) = \dim A - 1 = r - 1.$$

This applies to every irreducible component. $\square$

Exercise I.1.9

Permitted context.

Let k be algebraically closed. One may use the hypersurface-section theorem: if an affine variety W of dimension d is not contained in the zero set $H = V(f)$ of a nonzero polynomial, every irreducible component of $W \cap H$ has dimension $d - 1$. The irreducible components of a finite union are the maximal members among the irreducible components of the sets in that union.

Problem. Suppose an ideal $I \subseteq k[x_1, \dots, x_n]$ can be generated by r elements. Prove that every irreducible component of $V(I)$ has dimension at least $n - r$.

Solution. Write $I = (f_1, \dots, f_r)$, and argue by induction on r . For $r = 0$, the zero set is $\mathbb{A}_k^n$, of dimension n .

Let

$$X = V(f_1, \dots, f_{r-1}) = W_1 \cup \dots \cup W_m$$

be its irreducible-component decomposition. By induction, $\dim W_i \geq n - r + 1$ for every i . Since

$$V(I) = X \cap V(f_r) = \bigcup_{i=1}^m (W_i \cap V(f_r)),$$

every irreducible component Z of $V(I)$ is an irreducible component of some $W_i \cap V(f_r)$.

If $W_i \subseteq V(f_r)$, then $W_i \cap V(f_r) = W_i$, so

$$\dim Z = \dim W_i \geq n - r + 1.$$

Otherwise the hypersurface-section theorem gives

$$\dim Z = \dim W_i - 1 \geq n - r.$$

In either case $\dim Z \geq n - r$, which proves the assertion. $\square$

Exercise I.1.10

Permitted context.

The dimension of a nonempty topological space is the supremum of lengths n of chains $Z_0 \subsetneq \dots \subsetneq Z_n$ of nonempty irreducible closed subsets. Put $\dim \emptyset = -1$. A space is noetherian if every descending sequence of closed subsets stabilizes. A subset of an irreducible space that is nonempty and open is irreducible and dense; the closure of an irreducible subset is irreducible.

Problem. (a) Prove $\dim Y \leq \dim X$ for every subspace $Y \subseteq X$.

(b) For an open cover $X = \bigcup_i U_i$, prove $\dim X = \sup_i \dim U_i$.

(c) Give a space with a dense open subset of strictly smaller dimension.

(d) If X is irreducible of finite dimension and Y is closed with $\dim Y = \dim X$, prove $Y = X$.

(e) Construct a noetherian space of infinite dimension.

Solution. (a) Given a chain of irreducible closed subsets Z_j of Y , take their closures in X . They are irreducible closed subsets of X . Since $\overline{Z_j}^X \cap Y = Z_j$, all inclusions remain strict. Thus every chain length in Y occurs in X .

(b) Part (a) gives $\sup_i \dim U_i \leq \dim X$. Conversely, take a chain $Z_0 \subsetneq \dots \subsetneq Z_n$ in X , and choose U_i meeting Z_0 . Each $Z_j \cap U_i$ is nonempty, closed in U_i , and irreducible. It is dense in Z_j , so equality of any two such intersections would imply equality of their closures Z_j . They therefore give a strict chain of length n in U_i . Taking the supremum over all chains proves the other inequality.

(c) Let $X = \{\eta, s\}$, with closed subsets $\emptyset, \{s\}, X$. Both $\{s\}$ and X are irreducible, so $\dim X = 1$. The singleton $U = \{\eta\}$ is open and dense, but has dimension zero.

(d) Put $d = \dim X < \infty$. If $\dim Y = d$, a chain of length d exists in Y , since the supremum is over integers. Because Y is closed, its chain members are closed in X . If $Y \neq X$, adjoining the irreducible closed set X at the top gives a chain of length $d + 1$, a contradiction.

(e) Take a point η and disjoint finite sets

$$C_n = \{p_{n,1}, \dots, p_{n,n}\} \quad (n \geq 1).$$

Let $X = \{\eta\} \cup \bigcup_{n \geq 1} C_n$. Declare the proper closed subsets to be finite unions of initial segments $\{p_{n,1}, \dots, p_{n,j}\}$, with $0 \leq j \leq n$; also declare X closed. These sets are closed under finite unions and arbitrary intersections. For the latter, if some intersected set is proper, the intersection is contained in its finite support, and within each chain remains an initial segment. This defines a topology.

Every proper closed set is finite. A descending chain therefore stabilizes as soon as it leaves X , by decreasing cardinalities. Thus X is noetherian. The closure of $p_{n,j}$ is its initial segment, which is irreducible: any closed subset containing that point contains the whole segment. The nested initial segments of C_n give a chain of length $n - 1$, for every n . Consequently $\dim X = \infty$. $\square$

Exercise I.1.11

Permitted context.

Let k be algebraically closed. The vanishing ideal of a parametrized subset of affine space is the kernel of substitution into $k[t]$, because k is infinite. For a finitely generated domain over k , dimension equals the transcendence degree of its fraction field; in $k[x, y, z]$, $\text{ht } \mathfrak{p} + \dim k[x, y, z]/\mathfrak{p} = 3$. If an ideal I is generated by r elements, then $\dim_k I/\mathfrak{m}I \leq r$ for a maximal ideal with residue field k . The same statement holds after localization.

Problem. Let $Y = \{(t^3, t^4, t^5) : t \in k\} \subseteq \mathbb{A}^3$. Show that Y is a closed irreducible curve and that $I(Y)$ is a prime ideal of height two requiring at least three generators, even locally at the origin.

Solution. Let $\phi : k[x, y, z] \rightarrow k[t]$ send x, y, z to t^3, t^4, t^5 . The three polynomials

$$f_1 = y^2 - xz, \quad f_2 = yz - x^3, \quad f_3 = z^2 - x^2y$$

lie in its kernel. Set $J = (f_1, f_2, f_3)$. Using these relations, every monomial reduces modulo J to a $k[x]$ -linear combination of $1, y, z$. Indeed each replacement of y^2, yz, z^2 strictly decreases the total exponent of y, z , so the reductions terminate. The images of these normal forms are combinations of

$$t^{3a}, \quad t^{3a+4}, \quad t^{3a+5} \quad (a \geq 0),$$

whose exponents are pairwise distinct, as seen modulo three. They are linearly independent in $k[t]$. Thus the induced map on the quotient is injective, proving $\ker \phi = J$. It follows that J is prime.

The common zero set of J equals the parametrized set. If $x = 0$, its equations force $y = z = 0$. If $x \neq 0$, put $t = y/x$. Then

$$t^2 = z/x, \quad t^3 = yz/x^2 = x, \quad t^4 = y, \quad t^5 = z.$$

Hence every common zero has the required form. So $Y = V(J)$, $I(Y) = J$, and $A(Y) = k[t^3, t^4, t^5]$. Its fraction field is $k(t)$, because $t = t^4/t^3$, so Y has dimension one and J has height two.

Let $\mathfrak{m} = (x, y, z)$. Since $J \subseteq \mathfrak{m}^2$, we have $\mathfrak{m}J \subseteq \mathfrak{m}^3$. The quadratic initial forms of f_1, f_2, f_3 are

$$y^2 - xz, \quad yz, \quad z^2,$$

which are linearly independent. Thus no nonzero constant linear combination of the f_i belongs to $\mathfrak{m}J$. Their classes form a basis of $J/\mathfrak{m}J$, of dimension three. The same quotient computes $J_{\mathfrak{m}}/\mathfrak{m}_{\mathfrak{m}}J_{\mathfrak{m}}$, so neither J nor its localization can be generated by two elements. This gives the claimed failure of a local complete intersection at the origin. $\square$

Exercise I.1.12**Permitted context.**

Work with real points and the real Zariski topology. A nonempty space is irreducible if it is not the union of two proper closed subsets. A monic polynomial in x over $\mathbb{R}[y]$ that factors in $\mathbb{R}(y)[x]$ has its monic factors in $\mathbb{R}[y][x]$, by Gauss's lemma.

Problem. Find an irreducible polynomial in $\mathbb{R}[x, y]$ whose real zero set is nonempty and reducible.

Solution. Take

$$f = x^2 + (y^2 - 1)^2.$$

If it were reducible, its degree two and monicity in x , together with Gauss's lemma, would give a polynomial root $a(y) \in \mathbb{R}[y]$. That root would satisfy

$$a(y)^2 = -(y^2 - 1)^2.$$

At $y = 0$ this says $a(0)^2 = -1$, which is impossible over $\mathbb{R}$. A factor independent of x is also impossible because f is monic in x . Thus f is irreducible.

For real x, y , both summands of f are nonnegative. Their sum vanishes precisely when $x = 0$ and $y^2 = 1$. Therefore

$$V_{\mathbb{R}}(f) = \{(0, 1), (0, -1)\}.$$

Each singleton is algebraic and closed, so this set is the union of two proper nonempty closed subsets and is reducible. $\square$

I.2 Projective Varieties**Exercise I.2.1****Permitted context.**

Let k be algebraically closed and $S = k[x_0, \dots, x_n]$ with its standard grading. For a homogeneous ideal I , its affine zero set is a cone, and every nonzero point of that cone determines a point of the projective zero set $Z_+(I) \subseteq \mathbb{P}_k^n$. One may use the affine Hilbert Nullstellensatz.

Problem. Let $I \subseteq S$ be homogeneous, and let $f \in S$ be homogeneous of positive degree. Assume f vanishes at every point of $Z_+(I)$. Prove that $f^q \in I$ for some integer $q > 0$.

Solution. Let $C = V_{\mathbb{A}^{n+1}}(I)$ be the affine cone defined by I . If $a \in C$ is nonzero, its projective class $[a]$ belongs to $Z_+(I)$, so the hypothesis gives $f(a) = 0$. At the origin, $f(0) = 0$ because f has positive degree. Hence f vanishes on all of C .

The affine Nullstellensatz gives

$$f \in I(C) = \sqrt{I}.$$

By the definition of the radical, there is an integer $q > 0$ such that $f^q \in I$. This is the homogeneous Nullstellensatz. $\square$

Exercise I.2.2**Permitted context.**

Let k be algebraically closed, let $S = k[x_0, \dots, x_n]$, and write $S_+ = (x_0, \dots, x_n)$ for the irrelevant ideal. Let S_d denote the vector space of homogeneous forms of degree d . One may use the homogeneous Nullstellensatz.

Problem. For a homogeneous ideal $I \subseteq S$, prove that the following conditions are equivalent:

1. its projective zero set is empty;
2. $\sqrt{I}$ is either S or S_+ ;
3. I contains S_d for some $d > 0$.

Solution. Assume first that the projective zero set is empty. If $I = S$, then $\sqrt{I} = S$. Otherwise I is proper, and homogeneity gives $I \subseteq S_+$, hence $\sqrt{I} \subseteq S_+$. By the homogeneous Nullstellensatz, each homogeneous polynomial of positive degree that vanishes on the empty projective zero set belongs to $\sqrt{I}$. In particular, every x_i belongs to $\sqrt{I}$, so

$$S_+ \subseteq \sqrt{I} \subseteq S_+.$$

Thus $\sqrt{I} = S_+$, proving the second condition.

Suppose next that $\sqrt{I} = S$ or S_+ . In the first case $I = S$, so it contains S_d for every d . In the second case, for each i there is an integer $N_i > 0$ with $x_i^{N_i} \in I$. Choose

$$d > \sum_{i=0}^n (N_i - 1).$$

Every monomial of degree d is divisible by some $x_i^{N_i}$, and hence belongs to I . Since these monomials span S_d , we have $S_d \subseteq I$.

Finally, suppose $S_d \subseteq I$ for some $d > 0$. At any projective point at least one coordinate, say x_i , is nonzero. But $x_i^d \in S_d \subseteq I$, so that point cannot be a common zero of I . Therefore the projective zero set is empty, completing the cycle of implications. $\square$

Exercise I.2.3**Permitted context.**

Let k be algebraically closed and $S = k[x_0, \dots, x_n]$, graded by total degree. For homogeneous polynomials T , let $V_+(T)$ be their common projective zero set. For a subset $Y \subseteq \mathbb{P}^n$, let $I(Y)$ be the homogeneous ideal generated by homogeneous polynomials vanishing on Y . The irrelevant ideal is $S_+ = (x_0, \dots, x_n)$. We may use the affine Nullstellensatz and noetherianity of S . An ideal generated by homogeneous elements contains every homogeneous component of each of its elements; radicals and intersections of homogeneous ideals are homogeneous.

Problem. (a) Show $T_1 \subseteq T_2$ implies $V_+(T_1) \supseteq V_+(T_2)$.

(b) Show $Y_1 \subseteq Y_2$ implies $I(Y_1) \supseteq I(Y_2)$.

(c) Prove $I(Y_1 \cup Y_2) = I(Y_1) \cap I(Y_2)$.

(d) If $\mathfrak{a}$ is homogeneous and $V_+(\mathfrak{a}) \neq \emptyset$, prove $I(V_+(\mathfrak{a})) = \sqrt{\mathfrak{a}}$.

(e) Prove $V_+(I(Y)) = \overline{Y}$ for every projective subset Y .

Solution. (a) Every point annihilating every element of T_2 annihilates every element of its subset T_1 .

(b) A homogeneous polynomial vanishing on Y_2 vanishes on Y_1 ; passing to the ideals they generate gives the asserted inclusion.

(c) In each degree, vanishing on the union is equivalent to vanishing on both subsets. The two homogeneous ideals therefore have identical components in every degree and are equal. This also covers the empty subset, whose ideal is S .

(d) Put $Y = V_+(\mathfrak{a})$. Its nonemptiness implies that $\mathfrak{a}$ contains no nonzero constant. Its affine zero set in $\mathbb{A}^{n+1}$ is exactly the origin together with all nonzero representatives of points of Y . Every positive-degree homogeneous polynomial vanishing on Y therefore vanishes on this affine zero set; the affine Nullstellensatz puts it in $\sqrt{\mathfrak{a}}$. The only degree-zero polynomial vanishing on nonempty Y is zero. Thus $I(Y) \subseteq \sqrt{\mathfrak{a}}$. Conversely every homogeneous element whose power lies in $\mathfrak{a}$ vanishes on Y . Since the radical is homogeneous, its homogeneous components generate it, giving $\sqrt{\mathfrak{a}} \subseteq I(Y)$.

(e) The set $V_+(I(Y))$ is closed and contains Y , so it contains $\overline{Y}$. If $W = V_+(\mathfrak{b})$ is any closed set containing Y , each homogeneous element of $\mathfrak{b}$ vanishes on Y . Thus $\mathfrak{b} \subseteq I(Y)$, and part (a) gives $V_+(I(Y)) \subseteq W$. Intersecting all such W proves the reverse inclusion. $\square$

Exercise I.2.4**Permitted context.**

Let k be algebraically closed and $S = k[x_0, \dots, x_n]$, graded by total degree. For homogeneous polynomials T , let $V_+(T)$ be their common projective zero set. For a subset $Y \subseteq \mathbb{P}^n$, let $I(Y)$ be the homogeneous ideal generated by homogeneous polynomials vanishing on Y . The irrelevant ideal is $S_+ = (x_0, \dots, x_n)$. We may use the affine Nullstellensatz and noetherianity of S . An ideal generated by homogeneous elements contains every homogeneous component of each of its elements; radicals and intersections of homogeneous ideals are homogeneous.

For homogeneous $\mathfrak{a}$ with nonempty projective zero set, $I(V_+(\mathfrak{a})) = \sqrt{\mathfrak{a}}$. The projective zero set is empty precisely when $\sqrt{\mathfrak{a}} = S$ or S_+ . For every subset Y , $V_+(I(Y)) = \overline{Y}$. Primality of a homogeneous ideal can be checked on products of homogeneous elements.

Problem. (a) Establish the inclusion-reversing correspondence between projective algebraic sets and homogeneous radical ideals other than S_+ , including the empty set.

(b) Prove that an algebraic set Y is irreducible exactly when $I(Y)$ is prime.

(c) Prove that $\mathbb{P}^n$ is irreducible.

Solution. (a) For nonempty algebraic Y , its vanishing ideal is homogeneous and radical: if a power vanishes, the polynomial itself vanishes, and this can be applied to homogeneous components. It is not S_+ , since no projective point annihilates all coordinates. The two identities in the context show that I and V_+ are inverse on nonempty algebraic sets and homogeneous radical ideals other than S, S_+ . The empty set has ideal S , and $V_+(S) = \emptyset$. This extends the bijection as stated. Both operations reverse inclusion directly from their definitions.

(b) Suppose first that Y is irreducible. Its ideal is proper because $Y \neq \emptyset$. If homogeneous f, g satisfy $fg \in I(Y)$, then

$$Y = (Y \cap V_+(f)) \cup (Y \cap V_+(g)).$$

Irreducibility forces one term to be all of Y ; hence f or g lies in $I(Y)$. The homogeneous primality test therefore makes $I(Y)$ prime.

Conversely suppose $I(Y)$ is prime. In particular $Y \neq \emptyset$. If $Y = Y_1 \cup Y_2$ with both Y_i proper and closed in Y , then they are algebraic, since Y is closed. The correspondence in (a) gives $I(Y) \subsetneq I(Y_i)$. Choose homogeneous $f_i \in I(Y_i) \setminus I(Y)$. Their product vanishes on $Y_1 \cup Y_2$ but neither factor belongs to $I(Y)$, contradicting primality.

(c) A homogeneous polynomial vanishing on all projective points also vanishes on every nonzero vector; if it has positive degree it vanishes at zero as well. A polynomial vanishing on all of k^{n+1} is zero, by induction in the variables over the infinite field k . Nonzero constants do not vanish on projective space. Thus $I(\mathbb{P}^n) = (0)$, which is prime, and (b) applies. $\square$

Exercise I.2.5**Permitted context.**

Let k be algebraically closed and $S = k[x_0, \dots, x_n]$, graded by total degree. For homogeneous polynomials T , let $V_+(T)$ be their common projective zero set. For a subset $Y \subseteq \mathbb{P}^n$, let $I(Y)$ be the homogeneous ideal generated by homogeneous polynomials vanishing on Y . The irrelevant ideal is $S_+ = (x_0, \dots, x_n)$. We may use the affine Nullstellensatz and noetherianity of S . An ideal generated by homogeneous elements contains every homogeneous component of each of its elements; radicals and intersections of homogeneous ideals are homogeneous.

Every projective algebraic set Y satisfies $V_+(I(Y)) = Y$. A topological space is noetherian when descending chains of closed subsets stabilize.

Problem. (a) Prove that projective space is noetherian.

(b) Prove that every projective algebraic set is uniquely a finite union of irreducible closed subsets, none contained in another. Include the empty set.

Solution. (a) A descending chain of closed sets $Y_1 \supseteq Y_2 \supseteq \dots$ gives an ascending chain $I(Y_1) \subseteq I(Y_2) \subseteq \dots$. It stabilizes because S is noetherian. Applying V_+ shows that the original chain stabilizes.

(b) Suppose some nonempty closed set has no finite irreducible decomposition. By the descending chain condition, choose a minimal such set Y . It cannot be irreducible, so it is the union of two proper closed subsets. By minimality each of those is a finite union of irreducible closed sets, giving such a decomposition of Y , a contradiction. In a finite decomposition, remove any member contained in another.

For uniqueness, let $Y = \bigcup_{i=1}^r A_i = \bigcup_{j=1}^s B_j$ be two decompositions without containments. The irreducible set A_i is covered by the finitely many closed subsets $A_i \cap B_j$; it must lie in some B_j . In turn that B_j lies in some A_ℓ . Then $A_i \subseteq A_\ell$, so the no-containment condition forces $\ell = i$, and $A_i = B_j$. Interchanging the two decompositions proves equality of the collections. For $Y = \emptyset$, the unique collection is the empty one. $\square$

Exercise I.2.6**Permitted context.**

Let k be algebraically closed and $S = k[x_0, \dots, x_n]$, graded by total degree. For homogeneous polynomials T , let $V_+(T)$ be their common projective zero set. For a subset $Y \subseteq \mathbb{P}^n$, let $I(Y)$ be the homogeneous ideal generated by homogeneous polynomials vanishing on Y . The irrelevant ideal is $S_+ = (x_0, \dots, x_n)$. We may use the affine Nullstellensatz and noetherianity of S . An ideal generated by homogeneous elements contains every homogeneous component of each of its elements; radicals and intersections of homogeneous ideals are homogeneous.

The standard chart $U_i = \{x_i \neq 0\}$ is affine space with coordinates x_j/x_i , $j \neq i$. For a finitely generated k -domain, dimension equals the transcendence degree of its fraction field. The dimension of a topological space covered by open subsets is the supremum of their dimensions. A nonempty open subset of an irreducible space is dense.

Problem. Let $Y \subseteq \mathbb{P}^n$ be a projective variety and $R = S/I(Y)$. Prove $\dim R = \dim Y + 1$. For every nonempty affine chart $Y_i = Y \cap U_i$, identify its coordinate ring with $(R_{x_i})_0$, and prove $\dim Y_i = \dim Y$.

Solution. Choose i such that $Y_i \neq \emptyset$; then $x_i \neq 0$ in the graded domain R . The degree-zero subring

$$B = (R_{x_i})_0 = k[x_0/x_i, \dots, x_n/x_i]$$

is generated by the indicated ratios, omitting $x_i/x_i = 1$. Its relations are exactly the polynomials vanishing on the affine algebraic set Y_i . Indeed dehomogenization of an element of $I(Y)$ vanishes there. Conversely, homogenize a polynomial relation that vanishes on Y_i . The resulting homogeneous polynomial vanishes on the dense subset Y_i of Y , hence on Y , so belongs to $I(Y)$. This proves $B = A(Y_i)$.

Every homogeneous element of R_{x_i} of degree $e \in \mathbb{Z}$ is x_i^e times an element of B . The grading is a direct sum, so no nontrivial Laurent polynomial in x_i with coefficients in B can vanish. Consequently

$$R_{x_i} \cong B[x_i, x_i^{-1}], \quad \text{Frac}(R) = \text{Frac}(B)(x_i),$$

where x_i is transcendental over $\text{Frac}(B)$. The equality of fraction fields also uses that localization of a domain does not change its fraction field. Taking transcendence degrees gives $\dim R = \dim B + 1 = \dim Y_i + 1$. Thus every nonempty chart has the same dimension $\dim R - 1$. Since those charts form an open cover of Y , their supremum equals $\dim Y$, proving both claims. $\square$

Exercise I.2.7**Permitted context.**

Let k be algebraically closed and $S = k[x_0, \dots, x_n]$, graded by total degree. For homogeneous polynomials T , let $V_+(T)$ be their common projective zero set. For a subset $Y \subseteq \mathbb{P}^n$, let $I(Y)$ be the homogeneous ideal generated by homogeneous polynomials vanishing on Y . The irrelevant ideal is $S_+ = (x_0, \dots, x_n)$. We may use the affine Nullstellensatz and noetherianity of S . An ideal generated by homogeneous elements contains every homogeneous component of each of its elements; radicals and intersections of homogeneous ideals are homogeneous.

For a projective variety Z , every nonempty standard affine chart has dimension $\dim S(Z) - 1 = \dim Z$. Every nonempty open subset of an affine variety has the same dimension as that variety. The closure of an irreducible set is irreducible, and subspaces cannot have dimension greater than the ambient space.

Problem. (a) Prove $\dim \mathbb{P}^n = n$.

(b) If $Y \subseteq \mathbb{P}^n$ is a quasi-projective variety, prove $\dim Y = \dim \bar{Y}$.

Solution. (a) The homogeneous coordinate ring of projective space is $k[x_0, \dots, x_n]$, of dimension $n + 1$. The chart dimension formula therefore gives $\dim \mathbb{P}^n = n$.

(b) Put $Z = \bar{Y}$. It is a projective variety, and Y is a nonempty open subset of Z . Choose a standard chart meeting Y . Then $Y \cap U_i$ is a nonempty open subset of the affine variety $Z \cap U_i$. The two dimension statements in the context give

$$\dim(Y \cap U_i) = \dim(Z \cap U_i) = \dim Z.$$

Subspace monotonicity now gives $\dim Z \leq \dim Y \leq \dim Z$, proving equality. $\square$

Exercise I.2.8**Permitted context.**

Let k be algebraically closed and $S = k[x_0, \dots, x_n]$, graded by total degree. For homogeneous polynomials T , let $V_+(T)$ be their common projective zero set. For a subset $Y \subseteq \mathbb{P}^n$, let $I(Y)$ be the homogeneous ideal generated by homogeneous polynomials vanishing on Y . The irrelevant ideal is $S_+ = (x_0, \dots, x_n)$. We may use the affine Nullstellensatz and noetherianity of S . An ideal generated by homogeneous elements contains every homogeneous component of each of its elements; radicals and intersections of homogeneous ideals are homogeneous.

For a projective variety Y , $\dim S/I(Y) = \dim Y + 1$. For a prime $\mathfrak{p} \subset S$, $\text{ht } \mathfrak{p} + \dim S/\mathfrak{p} = n + 1$. The ring S is a unique factorization domain, in which each height-one prime is generated by an irreducible element. The homogeneous radical ideals with empty projective zero set are S, S_+ .

Problem. For $n \geq 1$, prove that a projective variety $Y \subseteq \mathbb{P}^n$ has dimension $n - 1$ exactly when it is the zero set of one irreducible homogeneous polynomial of positive degree.

Solution. Suppose $\dim Y = n - 1$. Then $\dim S/I(Y) = n$, so the prime $I(Y)$ has height one. It is therefore (f) for an irreducible polynomial f . Because this ideal is homogeneous, the top homogeneous component f_d belongs to (f) . It has the same total degree as f , so divisibility forces $f_d = cf$ for some nonzero scalar c . Thus f is homogeneous. It is nonconstant because $I(Y)$ is proper, and $Y = V_+(f)$.

Conversely, let f be irreducible, homogeneous, and of positive degree. Its ideal is a height-one prime, with quotient dimension n . It is neither S nor S_+ , the latter having quotient dimension zero. Thus $V_+(f)$ is nonempty. The projective Nullstellensatz gives $I(V_+(f)) = (f)$; its primality makes this zero set irreducible, and the dimension formula gives dimension $n - 1$. $\square$

Exercise I.2.9

Permitted context.

Work over an algebraically closed field k . For a projective algebraic set Y , $I(Y)$ denotes its homogeneous vanishing ideal. For homogeneous ideals with nonempty zero set, $I(V_+(\mathfrak{a})) = \sqrt{\mathfrak{a}}$. An algebraic set is irreducible exactly when its vanishing ideal is prime. For a projective variety, $\dim Y = \dim k[x_0, \dots, x_n]/I(Y) - 1$. Standard charts are affine spaces, and their topology is obtained by dehomogenizing polynomials.

For $f \in k[x_1, \dots, x_n]$ of degree d , write $f^h = x_0^d f(x_1/x_0, \dots, x_n/x_0)$. The chart $x_0 \neq 0$ identifies affine space with an open subset of projective space.

Problem. (a) For an affine variety $Y \subseteq \mathbb{A}^n$, prove that the homogeneous ideal of its projective closure is generated by f^h for all $f \in I(Y)$.

(b) Determine the ideals of the affine twisted cubic $\{(t, t^2, t^3) : t \in k\}$ and its projective closure. Show why homogenizing just a chosen generating set of the affine ideal need not generate the closure's ideal.

Solution. (a) Every f^h vanishes on Y and hence on its closure, since its projective zero set is closed. Conversely, let F be homogeneous of degree e and vanish on the closure. Then $f = F(1, x_1, \dots, x_n)$ belongs to $I(Y)$. If $F \neq 0$, its dehomogenization is nonzero, of some degree $d \leq e$, and

$$F = x_0^{e-d} f^h.$$

Thus every homogeneous element of the closure's ideal lies in the ideal generated by all these homogenizations, proving equality.

(b) Write the projective coordinates as w, x, y, z , with $w = 1$ the affine chart. Substitution gives the affine ideal

$$I_{\text{aff}} = (y - x^2, z - x^3);$$

division by these two relations identifies the quotient with $k[x]$. The projective ideal is

$$J = (wy - x^2, wz - xy, xz - y^2). \quad (1)$$

To prove this assertion directly, label w, x, y, z by indices 0, 1, 2, 3, and map them to s^3, s^2t, st^2, t^3 , respectively. The relations in (1) replace any product of two indices whose difference is at least two by the product of the two closer indices, preserving their sum. Each replacement strictly decreases the sum of squares of all indices in the monomial, so the process terminates. The resulting monomial uses at most two adjacent indices and is uniquely determined by its degree d and index sum j . Its image is $s^{3d-j}t^j$. These images are distinct monomials, hence linearly independent. Consequently J is exactly the substitution kernel and is prime.

For homogeneous F of degree d , vanishing at every $(1, t, t^2, t^3)$ is equivalent to the zero polynomial $F(s^3, s^2t, st^2, t^3)$: the latter is homogeneous, and its specialization $s = 1$ vanishes for every t in an infinite field. Thus J is the homogeneous ideal of the affine set and its closure.

Homogenizing the two displayed affine generators yields only

$$wy - x^2, \quad w^2z - x^3.$$

Both vanish along the whole line $w = x = 0$. But the point $[0 : 0 : 1 : 0]$ on this line does not satisfy $xz - y^2 = 0$. These two homogenizations therefore do not generate J , and even their projective zero set is too large. $\square$

Exercise I.2.10

Permitted context.

Work over an algebraically closed field k . For a projective algebraic set Y , $I(Y)$ denotes its homogeneous vanishing ideal. For homogeneous ideals with nonempty zero set, $I(V_+(\mathfrak{a})) = \sqrt{\mathfrak{a}}$. An algebraic set is irreducible exactly when its vanishing ideal is prime. For a projective variety, $\dim Y = \dim k[x_0, \dots, x_n]/I(Y) - 1$. Standard charts are affine spaces, and their topology is obtained by dehomogenizing polynomials.

An affine algebraic set is irreducible exactly when its vanishing ideal is prime, and its dimension is that of its coordinate ring. In a finite union of closed subsets, dimension is the maximum of their dimensions. A polynomial vanishing at every scalar multiple of a vector over an infinite field has each homogeneous component vanishing at that vector.

Problem. For nonempty projective algebraic $Y \subseteq \mathbb{P}^n$, let $C(Y) \subseteq \mathbb{A}^{n+1}$ consist of zero and all nonzero vectors representing points of Y .

- (a) Show that it is algebraic and has affine ideal $I(Y)$.
- (b) Show that $C(Y)$ is irreducible exactly when Y is.
- (c) Prove $\dim C(Y) = \dim Y + 1$, allowing reducible Y .

Solution. (a) The affine zero set of the homogeneous ideal $I(Y)$ is exactly $C(Y)$: away from zero its equations express membership in Y , and at zero every positive-degree equation vanishes. There is no nonzero constant in $I(Y)$ because $Y \neq \emptyset$. Thus the cone is closed. If $f = \sum f_d$ vanishes on $C(Y)$, fix a representative vector a of a point of Y and evaluate

$$0 = f(\lambda a) = \sum_d \lambda^d f_d(a) \quad (\lambda \in k).$$

Since k is infinite, all coefficients vanish. Therefore every f_d belongs to $I(Y)$, so $f \in I(Y)$. The reverse inclusion is immediate, proving equality of ideals.

(b) The affine and projective irreducibility criteria now both amount to primality of the same ideal $I(Y)$.

(c) Write $Y = Y_1 \cup \dots \cup Y_r$ as its finite nonempty irreducible components. Their cones are irreducible by (b); no one contains another, because such a containment would imply the corresponding containment after projectivizing nonzero vectors. Hence they are exactly the components of $C(Y)$. For each i , the coordinate-ring dimension formula gives

$$\dim C(Y_i) = \dim S/I(Y_i) = \dim Y_i + 1.$$

Taking the maximum over i proves the assertion. $\square$

Exercise I.2.11**Permitted context.**

Work over an algebraically closed field k . For a projective algebraic set Y , $I(Y)$ denotes its homogeneous vanishing ideal. For homogeneous ideals with nonempty zero set, $I(V_+(\mathfrak{a})) = \sqrt{\mathfrak{a}}$. An algebraic set is irreducible exactly when its vanishing ideal is prime. For a projective variety, $\dim Y = \dim k[x_0, \dots, x_n]/I(Y) - 1$. Standard charts are affine spaces, and their topology is obtained by dehomogenizing polynomials.

Problem. (a) Prove that a projective variety is an intersection of hyperplanes if and only if its homogeneous ideal is generated by linear forms.

(b) If such a variety has dimension r in $\mathbb{P}^n$, show that its ideal is minimally generated by $n - r$ linear forms.

(c) If linear varieties $Y, Z \subseteq \mathbb{P}^n$ have dimensions r, s , show that $r + s \geq n$ implies $Y \cap Z \neq \emptyset$. If the intersection is nonempty, show it is linear and has dimension at least $r + s - n$.

Solution. (a) If $I(Y)$ is generated by linear forms, its zero set is the intersection of their hyperplanes. Conversely, let Y be such an intersection, and choose a basis $\ell_1, \dots, \ell_c$ of the span of its defining linear forms. An invertible linear coordinate change makes them $x_0, \dots, x_{c-1}$. Their ideal is prime, since the quotient is the polynomial ring in the remaining $n+1-c$ variables. Because Y is nonempty, the projective Nullstellensatz gives $I(Y) = (\ell_1, \dots, \ell_c)$.

(b) The same quotient shows $\dim Y = (n + 1 - c) - 1 = n - c$, so $c = n - r$. These c independent linear forms generate the ideal minimally: any homogeneous generating collection must span its degree-one part, of dimension c . Even allowing nonhomogeneous generators cannot lower the number, because reduction modulo $S_+I(Y)$ leaves precisely that c -dimensional vector space.

(c) Let $V, W \subseteq k^{n+1}$ be the vector subspaces defining Y, Z . Their dimensions are $r + 1, s + 1$, and linear algebra gives

$$\dim(V \cap W) \geq (r + 1) + (s + 1) - (n + 1) = r + s - n + 1.$$

If $r + s \geq n$, this is positive, so its projectivization is nonempty. In every nonempty case

$$Y \cap Z = \mathbb{P}(V \cap W), \quad \dim(Y \cap Z) = \dim(V \cap W) - 1 \geq r + s - n.$$

The intersection is therefore a linear variety with the claimed bound. □

Exercise I.2.12**Permitted context.**

Work over an algebraically closed field k . For a projective algebraic set Y , $I(Y)$ denotes its homogeneous vanishing ideal. For homogeneous ideals with nonempty zero set, $I(V_+(\mathfrak{a})) = \sqrt{\mathfrak{a}}$. An algebraic set is irreducible exactly when its vanishing ideal is prime. For a projective variety, $\dim Y = \dim k[x_0, \dots, x_n]/I(Y) - 1$. Standard charts are affine spaces, and their topology is obtained by dehomogenizing polynomials.

Problem. Let $n, d > 0$, let α range over exponent vectors of total degree d in $n+1$ variables, and put $N = \binom{n+d}{d} - 1$. Define

$$v_d : \mathbb{P}^n \rightarrow \mathbb{P}^N, \quad [a] \mapsto [a^\alpha]_\alpha,$$

and $\theta : k[y_\alpha] \rightarrow k[x_0, \dots, x_n]$ by $\theta(y_\alpha) = x^\alpha$.

(a) Show that $\mathfrak{a} = \ker \theta$ is homogeneous and prime, with nonempty irreducible projective zero set.

(b) Prove $\text{im } v_d = V_+(\mathfrak{a})$.

(c) Prove that v_d is a homeomorphism onto its image.

(d) Identify the projective twisted cubic with $v_3(\mathbb{P}^1)$.

Solution. (a) The map multiplies degrees by d , so homogeneous components of different degrees cannot cancel in its image. Its kernel is therefore homogeneous. The image is a subring of a domain, so the kernel is prime. Substitution at any nonzero vector gives a projective common zero; at least one pure power a_i^d is nonzero. Thus its projective zero set is nonempty and irreducible.

(b) Substitution immediately gives $\text{im } v_d \subseteq V_+(\mathfrak{a})$. For the reverse inclusion, write $Y_i = y_{de_i}$ for a pure-power coordinate, where e_i is the i -th unit vector. The kernel contains the relations

$$y_\alpha^d = \prod_i Y_i^{\alpha_i}. \quad (1)$$

Thus at a projective common zero not all Y_i can vanish. Choose i with $Y_i \neq 0$, and normalize all coordinates by dividing by Y_i , so that $Y_i = 1$. Put $a_i = 1$ and $a_j = y_{(d-1)e_i + e_j}$ for $j \neq i$. There are further kernel relations

$$y_\alpha Y_i^{d-1} = \prod_j y_{(d-1)e_i + e_j}^{\alpha_j}, \quad (2)$$

because both monomials map to $x_i^{d(d-1)} x^\alpha$. After normalization (2) gives $y_\alpha = \prod_j a_j^{\alpha_j}$. The given point is therefore $v_d([a_0 : \dots : a_n])$. No extraction of d -th roots was used, so this also works when the characteristic divides d .

(c) Pulling back a homogeneous equation in the y_α gives a homogeneous polynomial in the x_i . Hence v_d is continuous. On the open set $Y_i \neq 0$, the inverse is given in the chart $x_i \neq 0$ by

$$\frac{x_j}{x_i} = \frac{y_{(d-1)e_i + e_j}}{Y_i} \quad (j \neq i).$$

These formulas prove injectivity, and are continuous in affine chart coordinates: inverse images of polynomial zero sets are zero sets obtained by clearing a nonvanishing denominator. They agree on overlaps, so they define a continuous inverse on the whole image. Thus v_d is a homeomorphism.

(d) With the coordinates ordered as s^3, s^2t, st^2, t^3 , the chart $s \neq 0$ of $v_3(\mathbb{P}^1)$ is $[1 : u : u^2 : u^3]$. It is the affine twisted cubic. It is dense in the image, since this chart is dense in $\mathbb{P}^1$ and (c) is a homeomorphism. The image is closed by (b), so it is precisely the projective closure. $\square$

Exercise I.2.13**Permitted context.**

Work over an algebraically closed field k . For a projective algebraic set Y , $I(Y)$ denotes its homogeneous vanishing ideal. For homogeneous ideals with nonempty zero set, $I(V_+(\mathfrak{a})) = \sqrt{\mathfrak{a}}$. An algebraic set is irreducible exactly when its vanishing ideal is prime. For a projective variety, $\dim Y = \dim k[x_0, \dots, x_n]/I(Y) - 1$. Standard charts are affine spaces, and their topology is obtained by dehomogenizing polynomials.

The degree-two Veronese map $v_2 : \mathbb{P}^2 \rightarrow Y \subseteq \mathbb{P}^5$ is a homeomorphism onto a projective variety. Every irreducible plane curve is $V_+(f)$ for an irreducible homogeneous polynomial of positive degree. The homomorphism $\theta : k[y_0, \dots, y_5] \rightarrow k[x_0, x_1, x_2]$ sends the six coordinates to the six quadratic monomials. Every even-degree monomial is a product of quadratic monomials.

Problem. For every irreducible closed curve Z on the Veronese surface $Y \subseteq \mathbb{P}^5$, construct a hypersurface V of $\mathbb{P}^5$ such that $V \cap Y = Z$ as sets.

Solution. The inverse image $W = v_2^{-1}(Z)$ is an irreducible closed curve in $\mathbb{P}^2$, because homeomorphisms preserve closedness, irreducibility, and dimension. Write $W = V_+(f)$, where f is homogeneous and irreducible of degree $m > 0$. Every monomial of f^2 , of degree $2m$, can be grouped into m quadratic monomials. Hence there is a homogeneous $G \in k[y_0, \dots, y_5]$ of degree m with $\theta(G) = f^2$. Its zero set meets Y exactly in Z , since $G(v_2(a)) = f(a)^2$.

If a hypersurface is required to be irreducible, factor $G = c \prod_j G_j$ into homogeneous irreducible factors. None has zero pullback, because the product pulls back to the nonzero polynomial f^2 . Unique factorization in $k[x_0, x_1, x_2]$ gives $\theta(G_j) = c_j f^{a_j}$ with $a_j \geq 1$: the exponent is positive because its degree is $2 \deg G_j > 0$. Thus $V = V_+(G_j)$, for any factor, is an irreducible hypersurface with $V \cap Y = Z$. The construction also applies in characteristic two. $\square$

Exercise I.2.14**Permitted context.**

Work over an algebraically closed field k . For a projective algebraic set Y , $I(Y)$ denotes its homogeneous vanishing ideal. For homogeneous ideals with nonempty zero set, $I(V_+(\mathfrak{a})) = \sqrt{\mathfrak{a}}$. An algebraic set is irreducible exactly when its vanishing ideal is prime. For a projective variety, $\dim Y = \dim k[x_0, \dots, x_n]/I(Y) - 1$. Standard charts are affine spaces, and their topology is obtained by dehomogenizing polynomials.

Problem. For nonnegative integers r, s , put $N = (r+1)(s+1) - 1$. Define the Segre map of sets

$$\sigma : \mathbb{P}^r \times \mathbb{P}^s \rightarrow \mathbb{P}^N, \quad ([a_i], [b_j]) \mapsto [a_i b_j]_{i,j}.$$

Prove that its image is a projective variety.

Solution. Rescaling either input rescales all output coordinates by a common nonzero scalar, so the map is well-defined. Let

$$\theta : k[z_{ij}] \rightarrow k[x_0, \dots, x_r, y_0, \dots, y_s], \quad z_{ij} \mapsto x_i y_j.$$

Its kernel $\mathfrak{a}$ is prime because its image is a domain, and homogeneous because a degree- d polynomial maps to one of total degree $2d$. Substitution gives $\text{im } \sigma \subseteq V_+(\mathfrak{a})$.

The kernel contains all two-by-two minors

$$z_{ij}z_{i'j'} - z_{ij'}z_{i'j}.$$

For a projective common zero, choose a nonzero entry $z_{i_0j_0}$. Set $a_i = z_{ij_0}$ and $b_j = z_{i_0j}/z_{i_0j_0}$. The minor relations imply $z_{ij} = a_ib_j$ for every pair i, j . Both vectors are nonzero since their indicated entries are nonzero. This point is therefore in the Segre image. Thus $\text{im } \sigma = V_+(\mathbf{a})$. It is nonempty, closed, and irreducible by primality, as required.

The same calculation also proves injectivity: the projective column and row factors of a nonzero rank-one matrix are uniquely determined. $\square$

Exercise I.2.15

Permitted context.

Work over an algebraically closed field k . For a projective algebraic set Y , $I(Y)$ denotes its homogeneous vanishing ideal. For homogeneous ideals with nonempty zero set, $I(V_+(\mathbf{a})) = \sqrt{\mathbf{a}}$. An algebraic set is irreducible exactly when its vanishing ideal is prime. For a projective variety, $\dim Y = \dim k[x_0, \dots, x_n]/I(Y) - 1$. Standard charts are affine spaces, and their topology is obtained by dehomogenizing polynomials.

Every proper closed subset of $\mathbb{P}^1$ is finite; in particular any two nonempty open subsets intersect. The Segre image consists of projective rank-one matrices. Its inverse on the chart with one nonzero entry is obtained by taking row and column ratios, which are continuous in affine charts.

Problem. Let $Q = V_+(xy - zw) \subseteq \mathbb{P}^3$.

- (a) Identify Q with the Segre image of $\mathbb{P}^1 \times \mathbb{P}^1$.
- (b) Describe two families of lines parametrized by $\mathbb{P}^1$, with different lines in the same family disjoint and every line in one family meeting each line of the other in one point.
- (c) Exhibit a different irreducible curve on Q , and show that the Segre bijection is not a homeomorphism from the product of the two Zariski topologies.

Solution. (a) Use the coordinates

$$([a_0 : a_1], [b_0 : b_1]) \mapsto [x : y : z : w] = [a_0b_0 : a_1b_1 : a_0b_1 : a_1b_0].$$

The equation $xy - zw = 0$ is the determinant of $\begin{pmatrix} x & z \\ w & y \end{pmatrix}$. Its nonzero matrices of determinant zero are exactly the rank-one matrices, hence exactly this Segre image. The equation is irreducible: as a polynomial in y , it is primitive and linear over $k[x, z, w]$, with coprime coefficients $x, -zw$, so Gauss's lemma applies. Thus Q is an irreducible surface.

(b) Fixing $[a]$ and varying $[b]$ gives a line L_a , since its coordinates span a two-dimensional vector subspace. Fixing $[b]$ gives the line M_b . Uniqueness of the two projective factors of a rank-one matrix shows that distinct L_a 's are disjoint, as are distinct M_b 's, and that

$$L_a \cap M_b = \{\sigma([a], [b])\}.$$

- (c) The diagonal pair $([s : t], [s : t])$ has image

$$[s^2 : t^2 : st : st].$$

Its image is $Q \cap V_+(z-w)$: a rank-one matrix is symmetric exactly when its row and column factors are proportional. In the plane $z = w$, this is the irreducible conic $xy = z^2$. For example it is the degree-two Veronese image of $\mathbb{P}^1$, so is an irreducible closed curve, and neither factor is constant. It is therefore none of the ruling lines in (b).

The diagonal $\Delta \subseteq \mathbb{P}^1 \times \mathbb{P}^1$ is dense for the product topology: every nonempty basic open $U \times V$ meets it, since $U \cap V \neq \emptyset$. It is a proper subset, hence not closed in that topology. Its image on Q is the closed conic just described. Therefore the Segre bijection from the product topological space cannot be a homeomorphism. $\square$

Exercise I.2.16

Permitted context.

Work over an algebraically closed field k . For a projective algebraic set Y , $I(Y)$ denotes its homogeneous vanishing ideal. For homogeneous ideals with nonempty zero set, $I(V_+(\mathfrak{a})) = \sqrt{\mathfrak{a}}$. An algebraic set is irreducible exactly when its vanishing ideal is prime. For a projective variety, $\dim Y = \dim k[x_0, \dots, x_n]/I(Y) - 1$. Standard charts are affine spaces, and their topology is obtained by dehomogenizing polynomials.

The projective twisted cubic in coordinates w, x, y, z is the closure of $[1 : t : t^2 : t^3]$. A prime ideal defines an irreducible algebraic set, and nonzero linear forms define hyperplanes.

Problem. (a) Show that the intersection in $\mathbb{P}^3$ of the quadrics

$$Q_1 = V_+(x^2 - yw), \quad Q_2 = V_+(xy - zw)$$

is the union of a twisted cubic and a line.

(b) For $C = V_+(x^2 - yz)$ and $L = V_+(y)$ in $\mathbb{P}^2$, show that their intersection is a single point P , but $I(C) + I(L) \neq I(P)$.

Solution. (a) On $w \neq 0$, normalize $w = 1$. The two equations then give $y = x^2$, $z = x^3$, exactly the affine twisted cubic. On $w = 0$, the first equation forces $x = 0$, and the second imposes no further restriction. These points form the line $M = V_+(w, x)$. The projective twisted cubic T lies on both quadrics, by substituting $[w : x : y : z] = [s^3 : s^2t : st^2 : t^3]$. We have therefore proved

$$Q_1 \cap Q_2 = T \cup M.$$

Neither curve contains the other: T has points with $w \neq 0$, whereas $[0 : 0 : 1 : 0] \in M$ is not on T . They meet at $[0 : 0 : 0 : 1]$. Both quadrics are irreducible by Gauss's lemma, applied to the primitive linear polynomials in w with coefficients $-y, x^2$ and $-z, xy$, respectively. Thus this is an intersection of varieties that is reducible.

(b) Setting $y = 0$ in the conic equation gives $x^2 = 0$. The only projective common zero is $P = [0 : 0 : 1]$, with ideal $I(P) = (x, y)$. The conic equation is irreducible by the same primitive-linear argument, so $I(C) = (x^2 - yz)$. Consequently

$$I(C) + I(L) = (x^2 - yz, y) = (x^2, y) \subsetneq (x, y).$$

The containment is strict because the image of x in $k[x, y, z]/(x^2, y) \cong k[x, z]/(x^2)$ is nonzero. Its square vanishes, explaining why the sum ideal has the right zero set but is not the vanishing ideal of that point. $\square$

Exercise I.2.17**Permitted context.**

Work over an algebraically closed field k . For a projective algebraic set Y , $I(Y)$ denotes its homogeneous vanishing ideal. For homogeneous ideals with nonempty zero set, $I(V_+(\mathfrak{a})) = \sqrt{\mathfrak{a}}$. An algebraic set is irreducible exactly when its vanishing ideal is prime. For a projective variety, $\dim Y = \dim k[x_0, \dots, x_n]/I(Y) - 1$. Standard charts are affine spaces, and their topology is obtained by dehomogenizing polynomials.

If an ideal of a polynomial ring is generated by q elements, every prime minimal over it has height at most q . For a projective variety $Y \subseteq \mathbb{P}^n$, $\text{ht } I(Y) = n - \dim Y$. A strict complete intersection has its homogeneous ideal generated by $n - \dim Y$ elements; a set-theoretic complete intersection is the common zero set of that many hypersurfaces.

In coordinates w, x, y, z , the projective twisted cubic T has ideal

$$I(T) = (wy - x^2, wz - xy, xz - y^2).$$

These generators follow from the parametrization $[s^3 : s^2t : st^2 : t^3]$: straightening pairs of indices of difference at least two gives one monomial for each degree and index sum, whose images are linearly independent.

Problem. (a) If a projective variety $Y \subseteq \mathbb{P}^n$ is the zero set of a homogeneous ideal generated by q elements, prove $\dim Y \geq n - q$.

(b) Prove that every strict complete intersection is a set-theoretic complete intersection.

(c) Show that the twisted cubic is not a strict complete intersection, but is the set-theoretic intersection of a quadric and a cubic surface.

(d) Explain the distinction illustrated by (c) in relation to the source's research question whether all irreducible projective space curves are set-theoretic intersections of two surfaces.

Solution. (a) Write $Y = V_+(\mathfrak{a})$. Since Y is nonempty, the projective Nullstellensatz gives $\sqrt{\mathfrak{a}} = I(Y)$, a prime ideal. It is the unique minimal prime over $\mathfrak{a}$. The height theorem gives $n - \dim Y = \text{ht } I(Y) \leq q$, as desired.

(b) If $I(Y)$ is generated by $n - r$ homogeneous polynomials, where $r = \dim Y$, then their common zero set is Y . If each hypersurface is required to be irreducible, choose from each generator an irreducible homogeneous factor vanishing on Y . Such a factor exists because $I(Y)$ is prime. The zero set of the chosen factor is contained in that of the original generator and contains Y . Their intersection is therefore still exactly Y . For $Y = \mathbb{P}^n$, use the empty intersection.

(c) The three displayed quadrics are linearly independent, and $I(T)$ has no degree-one elements, since the four parametrizing monomials are linearly independent. Put $\mathfrak{m} = (w, x, y, z)$. Then $\mathfrak{m}I(T)$ has no nonzero degree-two element, so the three quadrics remain independent in $I(T)/\mathfrak{m}I(T)$. This quotient cannot be spanned by two generators. Hence $I(T)$ cannot be generated by two polynomials.

On the other hand, take

$$Q = wy - x^2, \quad F = wz^2 - 2xyz + y^3.$$

Both vanish on T . At a common zero with $w \neq 0$, the equation $Q = 0$ gives $y = x^2/w$, and substitution in F gives

$$F = w \left(z - \frac{x^3}{w^2} \right)^2.$$

Thus $z = x^3/w^2$, so this point is on T . At $w = 0$, the equations force $x = 0$ and then $y = 0$, leaving the single point $[0 : 0 : 0 : 1]$, also on T . Therefore $V_+(Q) \cap V_+(F) = T$ set-theoretically, in every characteristic, including two. The quadric is irreducible. The cubic is irreducible as well: view it as a polynomial in w over $k[x, y, z]$. Its coefficients z^2 and $y^3 - 2xyz$ are coprime, so Gauss's lemma reduces irreducibility to that of a linear polynomial over the fraction field.

(d) Part (c) concerns two different requirements. Two equations can have radical ideal $I(T)$ even though $I(T)$ itself requires three generators. Thus failure of a strict complete intersection supplies no counterexample to the set-theoretic question. The general question asks for equations for every irreducible projective space curve; the explicit equations above establish only the twisted-cubic case. $\square$

I.3 Morphisms

Exercise I.3.1

Permitted context.

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

A projective variety has only constant global regular functions. A plane curve is the zero set of one irreducible polynomial, homogeneous in the projective case. A quotient of $k[x, y, z]$ by a proper ideal generated by two elements has dimension at least one. The homogeneous Nullstellensatz says that an empty projective zero set has affine cone consisting only of the origin. Conics mean irreducible degree-two curves. Affine linear changes put an affine conic into either a graph of a quadratic polynomial or $xy = 1$: the leading quadratic splits into either two different linear factors or a repeated factor. Polynomial rings are unique factorization domains.

Problem. (a) Classify affine plane conics up to isomorphism as $\mathbb{A}^1$ or $\mathbb{A}^1 \setminus \{0\}$.

(b) Show that no proper open subset of $\mathbb{A}^1$ is isomorphic to $\mathbb{A}^1$.

(c) Show that every projective plane conic is isomorphic to $\mathbb{P}^1$.

(d) Distinguish $\mathbb{A}^2$ and $\mathbb{P}^2$ topologically.

(e) Show that a variety which is both affine and projective up to isomorphism is a point.

Solution. (a) Projection onto the parameter coordinate identifies a quadratic graph with $\mathbb{A}^1$, with polynomial inverse. The maps $t \mapsto (t, t^{-1})$ and $(x, y) \mapsto x$ identify $xy = 1$ with $\mathbb{A}^1 \setminus \{0\}$. These are morphisms in both directions.

(b) A nonempty proper open subset is $U = \mathbb{A}^1 \setminus \{a_1, \dots, a_s\}$, where $s \geq 1$. The regular function $t - a_1$ is a nonconstant unit on U . An isomorphism would carry it to a nonconstant unit in $k[t]$, whose units are precisely $k^\times$, a contradiction. The empty open subset cannot be isomorphic to a nonempty variety.

(c) Choose a point on the conic and make it $[0 : 0 : 1]$. The equation is

$$z(ax + by) + q(x, y) = 0.$$

The linear form $ax + by$ is nonzero; otherwise the binary quadratic q splits over k , contradicting irreducibility. Change x, y so that this form is x . The coefficient of y^2 in q is nonzero, since otherwise the whole equation is divisible by x . Replacing z by $z + \alpha x + \beta y$, and rescaling, gives $xz = y^2$. No division by 2 was used. The morphism

$$[s : t] \longmapsto [s^2 : st : t^2]$$

has image this conic. On $x \neq 0$ its inverse is $[x : y]$; on $z \neq 0$ its inverse is $[y : z]$. These charts cover the conic and the inverses agree on their overlap. Thus it is an isomorphism in every characteristic.

(d) Any two projective plane curves meet. Otherwise the cone of their intersection would be just the origin, so the quotient by their two homogeneous equations would have dimension zero, contrary to the dimension bound in the context. But the two closed affine lines $x = 0$ and $x = 1$ are disjoint. A homeomorphism would send these to disjoint closed irreducible one-dimensional subsets of $\mathbb{P}^2$, hence to projective plane curves, giving a contradiction.

(e) The affine coordinate ring of such a variety is its ring of global regular functions. Projectivity makes that ring k , whose affine variety is one point. $\square$

Exercise I.3.2

Permitted context.

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

A proper closed subset of an irreducible curve is finite. Thus its Zariski topology on points is the cofinite topology.

Problem. (a) Prove that $t \mapsto (t^2, t^3)$ is a homeomorphic morphism from $\mathbb{A}^1$ onto $V(y^2 - x^3)$, but is not an isomorphism.

(b) In characteristic $p > 0$, prove the same assertions for the morphism $\mathbb{A}^1 \rightarrow \mathbb{A}^1$, $t \mapsto t^p$.

Solution. (a) Polynomial coordinates define a morphism. Away from the origin, a point of the cusp has $x \neq 0$, and $t = y/x$ satisfies $t^2 = x, t^3 = y$. The origin has the unique preimage 0. Hence the map is bijective, and it is a homeomorphism because both curves have the cofinite topology. Its pullback has image $k[t^2, t^3] \subset k[t]$. This image does not contain t , since its monomials have exponents zero or at least two. The pullback is not an isomorphism, so neither is the morphism.

(b) Algebraic closure gives a root of $T^p - a$ for each a . The identity $T^p - a = (T - b)^p$ for any such root makes it unique. Thus the polynomial morphism is bijective and again a homeomorphism. Its pullback $k[u] \rightarrow k[t]$, $u \mapsto t^p$, has image $k[t^p]$, which does not contain t . Here this is the k -morphism defined by that polynomial; it fixes constant functions under pullback. $\square$

Exercise I.3.3**Permitted context.**

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

Problem. Let $\varphi : X \rightarrow Y$ be a morphism.

- (a) Construct the induced local homomorphism $\varphi_P^* : \mathcal{O}_{\varphi(P), Y} \rightarrow \mathcal{O}_{P, X}$.
- (b) Prove that φ is an isomorphism exactly when it is a homeomorphism and all these local homomorphisms are isomorphisms.
- (c) If $\varphi(X)$ is dense in Y , prove that every φ_P^* is injective.

Solution. (a) A germ represented by f on a neighborhood V of $\varphi(P)$ is sent to the germ of $f \circ \varphi$ on $\varphi^{-1}(V)$. Agreement on a smaller neighborhood is preserved by inverse image, so this is well defined. It respects addition, multiplication, and k . The pulled-back value at P is $f(\varphi(P))$. Consequently the inverse image of the maximal ideal is exactly the maximal ideal at $\varphi(P)$, making it a local homomorphism.

(b) An inverse morphism supplies inverse maps on germs and on points. Conversely suppose the stated conditions hold, and write $\psi = \varphi^{-1}$ for the continuous inverse. For a regular function g on an open $U \subseteq X$ and $P \in U$, surjectivity on local rings gives a regular function h near $\varphi(P)$ whose pullback agrees with g near P . Shrink the latter neighborhood to $W \subseteq U$ inside the inverse image of the domain of h . Because φ is a homeomorphism, $\varphi(W)$ is open. On it, $g \circ \psi = h$. Thus $g \circ \psi$ is locally regular everywhere on $\varphi(U)$, hence regular. This proves that ψ is a morphism.

(c) Let a germ represented by f on V pull back to zero at P . Then $f \circ \varphi$ vanishes on a nonempty open subset of $\varphi^{-1}(V)$. The latter is irreducible, so a regular function zero on that dense open subset is zero everywhere there. The subset $\varphi(X) \cap V$ is dense in V . Thus f vanishes on a dense subset of V , and hence identically. Its germ is zero, as required. $\square$

Exercise I.3.4**Permitted context.**

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

For $d \geq 1$, the degree- d Veronese map has coordinates $y_\alpha = x^\alpha$, one for each multi-index α of total degree d . It is a bijection onto a closed projective subvariety V . Write e_i for the i -th coordinate vector.

Problem. Prove that the degree- d Veronese map $\nu_d : \mathbb{P}^n \rightarrow V$ is an isomorphism.

Solution. On $x_i \neq 0$, the ratios y_α/y_{de_i} are monomials in x_j/x_i , so the map is a morphism on these covering charts. On $V \cap \{y_{de_i} \neq 0\}$ recover the affine coordinates by

$$\frac{x_j}{x_i} = \frac{y_{(d-1)e_i + e_j}}{y_{de_i}}.$$

These are regular functions. The charts cover V because every image point has some $x_i^d \neq 0$. The displayed formula gives the set-theoretic inverse on each chart, so these inverse morphisms agree on overlaps and glue. This proves the assertion, including $d = 1$, without a separability or characteristic restriction. $\square$

Exercise I.3.5**Permitted context.**

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

The degree- d Veronese map is an isomorphism onto a closed subvariety of projective space. The complement of a hyperplane in $\mathbb{P}^N$ is isomorphic to $\mathbb{A}^N$. Closed subvarieties of affine space are affine.

Problem. If $H \subseteq \mathbb{P}^n$ is a hypersurface, prove that $\mathbb{P}^n \setminus H$ is affine.

Solution. Write $H = V_+(F)$, where F is nonzero homogeneous of positive degree d . In the degree- d Veronese coordinates, F becomes a nonzero linear form L . If V is the Veronese image, then

$$\nu_d(\mathbb{P}^n \setminus H) = V \cap (\mathbb{P}^N \setminus V_+(L)).$$

The right side is closed in the affine hyperplane complement, because V is closed in $\mathbb{P}^N$. It is therefore affine. Restricting the Veronese isomorphism proves the claim. $\square$

Exercise I.3.6**Permitted context.**

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

For $A = k[x, y]$, the principal open $D(g)$ has global regular-function ring A_g . The ring A is a unique factorization domain. Isomorphisms of affine coordinate rings correspond to isomorphisms of affine varieties.

Problem. Prove that $X = \mathbb{A}^2 \setminus \{(0, 0)\}$ is not affine.

Solution. The opens $D(x)$ and $D(y)$ cover X , so the regular functions on X , viewed in $k(x, y)$, form

$$\mathcal{O}(X) = A_x \cap A_y = A.$$

For the last equality, a reduced fraction in A_x has denominator dividing a power of x ; membership in A_y forces it also to divide a power of y . Unique factorization makes that denominator a unit.

The inclusion $j : X \rightarrow \mathbb{A}^2$ induces the identity isomorphism $A \rightarrow \mathcal{O}(X)$. If X were affine, the affine coordinate-ring equivalence would make j an isomorphism. But j misses the origin. This contradiction proves nonaffineness. $\square$

Exercise I.3.7**Permitted context.**

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

The complement of a projective hypersurface is affine. A projective variety has only constant global regular functions. An affine variety whose coordinate ring is k is a point. Every projective plane curve is a hypersurface.

Problem. (a) Show that two curves in $\mathbb{P}^2$ always meet.

(b) Show that a positive-dimensional projective variety $Y \subseteq \mathbb{P}^n$ meets every hypersurface H .

Solution. (b) If $Y \cap H$ were empty, Y would be a closed subvariety of the affine variety $\mathbb{P}^n \setminus H$; hence Y would be affine. Its global regular functions are k , since it is projective. Thus it would be a point, contradicting $\dim Y \geq 1$.

(a) Apply (b) with one curve as Y and the other as H . If the word curve allows a reducible algebraic set, choose a one-dimensional irreducible component from each and apply this result. $\square$

Exercise I.3.8**Permitted context.**

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

Problem. For different coordinate hyperplanes $H_i, H_j \subseteq \mathbb{P}^n$, prove that every regular function on $\mathbb{P}^n \setminus (H_i \cap H_j)$ is constant.

Solution. Relabel so that $i = 0, j = 1$, and put $U_\ell = \{x_\ell \neq 0\}$. The indicated open set is $U_0 \cup U_1$. On U_0 , write $t_\ell = x_\ell/x_0$. A regular function is a polynomial $P(t_1, \dots, t_n)$. On U_1 , use $s_0 = x_0/x_1$ and $s_\ell = x_\ell/x_1$ for $\ell \geq 2$. On the overlap,

$$t_1 = s_0^{-1}, \quad t_\ell = s_\ell s_0^{-1} \quad (\ell \geq 2).$$

A monomial $t_1^{a_1} \cdots t_n^{a_n}$ therefore becomes

$$s_0^{-(a_1 + \cdots + a_n)} s_2^{a_2} \cdots s_n^{a_n}.$$

Different exponent vectors give different Laurent monomials, so no cancellation can remove their negative powers of s_0 . For P to be polynomial in the U_1 coordinates, all nonzero-degree monomials must have coefficient zero. Thus P and the original function are constant. For $n = 1$ this is the same argument with no s_ℓ , $\ell \geq 2$. $\square$

Exercise I.3.9**Permitted context.**

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

For a k -algebra R and a maximal ideal $\mathfrak{m}$ with residue field k , the vector space $\mathfrak{m}/\mathfrak{m}^2$ and its dimension are invariants of that local k -algebra. Every maximal ideal of $k[s, t]$ has this dimension two.

Problem. Let $X = \mathbb{P}^1$ and let $Y \subseteq \mathbb{P}^2$ be its degree-two Veronese image. Although $X \cong Y$, show that their homogeneous coordinate rings are not isomorphic, even as ungraded k -algebras.

Solution. The rings are

$$S(X) = k[s, t], \quad S(Y) = k[a, b, c]/(ac - b^2).$$

The second equality follows from the parametrization $[s : t] \mapsto [s^2 : st : t^2]$: its image is exactly the irreducible conic $ac = b^2$. At the maximal ideal $\mathfrak{m} = (a, b, c)$ of the second ring, the relation has degree two, so it imposes no linear relation modulo $\mathfrak{m}^2$. Thus $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 3$. An isomorphism with $k[s, t]$ would give a maximal ideal in that polynomial ring with the same dimension, which is impossible. The Veronese map nevertheless gives $X \cong Y$; hence the homogeneous coordinate ring depends on the projective embedding. $\square$

Exercise I.3.10

Permitted context.

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

A subvariety is an irreducible locally closed subset with its induced variety structure. A regular function on a subvariety is locally the restriction of a quotient of ambient affine-chart polynomials with denominator nonzero at the point in question.

Problem. Let $\varphi : X \rightarrow Y$ be a morphism and let $X' \subseteq X, Y' \subseteq Y$ be subvarieties with $\varphi(X') \subseteq Y'$. Prove that $\varphi|_{X'} : X' \rightarrow Y'$ is a morphism.

Solution. The restricted map is continuous for the subspace topologies. Let g be regular on an open subset $V' \subseteq Y'$. Near any point $Q \in V'$, choose an ambient affine chart and write $g = a/b$ on a neighborhood in Y' , with $b(Q) \neq 0$. After shrinking an open neighborhood V of Q in Y , the same quotient defines a regular function h on V whose restriction agrees with g on $Y' \cap V \subseteq V'$. The pullback $h \circ \varphi$ is regular on $\varphi^{-1}(V)$. Restricting it to X' remains regular, and there it equals $g \circ \varphi|_{X'}$. This works near every point of the inverse image of V' . The pullback of g is therefore regular, proving the assertion. $\square$

Exercise I.3.11**Permitted context.**

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

Prime ideals in an affine coordinate ring correspond to irreducible closed subvarieties. The primes in $A_{\mathfrak{m}}$ correspond, by extension and contraction, to the primes of A contained in $\mathfrak{m}$. A nonempty open subset of an irreducible space is dense.

Problem. For $P \in X$, give a bijection between the prime ideals of $\mathcal{O}_{P,X}$ and the closed subvarieties of X containing P .

Solution. Choose an affine open neighborhood U of P , with ring A and maximal ideal $\mathfrak{m}_P$. A prime of $\mathcal{O}_{P,X} = A_{\mathfrak{m}_P}$ contracts to a prime $\mathfrak{p} \subseteq \mathfrak{m}_P$. The subvariety $V_U(\mathfrak{p})$ contains P ; take its closure in X .

Conversely, if $Z \subseteq X$ is closed irreducible and contains P , then $Z \cap U$ is nonempty, closed in U , and irreducible. Its prime ideal is contained in $\mathfrak{m}_P$; extend this ideal to $A_{\mathfrak{m}_P}$.

The two constructions are inverse: taking the closure in X and intersecting with U recovers a closed subset of U , and $\overline{Z \cap U} = Z$ by density. Extension and contraction of primes are inverse as well. Intrinsically, the prime associated to Z consists of germs whose restriction vanishes on a neighborhood of P in Z . This description also proves independence of the affine neighborhood. $\square$

Exercise I.3.12**Permitted context.**

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

For a finitely generated integral k -algebra A , dimension equals the transcendence degree of its fraction field, and $\text{ht } \mathfrak{p} + \dim A/\mathfrak{p} = \dim A$. Maximal ideals have residue field k . Localization identifies the primes of $A_{\mathfrak{p}}$ with the primes contained in $\mathfrak{p}$, so $\dim A_{\mathfrak{p}} = \text{ht } \mathfrak{p}$. Every nonempty open subset of a variety has its dimension and function field.

Problem. Prove that $\dim \mathcal{O}_{P,X} = \dim X$ for every point P of a variety X .

Solution. Choose an affine open neighborhood U of P , with coordinate ring A and maximal ideal $\mathfrak{m}$. Then $\mathcal{O}_{P,X} = A_{\mathfrak{m}}$. Since $A/\mathfrak{m} = k$ has dimension zero, the affine dimension formula gives

$$\dim \mathcal{O}_{P,X} = \text{ht } \mathfrak{m} = \dim A = \dim U = \dim X.$$

Here points are the closed points of the classical variety. The assertion is not a statement that every scheme-theoretic point has local-ring dimension $\dim X$. $\square$

Exercise I.3.13

Permitted context.

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

For a finitely generated integral k -algebra A , dimension equals the transcendence degree of its fraction field, and $\text{ht } \mathfrak{p} + \dim A/\mathfrak{p} = \dim A$. Maximal ideals have residue field k . Localization identifies the primes of $A_{\mathfrak{p}}$ with the primes contained in $\mathfrak{p}$, so $\dim A_{\mathfrak{p}} = \text{ht } \mathfrak{p}$. Every nonempty open subset of a variety has its dimension and function field.

For an irreducible locally closed subset Y , its closure has the same function field and dimension. The localization $A_{\mathfrak{p}}$ is local with residue field $\text{Frac}(A/\mathfrak{p})$. Eisenstein's criterion applies to polynomials over $k[t_1, \dots, t_d]$.

Problem. For a subvariety $Y \subseteq X$, consider pairs (U, f) , where $U \subseteq X$ is open, $U \cap Y \neq \emptyset$, and f is regular on U ; identify pairs whose functions agree on their overlap. Prove that their ring $\mathcal{O}_{Y,X}$ is local, has residue field $K(Y)$, and has dimension $\dim X - \dim Y$. Identify the cases $Y = \{P\}$ and $Y = X$, and explain why $K(Y)$ is not algebraically closed when $\dim Y > 0$.

Solution. Each representative defines a rational function on the irreducible variety X . Two representatives are equivalent exactly when they give the same rational function. Intersections of their domains still meet Y , since their intersections with irreducible Y are nonempty opens. Thus sum and product of representatives define a subring of $K(X)$.

Choose an affine open W meeting Y , put $A = A(W)$, and let $\mathfrak{p}$ be the prime ideal of $\overline{Y \cap W}$ in W . We claim that this subring is $A_{\mathfrak{p}}$. For $a/b \in A_{\mathfrak{p}}$, b does not vanish identically on $Y \cap W$. Thus $D_W(b)$ meets Y , and a/b is an allowed representative. Conversely take (U, f) and choose a point in $U \cap W \cap Y$. Near it $f = a/b$, with $a, b \in A$ and b nonzero at that point. Then $b \notin \mathfrak{p}$, and the rational function of this representative lies in $A_{\mathfrak{p}}$.

This identifies the maximal ideal with $\mathfrak{p}A_{\mathfrak{p}}$, and the residue field with

$$\text{Frac}(A/\mathfrak{p}) = K(Y).$$

The dimension formula gives

$$\dim \mathcal{O}_{Y,X} = \text{ht } \mathfrak{p} = \dim X - \dim Y.$$

For $Y = \{P\}$ this is the ordinary ring of germs at P . For $Y = X$, $\mathfrak{p} = 0$, and it is $K(X)$.

Finally let $d = \dim Y > 0$ and take a transcendence basis $t_1, \dots, t_d$ of $K(Y)/k$. The extension of $F = k(t_1, \dots, t_d)$ to $K(Y)$ is finite, say of degree e . For every integer $N > e$, the polynomial $T^N - t_1$ is irreducible over F by Eisenstein at t_1 . If $K(Y)$ were algebraically closed, it would contain a root, and hence an F -subextension of degree $N > e$, impossible. $\square$

Exercise I.3.14

Permitted context.

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

Problem. Let $H \cong \mathbb{P}^n$ be a hyperplane in $\mathbb{P}^{n+1}$ and $P \notin H$. Project a point $Q \neq P$ onto the intersection of the line PQ with H .

(a) Prove that this projection is a morphism.

(b) For the twisted cubic $[t : u] \mapsto [t^3 : t^2u : tu^2 : u^3]$, project from $[0 : 0 : 1 : 0]$ onto $z = 0$, and find its plane image.

Solution. (a) Choose coordinates with $H = \{x_{n+1} = 0\}$ and $P = [0 : \dots : 0 : 1]$. The map is

$$[x_0 : \dots : x_{n+1}] \mapsto [x_0 : \dots : x_n].$$

Away from P these coordinates are not all zero. On every chart $x_i \neq 0$, $i \leq n$, the target affine coordinates are the regular ratios x_j/x_i . These charts cover the domain, proving that the map is a morphism.

(b) The specified point is not on the twisted cubic: $x = w = 0$ would force $t = u = 0$. After dropping z , the parametrization becomes

$$[t : u] \mapsto [x : y : w] = [t^3 : t^2u : u^3].$$

It satisfies $y^3 = x^2w$. Conversely on $x \neq 0$, normalize $x = 1$; every point of this equation is $[1 : a : a^3]$, obtained from $[t : u] = [1 : a]$. On $x = 0$ the equation forces $y = 0$, leaving $[0 : 0 : 1]$, obtained from $[0 : 1]$. Thus the image is exactly $V_+(y^3 - x^2w)$. At $[0 : 0 : 1]$, the chart $w = 1$ has equation $x^2 = y^3$, the cuspidal cubic, with normalization parameter $x = v^3, y = v^2$. This description holds in characteristics two and three as well. $\square$

Exercise I.3.15**Permitted context.**

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

The polynomial functions on an affine variety are its coordinate ring. Tensoring vector spaces over a field preserves injections. Noether normalization gives a polynomial subalgebra of dimension $\dim A$ over which a finitely generated integral k -algebra A is a finite module. Integral finite extensions preserve dimension.

Problem. For affine varieties $X \subseteq \mathbb{A}^n$ and $Y \subseteq \mathbb{A}^m$, equip $X \times Y \subseteq \mathbb{A}^{n+m}$ with its induced algebraic-set structure.

- (a) Prove that it is irreducible.
- (b) Identify its coordinate ring with $A(X) \otimes_k A(Y)$.
- (c) Verify the categorical product property among varieties.
- (d) Prove $\dim(X \times Y) = \dim X + \dim Y$.

Solution. (a) The equations of X in the first coordinates and Y in the second show that $X \times Y$ is closed. Suppose $X \times Y = Z_1 \cup Z_2$ with Z_i closed, and put

$$X_i = \{x \in X : \{x\} \times Y \subseteq Z_i\}.$$

For fixed x , the fiber is isomorphic to irreducible Y , so it lies in one Z_i . Hence $X = X_1 \cup X_2$. To see X_i is closed, take each polynomial defining Z_i and express its restriction as $\sum_{\ell=1}^r a_\ell(x)b_\ell(y)$, choosing the finitely many b_ℓ linearly independent in $A(Y)$. The polynomial vanishes identically on $\{x\} \times Y$ exactly when all $a_\ell(x)$ vanish. Thus X_i is cut out by polynomial equations in X . Irreducibility of X gives $X = X_i$ for one i , and then $X \times Y = Z_i$.

- (b) The homomorphism

$$A(X) \otimes_k A(Y) \longrightarrow A(X \times Y), \quad a \otimes b \longmapsto ((x, y) \mapsto a(x)b(y))$$

is surjective because the coordinate functions generate the target. If an element in its kernel is written $\sum a_\ell \otimes b_\ell$ with the b_ℓ independent, evaluation at every x shows $a_\ell(x) = 0$ for every ℓ . Thus all $a_\ell = 0$; the homomorphism is injective.

(c) The projections are polynomial maps. Given morphisms $f : Z \rightarrow X$, $g : Z \rightarrow Y$, the only possible map is $h(z) = (f(z), g(z))$. Its affine coordinates are the regular coordinate functions of f and g , so it is a morphism to $\mathbb{A}^{n+m}$ with image in the closed subvariety $X \times Y$. It is therefore the required unique morphism.

(d) Choose normalization subalgebras $k[u_1, \dots, u_r] \subseteq A(X)$ and $k[v_1, \dots, v_s] \subseteq A(Y)$, with $r = \dim X$, $s = \dim Y$. Tensoring embeds $k[u_1, \dots, u_r, v_1, \dots, v_s]$ in the ring of (b). Tensor products of finite module generating sets show that the latter is finite over this polynomial subalgebra. Therefore its dimension is $r + s$, proving the formula. $\square$

Exercise I.3.16**Permitted context.**

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

The Segre embedding identifies $\mathbb{P}^n \times \mathbb{P}^m$ with the closed variety of rank-one matrices in $\mathbb{P}^{(n+1)(m+1)-1}$. Its affine chart where the (i, j) -entry is nonzero is the product of the standard affine charts $x_i \neq 0, y_j \neq 0$, with coordinates obtained by row and column ratios. Affine products are irreducible and satisfy the categorical product property. A quasi-projective variety is a locally closed irreducible subset of projective space.

Problem. For quasi-projective $X \subseteq \mathbb{P}^n$ and $Y \subseteq \mathbb{P}^m$, use the Segre embedding to give $X \times Y$ its induced structure.

- (a) Show that $X \times Y$ is quasi-projective.
- (b) Show that it is projective if both factors are projective.
- (c) Prove the categorical product property.

Solution. The Segre charts show that both projections are morphisms: their coordinates on each chart are the row and column ratios.

(a) Write $X = C \cap U$ and $Y = D \cap V$, with C, D closed and U, V open in their projective spaces. Inside the Segre variety, $X \times Y$ is the intersection of the closed set $p_1^{-1}(C) \cap p_2^{-1}(D)$ with the open set $p_1^{-1}(U) \cap p_2^{-1}(V)$, hence is locally closed.

Cover X and Y by nonempty affine opens X_a, Y_b . Their products are open in $X \times Y$ by the projections. The Segre chart structure on each agrees with the affine product structure: the charts on the ambient product have the ordinary coordinate ratios, and restriction gives those on the subvarieties. Thus each $X_a \times Y_b$ is irreducible. Any two such opens intersect, since any two nonempty opens of either factor intersect. An open cover by irreducible sets with pairwise nonempty intersections has irreducible union: two nonempty open subsets of the union meet by passing through a common covering set. Consequently $X \times Y$ is irreducible and quasi-projective.

(b) If X, Y are closed, their inverse images under the projections have closed intersection. Thus the product is closed in the projective Segre variety and is projective.

(c) The projections restrict to morphisms on $X \times Y$. Given $f : Z \rightarrow X$ and $g : Z \rightarrow Y$, define $h(z) = (f(z), g(z))$. The opens $f^{-1}(X_a) \cap g^{-1}(Y_b)$ cover Z . On each, the affine product property says that h is a morphism to $X_a \times Y_b$. These morphisms agree as maps on overlaps and glue to a morphism $Z \rightarrow X \times Y$. Its projections are f, g , and uniqueness follows pointwise. $\square$

Exercise I.3.17**Permitted context.**

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

A domain is normal if it is integrally closed in its fraction field. Polynomial rings over a field are unique factorization domains and therefore normal. Localizations of normal domains are normal: if z is integral over $S^{-1}A$, clearing denominators in a monic equation makes sz integral over A for some $s \in S$. For any domain A , $A = \bigcap_{\mathfrak{m} \text{ maximal}} A_{\mathfrak{m}}$ inside its fraction field. Indeed, if $z \notin A$, the proper ideal $\{a \in A : az \in A\}$ is contained in a maximal ideal at which z is still absent. A variety is normal at a point when its local ring is normal.

Every projective conic is isomorphic to $\mathbb{P}^1$. The integral closure of a finitely generated integral k -algebra in its fraction field is finite as a module over that algebra, and hence finitely generated over k .

Problem. (a) Prove that projective plane conics are normal.

(b) Prove that the projective quadrics $xy = zw$ and $xy = z^2$ in $\mathbb{P}^3$ are normal.

(c) Show that the affine cusp $y^2 = x^3$ is not normal.

(d) For affine Y , prove that Y is normal exactly when $A(Y)$ is normal.

(e) Construct the normalization $\pi : \tilde{Y} \rightarrow Y$ of an affine variety and show that every dominant morphism $f : Z \rightarrow Y$ from a normal variety has a unique morphism $\tilde{f} : Z \rightarrow \tilde{Y}$ satisfying $\pi\tilde{f} = f$.

Solution. (d) Put $A = A(Y)$. If A is normal, so is each localization $A_{\mathfrak{m}}$, proving normality at each point. Conversely, an element of $\text{Frac } A$ integral over A is integral over every $A_{\mathfrak{m}}$. If all these local rings are normal, it belongs to their intersection A . This proves the equivalence.

(a) The two standard affine charts of $\mathbb{P}^1$ have coordinate ring $k[t]$, a normal domain. Normality is a condition on local rings, so $\mathbb{P}^1$ and every variety isomorphic to it are normal.

(b) For $xy = zw$, each nonzero coordinate chart eliminates one other coordinate from the equation and is $\mathbb{A}^2$. The charts cover, and all their local rings are normal.

For $xy = z^2$, the charts $x \neq 0$ and $y \neq 0$ are likewise $\mathbb{A}^2$. The chart $w \neq 0$ has ring

$$R = k[x, y, z]/(xy - z^2) \cong k[s^2, t^2, st] \subseteq k[s, t] = B.$$

To verify the isomorphism, reduce every polynomial modulo $z^2 = xy$ to $a(x, y) + zb(x, y)$. The images of its monomials have respectively even-even and odd-odd exponent pairs in s, t , and are linearly independent. Thus the displayed surjection to the subring has zero kernel. The subring R consists of the polynomials all of whose monomials have even total degree. It satisfies

$$B \cap \text{Frac } R = R.$$

Indeed, if $b = p/q \in B$ with $p, q \in R$, $q \neq 0$, split $b = b_0 + b_1$ into its even- and odd-degree parts. In $qb = p$, the odd-degree part is $qb_1 = 0$, so $b_1 = 0$. If an element of $\text{Frac } R$ is integral over R , it is integral over normal B , hence lies in B , and the intersection identity puts it in R . Thus R is

normal in every characteristic. The three charts cover the quadric, since $x = y = w = 0$ also forces $z = 0$. Part (d) proves normality everywhere.

(c) The cusp has coordinate ring $A = k[t^2, t^3]$. The element $t = t^3/t^2 \in \text{Frac } A$ is integral, since it satisfies $T^2 - t^2 = 0$, but it is not in A . Thus (d) proves that the cusp is not normal. More precisely, it is not normal at its origin: if $t = a/b \in A_{(t^2, t^3)}$, the denominator b has a nonzero constant term. Then tb has a nonzero coefficient of t , whereas $a \in A$ has none, a contradiction.

(e) Let $A = A(Y)$, $K = \text{Frac } A$, and let $\tilde{A}$ be its integral closure in K . The finiteness theorem makes $\tilde{A}$ the coordinate ring of an affine variety $\tilde{Y}$. It is normal, so (d) makes $\tilde{Y}$ normal. The inclusion $A \subseteq \tilde{A}$ induces π .

Dominance of f gives an inclusion $K \hookrightarrow K(Z)$. For $b \in \tilde{A}$, its image is integral over $\mathcal{O}_{z,Z}$ for every $z \in Z$, since the coefficients of a monic equation for b come from A and pull back to regular functions. Normality of each local ring puts this rational function in every $\mathcal{O}_{z,Z}$. It is therefore regular on Z . We obtain a homomorphism $\tilde{A} \rightarrow \mathcal{O}(Z)$, and hence a morphism $\tilde{f} : Z \rightarrow \tilde{Y}$ with $\pi\tilde{f} = f$.

For uniqueness, write any $b \in \tilde{A} \subseteq K$ as c/a with $a, c \in A$, $a \neq 0$. Every lift must pull b back to $f^*(c)/f^*(a)$ on the nonempty open set where $f^*(a) \neq 0$; this set is nonempty by dominance. Regular functions equal on a dense open set are equal everywhere. Thus all coordinate pullbacks, and hence the lifts, agree. $\square$

Exercise I.3.18

Permitted context.

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

A domain is normal if it is integrally closed in its fraction field. Polynomial rings over a field are unique factorization domains and therefore normal. Localizations of normal domains are normal: if z is integral over $S^{-1}A$, clearing denominators in a monic equation makes sz integral over A for some $s \in S$. For any domain A , $A = \bigcap_{\mathfrak{m} \text{ maximal}} A_{\mathfrak{m}}$ inside its fraction field. Indeed, if $z \notin A$, the proper ideal $\{a \in A : az \in A\}$ is contained in a maximal ideal at which z is still absent. A variety is normal at a point when its local ring is normal.

If S is a homogeneous coordinate ring, the affine chart $x_i \neq 0$ has ring $(S_{x_i})_0$, the degree-zero subring of the graded localization. An affine variety is normal exactly when its coordinate ring is normal. Projective normality means integral closedness of S .

Problem. (a) Prove that a projectively normal variety is normal.

(b) For the twisted quartic

$$Y = \{[t^4 : t^3u : tu^3 : u^4] : [t : u] \in \mathbb{P}^1\} \subseteq \mathbb{P}^3,$$

prove that Y is normal but not projectively normal.

(c) Show that $Y \cong \mathbb{P}^1$, and conclude that projective normality depends on the embedding.

Solution. (a) Suppose S is normal and put $B = S_{x_i}$, $A = B_0$ for a nonempty chart. The localization B is normal. If $f \in \text{Frac } A$ is integral over A , it is integral over B , so $f \in B$. Write $f = p/q$ with $p, q \in A$, $q \neq 0$, and decompose f into homogeneous components in B . Since q has degree zero, $qf = p$ forces all nonzero-degree components of f to vanish. Hence $f \in B_0 = A$. Every chart ring is normal, so the variety is normal.

(b) Denote the target coordinates by x, y, z, w . The image above is closed: it is exactly the common zero set of

$$y^4 = x^3w, \quad z^4 = xw^3, \quad x^2z = y^3, \quad w^2y = z^3.$$

These relations hold on the parametrization. They force at least one of x, w to be nonzero. On $x \neq 0$, putting $a = y/x$ gives the point $[1 : a : a^3 : a^4]$; on $w \neq 0$, putting $b = z/w$ gives $[b^4 : b^3 : b : 1]$. These are precisely the two parametrizing charts. Their inverse maps to $\mathbb{P}^1$ are respectively $[t : u] = [x : y]$ and $[t : u] = [z : w]$. They are regular and agree on the overlap, proving $Y \cong \mathbb{P}^1$. The latter has normal affine charts $k[a]$ and $k[b]$, so Y is normal.

Its homogeneous coordinate ring is the image of the homogeneous substitution homomorphism, namely

$$R = k[t^4, t^3u, tu^3, u^4].$$

Indeed a homogeneous polynomial vanishes on all parametrized points exactly when its substituted polynomial vanishes on all of k^2 , equivalently when that polynomial is zero, since k is infinite. The element $v = t^2u^2$ belongs to $\text{Frac } R$, because $v = (t^3u)^2/t^4$. It is integral over R , since $v^2 = t^4u^4 \in R$. But $v \notin R$: in total degree four, R is spanned by the four displayed generators, none having exponent pair $(2, 2)$. Thus this embedding is not projectively normal.

(c) The inverse chart maps in (b) give $Y \cong \mathbb{P}^1$. In contrast, the standard embedding of $\mathbb{P}^1$ has normal homogeneous coordinate ring $k[t, u]$. This proves the embedding dependence. $\square$

Exercise I.3.19

Permitted context.

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

The Jacobian matrix of a polynomial map is the matrix of its formal partial derivatives, and satisfies the chain rule over any field. The units in $k[x_1, \dots, x_n]$ are the nonzero constants.

Problem. Let $\varphi : \mathbb{A}^n \rightarrow \mathbb{A}^n$ have polynomial coordinates $f_1, \dots, f_n$, and write $J_\varphi = \det(\partial f_i / \partial x_j)$.

(a) If φ is an automorphism, prove that $J_\varphi \in k^\times$.

(b) Discuss the converse research question posed in the source, including its necessary characteristic restriction.

Solution. (a) An inverse morphism ψ also has polynomial coordinates. The chain rule applied to $\psi \circ \varphi$ yields

$$D\psi(\varphi(x))D\varphi(x) = I_n.$$

Taking determinants gives

$$J_\psi(\varphi(x))J_\varphi(x) = 1$$

in the polynomial ring. Thus J_φ is a unit, and hence a nonzero constant.

(b) In characteristic $p > 0$, the converse is false. For example

$$(x_1, \dots, x_n) \mapsto (x_1 - x_1^p, x_2, \dots, x_n)$$

has Jacobian determinant 1, but is not injective: replacing x_1 by $x_1 + a$, $a \in \mathbb{F}_p$, leaves its image unchanged. In characteristic zero and dimension one, a polynomial with nonzero constant derivative is linear, so the converse does hold.

The source's research question, called the Jacobian conjecture, asks whether a polynomial map in characteristic zero with nonzero constant Jacobian determinant must be an automorphism, in dimensions at least two. Part (a) establishes the necessary condition. Sufficiency in dimension one and its failure in positive characteristic do not settle the characteristic-zero question in higher dimension. $\square$

Exercise I.3.20

Permitted context.

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

A noetherian normal domain R satisfies the height-one intersection theorem

$$R = \bigcap_{\text{ht } \mathfrak{q}=1} R_{\mathfrak{q}} \quad \text{inside } \text{Frac } R.$$

For an affine variety with ring A , $\text{ht } \mathfrak{p} + \dim A/\mathfrak{p} = \dim A$; localization preserves the chains of primes lying below a given prime. In particular all closed-point local rings have dimension equal to that of the variety. A positive-dimensional closed subvariety cannot consist of just one point.

Problem. Let $\dim Y \geq 2$, and suppose Y is normal at P .

- (a) Show that every regular function on $Y \setminus \{P\}$ extends to a regular function on Y .
- (b) Give a counterexample in dimension one.

Solution. (a) Let f be the given function, viewed as an element of $K(Y)$. Choose an affine open U containing P , put $A = A(U)$, and let $\mathfrak{m}$ correspond to P . The local ring $R = A_{\mathfrak{m}}$ is a noetherian normal domain of dimension $\dim Y \geq 2$.

Consider a height-one prime of R , corresponding to $\mathfrak{p} \subseteq \mathfrak{m}$ in A . It has $\text{ht } \mathfrak{p} = 1$. Consequently $V_U(\mathfrak{p})$ has dimension $\dim Y - 1 \geq 1$, and contains a point $Q \neq P$. The regularity of f at Q supplies a representation $f = a/b$ with $a, b \in A$ and $b(Q) \neq 0$. Thus $b \notin \mathfrak{p}$, and $f \in A_{\mathfrak{p}}$. This is exactly the localization of R at the selected height-one prime. The intersection theorem therefore implies $f \in R$.

Represent this germ by a regular function g on a neighborhood of P . As rational functions, $g = f$, so they agree on the punctured part of that neighborhood. Gluing g to the given function extends it to all of Y . The extension is unique because it agrees with f on the dense open $Y \setminus \{P\}$.

(b) On the normal curve $\mathbb{A}^1$, the function $1/t$ is regular away from 0, but is not in the local ring $k[t]_{(t)}$. Indeed $1/t = a/b$ with $b(0) \neq 0$ would imply $b = ta$, contradicting $b(0) \neq 0$. It has no regular extension across 0. $\square$

Exercise I.3.21

Permitted context.

Work over an algebraically closed field k ; varieties are irreducible and carry the Zariski topology. A regular function is locally a quotient of polynomials with nonvanishing denominator. A morphism is a continuous map pulling back regular functions to regular functions. Morphisms to an affine variety correspond contravariantly to k -algebra homomorphisms of its coordinate ring into the ring of global regular functions. Morphisms and regular functions can be checked and glued on open covers. The local ring at a point consists of germs of regular functions, with maximal ideal the germs vanishing there. On an affine variety with coordinate ring A , it is $A_{\mathfrak{m}}$. The function field is $\text{Frac } A$.

Products of varieties have their categorical product structure: a pair of morphisms to X, Y gives a morphism to $X \times Y$. A group variety has multiplication and inversion morphisms whose values give a group on points. The variety $\mathbb{G}_m = \mathbb{A}^1 \setminus \{0\}$ has ring $k[t, t^{-1}]$; its products have the analogous Laurent coordinate rings.

Problem. (a) Verify that $\mathbb{G}_a = \mathbb{A}^1$, with addition, is a group variety.

(b) Verify that $\mathbb{G}_m = \mathbb{A}^1 \setminus \{0\}$, with multiplication, is a group variety.

(c) Give $\text{Hom}(X, G)$ a natural group structure when G is a group variety.

(d) Identify $\text{Hom}(X, \mathbb{G}_a)$ with the additive group of $\mathcal{O}(X)$.

(e) Identify $\text{Hom}(X, \mathbb{G}_m)$ with the group $\mathcal{O}(X)^\times$.

Solution. (a) Addition $(a, b) \mapsto a + b$ and inversion $a \mapsto -a$ are polynomial morphisms. The field addition axioms give the group laws, with identity 0.

(b) Multiplication $(a, b) \mapsto ab$ is a morphism on $\mathbb{G}_m \times \mathbb{G}_m$. The inverse $a \mapsto a^{-1}$ is regular because $t^{-1} \in k[t, t^{-1}]$. The field multiplication axioms on $k^\times$ give the group laws, with identity 1.

(c) For morphisms $f, g : X \rightarrow G$, define

$$(fg)(x) = f(x)g(x).$$

This is a morphism: compose the paired map (f, g) with the multiplication morphism of G . The identity is the constant morphism with value e , and the inverse of f is the composition of f with inversion. Associativity and the other laws follow at each point from the laws in G . Precomposition with a morphism $X' \rightarrow X$ is a group homomorphism, expressing the naturality.

(d) A morphism $X \rightarrow \mathbb{A}^1$ is uniquely its coordinate regular function. Pointwise addition of morphisms gives the sum of these functions, identifying the group with $(\mathcal{O}(X), +)$.

(e) A morphism to $\mathbb{G}_m$ gives a regular function f with regular reciprocal, so $f \in \mathcal{O}(X)^\times$. Conversely a unit f has no zeros and defines a morphism to $\mathbb{A}^1$ with image in $\mathbb{G}_m$; locally its reciprocal is regular, giving the morphism to this open subvariety. Equivalently, it is the homomorphism $k[t, t^{-1}] \rightarrow \mathcal{O}(X)$, $t \mapsto f$. The two constructions are inverse and carry pointwise multiplication to multiplication of units. $\square$

I.4 Rational Maps

Exercise I.4.1

Permitted context.

Work over an algebraically closed field k . Varieties are irreducible. Regularity and the morphism property are local on the source. Two morphisms to a variety agreeing on a dense open subset agree everywhere on their common domain. A rational map is an equivalence class of morphisms on nonempty open subsets, with this agreement relation. Dominant rational maps correspond contravariantly to embeddings of function fields over k ; a function-field isomorphism is equivalent to birationality and to isomorphism of suitable nonempty open subsets. Every point has an affine open neighborhood.

Problem. Glue regular functions on two open subsets when they agree on the overlap. Deduce that a rational function has a largest open domain on which it is regular.

Solution. For $f \in \mathcal{O}(U)$, $g \in \mathcal{O}(V)$ agreeing on $U \cap V$, define h by these functions on $U \cup V$. Agreement makes this well defined. At each point it agrees on a neighborhood with either f or g , so is regular. The same reasoning glues any compatible family. Representatives of one rational function agree on their overlaps: their difference vanishes on a dense open subset, hence everywhere. They glue on the open union D of their domains. This union contains every other such domain and is the unique largest domain. $\square$

Exercise I.4.2

Permitted context.

Work over an algebraically closed field k . Varieties are irreducible. Regularity and the morphism property are local on the source. Two morphisms to a variety agreeing on a dense open subset agree everywhere on their common domain. A rational map is an equivalence class of morphisms on nonempty open subsets, with this agreement relation. Dominant rational maps correspond contravariantly to embeddings of function fields over k ; a function-field isomorphism is equivalent to birationality and to isomorphism of suitable nonempty open subsets. Every point has an affine open neighborhood.

Problem. Prove that a rational map $X \dashrightarrow Y$ has a unique largest open subset on which it is represented by a morphism.

Solution. Let D be the union of all representative domains. Representatives agree on overlaps by the agreement theorem for morphisms. They therefore glue to a map $D \rightarrow Y$. Its inverse images of open subsets are open on each member of the open cover, hence open. Its pullbacks of regular functions are regular locally, hence regular. Thus the glued map is a morphism. It represents the rational map and contains every other domain, proving the assertion. $\square$

Exercise I.4.3**Permitted context.**

Work over an algebraically closed field k . Varieties are irreducible. Regularity and the morphism property are local on the source. Two morphisms to a variety agreeing on a dense open subset agree everywhere on their common domain. A rational map is an equivalence class of morphisms on nonempty open subsets, with this agreement relation. Dominant rational maps correspond contravariantly to embeddings of function fields over k ; a function-field isomorphism is equivalent to birationality and to isomorphism of suitable nonempty open subsets. Every point has an affine open neighborhood.

The local rings of the affine plane are localizations of $k[u, v]$. In $k[u, v]_{(u, v)}$, neither v/u nor u/v is present: clearing a denominator nonzero at the origin contradicts unique factorization.

Problem. (a) Determine the regular domain of x_1/x_0 on $\mathbb{P}^2$.

(b) Determine its domain as a rational map to $\mathbb{P}^1$, using the usual inclusion $\mathbb{A}^1 \subseteq \mathbb{P}^1$.

Solution. (a) On $x_0 \neq 0$, this is the affine coordinate x_1/x_0 . At $x_0 = 0, x_1 \neq 0$, use $x_1 = 1$; the function is $1/u$ with $u = x_0/x_1$, absent from the local ring at $u = 0$. At $[0 : 0 : 1]$, use $u = x_0/x_2, v = x_1/x_2$; the function is v/u , excluded by the divisibility test. Its full domain is therefore $D_+(x_0)$.

(b) The map is $[x_0 : x_1 : x_2] \mapsto [x_0 : x_1]$, a morphism away from $P = [0 : 0 : 1]$. It maps $x_0 = 0, x_1 \neq 0$ to infinity. If it extended to P , one of the two target affine charts would contain the value. Pulling back its coordinate would make either v/u or u/v regular near P , impossible. Thus its largest domain is $\mathbb{P}^2 \setminus \{P\}$. $\square$

Exercise I.4.4**Permitted context.**

Work over an algebraically closed field k . Varieties are irreducible. Regularity and the morphism property are local on the source. Two morphisms to a variety agreeing on a dense open subset agree everywhere on their common domain. A rational map is an equivalence class of morphisms on nonempty open subsets, with this agreement relation. Dominant rational maps correspond contravariantly to embeddings of function fields over k ; a function-field isomorphism is equivalent to birationality and to isomorphism of suitable nonempty open subsets. Every point has an affine open neighborhood.

Every irreducible projective conic is isomorphic to $\mathbb{P}^1$. A curve is rational when its function field is $k(t)$. Gauss's lemma tests polynomial irreducibility over a fraction field.

Problem. (a) Show that projective conics are rational.

(b) Show that $y^2 = x^3$ is rational.

(c) For $Y = V_+(y^2z - x^2(x + z))$, show that projection from $[0 : 0 : 1]$ onto $z = 0$ is birational to $\mathbb{P}^1$.

Solution. (a) Its isomorphism to $\mathbb{P}^1$ is birational.

(b) Put $t = y/x$. The equation gives $t^2 = x, t^3 = y$, so the function field is $k(t)$.

(c) On $z = 1, x \neq 0$, put $t = y/x$. The equation gives

$$x = t^2 - 1, \quad y = t(t^2 - 1).$$

Projection is $[x : y] = [1 : t]$, with inverse on a nonempty open given by $t \mapsto [t^2 - 1 : t(t^2 - 1) : 1]$. The cubic is irreducible: a factor of $y^2 - x^2(x+1)$ over $k(x)$ would make $x+1$ a square there, contrary to its odd order at $x = -1$. Gauss's lemma and homogenization apply. Thus this is a birational map of varieties in every characteristic. In characteristic two the tangent cone is a repeated line; the designation *nodal* requires characteristic different from two. $\square$

Exercise I.4.5

Permitted context.

Work over an algebraically closed field k . Varieties are irreducible. Regularity and the morphism property are local on the source. Two morphisms to a variety agreeing on a dense open subset agree everywhere on their common domain. A rational map is an equivalence class of morphisms on nonempty open subsets, with this agreement relation. Dominant rational maps correspond contravariantly to embeddings of function fields over k ; a function-field isomorphism is equivalent to birationality and to isomorphism of suitable nonempty open subsets. Every point has an affine open neighborhood.

The quadric $xy = zw$ has rulings by lines, with distinct lines of one ruling disjoint. Any two projective plane curves intersect.

Problem. Prove that $Q = V_+(xy - zw) \subseteq \mathbb{P}^3$ is birational to $\mathbb{P}^2$, but not isomorphic to it.

Solution. On $x = 1$, its equation is $y = zw$, giving an open $\mathbb{A}^2$, also an open of $\mathbb{P}^2$. Hence the varieties are birational. An isomorphism would take two disjoint ruling lines to disjoint closed irreducible curves in $\mathbb{P}^2$, contradicting their intersection property. $\square$

Exercise I.4.6

Permitted context.

Work over an algebraically closed field k . Varieties are irreducible. Regularity and the morphism property are local on the source. Two morphisms to a variety agreeing on a dense open subset agree everywhere on their common domain. A rational map is an equivalence class of morphisms on nonempty open subsets, with this agreement relation. Dominant rational maps correspond contravariantly to embeddings of function fields over k ; a function-field isomorphism is equivalent to birationality and to isomorphism of suitable nonempty open subsets. Every point has an affine open neighborhood.

Problem. For $c : [x : y : z] \mapsto [yz : xz : xy]$:

- (a) Prove that c is birational and its own inverse.
- (b) Give open subsets on which it is an isomorphism.
- (c) Find the full domains of c, c^{-1} and describe their morphisms.

Solution. (a) On $xyz \neq 0$, the second iterate is $[x^2yz : xy^2z : xyz^2] = [x : y : z]$. Thus c is its own rational inverse.

(b) Take $U = V = \{xyz \neq 0\}$. The formula and its identical inverse are morphisms on this torus.

(c) The coordinates vanish simultaneously at just the three coordinate vertices. Elsewhere they define a morphism, contracting each punctured coordinate line to the opposite vertex. At the

vertex $x = 1, y = z = 0$, the expression is $[yz : z : y]$. An extension with first target coordinate nonzero would make $1/y, 1/z$ regular at the origin. With second coordinate nonzero it would make y/z regular; with third coordinate nonzero it would make z/y regular. Unique factorization in $k[y, z]_{(y, z)}$ excludes all three alternatives. Permuting coordinates handles the other vertices. Thus both full domains are the plane minus its three coordinate vertices. The contractions explain why these full-domain morphisms are not isomorphisms. $\square$

Exercise I.4.7

Permitted context.

Work over an algebraically closed field k . Varieties are irreducible. Regularity and the morphism property are local on the source. Two morphisms to a variety agreeing on a dense open subset agree everywhere on their common domain. A rational map is an equivalence class of morphisms on nonempty open subsets, with this agreement relation. Dominant rational maps correspond contravariantly to embeddings of function fields over k ; a function-field isomorphism is equivalent to birationality and to isomorphism of suitable nonempty open subsets. Every point has an affine open neighborhood.

Problem. If $\mathcal{O}_{Q,Y} \cong \mathcal{O}_{P,X}$ as k -algebras, prove that some open neighborhoods of P, Q are isomorphic, with P corresponding to Q .

Solution. Choose affine neighborhoods with rings A, B and write the isomorphism $\theta : B_{\mathfrak{n}} \rightarrow A_{\mathfrak{m}}$. It extends to a function-field isomorphism. Images of finitely many algebra generators of B are regular on a common neighborhood U_0 of P , defining a morphism $f : U_0 \rightarrow Y$. The isomorphism carries maximal ideal to maximal ideal and fixes k , so evaluation gives $f(P) = Q$. The inverse construction gives $g : V_0 \rightarrow X$ with $g(Q) = P$. Shrink within the original affine neighborhoods. Set $U_1 = U_0 \cap f^{-1}(V_0)$, $V_1 = V_0 \cap g^{-1}(U_0)$. The maps gf on U_1 and fg on V_1 induce identity function-field maps and therefore equal the identity on dense opens, hence everywhere. For $x \in U_1$, the equality $g(f(x)) = x \in U_0$ implies $f(x) \in V_1$; the analogous inclusion holds for g . Thus these restrictions are inverse neighborhood isomorphisms. $\square$

Exercise I.4.8**Permitted context.**

Work over an algebraically closed field k . Varieties are irreducible. Regularity and the morphism property are local on the source. Two morphisms to a variety agreeing on a dense open subset agree everywhere on their common domain. A rational map is an equivalence class of morphisms on nonempty open subsets, with this agreement relation. Dominant rational maps correspond contravariantly to embeddings of function fields over k ; a function-field isomorphism is equivalent to birationality and to isomorphism of suitable nonempty open subsets. Every point has an affine open neighborhood.

Noether normalization makes an affine d -dimensional coordinate ring finite integral over a polynomial ring in d variables. Lying over then gives a surjection on closed points over algebraically closed k . Proper closed subsets of an irreducible curve are finite. An infinite cardinal equals each of its positive finite powers.

Problem. (a) Prove that every positive-dimensional variety over k has cardinality $|k|$.
 (b) Prove that any two curves over k are homeomorphic.

Solution. (a) The field k is infinite. Projective space has finitely many affine charts, each of cardinality $|k|$, so an embedding in projective space gives $|X| \leq |k|$. Take an affine open U of dimension $d \geq 1$. Normalization and lying over give a surjection $U \rightarrow \mathbb{A}^d(k)$: the residue field above a maximal ideal is finite over k , hence k . Therefore $|X| \geq |U| \geq |k^d| = |k|$.

(b) The topology on a curve is cofinite. By (a) two curves have a bijection; it preserves finite sets in both directions and is therefore a homeomorphism. $\square$

Exercise I.4.9**Permitted context.**

Work over an algebraically closed field k . Varieties are irreducible. Regularity and the morphism property are local on the source. Two morphisms to a variety agreeing on a dense open subset agree everywhere on their common domain. A rational map is an equivalence class of morphisms on nonempty open subsets, with this agreement relation. Dominant rational maps correspond contravariantly to embeddings of function fields over k ; a function-field isomorphism is equivalent to birationality and to isomorphism of suitable nonempty open subsets. Every point has an affine open neighborhood.

Over a perfect field, finitely generated function fields are separably generated, and any generating set contains a separating transcendence basis. Finite separable extensions have distinct embeddings into an algebraic closure and satisfy the primitive element theorem. Finite integral extensions have closed surjective maps on prime spectra. A proper closed subset of a projective line is finite. Height-one primes in a polynomial ring are principal.

Problem. For projective $X \subseteq \mathbb{P}^n$ of dimension r , with $n \geq r + 2$, find a point outside X such that projection to a hyperplane is birational onto its image. Deduce that successive projections give a hypersurface in $\mathbb{P}^{r+1}$ birational to X .

Solution. Choose coordinates so every coordinate vertex lies outside X and X is not contained in $x_0 = 0$. These conditions can be imposed together: they are nonempty open conditions on invertible matrices, an irreducible variety. The vertex conditions are nonempty because X is proper, and the hyperplane condition is nonempty because X is nonempty. On $x_0 \neq 0$, the ratios $t_i = x_i/x_0$ generate $K = K(X)$. After permutation, $t_1, \dots, t_r$ are a separating basis. Choose two remaining generators β_1, β_2 , and let L be generated over k by all the other ratios. Then $K = L(\beta_1, \beta_2)$ is finite separable over L .

For all but finitely many $c \in k$, $\alpha = \beta_1 + c\beta_2$ generates K/L . Indeed distinct L -embeddings agree on this element for at most one c , since agreement on both β_i makes them equal. There are only finitely many embeddings. Projection retaining x_0 , the coordinates defining L , and the linear combination defining α , has point center P_c on the line joining the two remaining coordinate vertices. This line is not contained in X , so only finitely many of the distinct P_c belong to X . Since k is infinite, choose c avoiding both exceptional sets. Projection is then a morphism on X ; its image field is $L(\alpha) = K$, so it is birational to its image closure.

Its image is closed as well. Put the chosen center at the last coordinate vertex. An equation of X nonvanishing there has a nonzero x_n^d coefficient, and after scaling is monic in x_n . On each target chart $x_i \neq 0$, this makes x_n/x_i integral over the algebra of retained coordinate ratios. Thus the affine image on this chart is closed by lying over. These charts cover the target, so the projective image is closed. Iterate until the ambient dimension is $r + 1$; the image still has dimension r , and its height-one homogeneous ideal is principal. It is the required hypersurface. $\square$

Exercise I.4.10

Permitted context.

Work over an algebraically closed field k . Varieties are irreducible. Regularity and the morphism property are local on the source. Two morphisms to a variety agreeing on a dense open subset agree everywhere on their common domain. A rational map is an equivalence class of morphisms on nonempty open subsets, with this agreement relation. Dominant rational maps correspond contravariantly to embeddings of function fields over k ; a function-field isomorphism is equivalent to birationality and to isomorphism of suitable nonempty open subsets. Every point has an affine open neighborhood.

The blow-up of $\mathbb{A}^2$ at the origin is given by $xb = ya$ in $\mathbb{A}^2 \times \mathbb{P}^1$. The strict transform is the closure of the inverse image away from the origin. Irreducible curves have cofinite point topology.

Problem. Blow up $Y = V(y^2 - x^3)$ at the origin. Prove that its strict transform is $\mathbb{A}^1$, meets the exceptional curve in one point, and maps homeomorphically but not isomorphically to Y .

Solution. On $a = 1$, put $v = b/a$; then $y = xv$, and the total equation is $x^2(v^2 - x) = 0$. The strict transform is $x = v^2, y = v^3$, with coordinate ring $k[v]$. Its intersection with $x = 0$ is the single point $v = 0$. Scheme-theoretically this intersection is defined by $v^2 = 0$ and has length two; the intersection is not transverse. On $b = 1$, put $u = a/b$; then $x = yu$, and the strict transform is $1 - yu^3 = 0$. It has $u \neq 0$, so lies entirely in the first chart, with $v = u^{-1}$. Thus the first chart covers the transform. The projection is $v \mapsto (v^2, v^3)$, bijective with inverse $v = y/x$ off the origin and one point over the origin. Its bijection is a homeomorphism for the cofinite topologies. The pullback ring has image $k[v^2, v^3]$, not containing v ; hence it is not an isomorphism. $\square$

I.5 Nonsingular Varieties

Exercise I.5.1

Permitted context.

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

Assume $\text{char } k \neq 2$. A node has two distinct tangent directions, a cusp has local model $y^2 = x^3$, and a tacnode has two smooth branches with the same tangent and contact order two. An ordinary triple point has three distinct tangent directions. Real sketches refer to these equations over $\mathbb{R}$. In characteristic different from two every formal power-series unit has a square root after choosing a square root of its constant term; its remaining coefficients are determined recursively.

Problem. Find all singular points and sketch the real curves:

- (a) $x^2 = x^4 + y^4$.
- (b) $xy = x^6 + y^6$.
- (c) $x^3 = y^2 + x^4 + y^4$.
- (d) $x^2y + xy^2 = x^4 + y^4$. Identify the four types of singularity at the origin.

Solution. (a) For $f = x^2 - x^4 - y^4$, the derivatives are $2x(1 - 2x^2), -4y^3$. Thus a singular point has $y = 0$; its equation and first derivative then force $x = 0$. The leading term is x^2 , and $x^2(1 - x^2) = y^4$ gives the two formal smooth branches $x = \pm y^2$ + higher terms. The origin is a tacnode, with common tangent $x = 0$.

(b) The derivatives are $y - 6x^5, x - 6y^5$. In characteristic three these immediately force the origin. Otherwise a nonzero critical point has $xy = 6x^6 = 6y^6$; the equation then gives $xy = xy/3$, impossible since $2, 3 \neq 0$. If one coordinate is zero, the other is zero as well. The leading term xy makes the origin a node with the coordinate axes as tangents.

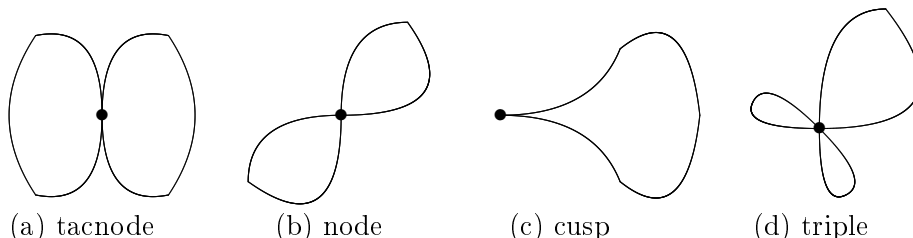
(c) The derivatives are $x^2(3 - 4x), -2y(1 + 2y^2)$. The origin is singular, with leading term $-y^2$, and

$$y^2(1 + y^2) = x^3(1 - x).$$

Taking square roots of units in the completed local ring converts this to a cusp $Y^2 = x^3$; its tangent is $y = 0$. The remaining critical candidates have $x = 3/4$ and $y^2 = -1/2$. The defining equation there equals $91/256$. Consequently there are exactly two additional singular points in characteristics 7 or 13, and none otherwise. At either extra point, the quadratic Taylor form is $-9u^2/8 + 2v^2$. Its two factors are distinct in these characteristics, so both extra points are nodes. Candidates with only one coordinate nonzero do not lie on the curve; in characteristic three $3/4 = 0$ adds no further candidate.

(d) Write $f = g_3 - g_4$, where $g_3 = xy(x + y)$ and $g_4 = x^4 + y^4$. At a singular point Euler's identity gives $3g_3 - 4g_4 = 0$, while $f = 0$ gives $g_3 = g_4$; hence both vanish. If $xy(x + y) = 0$, either a coordinate is zero or $y = -x$. The equation $x^4 + y^4 = 0$ then forces the origin, using $\text{char } k \neq 2$. Its three distinct tangents are $x = 0, y = 0, x + y = 0$, so it is an ordinary triple point.

Here are qualitative sketches over $\mathbb{R}$; a dot marks the origin in each panel. They illustrate the real branches, separately from the singularity assertions over general k :



The first has two lobes on opposite sides of the vertical tangent; the second has lobes in the first and third quadrants; the cusp has one lobe on the positive x -side. The triple point has one larger lobe in the first quadrant and two smaller lobes in the second and fourth. These real sketches do not represent the extra points that occur only in characteristics 7, 13. $\square$

Exercise I.5.2

Permitted context.

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

Assume $\text{char } k \neq 2$. A nondegenerate quadratic cone gives a conical double point. A transverse product of two distinct linear factors describes an ordinary double curve locally.

Problem. Find and describe the singular loci in $\mathbb{A}^3$:

- (a) $xy^2 = z^2$.
- (b) $x^2 + y^2 = z^2$.
- (c) $xy + x^3 + y^3 = 0$.

Solution. (a) The gradient is $(y^2, 2xy, -2z)$, so the singular locus is the line $y = z = 0$. At $(a, 0, 0)$, $a \neq 0$, the transverse quadratic form is $ay^2 - z^2$, with distinct factors. The parametrization $(u, v) \mapsto (u^2, v, uv)$ has two points above a general point of this singular line and only one above the origin. At the origin the branches coalesce in the displayed equation: this is the pinch point. Over $\mathbb{R}$, for $x > 0$ the two sheets are $z = \pm\sqrt{x}y$; at $x = 0$ they meet, while for $x < 0$ only the singular axis remains.

(b) The gradient $(2x, 2y, -2z)$ vanishes only at the origin. The equation is itself a nondegenerate quadratic form in three variables, giving a conical double point (the real circular cone).

(c) The derivatives are $y + 3x^2, x + 3y^2, 0$. In characteristic three they force $x = y = 0$. Otherwise at a nonzero critical pair, $xy = -3x^3 = -3y^3$, and the equation becomes $xy/3 = 0$, a contradiction. Thus the singular locus is the whole z -axis. The transverse lowest form is xy , two distinct lines. Since the equation is independent of z , the surface is the product of a nodal plane curve with an affine line, giving the double-line picture. $\square$

Exercise I.5.3**Permitted context.**

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

Problem. (a) For a plane curve, prove that multiplicity one at a point is equivalent to nonsingularity there.

(b) Compute the multiplicities at every singular point of the four curves $x^2 - x^4 - y^4 = 0$, $xy - x^6 - y^6 = 0$, $x^3 - y^2 - x^4 - y^4 = 0$, and $xy(x + y) - x^4 - y^4 = 0$, assuming characteristic different from two.

Solution. (a) Translate the point to the origin and expand $f = f_1 + f_2 + \cdots$, with f_i homogeneous of degree i . The coefficient vector of f_1 is $(f_x(0), f_y(0))$. Thus $f_1 \neq 0$ exactly when the Jacobian criterion says the point is nonsingular, and exactly when its multiplicity is one.

(b) At the origin the respective lowest forms are $x^2, xy, -y^2, xy(x + y)$. The multiplicities are therefore 2, 2, 2, 3. The derivative equations give no other singularities except on the third curve in characteristics 7, 13, where the two extra points have $x = 3/4, y^2 = -1/2$. After translation their quadratic forms are $-9u^2/8 + 2v^2$, nonzero, so their multiplicities are two. $\square$

Exercise I.5.4**Permitted context.**

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

The ambient local ring at a plane point is $R = k[x, y]_{(x, y)}$ after translation. Length of a finite-length R -module equals its dimension over k . The associated graded ring of R is $k[x, y]$. For a principal ideal (f) , its initial ideal is generated by the lowest homogeneous form of f , because orders add under multiplication. Distinct irreducible plane curves have no common one-dimensional component.

Problem. For distinct irreducible plane curves Y, Z through P , with local equations f, g , define $(Y \cdot Z)_P = \ell(R/(f, g))$.

(a) Prove finiteness and the inequality $(Y \cdot Z)_P \geq \mu_P(Y)\mu_P(Z)$.

(b) Show that all but finitely many lines through P meet Y with local multiplicity $\mu_P(Y)$.

(c) If a projective plane curve has degree d , prove that its total intersection multiplicity with a line not a component is d .

Solution. (a) The absence of a common component makes (f, g) primary to the maximal ideal after localization at P . Its quotient is zero-dimensional noetherian, hence Artinian and of finite length.

Put $m = \text{ord } f, n = \text{ord } g$ and $A = R/(f)$, with maximal ideal $\mathfrak{a}$. Since $\text{gr } A = k[x, y]/(f_m)$, direct counting gives

$$\ell(A/\mathfrak{a}^N) = mN - \frac{m(m-1)}{2} \quad (N \geq m).$$

Indeed the degree- j piece has dimension $j+1$ for $j < m$ and m for $j \geq m$. Multiplication by g induces

$$A/\mathfrak{a}^{N-n} \longrightarrow A/\mathfrak{a}^N \longrightarrow A/(g, \mathfrak{a}^N) \longrightarrow 0.$$

For large N , the last term is $A/(g)$. Its length is at least the difference of the first two lengths, which is mn , proving the bound without assuming the first map injective.

(b) A line of direction (a, b) is parametrized by (at, bt) . If $f_m(a, b) \neq 0$, the restricted series has order m , and the quotient of $k[t]_{(t)}$ has length m . Only the finitely many zeros of the nonzero binary form f_m in $\mathbb{P}^1$ are excluded.

(c) Restrict the degree- d homogeneous equation to the line $\mathbb{P}^1$. It is a nonzero binary form of degree d , whose linear factors over k , counted with multiplicity, number d . In each affine chart, the order of a root is exactly the local quotient length defining the intersection. Summing gives d . $\square$

Exercise I.5.5

Permitted context.

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

For a projective plane hypersurface, singularity is detected by vanishing of its homogeneous gradient at a point on the hypersurface. Two positive-degree projective plane curves meet; consequently a reducible or nonreduced plane equation has a singular point.

Problem. For each integer $d > 0$ and each characteristic, give a nonsingular projective plane curve of degree d .

Solution. If the characteristic does not divide d (including characteristic zero), take

$$F = x^d + y^d + z^d.$$

For $d > 1$, its gradient can vanish only when all coordinates vanish; for $d = 1$, the gradient is a nonzero constant vector.

If the characteristic divides d , then $d \geq 2$. Take

$$F = x^d + y^{d-1}z + xz^{d-1}.$$

Its derivative $F_x = z^{d-1}$ forces $z = 0$ at a possible singular point. For $d > 2$, $F_z = y^{d-1}$ there forces $y = 0$, and $F = 0$ then forces $x = 0$. For $d = 2$ in characteristic two, $F_z = y + x$; together with $z = 0, F = 0$ it again forces $x = y = 0$. No projective singular point exists. The intersection observation in the context proves irreducibility of each displayed equation, so these are nonsingular curves. $\square$

Exercise I.5.6**Permitted context.**

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

The blow-up of the plane at the origin has charts $y = xt$ and $x = ys$. The strict-transform equation is obtained by removing the largest common exceptional power from the substituted equation. It agrees with the original curve away from the origin. Assume characteristic different from two for the named node and tacnode. Here resolution means making the strict transform nonsingular. It does not require the total transform to have simple normal crossings; that embedded-resolution condition can require further blow-ups.

Problem. (a) Blow up the node $xy = x^6 + y^6$ and cusp $x^3 = y^2 + x^4 + y^4$ at the origin and show the points above the origin are nonsingular. Determine when their whole strict transforms are nonsingular.

(b) Prove that blowing up an ordinary double point gives two distinct nonsingular points above it.

(c) Show that blowing up the tacnode $x^2 = x^4 + y^4$ gives a node, which one more blow-up resolves.

(d) Resolve the triple cusp $y^3 = x^5$ by two blow-ups.

Solution. (a) For the node, the x -chart has equation $t - x^4(1 + t^6) = 0$. On the exceptional line $x = 0$, $t = 0$ and the t -derivative is one. The y -chart gives a second smooth point in the other tangent direction. The original curve has no singularity away from the origin, so this strict transform is nonsingular. For the cusp, the x -chart gives

$$x - t^2 - x^2 - x^2 t^4 = 0.$$

There is one point above the origin, $x = t = 0$, with x -derivative one. The other chart has no additional exceptional point. The whole strict transform is nonsingular except in characteristics 7, 13: the two other original nodes $x = 3/4, y^2 = -1/2$ are unaffected by this blow-up. This qualifies the global nonsingularity assertion for those fields.

(b) Choose coordinates so neither tangent is the vertical direction. If $f = f_2 + f_3 + \cdots$, the x -chart equation is

$$f_2(1, t) + x f_3(1, t) + \cdots = 0.$$

The two exceptional points are the distinct roots of $f_2(1, t)$. At each the t -derivative is nonzero, proving nonsingularity. The other chart gives no further point, by the choice of coordinates.

(c) In the y -chart, put $x = yu$. The equation becomes

$$u^2 - y^2(u^4 + 1) = 0.$$

Its unique exceptional point is $u = y = 0$, with leading form $u^2 - y^2$, two distinct lines. This is a node, resolved by (b). The x -chart adds no exceptional point.

(d) The original lowest degree is three. Putting $y = xt$ gives $t^3 = x^2$, a double cusp at the sole exceptional point. Blow that point up in the chart $x = ts$; the strict transform is $t = s^2$, a smooth curve. In each other chart there is no additional exceptional point. These calculations work in every characteristic; the final equation has a partial derivative equal to one. $\square$

Exercise I.5.7**Permitted context.**

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

Let $F(x, y, z)$ be homogeneous of degree $d > 1$, defining a nonsingular projective plane curve. The blow-up of affine space at its origin records the projective direction $[a : b : c]$; on $a \neq 0$ it has coordinates (x, v, w) with $y = xv, z = xw$.

Problem. Let $X = V(F) \subseteq \mathbb{A}^3$ and blow up its vertex O .

- (a) Show that O is its only singular point.
- (b) Show that the strict transform $\tilde{X}$ is nonsingular.
- (c) Identify the exceptional fiber with $Y = V_+(F)$.

Solution. (a) All first derivatives vanish at the origin since $d > 1$. At a nonzero point of the cone, projective nonsingularity says at least one partial derivative is nonzero. Thus just the vertex is singular.

(b) On the direction chart $a \neq 0$, substitution gives $x^d F(1, v, w)$; the strict transform is $F(1, v, w) = 0$. It is $Y \cap \{a \neq 0\}$ times $\mathbb{A}_x^1$, nonsingular by the affine Jacobian criterion. The charts $b \neq 0, c \neq 0$ give the same description and cover the transform.

(c) On the first chart the exceptional fiber has $x = 0$ and $F(1, v, w) = 0$. Its direction coordinates are $[1 : v : w]$. On overlaps these are exactly the usual transitions of Y , so the chart identifications glue to an isomorphism of the exceptional fiber with Y . $\square$

Exercise I.5.8**Permitted context.**

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

An affine r -dimensional variety in $\mathbb{A}^n$ is nonsingular where the Jacobian of generators of its ideal has rank $n - r$. For homogeneous f of degree d , Euler's identity is $\sum x_i f_{x_i} = df$, in every characteristic.

Problem. Let homogeneous $f_1, \dots, f_s$ generate the ideal of an r -dimensional $Y \subseteq \mathbb{P}^n$. Prove that $P = [a]$ is nonsingular exactly when $(f_{i, x_j}(a))$ has rank $n - r$, and show that its rank is independent of the representative a .

Solution. Replacing a by λa multiplies row i by $\lambda^{\deg f_i - 1}$, a nonzero scalar, preserving rank. Choose $a_0 \neq 0$ and normalize it to one. On this affine chart the equations are $g_i(t) = f_i(1, t)$; they generate

the affine ideal by localization and dehomogenization. Their Jacobian is the matrix formed by columns $1, \dots, n$ of the homogeneous Jacobian. Euler's identity on Y gives

$$f_{i,x_0}(a) = - \sum_{j=1}^n a_j f_{i,x_j}(a).$$

Thus column zero lies in the span of the other columns, and the two Jacobians have equal rank. The affine criterion proves the claim. No division by $\deg f_i$ occurs, so the proof includes degrees divisible by the characteristic. $\square$

Exercise I.5.9

Permitted context.

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

Two projective plane curves of positive degree intersect. The homogeneous gradient criterion detects nonsingularity on a projective plane hypersurface.

Problem. Let f be a nonzero homogeneous polynomial of positive degree in three variables. If its gradient is nonzero at every point of $V_+(f)$, prove that f is irreducible.

Solution. If $f = gh$ with both factors of positive degree, choose a projective common zero of g, h . The product rule $f_{x_i} = gh_{x_i} + hg_{x_i}$ makes every partial vanish there. This contradicts the hypothesis. The same argument includes repeated factors, since their common zero set is nonempty. Thus f is irreducible, and its projective zero set is nonsingular by the stated gradient criterion. $\square$

Exercise I.5.10

Permitted context.

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

At a closed point of a variety, the local-ring dimension equals the variety dimension. Nakayama's lemma identifies the minimum number of generators of the maximal ideal $\mathfrak{m}$ with $\dim_k \mathfrak{m}/\mathfrak{m}^2$, which is at least the ring dimension by the height theorem. Nonsingularity is equivalent to regularity of the local ring.

Problem. Define $T_P X = (\mathfrak{m}_P/\mathfrak{m}_P^2)^\vee$.

- Prove $\dim T_P X \geq \dim X$, with equality exactly at nonsingular points.
- Construct the natural linear map $T_P \varphi : T_P X \rightarrow T_{\varphi(P)} Y$ for a morphism $\varphi : X \rightarrow Y$.
- Show that the projection of $x = y^2$ onto the x -axis has zero tangent map at the origin.

Solution. (a) Duality preserves finite vector-space dimension. The inequality follows from the generator bound in the context. Equality is exactly the defining condition for the local ring to be regular, hence for the point to be nonsingular.

(b) The local homomorphism $\mathcal{O}_{\varphi(P),Y} \rightarrow \mathcal{O}_{P,X}$ sends maximal ideal to maximal ideal and its square into the square. It induces a k -linear map on their quotients; dualizing gives $T_P\varphi$. Functoriality follows from composition of these local homomorphisms.

(c) The coordinate ring of the parabola is $k[y]$, with $x = y^2$. Pullback sends the cotangent generator $x \bmod x^2$ of the axis to $y^2 \bmod (y)^2 = 0$. The cotangent map and its dual are zero, including in characteristic two. $\square$

Exercise I.5.11

Permitted context.

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

Assume $\text{char } k \neq 2$. For two equations in affine three-space, independent gradients give a nonsingular one-dimensional zero set locally. A plane curve with nonvanishing homogeneous gradient is irreducible.

Problem. Let $Y \subseteq \mathbb{P}^3$ be cut out by $x^2 - xz - yw = 0$, $yz - xw - zw = 0$, and put $P = [0 : 0 : 0 : 1]$. Show that projection from P identifies $Y \setminus \{P\}$ with $C \setminus \{S\}$, where

$$C = V_+(y^2z - x^3 + xz^2), \quad S = [1 : 0 : -1].$$

Prove that Y is an irreducible nonsingular curve. Explain why characteristic two must be excluded.

Solution. Eliminating w gives the displayed cubic equation. For a point $[x : y : z]$ of C , the desired value of w satisfies

$$yw = x(x - z), \quad (x + z)w = yz.$$

These are compatible precisely because it lies on C . On $y \neq 0$ use $w = x(x - z)/y$, and on $x + z \neq 0$ use $w = yz/(x + z)$, interpreted in homogeneous coordinates. They agree on the overlap and are regular inverse maps. The two opens cover $C \setminus \{S\}$. Over S the first equation would say $0 = 2$, so no point of $Y \setminus \{P\}$ maps there. This proves the claimed isomorphism.

The cubic C is nonsingular. On $z = 1$, its equation is $y^2 = x^3 - x$. A critical point has $y = 0, 3x^2 = 1$; the equation requires $x = 0, 1, -1$, none satisfying both conditions in characteristic different from two. At infinity its only point is $[0 : 1 : 0]$, where the z -partial is nonzero. Thus C is irreducible and nonsingular. Consequently $Y \setminus \{P\}$ is an irreducible nonsingular curve.

At P , set $w = 1$. The two gradients in x, y, z are $(0, -1, 0)$ and $(-1, 0, -1)$, linearly independent. Hence Y is a nonsingular curve near P ; in particular P is not an isolated component. It belongs to the closure of $Y \setminus \{P\}$, proving irreducibility of the whole Y and nonsingularity everywhere.

In characteristic two, the line $x = z, y = 0$ lies in Y , and its points other than P project to $S = [1 : 0 : 1]$. The proposed isomorphism fails. Thus the characteristic qualification is essential. $\square$

Exercise I.5.12**Permitted context.**

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

Assume $\text{char } k \neq 2$ and k algebraically closed. A nonzero quadratic form corresponds to a symmetric bilinear form; successive orthogonal decompositions diagonalize it. Nonzero scalars in k have square roots.

Problem. For a nonzero homogeneous quadratic f in $n + 1$ variables:

- (a) Put it in the form $x_0^2 + \cdots + x_r^2$ by linear coordinates.
- (b) Determine when it is irreducible.
- (c) In the irreducible case determine its singular locus.
- (d) Retain the irreducible case and assume $2 \leq r < n$. Describe the quadric as a cone with linear axis over a nonsingular quadric in $\mathbb{P}^r$.

Solution. (a) A nonzero symmetric bilinear form in characteristic different from two has a vector v with nonzero self-pairing: otherwise polarization would make the whole form zero. Split off its orthogonal one-dimensional summand and repeat on the complement. Rescale each nonzero diagonal coefficient by a square root. The number of nonzero coefficients is $r + 1$.

(b) With one coefficient the form is a square; with two it factors over k . Conversely a product of two linear forms has bilinear matrix of rank at most two, since its image lies in the span of their coefficient vectors. Thus rank at least three, equivalently $r \geq 2$, is exactly irreducibility.

(c) Its gradient is $(2x_0, \dots, 2x_r, 0, \dots, 0)$. The singular locus is $Z = V_+(x_0, \dots, x_r)$, of dimension $n - r - 1$. For $r = n$ it is empty (dimension convention -1); otherwise it is the stated linear space.

(d) Since $r \geq 2$, the base quadric below is nonempty and irreducible. In the complementary subspace $\mathbb{P}^r = \{x_{r+1} = \cdots = x_n = 0\}$, the same equation defines a nonsingular quadric Q' . Every point of the full quadric outside Z has nonzero first coordinate block representing a point of Q' , and its remaining block either is zero or represents a point of Z . It lies on their joining line. Conversely every such line satisfies the equation, since it involves only the first block. Including its endpoints gives exactly the cone with axis Z . $\square$

Exercise I.5.13**Permitted context.**

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

The normalization B of a finitely generated integral k -algebra A in its fraction field is a finite A -module. Integral closure commutes with localization (clear denominators in a monic integral equation). The support of a finite module M is $V(\text{Ann } M)$. An affine variety is normal at its maximal ideal precisely when that localization is integrally closed.

Problem. Without using existence of nonsingular points, prove that the nonnormal points of a variety form a proper closed subset.

Solution. On an affine open with ring A , let B be its normalization. By localization, $A_{\mathfrak{m}}$ is normal exactly when $B_{\mathfrak{m}} = A_{\mathfrak{m}}$, equivalently $(B/A)_{\mathfrak{m}} = 0$. The module B/A is finite, so the nonnormal set on this open is the closed set $V(\text{Ann}_A(B/A))$. Choose finitely many module generators of B and clear their denominators in $\text{Frac } A$. Their product yields nonzero $a \in A$ with $aB \subseteq A$. Thus the annihilator is nonzero, and its zero set is proper (or empty). These descriptions on affine opens agree because normality is intrinsic to the local ring. A subset closed on an open cover is closed globally. The nonempty normal open $D(a)$ in any one affine chart also proves that this global closed subset is proper. $\square$

Exercise I.5.14**Permitted context.**

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

Analytic isomorphism here means isomorphism of completed local k -algebras. For a plane equation of order m , its associated graded local ring is $k[x, y]/(f_m)$, with Hilbert function eventually equal to m . A formal substitution with invertible linear part is an automorphism. Weierstrass preparation replaces a series regular of order two in y , up to a unit, by $y^2 + a(x)y + b(x)$. Completions of reduced algebraic plane curve local rings are reduced. The number of branches and $\delta = \dim_k(\tilde{R}/R)$, with $\tilde{R}$ the normalization of a reduced complete curve ring, are ring invariants.

Problem. (a) Prove that analytically isomorphic plane singularities have the same multiplicity.

(b) If the leading form of $f \in k[[x, y]]$ is $g_s h_t$, where the homogeneous factors are coprime, lift the factorization to $f = gh$ with those leading forms.

(c) Show that all ordinary double points are analytically isomorphic, as are all ordinary triple points; explain the one-parameter variation of ordinary four-fold points.

(d) In characteristic different from two, classify a reduced plane double point as $y^2 = x^r$, $r \geq 2$, and prove uniqueness of r .

Solution. (a) An isomorphism of local rings preserves the maximal ideal, its powers, and their successive quotient dimensions. The eventual value of this Hilbert function is the order of the plane equation, so the multiplicities coincide.

(b) Put $m = s + t$. Fix the leading pieces g_s, h_t , whose product already equals the degree- m term of f . Construct higher homogeneous pieces successively, starting at $j = 1$. At degree $m + j$, the known lower pieces leave an error E_{m+j} , to be written

$$E_{m+j} = g_s h_{t+j} + h_t g_{s+j}.$$

The homogeneous map from forms of degrees $t + j, s + j$ onto forms of degree $s + t + j$ is surjective for $j \geq 0$. Indeed its kernel consists exactly of $(h_t q, -g_s q)$, with q of degree j , by coprimality. Its image dimension is $(t + j + 1) + (s + j + 1) - (j + 1) = s + t + j + 1$, the target dimension. Thus the correction exists at every step; the series converge in the formal topology and their product is f .

(c) Applying (b) repeatedly factors an ordinary multiple point into smooth branches with the prescribed distinct tangent lines. For two branches, their equations form formal coordinates, giving the ring $k[[x, y]]/(xy)$. For three, straighten the first two to $x = 0, y = 0$. The third has equation $xA + yB$, where A, B are units because its tangent is distinct from both axes. The change $x' = xA, y' = yB$ gives the form $x'y'(x' + y')$, up to a unit.

For four, already the cones

$$xy(x - y)(x - \lambda y) = 0, \quad \lambda \notin \{0, 1\},$$

give a one-parameter collection. An analytic isomorphism induces a linear isomorphism on cotangent spaces and therefore a projective transformation of their four unordered tangent directions. Cross-ratios imply that λ, μ can be equivalent only if

$$\mu \in \left\{ \lambda, 1 - \lambda, \lambda^{-1}, (1 - \lambda)^{-1}, \frac{\lambda}{\lambda - 1}, \frac{\lambda - 1}{\lambda} \right\}.$$

Conversely these changes arise by permuting four directions. Thus parameters modulo these finite identifications form a one-dimensional family of distinct analytic types; choosing one representative per orbit gives mutually nonisomorphic examples. This does not assert that every distinct raw value of λ gives a distinct unordered configuration, or classify the effect of all higher-order terms of an arbitrary ordinary four-fold point.

(d) Choose a y -direction where the quadratic leading form is nonzero. Preparation gives $y^2 + a(x)y + b(x)$, with $\text{ord } a \geq 1, \text{ord } b \geq 2$. Completing the square gives $y'^2 - c(x)$. Reducedness implies $c \neq 0$; otherwise the completed equation would be a square. Write $c = x^r u(x)$, with $r \geq 2$ and u a unit. It has a square root in $k[[x]]$, obtained coefficient by coefficient since two is invertible. Replacing y' by $y'/\sqrt{u}$ yields $y^2 = x^r$.

If $r = 2m + 1$, the normalization is $k[[t]]$ with $x = t^2, y = t^{2m+1}$. Its quotient by the original ring has basis $t, t^3, \dots, t^{2m-1}$, so $\delta = m$ and there is one branch. If $r = 2m$, the two branches are $y = \pm x^m$. The ring embeds in $k[[x]] \oplus k[[x]]$ as pairs congruent modulo x^m ; hence $\delta = m$ and there are two branches. In both cases $r = 2\delta - (\text{number of branches}) + 2$. These invariants prove uniqueness. The values $r = 2, 3, 4$ give respectively node, cusp, and tacnode. $\square$

Exercise I.5.15

Permitted context.

Work over an algebraically closed field k . For a reduced affine hypersurface $f = 0$, singular points are the common zeros of f and all its first partial derivatives. The multiplicity of a plane curve at the origin is the lowest nonzero total degree of its equation; the corresponding homogeneous form is its tangent cone. A regular local ring is characterized by equality of its dimension and the dimension of its cotangent space.

Nonzero degree- d ternary forms up to scalar form $\mathbb{P}^N$, $N = \binom{d+2}{2} - 1$. Projection of a closed subset of $\mathbb{P}^N \times \mathbb{P}^2$ to $\mathbb{P}^N$ is closed, by projective elimination. The gradient criterion is valid in every characteristic. For each $d > 0$ there is a smooth degree- d plane curve: use $x^d + y^d + z^d$ if the characteristic does not divide d , and $x^d + y^{d-1}z + xz^{d-1}$ otherwise.

Problem. (a) Describe the correspondence from $\mathbb{P}^N$ to plane zero sets defined by degree- d equations, including its failures of injectivity.

(b) Show that nonsingular irreducible degree- d curves correspond bijectively to a nonempty Zariski-open subset of $\mathbb{P}^N$.

Solution. (a) A nonzero coefficient vector defines a homogeneous form and its projective zero set; scalar multiples define the same set. Conversely every set admitting a degree- d equation is obtained. If $f = c \prod h_i^{a_i}$ is its factorization into distinct irreducible homogeneous forms, its zero set remembers exactly the factors h_i , not their positive multiplicities a_i . Thus fibers consist precisely of exponent choices satisfying $\sum a_i \deg h_i = d$, modulo a common scalar. For a squarefree f , all these exponents are forced to be one, so its point is uniquely determined.

(b) In parameter space times $\mathbb{P}^2$, impose the four equations

$$F = F_x = F_y = F_z = 0,$$

where F is the universal degree- d form. Their incidence set is closed and its projection is closed by elimination. The complement consists exactly of forms whose zero sets satisfy the nonvanishing-gradient condition. Such a form is irreducible: a nontrivial factorization would yield a common projective zero of two factors where all derivatives vanish. Consequently the complement parametrizes

precisely the stated curves, and (a) gives bijectivity. The explicit smooth examples in the context prove nonemptiness. The equation $F = 0$ must be retained in the incidence condition: when the characteristic divides d , Euler's identity does not recover it from the partial derivatives. $\square$

I.6 Nonsingular Curves

Exercise I.6.1

Permitted context.

Work over an algebraically closed field k . A nonsingular curve is an open subvariety of its unique nonsingular projective model. Rational maps from a nonsingular curve to a projective variety extend uniquely to morphisms. Function-field isomorphisms give birational maps and inverse isomorphisms of nonsingular projective models. Nonsingular curves are normal. Principal affine opens have localized coordinate rings, and localizations of unique factorization domains are unique factorization domains.

Problem. Let Y be a nonsingular rational curve not isomorphic to $\mathbb{P}^1$.

- (a) Show that Y is isomorphic to an open subset of $\mathbb{A}^1$.
- (b) Show that Y is affine.
- (c) Show that $A(Y)$ is a unique factorization domain.

Solution. (a) Its nonsingular projective model is $\mathbb{P}^1$, since its function field is $k(t)$. The open embedding in that model has nonempty complement. Send one missing point to infinity by a projective linear change of coordinates. Then Y lies openly in $\mathbb{A}^1$.

(b) A proper closed subset of $\mathbb{A}^1$ is finite, so the image is $D(h)$, where $h = \prod (t - a_i)$ ranges over the finite missing affine points; take $h = 1$ if none are missing. The graph $hu = 1$ realizes it as an affine variety.

(c) Its coordinate ring is $k[t, h^{-1}]$, a localization of the principal ideal domain $k[t]$, hence a unique factorization domain. $\square$

Exercise I.6.2**Permitted context.**

Work over an algebraically closed field k . A nonsingular curve is an open subvariety of its unique nonsingular projective model. Rational maps from a nonsingular curve to a projective variety extend uniquely to morphisms. Function-field isomorphisms give birational maps and inverse isomorphisms of nonsingular projective models. Nonsingular curves are normal. Principal affine opens have localized coordinate rings, and localizations of unique factorization domains are unique factorization domains.

Assume $\text{char } k \neq 2$. Gauss's lemma tests polynomial irreducibility over a fraction field. An affine variety is normal exactly when its coordinate ring is integrally closed. An affine nonsingular rational curve has a unique factorization coordinate ring, since it is a principal open subset of $\mathbb{A}^1$.

Problem. For $Y = V(y^2 - x^3 + x)$, put $A = k[x, y]/(y^2 - x^3 + x)$, $K = K(Y)$.

- (a) Prove that Y is nonsingular and A is a normal domain.
- (b) Show that $k[x]$ is polynomial and that A is its integral closure in K .
- (c) Define $\sigma(x) = x, \sigma(y) = -y$; prove that $N(a) = a\sigma(a)$ lies in $k[x]$, with $N(1) = 1$ and $N(ab) = N(a)N(b)$.
- (d) Determine the units, prove x, y irreducible, and show A is not a unique factorization domain.
- (e) Deduce that Y is not rational.

Solution. (a) The polynomial $x^3 - x$ is not a square in $k(x)$, because its order at $x = 0$ is one. Thus the quadratic in y is irreducible. A singular point would have $y = 0, 3x^2 = 1$, whereas the equation then requires $x = 0, \pm 1$; these cannot occur together when two is nonzero. This includes characteristic three. The curve is nonsingular, hence normal, so A is integrally closed.

(b) Division by the monic relation gives the free $k[x]$ -basis $1, y$. In particular $k[x]$ injects and has no polynomial relation. The element y is integral over it, so A is integral over $k[x]$. An element of K integral over $k[x]$ is also integral over A ; normality puts it in A . This proves the integral-closure statement.

(c) The substitution preserves the equation and squares to the identity. For the unique expression $a = f(x) + yg(x)$,

$$N(a) = f(x)^2 - (x^3 - x)g(x)^2.$$

It lies in $k[x]$, and the two norm identities follow from commutativity and the automorphism property.

(d) If both summands are nonzero, their degrees are respectively even and odd, so their leading terms cannot cancel. Thus for nonzero a ,

$$\deg N(a) = \max\{2 \deg f, 3 + 2 \deg g\},$$

omitting zero summands. A unit has norm a unit in $k[x]$, hence degree zero, so is a nonzero constant. Conversely those are units. Every nonunit has norm degree at least two. Since $N(x) = x^2$ and $N(y) = -(x^3 - x)$ have degrees two and three, neither can be a product of norms of two nonunits. Thus x, y are irreducible. But $x \mid y^2 = x(x - 1)(x + 1)$ and $x \nmid y$: the basis $1, y$ would require a polynomial coefficient $1/x$ if $y = xa$. Thus the irreducible x is not prime, excluding unique factorization.

(e) The affine curve is not $\mathbb{P}^1$, since it has nonconstant global regular functions. If it were rational, its nonsingularity and the open-subset result would make A a unique factorization domain, contradicting (d). $\square$

Exercise I.6.3**Permitted context.**

Work over an algebraically closed field k . A nonsingular curve is an open subvariety of its unique nonsingular projective model. Rational maps from a nonsingular curve to a projective variety extend uniquely to morphisms. Function-field isomorphisms give birational maps and inverse isomorphisms of nonsingular projective models. Nonsingular curves are normal. Principal affine opens have localized coordinate rings, and localizations of unique factorization domains are unique factorization domains.

Problem. Give counterexamples to extension of rational maps if (a) the nonsingular source has dimension at least two, or (b) the target need not be projective.

Solution. (a) The rational map $\mathbb{A}^2 \dashrightarrow \mathbb{P}^1$, $(x, y) \mapsto [x : y]$, has no extension at the origin. Either affine target chart at a proposed value would make y/x or x/y regular in $k[x, y]_{(x, y)}$, impossible by divisibility. The source is nonsingular of dimension two.

(b) The coordinate t gives a rational map $\mathbb{P}^1 \dashrightarrow \mathbb{A}^1$. At infinity it is $1/u$ in the parameter $u = 1/t$, absent from $k[u]_{(u)}$. Thus it cannot extend there, although the source is a nonsingular projective curve. $\square$

Exercise I.6.4**Permitted context.**

Work over an algebraically closed field k . A nonsingular curve is an open subvariety of its unique nonsingular projective model. Rational maps from a nonsingular curve to a projective variety extend uniquely to morphisms. Function-field isomorphisms give birational maps and inverse isomorphisms of nonsingular projective models. Nonsingular curves are normal. Principal affine opens have localized coordinate rings, and localizations of unique factorization domains are unique factorization domains.

A morphism from a projective variety has closed image (projective elimination applied to its closed graph). Proper closed subsets of an irreducible curve are finite.

Problem. Show that a nonconstant rational function on a nonsingular projective curve defines a surjective morphism to $\mathbb{P}^1$ with finite fibers.

Solution. Regard the function as a rational map to $\mathbb{P}^1$. The extension theorem supplies its morphism on the entire curve. Its image is irreducible and closed. A proper such subset of $\mathbb{P}^1$ is a single point, which would make the function constant. Thus the image is all of $\mathbb{P}^1$. Each fiber is closed and proper because the morphism is nonconstant, so is finite. Surjectivity also says it is nonempty. $\square$

Exercise I.6.5**Permitted context.**

Work over an algebraically closed field k . A nonsingular curve is an open subvariety of its unique nonsingular projective model. Rational maps from a nonsingular curve to a projective variety extend uniquely to morphisms. Function-field isomorphisms give birational maps and inverse isomorphisms of nonsingular projective models. Nonsingular curves are normal. Principal affine opens have localized coordinate rings, and localizations of unique factorization domains are unique factorization domains.

Varieties have closed diagonals. Projection $\mathbb{P}^n \times T \rightarrow T$ is closed for a variety T , by projective elimination on affine charts of T .

Problem. If a nonsingular projective curve X is a locally closed subvariety of a variety Y , prove that X is closed in Y .

Solution. The inclusion $i : X \rightarrow Y$ is a morphism, so its graph is closed in $X \times Y$: it is the inverse image of the diagonal of Y under $(x, y) \mapsto (i(x), y)$. Embed the projective X as a closed subset of $\mathbb{P}^n$. The graph is then closed in $\mathbb{P}^n \times Y$. Its projection to Y is closed by elimination, and is exactly the given subset X . This proof in fact uses only projectivity, not nonsingularity or the dimension-one hypothesis. $\square$

Exercise I.6.6**Permitted context.**

Work over an algebraically closed field k . A nonsingular curve is an open subvariety of its unique nonsingular projective model. Rational maps from a nonsingular curve to a projective variety extend uniquely to morphisms. Function-field isomorphisms give birational maps and inverse isomorphisms of nonsingular projective models. Nonsingular curves are normal. Principal affine opens have localized coordinate rings, and localizations of unique factorization domains are unique factorization domains.

An invertible two-by-two matrix acts on homogeneous coordinates of $\mathbb{P}^1$; scalar matrices act trivially. For relatively prime nonproportional $P, Q \in k[T]$, $P(T) - uQ(T)$ is irreducible in $k(u)[T]$: apply Gauss's lemma to this primitive polynomial linear in u .

Problem. (a) Show that $t \mapsto (at + b)/(ct + d)$, $ad - bc \neq 0$, gives an automorphism of $\mathbb{P}^1$.
 (b) Identify its automorphism group with the group of k -automorphisms of $k(t)$.
 (c) Show that all these automorphisms are fractional linear, so the projective linear group is the full automorphism group.

Solution. (a) Homogenizing gives $[T : U] \mapsto [aT + bU : cT + dU]$. The determinant condition prevents simultaneous zero coordinates. The inverse matrix supplies an inverse morphism.

(b) Pullback gives the contravariant correspondence. Every field automorphism and its inverse extend to inverse morphisms of the nonsingular projective model $\mathbb{P}^1$. To state a group isomorphism, send a geometric automorphism φ to $(\varphi^{-1})^*$; ordinary pullback alone reverses composition.

(c) A field automorphism sends t to a nonconstant $P(t)/Q(t)$ with coprime polynomials P, Q . Put $d = \max(\deg P, \deg Q)$. The polynomial $P(T) - uQ(T)$ has degree d in T and is irreducible over $k(u)$. Substituting the transcendental $u = P(t)/Q(t)$ proves $[k(t) : k(P(t)/Q(t))] = d$. For an automorphism this degree is one. Thus P, Q are linear or constant and give $(at + b)/(ct + d)$, with nonzero determinant because the ratio is nonconstant. Two matrices give the same function exactly when they differ by scalar, proving the asserted projective linear description. $\square$

Exercise I.6.7**Permitted context.**

Work over an algebraically closed field k . A nonsingular curve is an open subvariety of its unique nonsingular projective model. Rational maps from a nonsingular curve to a projective variety extend uniquely to morphisms. Function-field isomorphisms give birational maps and inverse isomorphisms of nonsingular projective models. Nonsingular curves are normal. Principal affine opens have localized coordinate rings, and localizations of unique factorization domains are unique factorization domains.

Every automorphism of $\mathbb{P}^1$ is fractional linear. Four unordered distinct points have cross-ratio defined up to the six values $\lambda, 1 - \lambda, \lambda^{-1}, (1 - \lambda)^{-1}, \lambda/(\lambda - 1), (\lambda - 1)/\lambda$. Fractional linear maps act transitively on ordered triples of distinct points.

Problem. If $\mathbb{A}^1 \setminus \{P_1, \dots, P_r\}$ and $\mathbb{A}^1 \setminus \{Q_1, \dots, Q_s\}$ are isomorphic, prove $r = s$. Is equality sufficient?

Solution. An isomorphism induces a function-field automorphism and hence extends uniquely to an automorphism of their projective completions $\mathbb{P}^1$. Its inverse extends too, so it bijects the missing sets, including the point at infinity on each side. Therefore $r + 1 = s + 1$, giving $r = s$.

Equality is not sufficient in general. For $r = s = 3$, take missing affine sets $\{0, 1, \lambda\}$, $\{0, 1, \mu\}$. Choose $\lambda \notin \{0, 1\}$ and choose μ outside $\{0, 1\}$ and its finite cross-ratio orbit, possible since k is infinite. The four missing projective points cannot be carried to one another by an automorphism, so the punctured curves are not isomorphic. For $r = s \leq 2$, equality does suffice, by the transitivity on at most three missing projective points. $\square$

I.7 Intersections in Projective Space**Exercise I.7.1****Permitted context.**

Work over an algebraically closed field k . The Hilbert polynomial of an r -dimensional projective variety has leading term $(\deg Y)m^r/r!$. Hilbert polynomials are additive in exact sequences of graded modules. For a reduced pure-dimensional algebraic set, degree is the sum of the degrees of its irreducible components. A proper hyperplane section has degree equal to that of the variety when component multiplicities are included; discarding multiplicities can only decrease degree.

Problem. (a) Compute the degree of the degree- d Veronese image of $\mathbb{P}^n$.
(b) Compute the degree of the Segre image of $\mathbb{P}^r \times \mathbb{P}^s$.

Solution. (a) Degree m in the Veronese coordinate ring consists of all forms of degree dm in $n + 1$ variables, because every such monomial factors into m degree- d monomials. Thus its Hilbert function is $\binom{dm+n}{n}$, whose leading coefficient is $d^n/n!$. The degree is d^n .

(b) Degree m in the Segre ring consists of all bihomogeneous forms of bidegree (m, m) , since each monomial pairs its m first-factor variables with its m second-factor variables. Its Hilbert function is $\binom{m+r}{r}\binom{m+s}{s}$. The leading coefficient is $1/(r!s!)$ and the dimension is $r + s$, so the degree is $(r + s)!/(r!s!) = \binom{r+s}{r}$. $\square$

Exercise I.7.2**Permitted context.**

Work over an algebraically closed field k . The Hilbert polynomial of an r -dimensional projective variety has leading term $(\deg Y)m^r/r!$. Hilbert polynomials are additive in exact sequences of graded modules. For a reduced pure-dimensional algebraic set, degree is the sum of the degrees of its irreducible components. A proper hyperplane section has degree equal to that of the variety when component multiplicities are included; discarding multiplicities can only decrease degree.

Define $p_a(Y) = (-1)^{\dim Y}(P_Y(0) - 1)$. The polynomial $\binom{m+n}{n}$ is the Hilbert polynomial of $\mathbb{P}^n$; binomial expressions below are evaluated as polynomials even at negative integers. Two homogeneous equations generating the ideal of a codimension-two complete intersection form a regular sequence.

Problem. Compute arithmetic genera:

- (a) $\mathbb{P}^n$.
- (b) A plane curve of degree d .
- (c) A degree- d hypersurface in $\mathbb{P}^n$.
- (d) A strict complete-intersection curve of surfaces of degrees a, b in $\mathbb{P}^3$.
- (e) The Segre product of projective varieties Y, Z of dimensions r, s , in terms of their arithmetic genera.

Solution. (a) Since $\binom{n}{n} = 1$, the answer is zero.

(b) Multiplication by the nonzero defining equation gives $P_Y(m) = \binom{m+2}{2} - \binom{m+2-d}{2}$. Thus $p_a(Y) = 1 - P_Y(0) = (d-1)(d-2)/2$.

(c) The same exact sequence gives

$$P_H(m) = \binom{m+n}{n} - \binom{m+n-d}{n}.$$

Consequently $p_a(H) = (-1)^n \binom{n-d}{n} = \binom{d-1}{n}$.

(d) The two regular-sequence exact sequences, equivalently their Koszul resolution, give

$$P_Y(m) = \binom{m+3}{3} - \binom{m+3-a}{3} - \binom{m+3-b}{3} + \binom{m+3-a-b}{3}.$$

Expanding at zero and using $p_a = 1 - P_Y(0)$ yields

$$p_a(Y) = 1 + \frac{ab(a+b-4)}{2}.$$

Ideal generation, not merely set-theoretic intersection, is the hypothesis needed for this Hilbert polynomial.

(e) The degree- m component of the Segre coordinate ring is the tensor product of the two degree- m components. Therefore $P_{Y \times Z}(m) = P_Y(m)P_Z(m)$. Substituting $P_Y(0) = 1 + (-1)^r p_a(Y)$ and the analogous formula for Z gives

$$p_a(Y \times Z) = p_a(Y)p_a(Z) + (-1)^s p_a(Y) + (-1)^r p_a(Z).$$

□

Exercise I.7.3**Permitted context.**

Work over an algebraically closed field k . The Hilbert polynomial of an r -dimensional projective variety has leading term $(\deg Y)m^r/r!$. Hilbert polynomials are additive in exact sequences of graded modules. For a reduced pure-dimensional algebraic set, degree is the sum of the degrees of its irreducible components. A proper hyperplane section has degree equal to that of the variety when component multiplicities are included; discarding multiplicities can only decrease degree.

If a plane curve is smooth at the origin, its local equation starts with a nonzero linear form. Intersection multiplicity with a line not a component is the order of its restricted equation. Homogeneous coordinates of a morphism may be given by forms of equal degree having no common zero on its domain.

Problem. For a plane curve Y , show that each nonsingular point has a unique tangent line with intersection multiplicity greater than one. Show that sending the point to this line is a morphism to the dual projective plane. Address the case when Y itself is a line.

Solution. Translate a smooth point to the affine origin and write $f = \ell + h$, where $\ell \neq 0$ is linear and h has order at least two. On a line with direction (a, b) , the restricted equation has linear coefficient $\ell(a, b)$. Its intersection multiplicity is one unless $\ell(a, b) = 0$, which selects exactly one line; on that line the order is greater than one. For a curve of degree greater than one the restricted equation cannot vanish identically, because that would make the line a component. If Y is a line, its tangent is itself: its intersection quotient has infinite length, while every other line through the point has multiplicity one.

For homogeneous equation F , the tangent line at a smooth point P has coefficients

$$[F_{x_0}(P) : F_{x_1}(P) : F_{x_2}(P)].$$

The coefficients have equal degree and are not all zero on $\text{Reg } Y$; Euler's identity shows the line passes through P , and dehomogenization identifies it with the line just found. Ratios on the three target charts are regular, so this is a morphism. Its image closure is the dual locus; for a line it is a point rather than a one-dimensional curve. $\square$

Exercise I.7.4**Permitted context.**

Work over an algebraically closed field k . The Hilbert polynomial of an r -dimensional projective variety has leading term $(\deg Y)m^r/r!$. Hilbert polynomials are additive in exact sequences of graded modules. For a reduced pure-dimensional algebraic set, degree is the sum of the degrees of its irreducible components. A proper hyperplane section has degree equal to that of the variety when component multiplicities are included; discarding multiplicities can only decrease degree.

The singular set of an irreducible plane curve is finite. The closure of an image of a curve under a morphism has dimension at most one. The tangent assignment on the smooth locus is a morphism. A line not equal to a plane curve of degree d has total intersection multiplicity d ; transverse smooth intersections have multiplicity one.

Problem. Show that a nonempty open set of lines in the dual plane meets a degree- d plane curve in exactly d distinct points.

Solution. For each singular point, lines through it form a projective line in the dual plane. The closure of all smooth-point tangent lines has dimension at most one. Their union is a finite union of proper closed subsets of the irreducible dual plane, hence is proper closed. Let U be its nonempty open complement. A line in U passes through no singular point and is nowhere tangent to the curve. If the curve itself is a line, that one dual point was also excluded. All intersections therefore have multiplicity one. Their total multiplicity is d , so there are exactly d distinct points. $\square$

Exercise I.7.5**Permitted context.**

Work over an algebraically closed field k . The Hilbert polynomial of an r -dimensional projective variety has leading term $(\deg Y)m^r/r!$. Hilbert polynomials are additive in exact sequences of graded modules. For a reduced pure-dimensional algebraic set, degree is the sum of the degrees of its irreducible components. A proper hyperplane section has degree equal to that of the variety when component multiplicities are included; discarding multiplicities can only decrease degree.

The multiplicity at $[0 : 0 : 1]$ is the lowest degree of a nonzero term of $F(x, y, 1)$. A binary homogeneous form factors into linear factors over k . Birational inverse maps on nonempty opens prove rationality.

Problem. (a) Prove that an irreducible plane curve of degree $d > 1$ has no point of multiplicity at least d .

(b) Prove that such a curve with a point of multiplicity $d - 1$ is rational.

Solution. (a) Put the point at $[0 : 0 : 1]$. Multiplicity at least d forces the degree- d equation to be $F = f_d(x, y)$. This binary form splits into d linear factors, contrary to irreducibility and $d > 1$.

(b) The equation now has the form

$$F = zf_{d-1}(x, y) + f_d(x, y), \quad f_{d-1} \neq 0.$$

Projection from the point sends $[x : y : z]$ to $[x : y]$. On the nonempty open $f_{d-1}(u, v) \neq 0$, its inverse is

$$[u : v] \longmapsto [uf_{d-1}(u, v) : vf_{d-1}(u, v) : -f_d(u, v)].$$

These degree- d coordinates satisfy $F = 0$, and direct substitution shows both compositions are the identity where defined. Thus the curve is birational to $\mathbb{P}^1$. $\square$

Exercise I.7.6

Permitted context.

Work over an algebraically closed field k . The Hilbert polynomial of an r -dimensional projective variety has leading term $(\deg Y)m^r/r!$. Hilbert polynomials are additive in exact sequences of graded modules. For a reduced pure-dimensional algebraic set, degree is the sum of the degrees of its irreducible components. A proper hyperplane section has degree equal to that of the variety when component multiplicities are included; discarding multiplicities can only decrease degree.

For a standard graded domain S over an infinite field, graded Noether normalization provides algebraically independent linear forms $l_0, \dots, l_r$ such that S is finite over $B = k[l_0, \dots, l_r]$. A finite graded module of rank e over this polynomial ring has Hilbert polynomial with leading coefficient $e/r!$: choose homogeneous generically independent generators; the remaining torsion modules have smaller dimension. Polynomial rings are integrally closed.

Problem. Prove that a reduced projective algebraic set of pure dimension r has degree one exactly when it is a linear variety.

Solution. For a linear $\mathbb{P}^r$, the coordinate ring is a polynomial ring in $r+1$ linear variables, of degree one. Conversely, additivity of degree and positivity of each component degree show that a pure-dimensional degree-one set has exactly one irreducible component. Its homogeneous coordinate ring S is a domain. Choose the normalization algebra B in the context. The Hilbert leading coefficient shows that $[\text{Frac } S : \text{Frac } B] = \deg Y = 1$. Thus $B \subseteq S \subseteq \text{Frac } B$ and S is integral over B . Integral closedness gives $S = B$. In particular all ambient degree-one coordinates restrict to linear combinations of $l_0, \dots, l_r$. Their linear relations define a linear $\mathbb{P}^r$ containing Y ; the coordinate-ring equality says that Y is exactly that linear space. $\square$

Exercise I.7.7**Permitted context.**

Work over an algebraically closed field k . The Hilbert polynomial of an r -dimensional projective variety has leading term $(\deg Y)m^r/r!$. Hilbert polynomials are additive in exact sequences of graded modules. For a reduced pure-dimensional algebraic set, degree is the sum of the degrees of its irreducible components. A proper hyperplane section has degree equal to that of the variety when component multiplicities are included; discarding multiplicities can only decrease degree.

A general codimension- r linear section of an r -dimensional projective variety consists of $\deg Y$ distinct smooth points. More generally, a finite intersection with a codimension- r linear subspace has at most $\deg Y$ distinct points: cut successively with a flag of hyperplanes, preserving proper intersection at each step, and use the degree bound for each cut. An irreducible cone with point vertex over Z has dimension $\dim Z + 1$ and degree $\deg Z$, since its coordinate ring is the polynomial extension of the coordinate ring of Z . The projective tangent space at a smooth point of an r -fold is a linear $\mathbb{P}^r$. Degree-one varieties are linear. Blowing up a smooth point has exceptional fiber $\mathbb{P}^{r-1}$ and resolves projection from that point. A projective dominant morphism of equal-dimensional varieties has finite fibers over a nonempty open subset of the target, by the dimension-of-fibers theorem.

Problem. Let $Y \subseteq \mathbb{P}^n$ have dimension r and degree $d > 1$, and let P be a smooth point. Let X be the closure of the union of lines PQ , $Q \in Y \setminus \{P\}$.

- (a) Prove that X is irreducible of dimension $r + 1$.
- (b) Prove that $\deg X < d$.

Solution. (a) Let Z be the closure of the projection of $Y \setminus \{P\}$ from P to a complementary hyperplane. It is irreducible, and X is exactly its cone with vertex P . The union-of-lines parametrization bounds $\dim X \leq r + 1$, while $Y \subseteq X$ gives $\dim X \geq r$. If $\dim X = r$, irreducibility makes $X = Y$. Then every line PQ is contained in Y , so its direction at P lies in $T_P Y$. Consequently Y lies in the projective tangent $\mathbb{P}^r$, and having the same dimension equals it. This gives degree one, a contradiction. Hence $\dim X = r + 1$, and $\dim Z = r$.

(b) The projection has zero-dimensional generic fiber, since $\dim Y = \dim Z$. More explicitly, the closed incidence graph in the blow-up at P is projective over Z ; over a nonempty open of Z its fibers are finite. Its exceptional image lies among the tangent directions at P , of dimension at most $r - 1$. Choose a general codimension- r linear subspace Λ of the target, so that $Z \cap \Lambda$ consists of $\deg Z$ distinct points, all in this finite-fiber open and outside the exceptional image. Their inverse images in $Y \setminus \{P\}$ are finite and nonempty. The span $L = \langle P, \Lambda \rangle$ has codimension r in $\mathbb{P}^n$, and $L \cap Y$ is these finite fibers together with P . Therefore it has at least $1 + \deg Z$ distinct points. The linear-section bound gives

$$1 + \deg Z \leq d.$$

Since $\deg X = \deg Z$ for a cone, the required strict inequality follows. $\square$

Exercise I.7.8**Permitted context.**

Work over an algebraically closed field k . The Hilbert polynomial of an r -dimensional projective variety has leading term $(\deg Y)m^r/r!$. Hilbert polynomials are additive in exact sequences of graded modules. For a reduced pure-dimensional algebraic set, degree is the sum of the degrees of its irreducible components. A proper hyperplane section has degree equal to that of the variety when component multiplicities are included; discarding multiplicities can only decrease degree.

A variety has a nonempty smooth locus. For a projective r -fold of degree greater than one, the cone swept out by lines through a smooth point has dimension $r+1$ and strictly smaller degree. A degree-one projective variety is linear. A codimension-one projective variety in a linear space has a principal homogeneous ideal, whose generator's degree equals the variety's degree.

Problem. Prove that a degree-two r -dimensional projective variety lies in a linear $\mathbb{P}^{r+1}$, where it is a quadric hypersurface.

Solution. Choose a smooth point and form the cone of joining lines. It contains Y , has dimension $r+1$, and has positive degree strictly less than two. Its degree is therefore one, so it is a linear space $L \cong \mathbb{P}^{r+1}$. The variety Y is a hypersurface in L . The generator of its homogeneous prime ideal has degree $\deg Y = 2$, giving the asserted quadric equation. There is no missing zero-dimensional case: over algebraically closed k an irreducible zero-dimensional variety is one point, of degree one. $\square$

Chapter II

II.1 Sheaves

Exercise II.1.1

Permitted context.

Let X be a topological space and A an abelian group. The constant presheaf assigns A to every nonempty open set, assigns 0 to the empty set, and uses the identity maps between nonempty opens and the unique restriction maps to the empty set. The constant sheaf $\mathcal{A}$ is the sheaf of locally constant A -valued functions. One may use the universal property of sheafification, the fact that sheafification preserves stalks, and the criterion that a morphism of sheaves is an isomorphism exactly when it is an isomorphism on every stalk.

Problem. Prove that the sheaf associated to the constant presheaf with value A is the constant sheaf $\mathcal{A}$.

Solution. There is a natural morphism from the constant presheaf to $\mathcal{A}$: an element $a \in A$ is sent on each nonempty open set to the constant function with value a . At a point $x \in X$, every germ of a locally constant function is represented on some neighborhood of x by a constant function. Consequently the induced map on the stalk at x is an isomorphism

$$A \longrightarrow \mathcal{A}_x.$$

Let A^+ be the sheafification of the constant presheaf. By the universal property, the natural morphism to $\mathcal{A}$ factors uniquely through a morphism

$$A^+ \longrightarrow \mathcal{A}.$$

Sheafification does not change stalks, so this morphism is an isomorphism on every stalk. The stalk criterion now shows that $A^+ \rightarrow \mathcal{A}$ is an isomorphism of sheaves. $\square$

Exercise II.1.2**Permitted context.**

Sheaves are sheaves of abelian groups on a fixed topological space X . The kernel of a sheaf morphism is computed sectionwise. Its image is the sheafification of the sectionwise image presheaf. Stalks are filtered colimits over neighborhoods, and filtered colimits of abelian groups preserve kernels, images, and exact sequences.

Problem. Let $\phi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves.

- (a) Prove that $(\ker \phi)_x = \ker(\phi_x)$ and $(\operatorname{im} \phi)_x = \operatorname{im}(\phi_x)$ for every $x \in X$.
- (b) Prove that ϕ is injective, respectively surjective, exactly when every map on stalks ϕ_x has that property.
- (c) Prove that a sequence of sheaves is exact exactly when its sequence of stalks is exact at every point.

Solution. (a) Taking the stalk of the sectionwise kernel gives

$$(\ker \phi)_x = \varinjlim_{x \in U} \ker(\mathcal{F}(U) \rightarrow \mathcal{G}(U)) \cong \ker(\mathcal{F}_x \rightarrow \mathcal{G}_x),$$

because filtered colimits are exact. For the image, first take the sectionwise image presheaf and then sheafify. Sheafification preserves stalks, and filtered colimits commute with images, so

$$(\operatorname{im} \phi)_x \cong \operatorname{im}(\phi_x).$$

(b) The morphism ϕ is injective exactly when $\ker \phi = 0$. A sheaf is zero exactly when all of its stalks are zero, and part (a) identifies these stalks with $\ker(\phi_x)$. This proves the injective assertion. Similarly, ϕ is surjective exactly when its cokernel is zero. Applying the same argument to the cokernel, or equivalently using the image calculation in part (a), proves the surjective assertion.

(c) For consecutive maps $\mathcal{F}^{i-1} \rightarrow \mathcal{F}^i \rightarrow \mathcal{F}^{i+1}$, exactness at $\mathcal{F}^i$ means

$$\operatorname{im}(\mathcal{F}^{i-1} \rightarrow \mathcal{F}^i) = \ker(\mathcal{F}^i \rightarrow \mathcal{F}^{i+1}).$$

Two subsheaves are equal exactly when their stalks agree. Part (a) identifies the two stalks in this equality with the image and kernel of the corresponding stalk maps. Thus exactness can be checked point by point. $\square$

Exercise II.1.3**Permitted context.**

For sheaves of abelian groups, a morphism is surjective if and only if it is surjective on every stalk. Equality of two germs means that their representatives agree after restriction to a sufficiently small neighborhood.

A nowhere-zero holomorphic function admits local logarithms, and a holomorphic differential has local primitives on small disks.

Problem. (a) Prove that a morphism $\phi : \mathcal{F} \rightarrow \mathcal{G}$ is surjective if and only if every section $s \in \mathcal{G}(U)$ can be lifted locally: there is an open cover $U = \bigcup_i U_i$ and sections $t_i \in \mathcal{F}(U_i)$ satisfying $\phi(t_i) = s|_{U_i}$.

(b) Give a surjective morphism of sheaves for which the map on sections over some open set is not surjective.

Solution. (a) Suppose ϕ is surjective. For each $x \in U$, surjectivity of ϕ_x gives a germ $\tau_x \in \mathcal{F}_x$ mapping to s_x . Choose an open neighborhood $V_x \subseteq U$ of x and a section $t_x \in \mathcal{F}(V_x)$ representing τ_x . The germs of $\phi(t_x)$ and $s|_{V_x}$ agree at x , so after shrinking to a neighborhood $U_x \subseteq V_x$, the two sections agree on U_x . The sets U_x cover U , and the restrictions $t_x|_{U_x}$ are the required local lifts.

Conversely, apply the local-lifting property to a representative of an arbitrary germ $s_x \in \mathcal{G}_x$. Some member of the resulting cover contains x , and the lift on that member produces a preimage of s_x in $\mathcal{F}_x$. Thus every ϕ_x is surjective, hence so is ϕ .

(b) Let $X = \mathbb{C}^\times$ with its usual complex topology. On X , consider the exponential morphism from the additive sheaf of holomorphic functions to the multiplicative sheaf of nowhere-vanishing holomorphic functions,

$$\exp : \mathcal{O}_X \longrightarrow \mathcal{O}_X^\times.$$

Every nowhere-vanishing holomorphic function has a holomorphic logarithm locally, so this is a surjective morphism of sheaves. The global section $z \mapsto z$ has no global holomorphic logarithm on $\mathbb{C}^\times$: integrating its logarithmic derivative around the unit circle gives

$$\frac{1}{2\pi i} \int_{|z|=1} \frac{dz}{z} = 1,$$

whereas the logarithmic derivative of an exponential e^h is dh and has integral zero around every closed curve. Hence the map on global sections is not surjective. $\square$

Exercise II.1.4**Permitted context.**

For a presheaf $\mathcal{P}$, write $\mathcal{P}^+$ for its associated sheaf. Sheafification preserves stalks. A morphism of sheaves is injective exactly when it is injective on every stalk.

The presheaves here take values in abelian groups.

Problem. (a) Let $\phi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of presheaves such that $\mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is injective for every open U . Prove that $\phi^+ : \mathcal{F}^+ \rightarrow \mathcal{G}^+$ is injective.

(b) Deduce that the image sheaf of a morphism of sheaves $\mathcal{F} \rightarrow \mathcal{G}$ is naturally a subsheaf of $\mathcal{G}$.

Solution. (a) The map on stalks $\phi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is injective. Indeed, if a germ represented by $s \in \mathcal{F}(U)$ maps to zero, then after restricting to some neighborhood $V \subseteq U$ of x , the section $\phi(s|_V)$ is zero. Injectivity of $\mathcal{F}(V) \rightarrow \mathcal{G}(V)$ implies $s|_V = 0$, so the original germ is zero. Because sheafification preserves stalks, ϕ^+ is injective on every stalk and therefore injective as a sheaf morphism.

(b) Let $\mathcal{I}$ be the presheaf defined by

$$\mathcal{I}(U) = \text{im}(\mathcal{F}(U) \rightarrow \mathcal{G}(U)).$$

The evident morphism $\mathcal{I} \rightarrow \mathcal{G}$ is injective on every open set. Part (a) therefore gives an injection $\mathcal{I}^+ \rightarrow \mathcal{G}^+$. Since $\mathcal{G}$ is already a sheaf, $\mathcal{G}^+ \cong \mathcal{G}$, while $\mathcal{I}^+$ is by definition the image sheaf. Hence the image sheaf is naturally a subsheaf of $\mathcal{G}$. $\square$

Exercise II.1.5

Permitted context.

A morphism of sheaves is injective or surjective exactly when every induced map on stalks is injective or surjective, respectively. A morphism of sheaves is an isomorphism exactly when it is an isomorphism on every stalk.

Problem. Prove that a morphism of sheaves is an isomorphism if and only if it is both injective and surjective.

Solution. Every isomorphism is injective and surjective. Conversely, let $\phi : \mathcal{F} \rightarrow \mathcal{G}$ be both injective and surjective. For every $x \in X$, the map

$$\phi_x : \mathcal{F}_x \longrightarrow \mathcal{G}_x$$

is both injective and surjective, hence is an isomorphism of abelian groups. The stalk criterion for isomorphisms then implies that ϕ is an isomorphism of sheaves.

Alternatively, local lifts give a direct construction of the inverse. For an open set U and a section $s \in \mathcal{G}(U)$, surjectivity provides an open cover $U = \bigcup_i U_i$ and sections $t_i \in \mathcal{F}(U_i)$ with $\phi(t_i) = s|_{U_i}$. On an overlap $U_i \cap U_j$, injectivity gives $t_i|_{U_i \cap U_j} = t_j|_{U_i \cap U_j}$. The sheaf axiom therefore glues the t_i to a unique section $t \in \mathcal{F}(U)$. This construction commutes with restriction and defines a morphism $\mathcal{G} \rightarrow \mathcal{F}$ inverse to ϕ . $\square$

Exercise II.1.6**Permitted context.**

If $\mathcal{F}' \subseteq \mathcal{F}$ is a subsheaf, the quotient sheaf $\mathcal{F}/\mathcal{F}'$ is the sheafification of $U \mapsto \mathcal{F}(U)/\mathcal{F}'(U)$, and

$$(\mathcal{F}/\mathcal{F}')_x \cong \mathcal{F}_x/\mathcal{F}'_x.$$

Exactness of sheaves can be checked on stalks.

Problem. (a) Prove that the canonical map $\mathcal{F} \rightarrow \mathcal{F}/\mathcal{F}'$ is surjective with kernel $\mathcal{F}'$.

(b) Conversely, given an exact sequence

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0,$$

prove that $\mathcal{F}'$ is identified with a subsheaf of $\mathcal{F}$ and $\mathcal{F}'' \cong \mathcal{F}/\mathcal{F}'$.

Solution. (a) At every point x , the canonical sequence is

$$0 \longrightarrow \mathcal{F}'_x \longrightarrow \mathcal{F}_x \longrightarrow \mathcal{F}_x/\mathcal{F}'_x \longrightarrow 0.$$

It is exact as a sequence of abelian groups. The displayed identification of the quotient stalk and the stalkwise exactness criterion therefore give an exact sequence of sheaves

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}/\mathcal{F}' \longrightarrow 0.$$

(b) Exactness at $\mathcal{F}'$ makes $\mathcal{F}' \rightarrow \mathcal{F}$ injective, and exactness at $\mathcal{F}$ identifies its image with $\ker(\mathcal{F} \rightarrow \mathcal{F}'')$. We may therefore identify $\mathcal{F}'$ with this subsheaf of $\mathcal{F}$. The map $\mathcal{F} \rightarrow \mathcal{F}''$ annihilates $\mathcal{F}'$, so the universal property of the quotient produces a morphism

$$\mathcal{F}/\mathcal{F}' \longrightarrow \mathcal{F}''.$$

On each stalk this is the canonical isomorphism

$$\mathcal{F}_x/\mathcal{F}'_x \longrightarrow \mathcal{F}''_x$$

coming from exactness. Hence it is an isomorphism of sheaves. □

Exercise II.1.7**Permitted context.**

For a morphism of sheaves $\phi : \mathcal{F} \rightarrow \mathcal{G}$, kernels, images, and cokernels have stalks equal to the corresponding kernels, images, and cokernels of ϕ_x . Quotient sheaves satisfy $(\mathcal{F}/\mathcal{K})_x = \mathcal{F}_x/\mathcal{K}_x$.

Problem. For a morphism $\phi : \mathcal{F} \rightarrow \mathcal{G}$, prove the natural isomorphisms

(a) $\operatorname{im} \phi \cong \mathcal{F}/\ker \phi$;

(b) $\operatorname{coker} \phi \cong \mathcal{G}/\operatorname{im} \phi$.

Solution. (a) The morphism $\mathcal{F} \rightarrow \text{im } \phi$ is surjective and has kernel $\ker \phi$, so it induces a natural map

$$\mathcal{F} / \ker \phi \longrightarrow \text{im } \phi.$$

At each point x , this is the usual first-isomorphism map

$$\mathcal{F}_x / \ker(\phi_x) \longrightarrow \text{im}(\phi_x),$$

which is an isomorphism. The induced sheaf morphism is therefore an isomorphism.

(b) By definition, the cokernel sheaf is the sheafification of the presheaf cokernel. The composite $\mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{G} / \text{im } \phi$ is zero, so it induces a natural morphism

$$\text{coker } \phi \longrightarrow \mathcal{G} / \text{im } \phi.$$

Its stalk at x is the canonical isomorphism

$$\text{coker}(\phi_x) = \mathcal{G}_x / \text{im}(\phi_x).$$

Hence the morphism is an isomorphism of sheaves. $\square$

Exercise II.1.8

Permitted context.

Sheaves are sheaves of abelian groups on X . For a morphism $\mathcal{F} \rightarrow \mathcal{G}$, the kernel sheaf is computed sectionwise:

$$(\ker(\mathcal{F} \rightarrow \mathcal{G}))(U) = \ker(\mathcal{F}(U) \rightarrow \mathcal{G}(U)).$$

Problem. For an open subset $U \subseteq X$, prove that the section functor $\Gamma(U, -)$ is left exact. Thus, from an exact sequence

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'',$$

deduce an exact sequence

$$0 \longrightarrow \Gamma(U, \mathcal{F}') \longrightarrow \Gamma(U, \mathcal{F}) \longrightarrow \Gamma(U, \mathcal{F}'').$$

Solution. Exactness of the sheaf sequence says that $\mathcal{F}'$ is the kernel of the morphism $\mathcal{F} \rightarrow \mathcal{F}''$. Since kernels of sheaves are computed sectionwise,

$$\Gamma(U, \mathcal{F}') = \ker(\Gamma(U, \mathcal{F}) \rightarrow \Gamma(U, \mathcal{F}'')).$$

The map $\Gamma(U, \mathcal{F}') \rightarrow \Gamma(U, \mathcal{F})$ is therefore injective, and its image is exactly the kernel of the following map. This is precisely the claimed exactness. No surjectivity onto $\Gamma(U, \mathcal{F}'')$ is asserted; local lifts of a section need not glue globally. $\square$

Exercise II.1.9**Permitted context.**

Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves of abelian groups on X . In an additive category, a biproduct is an object that satisfies the universal properties of both product and coproduct.

Problem. Prove that the presheaf

$$U \mapsto \mathcal{F}(U) \oplus \mathcal{G}(U)$$

is a sheaf and that it is simultaneously the direct product and direct sum of $\mathcal{F}$ and $\mathcal{G}$ in the category of sheaves of abelian groups.

Solution. Restrictions are defined componentwise. If pairs of sections (s_i, t_i) agree on all overlaps of an open cover, then the s_i glue uniquely to a section of $\mathcal{F}$, and the t_i glue uniquely to a section of $\mathcal{G}$. Their pair is the unique glued section. Thus the displayed presheaf is a sheaf; denote it by $\mathcal{F} \oplus \mathcal{G}$.

The component projections define morphisms

$$\mathcal{F} \oplus \mathcal{G} \longrightarrow \mathcal{F}, \quad \mathcal{F} \oplus \mathcal{G} \longrightarrow \mathcal{G}.$$

Given morphisms $\mathcal{H} \rightarrow \mathcal{F}$ and $\mathcal{H} \rightarrow \mathcal{G}$, pairing them sectionwise produces a unique morphism $\mathcal{H} \rightarrow \mathcal{F} \oplus \mathcal{G}$. Hence this sheaf is the product.

Similarly, the component inclusions define morphisms from $\mathcal{F}$ and $\mathcal{G}$ into $\mathcal{F} \oplus \mathcal{G}$. Given morphisms $\mathcal{F} \rightarrow \mathcal{H}$ and $\mathcal{G} \rightarrow \mathcal{H}$, their sectionwise sum produces the unique morphism $\mathcal{F} \oplus \mathcal{G} \rightarrow \mathcal{H}$ extending both. Hence it is also the coproduct, and therefore the direct sum. $\square$

Exercise II.1.10**Permitted context.**

Let $\{\mathcal{F}_i\}$ be a directed system of sheaves on X . Colimits of presheaves are computed sectionwise. If $\mathcal{P}$ is a presheaf and $\mathcal{G}$ is a sheaf, every presheaf morphism $\mathcal{P} \rightarrow \mathcal{G}$ factors uniquely through the sheafification $\mathcal{P}^+$.

All sheaves in this exercise are sheaves of abelian groups.

Problem. Define $\varinjlim \mathcal{F}_i$ to be the sheaf associated to the presheaf

$$U \mapsto \varinjlim_i \mathcal{F}_i(U).$$

Prove that this sheaf satisfies the universal property of the direct limit in the category of sheaves.

Solution. Let $\mathcal{P}$ be the sectionwise direct-limit presheaf. The canonical maps $\mathcal{F}_i(U) \rightarrow \mathcal{P}(U)$ commute with restriction and with the transition maps, so after sheafification they give a compatible family

$$\mathcal{F}_i \longrightarrow \mathcal{P}^+.$$

Now let $\mathcal{G}$ be a sheaf equipped with compatible morphisms $\psi_i : \mathcal{F}_i \rightarrow \mathcal{G}$. For every open U , the universal property of the direct limit of abelian groups gives a unique homomorphism

$$\mathcal{P}(U) = \varinjlim_i \mathcal{F}_i(U) \longrightarrow \mathcal{G}(U)$$

compatible with the $\psi_i(U)$. These homomorphisms commute with restriction, so they form a morphism of presheaves $\mathcal{P} \rightarrow \mathcal{G}$. Because $\mathcal{G}$ is a sheaf, this morphism factors uniquely through a sheaf morphism

$$\mathcal{P}^+ \longrightarrow \mathcal{G}.$$

The factorization is compatible with every ψ_i , and uniqueness follows first from the presheaf colimit and then from sheafification. Thus $\mathcal{P}^+$ is the direct limit in the category of sheaves. $\square$

Exercise II.1.11

Permitted context.

A topological space is noetherian if every ascending chain of open subsets stabilizes; equivalently, every open subset is quasi-compact. A filtered direct limit of abelian groups commutes with finite products and finite equalizers. For a finite open cover $U = \bigcup_{a=1}^n U_a$, the sheaf axiom for $\mathcal{F}$ says that

$$0 \longrightarrow \mathcal{F}(U) \longrightarrow \prod_a \mathcal{F}(U_a) \rightrightarrows \prod_{a,b} \mathcal{F}(U_a \cap U_b)$$

is an equalizer diagram.

Problem. Let $\{\mathcal{F}_i\}_{i \in I}$ be a filtered direct system of sheaves of abelian groups on a noetherian topological space X . Prove that the sectionwise direct-limit presheaf

$$\mathcal{P}(U) = \varinjlim_i \mathcal{F}_i(U)$$

already satisfies the sheaf axioms. Deduce that

$$\Gamma\left(X, \varinjlim_i \mathcal{F}_i\right) \cong \varinjlim_i \Gamma(X, \mathcal{F}_i).$$

Solution. First consider a finite open cover $U = U_1 \cup \cdots \cup U_n$. For every i , the sheaf axiom gives an equalizer diagram

$$0 \longrightarrow \mathcal{F}_i(U) \longrightarrow \prod_a \mathcal{F}_i(U_a) \rightrightarrows \prod_{a,b} \mathcal{F}_i(U_a \cap U_b).$$

Taking the filtered direct limit preserves this diagram because all products displayed here are finite. Hence $\mathcal{P}$ has unique gluing for every finite open cover.

Now let $U = \bigcup_{\alpha \in A} U_\alpha$ be an arbitrary cover and let $s_\alpha \in \mathcal{P}(U_\alpha)$ be compatible sections. Since U is quasi-compact, choose a finite subcover $U_{\alpha_1}, \dots, U_{\alpha_n}$. The finite-cover result produces a unique $s \in \mathcal{P}(U)$ whose restriction to each selected member is s_{α_a} .

For any α , the finite family

$$U_\alpha \cap U_{\alpha_1}, \dots, U_\alpha \cap U_{\alpha_n}$$

covers U_α . On each member of this cover, compatibility gives $s|_{U_\alpha \cap U_{\alpha_a}} = s_\alpha|_{U_\alpha \cap U_{\alpha_a}}$. Uniqueness for finite covers therefore implies $s|_{U_\alpha} = s_\alpha$. The same argument, applied to the difference of two putative gluings, proves uniqueness. Thus $\mathcal{P}$ is a sheaf.

Consequently no sheafification changes the sectionwise direct limit. Taking $U = X$ in its definition gives

$$\Gamma\left(X, \varinjlim_i \mathcal{F}_i\right) = \varinjlim_i \mathcal{F}_i(X) = \varinjlim_i \Gamma(X, \mathcal{F}_i).$$

□

Exercise II.1.12

Permitted context.

Limits of abelian groups are computed as compatible families inside products. A presheaf of abelian groups is a sheaf precisely when every compatible family of local sections has a unique gluing.

Problem. For an inverse system $\{\mathcal{F}_i\}$ of sheaves on a topological space X , prove that

$$U \mapsto \varprojlim_i \mathcal{F}_i(U)$$

is a sheaf. Show that the resulting sheaf, denoted $\varprojlim_i \mathcal{F}_i$, has the universal property of the inverse limit in the category of sheaves.

Solution. Write $\mathcal{L}(U) = \varprojlim_i \mathcal{F}_i(U)$, with restrictions defined componentwise. Suppose $U = \bigcup_\alpha U_\alpha$ and that $\ell_\alpha \in \mathcal{L}(U_\alpha)$ agree on overlaps. For every i , let $\ell_{\alpha,i} \in \mathcal{F}_i(U_\alpha)$ be the i -th component. The sheaf axiom for $\mathcal{F}_i$ gives a unique $\ell_i \in \mathcal{F}_i(U)$ restricting to all the $\ell_{\alpha,i}$.

If $i \rightarrow j$ is a transition arrow in the inverse system, the image of ℓ_i and the section ℓ_j have the same restriction to every U_α . They are therefore equal by uniqueness in $\mathcal{F}_j$. Thus the family $(\ell_i)_i$ is compatible and defines $\ell \in \mathcal{L}(U)$. Componentwise uniqueness proves uniqueness of ℓ , so $\mathcal{L}$ is a sheaf.

The projections $\mathcal{L} \rightarrow \mathcal{F}_i$ form a compatible cone. If $\mathcal{G}$ is a sheaf with a compatible cone $\psi_i : \mathcal{G} \rightarrow \mathcal{F}_i$, then for every open U the maps $\psi_i(U)$ induce a unique homomorphism

$$\mathcal{G}(U) \longrightarrow \varprojlim_i \mathcal{F}_i(U) = \mathcal{L}(U).$$

These homomorphisms commute with restriction because all the ψ_i do, and hence define a sheaf morphism $\mathcal{G} \rightarrow \mathcal{L}$. Its components are the prescribed maps, and componentwise uniqueness makes it the only such morphism. This is exactly the inverse-limit universal property. □

Exercise II.1.13**Permitted context.**

For a presheaf $\mathcal{F}$, its associated sheaf may be realized as the sheaf whose sections over U are functions t assigning to each $x \in U$ a germ $t(x) \in \mathcal{F}_x$, subject to the condition that every point has a neighborhood on which these germs are represented by one section of $\mathcal{F}$. The topology on a set receiving maps $\rho_a : T_a \rightarrow E$ that is strongest among those making every ρ_a continuous is characterized by: $W \subseteq E$ is open if and only if $\rho_a^{-1}(W)$ is open for every a .

Problem. Let $\mathcal{F}$ be a presheaf on X . Form

$$E(\mathcal{F}) = \coprod_{x \in X} \mathcal{F}_x$$

and let $\pi : E(\mathcal{F}) \rightarrow X$ send a germ in $\mathcal{F}_x$ to x . For $s \in \mathcal{F}(U)$, define $\bar{s} : U \rightarrow E(\mathcal{F})$ by $\bar{s}(x) = s_x$. Give $E(\mathcal{F})$ the strongest topology for which every such map $\bar{s}$ is continuous. Prove that the associated sheaf $\mathcal{F}^+$ is the sheaf of continuous sections of π . In particular, characterize when $\mathcal{F}$ itself is a sheaf in these terms.

Solution. For $s \in \mathcal{F}(U)$, put

$$B(s, U) = \{s_x : x \in U\} = \bar{s}(U).$$

For the specified topology, if an open W contains s_x , then $\bar{s}^{-1}(W)$ is an open neighborhood of x . The image of the restriction of s to that neighborhood is a set $B(s|_{\bar{s}^{-1}(W)}, \bar{s}^{-1}(W))$ contained in W . Thus the basic sets just defined refine every open of that topology.

We first show that these sets form a basis for the topology on $E(\mathcal{F})$. Given $t \in \mathcal{F}(V)$, the inverse image of $B(s, U)$ under $\bar{t}$ is

$$\{x \in U \cap V : t_x = s_x\}.$$

Equality of two germs holds on a neighborhood of the point where it holds, so this set is open. Hence every $B(s, U)$ is open by the definition of the topology. These sets cover $E(\mathcal{F})$, since every germ has a representative. If a germ e belongs to both $B(s, U)$ and $B(t, V)$, then s and t have equal germs at $\pi(e)$, and after shrinking to an open neighborhood $W \subseteq U \cap V$ their restrictions agree. Thus $e \in B(s|_W, W) \subseteq B(s, U) \cap B(t, V)$, proving the basis claim.

The projection π is continuous: for open $W \subseteq X$, the preimage $\pi^{-1}(W)$ has open inverse image under every map $\bar{s}$. Moreover, $\pi : B(s, U) \rightarrow U$ is a homeomorphism with inverse $\bar{s}$.

Let $t : U \rightarrow E(\mathcal{F})$ satisfy $\pi \circ t$ equal to the identity. If t is continuous and $x \in U$, choose a representative $s \in \mathcal{F}(V)$ of the germ $t(x)$, with $x \in V$. The basic open set $B(s, V)$ contains $t(x)$, so

$$W = t^{-1}(B(s, V))$$

is a neighborhood of x in U . For $y \in W$, both $t(y)$ and s_y lie over y in $B(s, V)$; hence $t(y) = s_y$. Thus t is locally represented by sections of $\mathcal{F}$.

Conversely, a section of π that is locally of the form $\bar{s}$ is continuous, because continuity is local on its domain and every $\bar{s}$ is continuous by construction. Therefore continuous sections of π are exactly the locally representable germ functions, which are the sections of $\mathcal{F}^+$ by the stated realization of sheafification. The identifications commute with restriction, so they give an isomorphism of sheaves.

The canonical map $\mathcal{F}(U) \rightarrow \mathcal{F}^+(U)$ sends s to $\bar{s}$. Hence $\mathcal{F}$ is a sheaf if and only if, for every open U , this map identifies $\mathcal{F}(U)$ with all continuous sections of π over U . $\square$

Exercise II.1.14**Permitted context.**

For a section $s \in \mathcal{F}(U)$, the equality $s_x = 0$ in the stalk $\mathcal{F}_x$ means that s restricts to zero on some neighborhood of x contained in U . Stalks commute with direct sums of sheaves. On a T_1 -space, the skyscraper sheaf with value $A \neq 0$ at a point p has nonzero stalk precisely at p .

Problem. For a sheaf $\mathcal{F}$ on X and a section $s \in \mathcal{F}(U)$, define

$$\text{Supp}(s) = \{x \in U : s_x \neq 0\}.$$

Prove that $\text{Supp}(s)$ is closed in U . The support of the sheaf is

$$\text{Supp}(\mathcal{F}) = \{x \in X : \mathcal{F}_x \neq 0\}.$$

Give an example showing that the latter set need not be closed.

Solution. If $x \notin \text{Supp}(s)$, then $s_x = 0$. By the definition of a stalk, there is an open neighborhood $V \subseteq U$ of x on which $s|_V = 0$. Every germ s_y with $y \in V$ is therefore zero, so

$$V \subseteq U \setminus \text{Supp}(s).$$

The complement of the support is open in U , and the support is closed.

For the second assertion, take $X = \mathbb{R}$ with its usual topology and, for every $q \in \mathbb{Q}$, let $\mathcal{S}_q$ be the skyscraper sheaf with value $\mathbb{Z}$ at q . Set

$$\mathcal{F} = \bigoplus_{q \in \mathbb{Q}} \mathcal{S}_q.$$

Because stalks commute with direct sums,

$$\mathcal{F}_x \cong \bigoplus_{q \in \mathbb{Q}} (\mathcal{S}_q)_x.$$

This group is $\mathbb{Z}$ when $x \in \mathbb{Q}$ and is zero when $x \notin \mathbb{Q}$. Thus $\text{Supp}(\mathcal{F}) = \mathbb{Q}$, which is not closed in $\mathbb{R}$. $\square$

Exercise II.1.15**Permitted context.**

Morphisms of sheaves of abelian groups may be added pointwise on sections. Sections of a sheaf that agree on an open cover are equal, and compatible local sections glue uniquely.

Problem. Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves of abelian groups on X . Show that

$$U \longmapsto \operatorname{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$$

is naturally a sheaf of abelian groups, with restriction given by restricting a sheaf morphism. This is the sheaf of local morphisms from $\mathcal{F}$ to $\mathcal{G}$.

Solution. For a fixed open U , define the sum of two morphisms by

$$(\varphi + \psi)_V(s) = \varphi_V(s) + \psi_V(s) \quad (V \subseteq U, s \in \mathcal{F}(V)).$$

The zero and additive inverses are defined in the same way. Naturality with respect to restrictions follows from that of φ and ψ , so this makes $\operatorname{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$ an abelian group. Restricting morphisms is a group homomorphism and gives a presheaf.

Let $U = \bigcup_{\alpha} U_{\alpha}$, and suppose morphisms

$$\varphi_{\alpha} : \mathcal{F}|_{U_{\alpha}} \longrightarrow \mathcal{G}|_{U_{\alpha}}$$

agree after restriction to every $U_{\alpha} \cap U_{\beta}$. For an open $V \subseteq U$ and $s \in \mathcal{F}(V)$, the sections

$$\varphi_{\alpha, V \cap U_{\alpha}}(s|_{V \cap U_{\alpha}}) \in \mathcal{G}(V \cap U_{\alpha})$$

agree on pairwise overlaps. They therefore glue uniquely to a section $\varphi_V(s) \in \mathcal{G}(V)$.

The uniqueness of these gluings shows that φ_V is a homomorphism, that the maps φ_V commute with restriction as V varies, and that the resulting sheaf morphism $\varphi : \mathcal{F}|_U \rightarrow \mathcal{G}|_U$ restricts to every φ_{α} . If two global morphisms have those restrictions, their values on every section agree locally and hence agree globally in $\mathcal{G}$. Thus the gluing is unique, proving the sheaf axiom. $\square$

Exercise II.1.16**Permitted context.**

A sheaf $\mathcal{F}$ of abelian groups is flasque if every restriction $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$, for open $V \subseteq U$, is surjective. A short exact sequence of sheaves is exact on stalks. In particular, an epimorphism of sheaves has local lifts of sections, while the kernel sheaf is computed sectionwise. Every set can be well ordered.

Problem. (a) Prove that a constant sheaf on an irreducible topological space is flasque.

(b) Let

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

be exact, with $\mathcal{F}'$ flasque. Prove that for every open U the induced sequence

$$0 \longrightarrow \mathcal{F}'(U) \longrightarrow \mathcal{F}(U) \longrightarrow \mathcal{F}''(U) \longrightarrow 0$$

is exact.

(c) In the sequence of part (b), assume both $\mathcal{F}'$ and $\mathcal{F}$ are flasque. Prove that $\mathcal{F}''$ is flasque.

(d) If $f : X \rightarrow Y$ is continuous and $\mathcal{F}$ is flasque on X , prove that $f_*\mathcal{F}$ is flasque on Y .

(e) For any sheaf $\mathcal{F}$, define

$$\mathcal{G}(U) = \prod_{x \in U} \mathcal{F}_x,$$

with restriction given by forgetting components. Thus a section of $\mathcal{G}$ is an arbitrary, not necessarily continuous, choice of a germ over every point. Prove that $\mathcal{G}$ is a flasque sheaf and that the germ maps give a natural injection $\mathcal{F} \rightarrow \mathcal{G}$.

Solution. (a) Let $\mathcal{A}$ be the constant sheaf associated to an abelian group A . Every nonempty open subset of an irreducible space is irreducible, hence connected. Therefore a locally constant map from a nonempty open set to A is constant, so

$$\mathcal{A}(U) = \begin{cases} A, & U \neq \emptyset, \\ 0, & U = \emptyset. \end{cases}$$

Every restriction is either the identity of A or the unique map to the zero group, and is consequently surjective.

(b) Left exactness on sections gives exactness through $\mathcal{F}(U)$; only surjectivity onto $\mathcal{F}''(U)$ remains. Take $s \in \mathcal{F}''(U)$. Stalkwise surjectivity gives an open cover $U = \bigcup_{\alpha < \kappa} U_\alpha$, indexed by an ordinal, and lifts $u_\alpha \in \mathcal{F}(U_\alpha)$ of $s|_{U_\alpha}$.

We construct compatible lifts on the increasing open sets

$$V_\alpha = \bigcup_{\beta < \alpha} U_\beta.$$

Start with $t_0 = 0$ on $V_0 = \emptyset$. At a limit ordinal, the previously constructed compatible sections glue uniquely on V_α . At a successor stage, suppose $t_\alpha \in \mathcal{F}(V_\alpha)$ has already been constructed and put $W = V_\alpha \cap U_\alpha$. The difference

$$t_\alpha|_W - u_\alpha|_W$$

maps to zero in $\mathcal{F}''(W)$. Exactness at $\mathcal{F}$ identifies it with a section $a \in \mathcal{F}'(W)$. Since $\mathcal{F}'$ is flasque, extend a to $\tilde{a} \in \mathcal{F}'(U_\alpha)$. The sections t_α and $u_\alpha + \tilde{a}$ now agree on W , so they glue to a lift on $V_{\alpha+1}$. Transfinite induction yields a lift $t_\kappa \in \mathcal{F}(U)$ of s , proving surjectivity.

(c) Let $V \subseteq U$ be open and $s \in \mathcal{F}''(V)$. Applying part (b) on the open subspace V lifts s to $t \in \mathcal{F}(V)$. Flasqueness of $\mathcal{F}$ extends t to a section $\tilde{t} \in \mathcal{F}(U)$. Its image in $\mathcal{F}''(U)$ restricts to s . Thus every restriction map of $\mathcal{F}''$ is surjective.

(d) For open $V \subseteq U \subseteq Y$, the restriction in the direct image is

$$(f_*\mathcal{F})(U) = \mathcal{F}(f^{-1}U) \longrightarrow \mathcal{F}(f^{-1}V) = (f_*\mathcal{F})(V).$$

It is surjective because $f^{-1}V \subseteq f^{-1}U$ are open in X .

(e) An arbitrary family of functions into the pointwise groups glues uniquely, point by point, so $U \mapsto \prod_{x \in U} \mathcal{F}_x$ is a sheaf. If $V \subseteq U$, a member of $\mathcal{G}(V)$ extends to $\mathcal{G}(U)$ by assigning zero to every point of $U \setminus V$. Thus $\mathcal{G}$ is flasque.

For each open U , define

$$\iota_U : \mathcal{F}(U) \longrightarrow \mathcal{G}(U), \quad s \longmapsto (s_x)_{x \in U}.$$

These homomorphisms commute with restriction. If $\iota_U(s) = 0$, every point $x \in U$ has a neighborhood on which s is zero. Those neighborhoods cover U , and uniqueness in the sheaf axiom gives $s = 0$. Hence ι is injective. $\square$

Exercise II.1.17

Permitted context.

For $P \in X$, a point Q lies in $\overline{\{P\}}$ precisely when every open neighborhood of Q contains P . The closure of a single point is irreducible. The constant sheaf associated to an abelian group A has section group A on each nonempty open subset of an irreducible space and zero on the empty set.

Problem. Let X be a topological space, $P \in X$, and A an abelian group. Define

$$i_P(A)(U) = \begin{cases} A, & P \in U, \\ 0, & P \notin U, \end{cases}$$

with the evident restriction maps. Verify that this is a sheaf and compute its stalk at every $Q \in X$. Then identify it with $i_*\mathcal{A}$, where $i : \overline{\{P\}} \rightarrow X$ is the inclusion and $\mathcal{A}$ is the constant sheaf associated to A on $\overline{\{P\}}$.

Solution. Let $U = \bigcup_\alpha U_\alpha$. If $P \notin U$, every group in the sheaf axiom is zero. If $P \in U$, at least one U_α contains P . A compatible family is then uniquely determined by its value in A on any member containing P , and that value gives its unique gluing in $i_P(A)(U) = A$. Thus $i_P(A)$ is a sheaf.

Its stalk at Q is

$$i_P(A)_Q = \varinjlim_{Q \in V} i_P(A)(V).$$

If $Q \notin \overline{\{P\}}$, some neighborhood V of Q omits P ; such neighborhoods are cofinal after intersecting with V , so the stalk is zero. If $Q \in \overline{\{P\}}$, every neighborhood of Q contains P . The directed system is then constantly A with identity transition maps, and the stalk is A . Hence

$$i_P(A)_Q \cong \begin{cases} A, & Q \in \overline{\{P\}}, \\ 0, & Q \notin \overline{\{P\}}. \end{cases}$$

Put $Z = \overline{\{P\}}$. Since Z is irreducible, $\mathcal{A}(W) = A$ for nonempty open $W \subseteq Z$, and $\mathcal{A}(\emptyset) = 0$. For open $U \subseteq X$,

$$(i_*\mathcal{A})(U) = \mathcal{A}(U \cap Z).$$

An open set U meets Z if and only if it contains P : if $Q \in U \cap Z$, the neighborhood U of Q must contain P . Consequently $(i_*\mathcal{A})(U)$ is A exactly when $P \in U$, and is zero otherwise. These identifications respect restrictions, giving $i_P(A) \cong i_*\mathcal{A}$. $\square$

Exercise II.1.18

Permitted context.

For a continuous map $f : X \rightarrow Y$, the direct image is $(f_*\mathcal{F})(V) = \mathcal{F}(f^{-1}V)$. For a sheaf $\mathcal{G}$ on Y , the inverse image $f^{-1}\mathcal{G}$ is the sheafification of the presheaf

$$U \mapsto \varinjlim_{V \supseteq f(U)} \mathcal{G}(V).$$

A morphism from a presheaf to a sheaf factors uniquely through sheafification.

Problem. For a continuous map $f : X \rightarrow Y$, construct natural morphisms

$$\varepsilon_{\mathcal{F}} : f^{-1}f_*\mathcal{F} \longrightarrow \mathcal{F} \quad \text{and} \quad \eta_{\mathcal{G}} : \mathcal{G} \longrightarrow f_*f^{-1}\mathcal{G}.$$

Use them to prove a natural bijection

$$\mathrm{Hom}_X(f^{-1}\mathcal{G}, \mathcal{F}) \cong \mathrm{Hom}_Y(\mathcal{G}, f_*\mathcal{F})$$

for sheaves $\mathcal{F}$ on X and $\mathcal{G}$ on Y .

Solution. For an open $U \subseteq X$, every open $V \subseteq Y$ containing $f(U)$ gives a restriction map

$$(f_*\mathcal{F})(V) = \mathcal{F}(f^{-1}V) \longrightarrow \mathcal{F}(U).$$

These maps are compatible as V shrinks, so they induce a morphism from the inverse-image presheaf of $f_*\mathcal{F}$ to $\mathcal{F}$. Because $\mathcal{F}$ is a sheaf, it factors uniquely through sheafification and gives

$$\varepsilon_{\mathcal{F}} : f^{-1}f_*\mathcal{F} \longrightarrow \mathcal{F}.$$

The construction uses only restriction maps, so it is natural in $\mathcal{F}$.

For open $V \subseteq Y$, a section $s \in \mathcal{G}(V)$ determines, by the canonical map into the inverse-image presheaf and then sheafification, a section of $f^{-1}\mathcal{G}$ over $f^{-1}V$. This defines

$$\eta_{\mathcal{G},V} : \mathcal{G}(V) \longrightarrow (f_*f^{-1}\mathcal{G})(V)$$

compatibly with restrictions, and hence a natural morphism $\eta_{\mathcal{G}} : \mathcal{G} \rightarrow f_*f^{-1}\mathcal{G}$.

These maps satisfy the two triangle identities

$$\varepsilon_{f^{-1}\mathcal{G}} \circ f^{-1}(\eta_{\mathcal{G}}) = \mathrm{id}_{f^{-1}\mathcal{G}}, \quad f_*(\varepsilon_{\mathcal{F}}) \circ \eta_{f_*\mathcal{F}} = \mathrm{id}_{f_*\mathcal{F}}.$$

Indeed, before sheafification a section representing a germ is first sent to the same section over its inverse-image open set and then restricted back; the result is its original germ. The first identity

follows after sheafification. The second is checked directly on $(f_*\mathcal{F})(V) = \mathcal{F}(f^{-1}V)$, where the composite is the identity restriction map.

Now define

$$\Phi(\alpha) = f_*(\alpha) \circ \eta_{\mathcal{G}}$$

for $\alpha : f^{-1}\mathcal{G} \rightarrow \mathcal{F}$, and

$$\Psi(\beta) = \varepsilon_{\mathcal{F}} \circ f^{-1}(\beta)$$

for $\beta : \mathcal{G} \rightarrow f_*\mathcal{F}$. Naturality of η and ε , followed by the two triangle identities, gives

$$\Psi(\Phi(\alpha)) = \alpha, \quad \Phi(\Psi(\beta)) = \beta.$$

Thus Φ and Ψ are inverse bijections. Every construction is natural in both sheaves, so f^{-1} is left adjoint to f_* . $\square$

Exercise II.1.19

Permitted context.

Let Z be closed in X , let $U = X \setminus Z$, and write $i : Z \rightarrow X$ and $j : U \rightarrow X$ for the inclusions. Direct image is defined by $(i_*\mathcal{F})(V) = \mathcal{F}(V \cap Z)$. Sheafification preserves stalks. A morphism of sheaves of abelian groups is an isomorphism, or a sequence of such sheaves is exact, if and only if this holds at every stalk.

Problem. (a) For a sheaf $\mathcal{F}$ on Z , show that

$$(i_*\mathcal{F})_x \cong \begin{cases} \mathcal{F}_x, & x \in Z, \\ 0, & x \notin Z. \end{cases}$$

(b) For a sheaf $\mathcal{F}$ on U , let $j_!\mathcal{F}$ be the sheafification of the presheaf

$$\mathcal{P}(V) = \begin{cases} \mathcal{F}(V), & V \subseteq U, \\ 0, & V \not\subseteq U. \end{cases}$$

Prove that its restriction to U is $\mathcal{F}$, that its stalk is $\mathcal{F}_x$ for $x \in U$ and zero for $x \notin U$, and that these properties characterize it up to a unique isomorphism compatible with the given identification over U .

(c) For a sheaf $\mathcal{F}$ on X , prove that there is a natural exact sequence

$$0 \longrightarrow j_!(\mathcal{F}|_U) \longrightarrow \mathcal{F} \longrightarrow i_*(\mathcal{F}|_Z) \longrightarrow 0.$$

Solution. (a) By definition,

$$(i_*\mathcal{F})_x = \varinjlim_{x \in V} \mathcal{F}(V \cap Z).$$

If $x \in Z$, the sets $V \cap Z$, as V ranges through neighborhoods of x in X , are exactly a cofinal system of neighborhoods of x in Z ; the limit is $\mathcal{F}_x$. If $x \notin Z$, the open neighborhood $X \setminus Z$ of x has empty intersection with Z . After intersecting all neighborhoods with it, the system is constantly $\mathcal{F}(\emptyset) = 0$, so the stalk is zero.

(b) If $x \in U$, neighborhoods of x contained in U are cofinal among all neighborhoods of x . On these neighborhoods $\mathcal{P}$ equals $\mathcal{F}$, and hence $\mathcal{P}_x = \mathcal{F}_x$. If $x \notin U$, no neighborhood of x is contained in U , so every term defining $\mathcal{P}_x$ is zero. Sheafification preserves these stalks. Also, $\mathcal{P}|_U = \mathcal{F}$ is already a sheaf, so $(j_!\mathcal{F})|_U \cong \mathcal{F}$.

For uniqueness, let $\mathcal{G}$ be a sheaf on X , supplied with an isomorphism $\theta : \mathcal{F} \rightarrow \mathcal{G}|_U$, and suppose $\mathcal{G}_x = 0$ outside U . Define a presheaf morphism $\mathcal{P} \rightarrow \mathcal{G}$: on $V \subseteq U$ use θ_V , and on every other V use the zero map. These maps commute with restriction, including a restriction from an open not contained in U to one contained in U , because the source section is then zero. The map factors uniquely through a sheaf morphism

$$j_!\mathcal{F} \longrightarrow \mathcal{G}.$$

It induces θ on stalks in U , and is the unique map between zero stalks outside U ; it is therefore an isomorphism on every stalk and hence an isomorphism of sheaves. The factorization property also gives uniqueness among isomorphisms compatible with θ .

(c) If $V \subseteq U$, restriction gives $\mathcal{F}(V) \rightarrow \mathcal{F}(V)$; together with the zero maps on opens not contained in U , these maps induce

$$j_!(\mathcal{F}|_U) \longrightarrow \mathcal{F}.$$

Restriction to Z , equivalently the unit for inverse and direct image, gives

$$\mathcal{F} \longrightarrow i_*(\mathcal{F}|_Z).$$

At $x \in U$, the resulting stalk sequence is

$$0 \longrightarrow \mathcal{F}_x \longrightarrow \mathcal{F}_x \longrightarrow 0 \longrightarrow 0,$$

where the middle map is the identity. At $x \in Z$, it is

$$0 \longrightarrow 0 \longrightarrow \mathcal{F}_x \longrightarrow \mathcal{F}_x \longrightarrow 0,$$

again with the nonzero map equal to the identity under the natural stalk identification. The sequence is exact at every stalk, hence exact as a sequence of sheaves. $\square$

Exercise II.1.20**Permitted context.**

For $s \in \mathcal{F}(V)$, its support is $\{x \in V : s_x \neq 0\}$. If $j : U \rightarrow X$ is the inclusion of an open subset, then

$$(j_*(\mathcal{F}|_U))(V) = \mathcal{F}(V \cap U).$$

Kernels of morphisms of sheaves of abelian groups are computed sectionwise.

Problem. Let Z be a closed subset of X and let $\mathcal{F}$ be a sheaf on X .

(a) Prove that

$$V \mapsto \{s \in \mathcal{F}(V) : \text{Supp}(s) \subseteq Z \cap V\}$$

is a sheaf. Denote it by $\mathcal{H}_Z^0(\mathcal{F})$.

(b) Put $U = X \setminus Z$ and let $j : U \rightarrow X$ be the inclusion. Prove that

$$0 \longrightarrow \mathcal{H}_Z^0(\mathcal{F}) \longrightarrow \mathcal{F} \longrightarrow j_*(\mathcal{F}|_U)$$

is exact. If $\mathcal{F}$ is flasque, prove in addition that the last map is surjective.

Solution. (a) Restriction preserves the support condition: if $W \subseteq V$, then

$$\text{Supp}(s|_W) \subseteq \text{Supp}(s) \cap W \subseteq Z \cap W.$$

Thus the displayed assignment is a presheaf. Suppose $V = \bigcup_{\alpha} V_{\alpha}$ and compatible sections $s_{\alpha} \in \mathcal{F}(V_{\alpha})$ all have support in $Z \cap V_{\alpha}$. They have a unique gluing $s \in \mathcal{F}(V)$. If $x \in V \setminus Z$, choose α with $x \in V_{\alpha}$. Then $(s_{\alpha})_x = 0$, so $s_x = 0$. Hence $\text{Supp}(s) \subseteq Z \cap V$, and the gluing belongs to the presheaf. Uniqueness is inherited from $\mathcal{F}$, proving the sheaf axiom.

(b) The natural morphism $\mathcal{F} \rightarrow j_*(\mathcal{F}|_U)$ is, on an open V , the restriction

$$\mathcal{F}(V) \longrightarrow \mathcal{F}(V \cap U).$$

Its kernel consists of sections whose restriction to $V \cap U$ is zero. This is equivalent to the vanishing of every germ at every point of $V \setminus Z$, and hence to $\text{Supp}(s) \subseteq Z \cap V$. The sectionwise kernel is therefore $\mathcal{H}_Z^0(\mathcal{F})$, which proves exactness.

If $\mathcal{F}$ is flasque, each displayed restriction map is surjective because $V \cap U \subseteq V$ are open. Thus the sheaf morphism $\mathcal{F} \rightarrow j_*(\mathcal{F}|_U)$ is surjective even on sections over every open set, and in particular is an epimorphism of sheaves. $\square$

Exercise II.1.21**Permitted context.**

Let X be a variety over an algebraically closed field k , equipped with its sheaf $\mathcal{O}_X$ of regular functions. Affine open subsets form a basis, and if Y is closed in an affine open W , restriction identifies $k[Y]$ with $k[W]/I(Y)$. A sequence of sheaves of abelian groups is exact if and only if it is exact on stalks. On $\mathbb{P}_k^1$, every global regular function is constant, every rational function has only finitely many poles, and $\Gamma(\mathbb{P}_k^1, \mathcal{O}) = k$. Sections on a noetherian space commute with filtered colimits (II.1.11).

Problem. Let X be a variety over k .

(a) For a closed subset $Y \subseteq X$, define

$$\mathcal{I}_Y(U) = \{f \in \mathcal{O}_X(U) : f(y) = 0 \text{ for every } y \in Y \cap U\}.$$

Show that $\mathcal{I}_Y$ is a sheaf of ideals in $\mathcal{O}_X$.

(b) If Y is a closed reduced algebraic subset and $i : Y \rightarrow X$ is the inclusion, prove

$$\mathcal{O}_X/\mathcal{I}_Y \cong i_*\mathcal{O}_Y.$$

(c) Take $X = \mathbb{P}_k^1$ and $Y = \{P, Q\}$ for distinct points. With $\mathcal{S} = i_{P*}\mathcal{O}_P \oplus i_{Q*}\mathcal{O}_Q$, show that

$$0 \longrightarrow \mathcal{I}_Y \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{S} \longrightarrow 0$$

is exact, but that the induced map $\Gamma(X, \mathcal{O}_X) \rightarrow \Gamma(X, \mathcal{S})$ is not surjective.

(d) Still on $X = \mathbb{P}_k^1$, let $K = k(X)$ and let $\mathcal{K}$ be the constant sheaf associated to the additive group of K . Construct a natural injection $\mathcal{O}_X \rightarrow \mathcal{K}$. For $I_P = K/\mathcal{O}_{X,P}$, prove

$$\mathcal{K}/\mathcal{O}_X \cong \bigoplus_{P \in X} i_P(I_P).$$

(e) Prove that in part (d) the sequence on global sections

$$0 \longrightarrow \Gamma(X, \mathcal{O}_X) \longrightarrow \Gamma(X, \mathcal{K}) \longrightarrow \Gamma(X, \mathcal{K}/\mathcal{O}_X) \longrightarrow 0$$

is exact.

Solution. (a) If $V \subseteq U$ and f vanishes on $Y \cap U$, then $f|_V$ vanishes on $Y \cap V$, so the assignment is a presheaf of ideals. Now let $U = \bigcup_{\alpha} U_{\alpha}$ and let compatible $f_{\alpha} \in \mathcal{I}_Y(U_{\alpha})$ be given. They glue uniquely in $\mathcal{O}_X$ to $f \in \mathcal{O}_X(U)$. If $y \in Y \cap U$, choose α with $y \in U_{\alpha}$; then $f(y) = f_{\alpha}(y) = 0$. Hence $f \in \mathcal{I}_Y(U)$. This proves the sheaf axiom, and the ideal property is immediate sectionwise.

(b) Restriction of regular functions defines

$$\rho : \mathcal{O}_X \longrightarrow i_*\mathcal{O}_Y.$$

Its sectionwise kernel is $\mathcal{I}_Y$. To check local surjectivity, take an affine open neighborhood $W \subseteq X$. Then $Y \cap W$ is closed in W , and the coordinate-ring description gives a surjection

$$\mathcal{O}_X(W) = k[W] \longrightarrow k[Y \cap W] = \mathcal{O}_Y(Y \cap W)$$

with kernel $I(Y \cap W) = \mathcal{I}_Y(W)$. Since affine opens form a basis, ρ is surjective on every stalk. Thus

$$0 \longrightarrow \mathcal{I}_Y \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_Y \longrightarrow 0$$

is exact, and taking the quotient gives the claimed isomorphism.

(c) The reduced algebraic set Y is the disjoint union of its two reduced points, so $i_*\mathcal{O}_Y \cong i_{P*}\mathcal{O}_P \oplus i_{Q*}\mathcal{O}_Q = \mathcal{S}$. Part (b) gives the asserted exact sequence. On global sections,

$$\Gamma(X, \mathcal{O}_X) = k, \quad \Gamma(X, \mathcal{S}) = k \oplus k,$$

and the map is $c \mapsto (c, c)$. For example, $(0, 1)$ is not in its image, so the map is not surjective. This exhibits the failure of the global-section functor to preserve epimorphisms.

(d) Every regular function on a nonempty open subset of the irreducible curve X determines a rational function. These inclusions commute with restriction and are injective, giving $\mathcal{O}_X \hookrightarrow \mathcal{K}$.

For a nonempty open V and $f \in \mathcal{K}(V) = K$, send f to its family of principal parts

$$([f]_P)_{P \in V}, \quad [f]_P \in I_P = K/\mathcal{O}_{X,P}.$$

Only the finitely many poles of f give nonzero entries, so this defines a section of $\bigoplus_P i_P(I_P)$. The construction commutes with restriction and gives a sheaf morphism

$$\delta : \mathcal{K} \longrightarrow \bigoplus_{P \in X} i_P(I_P).$$

At a point P , its map on stalks is the quotient map

$$K \longrightarrow K/\mathcal{O}_{X,P} = I_P :$$

all other skyscraper summands have zero stalk there. This map is surjective with kernel $\mathcal{O}_{X,P}$. The resulting sequence is therefore exact on every stalk:

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{K} \xrightarrow{\delta} \bigoplus_{P \in X} i_P(I_P) \longrightarrow 0.$$

Taking its cokernel identification proves the assertion.

(e) Sections on a noetherian space commute with filtered colimits by Exercise II.1.11. Applying this to the finite subsums gives

$$\Gamma(X, \mathcal{K}) = K, \quad \Gamma\left(X, \bigoplus_P i_P(I_P)\right) = \bigoplus_P I_P.$$

It remains only to prove that $K \rightarrow \bigoplus_P I_P$ is surjective. First fix $P = \infty$, choose the affine coordinate t on $X \setminus \{\infty\}$, and take a class in $K/\mathcal{O}_{X,\infty}$ represented by $a(t)/b(t)$. Euclidean division gives

$$\frac{a(t)}{b(t)} = q(t) + \frac{r(t)}{b(t)}, \quad \deg r < \deg b.$$

The second term is regular at ∞ , so the prescribed class is represented by the polynomial $q(t)$. This polynomial is regular at every finite point and therefore has no principal part except possibly at ∞ .

For an arbitrary P , apply the same argument after a projective linear change of coordinate carrying P to ∞ . Thus every element of the single summand I_P is the principal-part family of a rational function with no poles away from P . An element of $\bigoplus_P I_P$ has only finitely many nonzero components; summing the corresponding rational functions realizes that element. Hence the final map is surjective. Its kernel is $\Gamma(X, \mathcal{O}_X) = k$ by the exact sequence of part (d), completing the proof. $\square$

Exercise II.1.22**Permitted context.**

Compatible local sections of a sheaf glue uniquely. An isomorphism of sheaves may be checked locally on an open cover. Uniqueness of an object constructed from gluing data means uniqueness up to a unique isomorphism that respects the specified local identifications.

Problem. Let $X = \bigcup_i U_i$ be an open cover. Suppose a sheaf $\mathcal{F}_i$ is given on each U_i , together with isomorphisms

$$\varphi_{ij} : \mathcal{F}_i|_{U_i \cap U_j} \longrightarrow \mathcal{F}_j|_{U_i \cap U_j}$$

such that φ_{ii} is the identity and

$$\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$$

on every triple overlap. Prove that the $\mathcal{F}_i$ glue to a sheaf $\mathcal{F}$ on X : there are isomorphisms $\psi_i : \mathcal{F}|_{U_i} \rightarrow \mathcal{F}_i$ satisfying $\psi_j = \varphi_{ij} \circ \psi_i$ on overlaps. Prove the corresponding uniqueness statement.

Solution. For an open set $V \subseteq X$, define $\mathcal{F}(V)$ to be the group of families

$$(s_i)_i, \quad s_i \in \mathcal{F}_i(V \cap U_i),$$

that satisfy

$$\varphi_{ij}(s_i|_{V \cap U_i \cap U_j}) = s_j|_{V \cap U_i \cap U_j}$$

for every i, j . Restriction is componentwise.

This is a sheaf. Indeed, for a cover $V = \bigcup_\alpha V_\alpha$, a compatible family of elements of $\mathcal{F}(V_\alpha)$ can be glued in each component $\mathcal{F}_i$, producing $s_i \in \mathcal{F}_i(V \cap U_i)$. The transition equalities hold locally on the cover and hence globally by uniqueness in $\mathcal{F}_j$. Componentwise uniqueness proves uniqueness of the resulting family.

For $V \subseteq U_i$, projection to the i -th component defines

$$\psi_{i,V} : \mathcal{F}(V) \longrightarrow \mathcal{F}_i(V).$$

To construct its inverse, start with $s \in \mathcal{F}_i(V)$ and, for each j , set

$$s_j = \varphi_{ij}(s|_{V \cap U_j}) \in \mathcal{F}_j(V \cap U_j).$$

The cocycle identity implies that the family $(s_j)_j$ satisfies all compatibility relations. Since φ_{ii} is the identity, this construction is inverse to projection. It also commutes with restriction, so $\psi_i : \mathcal{F}|_{U_i} \rightarrow \mathcal{F}_i$ is an isomorphism. The defining compatibility relation gives $\psi_j = \varphi_{ij} \circ \psi_i$ on $U_i \cap U_j$.

Finally, let $\mathcal{G}$ be another sheaf with compatible isomorphisms $\theta_i : \mathcal{G}|_{U_i} \rightarrow \mathcal{F}_i$. For $s \in \mathcal{G}(V)$, the family

$$(\theta_i(s|_{V \cap U_i}))_i$$

is compatible with the φ_{ij} , and hence defines an element of $\mathcal{F}(V)$. These maps form a sheaf morphism $\Theta : \mathcal{G} \rightarrow \mathcal{F}$. On each U_i it is $\psi_i^{-1} \circ \theta_i$, so it is an isomorphism. Any morphism respecting the local identifications has these same restrictions on the cover $\{U_i\}$, and is therefore equal to Θ . This proves uniqueness up to a unique compatible isomorphism. $\square$

II.2 Schemes

Exercise II.2.1

Permitted context.

For a ring A and $f \in A$, write

$$D(f) = \{\mathfrak{p} \in \operatorname{Spec} A : f \notin \mathfrak{p}\}.$$

Prime ideals of A_f correspond to prime ideals of A disjoint from $\{1, f, f^2, \dots\}$. Principal open subsets form a basis of $\operatorname{Spec} A$, and $\mathcal{O}_{\operatorname{Spec} A}(D(g)) = A_g$.

Problem. Let $X = \operatorname{Spec} A$. Prove that the locally ringed space

$$(D(f), \mathcal{O}_X|_{D(f)})$$

is naturally isomorphic to $\operatorname{Spec} A_f$.

Solution. The localization map $A \rightarrow A_f$ induces a continuous map

$$\theta : \operatorname{Spec} A_f \longrightarrow \operatorname{Spec} A, \quad \mathfrak{q} \longmapsto \{a \in A : a/1 \in \mathfrak{q}\}.$$

Extension and contraction give inverse bijections between the prime ideals of A_f and the prime ideals of A not containing f , so the image of θ is $D(f)$.

For $g \in A$, the preimage of $D(g) \cap D(f)$ is $D(g/1) \subseteq \operatorname{Spec} A_f$. Conversely these opens form a basis on $\operatorname{Spec} A_f$. Thus θ is a homeomorphism onto $D(f)$.

It remains to compare the structure sheaves. On the corresponding basic opens there are natural isomorphisms

$$\mathcal{O}_X(D(f) \cap D(g)) = A_{fg} \cong (A_f)_{g/1} = \mathcal{O}_{\operatorname{Spec} A_f}(D(g/1)).$$

They are the canonical localization isomorphisms, so they commute with all restriction maps between basic opens. Hence they glue to an isomorphism between $\mathcal{O}_X|_{D(f)}$ and the structure sheaf of $\operatorname{Spec} A_f$. On a prime $\mathfrak{p} \in D(f)$, the induced stalk map is the natural isomorphism

$$A_{\mathfrak{p}} \cong (A_f)_{\mathfrak{p}A_f},$$

which is local. Therefore the homeomorphism with this sheaf isomorphism is an isomorphism of locally ringed spaces. $\square$

Exercise II.2.2**Permitted context.**

A locally ringed space is a scheme if every point has an open neighborhood isomorphic to an affine scheme. Principal opens $D(f)$ form a basis of an affine scheme $\text{Spec } A$, and with the restricted structure sheaf each $D(f)$ is isomorphic to $\text{Spec } A_f$.

Problem. Let $(X, \mathcal{O}_X)$ be a scheme and let $U \subseteq X$ be open. Prove that

$$(U, \mathcal{O}_X|_U)$$

is a scheme.

Solution. Fix $x \in U$. Since X is a scheme, there is an affine open neighborhood V of x , say $V \cong \text{Spec } A$. The set $U \cap V$ is open in V and contains x . Because principal opens form a basis, there is an $f \in A$ such that

$$x \in D(f) \subseteq U \cap V.$$

With the structure sheaf restricted from X , this neighborhood is the same locally ringed space as $D(f)$ inside $\text{Spec } A$, and hence is isomorphic to $\text{Spec } A_f$.

Thus every point of U has an affine open neighborhood for the restricted sheaf. This proves that $(U, \mathcal{O}_X|_U)$ is a scheme, called the open subscheme induced by X . $\square$

Exercise II.2.3**Permitted context.**

For a ring A , write $A_{\text{red}} = A/\sqrt{(0)}$. The nilradical is contained in every prime ideal, localization commutes with passage to the reduced quotient, and $\text{Spec } A_{\text{red}} \rightarrow \text{Spec } A$ is a homeomorphism. Sheafification preserves stalks, and a section of a sheaf is zero if all of its germs are zero.

Problem. A scheme X is called reduced if $\mathcal{O}_X(U)$ is a reduced ring for every open $U \subseteq X$.

- (a) Prove that X is reduced if and only if every local ring $\mathcal{O}_{X,x}$ is reduced.
- (b) Let $(\mathcal{O}_X)_{\text{red}}$ be the sheafification of

$$U \mapsto \mathcal{O}_X(U)_{\text{red}}.$$

Prove that $(X, (\mathcal{O}_X)_{\text{red}})$ is a scheme, denoted X_{red} , and that the natural morphism $X_{\text{red}} \rightarrow X$ is a homeomorphism on underlying spaces.

- (c) If X is reduced, prove that every morphism $f : X \rightarrow Y$ factors uniquely through $Y_{\text{red}} \rightarrow Y$.

Solution. (a) Suppose first that X is reduced. Given $x \in X$, choose an affine neighborhood $\text{Spec } A$. The ring $A = \mathcal{O}_X(\text{Spec } A)$ is reduced, and every localization $A_{\mathfrak{p}}$ is reduced: if $(a/s)^n = 0$, then some $t \notin \mathfrak{p}$ satisfies $ta^n = 0$, so $(ta)^n = 0$, whence $ta = 0$ and $a/s = 0$. Thus $\mathcal{O}_{X,x} \cong A_{\mathfrak{p}}$ is reduced.

Conversely, suppose every local ring is reduced. If $s \in \mathcal{O}_X(U)$ satisfies $s^n = 0$, then $(s_x)^n = 0$ in $\mathcal{O}_{X,x}$ for every $x \in U$. Hence $s_x = 0$ for every x , and the sheaf axiom gives $s = 0$. Therefore every $\mathcal{O}_X(U)$ is reduced.

(b) The assertion is local on X . Let $V = \text{Spec } A$ be an affine open. On a principal open $D(f) \subseteq V$,

$$\mathcal{O}_X(D(f))_{\text{red}} = (A_f)_{\text{red}} \cong (A_{\text{red}})_f.$$

These isomorphisms commute with restriction. Since the principal opens form a basis, after sheafification they identify

$$(V, (\mathcal{O}_X)_{\text{red}}|_V) \cong \text{Spec } A_{\text{red}}.$$

Thus the indicated locally ringed space is covered by affine schemes and is a scheme.

The quotient homomorphisms $A \rightarrow A_{\text{red}}$ induce, on these charts, morphisms $\text{Spec } A_{\text{red}} \rightarrow \text{Spec } A$. They are compatible on overlaps and therefore glue to $X_{\text{red}} \rightarrow X$. Every prime of A contains its nilradical, so contraction gives a homeomorphism $\text{Spec } A_{\text{red}} \cong \text{Spec } A$. These local homeomorphisms are the identity on the common underlying point set, and hence the global morphism is a homeomorphism on underlying spaces.

(c) Use the homeomorphism $Y_{\text{red}} \rightarrow Y$ to take the underlying continuous map of the desired factorization to be the map underlying f . For every open $V \subseteq Y$, the homomorphism

$$f_V^\# : \mathcal{O}_Y(V) \longrightarrow \mathcal{O}_X(f^{-1}V)$$

kills every nilpotent because its target is reduced by part (a) and the definition of a reduced scheme. It therefore factors uniquely through $\mathcal{O}_Y(V)_{\text{red}}$. These factorizations commute with restriction, so by the universal property of sheafification they give a unique sheaf morphism

$$(\mathcal{O}_Y)_{\text{red}} \longrightarrow f_* \mathcal{O}_X.$$

At $x \in X$, with $y = f(x)$, the resulting local-ring homomorphism is

$$\mathcal{O}_{Y,y}/\sqrt{(0)} \longrightarrow \mathcal{O}_{X,x}.$$

It is local: the inverse image of the maximal ideal of $\mathcal{O}_{X,x}$ is the maximal ideal of $\mathcal{O}_{Y,y}$ modulo its nilradical, because the original composite from $\mathcal{O}_{Y,y}$ was local. Hence the factorization is a morphism of schemes. Uniqueness follows from the uniqueness of every quotient factorization and of sheafification. $\square$

Exercise II.2.4

Permitted context.

A ring homomorphism $A \rightarrow B$ determines a unique morphism $\text{Spec } B \rightarrow \text{Spec } A$, and this gives a contravariant bijection for affine schemes. Morphisms of schemes whose restrictions agree on an open cover of the source glue uniquely.

Problem. Let A be a ring and X a scheme. Sending a morphism $f : X \rightarrow \text{Spec } A$ to its map on global sections gives

$$\alpha : \text{Hom}_{\text{Sch}}(X, \text{Spec } A) \longrightarrow \text{Hom}_{\text{Ring}}(A, \Gamma(X, \mathcal{O}_X)).$$

Prove that α is a bijection.

Solution. Let $\varphi : A \rightarrow \Gamma(X, \mathcal{O}_X)$ be a ring homomorphism. Choose an affine open cover $X = \bigcup_i U_i$, with $U_i = \text{Spec } B_i$. Restriction of global sections gives

$$A \xrightarrow{\varphi} \Gamma(X, \mathcal{O}_X) \longrightarrow \Gamma(U_i, \mathcal{O}_X) = B_i,$$

and hence a morphism $f_i : U_i \rightarrow \text{Spec } A$.

The morphisms f_i agree on overlaps. Indeed, cover $U_i \cap U_j$ by affine opens $W = \operatorname{Spec} C$. Both restrictions to W correspond to the same composite

$$A \xrightarrow{\varphi} \Gamma(X, \mathcal{O}_X) \longrightarrow \Gamma(W, \mathcal{O}_X) = C.$$

The affine correspondence makes the two restrictions equal on every such W , hence on $U_i \cap U_j$. The f_i therefore glue uniquely to a morphism $f_\varphi : X \rightarrow \operatorname{Spec} A$.

The map on global sections induced by f_φ is φ : after restriction to every U_i , both maps are the displayed composite, and sections agreeing on an open cover are equal. Thus α is surjective.

If $f, g : X \rightarrow \operatorname{Spec} A$ induce the same homomorphism on global sections, then on each affine $U_i = \operatorname{Spec} B_i$ their restrictions correspond to the same ring map

$$A \longrightarrow \Gamma(X, \mathcal{O}_X) \longrightarrow B_i.$$

Hence $f|_{U_i} = g|_{U_i}$ for every i , and gluing uniqueness gives $f = g$. Therefore α is injective as well. $\square$

Exercise II.2.5

Permitted context.

The prime ideals of $\mathbb{Z}$ are (0) and (p) for positive prime numbers p . A morphism from a scheme X to $\operatorname{Spec} A$ is naturally equivalent to a unital ring homomorphism $A \rightarrow \Gamma(X, \mathcal{O}_X)$.

Problem. Describe $\operatorname{Spec} \mathbb{Z}$ as a topological space, and prove that it is a final object in the category of schemes.

Solution. The points are the generic point (0) and one closed point (p) for each positive prime number p . If $n \neq 0$, then

$$V((n)) = \{(p) : p \text{ divides } n\},$$

which is finite; $V((0))$ is the entire space. Conversely, every finite set of closed points is $V((p_1 \cdots p_r))$. Thus the nonempty open sets are exactly the subsets containing (0) whose complements consist of finitely many closed points.

For any scheme X , the affine-target correspondence gives

$$\operatorname{Hom}_{\operatorname{Sch}}(X, \operatorname{Spec} \mathbb{Z}) \cong \operatorname{Hom}_{\operatorname{Ring}}(\mathbb{Z}, \Gamma(X, \mathcal{O}_X)).$$

There is exactly one unital homomorphism from $\mathbb{Z}$ to any unital ring, namely $n \mapsto n \cdot 1$. Hence there is exactly one morphism $X \rightarrow \operatorname{Spec} \mathbb{Z}$, proving that $\operatorname{Spec} \mathbb{Z}$ is final. $\square$

Exercise II.2.6**Permitted context.**

Prime ideals are proper ideals. Ring homomorphisms preserve the multiplicative identity; the zero ring has $0 = 1$. There is one continuous map from the empty space to any topological space, and local conditions on stalks are vacuous when the source has no points.

Problem. Describe the spectrum of the zero ring and prove that it is an initial object in the category of schemes.

Solution. The zero ring has no proper ideals, hence no prime ideals. Therefore

$$\mathrm{Spec}(0) = \emptyset.$$

Its structure sheaf is the unique sheaf of rings on the empty space, and the resulting locally ringed space is a scheme because the condition of having affine neighborhoods is vacuous.

Let X be any scheme. There is exactly one continuous map $f : \emptyset \rightarrow X$. For every open $V \subseteq X$,

$$(f_*\mathcal{O}_\emptyset)(V) = \mathcal{O}_\emptyset(\emptyset) = 0,$$

and there is exactly one unital homomorphism $\mathcal{O}_X(V) \rightarrow 0$. These maps form the unique possible morphism of sheaves $\mathcal{O}_X \rightarrow f_*\mathcal{O}_\emptyset$. There are no source stalks on which to check locality, so this is a morphism of locally ringed spaces. It is unique, and consequently $\mathrm{Spec}(0)$ is initial in the category of schemes. $\square$

Exercise II.2.7**Permitted context.**

For $x \in X$, write

$$k(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$$

for its residue field. The spectrum of a field K has one point and local ring K . A homomorphism of local rings is local precisely when the inverse image of the maximal ideal of the target is the maximal ideal of the source.

Problem. Let X be a scheme and K a field. Prove that giving a morphism $\mathrm{Spec} K \rightarrow X$ is equivalent to giving a point $x \in X$ together with an injective field homomorphism $k(x) \rightarrow K$.

Solution. Let $g : \mathrm{Spec} K \rightarrow X$ be a morphism and let x be the image of the unique point. The morphism on stalks is a local homomorphism

$$g_x^\# : \mathcal{O}_{X,x} \longrightarrow K.$$

The maximal ideal of K is zero, so locality says

$$\ker(g_x^\#) = \mathfrak{m}_x.$$

Consequently $g_x^\#$ factors through an injective homomorphism

$$k(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x \longrightarrow K.$$

Conversely, suppose $x \in X$ and an injection $\iota : k(x) \rightarrow K$ are given. The composite

$$\mathcal{O}_{X,x} \longrightarrow k(x) \xrightarrow{\iota} K$$

is a local homomorphism. Define the underlying continuous map from the one-point space $\operatorname{Spec} K$ to send its point to x . For an open $U \subseteq X$, map a section $s \in \mathcal{O}_X(U)$ to its germ s_x and then through the displayed homomorphism when $x \in U$; when $x \notin U$, the target section ring over the empty inverse image is the zero ring and the map is unique. These maps define a morphism of sheaves whose only map on source stalks is the specified local homomorphism. Hence they define a scheme morphism.

The two constructions recover both the image point and the stalk map, and are inverse to one another. $\square$

Exercise II.2.8

Permitted context.

For $x \in X$, the Zariski tangent space is

$$T_x X = \operatorname{Hom}_{k(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2, k(x)).$$

The ring of dual numbers $D = k[\epsilon]/(\epsilon^2)$ is local with maximal ideal (ϵ) , residue field k , and one prime ideal. A k -morphism induces homomorphisms of local k -algebras.

Problem. Let X be a scheme over a field k . Prove that k -morphisms

$$\operatorname{Spec} k[\epsilon]/(\epsilon^2) \longrightarrow X$$

are naturally equivalent to pairs consisting of a k -rational point $x \in X$ and a tangent vector in $T_x X$.

Solution. Let $h : \operatorname{Spec} D \rightarrow X$ be a k -morphism, and let x be the image of its unique point. Composing h with the closed point $\operatorname{Spec} k \rightarrow \operatorname{Spec} D$ gives a k -morphism $\operatorname{Spec} k \rightarrow X$ with image x . Equivalently, there is a k -embedding $k(x) \rightarrow k$. Since the structural map embeds k into $k(x)$ and the composite is the identity on k , this forces $k(x) = k$; thus x is k -rational.

The stalk map of h is a local k -algebra homomorphism

$$\varphi : \mathcal{O}_{X,x} \longrightarrow D.$$

Its reduction modulo ϵ is the residue map $\mathcal{O}_{X,x} \rightarrow k$. Therefore $\varphi(\mathfrak{m}_x) \subseteq (\epsilon)$. For $u \in \mathfrak{m}_x$ there is a unique scalar $\lambda(u) \in k$ with $\varphi(u) = \lambda(u)\epsilon$. The map λ is k -linear, and

$$\varphi(uv) = \lambda(u)\lambda(v)\epsilon^2 = 0 \quad (u, v \in \mathfrak{m}_x),$$

so it vanishes on $\mathfrak{m}_x^2$. It consequently defines an element

$$\bar{\lambda} \in \operatorname{Hom}_k(\mathfrak{m}_x/\mathfrak{m}_x^2, k) = T_x X.$$

Conversely, let x be k -rational and let $\bar{\lambda} \in T_x X$ be a tangent vector, viewed as a functional $\mathfrak{m}_x/\mathfrak{m}_x^2 \rightarrow k$. The structural copy of k in $\mathcal{O}_{X,x}$ splits the residue map, so every element is written uniquely as $c + u$, with $c \in k$ and $u \in \mathfrak{m}_x$. Define

$$\varphi(c + u) = c + \bar{\lambda}(u)\epsilon.$$

Because $\bar{\lambda}$ vanishes on $\mathfrak{m}_x^2$, this formula is multiplicative; it is plainly additive, preserves k , and is local. The correspondence between a point plus a local stalk homomorphism and a morphism from the one-point scheme $\operatorname{Spec} D$ now produces the desired k -morphism. The two constructions are inverse and natural. $\square$

Exercise II.2.9**Permitted context.**

A generic point of a closed subset Z is a point ζ for which $\overline{\{\zeta\}} = Z$. In $\text{Spec } A$, the closure of the point $\mathfrak{p}$ is $V(\mathfrak{p})$, and $V(I)$ is irreducible precisely when $\sqrt{I}$ is prime. Every nonempty open subset of an irreducible space is dense.

Problem. Prove that every nonempty irreducible closed subset of a scheme has a unique generic point.

Solution. Let $Z \subseteq X$ be nonempty, irreducible, and closed. Choose an affine open $U = \text{Spec } A$ meeting Z . The subset $Z \cap U$ is a nonempty irreducible closed subset of U , so

$$Z \cap U = V(\mathfrak{p})$$

for a unique prime ideal $\mathfrak{p} \subseteq A$. Let ζ be the corresponding point. Its closure inside U is $Z \cap U$.

Because $Z \cap U$ is a nonempty open subset of the irreducible space Z , it is dense in Z . The closure of ζ in X is contained in Z , since Z is closed and contains ζ , but it contains the closure of $Z \cap U$, because it contains $\overline{\{\zeta\}}^U = Z \cap U$. Therefore

$$\overline{\{\zeta\}}^X = Z,$$

so ζ is a generic point.

For uniqueness, suppose ζ' is another generic point of Z . Since $\zeta \in \overline{\{\zeta'\}}$, every open neighborhood of ζ , in particular U , contains ζ' . Inside $\text{Spec } A$, the two points have the same closure $Z \cap U$. Their corresponding prime ideals are therefore equal, since the closure of a prime $\mathfrak{q}$ is $V(\mathfrak{q})$. Thus $\zeta' = \zeta$. $\square$

Exercise II.2.10**Permitted context.**

The ring $\mathbb{R}[x]$ is a principal ideal domain. Its nonconstant monic irreducible polynomials are the linear polynomials $x - a$, with $a \in \mathbb{R}$, and the quadratic polynomials $(x - z)(x - \bar{z})$, with $z \in \mathbb{C} \setminus \mathbb{R}$, where z and $\bar{z}$ determine the same polynomial.

Problem. Describe $\text{Spec } \mathbb{R}[x]$, including its topology, and compare its points and topology with $\mathbb{R}$ and with $\mathbb{C}$.

Solution. There is one nonclosed point, the generic point (0) . Every other prime is maximal and is generated by a monic irreducible polynomial. Thus the closed points are

$$(x - a) \quad (a \in \mathbb{R})$$

and

$$((x - z)(x - \bar{z})) \quad (z \in \mathbb{C} \setminus \mathbb{R}),$$

with the same point obtained from z and $\bar{z}$. The first kind has residue field $\mathbb{R}$; the second has residue field $\mathbb{C}$.

For a nonzero polynomial f , the closed set $V(f)$ consists of the finitely many closed points corresponding to the distinct irreducible factors of f . The only closed set containing the generic point is the whole spectrum. Hence every nonempty open set contains (0) and all but finitely many closed points, and every such cofinite subset containing (0) is open.

The usual real line accounts only for the closed points of the form $(x - a)$; on this subset the induced Zariski topology is cofinite, not the usual Euclidean topology. All closed points together are naturally identified with the quotient set $\mathbb{C}/(z \sim \bar{z})$, while the spectrum has in addition the generic point (0) . Thus it is neither the usual real line nor the usual complex line as a topological space. $\square$

Exercise II.2.11

Permitted context.

For $k = \mathbb{F}_p$, the ring $k[x]$ is a principal ideal domain. For each $n \geq 1$, the polynomial $x^{p^n} - x$ is the product, without repetition, of all monic irreducible polynomials over k whose degrees divide n . Every finite field of order p^d is isomorphic to $\mathbb{F}_{p^d}$. We use the elementary Möbius inversion formula for sums over divisors of an integer.

Problem. Describe $\text{Spec } \mathbb{F}_p[x]$. Determine the residue field at each point and the number of points having each possible residue field.

Solution. The spectrum has the generic point (0) , whose residue field is the rational function field $\mathbb{F}_p(x)$. Its closed points are the ideals (f) , where $f \in \mathbb{F}_p[x]$ is monic and irreducible. If $\deg f = d$, then

$$k((f)) = \mathbb{F}_p[x]/(f) \cong \mathbb{F}_{p^d}.$$

As for every polynomial ring in one variable over a field, the nonempty open sets contain the generic point and omit only finitely many closed points.

Let $N_p(d)$ denote the number of monic irreducible polynomials of degree d . Factoring $x^{p^n} - x$ as in the context and comparing degrees gives

$$p^n = \sum_{d|n} d N_p(d).$$

Möbius inversion therefore yields

$$N_p(d) = \frac{1}{d} \sum_{e|d} \mu(e) p^{d/e}.$$

Consequently there are exactly $N_p(d)$ closed points whose residue field is isomorphic to $\mathbb{F}_{p^d}$, and there is one generic point with residue field $\mathbb{F}_p(x)$. $\square$

Exercise II.2.12**Permitted context.**

Sheaves on an open cover glue along sheaf isomorphisms satisfying the cocycle condition. A locally ringed space covered by open subspaces that are schemes is a scheme. An isomorphism of schemes identifies both the underlying open spaces and their structure sheaves by local homeomorphisms.

Problem. Let $\{X_i\}$ be a family of schemes. For $i \neq j$, let $U_{ij} \subseteq X_i$ be open and let $\varphi_{ij} : U_{ij} \rightarrow U_{ji}$ be an isomorphism. Assume $\varphi_{ji} = \varphi_{ij}^{-1}$, and on triple overlaps assume

$$\varphi_{ij}(U_{ij} \cap U_{ik}) = U_{ji} \cap U_{jk}, \quad \varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}.$$

Construct a scheme X and morphisms $\psi_i : X_i \rightarrow X$ such that: each ψ_i is an isomorphism onto an open subscheme; their images cover X ; the intersection of the i -th and j -th images is the image of U_{ij} ; and $\psi_i = \psi_j \circ \varphi_{ij}$ on U_{ij} . Explain the case in which all the U_{ij} are empty.

Solution. Set $U_{ii} = X_i$ and φ_{ii} equal to the identity. On the disjoint union of the underlying sets, declare

$$x \in X_i \sim y \in X_j \iff x \in U_{ij} \text{ and } y = \varphi_{ij}(x).$$

The inverse condition gives symmetry, and the triple-overlap condition gives transitivity, so this is an equivalence relation. It never identifies two distinct points of the same X_i .

Let X be the quotient set and let $\psi_i : X_i \rightarrow X$ be the canonical injections. Give X the topology in which $W \subseteq X$ is open if and only if every $\psi_i^{-1}(W)$ is open in X_i . Each $\psi_i(X_i)$ is open: its inverse image in X_j is U_{ji} . More generally, if $V \subseteq X_i$ is open, then

$$\psi_j^{-1}(\psi_i(V)) = \varphi_{ji}^{-1}(V \cap U_{ij}),$$

which is open. Hence ψ_i is an open continuous bijection from X_i to $\psi_i(X_i)$, and therefore a homeomorphism. The equivalence relation also gives

$$\psi_i(X_i) \cap \psi_j(X_j) = \psi_i(U_{ij}) = \psi_j(U_{ji})$$

and $\psi_i = \psi_j \circ \varphi_{ij}$ on U_{ij} .

Transport $\mathcal{O}_{X_i}$ by ψ_i to a sheaf on the open set $\psi_i(X_i)$. On the intersection of the i -th and j -th open sets, the scheme isomorphism φ_{ij} gives an isomorphism of these transported sheaves. The assumed cocycle identity for the φ_{ij} gives the cocycle identity for the sheaf isomorphisms. The sheaf-gluing theorem therefore produces a sheaf $\mathcal{O}_X$ whose restriction to $\psi_i(X_i)$ is identified with $\mathcal{O}_{X_i}$. These identifications make every ψ_i an isomorphism of locally ringed spaces onto an open subspace.

The images $\psi_i(X_i)$ cover X , and each is a scheme. Thus $(X, \mathcal{O}_X)$ is a scheme and has all the required properties. If every U_{ij} is empty for $i \neq j$, no distinct summands are identified; the construction is the disjoint union $\coprod_i X_i$. $\square$

Exercise II.2.13**Permitted context.**

A topological space is noetherian if its open subsets satisfy the ascending chain condition. A space is quasi-compact if every open cover has a finite subcover. Principal opens form a basis of $\text{Spec } A$, and

$$\bigcup_{\alpha} D(f_{\alpha}) = \text{Spec } A \iff (f_{\alpha})_{\alpha} = A.$$

Closed subsets of an affine spectrum are the sets $V(I)$.

Problem. (a) Prove that a topological space is noetherian if and only if every open subset is quasi-compact.

(b) Prove that every affine scheme is quasi-compact, but that its underlying space need not be noetherian.

(c) Prove that $\text{Spec } A$ is noetherian when A is a noetherian ring.

(d) Give an example of a nonnoetherian ring whose spectrum is nevertheless a noetherian topological space.

Solution. (a) Suppose X is noetherian and let $U = \bigcup_{\alpha} U_{\alpha}$ be an open cover of an open subset U . The finite unions of the U_{α} form a family of open sets. If no one of them equals U , successively adjoining a member that adds a new point produces a strictly ascending chain of open subsets, contrary to noetherianity. Hence finitely many U_{α} cover U .

Conversely, suppose every open subset is quasi-compact and let

$$U_1 \subseteq U_2 \subseteq \cdots$$

be an ascending chain. The open set $U = \bigcup_n U_n$ is covered by the U_n , so finitely many suffice. Since the cover is nested, $U = U_N$ for the largest index occurring, and the chain stabilizes.

(b) Let $\text{Spec } A = \bigcup_{\alpha} W_{\alpha}$ be an open cover. Refining by principal opens gives a cover by $D(f_{\lambda})$. Its complement is

$$V((f_{\lambda})_{\lambda}) = \emptyset,$$

so the ideal generated by all the f_{λ} is the unit ideal. An expression $1 = \sum_{\lambda \in F} a_{\lambda} f_{\lambda}$ uses only finitely many of them, and the corresponding $D(f_{\lambda})$, hence finitely many of the original W_{α} , cover the spectrum.

Affineness does not imply noetherianity of the underlying space. For

$$A = k[x_1, x_2, x_3, \dots],$$

the closed subsets

$$V(x_1) \supsetneq V(x_1, x_2) \supsetneq V(x_1, x_2, x_3) \supsetneq \cdots$$

form a strictly descending chain. Strictness follows because the prime ideal $(x_1, \dots, x_n)$ lies in the n -th set but not the next.

(c) A descending chain $V(I_1) \supseteq V(I_2) \supseteq \cdots$ may be written using radical ideals, in which case it corresponds to an ascending chain

$$\sqrt{I_1} \subseteq \sqrt{I_2} \subseteq \cdots.$$

If A is noetherian, this chain stabilizes, so the chain of closed sets stabilizes. Equivalently, the open subsets satisfy the ascending chain condition.

(d) Let

$$A = k[x_1, x_2, \dots] / (x_i x_j : i, j \geq 1),$$

and let $\mathfrak{m} = (x_1, x_2, \dots)$ denote the image of the indicated ideal. Then $\mathfrak{m}^2 = 0$, so every prime ideal contains $\mathfrak{m}$. Since $A/\mathfrak{m} \cong k$, the ideal $\mathfrak{m}$ itself is prime and is the only point of $\text{Spec } A$. The spectrum is therefore noetherian. But

$$(x_1) \subsetneq (x_1, x_2) \subsetneq \dots$$

is a strictly ascending chain of ideals, so A is not noetherian. $\square$

Exercise II.2.14

Permitted context.

For a nonnegatively graded ring S , let $S_+ = \bigoplus_{d>0} S_d$. The standard opens

$$D_+(f) = \{\mathfrak{p} \in \text{Proj } S : f \notin \mathfrak{p}\}, \quad f \in S_+ \text{ homogeneous},$$

form a basis, and $D_+(f) \cong \text{Spec}(S_f)_0$. A ring has empty spectrum if and only if it is the zero ring. Let $t(V)$ denote the scheme associated to a variety V .

Problem. (a) Prove that $\text{Proj } S = \emptyset$ if and only if every element of S_+ is nilpotent.

(b) Let $\varphi : S \rightarrow T$ be a degree-preserving homomorphism and set

$$U = \{\mathfrak{q} \in \text{Proj } T : \mathfrak{q} \not\supseteq \varphi(S_+)\}.$$

Show that U is open and that φ induces a natural morphism $U \rightarrow \text{Proj } S$.

(c) Suppose there is an integer d_0 such that $\varphi_d : S_d \rightarrow T_d$ is an isomorphism for every $d \geq d_0$. Prove that $U = \text{Proj } T$ and that the morphism in part (b) is an isomorphism.

(d) If V is a projective variety with homogeneous coordinate ring S , prove that $t(V) \cong \text{Proj } S$.

Solution. (a) If every element of S_+ is nilpotent, every prime ideal contains S_+ ; hence no homogeneous prime qualifies as a point of $\text{Proj } S$.

Conversely, suppose $\text{Proj } S$ is empty. For every homogeneous $f \in S_+$, the standard open

$$D_+(f) \cong \text{Spec}(S_f)_0$$

is empty, so $(S_f)_0$ is the zero ring. In particular $1 = 0$ in S_f , which says that some power of f is zero. Every homogeneous element of positive degree is therefore nilpotent. The nilpotent elements of a commutative ring form an ideal, so every finite sum of such homogeneous elements, and hence every element of S_+ , is nilpotent.

(b) One has

$$U = \bigcup_{\substack{f \in S_+ \\ f \text{ homogeneous}}} D_+(\varphi(f)),$$

so U is open. On each member of this cover, localization of φ gives a ring homomorphism

$$(S_f)_0 \longrightarrow (T_{\varphi(f)})_0$$

and thus a scheme morphism

$$D_+(\varphi(f)) \longrightarrow D_+(f).$$

These morphisms agree on pairwise intersections because further localization is functorial, so they glue to $U \rightarrow \text{Proj } S$. On points it is $\mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q})$; membership in U ensures that this homogeneous prime does not contain S_+ .

(c) If $\mathfrak{q} \in \text{Proj } T$ were outside U , it would contain $\varphi(S_+)$, and therefore T_d for every $d \geq d_0$. For any homogeneous $t \in T_+$, a sufficiently high power t^n has degree at least d_0 , so $t^n \in \mathfrak{q}$, and primality gives $t \in \mathfrak{q}$. This would imply $\mathfrak{q} \supseteq T_+$, a contradiction. Thus $U = \text{Proj } T$.

The opens $D_+(f)$ with f homogeneous of positive degree and $\deg f \geq d_0$ cover $\text{Proj } S$: replace any homogeneous element avoided by a given prime by a sufficiently high power. For such an f , of degree d , there are direct-limit descriptions

$$(S_f)_0 = \varinjlim (S_0 \xrightarrow{f} S_d \xrightarrow{f} S_{2d} \xrightarrow{f} \cdots)$$

and the analogous expression for T . The map between these systems is an isomorphism from a sufficiently late term onward, so it induces

$$(S_f)_0 \cong (T_{\varphi(f)})_0.$$

The morphism is consequently an isomorphism on an open cover, and hence is an isomorphism globally.

(d) Embed V in $\mathbb{P}_k^n$ and write

$$S = k[x_0, \dots, x_n]/I(V).$$

On the standard affine chart $x_i \neq 0$, the coordinate ring of $V \cap \mathbb{A}_i^n$ is obtained by dehomogenizing at x_i , namely

$$k[V \cap \mathbb{A}_i^n] \cong (S_{x_i})_0.$$

Therefore

$$t(V \cap \mathbb{A}_i^n) \cong \text{Spec}(S_{x_i})_0 = D_+(x_i) \subseteq \text{Proj } S.$$

On overlaps these identifications are the same localization identifications used in the constructions of both $t(V)$ and $\text{Proj } S$. They consequently glue, and the charts cover both schemes. This yields $t(V) \cong \text{Proj } S$. $\square$

Exercise II.2.15

Permitted context.

Let k be algebraically closed and let $t(V)$ be the scheme associated to a k -variety V . Affine charts of $t(V)$ are spectra of finitely generated reduced k -algebras. By the weak Nullstellensatz, the residue field at every maximal ideal of such an algebra is k . Every nonempty closed subset of a finite-type k -scheme has a closed point.

Problem. (a) Prove that a point $P \in t(V)$ is closed if and only if $k(P) = k$.

(b) If $f : X \rightarrow Y$ is a morphism of schemes over k and $k(P) = k$, prove that $k(f(P)) = k$.

(c) For k -varieties V, W , prove that the natural map

$$\text{Hom}_{\text{Var}}(V, W) \longrightarrow \text{Hom}_{\text{Sch}/k}(t(V), t(W))$$

is a bijection.

Solution. (a) If P is closed, choose an affine neighborhood $\text{Spec } A$ of P . Its corresponding prime $\mathfrak{p}$ is maximal, and the weak Nullstellensatz gives

$$k(P) = A/\mathfrak{p} = k.$$

Conversely, suppose $k(P) = k$, and let Q lie in the closure of $\{P\}$. Any affine open neighborhood $\text{Spec } B$ of Q also contains P . If $\mathfrak{p} \subseteq B$ represents P , then

$$k(P) = \text{Frac}(B/\mathfrak{p}) = k.$$

Since $B/\mathfrak{p}$ is a k -subalgebra of its fraction field k and contains k , it equals k . Thus $\mathfrak{p}$ is maximal. The specialization Q corresponds to a prime containing $\mathfrak{p}$, so it equals $\mathfrak{p}$, and $Q = P$. The closure of $\{P\}$ is therefore $\{P\}$.

(b) A morphism induces an injective homomorphism of residue fields

$$k(f(P)) \longrightarrow k(P) = k.$$

Because f is a k -morphism, this is a homomorphism of field extensions of k , and its image contains the structural copy of k . It follows that $k(f(P)) = k$.

(c) The construction t plainly preserves composition. A morphism of varieties is determined by its maps on affine coordinate rings, so if two such morphisms induce the same scheme morphism, they are equal. Thus the displayed map is injective.

For surjectivity, let $F : t(V) \rightarrow t(W)$ be a k -morphism. By parts (a) and (b), F sends the closed points corresponding to V into those corresponding to W , and hence determines a set map $g : V \rightarrow W$. We verify that it is regular locally. Around any closed point of $t(V)$, choose affine opens

$$\text{Spec } A \subseteq t(V), \quad \text{Spec } B \subseteq t(W)$$

such that $F(\text{Spec } A) \subseteq \text{Spec } B$. The restriction of F is induced by a k -algebra homomorphism $B \rightarrow A$. On closed points this is exactly the regular map between the corresponding affine varieties. These local maps agree because they all come from F , so they glue to a morphism of varieties $g : V \rightarrow W$.

The affine neighborhoods used above cover every closed point of $t(V)$. They in fact cover all of $t(V)$: otherwise their open union would have a nonempty closed complement, which would contain a closed point. On each of these affine opens, $t(g)$ and F arise from the same ring homomorphism $B \rightarrow A$. Hence they agree locally and therefore globally. Thus every F comes from a unique morphism g , proving bijectivity. $\square$

Exercise II.2.16**Permitted context.**

For $f \in \Gamma(X, \mathcal{O}_X)$, define

$$X_f = \{x \in X : f_x \notin \mathfrak{m}_x\}.$$

On an affine scheme $\text{Spec } B$, a section $g \in B$ is a unit in $B_{\mathfrak{p}}$ exactly when $g \notin \mathfrak{p}$, and

$$\Gamma(D(g), \mathcal{O}) = B_g.$$

A section whose germs vanish everywhere is zero.

Problem. (a) If $U = \text{Spec } B$ is an affine open of X and $\bar{f} = f|_U \in B$, prove

$$U \cap X_f = D(\bar{f}).$$

Deduce that X_f is open.

(b) Assume X is quasi-compact. Put $A = \Gamma(X, \mathcal{O}_X)$. If $a \in A$ restricts to zero on X_f , prove that $f^n a = 0$ for some $n > 0$.

(c) Suppose X has a finite affine open cover $\{U_i\}$ such that every $U_i \cap U_j$ is quasi-compact. If $b \in \Gamma(X_f, \mathcal{O}_{X_f})$, prove that $f^n b$ extends to a section in A for some $n > 0$.

(d) Under the hypothesis of part (c), prove

$$\Gamma(X_f, \mathcal{O}_{X_f}) \cong A_f.$$

Solution. (a) A point $x \in U$, represented by $\mathfrak{p} \in \text{Spec } B$, belongs to X_f precisely when $\bar{f}/1$ is a unit in $B_{\mathfrak{p}}$. This holds precisely when $\bar{f} \notin \mathfrak{p}$. Hence

$$U \cap X_f = D(\bar{f}).$$

Affine opens cover X , and the right side is open in U , so X_f is open.

(b) Choose a finite affine cover $X = \bigcup_{i=1}^r U_i$, possible because X is quasi-compact. Write

$$U_i = \text{Spec } B_i, \quad f_i = f|_{U_i}, \quad a_i = a|_{U_i}.$$

By part (a), $U_i \cap X_f = D(f_i)$. The hypothesis says that a_i maps to zero in $(B_i)_{f_i}$, so some $n_i > 0$ satisfies $f_i^{n_i} a_i = 0$ in B_i . If $n = \max_i n_i$, then $(f^n a)|_{U_i} = 0$ for every i . The sheaf axiom gives $f^n a = 0$ on X .

(c) Write $U_i = \text{Spec } B_i$. On $U_i \cap X_f = D(f_i)$, the section b is represented by an element of $(B_i)_{f_i}$. Since there are finitely many i , after increasing one integer n we may write

$$b|_{U_i \cap X_f} = \frac{a_i}{f_i^n} \quad \text{with } a_i \in B_i$$

for every i .

On $W_{ij} = U_i \cap U_j$, the section

$$d_{ij} = a_i|_{W_{ij}} - a_j|_{W_{ij}}$$

restricts to zero on $(W_{ij})_f = W_{ij} \cap X_f$, because both fractions represent b . The scheme W_{ij} is quasi-compact by hypothesis. Applying part (b) to W_{ij} , with f and d_{ij} restricted there, gives an exponent m_{ij} for which

$$f^{m_{ij}} d_{ij} = 0 \quad \text{on } W_{ij}.$$

There are only finitely many pairs. Taking m at least every m_{ij} , the sections $f^m a_i \in \mathcal{O}_X(U_i)$ agree on all overlaps. They glue to $a \in A$, and on X_f one has

$$a|_{X_f} = f^{n+m}b.$$

Thus a power of f times b extends globally.

(d) The section $f|_{X_f}$ is invertible: it is invertible in every stalk, so its local inverses glue. Restriction $A \rightarrow \Gamma(X_f, \mathcal{O}_{X_f})$ therefore factors through a homomorphism

$$\rho : A_f \longrightarrow \Gamma(X_f, \mathcal{O}_{X_f}).$$

If $\rho(a/f^r) = 0$, then $a|_{X_f} = 0$. Part (b), applicable because the finite affine cover makes X quasi-compact, gives $f^m a = 0$ for some m ; hence $a/f^r = 0$ in A_f . Thus ρ is injective. For any section b of the target, part (c) gives $f^n b = a|_{X_f}$ for some $a \in A$, so $b = \rho(a/f^n)$. Therefore ρ is an isomorphism. $\square$

Exercise II.2.17

Permitted context.

Morphisms of schemes may be glued on an open cover of the source. For $A = \Gamma(X, \mathcal{O}_X)$, the identity $A \rightarrow A$ determines a canonical morphism $X \rightarrow \text{Spec } A$. If X has a finite affine cover with quasi-compact pairwise intersections, then

$$\Gamma(X_f, \mathcal{O}_{X|X_f}) \cong A_f.$$

Problem. (a) Let $h : X \rightarrow Y$ be a morphism. Suppose Y has an open cover $\{U_i\}$ for which every induced morphism

$$h^{-1}(U_i) \longrightarrow U_i$$

is an isomorphism. Prove that h is an isomorphism.

(b) Prove that a scheme X is affine if and only if there are finitely many $f_1, \dots, f_r \in A = \Gamma(X, \mathcal{O}_X)$ such that every X_{f_i} is affine and $(f_1, \dots, f_r) = A$.

Solution. (a) Let $g_i : U_i \rightarrow h^{-1}(U_i) \subseteq X$ be the inverse of the displayed isomorphism, followed by the open immersion into X . On $U_i \cap U_j$, both g_i and g_j are inverses to the restriction of h ; uniqueness of an inverse makes them equal. They glue to a morphism $g : Y \rightarrow X$. The identities

$$h \circ g = \text{id}_Y, \quad g \circ h = \text{id}_X$$

hold on the respective open covers and hence globally. Thus h is an isomorphism.

(b) If X is affine, take $r = 1$ and $f_1 = 1$. Then $X_{f_1} = X$ and f_1 generates the unit ideal.

Conversely, suppose such f_i are given. Because they generate A , at every $x \in X$ at least one germ $(f_i)_x$ is a unit; otherwise all would belong to the maximal ideal of $\mathcal{O}_{X,x}$, as would their linear combination 1. Hence the affine opens X_{f_i} cover X . Moreover,

$$X_{f_i} \cap X_{f_j} = X_{f_i f_j}$$

is a principal open in the affine scheme X_{f_i} , and is therefore affine and quasi-compact. The global-section localization result from the context applies to this finite cover and gives

$$\Gamma(X_{f_i}, \mathcal{O}_X) = A_{f_i}.$$

Let $h : X \rightarrow \operatorname{Spec} A$ be the canonical morphism. The preimage of $D(f_i)$ is X_{f_i} . On this preimage, h is the affine morphism induced by the natural map

$$A_{f_i} \longrightarrow \Gamma(X_{f_i}, \mathcal{O}_X) = A_{f_i},$$

which is the identity under the displayed identification. Therefore

$$X_{f_i} \longrightarrow D(f_i)$$

is an isomorphism. Since the f_i generate the unit ideal, the $D(f_i)$ cover $\operatorname{Spec} A$. Part (a) now shows that h is an isomorphism, so X is affine. $\square$

Exercise II.2.18

Permitted context.

Let $\varphi : A \rightarrow B$ induce $f : \operatorname{Spec} B \rightarrow \operatorname{Spec} A$. Then

$$f^{-1}(D(a)) = D(\varphi(a)),$$

and the map of structure sheaves on this basic open is

$$A_a \longrightarrow B_{\varphi(a)}.$$

A morphism of sheaves is injective or surjective if and only if its stalk maps have that property. The nilradical of a ring is the intersection of all its prime ideals.

Problem. (a) For $a \in A$, prove that a is nilpotent if and only if $D(a) = \emptyset$.

(b) Prove that φ is injective if and only if

$$f^\# : \mathcal{O}_{\operatorname{Spec} A} \longrightarrow f_* \mathcal{O}_{\operatorname{Spec} B}$$

is injective. If these conditions hold, prove that f is dominant.

(c) If φ is surjective, prove that f is a homeomorphism onto a closed subset and that $f^\#$ is surjective.

(d) Conversely, suppose f is a homeomorphism onto a closed subset and $f^\#$ is surjective. Prove that φ is surjective.

Solution. (a) The set $D(a)$ is empty precisely when every prime ideal contains a . Since the intersection of all prime ideals is the nilradical, this is equivalent to a being nilpotent.

(b) Suppose φ is injective. For every $a \in A$, the localization map

$$A_a \longrightarrow B_{\varphi(a)}$$

is injective. Indeed, if the image of c/a^n is zero, then $\varphi(a)^m \varphi(c) = 0$ for some m ; injectivity gives $a^m c = 0$, so $c/a^n = 0$. Since the $D(a)$ form a basis, $f^\#$ is injective. Conversely, if $f^\#$ is injective, its map on sections over the whole affine scheme is

$$A \longrightarrow B,$$

namely φ , so φ is injective.

To prove dominance under these conditions, let $c \in A$ vanish at every point of $f(\operatorname{Spec} B)$. Then $\varphi(c)$ belongs to every prime of B , so it is nilpotent. Injectivity implies that c is nilpotent in A , and hence it vanishes at every point of $\operatorname{Spec} A$. Thus no proper closed subset contains the image, and f is dominant.

(c) Write $B = A/I$. Extension and contraction identify $\operatorname{Spec} B$ homeomorphically with the closed subset

$$V(I) \subseteq \operatorname{Spec} A.$$

At a point $\mathfrak{p} \in V(I)$, the relevant stalk map is the surjection

$$A_{\mathfrak{p}} \longrightarrow (A/I)_{\mathfrak{p}/I}.$$

At a point outside $V(I)$, the stalk of $f_*\mathcal{O}_{\operatorname{Spec} B}$ is zero, because that point has a neighborhood disjoint from the image. Every stalk map of $f^\#$ is therefore surjective, so $f^\#$ is surjective.

(d) Put $I = \ker \varphi$ and factor

$$A \twoheadrightarrow A' = A/I \hookrightarrow B.$$

Correspondingly, factor f as

$$\operatorname{Spec} B \xrightarrow{g} \operatorname{Spec} A' \xrightarrow{i} \operatorname{Spec} A.$$

By part (b), g is dominant. The closure of the image of f is $V(I)$; since the image is assumed closed, it equals $V(I)$. As i is a homeomorphism from $\operatorname{Spec} A'$ to $V(I)$ and f is a homeomorphism onto that subset, g is a homeomorphism.

The sheaf map for f factors as

$$\mathcal{O}_{\operatorname{Spec} A} \longrightarrow i_*\mathcal{O}_{\operatorname{Spec} A'} \longrightarrow i_*g_*\mathcal{O}_{\operatorname{Spec} B}.$$

The first arrow is surjective by part (c), and the composite is surjective by hypothesis. It follows stalkwise on $V(I)$ that the second arrow is surjective. The injection $A' \hookrightarrow B$, together with part (b), says that this second arrow is also injective after identifying $\operatorname{Spec} A'$ with its image. Thus

$$\mathcal{O}_{\operatorname{Spec} A'} \cong g_*\mathcal{O}_{\operatorname{Spec} B}.$$

Taking global sections yields $A' \cong B$, and the isomorphism is the injection induced by φ . Hence $A \rightarrow B$ is surjective. $\square$

Exercise II.2.19

Permitted context.

For $X = \operatorname{Spec} A$, one has $\Gamma(X, \mathcal{O}_X) = A$. Sections on disjoint open subsets glue uniquely. Every prime ideal of a product $A_1 \times A_2$ is of the form $\mathfrak{p}_1 \times A_2$ or $A_1 \times \mathfrak{p}_2$.

Problem. For a ring A , prove that the following conditions are equivalent:

- (i) $\operatorname{Spec} A$ is disconnected.
- (ii) There are nonzero $e_1, e_2 \in A$ such that

$$e_1^2 = e_1, \quad e_2^2 = e_2, \quad e_1 e_2 = 0, \quad e_1 + e_2 = 1.$$

- (iii) There are nonzero rings A_1, A_2 with $A \cong A_1 \times A_2$.

Solution. Assume (i). Write $X = \operatorname{Spec} A = U_1 \coprod U_2$, where both U_i are nonempty and open. On U_1 take the section 1 and on U_2 take the section 0. They glue to $e_1 \in \Gamma(X, \mathcal{O}_X) = A$. Interchanging 0 and 1 gives e_2 . All four identities in (ii) hold after restriction to both members of the cover and hence hold globally. Neither element is zero because it restricts to 1 on a nonempty open set. Thus (i) implies (ii).

Assume (ii). The ideals $A_1 = e_1 A$ and $A_2 = e_2 A$ are rings with identities e_1 and e_2 , respectively, and both are nonzero. Define

$$\Phi : A \longrightarrow A_1 \times A_2, \quad a \longmapsto (e_1 a, e_2 a).$$

Orthogonality makes Φ a unital ring homomorphism. Its inverse is

$$(u, v) \longmapsto u + v :$$

the identities $e_1 + e_2 = 1$ and $e_1 e_2 = 0$ verify both composites. Hence (ii) implies (iii).

Finally, if $A \cong A_1 \times A_2$ with both factors nonzero, then

$$\operatorname{Spec} A \cong \operatorname{Spec} A_1 \coprod \operatorname{Spec} A_2.$$

Both summands are nonempty and are open and closed, so the spectrum is disconnected. Thus (iii) implies (i). $\square$

II.3 First Properties of Schemes

Exercise II.3.1

Permitted context.

A morphism $f : X \rightarrow Y$ is locally of finite type if Y has an affine open cover $V_i = \operatorname{Spec} B_i$ such that every $f^{-1}(V_i)$ has an affine open cover $\operatorname{Spec} A_{ij}$ with each A_{ij} a finitely generated B_i -algebra. Localizing a finitely generated algebra at finitely many elements preserves finite generation. Base change to any localization of the base also preserves finite generation over that localized base. Principal opens form a basis on an affine scheme.

Problem. Prove that $f : X \rightarrow Y$ is locally of finite type if and only if, for every affine open $V = \operatorname{Spec} B \subseteq Y$, the open set $f^{-1}(V)$ can be covered by affine opens $U_j = \operatorname{Spec} A_j$ for which every A_j is a finitely generated B -algebra.

Solution. The stated condition immediately implies the definition by choosing any affine open cover of Y . Conversely, suppose the condition in the definition holds for a fixed affine cover $\{V_i\}$, and let $V = \operatorname{Spec} B$ be any affine open.

Fix $x \in f^{-1}(V)$ and choose i with $f(x) \in V_i$, together with one of the prescribed affine neighborhoods

$$U_0 = \operatorname{Spec} A_0 \subseteq f^{-1}(V_i)$$

of x , where A_0 is of finite type over $B_i = \Gamma(V_i, \mathcal{O}_Y)$. Inside $V \cap V_i$, choose a principal neighborhood

$$W = D_V(g)$$

of $f(x)$. It is affine, with coordinate ring B_g , and its inclusion in the affine scheme V_i corresponds to a homomorphism $B_i \rightarrow B_g$.

The open $U_0 \cap f^{-1}(W)$ is the fiber product of the affine schemes $U_0 \rightarrow V_i$ and $W \rightarrow V_i$. It is therefore affine with coordinate ring

$$A_0 \otimes_{B_i} B_g.$$

This ring is finitely generated over B_g , hence over B . It contains x and lies in $f^{-1}(V)$. As x was arbitrary, these affine neighborhoods cover $f^{-1}(V)$, proving the desired condition. $\square$

Exercise II.3.2

Permitted context.

A morphism $f : X \rightarrow Y$ is quasi-compact if Y has an affine open cover $\{V_i\}$ such that every $f^{-1}(V_i)$ is quasi-compact. Every quasi-compact scheme has a finite affine open cover. Principal opens form a basis in an affine scheme, and a finite union of quasi-compact subspaces is quasi-compact.

Problem. Prove that $f : X \rightarrow Y$ is quasi-compact if and only if $f^{-1}(V)$ is quasi-compact for every affine open $V \subseteq Y$.

Solution. Only the forward implication needs proof. Choose an affine cover $V_i = \text{Spec } B_i$ as in the definition. Since $f^{-1}(V_i)$ is quasi-compact, choose a finite affine cover

$$f^{-1}(V_i) = \bigcup_{j=1}^{r_i} U_{ij}.$$

For $b \in B_i$, the inverse image of $D(b)$ meets each U_{ij} in a principal open: it is defined there by the restriction of b . Hence

$$f^{-1}(D(b)) = \bigcup_{j=1}^{r_i} (U_{ij} \cap f^{-1}(D(b)))$$

is a finite union of affine, and therefore quasi-compact, opens.

Now let $V \subseteq Y$ be affine. The principal opens

$$D(b) \subseteq V_i \cap V$$

as i and $b \in B_i$ vary form an open cover of V . Since V is quasi-compact, finitely many, say $D(b_1), \dots, D(b_n)$, cover it. Their inverse images are quasi-compact by the first paragraph, and

$$f^{-1}(V) = \bigcup_{\ell=1}^n f^{-1}(D(b_\ell))$$

is consequently quasi-compact. The converse is immediate by taking an affine cover of Y . $\square$

Exercise II.3.3**Permitted context.**

A morphism is of finite type if the affine covers in the definition of locally finite type can be chosen finite over each member of an affine cover of the target. A morphism is locally of finite type, respectively quasi-compact, exactly when the corresponding condition holds over every affine open of the target.

We may use the following algebraic patching lemma. If $D(g_1), \dots, D(g_r)$ cover $\text{Spec } A$ and every A_{g_i} is a finitely generated B -algebra, then A is a finitely generated B -algebra.

Problem. (a) Prove that a morphism is of finite type if and only if it is locally of finite type and quasi-compact.

(b) Deduce that $f : X \rightarrow Y$ is of finite type if and only if, for every affine open $V = \text{Spec } B \subseteq Y$, the inverse image $f^{-1}(V)$ has a finite affine cover $\text{Spec } A_j$ with all A_j finitely generated over B .

(c) If f is of finite type, $V = \text{Spec } B \subseteq Y$ is affine, and $U = \text{Spec } A \subseteq f^{-1}(V)$ is any affine open, prove that A is a finitely generated B -algebra.

Solution. First note why the patching lemma in the context holds. Enlarge a finitely generated B -subalgebra $C \subseteq A$ so that it contains the g_i , coefficients of a relation $1 = \sum_i c_i g_i$, and numerators of finite generating sets for all A_{g_i} . Then $C_{g_i} = A_{g_i}$ for every i . For any $a \in A$, suitable powers $g_i^N a$ all lie in C . The elements g_i^N generate the unit ideal in C , so a C -linear combination of the $g_i^N a$ equals a . Thus $a \in C$, and $A = C$.

(a) A finite-type morphism is locally of finite type by definition, and the finite affine covers of inverse images show that it is quasi-compact. Conversely, suppose f is locally of finite type and quasi-compact. Choose an affine target cover on which the first condition gives affine source covers by finitely generated algebras. Each inverse image is quasi-compact, so finitely many members of its affine cover suffice. These are exactly the covers required for f to be of finite type.

(b) By the affine-independent criteria for local finite type and quasi-compactness, a finite-type morphism has both properties over every affine V . The locally finite-type affine cover of $f^{-1}(V)$ has a finite subcover because that inverse image is quasi-compact. This proves the forward direction; the reverse direction is the definition applied to any affine cover of Y .

(c) By part (b), write

$$f^{-1}(V) = \bigcup_{j=1}^r U_j, \quad U_j = \text{Spec } A_j,$$

with every A_j finitely generated over B . The intersections $U \cap U_j$ cover the affine scheme U . Refine this cover by principal opens of U , and use quasi-compactness to retain finitely many:

$$U = D(g_1) \cup \dots \cup D(g_m), \quad D(g_\ell) \subseteq U_{j(\ell)}.$$

We use a small consequence of the patching lemma. If an affine open $W = \text{Spec } C$ lies in $\text{Spec } A_{j(\ell)}$, then quasi-compactness of W gives a finite cover by principal opens $D_{A_{j(\ell)}}(h_q)$ contained in W . These same sets are the principal opens $D_C(h_q)$, and their rings are $(A_{j(\ell)})_{h_q}$, which are finitely generated over B . Patching shows that C is finitely generated over B . Applying this to $W = D(g_\ell)$ proves that every A_{g_ℓ} is a finitely generated B -algebra. The patching lemma now shows that A itself is finitely generated over B . $\square$

Exercise II.3.4**Permitted context.**

A morphism $f : X \rightarrow Y$ is finite if Y has an affine open cover $\text{Spec } B_i$ whose inverse images are affine $\text{Spec } A_i$, with every A_i finite as a B_i -module. Finiteness of a module is local on the base: if $D(b_1), \dots, D(b_r)$ cover $\text{Spec } B$ and every M_{b_i} is a finite B_{b_i} -module, then M is a finite B -module. We may use the affine criterion based on a finite cover by distinguished affine opens.

Problem. Prove that $f : X \rightarrow Y$ is finite if and only if, for every affine open $V = \text{Spec } B \subseteq Y$, its inverse image is affine,

$$f^{-1}(V) = \text{Spec } A,$$

with A finite as a B -module.

Solution. The stated condition plainly implies the definition. Conversely, assume finiteness is known over an affine cover of Y , and fix an arbitrary affine open $V = \text{Spec } B$.

As in the target-locality argument, choose finitely many principal opens $D(b_i) \subseteq V$, each lying in a member of the original cover, such that the $D(b_i)$ cover V . Put

$$W = f^{-1}(V), \quad W_i = f^{-1}(D(b_i)).$$

If $D_V(b_i) \subseteq V_j = \text{Spec } B_j$ is in the original cover, with $f^{-1}(V_j) = \text{Spec } A_j$, affine base change gives $W_i = \text{Spec}(A_j \otimes_{B_j} B_{b_i})$. The tensor product is finite over B_{b_i} . Thus

$$W_i = \text{Spec } A_i$$

with A_i finite over B_{b_i} . The pullbacks of the b_i are global functions on W , generate the unit ideal, and have $W_i = W_{b_i}$. Their pairwise intersections are distinguished opens in the affine schemes W_i , hence are affine and quasi-compact. The affineness criterion therefore gives

$$W = \text{Spec } A, \quad A = \Gamma(W, \mathcal{O}_W).$$

The localization theorem for global sections identifies

$$A_{b_i} \cong \Gamma(W_i, \mathcal{O}_W) = A_i.$$

Each A_{b_i} is a finite B_{b_i} -module. Since the $D(b_i)$ cover $\text{Spec } B$, finiteness of modules is local on the base, and A is a finite B -module. This proves the assertion for every affine V . $\square$

Exercise II.3.5**Permitted context.**

A morphism is quasi-finite here if all of its set-theoretic fibers are finite. Finite morphisms are stable under base change and composition. A finite-dimensional algebra over a field is Artinian and has only finitely many prime ideals. For an integral ring extension $A \subseteq B$, the lying-over theorem makes $\operatorname{Spec} B \rightarrow \operatorname{Spec} A$ surjective. The book uses quasi-finite here to mean finite fibers; the common modern definition also requires local finite type.

Problem. (a) Prove that every finite morphism is quasi-finite.

(b) Prove that every finite morphism is a closed map.

(c) Give a surjective morphism which is of finite type and quasi-finite but not finite.

Solution. (a) Let $f : X \rightarrow Y$ be finite and $y \in Y$. After restricting to an affine neighborhood of y , write

$$Y = \operatorname{Spec} A, \quad X = \operatorname{Spec} B$$

with B a finite A -module. The fiber is

$$X_y = \operatorname{Spec}(B \otimes_A k(y)).$$

The tensor product is finite-dimensional over $k(y)$, hence Artinian, and its spectrum is finite. Its underlying set is the set-theoretic fiber of f , so f is quasi-finite.

(b) It is enough to work over an affine open of Y . A closed subset $Z \subseteq X$, with its reduced closed subscheme structure, maps to Y by a finite morphism: the closed immersion $Z \rightarrow X$ and f are both finite. Thus it suffices to show that the image of a finite affine morphism is closed.

Let $A \rightarrow B$ be finite and put $I = \ker(A \rightarrow B)$. The induced injection $A/I \hookrightarrow B$ is integral. Lying over shows that $\operatorname{Spec} B \rightarrow \operatorname{Spec}(A/I)$ is surjective, while $\operatorname{Spec}(A/I) \rightarrow \operatorname{Spec} A$ has image $V(I)$. Hence the image is exactly the closed set $V(I)$. Applying this to $Z \rightarrow Y$ proves that $f(Z)$ is closed.

(c) Let

$$X = \mathbb{A}_k^1 \coprod (\mathbb{A}_k^1 \setminus \{0\})$$

and map both components to $Y = \mathbb{A}_k^1$ by their natural maps. The first component makes the morphism surjective. It is of finite type: one component is an isomorphism and the other is the quasi-compact open immersion $D(t) \rightarrow \operatorname{Spec} k[t]$. The fiber over 0 has one point and every other fiber has two, so the map is quasi-finite.

The second component is closed in the disjoint union X , but its image $\mathbb{A}_k^1 \setminus \{0\}$ is not closed in Y . By part (b), the morphism cannot be finite. $\square$

Exercise II.3.6**Permitted context.**

An integral scheme is nonempty, irreducible, and reduced. Every nonempty open subset of an irreducible space contains its generic point. If $U = \text{Spec } A$ is an affine open in an integral scheme, then A is a domain, and the generic point corresponds to the prime (0) .

Problem. Let X be an integral scheme with generic point ξ . Prove that $\mathcal{O}_{X,\xi}$ is a field. If $U = \text{Spec } A$ is any nonempty affine open of X , identify this field with the fraction field of A .

Solution. Every nonempty affine open $U = \text{Spec } A$ contains ξ . The ring A is a domain, and under the affine identification ξ corresponds to its zero prime. Therefore

$$\mathcal{O}_{X,\xi} \cong A_{(0)} = \text{Frac}(A).$$

This is a field. Since the left side is intrinsic to X , the fraction fields obtained from all nonempty affine opens are canonically identified. It is called the function field $K(X)$. $\square$

Exercise II.3.7**Permitted context.**

A dominant morphism between integral schemes sends the generic point to the generic point and induces an inclusion of function fields. A zero-dimensional domain of finite type over a field is a finite field extension. If a finite set of elements is algebraic over $\text{Frac}(A)$, then after inverting one nonzero element of A they are integral over the resulting localization. Integral finite-type algebras are finite modules. A finite-type algebra over a field is a Jacobson ring.

Problem. Let $f : X \rightarrow Y$ be a dominant, generically finite morphism of finite type between integral schemes. Prove that there is a dense open $V \subseteq Y$ for which

$$f^{-1}(V) \longrightarrow V$$

is finite.

Solution. Let ξ and η be the generic points of X and Y . Dominance gives $f(\xi) = \eta$. The generic fiber is a finite-type scheme over $K(Y)$ with finitely many points. It has only the point ξ . Indeed, an affine neighborhood of any point of that fiber also contains ξ , because X is irreducible. Its intersection with the generic fiber is the spectrum of a finite-type domain over $K(Y)$ with finitely many primes. A finite-type algebra over a field is Jacobson. In a domain with finitely many primes the intersection of its maximal ideals is zero; if every maximal ideal were nonzero, their product would be nonzero in that intersection. Thus the domain is a field, so its only prime is the one corresponding to ξ .

Choose an affine neighborhood $U = \text{Spec } B$ of ξ , lying over an affine $\text{Spec } A \subseteq Y$. The coordinate ring of its generic fiber is

$$(A \setminus \{0\})^{-1}B.$$

It is a zero-dimensional domain of finite type over $K(Y)$, hence a finite field extension. It is $K(X)$, so

$$[K(X) : K(Y)] < \infty.$$

We next shrink Y so that its whole inverse image lies in U . After shrinking the affine target around η , quasi-compactness of f gives a finite affine cover

$$f^{-1}(\operatorname{Spec} A) = W_1 \cup \cdots \cup W_r$$

in which U may be included as one member. Each nonempty $W_i \cap U$ contains ξ . Choose a nonzero $b_i \in \Gamma(W_i, \mathcal{O}_X)$ such that

$$\xi \in D_{W_i}(b_i) \subseteq W_i \cap U.$$

The element $b_i \in K(X)^\times$ is algebraic over $K(Y)$. If

$$b_i^n + c_{n-1}b_i^{n-1} + \cdots + c_0 = 0 \quad (c_j \in K(Y), c_0 \neq 0),$$

clearing denominators gives a nonzero $a_i \in A$ with $a_i \in b_i \Gamma(W_i, \mathcal{O}_X)$. Consequently

$$W_i \cap f^{-1}(D(a_i)) \subseteq D_{W_i}(b_i) \subseteq U.$$

For $a = \prod_i a_i \neq 0$, it follows from the cover by the W_i that

$$f^{-1}(D(a)) \subseteq U.$$

Replace A by A_a and B by B_a . We have now reduced to

$$Y = \operatorname{Spec} A, \quad X = \operatorname{Spec} B,$$

where $A \subseteq B$ are domains, B is a finitely generated A -algebra, and $\operatorname{Frac} B$ is finite over $\operatorname{Frac} A$.

Choose A -algebra generators $b_1, \dots, b_r$ of B . Each b_i is algebraic over $\operatorname{Frac} A$. Clearing the denominators in monic equations for all the b_i , there is a nonzero $a \in A$ such that every b_i is integral over A_a . Hence B_a is integral and of finite type over A_a , so it is a finite A_a -module. With $V = D(a) \subseteq \operatorname{Spec} A$, one has

$$f^{-1}(V) = \operatorname{Spec} B_a \longrightarrow \operatorname{Spec} A_a = V,$$

which is finite. Since $a \neq 0$ in the domain A , V is dense in Y . □

Exercise II.3.8**Permitted context.**

Localization commutes with integral closure: if A is a domain with fraction field K , $\tilde{A}$ is its integral closure in K , and $S \subseteq A$ is multiplicative, then

$$S^{-1}\tilde{A}$$

is the integral closure of $S^{-1}A$ in K . Localizations of an integrally closed domain are integrally closed. We may also use the finiteness theorem: the integral closure of a finitely generated domain over a field in a finite extension of its fraction field is finite as a module over that domain.

Problem. Let X be an integral scheme. For every nonempty affine open $U = \operatorname{Spec} A$, let $\tilde{A}$ be the integral closure of A in $K(X)$. Prove that the schemes $\tilde{U} = \operatorname{Spec} \tilde{A}$ glue to a normal integral scheme $\tilde{X}$, with a natural morphism $\nu : \tilde{X} \rightarrow X$. Prove the universal property: every dominant morphism from a normal integral scheme Z to X factors uniquely through ν . Finally, prove that ν is finite when X is of finite type over a field.

Solution. If $D(f) \subseteq U = \operatorname{Spec} A$, the localization fact in the context gives

$$\tilde{A}_f = (\tilde{A})_f$$

inside $K(X)$. Hence the normalizations constructed from two affine opens agree over every principal affine contained in their intersection. These principal opens cover the intersection, and the resulting identifications satisfy the cocycle condition because all are identities inside the same field $K(X)$. The schemes $\operatorname{Spec} \tilde{A}$ therefore glue to a scheme $\tilde{X}$.

The overlaps are covered by common distinguished affine neighborhoods, so the gluing identifies the fraction-field generic points. Each $\tilde{A}$ is a domain, all charts share that generic point, and their localizations are integrally closed. Thus $\tilde{X}$ is integral and all its local rings are integrally closed domains; it is normal. The inclusions $A \hookrightarrow \tilde{A}$ induce morphisms

$$\operatorname{Spec} \tilde{A} \longrightarrow \operatorname{Spec} A.$$

They agree on overlaps by the same localization identity, so they glue to $\nu : \tilde{X} \rightarrow X$.

Let $g : Z \rightarrow X$ be dominant, with Z normal and integral. Dominance induces an inclusion $K(X) \hookrightarrow K(Z)$. Fix $U = \operatorname{Spec} A \subseteq X$ and put $W = g^{-1}(U)$. If $\alpha \in \tilde{A}$, its image in $K(Z)$ is integral over the image of A . For every $z \in W$, it is therefore integral over the local ring $\mathcal{O}_{Z,z}$. Normality makes that local ring integrally closed in $K(Z)$, so the image of α lies in every $\mathcal{O}_{Z,z}$. A rational function on an integral scheme that lies in all local rings over an open set is a regular section there. We obtain a homomorphism

$$\tilde{A} \longrightarrow \Gamma(W, \mathcal{O}_Z),$$

and hence a morphism $W \rightarrow \operatorname{Spec} \tilde{A}$. These morphisms agree on overlaps and glue to $\tilde{g} : Z \rightarrow \tilde{X}$, with $\nu \circ \tilde{g} = g$. The map on function fields is forced by g , and regular maps into each affine chart are then forced on the dense generic point; thus the factorization is unique.

Finally suppose X is of finite type over a field. On $U = \operatorname{Spec} A$, the ring A is a finitely generated domain. Here the relevant field extension in the finiteness theorem is $K(X)/\operatorname{Frac} A$, which is the identity extension, so $\tilde{A}$ is a finite A -module. Therefore

$$\nu^{-1}(U) = \operatorname{Spec} \tilde{A} \longrightarrow U$$

is finite for every affine chart U , and ν is finite. □

Exercise II.3.9**Permitted context.**

Fiber products of affine schemes satisfy

$$\operatorname{Spec} B \times_{\operatorname{Spec} A} \operatorname{Spec} C \cong \operatorname{Spec}(B \otimes_A C).$$

Points of a localization correspond to primes disjoint from the multiplicative set.

Problem. (a) Show that

$$\mathbb{A}_k^1 \times_{\operatorname{Spec} k} \mathbb{A}_k^1 \cong \mathbb{A}_k^2,$$

but that the underlying point set is not the Cartesian product of the underlying point sets of the two affine lines, even when k is algebraically closed.

(b) Let s, t be indeterminates over k . Describe

$$\operatorname{Spec} k(s) \times_{\operatorname{Spec} k} \operatorname{Spec} k(t),$$

although each of the three displayed factor and base spaces has one point.

Solution. (a) The affine fiber-product formula gives

$$\operatorname{Spec} k[x] \times_{\operatorname{Spec} k} \operatorname{Spec} k[y] \cong \operatorname{Spec}(k[x] \otimes_k k[y]) \cong \operatorname{Spec} k[x, y] = \mathbb{A}_k^2.$$

On points, the two projections send a prime $\mathfrak{q} \subseteq k[x, y]$ to its contractions to $k[x]$ and $k[y]$. The distinct primes

$$(x - y) \quad \text{and} \quad (x - y^2)$$

both contract to (0) in each polynomial subring. Thus two different points of the scheme product map to the same pair of generic points. The underlying set cannot be the Cartesian product, regardless of whether k is algebraically closed.

(b) The product is

$$\operatorname{Spec} R, \quad R = k(s) \otimes_k k(t).$$

Equivalently,

$$R = S^{-1}k[s, t], \quad S = \{g(s)h(t) : 0 \neq g \in k[s], 0 \neq h \in k[t]\}.$$

Hence its points correspond exactly to primes $\mathfrak{p} \subseteq k[s, t]$ with

$$\mathfrak{p} \cap k[s] = (0), \quad \mathfrak{p} \cap k[t] = (0).$$

They include the generic point (0) and the height-one primes generated by irreducible polynomials involving both variables; there are no height-two primes with these contraction properties. Indeed, the residue field of a height-two prime is finite algebraic over k by Zariski's lemma, making both contractions nonzero. The distinct primes $(s - t^j)$, $j \geq 1$, all survive. Thus the product has infinitely many points even though both factors are one-point schemes. Its topology and structure sheaf are those of the displayed localization. $\square$

Exercise II.3.10**Permitted context.**

For $f : X \rightarrow Y$ and $y \in Y$, the scheme-theoretic fiber is

$$X_y = X \times_Y \operatorname{Spec} k(y).$$

If $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$, and y corresponds to $\mathfrak{p} \subseteq A$, then

$$X_y = \operatorname{Spec}(B \otimes_A k(\mathfrak{p})) = \operatorname{Spec}((A \setminus \mathfrak{p})^{-1}(B/\mathfrak{p}B)).$$

Problem. (a) Prove that the underlying space of X_y is homeomorphic to the set-theoretic fiber $f^{-1}(y)$ with its subspace topology.

(b) Let k be algebraically closed, let

$$X = \operatorname{Spec} k[s, t]/(s - t^2), \quad Y = \operatorname{Spec} k[s],$$

and let $f : X \rightarrow Y$ be induced by the inclusion $k[s] \rightarrow k[s, t]/(s - t^2)$. Assuming $\operatorname{char} k \neq 2$, describe the fibers over a nonzero $a \in k$, over 0, and over the generic point of Y . Also state what changes in characteristic 2.

Solution. (a) Work first with affine neighborhoods $Y = \operatorname{Spec} A$ and $X = \operatorname{Spec} B$. The primes of

$$(A \setminus \mathfrak{p})^{-1}(B/\mathfrak{p}B)$$

correspond to the primes $\mathfrak{q} \subseteq B$ which contain $\mathfrak{p}B$ and avoid the image of $A \setminus \mathfrak{p}$. Equivalently, they are exactly those satisfying $\mathfrak{q} \cap A = \mathfrak{p}$, namely the points of $f^{-1}(y)$. The quotient-spectrum correspondence gives the subspace topology on $V(\mathfrak{p}B)$, and localization then gives the subspace consisting of primes disjoint from $A \setminus \mathfrak{p}$. Thus the bijection is a homeomorphism. These affine homeomorphisms agree on overlaps and glue for general X and Y .

(b) For the closed point $s = a$, the fiber is

$$X_a \cong \operatorname{Spec} k[t]/(t^2 - a).$$

If $a \neq 0$ and $\operatorname{char} k \neq 2$, choose $\alpha \in k$ with $\alpha^2 = a$. The roots α and $-\alpha$ are distinct, so the Chinese remainder theorem gives

$$k[t]/(t^2 - a) \cong k \times k.$$

The fiber is two reduced points, both with residue field k . At $a = 0$, the ring is $k[t]/(t^2)$, a nonreduced local ring with one point.

At the generic point η , whose residue field is $k(s)$,

$$X_\eta \cong \operatorname{Spec}(k(s)[t]/(t^2 - s)).$$

The element s is not a square in $k(s)$, so $t^2 - s$ is irreducible. The fiber is one point with residue field a degree-two extension of $k(s)$.

In characteristic 2, the generic description remains a purely inseparable degree-two field extension, but every closed fiber is $k[t]/((t - \alpha)^2)$. Thus even for $a \neq 0$ it is a nonreduced one-point scheme; the asserted two-point conclusion requires the stated characteristic hypothesis. $\square$

Exercise II.3.11**Permitted context.**

An affine closed immersion has the form $\text{Spec}(A/I) \rightarrow \text{Spec } A$. Closed subschemes of a scheme correspond to quasi-coherent ideal sheaves, with a larger ideal giving a smaller closed subscheme. Arbitrary sums of quasi-coherent ideal sheaves are quasi-coherent. A closed immersion may be checked on an affine open cover of the target.

Problem. (a) Prove that a base change of a closed immersion is a closed immersion.

(b) If $Y \rightarrow X = \text{Spec } A$ is a closed immersion, prove that Y is affine and is isomorphic over X to $\text{Spec}(A/\mathfrak{a})$ for an ideal $\mathfrak{a} \subseteq A$.

(c) Give a closed subset $C \subseteq X$ its reduced induced subscheme structure. If $Y' \subseteq X$ is any closed subscheme with underlying space C , prove that $C_{\text{red}} \rightarrow X$ factors through $Y' \rightarrow X$.

(d) Prove that every morphism $f : Z \rightarrow X$ has a unique smallest closed subscheme of X through which it factors. This is the scheme-theoretic image. If Z is reduced, identify it with the reduced induced structure on $f(Z)$.

Solution. (a) The assertion is local on the new base. In affine coordinates, base change of

$$\text{Spec}(A/I) \longrightarrow \text{Spec } A$$

along $\text{Spec } B \rightarrow \text{Spec } A$ is

$$\text{Spec}((A/I) \otimes_A B) \cong \text{Spec}(B/IB) \longrightarrow \text{Spec } B,$$

which is a closed immersion.

(b) For every $y \in Y$, choose an affine open neighborhood in Y . Because the topology on Y is the subspace topology, there is a principal open $D(f) \subseteq X$ containing y such that $Y \cap D(f)$ lies in that affine neighborhood. It is a distinguished open there, and hence is affine. Since Y is closed in the quasi-compact space X , finitely many such $Y \cap D(f_i)$ cover Y .

The union of the corresponding $D(f_i)$ in X is an open neighborhood of Y in X . Its closed complement is quasi-compact and lies in $X \setminus Y$; add finitely many principal opens contained in $X \setminus Y$ to obtain a finite principal cover of all of X . The restrictions of the corresponding f_i to Y generate the unit ideal, and each distinguished open Y_{f_i} is affine, including the empty ones. The affine criterion shows that $Y = \text{Spec } B$.

The closed immersion is now induced by a ring homomorphism $A \rightarrow B$. Its underlying map is a homeomorphism onto a closed subset and its sheaf map is surjective, so the affine closed-immersion criterion makes $A \rightarrow B$ surjective. With $\mathfrak{a} = \ker(A \rightarrow B)$, this gives $Y \cong \text{Spec}(A/\mathfrak{a})$ over X .

(c) Work on an affine open $\text{Spec } A \subseteq X$. By part (b), the two closed subschemes have the forms

$$\text{Spec}(A/I), \quad \text{Spec}(A/I').$$

For the reduced induced structure, I is the radical ideal of functions vanishing on C . Equality of the underlying closed sets gives $\sqrt{I'} = I$, and in particular $I' \subseteq I$. The quotient map

$$A/I' \longrightarrow A/I$$

defines the required factorization locally. These maps are canonical and therefore agree on overlaps, giving the global factorization.

(d) For every closed subscheme $Y_\alpha \subseteq X$ through which f factors, let $\mathcal{I}_\alpha \subseteq \mathcal{O}_X$ be its ideal sheaf. Set

$$\mathcal{I} = \sum_{\alpha} \mathcal{I}_\alpha.$$

This is a quasi-coherent ideal sheaf and defines a closed subscheme $Y \subseteq X$. Each $\mathcal{I}_\alpha$ maps to zero under $\mathcal{O}_X \rightarrow f_*\mathcal{O}_Z$, so their sum does as well; hence f factors through Y . Since $\mathcal{I}_\alpha \subseteq \mathcal{I}$, the inclusion $Y \rightarrow X$ factors through every Y_α . This proves existence and the universal property, which also gives uniqueness.

Now suppose Z is reduced and put $C = \overline{f(Z)}$. Locally, a function vanishing on C pulls back to a function lying in every prime of every affine open of Z . It is therefore nilpotent, and hence zero because Z is reduced. Thus f factors through C_{red} . Every closed subscheme through which f factors has underlying space containing C ; part (c) shows that C_{red} factors through it. Hence C_{red} has the defining universal property of the scheme-theoretic image. $\square$

Exercise II.3.12

Permitted context.

A graded homomorphism $S \rightarrow T$ induces a morphism from

$$U = \{\mathfrak{q} \in \text{Proj } T : \mathfrak{q} \not\supseteq \varphi(S_+)\}$$

to $\text{Proj } S$. On a standard open it is induced by

$$(S_f)_0 \longrightarrow (T_{\varphi(f)})_0.$$

A morphism covered by affine closed immersions is a closed immersion.

Problem. (a) Let $\varphi : S \rightarrow T$ be a surjective degree-preserving homomorphism of graded rings. Prove that $U = \text{Proj } T$ and that the induced morphism

$$\text{Proj } T \longrightarrow \text{Proj } S$$

is a closed immersion.

(b) Let $I \subseteq S$ be homogeneous. Show that different homogeneous ideals can define the same closed subscheme of $\text{Proj } S$. Specifically, for an integer d_0 , put

$$I' = \bigoplus_{d \geq d_0} I_d.$$

Prove that I and I' define the same closed subscheme.

Solution. (a) Surjectivity in each degree gives $\varphi(S_+) = T_+$. No point of $\text{Proj } T$ contains T_+ , so $U = \text{Proj } T$. For every homogeneous $f \in S_+$, localization and then passage to degree zero give a surjection

$$(S_f)_0 \twoheadrightarrow (T_{\varphi(f)})_0.$$

Thus the morphism over $D_+(f)$ is an affine closed immersion. The standard opens cover $\text{Proj } S$, so the global morphism is a closed immersion.

(b) The quotients S/I and S/I' have identical graded pieces in all degrees $d \geq d_0$. More explicitly, on $D_+(f)$ an element of the ideal defining $\text{Proj}(S/I)$ has the form

$$\frac{a}{f^n}, \quad a \in I_{n \deg f}.$$

Multiplying numerator and denominator by a sufficiently high power of f makes the numerator have degree at least d_0 , without changing the fraction. It then belongs to I' . Hence I and I' induce the same ideal in every ring $(S_f)_0$. Their affine closed subschemes agree on the standard cover and therefore agree globally. $\square$

Exercise II.3.13

Permitted context.

A morphism is of finite type exactly when it is locally of finite type and quasi-compact. These properties may be checked over arbitrary affine opens of the target. A finitely generated algebra over a noetherian ring is noetherian, and fiber products of affine schemes correspond to tensor products.

Problem. Prove the following properties.

- (a) Every closed immersion is of finite type.
- (b) Every quasi-compact open immersion is of finite type.
- (c) A composite of finite-type morphisms is of finite type.
- (d) Finite-type morphisms are stable under base change.
- (e) If X and Y are of finite type over S , then $X \times_S Y$ is of finite type over S .
- (f) If $X \xrightarrow{f} Y \xrightarrow{g} Z$, f is quasi-compact, and $g \circ f$ is of finite type, then f is of finite type.
- (g) If $f : X \rightarrow Y$ is of finite type and Y is noetherian, then X is noetherian.

Solution. (a) Over $\text{Spec } A$, a closed immersion has inverse image $\text{Spec}(A/I)$. The quotient is a finitely generated A -algebra, so the morphism is of finite type.

(b) Let $j : U \rightarrow Y$ be a quasi-compact open immersion and $V = \text{Spec } A \subseteq Y$. The open $U \cap V$ is quasi-compact, so finitely many principal opens $D(f_i) \subseteq V$ cover it. Each has coordinate ring

$$A_{f_i} = A[t]/(f_i t - 1),$$

a finitely generated A -algebra. The affine criterion proves that j is of finite type.

(c) Work over an affine $\text{Spec } A$ in the final target. The first inverse image has a finite affine cover $\text{Spec } B_i$, with each B_i finitely generated over A . The inverse image of each of these has a finite affine cover $\text{Spec } C_{ij}$, with C_{ij} finitely generated over B_i . Each C_{ij} is finitely generated over A , and the combined cover is finite.

(d) Affine locally, base change replaces a finite-type map $A \rightarrow B$ by

$$C \longrightarrow B \otimes_A C.$$

If $b_1, \dots, b_n$ generate B over A , their tensors generate the latter algebra over C . The affine descriptions glue, proving stability under arbitrary base change.

(e) The projection $X \times_S Y \rightarrow Y$ is the base change of $X \rightarrow S$, hence is of finite type by part (d). Composing with the finite-type morphism $Y \rightarrow S$ and using part (c) proves the claim.

(f) It remains to show that f is locally of finite type. Work with affine neighborhoods

$$W = \operatorname{Spec} A \subseteq Z, \quad V = \operatorname{Spec} B \subseteq g^{-1}(W), \quad U = \operatorname{Spec} C \subseteq f^{-1}(V).$$

By the arbitrary-affine criterion for the finite-type morphism $g \circ f$, C is finitely generated over A . Thus the relevant ring maps have the form

$$A \longrightarrow B \longrightarrow C.$$

If C is generated over A by $c_1, \dots, c_n$, the same elements generate it over B , since the image of A is contained in the image of B . Thus f is locally of finite type. It is quasi-compact by hypothesis, so it is of finite type.

(g) A noetherian scheme has a finite affine cover $\operatorname{Spec} A_i$ with all A_i noetherian. Finite type and quasi-compactness give a finite affine cover of every $f^{-1}(\operatorname{Spec} A_i)$ by spectra of finitely generated A_i -algebras. Those algebras are noetherian by the Hilbert basis theorem. Hence X has a finite affine cover by noetherian rings and is a noetherian scheme. $\square$

Exercise II.3.14

Permitted context.

If A is a finitely generated algebra over a field k , the residue field at every maximal ideal is a finite extension of k . Conversely, for a prime $\mathfrak{p}$ of a finitely generated k -algebra, if $\operatorname{Frac}(A/\mathfrak{p})$ is algebraic over k , then $A/\mathfrak{p}$ is a field. Every nonzero ring has a maximal ideal.

Problem. Prove that the closed points of a scheme of finite type over a field are dense. Give an example showing that this need not hold for an arbitrary scheme.

Solution. Let X be of finite type over k , and let $U \subseteq X$ be a nonempty open set. Choose a nonempty affine open $V = \operatorname{Spec} A \subseteq U$. A maximal ideal of A gives a point $x \in V$ whose residue field $k(x)$ is finite over k .

We check that x is closed in all of X , not merely in V . Let $z \in \overline{\{x\}}$ and choose an affine neighborhood $\operatorname{Spec} B$ of z . It also contains x , since every neighborhood of a specialization contains its generizations. If $\mathfrak{p} \subseteq B$ represents x , then

$$\operatorname{Frac}(B/\mathfrak{p}) = k(x)$$

is algebraic over k . The fact in the context makes $B/\mathfrak{p}$ a field, so $\mathfrak{p}$ is maximal. The point z , represented by a prime containing $\mathfrak{p}$, must equal x . Thus x is closed in X . Every nonempty open contains such a point, proving density.

For a counterexample, take the spectrum of the discrete valuation ring

$$R = k[t]_{(t)}.$$

It has a generic point (0) and one closed point (t) . The nonempty open $D(t) = \{(0)\}$ contains no closed point, so the closed points are not dense. $\square$

Exercise II.3.15**Permitted context.**

Let X be of finite type over a field k . We may use these standard base-change facts.

First, a field extension is faithfully flat, so reducedness and irreducibility descend from X_L to X_K for extensions $L/K/k$. Second, if the base field is algebraically closed, every irreducible finite-type scheme remains irreducible after arbitrary field extension. Third, a reduced finite-type scheme over a perfect field is geometrically reduced. Finally, base change by an algebraic purely inseparable field extension is a universal homeomorphism.

Problem. Let $\bar{k}$, k_s , and k_p denote an algebraic closure, a separable closure, and a perfect closure of k , respectively.

(a) Prove the equivalence of:

$$X_{\bar{k}} \text{ is irreducible}; \quad X_{k_s} \text{ is irreducible}; \quad X_K \text{ is irreducible for every field extension } K/k.$$

(b) Prove the equivalence of:

$$X_{\bar{k}} \text{ is reduced}; \quad X_{k_p} \text{ is reduced}; \quad X_K \text{ is reduced for every field extension } K/k.$$

(c) Give an integral finite-type k -scheme which is neither geometrically irreducible nor geometrically reduced.

Solution. (a) The third condition implies the first, and the first implies the second by faithfully flat descent along $\bar{k}/k_s$. Conversely, $\bar{k}/k_s$ is purely inseparable. The projection

$$X_{\bar{k}} \longrightarrow X_{k_s}$$

is therefore a universal homeomorphism, so the second condition implies that $X_{\bar{k}}$ is irreducible.

It remains to pass from the first condition to an arbitrary K/k . Choose an algebraically closed field Ω containing both K and an embedded algebraic closure of k . Since an irreducible finite-type scheme over the algebraically closed field $\bar{k}$ stays irreducible after extending scalars, X_{Ω} is irreducible. The projection $X_{\Omega} \rightarrow X_K$ is surjective, and a continuous image of an irreducible space is irreducible. Hence X_K is irreducible.

(b) Again, the third condition implies the first, and the first descends to the second. Conversely, the field k_p is perfect. If X_{k_p} is reduced, the perfect-field result in the context says it remains reduced over $\bar{k}$, proving the first condition.

Assume now that $X_{\bar{k}}$ is reduced and let K/k be any extension. Embed K and $\bar{k}$ in a common algebraically closed extension Ω . A reduced finite-type scheme over the perfect field $\bar{k}$ is geometrically reduced, so X_{Ω} is reduced. Reducedness descends through the faithfully flat extension Ω/K , giving that X_K is reduced. This proves the third condition.

(c) Take

$$k = \mathbb{F}_3(t), \quad L = \mathbb{F}_9(t^{1/3}), \quad X = \operatorname{Spec} L.$$

The scheme X is integral and finite over k . Over an algebraic closure $\bar{k}$, the separable constant-field extension $\mathbb{F}_9/\mathbb{F}_3$ gives two factors, while the purely inseparable equation $u^3 - t$ becomes $(u - t^{1/3})^3$. Consequently

$$L \otimes_k \bar{k} \cong \bar{k}[\epsilon]/(\epsilon^3) \times \bar{k}[\epsilon]/(\epsilon^3).$$

Its spectrum is reducible and its ring is nonreduced. Thus X has neither geometric property. $\square$

Exercise II.3.16**Permitted context.**

In a noetherian topological space, descending chains of closed subsets stabilize. Equivalently, every nonempty collection of closed subsets has a minimal member.

Problem. Let X be noetherian and let $\mathcal{P}$ be a property of closed subsets. Suppose that whenever $\mathcal{P}$ holds for every proper closed subset of a closed set Y , it also holds for Y . Prove that $\mathcal{P}$ holds for every closed subset of X , in particular for X .

Solution. If the conclusion were false, the collection of closed subsets for which $\mathcal{P}$ fails would be nonempty. Choose a minimal member Y . Every proper closed subset of Y then satisfies $\mathcal{P}$, so the hypothesis implies that Y satisfies $\mathcal{P}$, a contradiction. Thus there is no counterexample. This argument is called noetherian induction. $\square$

Exercise II.3.17**Permitted context.**

A Zariski space is a noetherian topological space in which every nonempty irreducible closed subset has a unique generic point. In a noetherian space, every closed subset is a finite union of irreducible components. For a noetherian space X , let $t(X)$ be the set of its nonempty irreducible closed subsets, with closed subsets

$$t(C) = \{Z \in t(X) : Z \subseteq C\}$$

for closed $C \subseteq X$. Define $\alpha : X \rightarrow t(X)$ by $\alpha(x) = \overline{\{x\}}$.

Problem. (a) Prove that the underlying space of a noetherian scheme is a Zariski space.

(b) Prove that a minimal nonempty closed subset of a Zariski space is a single point.

(c) Prove that every Zariski space is T_0 .

(d) Prove that the generic point of an irreducible Zariski space lies in every nonempty open subset.

(e) Order points by declaring $x_1 \geq x_0$ when x_1 specializes to x_0 , meaning $x_0 \in \overline{\{x_1\}}$. Identify the minimal and maximal points. Prove that closed sets are stable under specialization and open sets under generization.

(f) Prove that $t(X)$ is a Zariski space for every noetherian X , and that X is a Zariski space if and only if $\alpha : X \rightarrow t(X)$ is a homeomorphism.

Solution. (a) The underlying space of a noetherian scheme is noetherian. Every nonempty irreducible closed subset of a scheme has a unique generic point, so it is a Zariski space.

(b) Let C be minimal among nonempty closed subsets. It is irreducible: a decomposition into two proper closed subsets would contradict minimality. Let η be its generic point. For any $x \in C$, the nonempty closed set $\overline{\{x\}}$ is contained in C , hence equals C . Both x and η are then generic points of C , so uniqueness gives $x = \eta$. Thus $C = \{\eta\}$.

(c) If distinct points x, y had the same closure, they would be two generic points of the same irreducible closed subset. Therefore their closures differ. If, for example, $y \notin \overline{\{x\}}$, then $X \setminus \overline{\{x\}}$ contains y but not x . This is the T_0 axiom.

(d) Let η be the generic point. If a nonempty open U omitted η , its closed complement would contain η , hence also $\overline{\{\eta\}} = X$. This would make U empty, a contradiction.

(e) The T_0 property makes specialization a partial order. A point is minimal exactly when it has no proper specialization, equivalently when its closure is a singleton; these are the closed points from part (b). If x is maximal, choose an irreducible component C containing x . Its generic point η_C generalizes x , so maximality forces $x = \eta_C$. Conversely, a proper generalization of η_C would have irreducible closure properly containing C , contradicting maximality of an irreducible component. Thus the maximal points are precisely the generic points of irreducible components.

If a closed set contains x , it contains $\overline{\{x\}}$, and hence every specialization of x . If an open set contains x and y is a generalization of x , every open neighborhood of x contains y ; hence the open set contains y .

(f) A descending chain

$$t(C_1) \supseteq t(C_2) \supseteq \cdots$$

gives a descending chain $C_1 \supseteq C_2 \supseteq \cdots$, because a closed set in a noetherian space is the union of its finitely many irreducible components. The latter chain stabilizes, so $t(X)$ is noetherian. Moreover, $t(C)$ is irreducible exactly when C is irreducible. In that case the point $C \in t(X)$ has closure $t(C)$, and it is the unique generic point. Hence $t(X)$ is a Zariski space.

For every closed $C \subseteq X$,

$$\alpha^{-1}(t(C)) = C,$$

so α is continuous. It is injective exactly when distinct points have distinct closures, and it is surjective exactly when every irreducible closed subset has a generic point. If X is a Zariski space, both hold; furthermore $\alpha(C) = t(C)$, so α is closed and is a homeomorphism. Conversely, if α is a homeomorphism, X is homeomorphic to the Zariski space $t(X)$, and is therefore itself a Zariski space. $\square$

Exercise II.3.18

Permitted context.

A constructible subset belongs to the Boolean algebra generated by the open subsets. A locally closed subset is an intersection of an open and a closed subset. In a Zariski space, closed subsets have finitely many irreducible components, and the generic point of an irreducible space lies in every nonempty open subset. The specialization criterion in (c) retains the Zariski-space hypothesis.

Problem. (a) Prove that a subset is constructible if and only if it is a finite disjoint union of locally closed subsets.

(b) In an irreducible Zariski space, prove that a constructible subset is dense if and only if it contains the generic point. Prove that a dense constructible subset contains a nonempty open subset.

(c) Prove that a subset is closed if and only if it is constructible and stable under specialization. State and prove the corresponding criterion for openness.

(d) If $f : X \rightarrow Y$ is continuous between Zariski spaces, prove that the inverse image of a constructible subset is constructible.

Solution. (a) Any particular constructible set is obtained by finitely many Boolean operations from finitely many open sets $U_1, \dots, U_n$. The disjoint atoms

$$\left(\bigcap_{i \in I} U_i \right) \cap \left(\bigcap_{j \notin I} (X \setminus U_j) \right), \quad I \subseteq \{1, \dots, n\},$$

partition X . Each atom is the intersection of an open set with a closed set and is therefore locally closed. The constructible set is the union of some of these finitely many atoms. Conversely, locally closed sets are constructible, and constructible sets are closed under finite unions.

(b) Let η be the generic point. Any subset containing η is dense because its closure contains $\overline{\{\eta\}} = X$.

Conversely, write a dense constructible subset as a finite union

$$C = \bigcup_{i=1}^r (U_i \cap F_i),$$

with U_i open and F_i closed. Refine the F_i into their irreducible components and discard empty intersections. The closure of each remaining piece is contained in its irreducible closed component. Since the finite union of these closures is X , irreducibility of X forces one component to be all of X . The corresponding piece is then a nonempty open subset of X . It contains η , so C contains both the generic point and a nonempty open subset.

(c) Closed subsets are constructible and stable under specialization. Conversely, suppose C has these two properties. Decompose C into finitely many locally closed pieces, and refine their closed parts into irreducible components. A nonempty piece open in an irreducible closed set F contains the generic point of F . Stability under specialization then makes C contain all of F . Consequently C is the finite union of these closed sets F , and is closed.

A subset is open if and only if it is constructible and stable under generization. This follows by applying the closed criterion to its complement: complements preserve constructibility and interchange the two stability conditions.

(d) The inverse image of a locally closed set $U \cap F$ is

$$f^{-1}(U) \cap f^{-1}(F),$$

again locally closed. Apply part (a) to a finite locally closed decomposition. □

Exercise II.3.19**Permitted context.**

We may use the following commutative-algebra lemma. Let $A \subseteq B$ be noetherian domains with B a finitely generated A -algebra. For every $0 \neq b \in B$, there is $0 \neq a \in A$ such that any homomorphism $A \rightarrow K$ to an algebraically closed field, with a mapping nonzero, extends to $B \rightarrow K$ with b mapping nonzero.

Open subschemes of noetherian schemes are quasi-compact, and quasi-compact open immersions and closed immersions are of finite type.

Problem. Let $f : X \rightarrow Y$ be a finite-type morphism of noetherian schemes. Prove Chevalley's theorem: the image of every constructible subset of X is constructible in Y , using these steps.

(a) Reduce to the case in which X and Y are affine integral noetherian schemes and f is dominant.

(b) In that case, use the lemma in the context to show that $f(X)$ contains a nonempty open subset of Y .

(c) Complete the proof by noetherian induction on the target.

(d) Give a morphism of varieties whose image is neither open nor closed.

Solution. (a) By the finite locally closed decomposition, it is enough to treat one locally closed subset $C \subseteq X$. Give its closure the reduced closed structure and C its open subscheme structure. Since X is noetherian, C is quasi-compact, and the composite

$$C \longrightarrow X \longrightarrow Y$$

is of finite type. Thus it suffices to prove that the image of an entire finite-type source is constructible.

Cover Y by finitely many affine opens and each inverse image by finitely many affine opens. Images and constructible sets may be united finitely, so reduce to X and Y affine. They have finitely many irreducible components. Give each component its reduced structure and treat its image separately, reducing to X integral. The closure of its image is an irreducible closed subset of Y ; replace Y by that closure with its reduced structure. A closed subscheme of an affine scheme is affine, and the resulting map is dominant. We have reached the stated case.

(b) Write $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$. Dominance and integrality make $A \rightarrow B$ injective. Apply the algebra lemma with $b = 1$, obtaining $0 \neq a \in A$. We claim

$$D(a) \subseteq f(X).$$

For $\mathfrak{p} \in D(a)$, embed $k(\mathfrak{p})$ in an algebraic closure K . The map $A \rightarrow k(\mathfrak{p}) \rightarrow K$ does not kill a , so it extends to a homomorphism $B \rightarrow K$. Its kernel is a prime $\mathfrak{q}$ whose contraction to A is $\mathfrak{p}$. Therefore $f(\mathfrak{q}) = \mathfrak{p}$. Since A is a domain and $a \neq 0$, $D(a)$ is a nonempty dense open subset of Y .

(c) Use noetherian induction on closed subsets of Y . The reductions in part (a), applied within any closed subset, show that an irreducible closed target either contains no dense image component or has a dense open subset U contained in the image. In the first case the closures of the images of the finitely many source components are proper closed subsets. Apply the induction hypothesis to the maps into these closures and take the finite union of their constructible images. In the latter case,

$$f(X) = U \cup f(X \setminus f^{-1}(U)).$$

The second source is closed in X , its image lies in the proper closed set $Y \setminus U$, and its morphism to that closed set is of finite type. The induction hypothesis makes the second image constructible. Thus $f(X)$ is constructible. Reducible targets are finite unions of their irreducible components, so the same conclusion holds there. Part (a) then gives the theorem for every constructible subset.

(d) Over an algebraically closed field, consider

$$\mathbb{A}^2 \longrightarrow \mathbb{A}^2, \quad (x, y) \longmapsto (x, xy).$$

Its image on variety points is

$$D(x) \cup \{(0, 0)\}.$$

It is not closed because it contains the dense open $D(x)$ but is not all of $\mathbb{A}^2$. It is not open because every open neighborhood of the origin meets the punctured line $\{(0, z) : z \neq 0\}$, which the image omits. $\square$

Exercise II.3.20

Permitted context.

Let A be a finitely generated domain over a field k . We may use the dimension results from affine algebraic geometry:

$$\dim A = \text{trdeg}_k \text{Frac } A, \quad \dim A_{\mathfrak{p}} = \text{ht } \mathfrak{p}, \quad \text{ht } \mathfrak{p} + \dim A/\mathfrak{p} = \dim A.$$

Every maximal ideal has height $\dim A$. Noether normalization gives a finite injection $k[x_1, \dots, x_d] \hookrightarrow A$, with $d = \dim A$. The dimension of a scheme is the supremum of the dimensions of the affine opens in any affine open cover.

Problem. Let X be an integral scheme of finite type over k .

- (a) For every closed point P , prove $\dim X = \dim \mathcal{O}_{X,P}$.
- (b) Prove

$$\dim X = \text{trdeg}_k K(X).$$

- (c) For a nonempty closed subset $Y \subseteq X$, prove

$$\text{codim}(Y, X) = \inf_{P \in Y} \dim \mathcal{O}_{X,P}.$$

- (d) For the nonempty closed subset in (c), prove

$$\dim Y + \text{codim}(Y, X) = \dim X.$$

- (e) If $U \subseteq X$ is nonempty and open, prove $\dim U = \dim X$.

(f) For a field extension k'/k , prove that every irreducible component of $X_{k'} = X \times_k k'$ has dimension $\dim X$.

Solution. (a) Choose an affine neighborhood $\text{Spec } A$ of P , represented by a maximal ideal $\mathfrak{m}$. Every nonempty affine open of X has fraction field $K(X)$, so the dimension formula in the context gives the same dimension $\text{trdeg}_k K(X)$ on every affine chart. A finite affine cover therefore shows

$$\dim X = \dim A.$$

Since maximal ideals of the finitely generated domain A have full height,

$$\dim \mathcal{O}_{X,P} = \dim A_{\mathfrak{m}} = \text{ht } \mathfrak{m} = \dim A = \dim X.$$

(b) On any nonempty affine open $\text{Spec } A$,

$$K(X) = \text{Frac } A.$$

The first equality in the context and part (a)'s affine-cover argument give

$$\dim X = \dim A = \text{trdeg}_k \text{Frac } A = \text{trdeg}_k K(X).$$

(c) Let $Y_1, \dots, Y_r$ be the irreducible components of Y , with generic points η_i . On an affine neighborhood, η_i corresponds to a prime $\mathfrak{p}_i$, and

$$\text{codim}(Y_i, X) = \text{ht } \mathfrak{p}_i = \dim \mathcal{O}_{X,\eta_i}.$$

If $P \in Y_i$, its prime contains $\mathfrak{p}_i$, so $\dim \mathcal{O}_{X,P} \geq \dim \mathcal{O}_{X,\eta_i}$. Hence the infimum over Y is attained at the generic point of a component and equals

$$\min_i \text{codim}(Y_i, X) = \text{codim}(Y, X).$$

(d) For each irreducible component Y_i , the height formula gives

$$\dim Y_i + \text{codim}(Y_i, X) = \dim X.$$

Now $\dim Y$ is the maximum of the $\dim Y_i$, while $\text{codim}(Y, X)$ is the minimum of their codimensions. The displayed equality shows that these extrema occur for the same components, and proves the formula for Y .

(e) The open U is noetherian and therefore quasi-compact, so it remains of finite type over k . It is integral and has the same generic point and function field as X . Part (b) gives

$$\dim U = \text{trdeg}_k K(U) = \text{trdeg}_k K(X) = \dim X.$$

(f) It is enough to work on an affine open $X = \text{Spec } A$, because every irreducible component meets the base change of an affine open and dimension is unchanged on nonempty open subsets. Put $d = \dim A$ and choose a finite injection

$$R = k[x_1, \dots, x_d] \hookrightarrow A.$$

After scalar extension,

$$R' = k'[x_1, \dots, x_d] \longrightarrow B = A \otimes_k k'$$

is finite.

We claim that every minimal prime $\mathfrak{q}$ of B contracts to (0) in R' . The finite R -module A is torsion-free, and it embeds in a finite free R -module after clearing denominators in $A \otimes_R \text{Frac } R$. Flat base change to R' embeds B in a finite free R' -module, so every nonzero element of R' is a non-zero-divisor on B . In a noetherian ring every element of a minimal prime is a zero-divisor. Thus $\mathfrak{q} \cap R' = (0)$.

It follows that $B/\mathfrak{q}$ is an integral finite extension of R' . Integral extensions preserve dimension, so

$$\dim B/\mathfrak{q} = \dim R' = d.$$

These quotients are the irreducible components of $X_{k'}$, and every one has dimension $\dim X$. $\square$

Exercise II.3.21**Permitted context.**

Let R be a discrete valuation ring with uniformizer π , fraction field K , and a coefficient field isomorphic to its residue field k . Then $\dim R[t] = 2$, while

$$R[t]_{\pi} = K[t]$$

has dimension 1. A principal prime of a domain has height at most one, and a nonzero principal prime here has height exactly one.

Problem. For $X = \operatorname{Spec} R[t]$, show that conclusions (a), (d), and (e) of Exercise II.3.20 fail when the finite-type-over-a-field hypothesis is removed.

Solution. Consider the principal ideal

$$\mathfrak{p} = (\pi t - 1) \subseteq R[t].$$

There is an isomorphism

$$R[t]/(\pi t - 1) \cong R[1/\pi] = K,$$

so $\mathfrak{p}$ is maximal and defines a closed point $P \in X$. It is a nonzero principal prime and hence has height one. Therefore

$$\dim \mathcal{O}_{X,P} = 1 \quad \text{but} \quad \dim X = 2,$$

contradicting conclusion (a).

Let $Y = \{P\}$. Then $\dim Y = 0$ and $\operatorname{codim}(Y, X) = 1$, so

$$\dim Y + \operatorname{codim}(Y, X) = 1 \neq 2 = \dim X.$$

This contradicts (d).

Finally, the nonempty open subset

$$U = D(\pi) = \operatorname{Spec} K[t]$$

has dimension 1, whereas X has dimension 2. Thus (e) also fails. □

Exercise II.3.22**Permitted context.**

Let $f : X \rightarrow Y$ be a dominant morphism of integral finite-type k -schemes, and put $e = \dim X - \dim Y$. We may use:

$$\mathrm{trdeg}_k k(x) = \dim X - \dim \mathcal{O}_{X,x}$$

for $x \in X$, and the analogous formula on Y . In a noetherian local ring of dimension d , there are d elements whose generated ideal has maximal-ideal radical. Krull's height theorem bounds the height of a minimal prime over an ideal generated by d elements by d . Finite morphisms and open immersions are stable under base change, and finite-type images of constructible sets are constructible.

Problem. (a) Let $Y' \subseteq Y$ be irreducible closed with generic point $\eta' \in f(X)$. If Z is an irreducible component of $f^{-1}(Y')$ whose image contains η' , prove

$$\mathrm{codim}(Z, X) \leq \mathrm{codim}(Y', Y).$$

(b) For $y \in f(X)$, prove that every irreducible component of X_y has dimension at least e .

(c) Prove that there is a dense open $U \subseteq X$ such that $\dim U_y = e$ for every $y \in f(U)$.

(d) For an integer h , let E_h consist of points $x \in X$ lying on an irreducible component of $X_{f(x)}$ of dimension at least h . Prove that $E_e = X$; that E_h is not dense for $h > e$; and that every E_h is closed.

(e) Let

$$C_h = \{y \in Y : \dim X_y = h\}.$$

Prove that every C_h is constructible and that C_e contains a dense open subset of Y .

Solution. (a) Let ζ be the generic point of Z . The hypothesis that $\eta' \in f(Z)$ forces $f(\zeta) = \eta'$. Work in affine neighborhoods and let $\mathfrak{p}$ and $\mathfrak{q}$ be the primes corresponding to η' and ζ . Since Z is an irreducible component of the inverse image, $\mathfrak{q}$ is minimal over $\mathfrak{p}\mathcal{O}_X$.

Put $d = \dim \mathcal{O}_{Y,\eta'}$. Choose $a_1, \dots, a_d$ in that local ring whose ideal has radical its maximal ideal. In $\mathcal{O}_{X,\zeta}$, the radical of the extended ideal is the maximal ideal: a smaller prime containing it would give a prime strictly below $\mathfrak{q}$ over $\mathfrak{p}$, contradicting the minimality of $\mathfrak{q}$. Krull's height theorem therefore gives

$$\dim \mathcal{O}_{X,\zeta} \leq d.$$

Using the codimension formula for generic points,

$$\mathrm{codim}(Z, X) = \dim \mathcal{O}_{X,\zeta} \leq \dim \mathcal{O}_{Y,\eta'} = \mathrm{codim}(Y', Y).$$

(b) Let T be an irreducible component of X_y , with generic point x . The closure $\overline{\{x\}}$ is an irreducible component of $f^{-1}(\overline{\{y\}})$ that dominates $\overline{\{y\}}$. Part (a) gives

$$\dim \mathcal{O}_{X,x} \leq \dim \mathcal{O}_{Y,y}.$$

Since T is an integral finite-type scheme over $k(y)$,

$$\begin{aligned} \dim T &= \mathrm{trdeg}_{k(y)} k(x) \\ &= \mathrm{trdeg}_k k(x) - \mathrm{trdeg}_k k(y) \\ &= (\dim X - \dim \mathcal{O}_{X,x}) - (\dim Y - \dim \mathcal{O}_{Y,y}) \\ &\geq \dim X - \dim Y = e. \end{aligned}$$

(c) Shrink to dense affine opens in X and Y , without changing their dimensions. Choose $t_1, \dots, t_e \in K(X)$ forming a transcendence basis over $K(Y)$; after shrinking once more, take them to be regular on X . They define a dominant, generically finite morphism

$$g : X \longrightarrow \mathbb{A}_Y^e.$$

By the generically finite result, there is a dense open $W \subseteq \mathbb{A}_Y^e$ such that

$$g^{-1}(W) \longrightarrow W$$

is finite. Dominance and closedness of a finite morphism allow W to be chosen so that this map is surjective. Put $U = g^{-1}(W)$.

For $y \in f(U)$, base change gives a finite surjection

$$U_y \longrightarrow W_y.$$

The latter is a nonempty open subset of $\mathbb{A}_{k(y)}^e$, so it has dimension e . A finite morphism makes every component of U_y have dimension at most e , while part (b), or its application to the open U , gives the reverse inequality. Thus $\dim U_y = e$.

(d) Part (b) immediately gives $E_e = X$, and also $E_h = X$ for $h \leq e$. For $h > e$, the open U from part (c) is disjoint from E_h : a component of X_y meeting U_y has a nonempty open subset inside a component of U_y , and therefore has dimension e . Hence E_h is not dense.

We prove closedness by noetherian induction on X . For $h > e$, put $F = X \setminus U$. A fiber component of dimension at least h cannot meet U , so

$$E_h(X \rightarrow Y) = E_h(F \rightarrow Y).$$

Write the reduced closed scheme F as the finite union of its irreducible components F_i , and let Y_i be the reduced closure of the image of F_i . Then $F_i \rightarrow Y_i$ is dominant and $\dim F_i < \dim X$. By induction, every $E_h(F_i \rightarrow Y_i)$ is closed in F_i , and

$$E_h(X \rightarrow Y) = \bigcup_i E_h(F_i \rightarrow Y_i).$$

This finite union is closed in X . Together with the case $h \leq e$, this proves closedness for all h .

(e) Put

$$F_h = \{y \in Y : \dim X_y \geq h\}.$$

By definition and part (d),

$$F_h = f(E_h).$$

Chevalley's theorem makes F_h constructible, and hence

$$C_h = F_h \setminus F_{h+1}$$

is constructible. The generic fiber has function field $K(X)$ over $K(Y)$, so

$$\dim X_\eta = \text{trdeg}_{K(Y)} K(X) = \dim X - \dim Y = e.$$

Thus the generic point η of Y belongs to C_e . A constructible subset of an irreducible Zariski space that contains the generic point contains a nonempty, hence dense, open subset. Therefore C_e contains a dense open subset of Y . $\square$

Exercise II.3.23**Permitted context.**

For affine k -varieties V and W ,

$$A(V \times_k W) \cong A(V) \otimes_k A(W).$$

The functor t sends an affine variety with coordinate ring A to $\text{Spec } A$. Products and the functor t are obtained globally by gluing their affine-chart constructions.

Problem. Let V, W be varieties over an algebraically closed field k . Prove that

$$t(V \times_k W) \cong t(V) \times_{\text{Spec } k} t(W).$$

Solution. First suppose V and W are affine varieties, with coordinate rings $A = A(V)$ and $B = A(W)$. Then

$$\begin{aligned} t(V \times_k W) &\cong \text{Spec}(A \otimes_k B) \\ &\cong \text{Spec } A \times_{\text{Spec } k} \text{Spec } B \\ &= t(V) \times_{\text{Spec } k} t(W). \end{aligned}$$

The isomorphism is natural in restrictions to affine open subsets.

For general V and W , the products of affine charts $V_i \times_k W_j$ cover $V \times_k W$, while $t(V_i) \times_{\text{Spec } k} t(W_j)$ cover the scheme fiber product. The affine isomorphisms above agree on all overlaps by their naturality, so they glue to the asserted global isomorphism. $\square$

II.4 Separated and Proper Morphisms**Exercise II.4.1****Permitted context.**

A morphism is proper if it is separated, of finite type, and universally closed. Finite morphisms are affine, are stable under arbitrary base change, and correspond locally to ring maps for which the target is a finite module. An integral extension satisfies lying over.

Problem. Prove that every finite morphism of schemes is proper.

Solution. Let $f : X \rightarrow Y$ be finite. It is of finite type. It is also separated: over an affine open $V = \text{Spec } A \subseteq Y$, write $f^{-1}(V) = \text{Spec } B$. The restriction of the diagonal is the morphism associated to the surjection

$$B \otimes_A B \longrightarrow B, \quad b_1 \otimes b_2 \longmapsto b_1 b_2,$$

so it is a closed immersion. Since this can be checked over an open cover of Y , the diagonal of f is a closed immersion.

It remains to prove universal closedness. Because every base change of f is finite, it is enough to show that a finite morphism is closed. Again work with a finite ring map $A \rightarrow B$. If $J \subseteq B$, set

$$I = \ker(A \longrightarrow B/J).$$

The induced injection $A/I \hookrightarrow B/J$ is integral, so lying over gives a surjection

$$\mathrm{Spec}(B/J) \longrightarrow \mathrm{Spec}(A/I).$$

Consequently the image of the closed subset $V(J) \subseteq \mathrm{Spec} B$ is exactly $V(I)$, hence is closed. Thus f , and every base change of f , is closed. Therefore f is proper. $\square$

Exercise II.4.2

Permitted context.

For two S -morphisms $f, g : X \rightarrow Y$, their equalizer is the inverse image of the diagonal under

$$(f, g) : X \longrightarrow Y \times_S Y.$$

If $Y \rightarrow S$ is separated, this equalizer is a closed subscheme of X . A closed subscheme whose underlying set is all of a reduced scheme is the whole scheme.

Problem. Let X be a reduced S -scheme and Y a separated S -scheme. Suppose two S -morphisms $f, g : X \rightarrow Y$ agree on an open dense subset of X . Prove that $f = g$. Give examples showing that the conclusion can fail if either X is nonreduced or Y is nonseparated.

Solution. Let

$$E = X \times_{(f,g), Y \times_S Y, \Delta_Y} Y.$$

Because $Y \rightarrow S$ is separated, Δ_Y is a closed immersion, so $E \hookrightarrow X$ is a closed immersion. The dense open set on which f and g agree is contained in $|E|$. Therefore $|E| = X$. On an affine open $\mathrm{Spec} A \subseteq X$, the ideal defining E has vanishing set all of $\mathrm{Spec} A$, and hence is contained in the nilradical of A . Since X is reduced, that ideal is zero. Thus $E = X$, which says precisely that $f = g$.

For failure without reducedness, take

$$X = \mathrm{Spec} k[x, \varepsilon]/(\varepsilon^2, x\varepsilon), \quad Y = \mathbb{A}_k^1,$$

and let $f, g : X \rightarrow Y$ correspond respectively to

$$t \longmapsto x, \quad t \longmapsto x + \varepsilon.$$

They are distinct, but on the dense open $D(x)$ one has $\varepsilon = 0$, so their restrictions agree.

For failure without separatedness, let Y be the affine line with doubled origin, obtained by gluing two copies Y_1, Y_2 of $\mathbb{A}_k^1$ along their punctured lines. The two open immersions

$$\mathbb{A}_k^1 = Y_i \longrightarrow Y$$

agree on $\mathbb{A}_k^1 \setminus \{0\}$ but send the origin to the two different origins. Here $X = \mathbb{A}_k^1$ is reduced, while Y is not separated. $\square$

Exercise II.4.3**Permitted context.**

The diagonal of a separated morphism is a closed immersion. Closed subschemes of affine schemes are affine, and a fiber product of affine schemes over an affine scheme is affine.

Problem. Let X be separated over an affine scheme S , and let U, V be affine open subsets of X . Prove that $U \cap V$ is affine. Give an example showing that separatedness is necessary.

Solution. The inverse image of $U \times_S V$ under the diagonal

$$\Delta_{X/S} : X \longrightarrow X \times_S X$$

is $U \cap V$: a morphism to X factors through this inverse image exactly when it factors through both U and V . Thus there is a cartesian square

$$\begin{array}{ccc} U \cap V & \longrightarrow & U \times_S V \\ \downarrow & & \downarrow \\ X & \xrightarrow{\Delta_{X/S}} & X \times_S X. \end{array}$$

The lower map is a closed immersion. Hence the upper map is a closed immersion. Since S, U, V are affine, $U \times_S V$ is affine, and therefore so is $U \cap V$.

For a counterexample, glue two copies U, V of $\mathbb{A}_k^2$ by the identity on $\mathbb{A}_k^2 \setminus \{(0, 0)\}$. The resulting scheme X is not separated and

$$U \cap V \cong \mathbb{A}_k^2 \setminus \{(0, 0)\}.$$

This intersection is not affine. Indeed, with coordinates x, y ,

$$\Gamma(\mathbb{A}_k^2 \setminus \{0\}, \mathcal{O}) = k[x, y]_x \cap k[x, y]_y = k[x, y]$$

inside $k(x, y)$. If the punctured plane were affine, its inclusion in $\mathbb{A}_k^2 = \text{Spec } k[x, y]$ would, after identifying both rings of global functions, be an isomorphism, which is impossible because the origin is missing. $\square$

Exercise II.4.4**Permitted context.**

The graph of a morphism to a separated S -scheme is a closed immersion. Proper morphisms are stable under base change and composition. For a quasi-compact morphism into a noetherian scheme, the underlying set of its scheme-theoretic image is the closure of its set-theoretic image.

Problem. Let $f : X \rightarrow Y$ be an S -morphism of separated schemes of finite type over a noetherian scheme S . Let $Z \subseteq X$ be a closed subscheme proper over S . Prove that $f(Z)$ is closed in Y , and that its scheme-theoretic image, with the image subscheme structure, is proper over S .

Solution. Write $q : Z \rightarrow Y$ for the restriction of f . Its graph gives a factorization

$$Z \xrightarrow{\Gamma_q} Z \times_S Y \xrightarrow{\text{pr}_2} Y.$$

Because $Y \rightarrow S$ is separated, Γ_q is a closed immersion. The second map is the base change of the proper morphism $Z \rightarrow S$. Thus q is proper. In particular it is closed, so $q(Z) = f(Z)$ is closed in Y .

Let $W \hookrightarrow Y$ be the scheme-theoretic image of q . Since $q(Z)$ is already closed, the underlying space of W is exactly $q(Z)$. To see this locally, for $V = \text{Spec } A \subseteq Y$, the subscheme $W \cap V$ is defined by

$$\ker(A \rightarrow \Gamma(q^{-1}(V), \mathcal{O}_Z)),$$

and $q(q^{-1}(V))$ is dense in that affine closed subscheme. Its global image is closed, so equality follows. Hence the induced map $p : Z \rightarrow W$ is surjective.

The morphism $W \rightarrow S$ is separated and of finite type because $W \hookrightarrow Y$ is a closed immersion. It remains to prove universal closedness. Let $T \rightarrow S$ be arbitrary and let $C \subseteq W_T$ be closed. Surjectivity is preserved by base change, so $p_T : Z_T \rightarrow W_T$ is surjective. Therefore the image of C in T equals the image of the closed subset $p_T^{-1}(C) \subseteq Z_T$. Since $Z_T \rightarrow T$ is closed, this image is closed. Thus every base change of $W \rightarrow S$ is closed, and $W \rightarrow S$ is proper. $\square$

Exercise II.4.5

Permitted context.

Let X be an integral scheme of finite type over a field k , with function field K . A valuation ring R of K/k has center $x \in X$ precisely when

$$\mathcal{O}_{X,x} \subseteq R, \quad \mathfrak{m}_R \cap \mathcal{O}_{X,x} = \mathfrak{m}_x.$$

Equivalently, the generic-point morphism $\text{Spec } K \rightarrow X$ extends to $\text{Spec } R \rightarrow X$ with closed point mapping to x .

We may use the valuative criteria and the following form of Chow's lemma: a separated integral finite-type k -scheme X admits a projective, surjective, birational morphism $X' \rightarrow X$, with X' quasi-projective over k . At a specialization one may choose a valuation ring of the relevant function field dominating the local domain there.

Problem. (a) If X is separated over k , prove that a valuation of K/k has at most one center on X .

- (b) If X is proper over k , prove that every valuation of K/k has exactly one center on X .
(c) Prove the converses of (a) and (b).
(d) If X is proper over an algebraically closed field k , prove that $\Gamma(X, \mathcal{O}_X) = k$.

Solution. (a) A center is the closed-point image of an extension in the diagram

$$\begin{array}{ccc} \text{Spec } K & \longrightarrow & X \\ \downarrow & & \downarrow \\ \text{Spec } R & \longrightarrow & \text{Spec } k. \end{array}$$

The uniqueness part of the valuative criterion for separatedness gives at most one such extension, hence at most one center.

(b) Properness gives both existence and uniqueness in the valuative criterion. The displayed diagram therefore has exactly one extension, so R has exactly one center on X .

(c) First suppose every valuation ring of K/k has at most one center. We show that the image of $\Delta : X \rightarrow X \times_k X$ is closed. It is irreducible with generic point $\Delta(\eta)$, where η is the generic point of X . Let z be a specialization of $\Delta(\eta)$. The specialization lemma supplies a valuation ring $R \subseteq K$ and a morphism

$$\mathrm{Spec} R \longrightarrow X \times_k X$$

whose generic point maps to $\Delta(\eta)$ and whose closed point maps to z . The two projections give two extensions of the same generic map $\mathrm{Spec} K \rightarrow X$. Their closed-point images are centers of R , so they are equal by hypothesis. Once the center is fixed, the extension is unique: on any affine neighborhood its ring map is the inclusion into $R \subseteq K$. Thus the two projections coincide, the morphism factors through Δ , and $z \in \Delta(X)$. Therefore $\Delta(X)$ is closed. Since a diagonal is an immersion, it is a closed immersion, and X is separated.

Now assume every valuation ring of K/k has exactly one center. We already know that X is separated. Apply Chow's lemma to obtain a projective, surjective, birational morphism

$$p : X' \longrightarrow X$$

with X' integral and quasi-projective. Embed X' as an open subscheme of an integral projective scheme $\overline{X}'$. We claim that $X' = \overline{X}'$. Otherwise choose $z \in \overline{X}' \setminus X'$ and a valuation ring R of K/k dominating $\mathcal{O}_{\overline{X}', z}$. By hypothesis R has a center on X . Properness of p then lifts this center uniquely to a center on X' . But $\overline{X}'$ is separated, so R cannot have both that center and the distinct center z on $\overline{X}'$. This contradiction proves the claim. Hence X' is projective, and in particular proper over k .

It remains to descend universal closedness along the surjection p . For any k -scheme T and any closed $C \subseteq X_T$, the inverse image $p_T^{-1}(C)$ is closed in the proper T -scheme X'_T . Its image in T is closed, and, because p_T is surjective, this image equals the image of C . Thus $X_T \rightarrow T$ is closed for every T . Since X is separated and of finite type, it is proper.

(d) A section $a \in \Gamma(X, \mathcal{O}_X)$ defines a morphism

$$u : X \longrightarrow \mathbb{A}_k^1 \hookrightarrow \mathbb{P}_k^1.$$

Because X is proper and $\mathbb{P}_k^1$ is separated over k , the graph factorization shows that $u : X \rightarrow \mathbb{P}_k^1$ is proper. Its image is therefore closed. It is irreducible because X is integral, and it cannot be all of $\mathbb{P}_k^1$, since it misses infinity. Hence its image is one closed point. As k is algebraically closed, that point is k -rational, so a is a constant in k . The reverse inclusion is automatic. $\square$

Exercise II.4.6

Permitted context.

A finitely generated algebra that is integral over its base ring is finite as a module. Surjective morphisms of schemes remain surjective after base change.

Problem. Let $f : X \rightarrow Y$ be a proper morphism of affine varieties over a field k . Prove that f is finite.

Solution. Write $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$. The A -algebra B is finitely generated because f is of finite type. We prove that each $b \in B$ is integral over A .

Base-changing f along $\operatorname{Spec} A[T] \rightarrow \operatorname{Spec} A$ gives a closed map

$$\operatorname{Spec} B[T] \longrightarrow \operatorname{Spec} A[T].$$

Inside its source consider

$$Z = V(bT - 1) = \operatorname{Spec}(B[T]/(bT - 1)).$$

Its image is closed. Put

$$J = \ker(A[T] \longrightarrow B[T]/(bT - 1)).$$

The image of Z is dense in $V(J)$ by the definition of J , and it is closed, hence it is all of $V(J)$. Thus $Z \rightarrow V(J)$ is surjective.

After base change by $V(T) \hookrightarrow \operatorname{Spec} A[T]$, the source is empty because $bT - 1$ and T generate the unit ideal. Surjectivity after base change therefore shows

$$\operatorname{Spec}(A[T]/(J, T)) = \emptyset, \quad \text{so} \quad J + (T) = A[T].$$

Consequently J contains a polynomial

$$F(T) = 1 + a_1T + \cdots + a_nT^n.$$

In B_b , substituting $T = b^{-1}$ gives

$$b^n + a_1b^{n-1} + \cdots + a_n = 0.$$

The equality in B_b means that, for some $N \geq 0$,

$$b^{N+n} + a_1b^{N+n-1} + \cdots + a_nb^N = 0$$

in B . This is a monic equation for b , so b is integral over A . Therefore B is both integral and finitely generated as an A -algebra, hence finite as an A -module. Thus f is finite. $\square$

Exercise II.4.7**Permitted context.**

For a complex scheme obtained by base change from $\mathbb{R}$, complex conjugation induces a semilinear involution. We may use elementary faithfully flat descent for the finite Galois extension $\mathbb{C}/\mathbb{R}$: affine schemes, morphisms, open immersions, finite type, and separatedness descend. If a complex algebra A has a conjugate-linear involution σ , then

$$A^\sigma \otimes_{\mathbb{R}} \mathbb{C} \xrightarrow{\sim} A,$$

because every $a \in A$ has the unique decomposition

$$a = \frac{a + \sigma(a)}{2} + i \frac{a - \sigma(a)}{2i}.$$

Problem. Let X be a separated finite-type scheme over $\mathbb{C}$, equipped with a semilinear involution σ . Assume that any two points of X lie in a common affine open.

(a) Prove that X has a unique separated finite-type real form X_0 , compatible with σ .

For such real forms prove:

(b) X_0 is affine if and only if X is affine.

(c) Morphisms $X_0 \rightarrow Y_0$ correspond bijectively to complex morphisms $X \rightarrow Y$ commuting with the involutions.

(d) If $X \cong \mathbb{A}_{\mathbb{C}}^1$, then $X_0 \cong \mathbb{A}_{\mathbb{R}}^1$.

(e) If $X \cong \mathbb{P}_{\mathbb{C}}^1$, then X_0 is isomorphic either to $\mathbb{P}_{\mathbb{R}}^1$ or to the conic

$$x_0^2 + x_1^2 + x_2^2 = 0$$

in $\mathbb{P}_{\mathbb{R}}^2$.

Solution. (a) For each $x \in X$, choose an affine open W containing both x and $\sigma(x)$. The intersection

$$U = W \cap \sigma(W)$$

is affine because X is separated, and it is invariant under σ . Thus invariant affine opens form a cover of X .

Write such an open as $U = \text{Spec } A$, and put $A_0 = A^\sigma$. The isomorphism in the context identifies

$$U \cong \text{Spec } A_0 \times_{\mathbb{R}} \mathbb{C}$$

and identifies σ with conjugation on the second factor. Intersections of two invariant affine opens are again invariant and affine. The corresponding real affine schemes glue: the overlap maps are obtained by faithfully flat descent from the complex open immersions. Denote the resulting real scheme by X_0 . Its base change to $\mathbb{C}$ is X , with the prescribed involution. Finite type and separatedness descend along the faithfully flat map $\text{Spec } \mathbb{C} \rightarrow \text{Spec } \mathbb{R}$.

The same affine calculation shows uniqueness. On every invariant affine, a compatible real form must be $\text{Spec}(A^\sigma)$, and these identifications agree on overlaps. Hence any two forms are uniquely isomorphic in a way inducing the given identification after complex base change.

(b) If X_0 is affine, its base change X is affine. Conversely, if $X = \text{Spec } A$, the whole scheme is an invariant affine, so part (a) constructs the form

$$\text{Spec}(A^\sigma).$$

Uniqueness identifies this form with X_0 , proving that X_0 is affine.

(c) Base change sends a real morphism to an equivariant complex morphism. Conversely, on invariant affine opens an equivariant map is given by a semilinear-compatible ring homomorphism, which restricts to a homomorphism between invariant real subrings. These restrictions glue and give a real morphism. Faithful flatness shows that the two constructions are inverse, so the correspondence is bijective.

(d) Choose a coordinate t on X . A semilinear automorphism of $\mathbb{C}[t]$ has the form

$$\sigma(t) = at + b, \quad a \in \mathbb{C}^\times, \quad b \in \mathbb{C}.$$

The condition $\sigma^2 = 1$ says

$$\bar{a}a = 1, \quad \bar{a}b + \bar{b} = 0.$$

Choose $c \in \mathbb{C}^\times$ with $c/\bar{c} = a$. Then $\bar{c}b$ is purely imaginary, so choose $d \in \mathbb{C}$ with $d - \bar{d} = \bar{c}b$. For $u = ct + d$ one obtains $\sigma(u) = u$. Hence

$$\mathbb{C}[t]^\sigma = \mathbb{R}[u],$$

and part (b) gives $X_0 \cong \mathbb{A}_{\mathbb{R}}^1$.

(e) First suppose σ fixes a closed complex point of $\mathbb{P}_{\mathbb{C}}^1$. Move it to infinity by a projective change of coordinates. Its affine complement is invariant. The coordinate change constructed in (d) is affine-linear, so it extends across infinity to a projective coordinate u satisfying $\sigma(u) = u$. In this coordinate the involution is ordinary conjugation on all of $\mathbb{P}_{\mathbb{C}}^1$, and hence $X_0 \cong \mathbb{P}_{\mathbb{R}}^1$.

Now suppose there is no fixed closed point. Choose a closed complex point p , and choose a coordinate taking p to 0 and $\sigma(p)$ to infinity. In this coordinate the action on complex points is

$$z \mapsto \frac{c}{\bar{z}}$$

for some $c \in \mathbb{R}^\times$. It has a fixed complex point exactly when $|z|^2 = c$, so the fixed-point-free case has $c < 0$. Rescaling the coordinate reduces the action to

$$z \mapsto -\frac{1}{\bar{z}}.$$

The morphism

$$\begin{array}{ccc} \mathbb{P}_{\mathbb{C}}^1 & \longrightarrow & V(x_0^2 + x_1^2 + x_2^2) \subseteq \mathbb{P}_{\mathbb{C}}^2, \\ [s : t] & \longmapsto & [s^2 - t^2 : i(s^2 + t^2) : 2st] \end{array}$$

is an isomorphism. Complex conjugation on the conic corresponds under it to

$$[s : t] \mapsto [-\bar{t} : \bar{s}],$$

which is $z \mapsto -1/\bar{z}$. Therefore its real descent is exactly the conic $x_0^2 + x_1^2 + x_2^2 = 0$. These two cases exhaust all involutions and give the asserted classification. $\square$

Exercise II.4.8**Permitted context.**

For $X \rightarrow Y \rightarrow Z$, the graph $X \rightarrow X \times_Z Y$ is a base change of $\Delta_{Y/Z}$, and is a closed immersion if $Y \rightarrow Z$ is separated.

Problem. Let $\mathcal{P}$ be a property such that:

- (a) closed immersions have $\mathcal{P}$;
- (b) composites of morphisms having $\mathcal{P}$ have $\mathcal{P}$;
- (c) base changes of morphisms having $\mathcal{P}$ have $\mathcal{P}$.

Under these assumptions, prove:

- (d) a product of morphisms having $\mathcal{P}$ has $\mathcal{P}$;
- (e) if $f : X \rightarrow Y$, $g : Y \rightarrow Z$, $g \circ f$ has $\mathcal{P}$, and g is separated, then f has $\mathcal{P}$;
- (f) if $f : X \rightarrow Y$ has $\mathcal{P}$, then the induced morphism $f_{\text{red}} : X_{\text{red}} \rightarrow Y_{\text{red}}$ has $\mathcal{P}$.

Solution. (d) Let $f : X \rightarrow Y$ and $f' : X' \rightarrow Y'$ be S -morphisms having $\mathcal{P}$. Factor their product as

$$X \times_S X' \longrightarrow Y \times_S X' \longrightarrow Y \times_S Y'.$$

The first arrow is the base change of f , and the second is the base change of f' . They have $\mathcal{P}$ by (c), so their composite has $\mathcal{P}$ by (b).

(e) Factor f as

$$X \xrightarrow{\gamma} X \times_Z Y \xrightarrow{\text{pr}_2} Y, \quad \gamma(x) = (x, f(x)).$$

The second map is the base change along g of $g \circ f : X \rightarrow Z$, and hence has $\mathcal{P}$. The first map is the base change of the diagonal $\Delta_g : Y \rightarrow Y \times_Z Y$; since g is separated, it is a closed immersion and has $\mathcal{P}$. Their composite f therefore has $\mathcal{P}$.

(f) The composite

$$X_{\text{red}} \hookrightarrow X \xrightarrow{f} Y$$

has $\mathcal{P}$ by (a) and (b). Because X_{red} is reduced, this composite factors uniquely as

$$X_{\text{red}} \xrightarrow{f_{\text{red}}} Y_{\text{red}} \hookrightarrow Y.$$

The second arrow is a closed immersion, hence separated. Apply (e) to this factorization: the composite has $\mathcal{P}$, so f_{red} has $\mathcal{P}$. $\square$

Exercise II.4.9**Permitted context.**

A morphism $X \rightarrow Y$ is projective if it factors as a closed immersion into $\mathbb{P}_Y^n$ followed by the projection. Projective morphisms are stable under base change. The Segre morphism is a closed immersion

$$\mathbb{P}_Z^m \times_Z \mathbb{P}_Z^n \hookrightarrow \mathbb{P}_Z^{(m+1)(n+1)-1}.$$

Problem. Prove that a composite of projective morphisms is projective. Deduce that projective morphisms satisfy properties (a)–(f) of Exercise II.4.8.

Solution. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be projective. Choose closed immersions

$$i : X \hookrightarrow \mathbb{P}_Y^m, \quad j : Y \hookrightarrow \mathbb{P}_Z^n.$$

Base-changing j along $\mathbb{P}_Z^m \rightarrow Z$ gives a closed immersion

$$\mathbb{P}_Y^m \hookrightarrow \mathbb{P}_Z^m \times_Z \mathbb{P}_Z^n.$$

Composing this with i , and then with the Segre embedding, gives a closed immersion

$$X \hookrightarrow \mathbb{P}_Z^m \times_Z \mathbb{P}_Z^n \hookrightarrow \mathbb{P}_Z^{(m+1)(n+1)-1}.$$

Its projection to Z is $g \circ f$, so $g \circ f$ is projective.

Now take $\mathcal{P}$ to mean projective. A closed immersion is projective by using $\mathbb{P}_Y^0 = Y$, so property (a) holds. The result just proved is property (b), and projectivity is stable under base change, giving (c). Exercise II.4.8 then supplies properties (d), (e), and (f). $\square$

Exercise II.4.10

Permitted context.

A finite-type scheme over a noetherian scheme is noetherian. A locally closed subscheme of a projective S -scheme is quasi-projective over S . A quasi-compact morphism has a scheme-theoretic image that can be computed locally on the target. Finite products of projective S -schemes are projective.

Problem. Let X be proper over a noetherian scheme S . Prove Chow's lemma: there are an S -projective scheme X' , a morphism $g : X' \rightarrow X$, and an open dense subset $U \subseteq X$ such that

$$g^{-1}(U) \longrightarrow U$$

is an isomorphism. Proceed by:

- (a) reducing to irreducible X ;
- (b) finding a finite cover $X = \bigcup U_i$ by S -quasi-projective opens, with open immersions $U_i \hookrightarrow P_i$ into projective S -schemes;
- (c) for $U = \bigcap U_i$, taking the scheme-theoretic closure X' of the map

$$U \longrightarrow X \times_S P_1 \times_S \cdots \times_S P_n$$

and proving that the projection $h : X' \rightarrow \prod_i P_i$ is a closed immersion;

- (d) proving $g^{-1}(U) \rightarrow U$ is an isomorphism.

Solution. (a) Let $Z_1, \dots, Z_r$ be the irreducible components of X . Give

$$V_i = X \setminus \bigcup_{j \neq i} Z_j$$

its original open subscheme structure, and let X_i be its scheme-theoretic closure in X . The underlying space of X_i is Z_i , and $X_i|_{V_i} = V_i$, including its nilpotents. Each X_i is irreducible and proper over S . Apply the irreducible case to obtain projective X'_i mapping isomorphically over a dense open $W_i \subseteq X_i$. Replacing W_i by $W_i \cap V_i$ makes it open in X . The disjoint union of these opens

is dense in X , and $X' = \coprod_i X'_i$, projective over S , maps isomorphically over their union. Thus the reduction preserves the required open subscheme even when X is generically nonreduced.

(b) The schemes S and X are noetherian and hence quasi-compact. Cover S by finitely many affine opens and refine a finite affine cover of X so that each member $U_i = \operatorname{Spec} B_i$ maps into some $\operatorname{Spec} A_i \subseteq S$. Since B_i is a finitely generated A_i -algebra, there is a closed immersion followed by an open immersion

$$U_i \hookrightarrow \mathbb{A}_{A_i}^{n_i} \hookrightarrow \mathbb{P}_{A_i}^{n_i}.$$

The last scheme is an open subscheme of $\mathbb{P}_S^{n_i}$. Hence U_i is locally closed in $\mathbb{P}_S^{n_i}$, and thus is open in its scheme-theoretic closure P_i . The scheme P_i is closed in $\mathbb{P}_S^{n_i}$, so it is projective over S . This gives the required finite quasi-projective cover.

(c) Assume X irreducible. Every nonempty U_i is dense, so

$$U = \bigcap_{i=1}^n U_i$$

is open dense. Put $P = \prod_{i=1}^n P_i$, let

$$j : U \longrightarrow X \times_S P$$

be the product of the inclusion $U \hookrightarrow X$ and the maps $U \hookrightarrow U_i \hookrightarrow P_i$, and let $X' \hookrightarrow X \times_S P$ be its scheme-theoretic image. Write $g : X' \rightarrow X$ and $h : X' \rightarrow P$ for the projections.

For each i , set

$$V_i = U_i \times_S \prod_{j \neq i} P_j \subseteq P.$$

We claim

$$h^{-1}(V_i) = g^{-1}(U_i)$$

and that h restricts there to a closed immersion into V_i . Indeed, over $U_i \times_S P$, the image of U lies in the graph of $U_i \hookrightarrow P_i$, which is closed because $P_i \rightarrow S$ is separated. Hence the scheme-theoretic closure is contained in that graph and projects as a closed subscheme of V_i . Conversely, over $X \times_S V_i$, the image lies in the graph of the open immersion $U_i \hookrightarrow X$. This graph is closed because $X \rightarrow S$ is separated. Thus a point of X' whose P_i -coordinate lies in U_i has its X -coordinate equal to it, proving the claim.

Since the U_i cover X , the image $h(X')$ is contained in $V = \bigcup_i V_i$, and $h : X' \rightarrow V$ is a closed immersion, as this is local on the target. On the other hand,

$$X' \hookrightarrow X \times_S P \longrightarrow P$$

is proper: the first map is closed and the second is the base change of the proper morphism $X \rightarrow S$. Hence $h(X')$ is closed in all of P . The two opens V and $P \setminus h(X')$ cover P ; on the first, h is a closed immersion, and on the second its source is empty. Therefore $h : X' \rightarrow P$ is a closed immersion. Since P is projective over S , X' is projective over S .

(d) Restrict the construction to $U \times_S P$. There the original map j is the graph of the morphism $U \rightarrow P$. Since $P \rightarrow S$ is separated, this graph is already closed. Its scheme-theoretic closure is therefore itself, and consequently

$$g^{-1}(U) \xrightarrow{\sim} U.$$

Together with (a)–(c), this proves Chow's lemma. □

Exercise II.4.11**Permitted context.**

We use the Krull–Akizuki theorem: the integral closure of a one-dimensional noetherian domain in a finite extension of its fraction field is noetherian of dimension one. An integrally closed one-dimensional noetherian local domain is a discrete valuation ring.

As customary in the valuative criterion, a field may be treated as the degenerate rank-zero valuation ring. If the term “discrete valuation ring” is required to have a nonzero maximal ideal, part (a) needs the minor qualification that the original local ring is not a field, except when the extension has positive transcendence degree. We also use the finite-type test for universal closedness: a finite-type morphism $T \rightarrow Y$ is universally closed if all $\mathbb{A}_Y^n \times_Y T \rightarrow \mathbb{A}_Y^n$ are closed. This test incorporates arbitrary base changes; it follows by finite-presentation approximation (Stacks Project, Lemma 32.14.2).

Problem. (a) Let $(\mathcal{O}, \mathfrak{m})$ be a noetherian local domain with fraction field K , and let L/K be a finitely generated field extension. Prove that there is a discrete valuation ring R of L dominating $\mathcal{O}$. Reduce first to L/K finite; then use a suitable generator x_1 of $\mathfrak{m}$, the ring

$$\mathcal{O}' = \mathcal{O} \left[\frac{x_2}{x_1}, \dots, \frac{x_n}{x_1} \right],$$

a prime minimal over (x_1) , and Krull–Akizuki.

(b) Let $f : X \rightarrow Y$ be a finite-type morphism of noetherian schemes. Prove that f is separated, respectively proper, if and only if the corresponding valuative criterion holds for all discrete valuation rings.

Solution. (a) Choose a transcendence basis $t_1, \dots, t_r$ for L/K . Then

$$\mathcal{O}_1 = \mathcal{O}[t_1, \dots, t_r]_{(\mathfrak{m}, t_1, \dots, t_r)}$$

is a noetherian local domain with fraction field $K(t_1, \dots, t_r)$, and it dominates $\mathcal{O}$. Since L is finite over this fraction field, replacing $\mathcal{O}$ by $\mathcal{O}_1$ reduces the problem to the case where L/K is finite.

Assume $\mathfrak{m} \neq 0$, and choose generators $\mathfrak{m} = (x_1, \dots, x_n)$. After renumbering, we may arrange that

$$x_1^{N-1} \notin \mathfrak{m}^N \quad \text{for every } N \geq 1.$$

Indeed, otherwise each x_i would satisfy the reverse inclusion for all sufficiently large N . Choosing one common N , a pigeonhole argument on monomials would give $\mathfrak{m}^d = \mathfrak{m}^{d+1}$ for some d ; Nakayama would imply $\mathfrak{m}^d = 0$, impossible in a domain with $\mathfrak{m} \neq 0$.

Form

$$\mathcal{O}' = \mathcal{O} \left[\frac{x_2}{x_1}, \dots, \frac{x_n}{x_1} \right] \subseteq K.$$

The ideal $x_1\mathcal{O}'$ is proper. Otherwise an expression for x_1^{-1} as a polynomial in the x_i/x_1 , after clearing denominators, would put x_1^{N-1} in $\mathfrak{m}^N$ for some N . Moreover,

$$\mathfrak{m} \subseteq x_1\mathcal{O}' \cap \mathcal{O}.$$

The contraction on the right is proper and $\mathfrak{m}$ is maximal, so equality holds.

Let $\mathfrak{p}$ be a prime minimal over $x_1\mathcal{O}'$. The noetherian ring $\mathcal{O}'$ is a domain, and the principal ideal theorem gives $\text{ht } \mathfrak{p} = 1$. Also $\mathfrak{p} \cap \mathcal{O} = \mathfrak{m}$. Thus

$$A = \mathcal{O}'_{\mathfrak{p}}$$

is a one-dimensional noetherian local domain dominating $\mathcal{O}$.

Let $\tilde{A}$ be the integral closure of A in L . Krull–Akizuki says that $\tilde{A}$ is a noetherian one-dimensional domain. Choose a maximal ideal $\mathfrak{n} \subseteq \tilde{A}$. By integrality its contraction is the maximal ideal of A . Therefore

$$R = \tilde{A}_{\mathfrak{n}}$$

dominates A , hence $\mathcal{O}$, and is an integrally closed one-dimensional noetherian local domain with fraction field L . It is therefore a discrete valuation ring.

If $\mathfrak{m} = 0$, the stated rank-zero convention handles the finite algebraic case. When L/K has positive transcendence degree, one may instead begin with $K[t_1, \dots, t_r]_{(t_1)}$ and apply the finite-extension argument, obtaining a nontrivial discrete valuation ring.

(b) Necessity follows immediately from the usual valuative criteria. For sufficiency, recall the specialization step in their proofs. If q lies in the closure of the image of a finite-type morphism $T \rightarrow Y$, with Y noetherian, choose a generic point ξ of an irreducible component of T whose image specializes to q . The image of

$$\mathcal{O}_{Y,q} \longrightarrow k(\xi)$$

is a noetherian local domain, and $k(\xi)$ is finitely generated over its fraction field. Part (a) therefore supplies a discrete valuation ring in $k(\xi)$ dominating that local domain. Its generic point maps to ξ and its closed point maps to q .

For separatedness, apply this argument to the diagonal into $X \times_Y X$. A point in its closure is reached by a DVR whose generic point lies in the diagonal. The two induced maps into X agree generically, so the uniqueness hypothesis makes them equal. Thus the diagonal has closed image, and a diagonal with closed image is a closed immersion.

For properness, fix $n \geq 0$ and put $T = \mathbb{A}_Y^n \times_Y X$. For every closed subscheme $Z \subseteq T$, the preceding specialization argument applies to $Z \rightarrow \mathbb{A}_Y^n$, whose target is noetherian. The DVR lifting hypothesis for f , together with the given map to $\mathbb{A}_Y^n$, gives a map to T . Its generic point lies in Z , so its whole image lies in Z : the defining ideal pulls back to zero in the domain R . Consequently the image of Z is closed. All these affine-space base changes are therefore closed. The universal-closedness test in the context now handles arbitrary, possibly nonnoetherian base changes. Combined with finite type and separatedness, it proves properness. $\square$

Exercise II.4.12**Permitted context.**

Let k be algebraically closed. A nontrivial valuation ring of a one-variable function field has a unique center on its nonsingular projective curve. On a nonsingular surface, the local ring at the generic point of a curve is a discrete valuation ring.

In part (b)(2), the local ring at the generic point of Y' is a discrete valuation ring provided X' is normal there, as is the case when X' is nonsingular or is obtained by blowing up nonsingular points. Without this normality condition, one must pass to the normalization; the unqualified statement is otherwise false.

Problem. Let k be algebraically closed.

(a) If K/k is finitely generated of transcendence degree one, prove that every valuation ring of K/k , other than K , is discrete and that these rings are the local rings of the abstract nonsingular projective curve C_K .

(b) Let K/k have transcendence degree two and let X be a complete nonsingular surface with function field K .

(1) For an irreducible curve $Y \subseteq X$ with generic point x_1 , show that $\mathcal{O}_{X,x_1}$ is a discrete valuation ring centered at x_1 .

(2) If $f : X' \rightarrow X$ is birational and an irreducible curve $Y' \subseteq X'$ is contracted to a closed point $x_0 \in X$, show that the local ring at the generic point of Y' gives a discrete valuation centered at x_0 , under the normality qualification in the context.

(3) Starting at a closed point x_0 , repeatedly blow up x_i and choose a closed point x_{i+1} on the new exceptional curve. Put

$$R_0 = \bigcup_{i \geq 0} \mathcal{O}_{X_i, x_i}.$$

Show that R_0 is local, that a valuation ring R of K dominating R_0 is a valuation ring of K/k centered at x_0 , and explain when such an R is discrete.

Solution. (a) Let C_K be the nonsingular projective curve with function field K , and let $R \neq K$ be a valuation ring of K/k . Properness gives a unique center $x \in C_K$. It cannot be the generic point, because then R would contain K and equal K . Thus x is closed. Since C_K is nonsingular, $\mathcal{O}_{C_K, x}$ is a discrete valuation ring, and R dominates it. An overring of a valuation ring inside its fraction field is a localization at a prime. A discrete valuation ring has only the primes (0) and its maximal ideal; domination rules out localization at (0) . Hence

$$R = \mathcal{O}_{C_K, x}.$$

Conversely each such local ring is a valuation ring of K/k , so the nontrivial valuation rings are precisely the closed points of C_K .

(b) We consider the three kinds of surface valuations in turn.

(1) The point x_1 has codimension one on the regular surface X . Hence $\mathcal{O}_{X, x_1}$ is a one-dimensional regular local domain and therefore a discrete valuation ring with fraction field K . Its center on X is x_1 by construction.

(2) Let η' be the generic point of Y' . Birationality identifies the function fields of X' and X with K . Under the stated normality hypothesis, $\mathcal{O}_{X', \eta'}$ is an integrally closed one-dimensional noetherian local domain, hence a discrete valuation ring of K/k . The local homomorphism

$$\mathcal{O}_{X, x_0} \longrightarrow \mathcal{O}_{X', \eta'}$$

sends $\mathfrak{m}_{x_0}$ into the maximal ideal, because the residue map factors through the constant map $Y' \rightarrow x_0$; its contraction is exactly $\mathfrak{m}_{x_0}$. Thus the valuation ring dominates $\mathcal{O}_{X,x_0}$, and its center on X is x_0 .

(3) The blowup map induces an inclusion

$$\mathcal{O}_{X_i, x_i} \subseteq \mathcal{O}_{X_{i+1}, x_{i+1}},$$

and the latter local ring dominates the former. Consequently

$$\mathfrak{m}_0 = \bigcup_{i \geq 0} \mathfrak{m}_{X_i, x_i}$$

is an ideal of R_0 . Every element outside $\mathfrak{m}_0$ is a unit in some, hence every later, local ring. Thus R_0 is local with maximal ideal $\mathfrak{m}_0$, and its fraction field is K .

If a valuation ring $R \subseteq K$ dominates R_0 , it dominates each $\mathcal{O}_{X_i, x_i}$. It contains k , and no nonzero element of k lies in its maximal ideal, so its valuation is trivial on k . It is therefore a valuation ring of K/k , and already its domination of $\mathcal{O}_{X, x_0}$ says that its center on X is x_0 .

Intrinsically, R is discrete exactly when its value group has a least positive element and is generated by that element; equivalently, R is noetherian. A principal maximal ideal alone does not suffice: the lexicographic valuation of $k(x, y)$ with $v(x) = (1, 0)$, $v(y) = (0, 1)$ has maximal ideal (y) but value group $\mathbb{Z}^2$. It dominates all the blowup local rings with parameters $y, x/y^i$, so it occurs in this construction. There is also a useful geometric formulation for this construction. A discrete R , after normalizing its value, has completion

$$\widehat{R} \cong \kappa(R)[[t]].$$

The inclusions of the local rings then give a dominant formal arc

$$\mathrm{Spec} \kappa(R)[[t]] \longrightarrow X$$

whose successive closed-point images lie over the x_i . Conversely, any such dominant formal arc, meaning that it induces an injection $K \hookrightarrow \kappa(R)((t))$, defines a discrete valuation ring

$$\{a \in K : \mathrm{ord}_t(a) \geq 0\}.$$

It dominates R_0 exactly when its successive centers are the chosen points. In the important residue-field- k case this is simply a dominant arc $\mathrm{Spec} k[[t]] \rightarrow X$.

This condition is genuinely restrictive. For example, on an affine chart with parameters x, y , a choice

$$x \longmapsto t, \quad y \longmapsto \phi(t),$$

where $\phi(t) \in tk[[t]]$ is transcendental over $k(t)$, gives an injective map $k(x, y) \hookrightarrow k((t))$ and hence a discrete example. In contrast, a monomial valuation with

$$v(x) = 1, \quad v(y) = \alpha$$

for irrational $\alpha > 0$ has value group $\mathbb{Z} + \alpha\mathbb{Z}$, which has no least positive element. Its successive centers give an infinite sequence of closed infinitely near points, but its valuation ring is not discrete. $\square$

II.5 Sheaves of Modules

Exercise II.5.1

Permitted context.

Let $(X, \mathcal{O}_X)$ be a ringed space. For an $\mathcal{O}_X$ -module $\mathcal{E}$, write

$$\mathcal{E}^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{O}_X).$$

A morphism of sheaves is an isomorphism if it is so on an open cover. On an open set where a finite locally free sheaf has basis $e_1, \dots, e_r$, its dual has the dual basis $e_1^\vee, \dots, e_r^\vee$.

Problem. Let $\mathcal{E}$ be finite locally free.

- (a) Prove that the evaluation map $\mathcal{E} \rightarrow (\mathcal{E}^\vee)^\vee$ is an isomorphism.
- (b) For every $\mathcal{O}_X$ -module $\mathcal{F}$, prove

$$\mathcal{H}om(\mathcal{E}, \mathcal{F}) \cong \mathcal{E}^\vee \otimes \mathcal{F}.$$

- (c) For all $\mathcal{F}, \mathcal{G}$, prove the natural adjunction

$$\mathcal{H}om(\mathcal{E} \otimes \mathcal{F}, \mathcal{G}) \cong \mathcal{H}om(\mathcal{F}, \mathcal{H}om(\mathcal{E}, \mathcal{G})).$$

- (d) If $f : X \rightarrow Y$ is a morphism of ringed spaces, $\mathcal{F}$ is an $\mathcal{O}_X$ -module, and $\mathcal{E}$ is finite locally free on Y , prove the projection formula

$$f_*(\mathcal{F} \otimes f^*\mathcal{E}) \cong f_*\mathcal{F} \otimes \mathcal{E}.$$

Solution. (a) Define the evaluation morphism by

$$e \mapsto (\lambda \mapsto \lambda(e)).$$

On an open set where $\mathcal{E} \cong \mathcal{O}_X^r$, it sends the basis e_i to the basis dual to $e_i^\vee$, and is therefore an isomorphism. Such opens cover X , so it is an isomorphism globally.

- (b) There is a natural morphism

$$\mathcal{E}^\vee \otimes \mathcal{F} \longrightarrow \mathcal{H}om(\mathcal{E}, \mathcal{F}), \quad \lambda \otimes s \mapsto (e \mapsto \lambda(e)s).$$

Where $\mathcal{E} \cong \mathcal{O}_X^r$, both sides identify with $\mathcal{F}^r$, and this morphism is the identity under those identifications. It is consequently a global isomorphism.

- (c) Given $u : \mathcal{E} \otimes \mathcal{F} \rightarrow \mathcal{G}$, send a local section s of $\mathcal{F}$ to the homomorphism

$$e \mapsto u(e \otimes s).$$

This gives a morphism $\mathcal{F} \rightarrow \mathcal{H}om(\mathcal{E}, \mathcal{G})$. Conversely, evaluate such a morphism on $\mathcal{E}$ to obtain $\mathcal{E} \otimes \mathcal{F} \rightarrow \mathcal{G}$. The two constructions are inverse on local sections and are compatible with restriction; they therefore give the stated natural bijection.

- (d) The adjunction map $f^{-1}\mathcal{E} \rightarrow f^{-1}f_*f^*\mathcal{E}$ and multiplication define a natural morphism

$$f_*\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{E} \longrightarrow f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{E}).$$

Restrict to an open $V \subseteq Y$ on which $\mathcal{E}|_V \cong \mathcal{O}_V^r$. Then

$$f^*\mathcal{E}|_{f^{-1}(V)} \cong \mathcal{O}_{f^{-1}(V)}^r,$$

and both sides of the displayed morphism restrict to $(f_*\mathcal{F}|_V)^r$. The map is the evident identity there. These V cover Y , so the projection morphism is an isomorphism. $\square$

Exercise II.5.2**Permitted context.**

If R is a discrete valuation ring with fraction field K , then $\operatorname{Spec} R = \{\eta, s\}$ has precisely the open subsets

$$\emptyset, \quad \{\eta\}, \quad X.$$

Moreover, $\mathcal{O}_X(X) = R$ and $\mathcal{O}_X(\{\eta\}) = K$.

Problem. Let $X = \operatorname{Spec} R$.

(a) Prove that an $\mathcal{O}_X$ -module is equivalent to the data of an R -module M , a K -vector space L , and a K -linear map

$$\rho : M \otimes_R K \longrightarrow L.$$

(b) Prove that the corresponding sheaf is quasi-coherent if and only if ρ is an isomorphism.

Solution. (a) For an $\mathcal{O}_X$ -module $\mathcal{F}$, put

$$M = \mathcal{F}(X), \quad L = \mathcal{F}(\{\eta\}).$$

Then M is an R -module, L is a K -vector space, and the restriction $M \rightarrow L$ is R -linear. It therefore extends uniquely to a K -linear map

$$\rho : M \otimes_R K \rightarrow L.$$

There are no other nontrivial restriction maps and no nontrivial open covers to check. Conversely, the given data define the values and the single restriction map of an $\mathcal{O}_X$ -module. These constructions are mutually inverse, also on morphisms.

(b) The quasi-coherent sheaf associated to M has

$$\widetilde{M}(X) = M, \quad \widetilde{M}(\{\eta\}) = M \otimes_R K.$$

There is a natural morphism $\widetilde{M} \rightarrow \mathcal{F}$, equal to the identity on X and to ρ on $\{\eta\}$. Hence it is an isomorphism exactly when ρ is. Since every quasi-coherent sheaf on the affine scheme X is associated to its module of global sections, this proves both directions. $\square$

Exercise II.5.3**Permitted context.**

For $X = \operatorname{Spec} A$ and an A -module M ,

$$\widetilde{M}(D(f)) = M_f.$$

The distinguished opens form a basis, and $\mathcal{F}(D(f))$ is naturally an A_f -module for every $\mathcal{O}_X$ -module $\mathcal{F}$.

Problem. Prove that for every A -module M and every $\mathcal{O}_X$ -module $\mathcal{F}$ there is a natural isomorphism

$$\operatorname{Hom}_A(M, \Gamma(X, \mathcal{F})) \cong \operatorname{Hom}_{\mathcal{O}_X}(\widetilde{M}, \mathcal{F}).$$

Solution. Taking global sections sends a sheaf morphism $\widetilde{M} \rightarrow \mathcal{F}$ to an A -linear map

$$M = \Gamma(X, \widetilde{M}) \longrightarrow \Gamma(X, \mathcal{F}).$$

Conversely, let $u : M \rightarrow \Gamma(X, \mathcal{F})$ be A -linear. For each $f \in A$, define

$$u_f : M_f \longrightarrow \mathcal{F}(D(f)), \quad \frac{m}{f^n} \longmapsto \frac{u(m)|_{D(f)}}{f^n}.$$

This is well-defined because the target is an A_f -module. The maps u_f commute with restriction to smaller distinguished opens. They therefore define a unique morphism $\widetilde{M} \rightarrow \mathcal{F}$, since the $D(f)$ form a basis.

The resulting sheaf morphism has global map u . Conversely, an $\mathcal{O}_X$ -linear sheaf morphism is forced on every $D(f)$ to be the localization of its map on X , so the first construction followed by the second returns it. The two assignments are inverse and are natural in both M and $\mathcal{F}$. $\square$

Exercise II.5.4

Permitted context.

On $U = \text{Spec } A$, the functor $M \mapsto \widetilde{M}$ is exact and commutes with arbitrary direct sums. Every module has a presentation by free modules. If A is noetherian, every finite A -module is finitely presented.

Problem. Prove that an $\mathcal{O}_X$ -module $\mathcal{F}$ is quasi-coherent if and only if locally it is the cokernel of a morphism of free sheaves. If X is noetherian, prove that $\mathcal{F}$ is coherent if and only if locally it is the cokernel of a morphism between free sheaves of finite rank.

Solution. Suppose first that $\mathcal{F}$ is quasi-coherent. Near any point choose an affine open $U = \text{Spec } A$ with $\mathcal{F}|_U \cong \widetilde{M}$. Choose a module presentation

$$A^{(I)} \longrightarrow A^{(J)} \longrightarrow M \longrightarrow 0.$$

Exactness and compatibility with direct sums give

$$\mathcal{O}_U^{(I)} \longrightarrow \mathcal{O}_U^{(J)} \longrightarrow \mathcal{F}|_U \longrightarrow 0.$$

Thus $\mathcal{F}$ has the required local presentation.

Conversely, free sheaves are quasi-coherent, and on an affine open the cokernel of

$$\mathcal{O}_U^{(I)} \longrightarrow \mathcal{O}_U^{(J)}$$

is the sheaf associated to the cokernel of $A^{(I)} \rightarrow A^{(J)}$. It is therefore quasi-coherent. Since quasi-coherence is local, $\mathcal{F}$ is quasi-coherent.

Now assume X noetherian. If $\mathcal{F}$ is coherent, then on an affine open it is associated to a finite module M . The noetherian property gives a finite presentation

$$A^m \longrightarrow A^n \longrightarrow M \longrightarrow 0,$$

and hence a presentation by finite-rank free sheaves. Conversely, the cokernel of a morphism between finite-rank free sheaves is locally associated to a finitely presented, hence finite, module. It is coherent. $\square$

Exercise II.5.5**Permitted context.**

For an affine morphism $f : \operatorname{Spec} B \rightarrow \operatorname{Spec} A$ and a B -module M , the direct image of $\widetilde{M}$ is the quasi-coherent $\mathcal{O}_{\operatorname{Spec} A}$ -module associated to M , viewed as an A -module. A module finite over a finite A -algebra is finite over A .

Problem. Let $f : X \rightarrow Y$ be a morphism of schemes.

- (a) Give an example in which $\mathcal{F}$ is coherent on X but $f_*\mathcal{F}$ is not coherent on Y , even though X, Y are varieties over a field.
- (b) Prove that every closed immersion is finite.
- (c) If f is finite between noetherian schemes and $\mathcal{F}$ is coherent on X , prove that $f_*\mathcal{F}$ is coherent on Y .

Solution. (a) Take the structure morphism

$$f : \mathbb{A}_k^1 \longrightarrow \operatorname{Spec} k$$

and $\mathcal{F} = \mathcal{O}_{\mathbb{A}_k^1}$. This sheaf is coherent, but

$$f_*\mathcal{F} = \Gamma(\mathbb{A}_k^1, \mathcal{O}) = k[t]$$

as a sheaf on the one-point scheme. It is not a finite k -module, so it is not coherent.

- (b) Let $i : Z \hookrightarrow Y$ be a closed immersion. For every affine $V = \operatorname{Spec} A \subseteq Y$, its inverse image is

$$i^{-1}(V) = \operatorname{Spec}(A/I)$$

for some ideal I . The A -module A/I is generated by 1, so the restriction of i over V is finite. Hence i is finite.

- (c) Work over $V = \operatorname{Spec} A \subseteq Y$. Because f is finite,

$$f^{-1}(V) = \operatorname{Spec} B$$

with B a finite A -algebra. The restriction of $\mathcal{F}$ is $\widetilde{M}$ for a finite B -module M . For every $a \in A$,

$$(f_*\mathcal{F})(D(a)) = \mathcal{F}(\operatorname{Spec} B_a) = M_a,$$

so $f_*\mathcal{F}|_V \cong \widetilde{M}$, with M regarded as an A -module. It is finite over A , because it is finite over the finite A -module B . Thus the direct image is coherent on each such V , and hence on Y . $\square$

Exercise II.5.6**Permitted context.**

For $X = \operatorname{Spec} A$, $\mathcal{F} = \widetilde{M}$, and $\mathfrak{p} \in X$, the stalk is $M_{\mathfrak{p}}$. If $\mathfrak{a} \subseteq A$, write

$$\Gamma_{\mathfrak{a}}(M) = \{m \in M : \mathfrak{a}^n m = 0 \text{ for some } n > 0\}.$$

For a closed set Z , the sheaf $\mathcal{H}_Z^0(\mathcal{F})$ consists of local sections whose support is contained in Z .

Problem. (a) For $m \in M$, prove

$$\operatorname{supp}(m) = V(\operatorname{Ann}(m)).$$

(b) If A is noetherian and M is finite, prove

$$\operatorname{Supp}(\widetilde{M}) = V(\operatorname{Ann}(M)).$$

(c) Deduce that the support of a coherent sheaf on a noetherian scheme is closed.

(d) If A is noetherian, $Z = V(\mathfrak{a})$, and $\mathcal{F} = \widetilde{M}$, prove

$$\widetilde{\Gamma_{\mathfrak{a}}(M)} \cong \mathcal{H}_Z^0(\mathcal{F}).$$

(e) On a noetherian scheme, prove that $\mathcal{H}_Z^0(\mathcal{F})$ is quasi-coherent when $\mathcal{F}$ is quasi-coherent, and coherent when $\mathcal{F}$ is coherent.

Solution. (a) The germ $m_{\mathfrak{p}}$ is zero exactly when some $s \notin \mathfrak{p}$ annihilates m . This is equivalent to $\operatorname{Ann}(m) \not\subseteq \mathfrak{p}$. Therefore the primes at which the germ is nonzero are precisely the primes containing $\operatorname{Ann}(m)$, which proves the formula.

(b) Let $m_1, \dots, m_r$ generate M . Then

$$M_{\mathfrak{p}} = 0 \iff (m_i)_{\mathfrak{p}} = 0 \text{ for every } i \iff \mathfrak{p} \notin \bigcup_i V(\operatorname{Ann}(m_i)).$$

Since

$$\operatorname{Ann}(M) = \bigcap_i \operatorname{Ann}(m_i),$$

the union on the right is $V(\operatorname{Ann}(M))$. Taking complements yields the claim.

(c) On every affine open $U = \operatorname{Spec} A$, a coherent sheaf is $\widetilde{M}$ for a finite module M , so part (b) shows that its support meets U in a closed subset of U . Closedness is local on the ambient space, hence the global support is closed.

(d) A section $m \in M$ has support contained in $V(\mathfrak{a})$ if and only if

$$V(\operatorname{Ann}(m)) \subseteq V(\mathfrak{a}),$$

or equivalently $\mathfrak{a} \subseteq \sqrt{\operatorname{Ann}(m)}$. Since $\mathfrak{a}$ is finitely generated, this holds exactly when $\mathfrak{a}^n m = 0$ for some n . Thus the two sides have the same global sections.

The identification is compatible with localization. More precisely, for every $f \in A$,

$$\Gamma_{\mathfrak{a}}(M)_f = \Gamma_{\mathfrak{a}_f}(M_f).$$

One inclusion is immediate. For the other, use finite generation of a power of $\mathfrak{a}$ to clear a common power of f from the annihilation relations. Applying the global calculation over $\operatorname{Spec} A_f$ now identifies the two sheaves on every distinguished open, proving the isomorphism.

(e) Locally write $Z = V(\mathfrak{a})$ and $\mathcal{F} = \widetilde{M}$. Part (d) expresses the subsheaf with support as the sheaf associated to the module $\Gamma_{\mathfrak{a}}(M)$, so it is quasi-coherent. If $\mathcal{F}$ is coherent, then M is finite. Because A is noetherian, its submodule $\Gamma_{\mathfrak{a}}(M)$ is finite, and the resulting sheaf is coherent. $\square$

Exercise II.5.7**Permitted context.**

For a finite module over a noetherian ring, kernels and cokernels of maps between finite modules are finite, and their supports are closed. Nakayama's lemma says that lifts of a basis of $M/\mathfrak{m}M$ generate a finite module M over a local ring.

Problem. Let X be noetherian and $\mathcal{F}$ coherent.

- (a) If $\mathcal{F}_x$ is a free $\mathcal{O}_{X,x}$ -module, prove that $\mathcal{F}$ is free on a neighborhood of x .
- (b) Prove that $\mathcal{F}$ is locally free if and only if all its stalks are free.
- (c) Prove that $\mathcal{F}$ is invertible if and only if there is a coherent sheaf $\mathcal{G}$ such that

$$\mathcal{F} \otimes \mathcal{G} \cong \mathcal{O}_X.$$

Solution. (a) Work on $U = \operatorname{Spec} A$, write $\mathcal{F}|_U = \widetilde{M}$, and let x correspond to $\mathfrak{p}$. Choose elements $m_1, \dots, m_r \in M$ whose images form a basis of $M_{\mathfrak{p}}$, after multiplying basis elements by denominators that are units in $A_{\mathfrak{p}}$. The map

$$u : A^r \longrightarrow M$$

is an isomorphism after localization at $\mathfrak{p}$. Its kernel and cokernel are finite and have supports not containing $\mathfrak{p}$. Some distinguished neighborhood $D(f)$ of $\mathfrak{p}$ avoids both supports. Hence u_f is an isomorphism, so $\mathcal{F}|_{D(f)}$ is free.

(b) One direction follows by taking stalks. Conversely, if every stalk is free, part (a) supplies a free neighborhood of every point. Thus $\mathcal{F}$ is locally free.

(c) If $\mathcal{F}$ is invertible, its dual is coherent and evaluation gives

$$\mathcal{F} \otimes \mathcal{F}^\vee \cong \mathcal{O}_X.$$

Conversely, suppose $\mathcal{F} \otimes \mathcal{G} \cong \mathcal{O}_X$. At x , put

$$B = \mathcal{O}_{X,x}, \quad P = \mathcal{F}_x, \quad Q = \mathcal{G}_x.$$

These are finite modules over the noetherian local ring $(B, \mathfrak{m})$, and $P \otimes_B Q \cong B$. Reducing modulo $\mathfrak{m}$ gives

$$(P/\mathfrak{m}P) \otimes_{B/\mathfrak{m}} (Q/\mathfrak{m}Q) \cong B/\mathfrak{m}.$$

Both vector spaces therefore have dimension one. Nakayama shows that P and Q are cyclic, say $P \cong B/I$ and $Q \cong B/J$. Then

$$B \cong P \otimes_B Q \cong B/(I + J).$$

The annihilators of the two ends are equal, so $I + J = 0$, and hence $I = J = 0$. Thus $P \cong B$ for every x . Part (b) now shows that $\mathcal{F}$ is locally free of rank one. $\square$

Exercise II.5.8**Permitted context.**

Let X be noetherian and $\mathcal{F}$ coherent. Define

$$\varphi(x) = \dim_{k(x)}(\mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} k(x)).$$

For a finite module over a local ring, this dimension is the minimal number of generators. A function is upper semicontinuous when each set $\{x : \varphi(x) \geq n\}$ is closed.

Problem. (a) Prove that φ is upper semicontinuous.

(b) If $\mathcal{F}$ is locally free and X is connected, prove that φ is constant.

(c) Conversely, if X is reduced and φ is constant, prove that $\mathcal{F}$ is locally free.

Solution. (a) Let $\varphi(x) = r$. On an affine neighborhood $\text{Spec } A$, write $\mathcal{F} = \widetilde{M}$ and let $\mathfrak{p}$ correspond to x . Lift a basis of $M_{\mathfrak{p}}/\mathfrak{p}M_{\mathfrak{p}}$. Nakayama and clearing denominators give, after shrinking to a distinguished neighborhood of $\mathfrak{p}$, a surjection

$$A^r \longrightarrow M.$$

Every fiber on this neighborhood is generated by at most r elements, so $\varphi \leq r$ there. Thus the set $\{x : \varphi(x) < n\}$ is open for every n , and its complement $\{x : \varphi(x) \geq n\}$ is closed.

(b) On an open set where $\mathcal{F} \cong \mathcal{O}_X^r$, the function φ equals r . Hence it is locally constant. A locally constant integer-valued function on a connected space is constant.

(c) Suppose $\varphi \equiv r$. On a reduced affine open $\text{Spec } A$, write $\mathcal{F} = \widetilde{M}$. As in (a), near any chosen point there is a surjection

$$A^r \longrightarrow M.$$

After shrinking, its kernel K is finite, so choose a finite presentation

$$A^m \xrightarrow{v} A^r \longrightarrow M \longrightarrow 0$$

whose image is K . For every prime $\mathfrak{q}$ in this open, the induced surjection

$$k(\mathfrak{q})^r \longrightarrow M \otimes_A k(\mathfrak{q})$$

is an isomorphism, since both sides have dimension r . Therefore the matrix $v \otimes k(\mathfrak{q})$ is zero. Every entry of the matrix v lies in every prime ideal of A , hence in the nilradical. Because A is reduced, all entries vanish. Thus $K = 0$, $M \cong A^r$, and $\mathcal{F}$ is free near the chosen point. It is therefore locally free. $\square$

Exercise II.5.9**Permitted context.**

Let S be generated by S_1 over S_0 , let M be graded, and put $X = \text{Proj } S$. Write

$$\Gamma_*(\mathcal{F}) = \bigoplus_{d \in \mathbb{Z}} \Gamma(X, \mathcal{F}(d)).$$

Under the finite-generation hypotheses on S in part (b), if a finite graded S -module N has $\tilde{N} = 0$, then $N_d = 0$ for all sufficiently large d : for finitely many degree-one generators f_i of S_+ , powers of the f_i kill a finite set of homogeneous generators of N . Under those same hypotheses, every coherent sheaf on X is a quotient of a finite sum of sufficiently negative twists.

Problem. (a) Construct a natural graded homomorphism

$$\alpha : M \longrightarrow \Gamma_*(\widetilde{M}).$$

(b) Suppose $S_0 = A$ is a finite-type k -algebra, S_1 is finite over A , and M is finite over S . Prove that

$$\alpha_d : M_d \longrightarrow \Gamma(X, \widetilde{M}(d))$$

is an isomorphism for every sufficiently large d .

(c) Retain the hypotheses on S in (b). Call two graded modules equivalent if their sufficiently high truncations are isomorphic, and call a module quasi-finitely generated if it is equivalent to a finite module. Prove that sheafification and Γ_* induce inverse equivalences between quasi-finitely generated graded modules modulo this equivalence and coherent sheaves on X .

Solution. (a) If $m \in M_d$, then on $D_+(f)$ its image in

$$(M(d)_f)_0$$

defines a section of $\widetilde{M}(d)$. These local sections agree on overlaps, so they glue to $\alpha_d(m)$. Multiplication of these fractions shows that the maps α_d form a natural graded S -homomorphism.

(b) Choose degree-one generators $f_1, \dots, f_r$ of S over A . For a graded module N , set

$$E(N)_d = \left\{ (n_i) \in \prod_i N_{d+1} : f_j n_i = f_i n_j \text{ for all } i, j \right\}.$$

There is a graded map

$$\beta : N \longrightarrow E(N), \quad n \longmapsto (f_1 n, \dots, f_r n).$$

When $N = M$, both M and $E(M)$ are finite S -modules because S is noetherian. On $D_+(f_i)$, the inverse to the sheafified map sends a compatible tuple to n_i/f_i . Hence $\tilde{\beta}$ is an isomorphism. Its finite kernel and cokernel have zero associated sheaf, so the observation in the context gives an integer d_0 such that

$$\beta_d : M_d \xrightarrow{\sim} E(M)_d \quad (d \geq d_0).$$

First α_d is injective for $d \geq d_0$. If a nonzero homogeneous m of such a degree has zero associated section, some power of S_+ kills m . Choose a nonzero homogeneous element in the last nonzero $S_+^q m$. It is killed by every f_i , contradicting injectivity of β in its degree. Identify $M_{\geq d_0}$ with its image.

Let $s \in H^0(X, \widetilde{M}(d))$, with $d \geq d_0$. Clearing denominators on the finite standard cover gives an N such that $S_+^N s$ lies in the image of M . More explicitly, for each i a sufficiently large $f_i^{q_i} s$ is represented by an element of M ; any degree- N monomial is divisible by one of these powers once $N > \sum_i (q_i - 1)$.

Descend on N . For any degree- $(N - 1)$ monomial h , put $t = hs$, and lift $f_i t$ uniquely to $m_i \in M_{d+N}$. The relations $f_j m_i = f_i m_j$ hold in M by the injectivity already proved. Surjectivity of β_{d+N-1} gives an m with $f_i m = m_i$ for all i . Then $\alpha(m) = t$, since their difference is zero on each $D_+(f_i)$. Thus $S_+^{N-1} s$ is also in the image. Continuing to $N = 0$ proves surjectivity of α_d .

(c) Define morphisms in the category of tails by

$$\mathrm{Hom}_{\mathrm{tail}}(M, N) = \varinjlim_q \mathrm{Hom}_{\mathrm{gr}, S}(M_{\geq q}, N_{\geq q}),$$

with transition maps given by restriction and composition on a common truncation. This specifies the category, rather than only an equivalence relation on objects.

Truncating a graded module does not change its associated sheaf. Part (b) says that for every finite M , the unit

$$M \longrightarrow \Gamma_*(\widetilde{M})$$

is an isomorphism in all sufficiently high degrees. The same holds for a quasi-finitely generated module after replacing it by an equivalent finite one.

Conversely, for every quasi-coherent $\mathcal{F}$ there is a natural isomorphism

$$\widetilde{\Gamma_*(\mathcal{F})} \cong \mathcal{F}.$$

If $\mathcal{F}$ is coherent, choose finitely many global sections of one twist $\mathcal{F}(n)$ that generate it. They generate a finite graded submodule

$$N \subseteq \Gamma_*(\mathcal{F})$$

whose associated sheaf is $\mathcal{F}$. Applying (b) to N shows that $N \rightarrow \Gamma_*(\mathcal{F})$ is an isomorphism in high degrees. Thus $\Gamma_*(\mathcal{F})$ is quasi-finitely generated. The unit and counit just described are isomorphisms in the quotient category, so the two functors are inverse equivalences. $\square$

Exercise II.5.10**Permitted context.**

Let $S = A[x_0, \dots, x_r]$ and $X = \text{Proj } S$. For a homogeneous ideal I , define

$$\bar{I} = \{s \in S : \text{for every } i \text{ some } n_i \text{ satisfies } x_i^{n_i} s \in I\}.$$

The closed subscheme defined by I has ideal sheaf $\tilde{I}$, and the opens $D_+(x_i)$ cover X . Here $\Gamma_{\geq 0}(\mathcal{F}) = \bigoplus_{d \geq 0} H^0(X, \mathcal{F}(d))$. For all $r \geq 0$, $\Gamma_{\geq 0}(\mathcal{O}_X) = S$, and sheafifying $\Gamma_{\geq 0}(\mathcal{F})$ recovers $\mathcal{F}$. Restricting to nonnegative degrees is essential when $r = 0$.

Problem. (a) Prove that $\bar{I}$ is a homogeneous ideal.

(b) Prove that two homogeneous ideals define the same closed subscheme of X if and only if their saturations agree.

(c) If $Y \subseteq X$ is closed, prove that $\Gamma_{\geq 0}(\mathcal{I}_Y)$ is saturated and is the largest homogeneous ideal defining Y .

(d) Deduce a bijection between saturated homogeneous ideals of S and closed subschemes of X .

Solution. (a) If $s, t \in \bar{I}$, then for each i one common sufficiently large power of x_i sends both s and t into I ; it therefore sends $s + t$ into I . If $a \in S$, any power that sends s into I also sends as into I . Hence $\bar{I}$ is an ideal.

Write $s = \sum_d s_d$ with s_d homogeneous. Since I is homogeneous, $x_i^{n_i} s \in I$ implies $x_i^{n_i} s_d \in I$ for every d . Thus every homogeneous component s_d lies in $\bar{I}$, so the ideal is homogeneous.

(b) For homogeneous $s \in S$,

$$s \in \bar{I} \iff s/1 \in I_{x_i} \text{ for every } i.$$

Consequently $\bar{I}$ is the intersection, inside S , of the contractions of the homogeneous localizations I_{x_i} .

Two ideals define the same closed subscheme precisely when

$$(I_1)_{(x_i)} = (I_2)_{(x_i)}$$

for every i . Since x_i is invertible in the full graded localization, its degree-zero part determines every homogeneous degree. Thus these equalities are equivalent to $(I_1)_{x_i} = (I_2)_{x_i}$ for every i , which by the preceding intersection formula is equivalent to $\bar{I}_1 = \bar{I}_2$.

(c) Put $J = \Gamma_{\geq 0}(\mathcal{I}_Y)$, viewed inside $\Gamma_{\geq 0}(\mathcal{O}_X)$. Suppose a homogeneous section s has the property that, for each i , some $x_i^{n_i} s$ belongs to J . On $D_+(x_i)$, the section x_i is invertible, so

$$s|_{D_+(x_i)} \in \mathcal{I}_Y(\deg s).$$

These opens cover X , and membership in a subsheaf is local. Therefore $s \in J$, proving that J is saturated.

If I defines Y , the natural map $\tilde{I} \rightarrow \mathcal{I}_Y$ sends each homogeneous element of I to a global twisted section of $\mathcal{I}_Y$. Hence $I \subseteq J$. Sheafification gives $\tilde{J} = \mathcal{I}_Y$, so J does define Y . Thus it is the largest ideal defining Y .

(d) Send a saturated ideal I to the subscheme defined by $\tilde{I}$, and send Y to $\Gamma_{\geq 0}(\mathcal{I}_Y)$. Part (c) says the second output is saturated. Part (b), together with the maximality in (c), says that starting from a saturated I returns I , while the ideal sheaf of the subscheme obtained from Y is again $\mathcal{I}_Y$. The two constructions are inverse. $\square$

Exercise II.5.11**Permitted context.**

For graded A -algebras S, T generated in degree one, their Segre product is

$$S \#_A T = \bigoplus_{n \geq 0} S_n \otimes_A T_n.$$

Degree-one standard affine charts cover all three Proj schemes.

Problem. For $X = \text{Proj } S$ and $Y = \text{Proj } T$, prove

$$\text{Proj}(S \#_A T) \cong X \times_{\text{Spec } A} Y.$$

Under this isomorphism prove

$$\mathcal{O}(1) \cong p_1^* \mathcal{O}_X(1) \otimes p_2^* \mathcal{O}_Y(1).$$

When degree-one generators x_i and y_j give projective embeddings of X and Y , show that the generators $x_i \otimes y_j$ give their Segre embedding.

Solution. Let $R = S \#_A T$. The products

$$D_+(f) \times_A D_+(g) = \text{Spec}(S_{(f)} \otimes_A T_{(g)})$$

for $f \in S_1, g \in T_1$, form an affine cover of $X \times_A Y$. There is a natural ring isomorphism

$$R_{(f \otimes g)} \xrightarrow{\sim} S_{(f)} \otimes_A T_{(g)}.$$

Indeed, a degree-zero fraction on the left gives degree-zero fractions in the two factors. Conversely, for a tensor of fractions on the right, multiply numerator and denominator by sufficiently high powers of f and g so that both denominators have the same exponent; it then comes from a power of $f \otimes g$.

These isomorphisms commute with further homogeneous localization. They therefore glue the affine charts of $\text{Proj } R$ to the displayed product, proving the first assertion.

On the same chart, the standard generator of $\mathcal{O}_{\text{Proj } R}(1)$ corresponds to the tensor product of the standard generators of $p_1^* \mathcal{O}_X(1)$ and $p_2^* \mathcal{O}_Y(1)$. The transition functions are likewise the products of the two transition functions. Hence the local identifications glue to

$$\mathcal{O}_{\text{Proj } R}(1) \cong p_1^* \mathcal{O}_X(1) \otimes p_2^* \mathcal{O}_Y(1).$$

If S_1 and T_1 are generated by $x_0, \dots, x_r$ and $y_0, \dots, y_s$, then R_1 is generated by $x_i \otimes y_j$. The resulting homogeneous coordinates on the product are exactly all pairwise products of the coordinates of the two factors. Thus the induced map to $\mathbb{P}_A^{(r+1)(s+1)-1}$ is the Segre embedding. $\square$

Exercise II.5.12**Permitted context.**

An invertible sheaf $\mathcal{L}$ on X is very ample relative to Y if there is an immersion

$$i : X \hookrightarrow \mathbb{P}_Y^n$$

such that $\mathcal{L} \cong i^*\mathcal{O}(1)$. Under the Segre embedding,

$$\mathcal{O}_{\mathbb{P}_Y^n \times_Y \mathbb{P}_Y^m}(1, 1) \cong p_1^*\mathcal{O}(1) \otimes p_2^*\mathcal{O}(1)$$

is the pullback of $\mathcal{O}(1)$.

Problem. (a) If $\mathcal{L}$ and $\mathcal{M}$ are very ample on X relative to Y , prove that $\mathcal{L} \otimes \mathcal{M}$ is very ample relative to Y .

(b) Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. If $\mathcal{L}$ is very ample for f and $\mathcal{M}$ is very ample for g , prove that

$$\mathcal{L} \otimes f^*\mathcal{M}$$

is very ample for $g \circ f$.

Solution. (a) Choose immersions

$$i : X \hookrightarrow \mathbb{P}_Y^n, \quad j : X \hookrightarrow \mathbb{P}_Y^m$$

realizing $\mathcal{L}$ and $\mathcal{M}$. The product map

$$(i, j) : X \longrightarrow \mathbb{P}_Y^n \times_Y \mathbb{P}_Y^m$$

is an immersion: factor through the closed graph of j in $X \times_Y \mathbb{P}_Y^m$, followed by the base change of i . Compose with the Segre closed immersion into $\mathbb{P}_Y^N$. The pullback of $\mathcal{O}(1)$ is

$$i^*\mathcal{O}(1) \otimes j^*\mathcal{O}(1) \cong \mathcal{L} \otimes \mathcal{M}.$$

This realizes the tensor product as very ample.

(b) Choose immersions

$$X \hookrightarrow \mathbb{P}_Y^n, \quad Y \hookrightarrow \mathbb{P}_Z^m$$

realizing $\mathcal{L}$ and $\mathcal{M}$. Base change of the second one gives an immersion

$$\mathbb{P}_Y^n \hookrightarrow \mathbb{P}_Z^n \times_Z \mathbb{P}_Z^m.$$

After composing with the first immersion and the Segre embedding, we obtain an immersion $X \hookrightarrow \mathbb{P}_Z^N$. The pulled-back hyperplane sheaf is

$$\mathcal{L} \otimes f^*\mathcal{M}.$$

Hence this sheaf is very ample relative to Z . □

Exercise II.5.13**Permitted context.**

Let S be generated by S_1 over S_0 . For $d > 0$, its d -th Veronese ring is

$$S^{(d)} = \bigoplus_{n \geq 0} S_{nd},$$

where S_{nd} is assigned new degree n . The degree- d monomials in degree-one generators generate $S^{(d)}$ in new degree one.

Problem. For $X = \text{Proj } S$, prove

$$\text{Proj } S^{(d)} \cong X$$

and prove that $\mathcal{O}(1)$ on the left corresponds to $\mathcal{O}_X(d)$. Show that the projective embedding defined by degree- d monomials is the d -uple embedding.

Solution. Let $f \in S_d = S_1^{(d)}$. The same element defines standard opens in both Proj schemes, and

$$(S_f^{(d)})_0 = (S_f)_0.$$

Indeed, a degree-zero fraction for the regraded ring has numerator in S_{nd} and denominator f^n , which is exactly a degree-zero fraction in the original grading. Degree- d monomials cover $\text{Proj } S$, because the degree-one generators generate the irrelevant ideal. The displayed affine isomorphisms agree after further localization and hence glue to

$$\text{Proj } S^{(d)} \xrightarrow{\sim} \text{Proj } S.$$

On $D_+(f)$, the standard trivialization of $\mathcal{O}_{\text{Proj } S^{(d)}}(1)$ uses the new degree-one element f . In the original grading that element has degree d , so the same transition functions are those of $\mathcal{O}_X(d)$. This proves the assertion about twisting sheaves.

If $x_0, \dots, x_r$ generate S_1 , the new degree-one generators are the monomials $x_0^{a_0} \cdots x_r^{a_r}$ with $\sum a_i = d$. The associated homogeneous-coordinate map lists all degree- d monomials, which is precisely the d -uple embedding. $\square$

Exercise II.5.14**Permitted context.**

Let k be algebraically closed and let $X \subseteq \mathbb{P}_k^r$ be connected and normal. Put

$$S = S(X), \quad S' = \bigoplus_{n \geq 0} \Gamma(X, \mathcal{O}_X(n)).$$

A connected normal noetherian scheme is integral. We use two standard graded facts: the integral closure of a graded domain is graded, and a Veronese subring of a normal graded domain is normal.

For the final criterion, normality must be accompanied by integrality or connectedness, since projective normality requires the homogeneous coordinate ring to be a domain. Over k , surjectivity in degree zero forces this connectedness.

Problem. (a) Prove that S is a domain and that S' is the integral closure of S in its fraction field.

(b) Prove that $S_n = S'_n$ for every sufficiently large n .

(c) Prove that $S^{(d)}$ is integrally closed for all sufficiently large d , and deduce that a sufficiently high d -uple embedding of X is projectively normal.

(d) In the same integral projective setting, prove that an embedding is projectively normal if and only if X is normal and every restriction map

$$\Gamma(\mathbb{P}_k^r, \mathcal{O}(n)) \longrightarrow \Gamma(X, \mathcal{O}_X(n))$$

is surjective for $n \geq 0$.

Solution. (b) We prove this first, since it supplies one ingredient for (a). The natural graded map

$$S \longrightarrow \Gamma_*(\tilde{S}) = \bigoplus_{n \in \mathbb{Z}} \Gamma(X, \mathcal{O}_X(n))$$

is the map of Exercise II.5.9. Its nonnegative target is S' , and that exercise shows that it is an isomorphism in all sufficiently large degrees. Thus

$$S_n = S'_n \quad (n \gg 0),$$

which proves (b).

(a) Since X is connected and normal, it is integral. Restriction of homogeneous polynomials gives an injection $S \hookrightarrow S'$: its kernel is exactly the saturated homogeneous ideal defining X . The ring S' is a domain, because a nonzero section of a line bundle is nonzero at the generic point, and the product of two such germs is nonzero. Hence S is a domain.

The eventual equality from (b) implies that S' lies in $\text{Frac } S$. Indeed, for homogeneous $u \in S'_m$, choose a nonzero homogeneous $a \in S$ of positive degree and a power a^q such that $\deg(a^q u)$ is in the equality range; then $a^q u \in S$. It also implies that S' is integral over S . For $m > 0$, a sufficiently high power of u belongs to $S_{qm} = S'_{qm}$, so u satisfies

$$T^q - u^q = 0.$$

In degree zero, properness and connectedness give $S'_0 = \Gamma(X, \mathcal{O}_X) = k = S_0$. Thus every homogeneous element, and hence every element, of S' is integral over S .

It remains to show that S' is integrally closed. The integral closure of a graded domain is graded, so it is enough to consider a homogeneous element $z \in \text{Frac } S'$ integral over S' . For every nonempty standard affine

$$U_i = X \cap D_+(x_i),$$

localization and the trivialization by x_i identify

$$S'_{x_i} \cong \Gamma(U_i, \mathcal{O}_X)[x_i, x_i^{-1}].$$

The affine ring $\Gamma(U_i, \mathcal{O}_X)$ is normal, and so is its Laurent polynomial ring. Hence $z \in S'_{x_i}$ for every i . If z has degree m , these local elements are compatible sections of $\mathcal{O}_X(m)$, and therefore glue to a global section. An integral homogeneous element cannot have negative degree in a nonnegatively graded domain, so $m \geq 0$, and $z \in S'_m$. Therefore S' is normal and is exactly the integral closure of S .

(c) Choose N such that $S_n = S'_n$ for $n \geq N$, and take $d \geq N$. Then

$$S^{(d)} = S'^{(d)},$$

including degree zero. A Veronese subring of the normal graded domain S' is normal: a homogeneous element of its fraction field integral over the Veronese is integral over S' , hence lies in S' , and its degree is divisible by d . Thus $S^{(d)}$ is integrally closed. Exercise II.5.13 identifies it with the homogeneous coordinate ring of the d -uple embedding, which is therefore projectively normal.

(d) Suppose first that X is normal and all restriction maps are surjective. The degree-zero map shows that $\Gamma(X, \mathcal{O}_X) = k$, so X is connected; normality then makes it integral. Surjectivity in every degree says precisely that

$$S = S'.$$

Part (a) now shows that this ring is an integrally closed domain, so the embedding is projectively normal.

Conversely, suppose S is an integrally closed domain. The standard affine charts of $X = \text{Proj } S$ have rings $(S_{x_i})_0$. Each is an integrally closed domain: an element of its fraction field integral over the degree-zero ring is integral over the normal localization S_{x_i} , and homogeneity puts it back in degree zero. Hence X is integral and normal. Applying (a), the ring S' is the integral closure of S in their common fraction field. Since S is already normal, $S = S'$, which says exactly that every displayed restriction map is surjective. $\square$

Exercise II.5.15**Permitted context.**

On a noetherian scheme, every open subset is quasi-compact and coherent sheaves form an abelian category. If $i : U \hookrightarrow X$ is a quasi-compact open immersion and $\mathcal{F}$ is quasi-coherent on U , then $i_*\mathcal{F}$ is quasi-coherent. A quasi-coherent sheaf is the union of subsheaves $\mathcal{F}_\lambda$ when every local section belongs locally, and hence on each quasi-compact open globally, to one member of the directed family.

Problem. Let X be noetherian.

- (a) If X is affine, prove that every quasi-coherent sheaf is the union of its coherent subsheaves.
- (b) If X is affine, $U \subseteq X$ is open, and $\mathcal{F}$ is coherent on U , prove that $\mathcal{F}$ extends to a coherent sheaf on X .
- (c) In (b), suppose $\mathcal{F} \subseteq \mathcal{G}|_U$ for a quasi-coherent $\mathcal{G}$ on X . Prove that the extension can be chosen as a coherent subsheaf of $\mathcal{G}$.
- (d) Prove (c) for an arbitrary noetherian X . Deduce the extension theorem by taking $\mathcal{G} = i_*\mathcal{F}$.
- (e) Prove that on any noetherian scheme every quasi-coherent sheaf is the union of its coherent subsheaves.

Solution. (a) Write $X = \operatorname{Spec} A$ and $\mathcal{F} = \widetilde{M}$. The module M is the directed union of its finite submodules N . Since A is noetherian, $\widetilde{N}$ is coherent. On a distinguished open,

$$\widetilde{M}(D(f)) = M_f = \bigcup_N N_f.$$

Every open of the noetherian space X is quasi-compact and hence has a finite distinguished cover. Filtered unions commute with the finite equalizers expressing the sheaf gluing condition on such a cover. Therefore

$$\mathcal{F}(V) = \bigcup_N \widetilde{N}(V)$$

for every open V , as required.

(b) Let $i : U \hookrightarrow X$. The sheaf $i_*\mathcal{F}$ is quasi-coherent. By (a) it is the directed union of coherent subsheaves. Choose a finite affine cover of the quasi-compact open U on which $\mathcal{F}$ has finitely many generators. All those finitely many generators lie in one coherent subsheaf $\mathcal{F}'$ of $i_*\mathcal{F}$, after enlarging within the directed family. Since $\mathcal{F}'|_U \subseteq \mathcal{F}$ and contains local generating sets, equality holds:

$$\mathcal{F}'|_U = \mathcal{F}.$$

(c) Form the fiber product of quasi-coherent sheaves

$$\mathcal{H} = \mathcal{G} \times_{i_*i^*\mathcal{G}} i_*\mathcal{F}.$$

It is a quasi-coherent subsheaf of $\mathcal{G}$, and restriction to U identifies $\mathcal{H}|_U$ with $\mathcal{F}$. Apply the argument of (b), now to the union of coherent subsheaves of $\mathcal{H}$. A coherent member containing finite local generators of $\mathcal{F}$ restricts exactly to $\mathcal{F}$, and its inclusion in $\mathcal{H} \subseteq \mathcal{G}$ gives the desired extension.

(d) Choose a finite affine cover

$$X = X_1 \cup \cdots \cup X_m.$$

Starting with $\mathcal{F} \subseteq \mathcal{G}|_U$, proceed inductively. Suppose a coherent subsheaf has been constructed on

$$W_i = U \cup X_1 \cup \cdots \cup X_i.$$

On the open $W_i \cap X_{i+1}$ it is a coherent subsheaf of $\mathcal{G}|_{X_{i+1}}$. Part (c), applied to the affine scheme X_{i+1} , extends it to a coherent subsheaf there. The two subsheaves agree on the overlap as subsheaves of $\mathcal{G}$, so they glue to a coherent subsheaf on W_{i+1} . After finitely many steps this gives $\mathcal{F}' \subseteq \mathcal{G}$ on X , with $\mathcal{F}'|_U = \mathcal{F}$.

For the announced extension theorem, $i_*\mathcal{F}$ is quasi-coherent and restricts to $\mathcal{F}$; take $\mathcal{G} = i_*\mathcal{F}$.

(e) Let $\mathcal{G}$ be quasi-coherent, let $V \subseteq X$ be open, and let $s \in \mathcal{G}(V)$. The image of $\mathcal{O}_V \rightarrow \mathcal{G}|_V$, $1 \mapsto s$, is coherent because V is noetherian. By (d) it extends to a coherent subsheaf of $\mathcal{G}$ containing s . The sums of finitely many such coherent subsheaves are coherent, so they form a directed family. Every local section lies in one of them, proving that their union is $\mathcal{G}$. $\square$

Exercise II.5.16

Permitted context.

Tensor, symmetric, and exterior powers of sheaves are obtained by sheafifying the corresponding module constructions. Exact sequences of locally free sheaves split locally, because a local basis of the quotient can be lifted. Pullback is a tensor functor and a left adjoint, so it commutes with tensor products and cokernels.

Problem. Let $\mathcal{F}$ be an $\mathcal{O}_X$ -module.

(a) If $\mathcal{F}$ is locally free of rank n , compute the ranks of $T^r\mathcal{F}$, $S^r\mathcal{F}$, and $\bigwedge^r\mathcal{F}$.

(b) If $\mathcal{F}$ is locally free of rank n and $0 \leq r \leq n$, prove that wedge multiplication is a perfect pairing

$$\bigwedge^r \mathcal{F} \otimes \bigwedge^{n-r} \mathcal{F} \longrightarrow \bigwedge^n \mathcal{F}.$$

(c) For an exact sequence of locally free sheaves

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0,$$

construct a filtration of $S^r\mathcal{F}$ with quotients

$$S^p\mathcal{F}' \otimes S^{r-p}\mathcal{F}''.$$

(d) Prove the analogous statement for exterior powers and deduce

$$\bigwedge^n \mathcal{F} \cong \bigwedge^{n'} \mathcal{F}' \otimes \bigwedge^{n''} \mathcal{F}'', \quad n = n' + n''.$$

(e) Prove that pullback commutes with tensor, symmetric, and exterior powers.

Solution. (a) The assertion is local, so take $\mathcal{F} \cong \mathcal{O}_X^n$ with basis $e_1, \dots, e_n$. Pure tensors give a basis of $T^r\mathcal{F}$; nondecreasing r -tuples give a basis of $S^r\mathcal{F}$; and strictly increasing r -tuples give a basis of $\bigwedge^r\mathcal{F}$. Their ranks are respectively

$$n^r, \quad \binom{n+r-1}{r}, \quad \binom{n}{r}.$$

(b) Again work in a local basis. For every r -element subset $I \subseteq \{1, \dots, n\}$, the basis vector e_I wedges to $\pm e_1 \wedge \dots \wedge e_n$ with e_{I^c} , and wedges to zero with every other complementary-degree basis

vector. The matrix of the pairing is therefore a signed permutation matrix. Hence it induces an isomorphism

$$\bigwedge^r \mathcal{F} \cong \left(\bigwedge^{n-r} \mathcal{F} \right)^\vee \otimes \bigwedge^n \mathcal{F}.$$

For $n = 2, r = 1$, this is $\mathcal{F} \cong \mathcal{F}^\vee \otimes \bigwedge^2 \mathcal{F}$.

(c) Define $F^p \subseteq S^r \mathcal{F}$ to be the image of

$$S^p \mathcal{F}' \otimes S^{r-p} \mathcal{F} \longrightarrow S^r \mathcal{F}.$$

Then

$$S^r \mathcal{F} = F^0 \supseteq F^1 \supseteq \dots \supseteq F^r \supseteq F^{r+1} = 0.$$

Locally split $\mathcal{F} \cong \mathcal{F}' \oplus \mathcal{F}''$. The standard decomposition

$$S^r(\mathcal{F}' \oplus \mathcal{F}'') \cong \bigoplus_{i=0}^r S^i \mathcal{F}' \otimes S^{r-i} \mathcal{F}''$$

identifies F^p with the sum of the terms for $i \geq p$. Consequently

$$F^p / F^{p+1} \cong S^p \mathcal{F}' \otimes S^{r-p} \mathcal{F}''.$$

This description is independent of the local splitting, so the local isomorphisms glue.

(d) Define the filtration in the same way using wedge multiplication. Under a local splitting,

$$\bigwedge^r(\mathcal{F}' \oplus \mathcal{F}'') \cong \bigoplus_{i=0}^r \bigwedge^i \mathcal{F}' \otimes \bigwedge^{r-i} \mathcal{F}'',$$

and the same argument gives the required quotients. For $r = n' + n''$, all the associated graded pieces vanish except the one with $i = n'$. The filtration therefore gives the asserted canonical determinant isomorphism.

(e) There are natural maps

$$f^*(\mathcal{F}^{\otimes r}) \longrightarrow (f^* \mathcal{F})^{\otimes r},$$

which are isomorphisms by associativity of tensor product. Symmetric and exterior powers are cokernels of the permutation and alternating relations in tensor powers. Since f^* commutes with cokernels, the tensor isomorphism descends to natural isomorphisms

$$f^*(S^r \mathcal{F}) \cong S^r(f^* \mathcal{F}), \quad f^*\left(\bigwedge^r \mathcal{F}\right) \cong \bigwedge^r(f^* \mathcal{F}).$$

□

Exercise II.5.17**Permitted context.**

A morphism $f : X \rightarrow Y$ is affine when Y has an affine open cover whose inverse images are affine. We may use the affine communication lemma: if a scheme over $\operatorname{Spec} A$ is covered by finitely many affine principal opens defined by elements generating the unit ideal of A , then the scheme is affine. An affine structural morphism need not have an affine total space unless its base is affine.

- Problem.** (a) Prove that f is affine if and only if $f^{-1}(V)$ is affine for every affine open $V \subseteq Y$.
 (b) Prove that affine morphisms are quasi-compact and separated, and that finite morphisms are affine.
 (c) For a quasi-coherent $\mathcal{O}_Y$ -algebra $\mathcal{A}$, construct a unique Y -scheme

$$\operatorname{Spec}_Y \mathcal{A}$$

whose inverse image over every affine $V \subseteq Y$ is $\operatorname{Spec} \mathcal{A}(V)$, compatibly with restriction.

- (d) Prove that the structural morphism of this relative spectrum is affine and

$$f_* \mathcal{O}_{\operatorname{Spec}_Y \mathcal{A}} \cong \mathcal{A}.$$

Conversely, prove that for affine $f : X \rightarrow Y$, $\mathcal{A} = f_* \mathcal{O}_X$ is quasi-coherent and $X \cong \operatorname{Spec}_Y \mathcal{A}$.

- (e) For affine f and $\mathcal{A} = f_* \mathcal{O}_X$, prove that f_* is an equivalence from quasi-coherent $\mathcal{O}_X$ -modules to quasi-coherent $\mathcal{A}$ -modules.

Solution. (a) Only one direction needs proof. Assume an affine cover $\{V_i\}$ of Y has affine inverse images, and let $V = \operatorname{Spec} A \subseteq Y$ be arbitrary. Refine the cover $\{V \cap V_i\}$ by distinguished opens of V . Quasi-compactness of V gives finitely many

$$D(a_1), \dots, D(a_m)$$

covering V , each contained in some V_i . Their inverse images are affine: each is the fiber product of the two affine schemes $f^{-1}(V_i)$ and $D(a_j)$ over the affine scheme V_i . Inside $f^{-1}(V)$ they are the principal opens defined by the pullbacks of the a_j . The a_j generate the unit ideal of A , so the affine communication lemma applied to $f^{-1}(V)$ shows that it is affine.

(b) The inverse image of every affine open is affine and therefore quasi-compact; hence f is quasi-compact. Separatedness is local on the base. Over $V = \operatorname{Spec} A$, write

$$f^{-1}(V) = \operatorname{Spec} B.$$

The diagonal is defined by the surjection

$$B \otimes_A B \longrightarrow B,$$

so it is a closed immersion. Thus f is separated. A finite morphism is affine by its local definition.

- (c) Cover Y by affine opens $V_i = \operatorname{Spec} A_i$, and put

$$X_i = \operatorname{Spec} \mathcal{A}(V_i).$$

If $D(a) \subseteq V_i$, quasi-coherence gives

$$\mathcal{A}(D(a)) \cong \mathcal{A}(V_i)_a,$$

so $\text{Spec } \mathcal{A}(D(a))$ is the corresponding principal open of X_i . Covering every overlap $V_i \cap V_j$ by such distinguished opens produces canonical identifications between open pieces of X_i and X_j . They satisfy the cocycle condition because all maps are restriction maps of $\mathcal{A}$. Gluing yields a scheme X and a morphism

$$f : X = \mathbf{Spec}_Y \mathcal{A} \longrightarrow Y.$$

The construction gives the required description on every affine open by (a). That description also forces all the gluing maps, proving uniqueness up to unique Y -isomorphism.

(d) The construction and (a) show that f is affine. On an affine $V \subseteq Y$,

$$(f_* \mathcal{O}_X)(V) = \Gamma(\text{Spec } \mathcal{A}(V), \mathcal{O}) = \mathcal{A}(V),$$

compatibly with localization. These identifications give $f_* \mathcal{O}_X \cong \mathcal{A}$.

Conversely, for affine f , write over $V = \text{Spec } A$

$$f^{-1}(V) = \text{Spec } B.$$

Then $f_* \mathcal{O}_X|_V \cong \tilde{B}$, so $\mathcal{A} = f_* \mathcal{O}_X$ is a quasi-coherent algebra. The affine charts of $\mathbf{Spec}_Y \mathcal{A}$ are $\text{Spec } B$, with the same localization maps as those of X ; they glue to a canonical isomorphism $X \cong \mathbf{Spec}_Y \mathcal{A}$.

(e) On $V = \text{Spec } A$ as above, a quasi-coherent sheaf on $f^{-1}(V) = \text{Spec } B$ is the same thing as a B -module. Its direct image is that module viewed as a module over $\mathcal{A}|_V = \tilde{B}$. Conversely, a quasi-coherent $\mathcal{A}$ -module restricts on V to a B -module and hence defines a quasi-coherent sheaf on $\text{Spec } B$. Localization of modules shows that these local constructions agree on overlaps. They glue to mutually inverse functors, and the local affine equivalences show that the unit and counit are isomorphisms. $\square$

Exercise II.5.18

Permitted context.

A geometric vector bundle of rank n over Y is locally $\mathbb{A}_Y^n$, with transition automorphisms linear in the fiber coordinates. For a locally free sheaf $\mathcal{E}$, put

$$\mathbf{V}(\mathcal{E}) = \mathbf{Spec}_Y(\text{Sym } \mathcal{E}).$$

The symmetric algebra has the universal property that $\mathcal{O}_Y$ -algebra maps $\text{Sym } \mathcal{E} \rightarrow \mathcal{B}$ correspond to $\mathcal{O}_Y$ -linear maps $\mathcal{E} \rightarrow \mathcal{B}$.

Problem. (a) Prove that $\mathbf{V}(\mathcal{E})$ is a geometric vector bundle of rank n , independent up to isomorphism of the local bases chosen.

(b) If $f : X \rightarrow Y$ is a geometric vector bundle, prove that its sheaf of sections $\mathcal{S}(X/Y)$ is naturally a locally free $\mathcal{O}_Y$ -module of rank n .

(c) For $X = \mathbf{V}(\mathcal{E})$, prove naturally that

$$\mathcal{S}(X/Y) \cong \mathcal{E}^\vee.$$

(d) Deduce a bijection between isomorphism classes of rank- n locally free sheaves and rank- n geometric vector bundles over Y .

Solution. (a) On an open U where $\mathcal{E}|_U \cong \mathcal{O}_U^n$,

$$\mathrm{Sym}(\mathcal{E}|_U) \cong \mathcal{O}_U[x_1, \dots, x_n],$$

so

$$\mathbf{V}(\mathcal{E})|_U \cong \mathbb{A}_U^n.$$

On an overlap, changing a basis of $\mathcal{E}$ by an invertible matrix changes the generators x_i by the corresponding linear substitution. Thus the transition functions are linear. A different choice of bases merely changes the local trivializations by such linear maps and hence gives an isomorphic vector bundle.

(b) Over a trivializing open U , a section of $\mathbb{A}_U^n \rightarrow U$ is an n -tuple of sections of $\mathcal{O}_U$. Therefore

$$\mathcal{S}(X/Y)|_U \cong \mathcal{O}_U^n.$$

Addition and scalar multiplication are defined coordinatewise. The transition matrices of the vector bundle are linear, so they preserve these operations; consequently the local module structures glue to a natural $\mathcal{O}_Y$ -module structure. The displayed local identifications prove that it is locally free of rank n .

(c) For every open $V \subseteq Y$, a section of $\mathbf{V}(\mathcal{E})$ over V is a V -morphism

$$V \longrightarrow \mathbf{Spec}_V \mathrm{Sym}(\mathcal{E}|_V).$$

By relative affine duality and the universal property of the symmetric algebra, these are naturally equivalent to

$$\mathrm{Hom}_{\mathcal{O}_V\text{-alg}}(\mathrm{Sym}(\mathcal{E}|_V), \mathcal{O}_V) \cong \mathrm{Hom}_{\mathcal{O}_V}(\mathcal{E}|_V, \mathcal{O}_V).$$

The latter is $\Gamma(V, \mathcal{E}^\vee)$. Compatibility with restriction and with the module operations gives the claimed isomorphism of sheaves.

(d) Starting with $\mathcal{E}$, parts (a)–(c) produce the bundle $\mathbf{V}(\mathcal{E})$, whose sheaf of sections is $\mathcal{E}^\vee$. Dualizing again recovers $\mathcal{E}$, because finite locally free sheaves are canonically isomorphic to their double duals.

Conversely, start with a geometric vector bundle $X \rightarrow Y$, put $\mathcal{S} = \mathcal{S}(X/Y)$, and form $\mathbf{V}(\mathcal{S}^\vee)$. On every bundle chart both schemes are $\mathbb{A}_U^n$, and the transition matrices agree by the construction of $\mathcal{S}$. The local identifications therefore glue to a vector-bundle isomorphism

$$X \cong \mathbf{V}(\mathcal{S}^\vee).$$

The two constructions are inverse on isomorphism classes. □

II.6 Divisors

Exercise II.6.1

Permitted context.

Let $n \geq 1$. Condition $(*)$ means that a scheme is noetherian, integral, separated, and regular in codimension one. For such a scheme the Weil divisor class group is defined. We use $K = K(X)$ and write

$$p : X \times \mathbb{P}^n \longrightarrow X$$

for the projection. The standard affine-bundle calculation gives $\mathrm{Cl}(X \times \mathbb{A}^n) \cong \mathrm{Cl}(X)$. The product with projective space means $\mathbb{P}_X^n$, with the indicated structural base.

Problem. Prove that, if X satisfies $(*)$, then so does $X \times \mathbb{P}^n$, and prove a natural decomposition

$$\mathrm{Cl}(X \times \mathbb{P}^n) \cong \mathrm{Cl}(X) \oplus \mathbb{Z}.$$

Identify a generator of the second summand.

Solution. Put $Y = X \times \mathbb{P}^n$. It is noetherian and separated because p is projective and X has those properties. If $W = \mathrm{Spec} A \subseteq X$ is nonempty affine, the inverse image of W is covered by spectra of polynomial rings over the domain A . These charts have the same generic point and glue along nonempty opens, so Y is integral.

We check regularity in codimension one. Let $y \in Y$ have codimension one and let $x = p(y)$. Since p is flat, the dimension formula leaves two possibilities. If x is the generic point of X , then y is a codimension-one point of $Y_\eta = \mathbb{P}_K^n$, whose local ring is a discrete valuation ring. Otherwise x has codimension one in X and y is the generic point of the fiber over $\overline{\{x\}}$. On a standard affine chart its local ring is

$$\mathcal{O}_{X,x}[t_1, \dots, t_n]_{(\pi)},$$

where $\mathcal{O}_{X,x}$ is a discrete valuation ring with uniformizer π . This localization is again a discrete valuation ring. Hence Y satisfies $(*)$.

Let $H = X \times \mathbb{P}^{n-1}$ be a relative hyperplane. We now describe the class group in a way that also proves there are no hidden relations. Restrict a divisor to the generic fiber and take its class:

$$r : \mathrm{Cl}(Y) \longrightarrow \mathrm{Cl}(\mathbb{P}_K^n) \cong \mathbb{Z}.$$

This is well defined, and $r([H]) = 1$.

Pullback of prime divisors gives a homomorphism

$$p^* : \mathrm{Cl}(X) \longrightarrow \mathrm{Cl}(Y).$$

We claim that

$$0 \longrightarrow \mathrm{Cl}(X) \xrightarrow{p^*} \mathrm{Cl}(Y) \xrightarrow{r} \mathbb{Z} \longrightarrow 0 \quad (1)$$

is exact. Surjectivity on the right follows from the relative hyperplane. Suppose $r([D]) = 0$. On $\mathbb{P}_K^n$, the divisor $D|_{Y_\eta}$ is therefore the divisor of some $f \in K(\mathbb{P}_K^n)^*$. Subtracting $\mathrm{div}_Y(f)$ from D removes every horizontal component. Every remaining prime divisor is vertical; flatness and the dimension formula show that it is $p^{-1}(Z)$ for a unique prime divisor $Z \subseteq X$. Thus D differs from p^*E by a principal divisor.

It remains to see that p^* is injective. If $p^*E = \operatorname{div}_Y(g)$, then the restriction of g to $\mathbb{P}_K^n$ has zero divisor. A rational function on projective space with neither zeros nor poles is an element of K^* . Hence $g \in K(X)^*$, and comparison at the vertical discrete valuations gives $E = \operatorname{div}_X(g)$. This proves (1).

Finally, $1 \mapsto [H]$ splits r . Consequently

$$\operatorname{Cl}(X \times \mathbb{P}^n) \cong p^* \operatorname{Cl}(X) \oplus \mathbb{Z}[H],$$

as required. $\square$

Exercise II.6.2

Permitted context.

Let k be algebraically closed and let $X \subseteq \mathbb{P}_k^n$ be an integral closed subvariety that is regular in codimension one and has positive dimension. If $D = \sum m_i Y_i$ is a Weil divisor on X , set

$$\deg D = \sum_i m_i \deg Y_i.$$

At the generic point of each prime divisor Y_i , the local ring of X is a discrete valuation ring; its valuation is denoted v_{Y_i} .

Problem. (a) If a hypersurface $V \subseteq \mathbb{P}^n$ does not contain X , define an intersection divisor $V \cdot X$ by restricting a local equation of V to X and taking its order along every component of $V \cap X$. Prove that this is well defined and extends linearly to ambient divisors having no component that contains X .

(b) Prove that intersection with X carries principal ambient divisors to principal divisors on X , and hence gives a homomorphism $\operatorname{Cl}(\mathbb{P}^n) \rightarrow \operatorname{Cl}(X)$.

(c) Identify the coefficient of a component Y in $V \cdot X$ with the intersection multiplicity $i(X, V; Y)$. Deduce

$$\deg(D \cdot X) = \deg(D) \deg(X).$$

(d) Show that every principal divisor on X is obtained by intersecting X with a principal divisor on $\mathbb{P}^n$. Deduce that degree descends to $\operatorname{Cl}(X) \rightarrow \mathbb{Z}$, and prove that the diagram

$$\begin{array}{ccc} \operatorname{Cl}(\mathbb{P}^n) & \longrightarrow & \operatorname{Cl}(X) \\ \deg \downarrow & & \downarrow \deg \\ \mathbb{Z} & \xrightarrow{m \mapsto m \deg X} & \mathbb{Z} \end{array}$$

commutes. In particular, prove that the top map is injective.

Solution. (a) Let Y be an irreducible component of $V \cap X$, with generic point η_Y . The projective dimension theorem makes Y a prime divisor on X . Choose an open neighborhood U of η_Y in $\mathbb{P}^n$ on which V has one equation f . Define

$$n_Y = v_Y(f|_X).$$

If f' is another local equation, then near η_Y one has $f' = uf$, where u is a unit. Its restriction to X is still a unit, so $v_Y(f'|_X) = v_Y(f|_X)$. Thus

$$V \cdot X = \sum_{Y \subseteq V \cap X} n_Y Y$$

is independent of all choices. Valuations are additive under multiplication of equations, which gives the asserted linear extension.

(b) Let $D = \operatorname{div}_{\mathbb{P}^n}(g)$, with no component containing X . At the generic point η_X , the function g is a unit in the ambient local ring, so it has a nonzero image $\bar{g} \in K(X)^*$. Near the generic point of a prime divisor $Y \subseteq X$, write the ambient divisor of g as a sum of local equations of its prime components. Restricting this factorization to X and applying v_Y gives

$$\operatorname{coeff}_Y(D \cdot X) = v_Y(\bar{g}).$$

Therefore

$$D \cdot X = \operatorname{div}_X(\bar{g}). \quad (1)$$

Every class on $\mathbb{P}^n$ has a representative supported on hyperplanes that do not contain X . Equation (1) shows that changing such a representative by a principal divisor does not change the class on X , so the map on class groups is well defined.

(c) Work in the local ring $R = \mathcal{O}_{X, \eta_Y}$. This is a discrete valuation ring, and the restriction $\bar{f}$ of a local equation of V is nonzero. The local definition of intersection multiplicity gives

$$i(X, V; Y) = \operatorname{length}_R(R/(\bar{f})).$$

If $\bar{f} = u\pi^a$, with u a unit and π a uniformizer, then this length is $a = v_Y(\bar{f}) = n_Y$. Hence the divisor coefficients are precisely the intersection multiplicities.

The generalized Bezout theorem now yields, for an irreducible hypersurface V ,

$$\deg(V \cdot X) = \sum_Y i(X, V; Y) \deg Y = \deg V \deg X.$$

Linearity gives the displayed formula for every allowed divisor D .

(d) Take $h \in K(X)^*$. The residue map

$$\mathcal{O}_{\mathbb{P}^n, \eta_X} \longrightarrow \mathcal{O}_{X, \eta_X} = K(X)$$

is surjective, so choose a lift $g \in \mathcal{O}_{\mathbb{P}^n, \eta_X}$ of h . Because its residue is nonzero, g is a unit at η_X ; thus no component of $\operatorname{div}_{\mathbb{P}^n}(g)$ contains X . The same valuation calculation as in (1) gives

$$\operatorname{div}_X(h) = \operatorname{div}_{\mathbb{P}^n}(g) \cdot X.$$

Part (c) therefore implies that every principal divisor on X has degree zero. Degree consequently defines a homomorphism on $\operatorname{Cl}(X)$, and part (c) is exactly the commutativity of the stated diagram. Since $\operatorname{Cl}(\mathbb{P}^n) \cong \mathbb{Z}$ by degree and $\deg X > 0$, multiplication by $\deg X$ is injective. The top horizontal map must therefore be injective as well. $\square$

Exercise II.6.3**Permitted context.**

Let $V \subseteq \mathbb{P}_k^n$ be an integral projective variety of positive dimension, regular in codimension one. Let $X = C(V) \subseteq \mathbb{A}_k^{n+1}$ be its affine cone, $\bar{X} \subseteq \mathbb{P}_k^{n+1}$ its projective cone, and P the vertex. Write H for a hyperplane of $\mathbb{P}^n$ not containing V . All class groups below are Weil divisor class groups. In (d), remove Laurent monomial units so that g, h are ordinary polynomials in the generic fiber coordinate, normalized to constant coefficient one.

Problem. (a) Show that projection away from P makes $\bar{X} \setminus \{P\} \rightarrow V$ a Zariski-locally trivial $\mathbb{A}^1$ -bundle, and prove

$$\mathrm{Cl}(V) \cong \mathrm{Cl}(\bar{X} \setminus \{P\}) \cong \mathrm{Cl}(\bar{X}).$$

(b) Under this identification, prove that the divisor at infinity $V \subseteq \bar{X}$ corresponds to the hyperplane-section class $[V \cdot H]$. Deduce an exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{1 \mapsto [V \cdot H]} \mathrm{Cl}(V) \longrightarrow \mathrm{Cl}(X) \longrightarrow 0.$$

(c) Prove that the homogeneous coordinate ring $S(V)$ is a unique factorization domain exactly when V is projectively normal and $\mathrm{Cl}(V) \cong \mathbb{Z}$, generated by the hyperplane class.

(d) Prove that localization at the vertex induces an isomorphism

$$\mathrm{Cl}(X) \xrightarrow{\sim} \mathrm{Cl}(\mathrm{Spec} \mathcal{O}_{X,P}).$$

Solution. (a) Use homogeneous coordinates $[x_0 : \cdots : x_n : z]$, with $P = [0 : \cdots : 0 : 1]$ and with V in $z = 0$. On $U_i = V \cap D_+(x_i)$, every point of the projective cone away from P has coordinates

$$([x_0 : \cdots : x_n], z/x_i),$$

which gives an isomorphism

$$\pi^{-1}(U_i) \cong U_i \times \mathbb{A}^1.$$

On overlaps the fiber coordinate changes by multiplication by a unit, so these charts form the total space of a line bundle.

The affine-bundle class-group argument applies chartwise and at the generic fiber: pullback gives

$$\pi^* : \mathrm{Cl}(V) \xrightarrow{\sim} \mathrm{Cl}(\bar{X} \setminus \{P\}). \quad (1)$$

Indeed, a divisor can first be made vertical by subtracting a principal divisor on the generic $\mathbb{A}^1$, after which it is the pullback of a divisor on V ; the same generic-fiber argument proves injectivity. Since $\dim V \geq 1$, the point P has codimension at least two in $\bar{X}$. Removing it changes neither prime divisors nor principal-divisor relations. Thus the second map in (1), from $\mathrm{Cl}(\bar{X})$, is also an isomorphism.

(b) Let $L \subseteq \mathbb{P}^{n+1}$ be the hyperplane obtained by joining H to P , and let $L_\infty = (z = 0)$. The two ambient hyperplanes are linearly equivalent. Their intersections with $\bar{X}$ are respectively $\pi^*(V \cdot H)$ and the divisor at infinity V . Hence

$$[V] = \pi^*[V \cdot H] \quad \text{in } \mathrm{Cl}(\bar{X}). \quad (2)$$

The complement of the divisor at infinity is the affine cone X . The localization sequence for class groups and (2) give

$$\mathbb{Z}[V.H] \longrightarrow \mathrm{Cl}(V) \longrightarrow \mathrm{Cl}(X) \longrightarrow 0.$$

The first map is injective: the degree homomorphism of Exercise II.6.2 sends $[V.H]$ to the positive integer $\deg V$. This proves the asserted short exact sequence.

(c) The cone is $X = \mathrm{Spec} S(V)$. A noetherian domain is a unique factorization domain if and only if it is normal and its divisor class group is zero. Normality of $S(V)$ is precisely projective normality of the given embedding. By (b),

$$\mathrm{Cl}(X) = 0 \iff \mathrm{Cl}(V) = \mathbb{Z}[V.H],$$

and the injection in (b) says that this cyclic group is infinite. Combining the two criteria proves (c).

(d) For a normal noetherian domain A , localization induces a surjection

$$\mathrm{Cl}(A) \longrightarrow \mathrm{Cl}(A_{\mathfrak{m}})$$

whose kernel is generated by height-one primes not contained in $\mathfrak{m}$. Here $A = S(V)$ and $\mathfrak{m} = A_+$ is the vertex. The same statement remains valid under the present regular-in-codimension-one hypothesis because the argument uses only the codimension-one valuation rings. It is therefore enough to show that every prime divisor of X avoiding P is principal.

We record the graded argument. Let D be such a prime divisor. On the punctured cone

$$X^\circ = X \setminus \{P\} \longrightarrow V$$

it must dominate V : every divisor lying over a proper divisor of V is a cone and contains P . Over the generic point of V , the punctured cone is $\mathrm{Spec} K[t, t^{-1}]$, after choosing a basis of the one-dimensional generic fiber. Thus D is defined there by an irreducible Laurent polynomial g , which we normalize to have constant coefficient 1.

Because D avoids the vertex, its prime ideal contains an element with nonzero constant term. After scaling, write it as

$$f = 1 + f_1 + \cdots + f_a, \quad f_i \in A_i.$$

In $K[t, t^{-1}]$, the polynomial g divides f ; write $f = gh$, normalizing h also to have constant coefficient 1. For a prime divisor $Z \subseteq V$, the valuation along its cone is the Gauss extension of v_Z : after trivializing $\mathcal{O}_V(1)$ at Z , it sends a Laurent polynomial to the minimum valuation of its coefficients. All coefficients f_i are regular at Z , while the constant coefficient is 1, so

$$v_{C(Z)}(f) = 0.$$

Both g and h have constant coefficient 1, hence their Gauss valuations are at most zero. Since their sum is zero, both are zero. Thus g has no vertical zero or pole. On the generic fiber, irreducibility gives valuation one along D and zero along every other horizontal prime. Consequently

$$\mathrm{div}_X(g) = D.$$

Every generator of the localization kernel is therefore zero, so the surjection on class groups is injective and hence an isomorphism. $\square$

Exercise II.6.4**Permitted context.**

Let k be a field of characteristic different from 2, put

$$R = k[x_1, \dots, x_n],$$

and let $f \in R$ be nonconstant and square-free. Thus every irreducible factor of f occurs with exponent one.

Problem. Prove that

$$A = R[z]/(z^2 - f)$$

is integrally closed in its fraction field.

Solution. Because f is square-free and nonconstant, it is not a square in $F = \text{Frac } R$. Hence $T^2 - f$ is irreducible over F , and Gauss's lemma shows that A is a domain. Its fraction field is

$$L = F(z), \quad z^2 = f,$$

and every element of L has a unique expression $a + bz$ with $a, b \in F$.

Let $\alpha = a + bz \in L$ be integral over A . Since A is finite and integral over R , transitivity makes α integral over R . The nontrivial F -automorphism of L sends z to $-z$ and preserves A , so $\bar{\alpha} = a - bz$ is integral over R as well. Therefore

$$\text{Tr}_{L/F}(\alpha) = 2a, \quad \text{N}_{L/F}(\alpha) = a^2 - b^2f$$

are integral over R . They belong to F , and R is a unique factorization domain, hence integrally closed. Thus

$$2a \in R, \quad a^2 - b^2f \in R.$$

The characteristic assumption gives $a \in R$, and consequently $b^2f \in R$.

It remains to show $b \in R$. For every irreducible $p \in R$, let v_p be its exponent valuation. From $b^2f \in R$ we have

$$2v_p(b) + v_p(f) \geq 0.$$

Square-freeness says $v_p(f)$ is either 0 or 1. If the integer $v_p(b)$ were negative, the left side would be at most -1 , a contradiction. Hence $v_p(b) \geq 0$ for every irreducible p , so $b \in R$. We have proved $\alpha = a + bz \in A$; therefore A is integrally closed. $\square$

Exercise II.6.5**Permitted context.**

Let k be algebraically closed of characteristic different from 2, and let $2 \leq r \leq n$. Consider the affine and projective quadrics

$$X = \operatorname{Spec} k[x_0, \dots, x_n]/(x_0^2 + \dots + x_r^2), \quad Q = V_+(x_0^2 + \dots + x_r^2) \subseteq \mathbb{P}_k^n.$$

The unused variables $x_{r+1}, \dots, x_n$ make these quadrics cones when $r < n$.

Problem. (a) Prove that X is normal.

(b) After a linear change of coordinates, write the equation as

$$x_0x_1 = x_2^2 + \dots + x_r^2.$$

Compute $\operatorname{Cl}(X)$, obtaining respectively $\mathbb{Z}/2\mathbb{Z}$, $\mathbb{Z}$, and 0 when $r = 2$, $r = 3$, and $r \geq 4$.

(c) Compute $\operatorname{Cl}(Q)$. Show that it is $\mathbb{Z}$, with a hyperplane equal to twice a generator, for $r = 2$; that it is $\mathbb{Z} \oplus \mathbb{Z}$ for $r = 3$; and that it is $\mathbb{Z}$, generated by a hyperplane section, for $r \geq 4$.

(d) For $r \geq 4$, prove Klein's theorem: every prime divisor $Y \subseteq Q$ is the scheme-theoretic intersection of Q with an irreducible hypersurface of $\mathbb{P}^n$, with multiplicity one.

Solution. (a) Regard the equation as

$$x_0^2 = -(x_1^2 + \dots + x_r^2).$$

The polynomial on the right is square-free when $r \geq 2$: in the case $r = 2$ it is a product of two distinct linear forms, and for $r \geq 3$ it has no repeated irreducible factor. Exercise II.6.4 therefore shows that the coordinate ring of X is integrally closed.

(b) Choose $i \in k$ with $i^2 = -1$. Replacing x_0, x_1 by $x_0 + ix_1, x_0 - ix_1$, and rescaling, gives the stated equation. Put

$$A = k[x_0, \dots, x_n]/(x_0x_1 - q), \quad q = x_2^2 + \dots + x_r^2.$$

After inverting x_0 ,

$$A_{x_0} \cong k[x_0, x_0^{-1}, x_2, \dots, x_n],$$

a unique factorization domain whose units are $k^*x_0^{\mathbb{Z}}$. The divisor localization sequence says that $\operatorname{Cl}(A)$ is generated by the prime components of $V(x_0)$, with precisely the relations supplied by these units.

If $r = 2$, the scheme $V(x_0)$ has the single prime component $D = V(x_0, x_2)$, occurring twice, so $\operatorname{div}(x_0) = 2D$. Hence

$$\operatorname{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}.$$

If $r = 3$, write $q = (x_2 + ix_3)(x_2 - ix_3)$. There are two components D_+ and D_- , and $\operatorname{div}(x_0) = D_+ + D_-$. Thus

$$\operatorname{Cl}(X) \cong \mathbb{Z}^2/\mathbb{Z}(1, 1) \cong \mathbb{Z}.$$

For $r \geq 4$, the quadratic form q in at least three variables is irreducible. Consequently $V(x_0)$ is one reduced prime divisor D , and $\operatorname{div}(x_0) = D$; its class generates and is zero. Therefore $\operatorname{Cl}(X) = 0$.

(c) The affine quadric X is the affine cone over Q . Exercise II.6.3 gives

$$0 \longrightarrow \mathbb{Z}[H] \longrightarrow \operatorname{Cl}(Q) \longrightarrow \operatorname{Cl}(X) \longrightarrow 0, \quad (1)$$

where H is a hyperplane section.

For $r = 2$, first take $n = 2$. The smooth conic is $\mathbb{P}^1$, so its class group is $\mathbb{Z}$, generated by a point, while H has degree 2. Increasing n forms successive projective cones, and Exercise II.6.3(a) leaves the class group unchanged. Thus the same conclusion holds for every $n \geq 2$.

For $r = 3$, the nondegenerate quadric surface in $\mathbb{P}^3$ is $\mathbb{P}^1 \times \mathbb{P}^1$ under the Segre embedding. Its class group is $\mathbb{Z} \oplus \mathbb{Z}$, generated by the two rulings, and H has class $(1, 1)$. Successive projective cones again do not change the class group. Finally, if $r \geq 4$, part (b) and (1) show directly that $\text{Cl}(Q) = \mathbb{Z}[H]$.

(d) Let

$$B = k[x_0, \dots, x_n]/(x_0^2 + \dots + x_r^2)$$

be the homogeneous coordinate ring of Q . By (a) it is normal, and by (b) its class group is zero for $r \geq 4$. Hence B is a unique factorization domain.

The cone over a prime divisor $Y \subseteq Q$ corresponds to a homogeneous height-one prime $\mathfrak{p} \subseteq B$. Factoriality gives $\mathfrak{p} = (g)$. Since the ideal is homogeneous, its generator may be chosen homogeneous; primality makes g irreducible. Lift g to a homogeneous polynomial $G \in k[x_0, \dots, x_n]$. The lift is irreducible: a nontrivial homogeneous factorization of G would give a nontrivial factorization of g in B , because no positive-degree homogeneous factor can become a unit there. Thus $V(G)$ is an irreducible hypersurface, and

$$Q \cap V(G) = \text{Proj}(B/(g)) = Y$$

scheme-theoretically. Since (g) is prime, the multiplicity is one. $\square$

Exercise II.6.6

Permitted context.

Work over an algebraically closed field k of characteristic different from two. Let

$$E : y^2z = x^3 - xz^2$$

be the smooth plane cubic with origin $O = [0 : 1 : 0]$. Its group law is the one obtained from

$$E \longrightarrow \text{Cl}^0(E), \quad P \longmapsto [P - O].$$

Intersections with a line are always counted with multiplicity. When we say that a point is annihilated by m , the identity is included; every nonidentity point so obtained has exact order m in the cases below.

Problem. (a) Prove that a line cuts out the divisor $P + Q + R$ exactly when $P + Q + R = O$.

(b) Prove that $2P = O$ exactly when the tangent line at P passes through O .

(c) Prove that $3P = O$ exactly when P is an inflection point.

(d) Over $\mathbb{C}$, prove that the points whose homogeneous coordinates can be chosen in $\mathbb{Q}$ form a subgroup, and determine that subgroup explicitly.

Solution. (a) The line at infinity $z = 0$ cuts out the divisor $3O$ on E . If a line L cuts out $P + Q + R$, the quotient of equations of L and $z = 0$ is a rational function on E , so

$$P + Q + R \sim 3O.$$

This is equivalent to $[P - O] + [Q - O] + [R - O] = 0$, hence to $P + Q + R = O$.

Conversely, suppose $P + Q + R = O$. Then the effective divisor $P + Q + R$ is linearly equivalent to $3O$. Hence there is a rational function h with $\operatorname{div}(h) + 3O = P + Q + R$. In particular $h \in L(3O)$. On $E \setminus \{O\}$, write the affine coordinates as x, y . Its coordinate ring is

$$k[x, y]/(y^2 - x^3 + x),$$

so every regular function has a unique form $p(x) + yq(x)$. Since $\operatorname{ord}_O(x) = -2$ and $\operatorname{ord}_O(y) = -3$, the pole bound forces p to have degree at most one and q to be constant. The space $L(3O)$ is therefore spanned by $1, x, y$. The homogenization of $h = a + bx + cy$ is a line section whose divisor is $\operatorname{div}(h) + 3O = P + Q + R$, proving the converse.

(b) Let the tangent at P meet E a third time at R . Part (a), with multiplicity, gives

$$2P + R = O.$$

The tangent passes through O exactly when $R = O$, and this is equivalent to $2P = O$.

(c) With the same notation, P is an inflection point exactly when $R = P$. By $2P + R = O$, this is equivalent to $3P = O$.

(d) The inverse map and the chord-tangent construction are defined over $\mathbb{Q}$. More explicitly, a rational line through two rational points has rational coefficients, and after two rational roots are removed from its intersection polynomial with E , the third root is rational. Tangents are handled by the same calculation. Thus $E(\mathbb{Q})$ is a subgroup.

We show that it has no points beyond

$$O, \quad (0, 0), \quad (1, 0), \quad (-1, 0) \tag{1}$$

on the affine chart $z = 1$. Suppose $P = (x, y) \in E(\mathbb{Q})$ has $y \neq 0$. Then

$$a = \frac{x^2 - 1}{y}, \quad b = \frac{2x}{y}, \quad c = \frac{x^2 + 1}{y}$$

satisfy

$$a^2 + b^2 = c^2, \quad \frac{ab}{2} = 1,$$

up to changing signs of the sides. They would therefore give a rational right triangle of area 1.

For completeness, Fermat's descent rules this out. If a rational right triangle has square area, clear denominators and remove the common divisor to obtain a primitive integral right triangle of square area, chosen with minimal hypotenuse. Write its sides as

$$2mn, \quad m^2 - n^2, \quad m^2 + n^2,$$

where $m > n$, $\gcd(m, n) = 1$, and m, n have opposite parity. Its area is

$$mn(m + n)(m - n).$$

The four displayed factors are pairwise coprime, so their product can be a square only if

$$m = e^2, \quad n = f^2, \quad m + n = g^2, \quad m - n = h^2.$$

The even parameter must be n . Indeed, if m were even, then $n = f^2$ would be odd, whereas $g^2 - h^2 = 2n$ is impossible: the difference of two odd squares is divisible by 8, while $2n$ is not. Thus f is even. Set

$$u = \frac{g + h}{2}, \quad v = \frac{g - h}{2}.$$

Now

$$u^2 + v^2 = e^2, \quad \frac{uv}{2} = \left(\frac{f}{2}\right)^2.$$

This is a smaller integral right triangle with square area and hypotenuse $e < m^2 + n^2$, contradicting minimality. Hence no rational right triangle has area 1, and a rational point on E must have $y = 0$.

The equation then gives $x(x-1)(x+1) = 0$, producing exactly the three affine points in (1), together with O . Each nonzero affine point is annihilated by 2, so

$$E(\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}.$$

□

Exercise II.6.7

Permitted context.

Let k be algebraically closed of characteristic different from 2, and let

$$X : y^2z = x^3 + x^2z$$

be the nodal plane cubic. Its node is $P = [0 : 0 : 1]$. We write $\text{CaCl}^0(X)$ for Cartier divisors of degree zero modulo principal Cartier divisors, and identify $\mathbb{G}_m(k)$ with k^* . For a required branch-value ratio 1, use the constant function 1; fractional linear interpolation is needed only for unequal values.

Problem. Compute the degree-zero Cartier divisor class group of X , and prove naturally that

$$\text{CaCl}^0(X) \cong \mathbb{G}_m(k).$$

Solution. On the chart $z = 1$, put $t = y/x$ in the function field. The equation gives

$$x = t^2 - 1, \quad y = t(t^2 - 1),$$

so $K(X) = k(t)$. The normalization is $\mathbb{P}^1$, and its map to X identifies the two points $t = 1$ and $t = -1$ to form the node. More precisely,

$$\mathcal{O}_{X,P} = k[t^2 - 1, t(t^2 - 1)]_{(t^2-1, t(t^2-1))} \subseteq k(t).$$

Inside the semilocal normalization ring at $t = 1, -1$, this is exactly the set of rational functions that are regular at both points and have the same value there. Indeed, the inclusion from left to right is immediate. Conversely, if $a(t)/b(t)$ has equal values at 1 and -1 , multiply numerator and denominator by $b(-t)$; both resulting polynomials have equal values at the two points, hence belong to $k + (t^2 - 1)k[t]$.

Every Cartier divisor is linearly equivalent to one supported away from P : subtract a rational function that is a local equation for the divisor at P . Two degree-zero divisors supported in $X \setminus \{P\}$ represent the same Cartier class precisely when their difference is the divisor of an element of $\mathcal{O}_{X,P}^*$. Under normalization we therefore obtain

$$\text{CaCl}^0(X) \cong \frac{\text{Div}^0(\mathbb{P}^1 \setminus \{1, -1\})}{\{\text{div}(u) : u(1) = u(-1) \neq 0\}}. \quad (1)$$

For a degree-zero divisor D in the numerator, choose $r \in k(t)^*$ with $D = \operatorname{div}_{\mathbb{P}^1}(r)$. Because the support avoids 1 and -1 , the two values $r(1), r(-1)$ are defined and nonzero. The function r is unique up to a nonzero scalar, so

$$\Phi(D) = \frac{r(1)}{r(-1)} \in k^*$$

is well defined and multiplicative. Its kernel consists exactly of the divisors in the denominator of (1). It is surjective: for any $\lambda \in k^*$, a fractional linear function can be chosen with neither a zero nor a pole at ± 1 and with ratio of values λ . Thus Φ induces the required natural isomorphism

$$\operatorname{CaCl}^0(X) \xrightarrow{\sim} k^* = \mathbb{G}_m(k).$$

□

Exercise II.6.8

Permitted context.

The group law on $\operatorname{Pic}(Y)$ is tensor product. For a morphism $f : X \rightarrow Y$, pullback of an $\mathcal{O}_Y$ -module is

$$f^*\mathcal{F} = f^{-1}\mathcal{F} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$

On a nonsingular integral scheme, divisor classes and invertible sheaves correspond by $D \mapsto \mathcal{O}(D)$.

Problem. (a) Prove that pullback induces a homomorphism

$$f^* : \operatorname{Pic}(Y) \longrightarrow \operatorname{Pic}(X).$$

(b) If f is a finite morphism of nonsingular curves, prove that this homomorphism agrees, under the divisor–line–bundle correspondence, with pullback of divisors including ramification multiplicities.

(c) Let $X \subseteq \mathbb{P}_k^n$ be an integral locally factorial closed subscheme. Prove that restriction of invertible sheaves along the inclusion agrees with the intersection homomorphism $\operatorname{Cl}(\mathbb{P}^n) \rightarrow \operatorname{Cl}(X)$ of Exercise II.6.2.

Solution. (a) An invertible sheaf is locally isomorphic to the structure sheaf, so its pullback is again invertible. There are natural isomorphisms

$$f^*(\mathcal{L} \otimes \mathcal{M}) \cong f^*\mathcal{L} \otimes f^*\mathcal{M}, \quad f^*(\mathcal{L}^{-1}) \cong (f^*\mathcal{L})^{-1}.$$

They show that the isomorphism class of $f^*\mathcal{L}$ depends only on that of $\mathcal{L}$, preserves the tensor-product law, and preserves inverses. Hence it defines the claimed group homomorphism.

(b) It is enough to treat a prime divisor $P \in Y$. Let t be a uniformizer at P . For every $Q \in X$ above P , the local rings are discrete valuation rings and

$$v_Q(f^\#t) = e(Q/P),$$

the ramification index. Thus the pulled-back Cartier divisor has associated Weil divisor

$$f^*P = \sum_{Q \mapsto P} e(Q/P)Q.$$

Locally $\mathcal{O}_Y(P)$ is the fractional ideal generated by t^{-1} . Tensoring with $\mathcal{O}_X$ makes it the fractional ideal generated by $(f^\#t)^{-1}$, whose divisor is the displayed pullback. Therefore, for every divisor D ,

$$f^*\mathcal{O}_Y(D) \cong \mathcal{O}_X(f^*D).$$

This proves compatibility on class groups.

(c) Local factoriality identifies Weil and Cartier divisors on X . Choose a hyperplane H not containing X . Its defining linear form trivializes $\mathcal{O}_{\mathbb{P}^n}(H)$ away from H ; after restriction to X , the same local equation defines the Cartier divisor $H \cdot X$. Consequently

$$\mathcal{O}_{\mathbb{P}^n}(H)|_X \cong \mathcal{O}_X(H \cdot X).$$

Both homomorphisms in question are additive, and $\text{Cl}(\mathbb{P}^n) = \mathbb{Z}[H]$. Agreement on H therefore proves agreement for every class. $\square$

Exercise II.6.9

Permitted context.

Let X be an integral projective curve over an algebraically closed field k , let $\pi : \tilde{X} \rightarrow X$ be its normalization, and for $P \in X$ write

$$\tilde{\mathcal{O}}_P$$

for the integral closure of $\mathcal{O}_{X,P}$ in $K(X)$. This is a semilocal ring when P has more than one branch. Only finitely many of the local quotients below are nonzero. An invertible module over a semilocal ring is free. Thus for a finite normalization π , a line bundle becomes trivial above a sufficiently small neighborhood of any point, and $R^1\pi_*\mathcal{O}_{\tilde{X}}^* = 0$. The Leray sequence then identifies the two first cohomology groups used below.

Problem. (a) Prove the exact sequence

$$0 \longrightarrow \bigoplus_{P \in X} \tilde{\mathcal{O}}_P^* / \mathcal{O}_{X,P}^* \longrightarrow \text{Pic}(X) \xrightarrow{\pi^*} \text{Pic}(\tilde{X}) \longrightarrow 0.$$

(b) Apply it to prove

$$0 \longrightarrow \mathbb{G}_a(k) \longrightarrow \text{Pic}(X) \longrightarrow \mathbb{Z} \longrightarrow 0$$

for a cuspidal plane cubic, and

$$0 \longrightarrow \mathbb{G}_m(k) \longrightarrow \text{Pic}(X) \longrightarrow \mathbb{Z} \longrightarrow 0$$

for a nodal plane cubic.

Solution. (a) Inside the constant sheaf $K(X)^*$, there is an exact sequence of sheaves of abelian groups

$$1 \longrightarrow \mathcal{O}_X^* \longrightarrow \pi_*\mathcal{O}_{\tilde{X}}^* \longrightarrow \mathcal{Q} \longrightarrow 1. \quad (1)$$

Localization commutes with integral closure for this finite normalization, so

$$\mathcal{Q}_P \cong \tilde{\mathcal{O}}_P^* / \mathcal{O}_{X,P}^*.$$

The normalization is an isomorphism over the regular locus, whose complement is a finite set. Hence $\mathcal{Q}$ is the finite direct sum of the corresponding skyscraper sheaves. In particular,

$$H^0(X, \mathcal{Q}) = \bigoplus_{P \in X} \tilde{\mathcal{O}}_P^* / \mathcal{O}_{X,P}^*, \quad H^1(X, \mathcal{Q}) = 0. \quad (2)$$

Both X and $\tilde{X}$ are integral and projective over k , so their global units are k^* . The map between these two unit groups is an isomorphism. Taking cohomology in (1), using (2) and

$$H^1(X, \pi_* \mathcal{O}_{\tilde{X}}^*) = H^1(\tilde{X}, \mathcal{O}_{\tilde{X}}^*),$$

therefore gives exactly

$$0 \longrightarrow H^0(X, \mathcal{Q}) \longrightarrow \text{Pic}(X) \xrightarrow{\pi^*} \text{Pic}(\tilde{X}) \longrightarrow 0.$$

(b) The normalization of either irreducible cubic is $\mathbb{P}^1$, whose Picard group is $\mathbb{Z}$. At the cusp one may choose a parameter t for which

$$\mathcal{O}_{X,P} = k[t^2, t^3]_{(t^2, t^3)}, \quad \tilde{\mathcal{O}}_P = k[t]_{(t)}.$$

The conductor contains $t^2 k[t]_{(t)}$. Consequently the quotient of unit groups is measured by the coefficient of t :

$$\tilde{\mathcal{O}}_P^* / \mathcal{O}_{X,P}^* \xrightarrow{\sim} (k, +), \quad u \longmapsto \frac{u'(0)}{u(0)}.$$

The map is a homomorphism by the product rule, is onto using $1 + ct$, and has kernel precisely the units with no linear term, which are the units of $\mathcal{O}_{X,P}$. Part (a) gives the cuspidal sequence.

At the node the normalization has two points above P . With a parameter t taking the distinct values 0 and 1 there, the local ring consists of functions regular at both points and having equal values. Thus

$$\tilde{\mathcal{O}}_P^* / \mathcal{O}_{X,P}^* \xrightarrow{\sim} k^*, \quad u \longmapsto \frac{u(0)}{u(1)}.$$

Its kernel is exactly the local unit group downstairs, and surjectivity follows by using the polynomial $u(t) = \lambda + (1 - \lambda)t$ for the ratio $\lambda \in k^*$. It is a unit at both branch points; for $\lambda = 1$ it is constant. Applying (a) again gives the nodal sequence. $\square$

Exercise II.6.10**Permitted context.**

For a noetherian scheme X , let $K(X)$ be the abelian group generated by symbols $[\mathcal{F}]$ for coherent $\mathcal{O}_X$ -modules, subject to

$$[\mathcal{F}] = [\mathcal{F}'] + [\mathcal{F}'']$$

for every exact sequence $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$. This is the Grothendieck group of coherent sheaves, often denoted $G_0(X)$. The category of coherent sheaves on an open $U \subseteq X$ is the Serre quotient of $\text{Coh}(X)$ by coherent sheaves supported in $X \setminus U$. Morphisms in this quotient allow kernels and cokernels supported in that complement.

Problem. (a) Compute $K(\mathbb{A}_k^1)$.

(b) If X is integral, prove that generic rank induces a surjective homomorphism

$$\text{rank} : K(X) \longrightarrow \mathbb{Z}.$$

(c) For a closed subscheme $i : Y \hookrightarrow X$, with open complement $j : U \hookrightarrow X$, prove exactness of

$$K(Y) \xrightarrow{i_*} K(X) \xrightarrow{j^*} K(U) \longrightarrow 0.$$

Solution. (a) Write $\mathbb{A}_k^1 = \text{Spec } k[t]$. Every finitely generated $k[t]$ -module M admits an exact sequence

$$0 \longrightarrow k[t]^s \longrightarrow k[t]^r \longrightarrow M \longrightarrow 0,$$

because the kernel of a map between finite free modules over the principal ideal domain $k[t]$ is finite free. Therefore

$$[M] = (r - s)[k[t]],$$

so $K(\mathbb{A}_k^1)$ is cyclic. Generic rank sends $[k[t]]$ to 1, and hence proves

$$K(\mathbb{A}_k^1) \cong \mathbb{Z}.$$

(b) Let η be the generic point and put $K_\eta = \mathcal{O}_{X,\eta}$, the function field. Taking the stalk at η is exact, so any exact sequence of coherent sheaves gives an exact sequence of finite-dimensional K_η -vector spaces. Dimension is additive, and hence

$$[\mathcal{F}] \longmapsto \dim_{K_\eta} \mathcal{F}_\eta$$

respects all defining relations of $K(X)$. It is onto because $\mathcal{O}_X$ has rank one.

(c) Both maps are well defined: extension by zero along a closed immersion and restriction to an open subscheme are exact and preserve coherence. Their composite is zero. Moreover, every coherent sheaf on U extends to a coherent sheaf on the noetherian scheme X , so j^* is surjective on Grothendieck groups.

Let $\text{Coh}_Y(X)$ be the full subcategory of coherent sheaves whose set-theoretic support lies in Y . It is a Serre subcategory: it is closed under subobjects, quotients, and extensions. Restriction realizes

$$\text{Coh}(U) \cong \text{Coh}(X) / \text{Coh}_Y(X). \quad (1)$$

Essential surjectivity is the extension theorem just used. A morphism on U extends after replacing its source and target by subsheaves with quotient supported on Y ; taking the coherent closure of its graph gives the morphism statement in (1).

For any abelian category $\mathcal{A}$ and Serre subcategory $\mathcal{B}$, the generators-and-relations definition gives an exact sequence

$$K_0(\mathcal{B}) \longrightarrow K_0(\mathcal{A}) \longrightarrow K_0(\mathcal{A}/\mathcal{B}) \longrightarrow 0. \quad (2)$$

Indeed, objects of the quotient have the same generators, while a sequence that becomes exact in the quotient can be corrected by kernels and cokernels belonging to $\mathcal{B}$. Applying (2) to (1), the kernel of $K(X) \rightarrow K(U)$ is generated by classes of coherent sheaves supported on Y .

It remains to see that all these classes come from $K(Y)$. Let $\mathcal{I}$ define Y , and let $\mathcal{F}$ be coherent with support in Y . Noetherianity and the support condition imply $\mathcal{I}^N \mathcal{F} = 0$ for some N . The finite filtration

$$\mathcal{F} \supseteq \mathcal{I}\mathcal{F} \supseteq \cdots \supseteq \mathcal{I}^N \mathcal{F} = 0$$

has quotients annihilated by $\mathcal{I}$, hence coherent $\mathcal{O}_Y$ -modules. Additivity expresses $[\mathcal{F}]$ as a sum of their extensions by zero. Thus the kernel is exactly the image of $K(Y)$, proving the sequence. $\square$

Exercise II.6.11**Permitted context.**

Let X be a nonsingular integral curve over an algebraically closed field. Write $K(X)$ for the Grothendieck group of coherent sheaves, and $[\mathcal{F}]$ for the class of $\mathcal{F}$. On this regular curve, $\text{Cl}(X) = \text{Pic}(X)$, and every local ring at a closed point is a discrete valuation ring.

Problem. (a) For a divisor $D = \sum n_P P$, define

$$\psi(D) = \sum_P n_P [k(P)] \in K(X).$$

If D is effective, prove $\psi(D) = [\mathcal{O}_D]$. Show that ψ depends only on the linear-equivalence class and hence defines $\psi : \text{Pic}(X) \rightarrow K(X)$.

(b) Prove that every coherent sheaf has a locally free resolution

$$0 \longrightarrow \mathcal{E}_1 \longrightarrow \mathcal{E}_0 \longrightarrow \mathcal{F} \longrightarrow 0.$$

Define

$$\det \mathcal{F} = \bigwedge^{r_0} \mathcal{E}_0 \otimes \left(\bigwedge^{r_1} \mathcal{E}_1 \right)^{-1}.$$

Prove independence of the resolution, additivity in short exact sequences, and

$$\det(\psi(D)) \cong \mathcal{O}_X(D).$$

(c) If $\mathcal{F}$ has rank r , construct a divisor D , a torsion sheaf $\mathcal{T}$, and an exact sequence

$$0 \longrightarrow \mathcal{O}_X(D)^{\oplus r} \longrightarrow \mathcal{F} \longrightarrow \mathcal{T} \longrightarrow 0.$$

Deduce that $[\mathcal{F}] - r[\mathcal{O}_X]$ lies in the image of ψ .

(d) Use rank, determinant, and ψ to prove

$$K(X) \cong \text{Pic}(X) \oplus \mathbb{Z}.$$

Solution. (a) At a point P , the local ring is a discrete valuation ring. If $D = \sum n_P P$ is effective, the stalk of $\mathcal{O}_D$ at P has length n_P and a filtration whose successive quotients are $k(P)$. Additivity in $K(X)$ gives

$$[\mathcal{O}_D] = \sum_P n_P [k(P)] = \psi(D). \quad (1)$$

The ideal sequence of an effective divisor is

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0,$$

so

$$\psi(D) = [\mathcal{O}_X] - [\mathcal{O}_X(-D)]. \quad (2)$$

If $D \sim D'$, choose an effective divisor E such that both $D + E$ and $D' + E$ are effective. The corresponding invertible sheaves $\mathcal{O}_X(-D - E)$ and $\mathcal{O}_X(-D' - E)$ are isomorphic, so (2) gives

$\psi(D + E) = \psi(D' + E)$, and cancellation of $\psi(E)$ gives $\psi(D) = \psi(D')$. Thus ψ descends to the Picard group.

(b) A nonsingular curve is regular and quasi-projective. By clearing poles of finitely many generators on a finite affine cover, one gets a surjection from a finite direct sum of invertible sheaves $\mathcal{E}_0 \twoheadrightarrow \mathcal{F}$. Let $\mathcal{E}_1$ be its kernel. At each closed point, $(\mathcal{E}_1)_P$ is a submodule of a finite free module over a discrete valuation ring, hence is free. Coherence then makes $\mathcal{E}_1$ locally free. This gives the required resolution.

Suppose two such resolutions use $(\mathcal{E}_1, \mathcal{E}_0)$ and $(\mathcal{E}'_1, \mathcal{E}'_0)$. Their fiber product over $\mathcal{F}$ is a locally free sheaf $\mathcal{E}$, fitting into

$$0 \rightarrow \mathcal{E}_1 \rightarrow \mathcal{E} \rightarrow \mathcal{E}'_0 \rightarrow 0, \quad 0 \rightarrow \mathcal{E}'_1 \rightarrow \mathcal{E} \rightarrow \mathcal{E}_0 \rightarrow 0.$$

Both sequences split locally because their quotients are locally free. Multiplicativity of top exterior powers gives

$$\det \mathcal{E}_1 \otimes \det \mathcal{E}'_0 \cong \det \mathcal{E} \cong \det \mathcal{E}'_1 \otimes \det \mathcal{E}_0,$$

which proves independence of the resolution.

Now let $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ be exact. Choose a locally free surjection $\mathcal{E} \twoheadrightarrow \mathcal{F}$, and put

$$\mathcal{K} = \ker(\mathcal{E} \rightarrow \mathcal{F}), \quad \mathcal{K}'' = \ker(\mathcal{E} \rightarrow \mathcal{F}'').$$

Both kernels are locally free, and there are resolutions

$$0 \rightarrow \mathcal{K} \rightarrow \mathcal{K}'' \rightarrow \mathcal{F}' \rightarrow 0, \quad 0 \rightarrow \mathcal{K} \rightarrow \mathcal{E} \rightarrow \mathcal{F} \rightarrow 0, \quad 0 \rightarrow \mathcal{K}'' \rightarrow \mathcal{E} \rightarrow \mathcal{F}'' \rightarrow 0.$$

Substitution in the definition yields

$$\det \mathcal{F} \cong \det \mathcal{F}' \otimes \det \mathcal{F}''.$$

Thus determinant defines a homomorphism $\det : K(X) \rightarrow \text{Pic}(X)$.

For effective D , the ideal sequence above is a locally free resolution of $\mathcal{O}_D$, so

$$\det(\psi(D)) = \det \mathcal{O}_D \cong \mathcal{O}_X(D).$$

Additivity extends this identity to every divisor.

(c) Choose a basis $s_1, \dots, s_r$ of the generic fiber $\mathcal{F}_\eta$. These are rational sections of $\mathcal{F}$. Their poles occur at finitely many closed points, so a sufficiently large effective divisor E clears all of them and produces a map

$$\mathcal{O}_X(-E)^{\oplus r} \rightarrow \mathcal{F}$$

that is an isomorphism at the generic point. Its kernel has generic rank zero but is a subsheaf of a locally free sheaf, which is torsion-free; hence the kernel is zero. Its cokernel $\mathcal{T}$ has generic stalk zero and is therefore a torsion sheaf. Taking $D = -E$ proves the asserted sequence.

For the divisor $D = -E$ just constructed, equation (2) for the effective E gives

$$[\mathcal{O}_X(D)] - [\mathcal{O}_X] = \psi(D). \tag{3}$$

Moreover, a torsion sheaf has finite length at finitely many points, so a composition series gives

$$[\mathcal{T}] = \psi \left(\sum_P \text{length}_{\mathcal{O}_{X,P}}(\mathcal{T}_P) P \right).$$

The sequence in (c) and (3) now show that

$$[\mathcal{F}] - r[\mathcal{O}_X] = r\psi(D) + [\mathcal{T}] \in \text{im } \psi.$$

(d) Define

$$\Theta : K(X) \longrightarrow \text{Pic}(X) \oplus \mathbb{Z}, \quad \alpha \longmapsto (\det \alpha, \text{rank } \alpha),$$

and

$$\Psi : \text{Pic}(X) \oplus \mathbb{Z} \longrightarrow K(X), \quad (\mathcal{O}_X(D), n) \longmapsto \psi(D) + n[\mathcal{O}_X].$$

The map Ψ is well defined by (a). Part (b) and the fact that $\psi(D)$ has rank zero show $\Theta\Psi = \text{id}$. Part (c) says every class in $K(X)$ is $n[\mathcal{O}_X] + \psi(D)$; applying determinant identifies $\mathcal{O}_X(D)$ with its first component. Hence $\Psi\Theta = \text{id}$ as well, proving the claimed isomorphism. $\square$

Exercise II.6.12

Permitted context.

Let X be a complete nonsingular integral curve over an algebraically closed field. Exercise II.6.11 gives

$$K(X) \cong \text{Pic}(X) \oplus \mathbb{Z}$$

through determinant and rank.

Problem. Prove that there is a unique integer-valued degree for coherent sheaves on X satisfying all three properties:

- (1) $\deg \mathcal{O}_X(D) = \deg D$ for every divisor D ;
- (2) if $\mathcal{T}$ is torsion, then

$$\deg \mathcal{T} = \sum_{P \in X} \text{length}_{\mathcal{O}_{X,P}}(\mathcal{T}_P);$$

- (3) degree is additive in every short exact sequence of coherent sheaves.

Solution. Additivity means precisely that the desired function must factor through a homomorphism

$$\deg : K(X) \longrightarrow \mathbb{Z}.$$

Under the decomposition of Exercise II.6.11, define

$$\deg(\alpha) = \deg(\det \alpha). \tag{1}$$

Here the degree on the right is the usual degree homomorphism $\text{Pic}(X) \rightarrow \mathbb{Z}$. Since determinant is multiplicative in short exact sequences, (1) is additive.

For a line bundle $\mathcal{O}_X(D)$, its determinant is itself, so (1) gives property (1). If $\mathcal{T}$ is torsion, a composition series over the discrete valuation rings gives

$$[\mathcal{T}] = \sum_P \text{length}_{\mathcal{O}_{X,P}}(\mathcal{T}_P)[k(P)].$$

Exercise II.6.11 identifies $\det[k(P)]$ with $\mathcal{O}_X(P)$, of degree one. Formula (1) therefore gives property (2).

Finally, uniqueness follows from the same decomposition. Every class has the form

$$\psi(D) + r[\mathcal{O}_X].$$

Properties (1) and (2) force its degree to be $\deg D$, while property (1) with $D = 0$ forces $\deg \mathcal{O}_X = 0$. Additivity then determines the value on every class, so no other degree function can satisfy the three conditions. $\square$

II.7 Projective Morphisms

Exercise II.7.1

Permitted context.

Let $(X, \mathcal{O}_X)$ be a locally ringed space. An invertible sheaf has stalk isomorphic to $\mathcal{O}_{X,x}$ at every point x .

Problem. Prove that every surjective morphism

$$\mathcal{L} \longrightarrow \mathcal{M}$$

between invertible $\mathcal{O}_X$ -modules is an isomorphism.

Solution. It is enough to check the morphism on stalks. Fix $x \in X$, choose bases of the free rank-one modules $\mathcal{L}_x$ and $\mathcal{M}_x$, and identify the stalk map with multiplication by some $a \in \mathcal{O}_{X,x}$. Surjectivity gives $b \in \mathcal{O}_{X,x}$ with $ab = 1$. Thus a is a unit and multiplication by a is an isomorphism. Every stalk map is therefore an isomorphism, so the original sheaf morphism is an isomorphism. $\square$

Exercise II.7.2

Permitted context.

Let X be a k -scheme and $\mathcal{L}$ an invertible sheaf. Suppose

$$s_0, \dots, s_n, \quad t_0, \dots, t_m$$

are two base-point-free lists of global sections, both spanning the same vector subspace $V \subseteq \Gamma(X, \mathcal{L})$, with $n \leq m$. They define morphisms $\phi : X \rightarrow \mathbb{P}_k^n$ and $\psi : X \rightarrow \mathbb{P}_k^m$. When $m = n$, the projection center of dimension -1 means the empty set.

Problem. Show that, after an automorphism of $\mathbb{P}^n$, the morphism ϕ is the composite of ψ with a linear projection

$$\mathbb{P}^m \setminus L \longrightarrow \mathbb{P}^n,$$

where L is a linear subspace of dimension $m - n - 1$, and prove that $\psi(X) \cap L = \emptyset$.

Solution. Let

$$q_s : k^{n+1} \twoheadrightarrow V, \quad q_t : k^{m+1} \twoheadrightarrow V$$

send the standard bases to the two lists of sections. Choose complements to their kernels. Since

$$\dim \ker q_t = m + 1 - \dim V \geq n + 1 - \dim V = \dim \ker q_s,$$

we can lift q_s through q_t by an injective linear map

$$B : k^{n+1} \hookrightarrow k^{m+1}, \quad q_t B = q_s. \tag{1}$$

Equivalently, the s_i can be written as linear combinations

$$s_i = \sum_{j=0}^m b_{ji} t_j$$

whose $n + 1$ coefficient vectors are linearly independent. A change of basis in k^{n+1} accounts for the allowed automorphism of the target $\mathbb{P}^n$.

Let $\ell_0, \dots, \ell_n$ be the corresponding independent linear forms on the homogeneous coordinates of $\mathbb{P}^m$. Their common zero locus

$$L = V_+(\ell_0, \dots, \ell_n)$$

has dimension $m - n - 1$, and the tuple $[\ell_0 : \dots : \ell_n]$ defines the linear projection away from L . Pulling these forms back along ψ gives exactly the sections $s_0, \dots, s_n$. They generate $\mathcal{L}$ at every point, so they never vanish simultaneously; hence $\psi(X) \cap L = \emptyset$. The projected morphism is therefore defined on all of $\psi(X)$, and its homogeneous coordinates are those defining ϕ . This proves the assertion. $\square$

Exercise II.7.3

Permitted context.

Let $n \geq 1$ and let $\phi : \mathbb{P}_k^n \rightarrow \mathbb{P}_k^m$ be a morphism. If the image lies in a proper linear subspace of $\mathbb{P}^m$, a factorization by a linear projection naturally ends with the inclusion of that linear span.

Thus the usual direct projection statement is read after replacing $\mathbb{P}^m$ by the linear span of the image.

Problem. (a) Prove that either ϕ is constant, or

$$m \geq n, \quad \dim \phi(\mathbb{P}^n) = n.$$

(b) In the nonconstant case, prove that there is a unique integer $d \geq 1$ such that ϕ is induced by base-point-free degree d forms. Factor it through the d -uple Veronese embedding, a linear projection, and a change of projective coordinates, with the linear-span qualification stated above. Prove also that ϕ is finite and hence has finite fibers.

Solution. (a) The morphism is represented by an invertible sheaf

$$\phi^* \mathcal{O}_{\mathbb{P}^m}(1) \cong \mathcal{O}_{\mathbb{P}^n}(d)$$

and $m + 1$ global sections $p_0, \dots, p_m$ having no common zero. There are no nonzero sections when $d < 0$. If $d = 0$, all sections are constants and the morphism is constant. We henceforth assume $d > 0$.

Put $S = k[y_0, \dots, y_n]$. Since the homogeneous ideal $I = (p_0, \dots, p_m)$ has empty projective zero locus, its radical contains $S_+ = (y_0, \dots, y_n)$. Consequently

$$S_q \subseteq I \quad (q \geq N) \tag{1}$$

for some N . On the target chart $D_+(T_i)$, the inverse image is

$$D_+(p_i) = \text{Spec}(S_{p_i})_0.$$

Let

$$B_i = k[p_0/p_i, \dots, p_m/p_i] \subseteq (S_{p_i})_0.$$

If $h \in S_{qd}$ and $qd \geq N$, (1) gives homogeneous $g_j \in S_{(q-1)d}$ such that $h = \sum_j p_j g_j$. Hence

$$\frac{h}{p_i^q} = \sum_j \frac{p_j}{p_i} \frac{g_j}{p_i^{q-1}}. \tag{2}$$

Repeated use of (2) shows that $(S_{p_i})_0$ is generated as a B_i -module by the finitely many fractions

$$\frac{y^\alpha}{p_i^q}, \quad |\alpha| = qd < N.$$

Thus every affine restriction of ϕ is finite, and ϕ is a finite morphism.

Its image is therefore closed, and the induced finite dominant map from $\mathbb{P}^n$ to that image preserves dimension. Hence $\dim \phi(\mathbb{P}^n) = n$, which forces $m \geq n$. This also proves the finite-fiber assertion needed in (b).

(b) For the factorization, let

$$N_d = \binom{n+d}{d} - 1.$$

All degree- d monomials define the Veronese closed immersion

$$v_d : \mathbb{P}^n \hookrightarrow \mathbb{P}^{N_d}.$$

The forms p_i are linear combinations of those monomials. If $W \subseteq H^0(\mathbb{P}^n, \mathcal{O}(d))$ is their span and $\dim W = r$, these combinations define a linear projection

$$\mathbb{P}^{N_d} \setminus L \longrightarrow \mathbb{P}^{r-1}$$

whose center misses the Veronese image, because the p_i have no common zero. After choosing a basis of W , the original map is this projection followed by the linear inclusion $\mathbb{P}^{r-1} \hookrightarrow \mathbb{P}^m$ and an automorphism of $\mathbb{P}^m$. When the image spans $\mathbb{P}^m$, one has $r = m + 1$, so this is the stated direct linear projection to $\mathbb{P}^m$.

Finally, d is unique because it is determined by the isomorphism class of $\phi^* \mathcal{O}_{\mathbb{P}^m}(1)$ in $\text{Pic}(\mathbb{P}^n) \cong \mathbb{Z}$. $\square$

Exercise II.7.4

Permitted context.

Let A be noetherian. We use the criterion that an invertible sheaf on a finite-type A -scheme is ample exactly when some positive power gives an immersion into a projective space over A .

The affine line with doubled origin is obtained by gluing two copies $U, V \cong \mathbb{A}_k^1$ by the identity on $U \cap V \cong \mathbb{G}_m$.

Problem. (a) Prove that a finite-type A -scheme possessing an ample invertible sheaf is separated over A .

(b) For the affine line with doubled origin, compute the Picard group and determine exactly which invertible sheaves are generated by global sections. Use this computation directly to prove that the scheme has no ample invertible sheaf.

Solution. (a) If $\mathcal{L}$ is ample, the cited criterion supplies an immersion

$$X \longrightarrow \mathbb{P}_A^N.$$

An immersion is separated, projective space is separated over A , and a composite of separated morphisms is separated. Hence $X \rightarrow \text{Spec } A$ is separated.

(b) Every invertible sheaf on either U or V is trivial. After choosing trivializations, gluing is multiplication on $\mathbb{G}_m$ by a unit of

$$k[x, x^{-1}]^* = k^* x^{\mathbb{Z}}.$$

Changing either trivialization multiplies this unit by an element of k^* . Therefore

$$\mathrm{Pic}(X) \cong k[x, x^{-1}]^* / (k[x]^* k[x]^*) \cong \mathbb{Z}.$$

Let $\mathcal{L}_r$ denote the class represented by x^r .

Choose local bases so that $e_U = x^r e_V$ on the overlap. A global section is a pair $f e_U, g e_V$, with $f, g \in k[x]$, satisfying

$$g = x^r f \quad \text{in } k[x, x^{-1}].$$

If $r > 0$, every such g vanishes at the origin in V , so global sections do not generate the fiber at that origin. If $r < 0$, every f vanishes at the origin in U . For $r = 0$, the sheaf is trivial and is generated by 1. Thus

$$\mathcal{L}_r \text{ is globally generated} \iff r = 0. \quad (1)$$

Suppose $\mathcal{L}_r$ were ample. Apply the global-generation definition of ampleness to $\mathcal{L}_1$. It would require

$$\mathcal{L}_1 \otimes \mathcal{L}_r^{\otimes n} = \mathcal{L}_{1+nr}$$

to be globally generated for every sufficiently large n . By (1), this would force $1 + nr = 0$ for every sufficiently large n , which is impossible. Hence no invertible sheaf on the doubled line is ample. $\square$

Exercise II.7.5

Permitted context.

Let X be noetherian and let $\mathcal{L}, \mathcal{M}$ be invertible sheaves. In parts (d) and (e), assume that X is of finite type over a noetherian ring A , and interpret very ample relative to $\mathrm{Spec} A$, using an immersion into a single $\mathbb{P}_A^N$ to define very ampleness.

We repeatedly use the elementary fact that the tensor product of two globally generated sheaves is globally generated: tensor products of generating global sections generate every stalk.

Problem. Prove:

- (a) if $\mathcal{L}$ is ample and $\mathcal{M}$ is globally generated, then $\mathcal{L} \otimes \mathcal{M}$ is ample;
- (b) if $\mathcal{L}$ is ample and $\mathcal{M}$ is arbitrary, then $\mathcal{M} \otimes \mathcal{L}^n$ is ample for all sufficiently large n ;
- (c) the tensor product of two ample invertible sheaves is ample;
- (d) if $\mathcal{L}$ is very ample and $\mathcal{M}$ is globally generated, then $\mathcal{L} \otimes \mathcal{M}$ is very ample;
- (e) if $\mathcal{L}$ is ample, then $\mathcal{L}^n$ is very ample for every sufficiently large n .

Solution. (a) Let $\mathcal{F}$ be coherent. For $q \gg 0$, $\mathcal{F} \otimes \mathcal{L}^q$ is globally generated. So is $\mathcal{M}^q$, because $\mathcal{M}$ is globally generated. Therefore

$$\mathcal{F} \otimes (\mathcal{L} \otimes \mathcal{M})^q \cong (\mathcal{F} \otimes \mathcal{L}^q) \otimes \mathcal{M}^q$$

is globally generated for $q \gg 0$. This is the global-generation criterion for ampleness.

(b) Ampleness of $\mathcal{L}$, applied to $\mathcal{M}$, gives an N such that $\mathcal{M} \otimes \mathcal{L}^q$ is globally generated for every $q \geq N$. For $n \geq N + 1$, write

$$\mathcal{M} \otimes \mathcal{L}^n = \mathcal{L} \otimes (\mathcal{M} \otimes \mathcal{L}^{n-1}).$$

Part (a) makes this sheaf ample.

(c) For a coherent $\mathcal{F}$, the sheaves $\mathcal{F} \otimes \mathcal{L}^q$ and $\mathcal{M}^q$ are both globally generated once q is large. Their tensor product is $\mathcal{F} \otimes (\mathcal{L} \otimes \mathcal{M})^q$, so the same criterion proves that $\mathcal{L} \otimes \mathcal{M}$ is ample.

(d) Choose sections $s_0, \dots, s_r$ realizing a very ample immersion for $\mathcal{L}$, and finite generating sections $t_0, \dots, t_s$ for $\mathcal{M}$. They define an immersion and a morphism

$$i : X \longrightarrow \mathbb{P}_A^r, \quad j : X \longrightarrow \mathbb{P}_A^s.$$

The products $s_a t_b$ define the composite

$$X \xrightarrow{(i,j)} \mathbb{P}_A^r \times_A \mathbb{P}_A^s \xrightarrow{\text{Segre}} \mathbb{P}_A^{(r+1)(s+1)-1}. \quad (1)$$

The map (i, j) is an immersion: it factors through the closed graph of j in $X \times_A \mathbb{P}_A^s$, followed by the base change of the immersion i . The Segre map is a closed immersion, so (1) is an immersion. Its pulled-back $\mathcal{O}(1)$ is $\mathcal{L} \otimes \mathcal{M}$, proving very ampleness.

(e) Some power $\mathcal{L}^a$ is very ample by the ample-sheaf criterion. Also, there is b such that $\mathcal{L}^q$ is globally generated for every $q \geq b$. If $n \geq a + b$, then

$$\mathcal{L}^n = \mathcal{L}^a \otimes \mathcal{L}^{n-a},$$

and part (d) makes it very ample. □

Exercise II.7.6

Permitted context.

Let X be a nonsingular projective integral variety over an algebraically closed field, let D be a divisor, and put $\mathcal{L} = \mathcal{O}_X(D)$. We use the convention

$$\dim |qD| = h^0(X, \mathcal{L}^q) - 1,$$

so an empty linear system has dimension -1 .

Problem. (a) If D is very ample and gives an embedding $X \hookrightarrow \mathbb{P}^N$, prove that

$$\dim |qD| = P_X(q) - 1$$

for all sufficiently large q , where P_X is the Hilbert polynomial of this embedded variety.

(b) If the class of D has exact order r in $\text{Pic}(X)$, prove

$$\dim |qD| = \begin{cases} 0, & r \text{ divides } q, \\ -1, & r \text{ does not divide } q. \end{cases}$$

Thus the function is periodic with period r .

Solution. (a) Under the given embedding, $\mathcal{L} \cong \mathcal{O}_X(1)$, and hence $\mathcal{L}^q \cong \mathcal{O}_X(q)$. If S is the homogeneous coordinate ring of X , the natural map

$$S_q \longrightarrow \Gamma(X, \mathcal{O}_X(q))$$

is an isomorphism for $q \gg 0$. In the same range the Hilbert function $\dim_k S_q$ equals $P_X(q)$. Therefore

$$\dim |qD| = h^0(X, \mathcal{O}_X(q)) - 1 = P_X(q) - 1.$$

(b) If r divides q , then $\mathcal{L}^q \cong \mathcal{O}_X$. Since X is projective and integral, $\Gamma(X, \mathcal{O}_X) = k$, so the linear system has dimension zero.

Conversely, suppose $s \neq 0$ is a global section of $\mathcal{L}^q$. The line bundle $\mathcal{L}^q$ is torsion. If its order is a , then, after choosing a trivialization $(\mathcal{L}^q)^a \cong \mathcal{O}_X$, the tensor power s^a is a nonzero global regular function. It is therefore a nonzero constant. Thus s vanishes nowhere and trivializes $\mathcal{L}^q$. Exact order r now implies $r \mid q$. Hence there are no nonzero sections when $r \nmid q$, giving dimension -1 . $\square$

Exercise II.7.7

Permitted context.

Let k be algebraically closed. In part (b) assume $\text{char } k \neq 2$, which is needed for the stated five-dimensional subsystem to be an immersion. On $\mathbb{P}_k^2$, with coordinates x, y, z , the complete system of conics is $H^0(\mathcal{O}_{\mathbb{P}^2}(2))$.

Problem. (a) Show that the six quadratic monomials give the Veronese closed immersion $\mathbb{P}^2 \hookrightarrow \mathbb{P}^5$.

(b) Prove that the five sections

$$x^2, \quad y^2, \quad z^2, \quad y(x-z), \quad (x-y)z$$

give a closed immersion $\mathbb{P}^2 \hookrightarrow \mathbb{P}^4$.

(c) Fix $P \in \mathbb{P}^2(k)$. Show that the conics through P embed $\mathbb{P}^2 \setminus \{P\}$ in $\mathbb{P}^4$, extend to a closed immersion of $\tilde{X} = \text{Bl}_P \mathbb{P}^2$, and make $\tilde{X}$ a degree-three surface. Prove that strict transforms of lines through P are mutually disjoint projective lines covering $\tilde{X}$.

Solution. (a) The complete linear system $|\mathcal{O}_{\mathbb{P}^2}(2)|$ gives the second Veronese map

$$[x : y : z] \longmapsto [x^2 : y^2 : z^2 : xy : xz : yz].$$

The general d -uple embedding theorem, with $d = 2$, says that this is a closed immersion.

(b) View the map as projection of the Veronese surface. In coordinates

$$[X_0 : X_1 : X_2 : X_3 : X_4 : X_5] = [x^2 : y^2 : z^2 : xy : xz : yz],$$

the five sections are

$$X_0, \quad X_1, \quad X_2, \quad X_3 - X_5, \quad X_4 - X_5.$$

Thus the center of projection is

$$c = [0 : 0 : 0 : 1 : 1 : 1] \in \mathbb{P}^5.$$

Under the symmetric-matrix description of the quadratic Veronese embedding, its secant variety consists of symmetric matrices of rank at most 2. The point c corresponds, up to harmless nonzero scalars in the off-diagonal entries, to

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix},$$

whose determinant is 2. Hence c is outside the secant variety when $\text{char } k \neq 2$.

Projection from a point outside the secant variety restricts to a closed immersion: it separates every length-two subscheme, since failure to separate two points or a tangent vector would put the center on the corresponding secant or tangent line. Therefore the five displayed sections give a closed immersion into $\mathbb{P}^4$.

(c) Move P to $[0 : 0 : 1]$. The conics through P are spanned by

$$x^2, \quad xy, \quad y^2, \quad xz, \quad yz. \quad (1)$$

The blowup has the incidence presentation

$$\tilde{X} = \{([x : y : z], [s : t]) \in \mathbb{P}^2 \times \mathbb{P}^1 : xt = ys\}. \quad (2)$$

Away from P , the second coordinate is the direction $[s : t] = [x : y]$; over P , it records the exceptional direction. The Segre embedding of the product in (2) has coordinates

$$xs, \quad xt, \quad ys, \quad yt, \quad zs, \quad zt.$$

On $\tilde{X}$, the relation $xt = ys$ places the image in a linear $\mathbb{P}^4$, and substituting $[s : t] = [x : y]$ recovers, up to order, exactly the five sections in (1). Since (2) and the Segre map are closed immersions, the induced map $\tilde{X} \hookrightarrow \mathbb{P}^4$ is a closed immersion. On the complement of the exceptional curve it is the map defined by the conics through P .

Let H be the pullback of a line and E the exceptional curve. The hyperplane class of this embedding is $2H - E$. Using

$$H^2 = 1, \quad H \cdot E = 0, \quad E^2 = -1,$$

its degree is

$$(2H - E)^2 = 4 - 1 = 3.$$

A line through P has strict-transform class $H - E$, and

$$(2H - E) \cdot (H - E) = 1.$$

It therefore maps isomorphically to a line in $\mathbb{P}^4$. Distinct such transforms are disjoint because their only intersection downstairs was P , and they meet E at distinct direction points. Every point of (2) has a direction $[s : t]$, so these lines cover $\tilde{X}$. Thus the embedded cubic surface is ruled. $\square$

Exercise II.7.8**Permitted context.**

Let X be noetherian, let $\mathcal{E}$ be a finite locally free $\mathcal{O}_X$ -module, and use the quotient convention

$$\mathbb{P}(\mathcal{E}) = \mathbf{Proj}_X(\mathrm{Sym} \mathcal{E}).$$

If $\pi : \mathbb{P}(\mathcal{E}) \rightarrow X$, there is a tautological surjection

$$\pi^* \mathcal{E} \twoheadrightarrow \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1).$$

Problem. Prove a natural bijection between sections $\sigma : X \rightarrow \mathbb{P}(\mathcal{E})$ of π and isomorphism classes of surjections

$$\mathcal{E} \twoheadrightarrow \mathcal{L}$$

with $\mathcal{L}$ invertible.

Solution. Given a section σ , pull back the tautological quotient:

$$\mathcal{E} \cong \sigma^* \pi^* \mathcal{E} \twoheadrightarrow \sigma^* \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1).$$

The target $\mathcal{L} = \sigma^* \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ is invertible, so this produces a quotient line bundle on X .

Conversely, a surjection $\mathcal{E} \twoheadrightarrow \mathcal{L}$ induces a surjective map of graded algebras

$$\mathrm{Sym} \mathcal{E} \longrightarrow \mathrm{Sym} \mathcal{L}.$$

The relative- Proj universal property gives an X -morphism

$$\sigma : X \cong \mathbb{P}(\mathcal{L}) \longrightarrow \mathbb{P}(\mathcal{E}).$$

Because it is an X -morphism, $\pi\sigma = \mathrm{id}_X$.

These constructions commute with restriction to open sets. Locally, where $\mathcal{E} \cong \mathcal{O}_X^{r+1}$, they are the familiar bijection between a point of $\mathbb{P}^r$ and a rank-one quotient of the free module. The local constructions are inverse and hence so are the global ones. Replacing a quotient by an isomorphic quotient does not change the section, giving the claimed natural bijection. $\square$

Exercise II.7.9**Permitted context.**

Let X be a connected regular noetherian scheme and let $\mathcal{E}$ be finite locally free of rank $r \geq 2$. Write $\pi : \mathbb{P}(\mathcal{E}) \rightarrow X$, using the quotient convention. Connectedness is necessary for a single $\mathbb{Z}$ -summand; for a disconnected regular noetherian scheme, the result applies on each open-and-closed connected component and the integer is a locally constant function on X .

Problem. (a) Prove

$$\mathrm{Pic}(\mathbb{P}(\mathcal{E})) \cong \mathrm{Pic}(X) \oplus \mathbb{Z}, \quad (\mathcal{M}, d) \mapsto \pi^* \mathcal{M} \otimes \mathcal{O}_{\mathbb{P}(\mathcal{E})}(d).$$

(b) For another finite locally free sheaf $\mathcal{E}'$, prove that

$$\mathbb{P}(\mathcal{E}) \cong_X \mathbb{P}(\mathcal{E}')$$

if and only if $\mathcal{E}' \cong \mathcal{E} \otimes \mathcal{M}$ for some invertible sheaf $\mathcal{M}$ on X .

Solution. (a) First let $U = \mathrm{Spec} A$ be connected, regular, and affine, and suppose $\mathcal{E}|_U$ is free. Then $\mathbb{P}(\mathcal{E})|_U \cong \mathbb{P}_A^{r-1}$. Both U and $\mathbb{P}_A^{r-1}$ are regular and hence locally factorial. The class-group computation for projective space over an integral regular base, together with $\mathrm{Pic} = \mathrm{Cl}$, gives

$$\mathrm{Pic}(\mathbb{P}_A^{r-1}) \cong \mathrm{Pic}(A) \oplus \mathbb{Z}, \tag{1}$$

where the second generator is $\mathcal{O}(1)$.

Cover X by connected affine opens U_i trivializing $\mathcal{E}$. For $\mathcal{N} \in \mathrm{Pic}(\mathbb{P}(\mathcal{E}))$, equation (1) gives

$$\mathcal{N}|_{\pi^{-1}(U_i)} \cong \pi^* \mathcal{M}_i \otimes \mathcal{O}(k_i).$$

On a fiber over a point of $U_i \cap U_j$, both expressions have degree k_i and k_j ; hence these integers agree whenever the overlap is nonempty. Connectedness makes them a single integer k .

Put $\mathcal{N}_0 = \mathcal{N} \otimes \mathcal{O}(-k)$. Locally it is a pullback from U_i . Therefore

$$\mathcal{M} = \pi_* \mathcal{N}_0$$

is invertible, and the adjunction map $\pi^* \mathcal{M} \rightarrow \mathcal{N}_0$ is an isomorphism; both assertions can be checked on the trivializing opens, using $\pi_* \mathcal{O}_{\mathbb{P}_{U_i}^{r-1}} = \mathcal{O}_{U_i}$. This proves surjectivity of the displayed homomorphism.

For uniqueness, if $\pi^* \mathcal{M} \otimes \mathcal{O}(k) \cong \mathcal{O}$, restriction to any fiber gives $k = 0$. Pushing forward then gives $\mathcal{M} \cong \mathcal{O}_X$. Thus the homomorphism is an isomorphism.

(b) Let $f : \mathbb{P}(\mathcal{E}) \xrightarrow{\sim} \mathbb{P}(\mathcal{E}')$ be an X -isomorphism. Part (a) gives

$$f^* \mathcal{O}_{\mathbb{P}(\mathcal{E}')} (1) \cong \pi^* \mathcal{M} \otimes \mathcal{O}_{\mathbb{P}(\mathcal{E})} (k).$$

On every geometric fiber, f is an automorphism of projective space and hence pulls $\mathcal{O}(1)$ back to $\mathcal{O}(1)$. Thus $k = 1$. Since

$$\pi_* \mathcal{O}_{\mathbb{P}(\mathcal{E})} (1) \cong \mathcal{E}, \quad \pi'_* \mathcal{O}_{\mathbb{P}(\mathcal{E}')} (1) \cong \mathcal{E}',$$

pushforward and the projection formula yield

$$\mathcal{E}' \cong \pi_* f^* \mathcal{O}_{\mathbb{P}(\mathcal{E}')} (1) \cong \mathcal{M} \otimes \mathcal{E}.$$

Conversely, if $\mathcal{E}' \cong \mathcal{E} \otimes \mathcal{M}$, then tensoring the degree- d part of $\mathrm{Sym} \mathcal{E}$ by $\mathcal{M}^d$ does not change relative Proj. Hence

$$\mathbb{P}(\mathcal{E} \otimes \mathcal{M}) \cong_X \mathbb{P}(\mathcal{E}),$$

which proves the converse. □

Exercise II.7.10**Permitted context.**

Let X be noetherian. A transition automorphism of $\mathbb{P}_U^n$ is called linear if, locally on U , it is induced by an invertible linear change of the degree-one homogeneous coordinates. Scalar matrices induce the same projective automorphism.

Problem. (a) Define a projective n -space bundle over X by Zariski-local trivializations with linear transition automorphisms.

(b) Prove that $\mathbb{P}(\mathcal{E})$ is such a bundle whenever $\mathcal{E}$ is locally free of rank $n + 1$.

(c) If X is regular, prove that every projective n -space bundle $P \rightarrow X$ is isomorphic to $\mathbb{P}(\mathcal{E})$ for some locally free $\mathcal{E}$. Give a weaker sufficient hypothesis.

(d) For regular X , deduce a bijection between isomorphism classes of projective n -space bundles and rank- $n + 1$ locally free sheaves modulo

$$\mathcal{E} \sim \mathcal{E} \otimes \mathcal{M}, \quad \mathcal{M} \in \text{Pic}(X).$$

Solution. If $n = 0$, a $\mathbb{P}^0$ -bundle is the base X itself. Every rank-one locally free sheaf has this projectivization, and any two differ by tensoring with an invertible sheaf. This proves the assertion in that case. Assume henceforth $n \geq 1$.

(a) A projective n -space bundle is a morphism $\pi : P \rightarrow X$ with an open cover $X = \bigcup U_i$ and U_i -isomorphisms

$$\pi^{-1}(U_i) \cong U_i \times \mathbb{P}^n$$

such that every transition map over $U_i \cap U_j$ is locally induced by an element of $\text{GL}_{n+1}(\mathcal{O}_X)$. Composition and inversion of coordinate changes preserve this condition, so it is independent of refinements and defines a gluing category.

(b) Choose frames of $\mathcal{E}$ on an open cover. They identify $\mathbb{P}(\mathcal{E})|_{U_i}$ with $\mathbb{P}_{U_i}^n$. Changes of frame are invertible $(n + 1)$ -by- $(n + 1)$ matrices, and their projectivizations are exactly the required linear transition maps.

(c) Work on one connected component; a connected regular scheme is integral. Choose a dense open $U \subseteq X$ over which P is trivial, and let $\mathcal{O}(1)$ be the usual invertible sheaf on $P_U = \pi^{-1}(U)$. The total space P is regular, because it is locally a projective space over X , and hence it is locally factorial. Represent $\mathcal{O}(1)$ by a Cartier divisor on P_U , take the closures in P of its prime components, and use local factoriality to obtain a Cartier divisor on P . Its invertible sheaf is an extension $\mathcal{L}$ of $\mathcal{O}_{P_U}(1)$.

Over a trivializing open $V \subseteq X$, the projective-space Picard calculation gives

$$\mathcal{L}|_{P_V} \cong \pi^* \mathcal{M}_V \otimes \mathcal{O}_{\mathbb{P}_V^n}(a_V).$$

On the dense overlap with U , its fiber degree is 1, so $a_V = 1$ on every component meeting U . Thus

$$\mathcal{E} = \pi_* \mathcal{L}$$

is locally free of rank $n + 1$. The adjunction evaluation map

$$\pi^* \mathcal{E} \twoheadrightarrow \mathcal{L}$$

is surjective: on P_V it is, after the harmless factor $\mathcal{M}_V$, the standard evaluation $\mathcal{O}_V^{n+1} \rightarrow \mathcal{O}_{\mathbb{P}_V^n}(1)$. The quotient universal property produces

$$g : P \rightarrow \mathbb{P}(\mathcal{E}).$$

On every V , this is the standard isomorphism $\mathbb{P}_V^n \rightarrow \mathbb{P}(\mathcal{O}_V^{n+1} \otimes \mathcal{M}_V)$. Hence g is globally an isomorphism.

The proof used local factoriality, rather than the full force of regularity: it extended the divisor and used the projective-space Picard calculation. Thus a noetherian locally factorial base is a weaker sufficient hypothesis.

(d) Part (b) constructs a bundle from every locally free sheaf, and part (c) proves surjectivity on isomorphism classes. Exercise II.7.9 shows that

$$\mathbb{P}(\mathcal{E}) \cong_X \mathbb{P}(\mathcal{E}') \iff \mathcal{E}' \cong \mathcal{E} \otimes \mathcal{M}$$

for an invertible $\mathcal{M}$. This is precisely the stated bijection. $\square$

Exercise II.7.11

Permitted context.

For a coherent ideal sheaf $\mathcal{I}$ on a noetherian scheme X , write

$$\mathrm{Bl}_{\mathcal{I}} X = \mathbf{Proj}_X \left(\bigoplus_{n \geq 0} \mathcal{I}^n \right).$$

In part (c), start with a morphism $f : Z \rightarrow X$ represented as such a blowup by the theorem being strengthened.

Problem. (a) Prove that blowing up $\mathcal{I}^d$, for any $d \geq 1$, gives the same X -scheme as blowing up $\mathcal{I}$.
 (b) If $\mathcal{J}$ is an invertible ideal sheaf, prove

$$\mathrm{Bl}_{\mathcal{I}\mathcal{J}} X \cong_X \mathrm{Bl}_{\mathcal{I}} X.$$

(c) Suppose X is regular, and let $U \subseteq X$ be the largest open set over which f is an isomorphism. Show that the blowup ideal may be chosen so that its zero scheme has support exactly $X \setminus U$.

Solution. (a) The Rees algebra of $\mathcal{I}^d$ is

$$\bigoplus_{n \geq 0} \mathcal{I}^{dn},$$

the d -th Veronese subalgebra of $\bigoplus_{n \geq 0} \mathcal{I}^n$. Relative Proj is unchanged on passing to a positive Veronese subalgebra. Hence the two blowups are canonically isomorphic over X .

(b) Since $\mathcal{J}$ is invertible,

$$(\mathcal{I}\mathcal{J})^n \cong \mathcal{I}^n \otimes \mathcal{J}^n.$$

Degree-wise tensoring a graded algebra by the powers of an invertible sheaf does not change relative Proj. Equivalently, on an open where $\mathcal{J}$ is generated by one nonzerodivisor a , multiplication by a^n identifies the degree- n pieces, and these local Proj isomorphisms agree on overlaps. This proves (b).

(c) Begin with $Z \cong \mathrm{Bl}_{\mathcal{I}} X$. The tautological pullback of $\mathcal{I}$ to the blowup is invertible. Over U , where the blowup is the identity, this says that $\mathcal{I}|_U$ is an invertible ideal. Thus

$$\mathcal{I}|_U = \mathcal{O}_U(-D)$$

for an effective Cartier divisor D on U .

Work componentwise, so X is integral. Take the closures of the prime components of D in X . Regularity makes the resulting effective Weil divisor $\overline{D}$ Cartier. We claim

$$\mathcal{I} \subseteq \mathcal{O}_X(-\overline{D}). \quad (1)$$

This is local. In a regular local ring, which is a unique factorization domain, each element of $\mathcal{I}$ has along every height-one component of $\overline{D}$ at least the order prescribed by D , because equality holds at that component's generic point in U . Unique factorization then gives divisibility by a local equation of $\overline{D}$, proving (1).

Set

$$\mathcal{K} = \mathcal{I} \otimes \mathcal{O}_X(\overline{D}).$$

By (1), $\mathcal{K}$ is a coherent ideal sheaf, and $\mathcal{K}|_U = \mathcal{O}_U$. Moreover

$$\mathcal{I} = \mathcal{K} \mathcal{O}_X(-\overline{D}),$$

where $\mathcal{O}_X(-\overline{D})$ is an invertible ideal. Part (b) therefore gives

$$\mathrm{Bl}_{\mathcal{K}} X \cong_X \mathrm{Bl}_{\mathcal{I}} X \cong_X Z.$$

The support of $\mathcal{O}_X/\mathcal{K}$ lies in $X \setminus U$. If it omitted a point $x \in X \setminus U$, then $\mathcal{K} = \mathcal{O}_X$ on a neighborhood of x , and its blowup would be the identity there. This would enlarge U , contrary to maximality. Hence the support is exactly $X \setminus U$. $\square$

Exercise II.7.12

Permitted context.

Let X be noetherian, and let $Y, Z \subseteq X$ be closed subschemes with ideal sheaves $\mathcal{I}$ and $\mathcal{J}$. Assume neither contains the other. Their scheme-theoretic intersection is defined by $\mathcal{K} = \mathcal{I} + \mathcal{J}$.

Problem. Let

$$\tilde{X} = \mathrm{Bl}_{Y \cap Z} X.$$

Prove that the strict transforms $\tilde{Y}$ and $\tilde{Z}$ are disjoint.

Solution. The question is local on X . On an affine open write the three ideals as $I, J, K = I + J$, and let

$$R = \bigoplus_{d \geq 0} K^d$$

be the Rees algebra. The strict transform of Y is cut out in $\mathrm{Proj} R$ by the homogeneous ideal whose degree- d piece is

$$K^d \cap I. \quad (1)$$

Indeed, its homogeneous coordinate algebra is the image of R in the Rees algebra of the induced ideal on A/I :

$$\bigoplus_{d \geq 0} \frac{K^d + I}{I} \cong \bigoplus_{d \geq 0} \frac{K^d}{K^d \cap I}.$$

The analogous ideal for $\tilde{Z}$ has degree- d piece $K^d \cap J$.

For every $d \geq 1$,

$$K^d = (I + J)^d \subseteq (K^d \cap I) + (K^d \cap J). \quad (2)$$

To see this, expand a product of d factors from $I + J$: a term containing an I -factor belongs to $I \cap K^d$, while the remaining all- J term belongs to $J \cap K^d$. The reverse inclusion in (2) is automatic, so equality holds.

Thus the sum of the two homogeneous ideals defining the strict transforms contains every positive-degree piece of R , namely the irrelevant ideal. No homogeneous prime representing a point of $\text{Proj } R$ can contain that irrelevant ideal. Consequently the scheme-theoretic intersection $\tilde{Y} \cap \tilde{Z}$ is empty. $\square$

Exercise II.7.13

Permitted context.

Let k be algebraically closed with $\text{char } k \neq 2$, and let

$$C : y^2 z = x^3 + x^2 z$$

be the nodal cubic. Its smooth locus is $\mathbb{G}_m$. Multiplication by $a \in k^*$ on that group extends to an automorphism $\varphi_a : C \rightarrow C$.

Glue

$$X_0 = C \times (\mathbb{P}^1 \setminus \{\infty\}), \quad X_\infty = C \times (\mathbb{P}^1 \setminus \{0\})$$

over $C \times \mathbb{G}_m$ by

$$(P, u) \longmapsto (\varphi_u(P), u).$$

The second projections glue to $\pi : X \rightarrow \mathbb{P}^1$. For the finite normalization, local inverse images are spectra of semilocal rings with trivial Picard groups. Hence $R^1 \nu_* \mathcal{O}_{\tilde{Y}}^* = 0$, and the Leray sequence justifies the first-cohomology comparison for the unit sheaves.

Problem. (a) Prove that π is proper and that X is a complete variety.

(b) Compute

$$\text{Pic}(C \times \mathbb{A}^1) \cong k^* \oplus \mathbb{Z}, \quad \text{Pic}(C \times \mathbb{G}_m) \cong k^* \oplus \mathbb{Z} \oplus \mathbb{Z}.$$

(c) Choose these coordinates so that restriction is

$$(t, n) \longmapsto (t, 0, n)$$

and pullback by $\varphi(P, u) = (\varphi_u(P), u)$ is

$$(t, d, n) \longmapsto (t, d + n, n).$$

(d) Prove that every invertible sheaf on X restricts to degree zero on $C \times \{0\}$. Deduce that X is proper but not projective and that π is not projective.

Solution. (a) Properness is local on the base in the Zariski topology. Over each of the two standard affine opens of $\mathbb{P}^1$, the morphism π is the projection

$$C \times \mathbb{A}^1 \longrightarrow \mathbb{A}^1,$$

which is proper because it is the base change of the proper curve $C \rightarrow \operatorname{Spec} k$. Hence π is proper. The composite

$$X \xrightarrow{\pi} \mathbb{P}^1 \longrightarrow \operatorname{Spec} k$$

is proper, so X is complete. The pieces are integral schemes of finite type and are glued along a dense open, so X is an integral variety.

(b) Choose a coordinate on the normalization $\nu : \mathbb{P}^1 \rightarrow C$ for which 0 and ∞ are the two points above the node. For $T = \mathbb{A}^1$ or $\mathbb{G}_m$, put

$$Y = C \times T, \quad \tilde{Y} = \mathbb{P}^1 \times T.$$

The normalization map gives an exact sequence of unit sheaves on Y :

$$1 \longrightarrow \mathcal{O}_Y^* \longrightarrow \nu_* \mathcal{O}_{\tilde{Y}}^* \longrightarrow i_* \mathcal{O}_T^* \longrightarrow 1, \quad (1)$$

where $i : T \rightarrow Y$ is the node section. The last arrow takes the ratio of the values on the two branches. Its kernel consists exactly of units descending to the node, and it is locally surjective.

The global units on Y and $\tilde{Y}$ are both the units on T ; their branch ratio is 1. Moreover

$$H^1(T, \mathcal{O}_T^*) = \operatorname{Pic}(T) = 0.$$

The cohomology sequence of (1) therefore becomes

$$0 \longrightarrow \Gamma(T, \mathcal{O}_T^*) \longrightarrow \operatorname{Pic}(C \times T) \longrightarrow \operatorname{Pic}(\mathbb{P}^1 \times T) \longrightarrow 0. \quad (2)$$

The last group is $\mathbb{Z}$, measured by degree on the $\mathbb{P}^1$ -fiber. A divisor $q \times T$, for a smooth point $q \in C$, splits (2). Since

$$\Gamma(\mathbb{A}^1, \mathcal{O}^*) = k^*, \quad \Gamma(\mathbb{G}_m, \mathcal{O}^*) = k^* u^{\mathbb{Z}},$$

the two asserted Picard-group formulas follow.

(c) Sequence (2) also gives concrete coordinates. A line bundle on $C \times T$ is specified by its degree n on the normalized $\mathbb{P}^1$ -fiber and by a branch-gluing unit $\lambda(u) \in \Gamma(T, \mathcal{O}_T^*)$. For $T = \mathbb{A}^1$, write $\lambda = t \in k^*$, giving (t, n) . For $T = \mathbb{G}_m$, write uniquely

$$\lambda(u) = tu^d, \quad t \in k^*, \quad d \in \mathbb{Z},$$

giving (t, d, n) . Restriction from $\mathbb{A}^1$ to $\mathbb{G}_m$ leaves a constant gluing unit constant, and hence is

$$(t, n) \longmapsto (t, 0, n).$$

Multiplication by u on the normalized $\mathbb{P}^1$ fixes 0, ∞ . On a degree- n line bundle it changes the ratio of the standard trivializations at those two points by u^n . Thus pullback by the gluing automorphism replaces $\lambda(u) = tu^d$ by tu^{d+n} , while preserving the degree. With this consistent choice of branch order,

$$\varphi^*(t, d, n) = (t, d + n, n).$$

(d) Let an invertible sheaf $\mathcal{L}$ on X have classes (t_0, n_0) and (t_∞, n_∞) on X_0 and X_∞ . On the overlap, the gluing of X requires

$$(t_0, 0, n_0) = \varphi^*(t_\infty, 0, n_\infty).$$

Part (c) turns this into

$$(t_0, 0, n_0) = (t_\infty, n_\infty, n_\infty).$$

Consequently

$$n_0 = n_\infty = 0. \quad (3)$$

In particular, $\mathcal{L}|_{C \times \{0\}}$ has degree zero.

If X were projective, it would possess a very ample invertible sheaf. Its restriction to the closed fiber $C \times \{0\}$ would still be very ample, but (3) makes its pullback to the normalization $\mathbb{P}^1$ a degree-zero line bundle, hence $\mathcal{O}_{\mathbb{P}^1}$. All its pulled-back global sections are constant, so the induced map on $\mathbb{P}^1$ is constant, contradicting a closed immersion of the nontrivial curve C . Thus X is not projective. Finally, if π were projective, composing it with the projective morphism $\mathbb{P}^1 \rightarrow \operatorname{Spec} k$ would make X projective. Hence π is not projective. $\square$

Exercise II.7.14

Permitted context.

For part (b), condition $(\dagger)$ supplies a quasi-coherent graded $\mathcal{O}_X$ -algebra $\mathcal{S} = \bigoplus_{d \geq 0} \mathcal{S}_d$, with $\mathcal{S}_0 = \mathcal{O}_X$, generated in degree one by the coherent sheaf $\mathcal{S}_1$, and such that the associated $\mathcal{O}_P(1)$ on $P = \mathbf{Proj}_X \mathcal{S}$ is invertible. Ampleness of $\mathcal{L}$ is understood relative to $f : X \rightarrow Y$. Throughout, very ample relative to a base means realized by an immersion into a single finite-dimensional projective space over that base. In (b) the base is required to be affine and noetherian.

Problem. (a) Give a noetherian X and a finite locally free $\mathcal{E}$ for which $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ is not very ample relative to X .

(b) Let $Y = \operatorname{Spec} A$ with A noetherian, let $f : X \rightarrow Y$ be of finite type, let $\mathcal{L}$ be f -ample, and let $\pi : P \rightarrow X$ be as in the context. Prove that

$$\mathcal{O}_P(1) \otimes \pi^* \mathcal{L}^n$$

is very ample relative to Y for every sufficiently large n .

Solution. (a) Take

$$X = \mathbb{P}_k^1, \quad \mathcal{E} = \mathcal{O}_X(-1)^{\oplus 2}.$$

Twisting a vector bundle by a line bundle does not change its projectivization, so

$$\mathbb{P}(\mathcal{E}) \cong \mathbb{P}(\mathcal{O}_X^{\oplus 2}) \cong \mathbb{P}^1 \times \mathbb{P}^1.$$

Under this isomorphism, with the base as the first factor,

$$\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1) \cong \mathcal{O}_{\mathbb{P}^1 \times \mathbb{P}^1}(-1, 1).$$

It has no nonzero global sections. A relatively very ample sheaf is the pullback of $\mathcal{O}(1)$ under an immersion into some $\mathbb{P}_X^N$, and is therefore generated by its defining sections. Hence this $\mathcal{O}(1)$ is not very ample relative to X .

(b) Choose $a > 0$ such that $\mathcal{L}^a$ is very ample relative to Y . Such an a exists by the ample-sheaf criterion. By ampleness again, for every sufficiently large q , the coherent sheaf

$$\mathcal{S}_1 \otimes \mathcal{L}^q$$

is generated by finitely many sections. A resulting surjection

$$\mathcal{O}_X^{N+1} \twoheadrightarrow \mathcal{S}_1 \otimes \mathcal{L}^q$$

extends, because $\mathcal{S}$ is generated in degree one, to a surjection onto the degreewise twisted graded algebra

$$\bigoplus_{d \geq 0} \mathcal{S}_d \otimes \mathcal{L}^{qd}.$$

Relative Proj of this algebra is still P , and its twisting sheaf is

$$\mathcal{O}_P(1) \otimes \pi^* \mathcal{L}^q.$$

The graded surjection therefore gives a closed immersion

$$P \hookrightarrow \mathbb{P}_X^N$$

realizing this sheaf. It is very ample relative to X .

Now let $n \geq a$ and put $q = n - a$. Once n is large, so is q , and the preceding paragraph applies. The composition rule for very ample sheaves, applied to $P \xrightarrow{\pi} X \xrightarrow{f} Y$, says that

$$(\mathcal{O}_P(1) \otimes \pi^* \mathcal{L}^q) \otimes \pi^*(\mathcal{L}^a) = \mathcal{O}_P(1) \otimes \pi^* \mathcal{L}^n$$

is very ample relative to Y . This holds for every sufficiently large n . The affine-base hypothesis cannot be dropped with this definition. For $f = \text{id}_{\mathbb{P}^1}$, $\mathcal{L} = \mathcal{O}_{\mathbb{P}^1}$, and $\mathcal{S} = \text{Sym}(\mathcal{O}(-1)^{\oplus 2})$, the sheaf in (b) is $\mathcal{O}(-1, 1)$ for every n . The identity makes $\mathcal{L}$ relatively ample, but (a) proves that this sheaf is never relatively very ample. $\square$

II.8 Differentials

Exercise II.8.1

Permitted context.

Write $\kappa(B) = B/\mathfrak{m}$ for the residue field of a local ring $(B, \mathfrak{m})$. We may use the conormal sequence for $k \rightarrow B \rightarrow \kappa(B)$, the equality

$$\dim_L \Omega_{L/k} = \text{trdeg}_k L$$

when the finitely generated field extension L/k is separably generated, and the dimension formula for an integral algebra of finite type over a field. We also use the relative coefficient-field theorem: if a complete equicharacteristic local ring contains k and its residue field is separably generated over k , it has a coefficient field containing k .

Problem. (a) Let $(B, \mathfrak{m})$ be a local ring containing a field k , and suppose $\kappa = \kappa(B)$ is separably generated over k . Prove that the conormal sequence is short exact:

$$0 \longrightarrow \mathfrak{m}/\mathfrak{m}^2 \longrightarrow \Omega_{B/k} \otimes_B \kappa \longrightarrow \Omega_{\kappa/k} \longrightarrow 0.$$

(b) Suppose in addition that k is perfect and that B is a localization of a finitely generated k -algebra. Prove that B is regular if and only if $\Omega_{B/k}$ is free of rank

$$\dim B + \text{trdeg}_k \kappa.$$

(c) Let X be an irreducible finite-type scheme of dimension n over a perfect field. For every point $x \in X$, prove that $\mathcal{O}_{X,x}$ is regular exactly when $(\Omega_{X/k})_x$ is free of rank n .

(d) Deduce that the regular points of a variety over an algebraically closed field form an open dense subset.

Solution. (a) Put $\overline{B} = B/\mathfrak{m}^2$ and $\overline{\mathfrak{m}} = \mathfrak{m}/\mathfrak{m}^2$. This is a complete local ring whose maximal ideal is square-zero. The coefficient-field theorem supplies a subfield $K \subseteq \overline{B}$, containing the image of k , such that $K \rightarrow \kappa$ is an isomorphism. Consequently every element of $\overline{B}$ has a unique expression

$$\lambda + u, \quad \lambda \in K, \quad u \in \overline{\mathfrak{m}}. \quad (1)$$

Let $h : \mathfrak{m}/\mathfrak{m}^2 \rightarrow \kappa$ be any κ -linear map. Define

$$D : B \longrightarrow \kappa, \quad D(b) = h(u),$$

where the image of b in $\overline{B}$ is written as in (1). This map vanishes on k . If $b = \lambda + u$ and $c = \mu + v$ modulo $\mathfrak{m}^2$, then $uv = 0$, so

$$D(bc) = h(\lambda v + \mu u) = \overline{b}D(c) + \overline{c}D(b).$$

Thus D is a k -derivation, and its restriction to $\mathfrak{m}/\mathfrak{m}^2$ is h .

Dualizing the first map in the conormal sequence identifies its dual with the restriction map

$$\text{Der}_k(B, \kappa) \longrightarrow \text{Hom}_\kappa(\mathfrak{m}/\mathfrak{m}^2, \kappa).$$

The construction shows that this map is surjective. Linear functionals separate vectors, hence the conormal map is injective. The other two terms were already right exact, proving (a).

(b) Set

$$r = \dim B + \operatorname{trdeg}_k \kappa.$$

If $\Omega_{B/k}$ is free of rank r , part (a) and the field differential formula give

$$\dim_{\kappa} \mathfrak{m}/\mathfrak{m}^2 = r - \dim_{\kappa} \Omega_{\kappa/k} = \dim B.$$

This is precisely the regular-local-ring criterion.

Conversely, suppose B is regular. Then B is a domain; write $K = \operatorname{Frac}(B)$. Part (a) gives

$$\dim_{\kappa}(\Omega_{B/k} \otimes_B \kappa) = r. \quad (2)$$

Because k is perfect, K/k is separably generated. Localization of differentials and the dimension formula give

$$\dim_K(\Omega_{B/k} \otimes_B K) = \operatorname{trdeg}_k K = \dim B + \operatorname{trdeg}_k \kappa = r. \quad (3)$$

The module $\Omega_{B/k}$ is finite. Lift a basis in (2) and use Nakayama's lemma to obtain a surjection $B^r \twoheadrightarrow \Omega_{B/k}$. After tensoring with K , (3) makes this an isomorphism. Its kernel therefore becomes zero over K . That kernel lies in the torsion-free module B^r , so it is zero. Hence $\Omega_{B/k} \cong B^r$.

(c) Put $B = \mathcal{O}_{X,x}$ and $\kappa = \kappa(x)$. Since X_{red} is integral of dimension n and has the same local dimensions and residue fields as X , the dimension formula applied to X_{red} says

$$\dim B + \operatorname{trdeg}_k \kappa = n.$$

Moreover, differentials commute with localization:

$$(\Omega_{X/k})_x \cong \Omega_{B/k}.$$

Part (b) now gives the asserted equivalence.

(d) By (c), the regular locus is the locus on which the coherent sheaf $\Omega_{X/k}$ is locally free of rank $n = \dim X$. This locus is open. At the generic point η ,

$$(\Omega_{X/k})_{\eta} = \Omega_{k(X)/k}$$

is an n -dimensional $k(X)$ -vector space, because an algebraically closed field is perfect. Thus the open locus contains η , and is therefore dense. $\square$

Exercise II.8.2**Permitted context.**

A variety is integral and of finite type over an algebraically closed field k . A family of global sections generates a locally free sheaf if its evaluation map is surjective. We may use that the image of a finite-type morphism is constructible and that the dimension of the closure of an image is at most the dimension of its source.

Problem. Let X be an n -dimensional variety, let $\mathcal{E}$ be locally free of rank $r > n$, and let $V \subseteq \Gamma(X, \mathcal{E})$ generate $\mathcal{E}$. Prove that some $s \in V$ is nonzero in every fiber $\mathcal{E}_x/\mathfrak{m}_x\mathcal{E}_x$. Deduce an exact sequence

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}' \longrightarrow 0$$

in which $\mathcal{E}'$ is locally free.

Solution. First replace V by a finite-dimensional subspace that still generates $\mathcal{E}$. Indeed, at each point finitely many members of V span the fiber. The same remains true on an open neighborhood, and quasi-compactness of X gives finitely many such neighborhoods and sections. Let $m = \dim_k V$.

The evaluation map is a surjection of vector bundles

$$V \otimes_k \mathcal{O}_X \twoheadrightarrow \mathcal{E}.$$

Its kernel $\mathcal{K}$ is locally free of rank $m - r$, since the surjection splits locally. Inside $X \times \mathbb{A}(V)$, consider the incidence scheme

$$Z = \{(x, s) : s(x) = 0\}.$$

It is the total space of $\mathcal{K}$, and hence

$$\dim Z = n + m - r < m = \dim \mathbb{A}(V).$$

For the projection $p : Z \rightarrow \mathbb{A}(V)$, the closure of $p(Z)$ therefore has dimension strictly less than m . Choose a k -rational point s outside that proper closed subset. Such a point exists because k is algebraically closed. No point of Z lies over s , which says exactly that

$$s_x \notin \mathfrak{m}_x\mathcal{E}_x \quad \text{for every } x \in X.$$

The section defines $\mathcal{O}_X \rightarrow \mathcal{E}$. At a point x , trivialize $\mathcal{E}_x \cong \mathcal{O}_{X,x}^r$. At least one coordinate of s_x is a unit, so elementary changes of basis send s_x to the first standard basis vector. Consequently the stalk map is a split injection and its cokernel is free of rank $r - 1$. These stalkwise cokernels form a locally free sheaf $\mathcal{E}'$, giving the displayed exact sequence. $\square$

Exercise II.8.3**Permitted context.**

For a nonsingular d -dimensional variety T , its canonical sheaf is $\omega_T = \bigwedge^d \Omega_{T/k}$, and $p_g(T) = \dim_k \Gamma(T, \omega_T)$. A smooth plane cubic has $\omega \cong \mathcal{O}$ and arithmetic genus one. For projective curves C, D , the arithmetic-genus product formula specializes to

$$p_a(C \times D) = p_a(C)p_a(D) - p_a(C) - p_a(D).$$

Problem. (a) For S -schemes X, Y , prove

$$\Omega_{X \times_S Y/S} \cong p_1^* \Omega_{X/S} \oplus p_2^* \Omega_{Y/S}.$$

(b) If X, Y are nonsingular varieties over k , deduce

$$\omega_{X \times_k Y} \cong p_1^* \omega_X \otimes p_2^* \omega_Y.$$

(c) If Y is a nonsingular plane cubic and $T = Y \times Y$, show that $p_g(T) = 1$ whereas $p_a(T) = -1$.

Solution. (a) The assertion is local on all three schemes. For ring maps $A \rightarrow B$ and $A \rightarrow C$, and for any $B \otimes_A C$ -module M , restriction to the two factors gives a natural bijection

$$\mathrm{Der}_A(B \otimes_A C, M) \longrightarrow \mathrm{Der}_A(B, M) \oplus \mathrm{Der}_A(C, M).$$

Its inverse sends (D_B, D_C) to

$$D(b \otimes c) = cD_B(b) + bD_C(c).$$

The formula is balanced over A , satisfies the Leibniz rule, and is forced on pure tensors. The universal property of differentials therefore yields

$$\Omega_{(B \otimes_A C)/A} \cong (\Omega_{B/A} \otimes_B (B \otimes_A C)) \oplus (\Omega_{C/A} \otimes_C (B \otimes_A C)).$$

Sheafifying gives the required formula.

(b) Put $a = \dim X$ and $b = \dim Y$. Part (a) gives a locally free sheaf of rank $a + b$:

$$\Omega_{X \times Y/k} \cong p_1^* \Omega_{X/k} \oplus p_2^* \Omega_{Y/k}.$$

The differential criterion shows that the product is nonsingular. Taking its top exterior power and using $\det(E \oplus F) = \det E \otimes \det F$, we obtain

$$\omega_{X \times Y} \cong p_1^* \omega_X \otimes p_2^* \omega_Y.$$

(c) Since $\omega_Y \cong \mathcal{O}_Y$, part (b) gives $\omega_T \cong \mathcal{O}_T$. The projective integral variety T has only constant global regular functions, so

$$p_g(T) = \dim_k \Gamma(T, \mathcal{O}_T) = 1.$$

On the other hand, $p_a(Y) = 1$, and the stated product formula gives

$$p_a(T) = 1 \cdot 1 - 1 - 1 = -1.$$

□

Exercise II.8.4**Permitted context.**

Let $S = k[x_0, \dots, x_n]$. A closed subscheme $Y \subseteq \mathbb{P}_k^n$, with saturated homogeneous ideal I_Y , is a strict complete intersection if I_Y is generated by $r = \text{codim}(Y, \mathbb{P}^n)$ elements. We may use the following facts: a regular sequence quotient of S is Cohen–Macaulay and unmixed; Serre’s conditions R_1 and S_2 characterize normality; the homogeneous coordinate ring is normal exactly when the embedding is projectively normal; and Bertini’s theorem applies to the complete linear systems of degree- d hypersurfaces. We may also use the arithmetic-genus formulas obtained from the Hilbert polynomial for hypersurfaces and complete-intersection curves. Binomial coefficients $\binom{a}{b}$ below are understood to be zero when $a < b$. The ground field k is algebraically closed.

Problem. (a) If $Y \subseteq \mathbb{P}^n$ has codimension r , prove that it is a strict complete intersection if and only if it is the scheme-theoretic intersection of r hypersurfaces.

(b) Prove that a normal complete intersection of positive dimension is projectively normal.

(c) Under the hypotheses of (b), prove that

$$H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(\ell)) \longrightarrow H^0(Y, \mathcal{O}_Y(\ell))$$

is surjective for every $\ell \geq 0$, and deduce that Y is connected.

(d) Given $d_1, \dots, d_r \geq 1$ with $r < n$, construct smooth hypersurfaces H_i of degrees d_i whose intersection is smooth, irreducible, and of codimension r .

(e) For the smooth complete intersection Y in (d), prove

$$\omega_Y \cong \mathcal{O}_Y(d_1 + \dots + d_r - n - 1).$$

(f) If $n \geq 2$ and $Y \subseteq \mathbb{P}^n$ is a smooth hypersurface of degree d , prove

$$p_g(Y) = \binom{d-1}{n}.$$

Show that this equals the arithmetic genus. In particular, recover $(d-1)(d-2)/2$ for a smooth plane curve.

(g) If a smooth curve $Y \subseteq \mathbb{P}^3$ is the intersection of smooth surfaces of degrees d, e , prove

$$p_g(Y) = \frac{1}{2}de(d+e-4) + 1,$$

and show again that the geometric and arithmetic genera agree.

Solution. (a) Suppose first that I_Y can be generated by r elements. The graded Nakayama lemma says that a basis of the graded vector space I_Y/S_+I_Y lifts to a minimal homogeneous generating set. Its cardinality is at most r , while the height theorem says that at least $r = \text{ht } I_Y$ generators are necessary. Thus there are homogeneous generators $f_1, \dots, f_r$. The hypersurfaces $H_i = V_+(f_i)$ then satisfy

$$\mathcal{I}_Y = \mathcal{I}_{H_1} + \dots + \mathcal{I}_{H_r},$$

so their scheme-theoretic intersection is Y .

Conversely, suppose $Y = H_1 \cap \dots \cap H_r$ as schemes. A hypersurface in $\mathbb{P}^n$ has invertible ideal sheaf $\mathcal{O}_{\mathbb{P}^n}(-d_i)$, and its inclusion into $\mathcal{O}_{\mathbb{P}^n}$ is given by a homogeneous polynomial f_i of degree d_i . Put $J = (f_1, \dots, f_r)$. The codimension assumption gives $\text{ht } J = r$; hence the f_i form a regular

sequence in the Cohen–Macaulay ring S . It follows that S/J is unmixed and every associated prime has height r . The irrelevant ideal S_+ , of height $n+1 > r$, is therefore not associated to S/J . Thus S/J has no S_+ -torsion, which is equivalent to J being saturated. The ideal sheaf equality in the hypothesis says $I_Y = J^{\text{sat}}$, so in fact $I_Y = J$. Therefore Y is a strict complete intersection.

(b) Let $A = S/I_Y$, the homogeneous coordinate ring. Part (a) makes A a complete-intersection ring, so it satisfies S_2 . The punctured affine cone

$$\text{Spec } A \setminus \{A_+\} \longrightarrow Y$$

is locally a product with $\mathbb{G}_m$. Since Y is normal, the punctured cone is normal and in particular regular in codimension one. The vertex has codimension $\dim A = \dim Y + 1 \geq 2$, so it introduces no codimension-one point. Hence A satisfies R_1 as well as S_2 , and is normal by Serre's criterion. Every irreducible component of this affine cone contains its vertex. Distinct components of a normal noetherian scheme are disjoint, so there is only one. Thus A is a domain, proving projective normality of the embedding.

(c) The projective-normality criterion identifies A with the section ring of $\mathcal{O}_Y(1)$. Degree by degree, this says that

$$S_\ell = H^0(\mathbb{P}^n, \mathcal{O}(\ell)) \longrightarrow H^0(Y, \mathcal{O}_Y(\ell))$$

is surjective for every $\ell \geq 0$. At $\ell = 0$ it gives $H^0(Y, \mathcal{O}_Y) = k$. A disconnection of Y would produce a nonconstant idempotent global function, so Y is connected.

(d) Begin with $Y_0 = \mathbb{P}^n$. Suppose smooth irreducible Y_{i-1} of dimension $n-i+1$ has been constructed. Because $i \leq r < n$, this dimension is at least two. Apply Bertini to the degree- d_i complete linear system, equivalently to hyperplanes after the d_i -uple embedding. There is a nonempty open set of degree- d_i forms whose zero hypersurface is smooth in $\mathbb{P}^n$, and another nonempty open set whose restriction to Y_{i-1} has smooth irreducible zero locus of codimension one. Choose a form in their intersection and call its hypersurface H_i . Then $Y_i = Y_{i-1} \cap H_i$ has all the required properties. Induction through $i = r$ constructs $Y = Y_r$.

(e) Let W be smooth and let $Z = W \cap H$ be a smooth section by a hypersurface of degree d . The conormal sequence is

$$0 \longrightarrow \mathcal{O}_Z(-d) \longrightarrow \Omega_{W/k}|_Z \longrightarrow \Omega_{Z/k} \longrightarrow 0.$$

Taking determinants gives the adjunction formula

$$\omega_Z \cong (\omega_W \otimes \mathcal{O}_W(d))|_Z. \quad (1)$$

Starting with $\omega_{\mathbb{P}^n} = \mathcal{O}_{\mathbb{P}^n}(-n-1)$ and applying (1) successively to the hypersurfaces in (d) yields

$$\omega_Y \cong \mathcal{O}_Y(d_1 + \cdots + d_r - n - 1).$$

(f) Set $q = d - n - 1$. By (e), $\omega_Y \cong \mathcal{O}_Y(q)$. If $q < 0$, an anti-ample line bundle on the positive-dimensional projective integral variety Y has no nonzero section, so $p_g(Y) = 0 = \binom{d-1}{n}$. If $q \geq 0$, part (c) makes the restriction map in degree q surjective. Its kernel is $(I_Y)_q$, which is zero because I_Y is generated in degree $d > q$. Therefore

$$p_g(Y) = \dim_k S_q = \binom{q+n}{n} = \binom{d-1}{n}.$$

The complete-intersection Hilbert-polynomial calculation from the arithmetic-genus formula gives the same binomial coefficient. Hence $p_g(Y) = p_a(Y)$. For $n = 2$, their common value is $\binom{d-1}{2} = (d-1)(d-2)/2$.

(g) Adjunction gives

$$\omega_Y \cong \mathcal{O}_Y(q), \quad q = d + e - 4.$$

If $q < 0$, then (d, e) , up to order, is $(1, 1)$ or $(1, 2)$; both the asserted formula and $p_g(Y)$ are zero. Now assume $q \geq 0$. If F, G are the defining forms, their Koszul resolution gives

$$0 \longrightarrow S(-d-e) \longrightarrow S(-d) \oplus S(-e) \longrightarrow I_Y \longrightarrow 0.$$

In degree q , the left term is $S_{-4} = 0$, and hence

$$\dim_k(I_Y)_q = \binom{e-1}{3} + \binom{d-1}{3}.$$

Using the surjectivity in (c), we obtain

$$\begin{aligned} p_g(Y) &= \binom{d+e-1}{3} - \binom{e-1}{3} - \binom{d-1}{3} \\ &= \frac{1}{2}de(d+e-4) + 1. \end{aligned}$$

This is also the arithmetic genus furnished by the Hilbert-polynomial formula for a complete intersection of degrees d, e , so the two genera agree. $\square$

Exercise II.8.5

Permitted context.

Let X be a nonsingular variety and let $Y \subseteq X$ be a nonsingular subvariety of codimension $r \geq 2$. Write

$$\pi : \tilde{X} = \text{Bl}_Y X \longrightarrow X, \quad E = \pi^{-1}(Y).$$

The blowup is nonsingular, E is the projective bundle associated to the conormal bundle of Y , and $\mathcal{O}_{\tilde{X}}(-E)|_{\mathbb{P}^{r-1}} \cong \mathcal{O}_{\mathbb{P}^{r-1}}(1)$ on every fiber. On a regular scheme, the Picard and divisor class groups agree.

Problem. (a) Prove that pullback and the exceptional divisor induce an isomorphism

$$\text{Pic}(X) \oplus \mathbb{Z} \longrightarrow \text{Pic}(\tilde{X}), \quad (\mathcal{L}, m) \longmapsto \pi^* \mathcal{L} \otimes \mathcal{O}_{\tilde{X}}(mE).$$

(b) Prove the canonical-sheaf formula

$$\omega_{\tilde{X}} \cong \pi^* \omega_X \otimes \mathcal{O}_{\tilde{X}}((r-1)E).$$

Solution. (a) Put $U = X \setminus Y$. The blowup identifies $\tilde{X} \setminus E$ with U . Because Y has codimension at least two, deleting it does not change the divisor class group of the regular scheme X , and hence

$$\text{Pic}(X) \cong \text{Cl}(X) \xrightarrow{\sim} \text{Cl}(U).$$

The localization sequence for divisor class groups on $\tilde{X}$, together with the irreducibility of the projective bundle $E \rightarrow Y$, gives an exact sequence

$$\mathbb{Z}[E] \longrightarrow \text{Pic}(\tilde{X}) \longrightarrow \text{Pic}(U) \longrightarrow 0. \quad (1)$$

Under the identification $\text{Pic}(U) \cong \text{Pic}(X)$, the composite of π^* with the second map in (1) is the identity. Thus (1) is split and only injectivity of its first map remains.

Restrict $\mathcal{O}_{\tilde{X}}(mE)$ to a fiber $F \cong \mathbb{P}^{r-1}$ of $E \rightarrow Y$. The blowup description gives

$$\mathcal{O}_{\tilde{X}}(mE)|_F \cong \mathcal{O}_F(-m).$$

This line bundle is trivial only for $m = 0$. Hence $[E]$ has infinite order, the first map in (1) is injective, and the asserted direct-sum decomposition follows.

(b) The formula may be checked etale-locally at a general point of Y . Choose smooth coordinates $x_1, \dots, x_n$ in which Y is given by $x_1 = \dots = x_r = 0$. On the blowup chart in which the first homogeneous coordinate is nonzero, use coordinates

$$x_1, \quad u_2 = \frac{x_2}{x_1}, \dots, u_r = \frac{x_r}{x_1}, \quad x_{r+1}, \dots, x_n.$$

Here E is defined by $x_1 = 0$, and direct differentiation gives, up to sign,

$$\pi^*(dx_1 \wedge \dots \wedge dx_n) = x_1^{r-1} dx_1 \wedge du_2 \wedge \dots \wedge du_r \wedge dx_{r+1} \wedge \dots \wedge dx_n. \quad (2)$$

Thus the natural map $\pi^*\omega_X \rightarrow \omega_{\tilde{X}}$ vanishes to order $r-1$ along E . Away from E , π is an isomorphism, so (2) has no other zero divisor. Therefore its divisor is exactly $(r-1)E$, which is equivalent to

$$\omega_{\tilde{X}} \cong \pi^*\omega_X \otimes \mathcal{O}_{\tilde{X}}((r-1)E).$$

□

Exercise II.8.6

Permitted context.

Let k be algebraically closed, let A be a finitely generated k -algebra with $\text{Spec } A$ nonsingular, and let

$$0 \longrightarrow I \longrightarrow B' \longrightarrow B \longrightarrow 0$$

be an exact sequence of k -algebras with $I^2 = 0$. The action of B' on I factors through B , so a map $f : A \rightarrow B$ makes I an A -module. For a nonsingular affine variety, $\Omega_{A/k}$ is a finite projective A -module and the conormal sequence of a polynomial presentation is left exact.

Problem. Given a k -algebra map $f : A \rightarrow B$, prove that it lifts to a k -algebra map $g : A \rightarrow B'$. Establish this in the following steps.

(a) If $g, g' : A \rightarrow B'$ are two lifts, prove that $g - g'$ is a k -derivation $A \rightarrow I$. Conversely, show that adding any such derivation to a lift produces another lift.

(b) Present $A = P/J$, where $P = k[x_1, \dots, x_N]$. Construct a lift $h : P \rightarrow B'$ of $P \rightarrow A \xrightarrow{f} B$, and show that its restriction induces an A -linear map

$$\bar{h} : J/J^2 \longrightarrow I.$$

(c) Use the conormal sequence and (a)–(b) to modify h so that it kills J , and thereby obtain the desired lift from A .

Solution. (a) Since the two maps have the same composite with $B' \rightarrow B$, their difference $\theta = g - g'$ takes values in I . It is additive and vanishes on k . For $a, b \in A$,

$$\begin{aligned}\theta(ab) &= g(a)g(b) - g'(a)g'(b) \\ &= g(a)\theta(b) + g'(b)\theta(a) \\ &= a\theta(b) + b\theta(a).\end{aligned}$$

The last equality uses the A -module structure on I . Thus $\theta \in \text{Der}_k(A, I)$, equivalently $\theta \in \text{Hom}_A(\Omega_{A/k}, I)$.

Conversely, let $\theta : A \rightarrow I$ be a k -derivation and set $g_\theta = g + \theta$. Because $I^2 = 0$,

$$g_\theta(a)g_\theta(b) = g(ab) + a\theta(b) + b\theta(a) = g_\theta(ab).$$

Also $\theta(1) = 0$, so g_θ is a unital k -algebra map, and it still lifts f .

(b) Choose arbitrary lifts in B' of the elements $f(\bar{x}_i) \in B$. The universal property of the polynomial ring gives a homomorphism $h : P \rightarrow B'$ lifting the prescribed map $P \rightarrow B$. Since the latter kills J , one has $h(J) \subseteq I$, and therefore $h(J^2) \subseteq I^2 = 0$. Thus restriction induces an additive map $\bar{h} : J/J^2 \rightarrow I$.

If $p \in P$ and $j \in J$, then $h(pj) = h(p)h(j)$. On I , multiplication by $h(p)$ depends only on the image of p in A , because elements of I annihilate I . This proves that $\bar{h}$ is A -linear.

(c) Nonsingularity gives the exact conormal sequence

$$0 \longrightarrow J/J^2 \longrightarrow \Omega_{P/k} \otimes_P A \longrightarrow \Omega_{A/k} \longrightarrow 0. \quad (1)$$

Since $\Omega_{A/k}$ is projective, applying $\text{Hom}_A(-, I)$ to (1) gives a surjection

$$\text{Hom}_A(\Omega_{P/k} \otimes_P A, I) \twoheadrightarrow \text{Hom}_A(J/J^2, I). \quad (2)$$

The source of (2) is naturally $\text{Hom}_P(\Omega_{P/k}, I)$, the module of k -derivations $P \rightarrow I$. Choose a derivation θ whose restriction in (2) is $\bar{h}$.

Regard I as an ideal of B' and put $h' = h - \theta$. The square-zero calculation in (a), applied to P , shows that h' is again a k -algebra homomorphism lifting $P \rightarrow B$. For every $j \in J$, the choice of θ gives $\theta(j) = h(j)$, so $h'(j) = 0$. Hence h' factors uniquely through $P/J = A$, producing the required lift $g : A \rightarrow B'$. $\square$

Exercise II.8.7**Permitted context.**

An infinitesimal extension of a finite-type k -scheme X by a coherent sheaf $\mathcal{F}$ is a scheme X' with a square-zero ideal $\mathcal{I}$ such that $(X', \mathcal{O}_{X'}/\mathcal{I}) \cong (X, \mathcal{O}_X)$ and $\mathcal{I} \cong \mathcal{F}$ as $\mathcal{O}_X$ -modules. Its trivial extension has sheaf of rings $\mathcal{O}_X \oplus \mathcal{F}$, with multiplication

$$(a, u)(b, v) = (ab, av + bu).$$

We may use the infinitesimal lifting property for smooth affine k -schemes and exactness of the augmented Čech complex of a quasi-coherent sheaf for a finite principal-open cover of an affine scheme.

Problem. Let X be affine and smooth over k , and let $\mathcal{F}$ be coherent on X . Prove that every infinitesimal extension of X by $\mathcal{F}$ is isomorphic, as an extension, to the trivial one.

Solution. Write $X = \operatorname{Spec} A$, and let $(X', \mathcal{I})$ be such an extension. A square-zero closed immersion is a homeomorphism on underlying spaces, so we identify the spaces of X and X' .

Choose a finite principal-open cover $U_i = D(f_i)$ of X such that the corresponding opens U'_i of X' are affine. To obtain one, start with affine opens of X' , refine their reductions by principal opens of the affine scheme X , and lift each defining function; its nonvanishing locus in the affine thickening is affine. On each member of this cover we have a square-zero extension of rings

$$0 \longrightarrow \mathcal{F}(U_i) \longrightarrow \mathcal{O}_{X'}(U'_i) \longrightarrow \mathcal{O}_X(U_i) \longrightarrow 0.$$

Since U_i is smooth and affine, the infinitesimal lifting property provides a k -algebra section s_i of the last arrow.

On $U_{ij} = U_i \cap U_j$, part (a) of the lifting property identifies

$$c_{ij} = s_i - s_j$$

with a k -derivation $\mathcal{O}_X|_{U_{ij}} \rightarrow \mathcal{F}|_{U_{ij}}$. Equivalently, c_{ij} is a section on U_{ij} of

$$\mathcal{D} = \mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{F}).$$

Because X is nonsingular, $\Omega_{X/k}$ is finite locally free, so $\mathcal{D}$ is quasi-coherent. The identities among the differences make (c_{ij}) a Čech one-cocycle. Exactness of the Čech complex on the affine scheme X gives sections $d_i \in \mathcal{D}(U_i)$ such that

$$c_{ij} = d_i - d_j.$$

By the converse assertion in the lifting property, $s'_i = s_i - d_i$ is still a ring-theoretic splitting. The displayed identity says that the s'_i agree on all overlaps, so they glue to a morphism of sheaves of k -algebras

$$s : \mathcal{O}_X \longrightarrow \mathcal{O}_{X'}$$

splitting $\mathcal{O}_{X'} \twoheadrightarrow \mathcal{O}_X$.

Now define

$$\Phi : \mathcal{O}_X \oplus \mathcal{F} \longrightarrow \mathcal{O}_{X'}, \quad (a, u) \longmapsto s(a) + u.$$

It is an additive isomorphism because s splits the quotient and $\mathcal{F} = \mathcal{I}$ is its kernel. Moreover, $\mathcal{I}^2 = 0$ and its $\mathcal{O}_X$ -module structure is induced through the quotient, so

$$\Phi(a, u)\Phi(b, v) = \Phi(ab, av + bu).$$

Thus Φ is an isomorphism of sheaves of rings and identifies $(X', \mathcal{I})$ with the trivial extension. $\square$

Exercise II.8.8**Permitted context.**

For a nonsingular projective variety X , define

$$P_m(X) = \dim_k H^0(X, \omega_X^{\otimes m}) \quad (m > 0), \quad h^{q,0}(X) = \dim_k H^0(X, \Omega_{X/k}^q),$$

where $\Omega_{X/k}^q = \bigwedge^q \Omega_{X/k}$. We may use two facts from the rational-map extension argument: a rational map from a nonsingular variety to a projective variety is defined outside a subset of codimension at least two; and sections of a locally free sheaf on a nonsingular variety extend uniquely across such a subset.

Problem. Prove that every plurigenus P_m and every number $h^{q,0}$ is a birational invariant among nonsingular projective varieties.

Solution. Let X and X' be birational nonsingular projective varieties, and choose dense opens $U \subseteq X$ and $U' \subseteq X'$ that are isomorphic. Regard this isomorphism as a rational map $\varphi : X \dashrightarrow X'$. There is an open $V \subseteq X$, containing U , such that $X \setminus V$ has codimension at least two and φ extends to a morphism $f : V \rightarrow X'$.

Functoriality of differentials gives

$$f^* \Omega_{X'/k}^q \longrightarrow \Omega_{V/k}^q.$$

Likewise, taking the top exterior power of the differential and then the m -th tensor power gives

$$f^*(\omega_{X'}^{\otimes m}) \longrightarrow \omega_V^{\otimes m}.$$

Pulling back global sections and using extension across $X \setminus V$ therefore defines linear maps

$$H^0(X', \Omega_{X'/k}^q) \longrightarrow H^0(X, \Omega_{X/k}^q) \quad (1)$$

and

$$H^0(X', \omega_{X'}^{\otimes m}) \longrightarrow H^0(X, \omega_X^{\otimes m}). \quad (2)$$

Both are injective. Indeed, on U the map f is the chosen isomorphism $U \cong U'$. A section in the kernel consequently vanishes on the dense open U' , and a section of a locally free sheaf that vanishes on a dense open of an integral scheme is zero. It follows from (1)–(2) that

$$h^{q,0}(X') \leq h^{q,0}(X), \quad P_m(X') \leq P_m(X).$$

Apply the same construction to the inverse birational map $X' \dashrightarrow X$. The reverse inequalities follow, proving

$$h^{q,0}(X') = h^{q,0}(X), \quad P_m(X') = P_m(X)$$

for every permitted q and every $m > 0$. □

II.9 Formal Schemes

Exercise II.9.1

Permitted context.

Let k be algebraically closed, let $X = \mathbb{P}_k^n$, and let $Y \subseteq X$ be a connected, nonsingular, positive-dimensional closed subvariety with ideal sheaf $\mathcal{I}$. Its formal completion has underlying space Y and structure sheaf

$$\mathcal{O}_{\widehat{X}} = \varprojlim_{r \geq 1} \mathcal{O}_X / \mathcal{I}^r.$$

We may use the Euler sequence on projective space, the exact conormal sequence for the regular immersion $Y \hookrightarrow X$, and

$$\mathcal{I}^r / \mathcal{I}^{r+1} \cong \mathrm{Sym}_{\mathcal{O}_Y}^r(\mathcal{I} / \mathcal{I}^2).$$

Problem. (a) Prove that there is an injection

$$\mathcal{I} / \mathcal{I}^2 \hookrightarrow \mathcal{O}_Y(-1)^{\oplus(n+1)}.$$

(b) Prove that

$$H^0(Y, \mathcal{I}^r / \mathcal{I}^{r+1}) = 0 \quad (r \geq 1).$$

(c) Using the successive infinitesimal-neighborhood sequences, prove

$$H^0(Y, \mathcal{O}_X / \mathcal{I}^r) = k \quad (r \geq 1).$$

(d) Deduce that every formal-regular function along Y is constant:

$$\Gamma(\widehat{X}, \mathcal{O}_{\widehat{X}}) = k.$$

Solution. (a) Since both X and Y are nonsingular, the closed immersion is regular and its conormal sequence begins

$$0 \longrightarrow \mathcal{I} / \mathcal{I}^2 \longrightarrow \Omega_{X/k}|_Y. \quad (1)$$

The Euler sequence begins

$$0 \longrightarrow \Omega_{X/k} \longrightarrow \mathcal{O}_X(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Tensoring with $\mathcal{O}_Y$ preserves injectivity on the left because the last term $\mathcal{O}_X$ is locally free. Restricting the resulting injection to Y and composing with (1) gives

$$\mathcal{I} / \mathcal{I}^2 \hookrightarrow \mathcal{O}_Y(-1)^{\oplus(n+1)}.$$

(b) Let $\mathcal{Q}$ be the quotient of the injection in (a). The conormal and Euler sequences give

$$0 \longrightarrow \Omega_{Y/k} \longrightarrow \mathcal{Q} \longrightarrow \mathcal{O}_Y \longrightarrow 0.$$

Its end terms are locally free, so $\mathcal{Q}$ is locally free. It is this fact about the quotient that makes the injection in (a) locally split. Taking the r -th symmetric power therefore gives an injection

$$\mathcal{I}^r / \mathcal{I}^{r+1} \cong \mathrm{Sym}^r(\mathcal{I} / \mathcal{I}^2) \hookrightarrow \mathrm{Sym}^r(\mathcal{O}_Y(-1)^{\oplus(n+1)}) \cong \mathcal{O}_Y(-r)^{\oplus N_r} \quad (2)$$

for a positive integer N_r .

It remains to show $H^0(Y, \mathcal{O}_Y(-r)) = 0$. A connected nonsingular variety is integral: distinct irreducible components of a regular scheme are disjoint, and connectedness leaves only one. Choose a hyperplane H that does not contain Y . Since $\dim Y > 0$, the projective dimension theorem gives $H \cap Y \neq \emptyset$. Multiplication by an equation of rH is an injection

$$\mathcal{O}_Y(-r) \hookrightarrow \mathcal{O}_Y.$$

A global section maps to a global regular function on the projective integral variety Y , hence to a constant in k . That function vanishes on $H \cap Y$, so the constant is zero. The original section is therefore zero. Taking global sections in (2) proves (b).

(c) For every $r \geq 1$ there is an exact sequence on Y

$$0 \longrightarrow \mathcal{I}^r / \mathcal{I}^{r+1} \longrightarrow \mathcal{O}_X / \mathcal{I}^{r+1} \longrightarrow \mathcal{O}_X / \mathcal{I}^r \longrightarrow 0. \quad (3)$$

By (b), taking global sections makes the last map in (3) injective:

$$H^0(Y, \mathcal{O}_X / \mathcal{I}^{r+1}) \hookrightarrow H^0(Y, \mathcal{O}_X / \mathcal{I}^r).$$

For $r = 1$, the right side is $H^0(Y, \mathcal{O}_Y) = k$. At every level the constant sections give a copy of k in the group on the left. Induction therefore squeezes every group between k and k , proving the assertion.

(d) Inverse limits of sheaves are computed sectionwise. Hence

$$\begin{aligned} \Gamma(\widehat{X}, \mathcal{O}_{\widehat{X}}) &= \varprojlim_r \Gamma(Y, \mathcal{O}_X / \mathcal{I}^r) \\ &= \varprojlim_r k = k. \end{aligned}$$

The transition maps on the final inverse system are the identity on constant sections. $\square$

Exercise II.9.2

Permitted context.

Retain the hypotheses of Exercise II.9.1: $X = \mathbb{P}_k^n$ and $Y \subseteq X$ is connected, nonsingular, and positive-dimensional. If $\mathcal{I}$ is its ideal sheaf and

$$Y_r = (Y, \mathcal{O}_X / \mathcal{I}^r),$$

then $\Gamma(Y_r, \mathcal{O}_{Y_r}) = k$ for every $r \geq 1$. We may use the Krull intersection theorem for a noetherian domain.

Problem. Let $f : X \rightarrow Z$ be a morphism of k -varieties. If $f(Y)$ is the single point $P \in Z$, prove that $f(X) = \{P\}$.

Solution. Choose an affine neighborhood $W = \operatorname{Spec} B$ of P , and let $\mathfrak{m}_P \subseteq B$ be its maximal ideal. The underlying space of every Y_r is Y , whose image is P , so the restriction $Y_r \rightarrow Z$ factors through the open set W . It corresponds to a k -algebra map

$$B \longrightarrow \Gamma(Y_r, \mathcal{O}_{Y_r}) = k.$$

After reduction to Y , this map is evaluation at P . It is therefore the quotient $B \rightarrow B/\mathfrak{m}_P = k$. Thus every infinitesimal neighborhood Y_r maps scheme-theoretically to P .

The open set $f^{-1}(W)$ contains Y . Choose a nonempty affine open $U = \operatorname{Spec} A$ inside it that meets Y , and let $I \subsetneq A$ define $Y \cap U$. Since U is an open subscheme of projective space, A is a noetherian domain. If $\varphi : B \rightarrow A$ is the map induced by $f|_U$, the conclusion for all Y_r says

$$\varphi(\mathfrak{m}_P) \subseteq I^r \quad \text{for every } r \geq 1.$$

The Krull intersection theorem gives

$$\bigcap_{r \geq 1} I^r = 0,$$

and consequently $\varphi(\mathfrak{m}_P) = 0$. Hence $f|_U$ factors through the reduced point P .

Let $c_P : X \rightarrow Z$ be the constant morphism with value P . A variety is separated, so the equalizer of f and c_P is closed in X . It contains the nonempty open U , which is dense because X is integral. The equalizer is therefore all of X , and $f = c_P$. $\square$

Exercise II.9.3

Permitted context.

Let $\mathfrak{X}$ be an affine noetherian formal scheme with an ideal of definition $\mathfrak{I}$. A coherent sheaf $\mathfrak{G}$ is complete:

$$\mathfrak{G} \cong \varprojlim_n \mathfrak{G}/\mathfrak{I}^n \mathfrak{G},$$

and on an affine formal open its quotient sections form the surjective system $M/\mathfrak{I}^n M$. We may also use the affine lifting lemma: in an exact sequence of sheaves on an affine scheme, global sections are exact if the kernel is quasi-coherent, even when the other two sheaves are only sheaves of modules.

Problem. Let

$$0 \longrightarrow \mathfrak{F}' \longrightarrow \mathfrak{F} \longrightarrow \mathfrak{F}'' \longrightarrow 0 \quad (1)$$

be exact, and suppose $\mathfrak{F}'$ is coherent.

(a) For an affine formal open $\mathfrak{U} \subseteq \mathfrak{X}$ and $n > 0$, prove exactness of

$$\begin{aligned} 0 \longrightarrow \Gamma(\mathfrak{U}, \mathfrak{F}'/\mathfrak{I}^n \mathfrak{F}') &\longrightarrow \Gamma(\mathfrak{U}, \mathfrak{F}/\mathfrak{I}^n \mathfrak{F}) \\ &\longrightarrow \Gamma(\mathfrak{U}, \mathfrak{F}'') \longrightarrow 0. \end{aligned} \quad (2)$$

(b) Pass to the inverse limit, prove

$$\mathfrak{F} \cong \varprojlim_n \mathfrak{F}/\mathfrak{I}^n \mathfrak{F},$$

and deduce that applying global sections to (1) gives a short exact sequence.

Solution. (a) Regard $\mathfrak{I}^n \mathfrak{F}'$ as a subsheaf of $\mathfrak{F}$ through the first map in (1). Quotienting gives an exact sequence

$$0 \longrightarrow \mathfrak{F}'/\mathfrak{I}^n \mathfrak{F}' \longrightarrow \mathfrak{F}/\mathfrak{I}^n \mathfrak{F} \longrightarrow \mathfrak{F}'' \longrightarrow 0. \quad (3)$$

The first term is coherent and is annihilated by $\mathfrak{I}^n$. It is therefore a quasi-coherent sheaf on the ordinary affine scheme

$$U_n = (\mathfrak{U}, \mathcal{O}_{\mathfrak{U}}/\mathfrak{I}^n).$$

All sheaves in (3) have the same underlying topological space. Only its kernel is asserted to be a module on U_n . The vanishing $H^1(U_n, \mathfrak{F}'/\mathcal{I}^n \mathfrak{F}') = 0$ therefore proves (2) by the long exact sequence of abelian sheaves. Concretely, local lifts of a section of $\mathfrak{F}''$ differ by sections of the quasi-coherent first term, and affine exactness removes that Čech one-cocycle so that the lifts glue.

(b) For a fixed affine $\mathfrak{U}$, abbreviate the three terms of (2) by A_n, B_n, C . Coherence of $\mathfrak{F}'$ identifies

$$A_n = M/\mathcal{I}^n M$$

for a finite module M , so the transition maps $A_{n+1} \rightarrow A_n$ are surjective. Taking inverse limits of

$$0 \longrightarrow A_n \longrightarrow B_n \longrightarrow C \longrightarrow 0$$

is consequently exact. For surjectivity on the right, one may choose a lift in B_1 and recursively alter a lift in B_{n+1} by an element of A_{n+1} so that it restricts to the lift already chosen in B_n . Hence

$$0 \longrightarrow \Gamma(\mathfrak{U}, \mathfrak{F}') \longrightarrow \varprojlim_n \Gamma(\mathfrak{U}, \mathfrak{F}'/\mathcal{I}^n \mathfrak{F}') \longrightarrow \Gamma(\mathfrak{U}, \mathfrak{F}'') \longrightarrow 0. \quad (4)$$

Set

$$\mathfrak{G} = \varprojlim_n \mathfrak{F}'/\mathcal{I}^n \mathfrak{F}'.$$

Limits of sheaves are computed sectionwise, so (4), for all affine formal opens in a basis, gives an exact sequence

$$0 \longrightarrow \mathfrak{F}' \longrightarrow \mathfrak{G} \longrightarrow \mathfrak{F}'' \longrightarrow 0. \quad (5)$$

There is a natural morphism from (1) to (5): it is the completeness isomorphism on $\mathfrak{F}'$, the map $\mathfrak{F} \rightarrow \mathfrak{G}$ in the middle, and the identity on $\mathfrak{F}''$. On every stalk, the short five lemma makes the middle map an isomorphism. Thus

$$\mathfrak{F} \cong \varprojlim_n \mathfrak{F}'/\mathcal{I}^n \mathfrak{F}'.$$

Finally take $\mathfrak{U} = \mathfrak{X}$ in (4) and use this isomorphism. The resulting sequence is

$$0 \longrightarrow \Gamma(\mathfrak{X}, \mathfrak{F}') \longrightarrow \Gamma(\mathfrak{X}, \mathfrak{F}) \longrightarrow \Gamma(\mathfrak{X}, \mathfrak{F}'') \longrightarrow 0,$$

as required. □

Exercise II.9.4**Permitted context.**

On an affine noetherian formal scheme $\mathfrak{U} = \mathrm{Spf} A$, the functor

$$M \mapsto M^\Delta$$

is an exact equivalence from finite A -modules to coherent $\mathcal{O}_{\mathfrak{U}}$ -modules, with quasi-inverse global sections. We may use Exercise II.9.3: global sections are exact when the kernel in a short exact sequence is coherent.

Problem. Let $\mathfrak{X}$ be a noetherian formal scheme. If

$$0 \longrightarrow \mathfrak{F}' \longrightarrow \mathfrak{F} \longrightarrow \mathfrak{F}'' \longrightarrow 0$$

is exact and both $\mathfrak{F}'$ and $\mathfrak{F}''$ are coherent, prove that $\mathfrak{F}$ is coherent.

Solution. Coherence is local, so take $\mathfrak{X} = \mathrm{Spf} A$. Put

$$M' = \Gamma(\mathfrak{X}, \mathfrak{F}'), \quad M = \Gamma(\mathfrak{X}, \mathfrak{F}), \quad M'' = \Gamma(\mathfrak{X}, \mathfrak{F}'').$$

Exercise II.9.3 gives a short exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0. \quad (1)$$

The outer modules are finite over A , by the affine equivalence. Lifting a finite generating set of M'' to M and adjoining a finite generating set of M' shows that M is finite.

Apply the exact functor $(\cdot)^\Delta$ to (1). There is a natural morphism from the resulting sequence

$$0 \longrightarrow (M')^\Delta \longrightarrow M^\Delta \longrightarrow (M'')^\Delta \longrightarrow 0$$

to the original sheaf sequence. The two outer vertical maps are isomorphisms because $\mathfrak{F}'$ and $\mathfrak{F}''$ are coherent. The short five lemma, applied on stalks, makes

$$M^\Delta \xrightarrow{\sim} \mathfrak{F}$$

an isomorphism. Since M is finite, $\mathfrak{F}$ is coherent. The same argument on each affine formal open proves the result on $\mathfrak{X}$. $\square$

Exercise II.9.5**Permitted context.**

A noetherian formal scheme is quasi-compact and has a finite affine formal open cover. On $\mathfrak{U} = \mathrm{Spf} A$, coherent sheaves are exactly the sheaves M^Δ associated to finite A -modules, and this correspondence is faithful.

Problem. Let $\mathfrak{F}$ be a coherent sheaf on a noetherian formal scheme. If its global sections generate it, prove that finitely many of those global sections already generate it.

Solution. Let $\{s_\alpha\}_{\alpha \in \Lambda}$ be a generating family and choose a finite affine formal open cover

$$\mathfrak{X} = \mathfrak{U}_1 \cup \cdots \cup \mathfrak{U}_t.$$

Write

$$\mathfrak{F}|_{\mathfrak{U}_i} \cong M_i^\Delta$$

for a finite module M_i over $A_i = \Gamma(\mathfrak{U}_i, \mathcal{O}_{\mathfrak{X}})$. The restrictions $s_\alpha|_{\mathfrak{U}_i}$ correspond to elements $m_{\alpha,i} \in M_i$.

Let $N_i \subseteq M_i$ be the submodule generated by all the $m_{\alpha,i}$. Since the original sections generate the sheaf, the associated sheaf of M_i/N_i is zero. The quotient is a finite A_i -module, so faithfulness of the affine equivalence gives $M_i/N_i = 0$. Thus the elements $m_{\alpha,i}$ generate M_i . Because M_i is finite, some finite subfamily, indexed by $\Lambda_i \subseteq \Lambda$, already generates it.

The union $\Lambda_1 \cup \cdots \cup \Lambda_t$ is finite. The corresponding global sections generate $\mathfrak{F}$ on every member of the cover, and hence on all of $\mathfrak{X}$. $\square$

Exercise II.9.6

Permitted context.

Let $\mathfrak{X}$ be a noetherian formal scheme with ideal of definition $\mathfrak{I}$, and put

$$Y_n = (|\mathfrak{X}|, \mathcal{O}_{\mathfrak{X}}/\mathfrak{I}^n), \quad n \geq 1.$$

Assume that the inverse system of rings

$$A_n = \Gamma(Y_n, \mathcal{O}_{Y_n})$$

satisfies the Mittag-Leffler condition. We may use the formal coherence theorem: a compatible inverse system of coherent sheaves $\mathcal{F}_n$ on Y_n , with $\mathcal{F}_{n+1} \otimes_{\mathcal{O}_{Y_n}} \cong \mathcal{F}_n$, has a coherent limit on $\mathfrak{X}$ whose reduction to Y_n is $\mathcal{F}_n$. We also use the elementary fact that a nonempty Mittag-Leffler inverse sequence of sets has a nonempty inverse limit. In (d), all the projective schemes Y_n and their transition maps are over the same field k .

Problem. Prove that reduction induces an isomorphism

$$\mathrm{Pic}(\mathfrak{X}) \xrightarrow{\sim} \varprojlim_n \mathrm{Pic}(Y_n)$$

by establishing the following points.

- (a) Show that the inverse system of unit groups A_n^* is Mittag-Leffler.
- (b) If a coherent $\mathfrak{F}$ has $\mathfrak{F}/\mathfrak{I}^n \mathfrak{F} \cong \mathcal{O}_{Y_n}$ for every n , prove $\mathfrak{F} \cong \mathcal{O}_{\mathfrak{X}}$. Deduce injectivity of the Picard-group map.
- (c) Starting from a compatible family of invertible sheaf classes on the Y_n , construct an invertible sheaf on $\mathfrak{X}$ that reduces to that family.
- (d) Verify the ring Mittag-Leffler hypothesis when $\mathfrak{X}$ is affine, and also when every Y_n is projective over a field.

Solution. (a) For $m \geq n$, let $\rho_{mn} : A_m \rightarrow A_n$ be restriction. Its kernel consists of sections lying in the nilpotent ideal $\mathfrak{I}^n/\mathfrak{I}^m$; hence that kernel is nilpotent. More importantly, nilpotence is valid sheafwise: a section of $\mathcal{O}_{Y_m}$ whose restriction to Y_n is a unit is locally a unit, and therefore has a unique global inverse. It follows that

$$A_m^* = \rho_{mn}^{-1}(A_n^*)$$

and consequently

$$\mathrm{im}(A_m^* \rightarrow A_n^*) = \mathrm{im}(A_m \rightarrow A_n) \cap A_n^*. \quad (1)$$

For fixed n , the ring images on the right stabilize as m increases by hypothesis. Their intersections with A_n^* then stabilize as well. Thus (A_n^*) is Mittag-Leffler.

(b) Put $\mathfrak{F}_n = \mathfrak{F}/\mathfrak{I}^n \mathfrak{F}$, and let

$$T_n = \mathrm{Isom}_{\mathcal{O}_{Y_n}}(\mathfrak{F}_n, \mathcal{O}_{Y_n}).$$

Each T_n is nonempty and is a torsor under

$$\mathrm{Aut}(\mathcal{O}_{Y_n}) = A_n^*.$$

Restriction gives an inverse system of the T_n . For fixed n , the image of $T_m \rightarrow T_n$ is a coset under the image of $A_m^* \rightarrow A_n^*$. By (a), the latter subgroups eventually stabilize. The former images are nested nonempty cosets, so they stabilize too. Hence (T_n) is a nonempty Mittag-Leffler system of sets and has a compatible element $(\varphi_n) \in \varprojlim T_n$.

Coherence gives the completeness isomorphisms

$$\mathfrak{F} \cong \varprojlim_n \mathfrak{F}_n, \quad \mathcal{O}_{\mathfrak{X}} \cong \varprojlim_n \mathcal{O}_{Y_n}.$$

Taking the inverse limit of the compatible φ_n therefore gives

$$\mathfrak{F} \xrightarrow{\sim} \mathcal{O}_{\mathfrak{X}}.$$

If an invertible sheaf on $\mathfrak{X}$ maps to zero in every $\mathrm{Pic}(Y_n)$, its reductions satisfy the hypothesis just considered, so it is trivial. The natural Picard-group map is injective.

(c) Let (ξ_n) represent an element of the inverse limit of the Picard groups. Choose an invertible sheaf $\mathcal{L}_n$ in the class ξ_n . Compatibility of the classes permits choices of isomorphisms

$$\alpha_n : \mathcal{L}_{n+1} \otimes_{\mathcal{O}_{Y_{n+1}}} \mathcal{O}_{Y_n} \xrightarrow{\sim} \mathcal{L}_n.$$

Compose the quotient map $\mathcal{L}_{n+1} \rightarrow \mathcal{L}_{n+1} \otimes_{\mathcal{O}_{Y_{n+1}}} \mathcal{O}_{Y_n}$ with α_n , and use composites to define $\mathcal{L}_m \rightarrow \mathcal{L}_n$ for every $m \geq n$. This is an inverse system satisfying the hypotheses of the formal coherence theorem. Consequently

$$\mathfrak{L} = \varprojlim_n \mathcal{L}_n$$

is coherent and $\mathfrak{L}/\mathfrak{I}^n \mathfrak{L} \cong \mathcal{L}_n$.

We prove that $\mathfrak{L}$ is invertible without allowing the trivializing neighborhood to depend on n . Fix a point, choose an affine formal neighborhood $\mathfrak{U} = \mathrm{Spf} A$, and write

$$\mathfrak{L}|_{\mathfrak{U}} = M^{\Delta}$$

for a finite A -module M . Let $J \subseteq A$ be the ideal of definition induced by $\mathfrak{I}$. Shrink $\mathfrak{U}$ once so that the line bundle M/JM on $\mathrm{Spec}(A/J)$ is free. Choose a basis $\bar{s}$ and lift it to $s \in M$.

For every n , consider

$$A/J^n \longrightarrow M/J^n M, \quad 1 \longmapsto s. \quad (2)$$

The target is the module of the line bundle $\mathcal{L}_n|_{\mathfrak{U}}$, so it is locally free of rank one. In a local trivialization, (2) is multiplication by an element $c \in A/J^n$. Its reduction modulo J is a unit because $\bar{s}$ is a basis. Since J/J^n is nilpotent, c is a unit: if $cd = 1 - e$ with $e \in J/J^n$, a finite geometric series in e supplies an inverse. Thus (2) is an isomorphism at every point, and hence globally.

The adic ring A and its finite module M are complete, so taking inverse limits in (2) gives

$$A \xrightarrow{\sim} M.$$

Therefore $\mathfrak{L}$ is free of rank one on the fixed neighborhood $\mathfrak{U}$. It is an invertible sheaf on $\mathfrak{X}$, and its reductions represent the prescribed ξ_n . This proves surjectivity.

(d) If $\mathfrak{X} = \mathrm{Spf} A$ with ideal of definition J , then

$$Y_n = \mathrm{Spec}(A/J^n), \quad \Gamma(Y_n, \mathcal{O}_{Y_n}) = A/J^n.$$

The transition maps are surjective, so the system is Mittag-Leffler.

If every Y_n is projective over a field k , each $A_n = H^0(Y_n, \mathcal{O}_{Y_n})$ is a finite-dimensional k -vector space. For fixed n , the images of $A_m \rightarrow A_n$ form a descending chain of subspaces of the finite-dimensional space A_n , so they stabilize. This is the Mittag-Leffler condition. $\square$

Chapter III

III.2 Cohomology of Sheaves

Exercise III.2.1

Permitted context.

For a locally closed subset $T \subseteq X$, write $\mathbb{Z}_T$ for the constant sheaf $\mathbb{Z}$ on T , extended by zero to X . A constant sheaf on an irreducible topological space is flasque, as is its direct image under a closed immersion. A flasque resolution computes sheaf cohomology.

Problem. (a) Let $X = \mathbb{A}_k^1$, where k is infinite. For distinct closed points P, Q , put $U = X \setminus \{P, Q\}$. Prove

$$H^1(X, \mathbb{Z}_U) \neq 0.$$

(b) More generally, in $X = \mathbb{A}_k^n$ take $n + 1$ affine hyperplanes in general position, let Y be their union, and put $U = X \setminus Y$. Prove

$$H^n(X, \mathbb{Z}_U) \neq 0.$$

Solution. (a) Let $V = \{P, Q\}$. There is an exact sequence

$$0 \longrightarrow \mathbb{Z}_U \longrightarrow \mathbb{Z}_X \longrightarrow \mathbb{Z}_V \longrightarrow 0. \quad (1)$$

The affine line is irreducible, so $\mathbb{Z}_X$ is flasque. The sheaf $\mathbb{Z}_V$ is a finite sum of skyscraper sheaves and is also flasque. Thus (1) is a flasque resolution. On global sections its nonzero map is the diagonal map, and hence

$$H^1(X, \mathbb{Z}_U) \cong \operatorname{coker}(\mathbb{Z} \longrightarrow \mathbb{Z} \oplus \mathbb{Z}) \cong \mathbb{Z}.$$

(b) Write the hyperplanes as $H_0, \dots, H_n$, chosen so that every intersection of $r \leq n$ of them is a nonempty affine linear space of codimension r , while $\bigcap_{i=0}^n H_i = \emptyset$. For a nonempty index set I , put $H_I = \bigcap_{i \in I} H_i$. Alternating restriction maps give an exact complex

$$\begin{aligned} 0 \longrightarrow \mathbb{Z}_U \longrightarrow \mathbb{Z}_X \longrightarrow \bigoplus_{|I|=1} \mathbb{Z}_{H_I} \longrightarrow \bigoplus_{|I|=2} \mathbb{Z}_{H_I} \\ \longrightarrow \cdots \longrightarrow \bigoplus_{|I|=n} \mathbb{Z}_{H_I} \longrightarrow 0. \end{aligned} \quad (2)$$

Exactness may be checked at a point x . If $x \in U$, the first map is an isomorphism on the stalk. Otherwise, the indices of the hyperplanes through x form a nonempty set, and the stalk complex is the augmented cochain complex of a simplex, hence exact.

Every nonempty H_I is irreducible. All terms following $\mathbb{Z}_U$ in (2) are therefore flasque, so (2) computes cohomology. Taking global sections produces

$$\mathbb{Z} \longrightarrow \mathbb{Z}^{\binom{n+1}{1}} \longrightarrow \cdots \longrightarrow \mathbb{Z}^{\binom{n+1}{n}} \longrightarrow 0. \quad (3)$$

If one appends the final group $\mathbb{Z}$, indexed by the full set $\{0, \dots, n\}$, this is the exact augmented cochain complex of the n -simplex. Its last differential onto $\mathbb{Z}$ is surjective. Consequently the cokernel of the penultimate map in (3) is $\mathbb{Z}$, and

$$H^n(X, \mathbb{Z}_U) \cong \mathbb{Z} \neq 0.$$

□

Exercise III.2.2**Permitted context.**

Let $X = \mathbb{P}_k^1$ over an algebraically closed field, let $\mathcal{K}$ be its sheaf of rational functions, and regard $\mathcal{O}_X \subseteq \mathcal{K}$. We use the elementary partial-fraction decomposition of rational functions.

Problem. Prove that

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{K} \longrightarrow \mathcal{K}/\mathcal{O}_X \longrightarrow 0$$

is a flasque resolution. Deduce that

$$H^i(X, \mathcal{O}_X) = 0 \quad (i > 0).$$

Solution. Every nonempty open subset of the integral curve X has the same function field $K = k(X)$. Thus $\mathcal{K}$ is the constant sheaf with value K , and it is flasque.

Taking the principal part at each closed point gives a sheaf isomorphism

$$\mathcal{K}/\mathcal{O}_X \cong \bigoplus_{p \in X} i_{p*}(K/\mathcal{O}_{X,p}). \quad (1)$$

Indeed, this can be checked on stalks; a rational function has poles at only finitely many points. Each summand is a skyscraper sheaf. On the noetherian space X , sections of the direct sum over an open set involve only finitely many summands, so restrictions remain surjective. Hence the sheaf in (1) is flasque.

The displayed sequence is therefore a length-one flasque resolution. It gives $H^i(X, \mathcal{O}_X) = 0$ for $i \geq 2$, while

$$H^1(X, \mathcal{O}_X) \cong \operatorname{coker} \left(K \longrightarrow \bigoplus_{p \in X} K/\mathcal{O}_{X,p} \right).$$

Choose an affine coordinate t . At a finite point $t = a$, write the specified principal part as a finite sum of $c_{a,j}/(t-a)^j$. At infinity write it as a polynomial in t with zero constant term. Adding these finitely many expressions realizes all the prescribed principal parts and has no poles elsewhere. The principal-parts map is therefore surjective. Its cokernel is zero, proving the assertion for every $i > 0$. $\square$

Exercise III.2.3**Permitted context.**

Let Y be closed in a topological space X . For a sheaf $\mathcal{F}$ of abelian groups, define

$$\Gamma_Y(X, \mathcal{F}) = \{s \in \Gamma(X, \mathcal{F}) : \text{supp}(s) \subseteq Y\}.$$

Write $H_Y^i(X, -)$ for the right derived functors of $\Gamma_Y(X, -)$. Injective sheaves are flasque, restrictions of flasque sheaves are flasque, and flasque resolutions compute ordinary sheaf cohomology.

Problem. (a) Prove that $\Gamma_Y(X, -)$ is left exact.

(b) If

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$$

is exact and $\mathcal{F}'$ is flasque, prove that applying $\Gamma_Y(X, -)$ gives a short exact sequence.

(c) Prove that every flasque sheaf is acyclic for $\Gamma_Y(X, -)$.

(d) If $\mathcal{F}$ is flasque and $U = X \setminus Y$, prove

$$0 \rightarrow \Gamma_Y(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{F}) \rightarrow \Gamma(U, \mathcal{F}|_U) \rightarrow 0.$$

(e) Construct the natural long exact sequence

$$\begin{aligned} 0 \rightarrow H_Y^0(X, \mathcal{F}) \rightarrow H^0(X, \mathcal{F}) \rightarrow H^0(U, \mathcal{F}|_U) \\ \rightarrow H_Y^1(X, \mathcal{F}) \rightarrow H^1(X, \mathcal{F}) \rightarrow H^1(U, \mathcal{F}|_U) \rightarrow \cdots \end{aligned}$$

(f) If $V \subseteq X$ is open and contains Y , prove the excision isomorphisms

$$H_Y^i(X, \mathcal{F}) \cong H_Y^i(V, \mathcal{F}|_V)$$

functorially in $\mathcal{F}$ and for all i .

Solution. (a) The functor of global sections is left exact, and support can be tested on germs. Thus, for a morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$,

$$\begin{aligned} \Gamma_Y(X, \ker \varphi) &= \{s \in \Gamma(X, \mathcal{F}) : \varphi(s) = 0, s|_{X \setminus Y} = 0\} \\ &= \ker(\Gamma_Y(X, \mathcal{F}) \rightarrow \Gamma_Y(X, \mathcal{G})). \end{aligned}$$

This proves left exactness.

(b) Let $s'' \in \Gamma_Y(X, \mathcal{F}'')$. A short exact sequence with flasque kernel is exact on global sections, so choose $t \in \Gamma(X, \mathcal{F})$ mapping to s'' . On $U = X \setminus Y$, the restriction $t|_U$ maps to zero and therefore comes from some $a' \in \Gamma(U, \mathcal{F}')$. Flasqueness extends a' to $\tilde{a}' \in \Gamma(X, \mathcal{F}')$. Then

$$t - \tilde{a}'$$

still maps to s'' and vanishes on U , so it is supported in Y . This proves surjectivity; exactness at the other two terms follows from (a).

(c) Embed a flasque sheaf $\mathcal{F}$ into an injective sheaf $\mathcal{I}$, and let $\mathcal{G} = \mathcal{I}/\mathcal{F}$. Both $\mathcal{I}$ and $\mathcal{F}$ are flasque, and the usual restriction-lifting argument shows that $\mathcal{G}$ is flasque. Part (b) makes

$$\Gamma_Y(X, \mathcal{I}) \rightarrow \Gamma_Y(X, \mathcal{G})$$

surjective. The derived-functor sequence therefore gives $H_Y^1(X, \mathcal{F}) = 0$, and, for $i \geq 2$,

$$H_Y^i(X, \mathcal{F}) \cong H_Y^{i-1}(X, \mathcal{G}).$$

Repeating this argument and inducting on i proves $H_Y^i(X, \mathcal{F}) = 0$ for every $i > 0$.

(d) The kernel of restriction $\Gamma(X, \mathcal{F}) \rightarrow \Gamma(U, \mathcal{F}|_U)$ consists exactly of the sections supported in Y . The restriction map is surjective because $\mathcal{F}$ is flasque. This gives the claimed short exact sequence.

(e) Choose a flasque resolution $\mathcal{F} \rightarrow \mathcal{I}^\bullet$. By (c), it computes cohomology with supports; it also computes ordinary cohomology on X and, after restriction, on U . Part (d), degree by degree, gives a short exact sequence of complexes

$$0 \rightarrow \Gamma_Y(X, \mathcal{I}^\bullet) \rightarrow \Gamma(X, \mathcal{I}^\bullet) \rightarrow \Gamma(U, \mathcal{I}^\bullet|_U) \rightarrow 0.$$

The associated cohomology sequence is exactly the displayed long exact sequence.

(f) Use the same flasque resolution. Its restriction to V is a flasque resolution of $\mathcal{F}|_V$, and hence is acyclic for sections supported in Y by (c). Restriction induces a natural isomorphism

$$\Gamma_Y(X, \mathcal{I}^q) \xrightarrow{\sim} \Gamma_Y(V, \mathcal{I}^q|_V) \quad (1)$$

in every degree. To see surjectivity in (1), glue a section on V supported in Y with the zero section on $X \setminus Y$; the two agree on $V \setminus Y$. Injectivity is immediate. Thus the two complexes computing supported cohomology are isomorphic, and taking cohomology proves excision and its functoriality. $\square$

Exercise III.2.4

Permitted context.

For a closed subset $Y \subseteq X$, write $H_Y^i(X, \mathcal{F})$ for cohomology with support in Y . Flasque sheaves are acyclic for this functor, so flasque resolutions may be used to compute it.

Problem. For closed subsets $Y_1, Y_2 \subseteq X$, construct the Mayer–Vietoris sequence

$$\begin{aligned} \cdots \rightarrow H_{Y_1 \cap Y_2}^i(X, \mathcal{F}) &\rightarrow H_{Y_1}^i(X, \mathcal{F}) \oplus H_{Y_2}^i(X, \mathcal{F}) \\ &\rightarrow H_{Y_1 \cup Y_2}^i(X, \mathcal{F}) \rightarrow H_{Y_1 \cap Y_2}^{i+1}(X, \mathcal{F}) \rightarrow \cdots \end{aligned}$$

Solution. If $\mathcal{I}$ is flasque, there is a short exact sequence

$$\begin{aligned} 0 \longrightarrow \Gamma_{Y_1 \cap Y_2}(X, \mathcal{I}) &\longrightarrow \Gamma_{Y_1}(X, \mathcal{I}) \oplus \Gamma_{Y_2}(X, \mathcal{I}) \\ &\longrightarrow \Gamma_{Y_1 \cup Y_2}(X, \mathcal{I}) \longrightarrow 0, \end{aligned} \quad (1)$$

where the first map is $t \mapsto (t, -t)$ and the second is addition. Only surjectivity needs proof. Let s be supported in $Y_1 \cup Y_2$. On

$$W = (X \setminus Y_1) \cup (X \setminus Y_2) = X \setminus (Y_1 \cap Y_2)$$

glue the zero section on $X \setminus Y_1$ to $s|_{X \setminus Y_2}$. They agree on $X \setminus (Y_1 \cup Y_2)$. Since $\mathcal{I}$ is flasque, the resulting section on W extends to a global section s_1 . It vanishes off Y_1 , while $s - s_1$ vanishes off Y_2 . Thus $s = s_1 + (s - s_1)$, proving surjectivity in (1).

Choose a flasque resolution $\mathcal{F} \rightarrow \mathcal{I}^\bullet$. Applying (1) in every degree gives a short exact sequence of complexes. Its long exact cohomology sequence is the asserted Mayer–Vietoris sequence. $\square$

Exercise III.2.5**Permitted context.**

Let X be a Zariski space: it is noetherian, and every irreducible closed subset has a unique generic point. If $P \in X$ is closed, put

$$X_P = \{Q \in X : P \in \overline{\{Q\}}\}$$

with its subspace topology, and let $j : X_P \rightarrow X$ be the inclusion. For a sheaf $\mathcal{F}$, write $\mathcal{F}_P = j^{-1}\mathcal{F}$. The symbol P in H_P^i denotes the closed support $\{P\}$.

Problem. Prove, naturally for every sheaf $\mathcal{F}$ and every i , that

$$H_P^i(X, \mathcal{F}) \cong H_P^i(X_P, \mathcal{F}_P).$$

Solution. We first record a topological description of inverse image in this situation. If $W \subseteq X_P$ is open, then

$$(j^{-1}\mathcal{F})(W) = \varinjlim_{W \subseteq V \cap X_P} \Gamma(V, \mathcal{F}), \quad (1)$$

where V runs through open subsets of X . In other words, the usual presheaf defining $j^{-1}\mathcal{F}$ is already a sheaf.

Here is the needed shrinking argument. Every open subset of the noetherian space X_P is quasi-compact, so gluing reduces to a finite cover. Suppose ambient representatives agree on an ambient neighborhood N of an overlap. The closed set $X \setminus N$ has finitely many irreducible components. If η is the generic point of one component and does not belong to an open $W \subseteq X_P$, then $\overline{\{\eta\}} \cap W = \emptyset$: otherwise η would lie in X_P , and stability of W under generization would put η in W . For each component, remove it from one of the two ambient neighborhoods whose corresponding opens do not contain its generic point. The shrunken ambient neighborhoods still contain the given opens in X_P , but now meet inside N . Repeating this for the finitely many pairs lets the representatives glue. The same argument proves uniqueness, establishing (1).

If $\mathcal{I}$ is flasque on X , formula (1) shows that $j^{-1}\mathcal{I}$ is flasque on X_P . Indeed, a section on a smaller open is represented on some ambient open V ; extend that representative to X by flasqueness, then restrict it to the larger open of X_P .

We next compare the degree-zero functors. Since an open subset of X contains P exactly when it contains all of X_P , (1) gives

$$\Gamma(X_P, \mathcal{F}_P) = \varinjlim_{P \in V} \Gamma(V, \mathcal{F}) = \mathcal{F}_P^{\text{stalk}}. \quad (2)$$

We claim that the subgroups supported at P agree:

$$\Gamma_P(X, \mathcal{F}) \xrightarrow{\sim} \Gamma_P(X_P, \mathcal{F}_P). \quad (3)$$

Injectivity is immediate. For surjectivity, represent an element on the right by $s \in \Gamma(V, \mathcal{F})$, where V contains X_P . Let $S \subseteq V$ be the support of s . The support condition on X_P says

$$S \cap X_P \subseteq \{P\}.$$

The noetherian closed set S has finitely many irreducible components. If such a component contains P , its generic point lies in X_P , so the component must be $\{P\}$. Remove all other components from

V . On the resulting neighborhood V' of P , the section is supported in $\{P\}$. It glues with zero on $X \setminus \{P\}$ to a global section on X supported at P . This proves (3).

Finally, take an injective resolution $\mathcal{F} \rightarrow \mathcal{I}^\bullet$. Inverse image is exact, so $\mathcal{F}_P \rightarrow j^{-1}\mathcal{I}^\bullet$ is a resolution. Its terms are flasque by the preceding argument and hence acyclic for sections supported at P . The isomorphisms (3), applied degree by degree, identify the two complexes

$$\Gamma_P(X, \mathcal{I}^\bullet) \quad \text{and} \quad \Gamma_P(X_P, j^{-1}\mathcal{I}^\bullet).$$

Their cohomology groups are the two sides in the statement, which proves the natural isomorphism. $\square$

Exercise III.2.6

Permitted context.

For an open immersion $u : U \rightarrow X$, let $\mathbb{Z}_U = u_!\mathbb{Z}$. The sheaves $\mathbb{Z}_U$, as U runs through the open subsets of X , form a family of generators for sheaves of abelian groups. We may use the corresponding Baer criterion: a sheaf $\mathcal{I}$ is injective if every map $\mathcal{R} \rightarrow \mathcal{I}$, for every subsheaf $\mathcal{R} \subseteq \mathbb{Z}_U$, extends across $\mathcal{R} \subseteq \mathbb{Z}_U$. A space and its sobrification have the same lattice of open sets and equivalent sheaf categories. The sobrification consists of nonempty irreducible closed subsets, with opens defined by meeting an original open. It is noetherian when the original space is, and every irreducible closed subset has a generic point.

Problem. Let X be a noetherian topological space and $(\mathcal{I}_\alpha)$ a filtered direct system of injective sheaves of abelian groups. Prove that

$$\varinjlim_{\alpha} \mathcal{I}_\alpha$$

is injective.

Solution. We use a finiteness consequence of the noetherian hypothesis: every subsheaf of a finite sum of sheaves $\mathbb{Z}_U$ is finitely presented. We can replace X by its sobrification; this leaves the sheaf category and the assertion unchanged. To prove the noetherian-object assertion, let $\mathcal{R}_1 \subseteq \mathcal{R}_2 \subseteq \cdots$ be subsheaves of a fixed finite sum $\mathcal{G}$ of the $\mathbb{Z}_U$. On a dense open of each irreducible component, disjoint from the other components, $\mathcal{G}$ is a constant sheaf of finite rank. The union of the generic stalks of the $\mathcal{R}_i$ is a subgroup H of a finite-rank free abelian group, so it is finitely generated and occurs at some finite stage. Represent those generators on a smaller dense open. On an irreducible open, sections of the ambient constant sheaf inject into its generic stalk. Therefore every later $\mathcal{R}_i$ equals the same constant subgroup H on this open; generic generation suffices here because of the embedding in the constant ambient sheaf.

The complement is a proper closed subset. Restriction to it sends $\mathbb{Z}_U$ to extension by zero of the constant sheaf on its intersection with U , and is exact. Noetherian induction on closed subsets therefore makes the restricted chain stationary. Combining the open and closed parts proves that the original chain is stationary on X . Thus the finite sums $\mathcal{G}$ are noetherian objects. They are also finitely presented: by adjunction, $\text{Hom}(\mathbb{Z}_U, \mathcal{G}) = \Gamma(U, \mathcal{G})$, and sections over U commute with filtered colimits because all opens of a noetherian space, and all their pairwise intersections, are quasi-compact. After choosing finitely many generators of a subsheaf, its relation sheaf is again a subsheaf of a finite sum of noetherian objects and is finitely generated. This gives a finite presentation and proves the assertion. Equivalently, for such a sheaf $\mathcal{R}$,

$$\text{Hom} \left(\mathcal{R}, \varinjlim_{\alpha} \mathcal{G}_{\alpha} \right) \cong \varinjlim_{\alpha} \text{Hom}(\mathcal{R}, \mathcal{G}_{\alpha}) \quad (2)$$

for every filtered system.

Put $\mathcal{I} = \varinjlim_{\alpha} \mathcal{I}_{\alpha}$. To apply the Baer criterion, take $\mathcal{R} \subseteq \mathbb{Z}_U$ and a morphism $f : \mathcal{R} \rightarrow \mathcal{I}$. By this finiteness and (2), f factors through some stage:

$$\mathcal{R} \xrightarrow{f_{\alpha}} \mathcal{I}_{\alpha} \longrightarrow \mathcal{I}.$$

The sheaf $\mathcal{I}_{\alpha}$ is injective, so f_{α} extends to a morphism

$$\mathbb{Z}_U \longrightarrow \mathcal{I}_{\alpha}.$$

Composing with $\mathcal{I}_{\alpha} \rightarrow \mathcal{I}$ extends f . The Baer criterion now shows that $\mathcal{I}$ is injective. $\square$

Exercise III.2.7

Permitted context.

Give S^1 its usual topology. Let $\mathbb{Z}$ denote the constant sheaf, let $\mathcal{R}$ be the sheaf of germs of continuous real-valued functions, and let $\mathcal{T}$ be the sheaf of germs of continuous S^1 -valued functions. Finite open covers of the circle admit subordinate partitions of unity.

Problem. (a) Compute $H^1(S^1, \mathbb{Z})$ from the derived-functor definition of sheaf cohomology.
 (b) Prove that $H^1(S^1, \mathcal{R}) = 0$.

Solution. (b) Consider an exact sequence

$$0 \longrightarrow \mathcal{R} \longrightarrow \mathcal{I} \longrightarrow \mathcal{Q} \longrightarrow 0 \quad (1)$$

with $\mathcal{I}$ injective. Let $q \in \Gamma(S^1, \mathcal{Q})$. Choose a finite open cover (U_i) and lifts $t_i \in \mathcal{I}(U_i)$. On a double intersection put

$$v_{ij} = t_i - t_j \in \mathcal{R}(U_i \cap U_j).$$

These sections satisfy

$$v_{ij} - v_{ik} + v_{jk} = 0. \quad (2)$$

Choose a partition of unity (φ_k) subordinate to the cover. On U_i , set

$$w_i = \sum_k \varphi_k v_{ik}. \quad (3)$$

The product in (3), initially defined on $U_i \cap U_k$, extends by zero across the complement of the support of φ_k , so w_i is a well-defined continuous function. From (2) and $\sum_k \varphi_k = 1$, one gets

$$v_{ij} = w_i - w_j.$$

Consequently the local sections $t_i - w_i$ agree on overlaps and glue to a global section of $\mathcal{I}$ lifting q . Thus $\Gamma(S^1, \mathcal{I}) \rightarrow \Gamma(S^1, \mathcal{Q})$ is surjective. The long exact sequence of (1), together with injectivity of $\mathcal{I}$, now gives

$$H^1(S^1, \mathcal{R}) = 0.$$

This proves (b).

(a) The local existence of a continuous argument gives an exact exponential sequence of sheaves

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{R} \xrightarrow{f \mapsto \exp(2\pi i f)} \mathcal{T} \longrightarrow 0. \quad (4)$$

Using (b), the long exact sequence of (4) identifies

$$H^1(S^1, \mathbb{Z}) \cong \frac{C(S^1, S^1)}{\{\exp(2\pi i f) : f \in C(S^1, \mathbb{R})\}}. \quad (5)$$

The winding-number map

$$\deg : C(S^1, S^1) \longrightarrow \mathbb{Z}$$

is surjective, since $z \mapsto z^d$ has degree d . Its kernel consists exactly of maps admitting a continuous lift $S^1 \rightarrow \mathbb{R}$ through $\mathbb{R} \rightarrow S^1$, and these are precisely the maps in the denominator of (5). Therefore degree induces an isomorphism

$$H^1(S^1, \mathbb{Z}) \cong \mathbb{Z}.$$

□

III.3 Cohomology of a Noetherian Affine Scheme

Exercise III.3.1

Permitted context.

For a noetherian scheme X , the reduction $i : X_{\text{red}} \rightarrow X$ is defined by the nilradical ideal sheaf $\mathcal{N}$. We may use Serre's criterion in the following form: X is affine if and only if $H^1(X, \mathcal{J}) = 0$ for every coherent ideal sheaf $\mathcal{J} \subseteq \mathcal{O}_X$. Higher cohomology of a quasi-coherent sheaf on a noetherian affine scheme vanishes.

Problem. Let X be a noetherian scheme. Prove that X is affine if and only if X_{red} is affine.

Solution. If $X = \text{Spec } A$, then

$$X_{\text{red}} = \text{Spec}(A/\sqrt{(0)}),$$

so one implication is immediate.

Conversely, suppose that X_{red} is affine. The nilradical $\mathcal{N}$ is coherent and nilpotent. Indeed, this is true on every noetherian affine open, and a finite affine cover gives one exponent r such that $\mathcal{N}^r = 0$.

Let $\mathcal{J} \subseteq \mathcal{O}_X$ be any coherent ideal sheaf. It has the finite filtration

$$\mathcal{J} \supseteq \mathcal{N}\mathcal{J} \supseteq \cdots \supseteq \mathcal{N}^r \mathcal{J} = 0. \quad (1)$$

Each quotient

$$\mathcal{Q}_q = \mathcal{N}^q \mathcal{J} / \mathcal{N}^{q+1} \mathcal{J}$$

is coherent and is annihilated by $\mathcal{N}$. Hence $\mathcal{Q}_q = i_* \mathcal{G}_q$ for a coherent sheaf $\mathcal{G}_q$ on X_{red} . A closed immersion does not change cohomology of a sheaf under direct image, and therefore

$$H^1(X, \mathcal{Q}_q) = H^1(X_{\text{red}}, \mathcal{G}_q) = 0. \quad (2)$$

Starting with the zero term of (1) and using the long exact sequence for each successive quotient, (2) gives $H^1(X, \mathcal{J}) = 0$. Since this holds for every coherent ideal sheaf, Serre's criterion shows that X is affine. □

Exercise III.3.2**Permitted context.**

A noetherian scheme has finitely many irreducible components. We may use Serre's criterion: a noetherian scheme X is affine if and only if $H^1(X, \mathcal{J}) = 0$ for every coherent ideal sheaf $\mathcal{J} \subseteq \mathcal{O}_X$.

Problem. Let X be a reduced noetherian scheme. Prove that X is affine if and only if every irreducible component of X , with its reduced induced scheme structure, is affine.

Solution. If X is affine, every irreducible component is the closed subscheme defined by a minimal prime, and hence is affine.

For the converse, write the irreducible components as $C_1, \dots, C_r$, and let $\mathcal{I}_i$ be the ideal sheaf of C_i . Since X is reduced,

$$\mathcal{I}_1 \cdots \mathcal{I}_r \subseteq \bigcap_{i=1}^r \mathcal{I}_i = 0. \quad (1)$$

Put $\mathcal{K}_0 = \mathcal{O}_X$ and $\mathcal{K}_i = \mathcal{I}_1 \cdots \mathcal{I}_i$.

Fix a coherent ideal sheaf $\mathcal{J}$. Equation (1) gives a finite filtration

$$0 = \mathcal{J}\mathcal{K}_r \subseteq \mathcal{J}\mathcal{K}_{r-1} \subseteq \cdots \subseteq \mathcal{J}\mathcal{K}_0 = \mathcal{J}. \quad (2)$$

The quotient

$$\mathcal{Q}_i = \mathcal{J}\mathcal{K}_{i-1} / \mathcal{J}\mathcal{K}_i$$

is coherent and is annihilated by $\mathcal{I}_i$. If $j_i : C_i \rightarrow X$ denotes the closed immersion, then $\mathcal{Q}_i = (j_i)_* \mathcal{G}_i$ for a coherent sheaf $\mathcal{G}_i$ on C_i . By hypothesis C_i is affine, so

$$H^1(X, \mathcal{Q}_i) = H^1(C_i, \mathcal{G}_i) = 0. \quad (3)$$

The long exact sequences associated to (2), read successively from left to right, now imply $H^1(X, \mathcal{J}) = 0$. Serre's criterion therefore proves that X is affine. $\square$

Exercise III.3.3**Permitted context.**

Let A be noetherian and $\mathfrak{a} \subseteq A$ an ideal. For an A -module M , set

$$\Gamma_{\mathfrak{a}}(M) = \{m \in M : \mathfrak{a}^n m = 0 \text{ for some } n \geq 1\}.$$

If $X = \operatorname{Spec} A$ and $Y = V(\mathfrak{a})$, an injective A -module I gives a flasque sheaf $\widetilde{I}$ on X .

Problem. (a) Prove that $\Gamma_{\mathfrak{a}}$ is a left-exact functor, and denote its right derived functors by $H_{\mathfrak{a}}^i$.
 (b) For every A -module M , prove the natural isomorphism

$$H_{\mathfrak{a}}^i(M) \cong H_Y^i(X, \widetilde{M}).$$

(c) Prove that every local-cohomology module is $\mathfrak{a}$ -power torsion:

$$\Gamma_{\mathfrak{a}}(H_{\mathfrak{a}}^i(M)) = H_{\mathfrak{a}}^i(M).$$

Solution. (a) The displayed rule defines a submodule and is functorial in M . If $u : M \rightarrow N$, then

$$\Gamma_{\mathfrak{a}}(\ker u) = \ker u \cap \Gamma_{\mathfrak{a}}(M) = \ker(\Gamma_{\mathfrak{a}}(M) \rightarrow \Gamma_{\mathfrak{a}}(N)).$$

Thus $\Gamma_{\mathfrak{a}}$ preserves kernels and is left exact. Its right derived functors in the category of A -modules are, by definition, the modules $H_{\mathfrak{a}}^i(-)$.

(b) For every module L there is a natural equality

$$\Gamma_{\mathfrak{a}}(L) = \Gamma_Y(X, \widetilde{L}). \quad (1)$$

Indeed, if $\mathfrak{a} = (a_1, \dots, a_s)$, the section defined by $\ell \in L$ vanishes on every $D(a_j)$ if and only if suitable powers of all the a_j kill ℓ ; a sufficiently large power of $\mathfrak{a}$ then kills ℓ .

Choose an injective resolution $M \rightarrow I^{\bullet}$. Sheafification on an affine scheme is exact, so $\widetilde{M} \rightarrow \widetilde{I}^{\bullet}$ is a resolution. Its terms are flasque, hence are acyclic for sections with support. Applying (1) degree by degree gives an isomorphism of complexes

$$\Gamma_{\mathfrak{a}}(I^{\bullet}) \cong \Gamma_Y(X, \widetilde{I}^{\bullet}).$$

Taking cohomology proves the asserted natural isomorphism.

(c) Each term $\Gamma_{\mathfrak{a}}(I^q)$ is $\mathfrak{a}$ -power torsion. The class of such modules is closed under submodules and quotients: the exponent may depend on the element, which is exactly what the definition allows. Therefore every cohomology module of the complex $\Gamma_{\mathfrak{a}}(I^{\bullet})$, being a subquotient of one of its terms, is again $\mathfrak{a}$ -power torsion. This is precisely the displayed equality in (c). $\square$

Exercise III.3.4**Permitted context.**

Let A be noetherian, $\mathfrak{a} \subseteq A$ an ideal, and M an A -module. The number $\text{depth}_{\mathfrak{a}} M$ is the supremum of the lengths of M -regular sequences whose terms lie in $\mathfrak{a}$. For a finite module, the zero divisors on M form the union of the finitely many primes in $\text{Ass}_A(M)$. Every $H_{\mathfrak{a}}^i(M)$ is $\mathfrak{a}$ -power torsion. For the finite modules in this exercise, require $M \neq 0$ and $\mathfrak{a}M \neq M$. Regular sequences are required to leave a nonzero quotient. These conditions remove the degenerate empty-support case from the depth comparison.

Problem. (a) Prove that $\text{depth}_{\mathfrak{a}} M \geq 1$ implies $\Gamma_{\mathfrak{a}}(M) = 0$. Prove the converse when M is finitely generated.

(b) If M is finitely generated, prove for every $n \geq 0$ that the following conditions are equivalent:

$$\text{depth}_{\mathfrak{a}} M \geq n, \quad H_{\mathfrak{a}}^i(M) = 0 \quad (0 \leq i < n).$$

Solution. (a) Choose an M -regular element $x \in \mathfrak{a}$. If $\mathfrak{a}^r m = 0$, then $x^r m = 0$; injectivity of successive multiplications by x gives $m = 0$. Thus $\Gamma_{\mathfrak{a}}(M) = 0$.

Conversely, suppose that M is finite and $\Gamma_{\mathfrak{a}}(M) = 0$. No associated prime of M can contain $\mathfrak{a}$: if $\mathfrak{p} = \text{Ann}(m)$ contained $\mathfrak{a}$, then the nonzero element m would lie in $\Gamma_{\mathfrak{a}}(M)$. Finite prime avoidance therefore gives

$$x \in \mathfrak{a} \setminus \bigcup_{\mathfrak{p} \in \text{Ass}_A(M)} \mathfrak{p}.$$

Such an x is not a zero divisor on M , so it is a regular sequence of length one.

(b) We argue by induction on n ; the assertion for $n = 0$ is empty, and the case $n = 1$ is (a), since $H_{\mathfrak{a}}^0(M) = \Gamma_{\mathfrak{a}}(M)$.

Assume the equivalence known with $n - 1$ in place of n . First suppose that $\text{depth}_{\mathfrak{a}} M \geq n$. Choose a regular sequence beginning with $x \in \mathfrak{a}$, and put $\overline{M} = M/xM$. Then

$$0 \longrightarrow M \xrightarrow{x} M \longrightarrow \overline{M} \longrightarrow 0 \quad (1)$$

is exact, while both $\text{depth}_{\mathfrak{a}} M \geq n - 1$ and $\text{depth}_{\mathfrak{a}} \overline{M} \geq n - 1$. The induction hypothesis gives

$$H_{\mathfrak{a}}^i(M) = H_{\mathfrak{a}}^i(\overline{M}) = 0 \quad (i < n - 1).$$

At the next degree, the long exact sequence of (1) shows that multiplication by x on $H_{\mathfrak{a}}^{n-1}(M)$ is injective. This module is $\mathfrak{a}$ -power torsion, so every one of its elements is killed by a power of x . An injective nilpotent action is possible only on the zero module. Hence $H_{\mathfrak{a}}^{n-1}(M) = 0$ as well.

Conversely, suppose $H_{\mathfrak{a}}^i(M) = 0$ for $i < n$. When $n > 0$, part (a) supplies an M -regular $x \in \mathfrak{a}$. In the long exact sequence of (1), for $i < n - 1$ the groups on either side of $H_{\mathfrak{a}}^i(\overline{M})$ are $H_{\mathfrak{a}}^i(M)$ and $H_{\mathfrak{a}}^{i+1}(M)$, both zero. Therefore

$$H_{\mathfrak{a}}^i(\overline{M}) = 0 \quad (i < n - 1).$$

By induction, $\overline{M}$ has an $\overline{M}$ -regular sequence of length $n - 1$ in $\mathfrak{a}$. Prepending x gives an M -regular sequence of length n , completing the induction. $\square$

Exercise III.3.5**Permitted context.**

Let P be a closed point of a noetherian scheme X . We may use the excision and localization isomorphisms for cohomology with support at P , as well as the exact sequence

$$0 \rightarrow H_P^0(W, \mathcal{F}) \rightarrow H^0(W, \mathcal{F}) \rightarrow H^0(W \setminus \{P\}, \mathcal{F}) \rightarrow H_P^1(W, \mathcal{F}) \rightarrow H^1(W, \mathcal{F}).$$

Problem. Prove that the following conditions are equivalent:

- (i) $\text{depth } \mathcal{O}_{X,P} \geq 2$.
- (ii) For every open neighborhood W of P , every regular function on $W \setminus \{P\}$ extends uniquely to a regular function on W .

Solution. Choose an affine neighborhood $V = \text{Spec } A$ of P , and let $\mathfrak{m}$ be the corresponding maximal ideal. By the local-cohomology description of depth,

$$\text{depth } \mathcal{O}_{X,P} \geq 2 \iff H_{\mathfrak{m}A_{\mathfrak{m}}}^0(A_{\mathfrak{m}}) = H_{\mathfrak{m}A_{\mathfrak{m}}}^1(A_{\mathfrak{m}}) = 0. \quad (1)$$

The comparison between algebraic local cohomology and cohomology with supports, followed by localization to $\text{Spec } A_{\mathfrak{m}}$, turns (1) into

$$H_P^0(V, \mathcal{O}_V) = H_P^1(V, \mathcal{O}_V) = 0. \quad (2)$$

Since V is affine, $H^1(V, \mathcal{O}_V) = 0$. The support exact sequence therefore shows that (2) is equivalent to the restriction map

$$\Gamma(V, \mathcal{O}_V) \longrightarrow \Gamma(V \setminus \{P\}, \mathcal{O}_V) \quad (3)$$

being an isomorphism. Thus condition (i) is equivalent to (3) for every sufficiently small affine neighborhood V of P .

It remains only to pass from affine neighborhoods to arbitrary ones. Let W be any neighborhood of P , choose an affine $V \subseteq W$, and take $s \in \Gamma(W \setminus \{P\}, \mathcal{O}_X)$. If (3) is an isomorphism, $s|_{V \setminus \{P\}}$ has a unique extension to V . That extension and s agree on their overlap and hence glue over the cover

$$W = V \cup (W \setminus \{P\}).$$

The same cover proves uniqueness. Conversely, (ii) applies in particular to every affine neighborhood V , so (3) is an isomorphism and (1) follows. This proves the equivalence. $\square$

Exercise III.3.6**Permitted context.**

Let X be a noetherian scheme. Choose a finite affine open cover $X = \bigcup_{i=1}^r U_i$, write $U_i = \operatorname{Spec} A_i$, and let $j_i : U_i \rightarrow X$ be the inclusions. In the standard construction, a quasi-coherent sheaf $\mathcal{F}$ is embedded into

$$\mathcal{G} = \bigoplus_{i=1}^r (j_i)_* \tilde{I}_i,$$

where each I_i is an injective A_i -module containing the module corresponding to $\mathcal{F}|_{U_i}$.

Problem. (a) Prove that $\mathcal{G}$ is injective in the category $\operatorname{Qco}(X)$. Conclude that this category has enough injectives.

(b) Prove that every injective object of $\operatorname{Qco}(X)$ is flasque as a sheaf.

(c) Prove that the derived functors of global sections computed in $\operatorname{Qco}(X)$ agree with ordinary sheaf cohomology.

Solution. (a) Because X is noetherian, each j_i is quasi-compact, so $(j_i)_*$ carries quasi-coherent sheaves to quasi-coherent sheaves. For $\mathcal{E} \in \operatorname{Qco}(X)$, adjunction gives

$$\operatorname{Hom}_X(\mathcal{E}, (j_i)_* \tilde{I}_i) \cong \operatorname{Hom}_{U_i}(\mathcal{E}|_{U_i}, \tilde{I}_i). \quad (1)$$

Restriction to U_i is exact. Under the equivalence $\operatorname{Qco}(U_i) \simeq \operatorname{Mod}(A_i)$, the sheaf $\tilde{I}_i$ corresponds to the injective module I_i . Consequently the functor in (1) is exact, and $(j_i)_* \tilde{I}_i$ is injective in $\operatorname{Qco}(X)$. The cover is finite, so its direct sum is a finite biproduct of injectives and is injective. Since the standard construction embeds every quasi-coherent $\mathcal{F}$ into such a $\mathcal{G}$, the category has enough injectives.

(b) Each $\tilde{I}_i$ is flasque on U_i , and direct image under an open immersion preserves flasqueness. Hence the sheaf $\mathcal{G}$ above is flasque. Apply the construction to an injective object $\mathcal{I} \in \operatorname{Qco}(X)$, obtaining

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{G}.$$

Injectivity of $\mathcal{I}$ splits this monomorphism inside $\operatorname{Qco}(X)$, so $\mathcal{G} \cong \mathcal{I} \oplus \mathcal{K}$. A direct summand of a flasque sheaf is flasque: projecting a lift in $\mathcal{G}(W)$ onto $\mathcal{I}(W)$ lifts any section of $\mathcal{I}(V)$ for $V \subseteq W$. Thus $\mathcal{I}$ is flasque.

(c) Resolve $\mathcal{F} \in \operatorname{Qco}(X)$ by injectives in that category. By (b), every term of this resolution is flasque as an ordinary $\mathcal{O}_X$ -module. Applying $\Gamma(X, -)$ and taking cohomology therefore computes both the derived functors in $\operatorname{Qco}(X)$ and ordinary sheaf cohomology. The two constructions consequently agree, naturally in $\mathcal{F}$. $\square$

Exercise III.3.7**Permitted context.**

Let A be noetherian, $X = \operatorname{Spec} A$, $\mathfrak{a} \subseteq A$, and $U = X \setminus V(\mathfrak{a})$. The transition map in the system $\operatorname{Hom}_A(\mathfrak{a}^n, M)$ is restriction along $\mathfrak{a}^{n+1} \subseteq \mathfrak{a}^n$.

Problem. (a) Prove Deligne's formula

$$\Gamma(U, \widetilde{M}) \cong \varinjlim_n \operatorname{Hom}_A(\mathfrak{a}^n, M)$$

for every A -module M .

(b) Use the formula to prove that $\widetilde{I}$ is flasque whenever I is an injective A -module.

Solution. (a) Since $\widetilde{\mathfrak{a}^n}|_U = \mathcal{O}_U$, restriction defines compatible maps

$$\operatorname{Hom}_A(\mathfrak{a}^n, M) \longrightarrow \Gamma(U, \widetilde{M}),$$

and hence a natural map

$$\Phi : \varinjlim_n \operatorname{Hom}_A(\mathfrak{a}^n, M) \longrightarrow \Gamma(U, \widetilde{M}). \quad (1)$$

We first prove injectivity. Suppose $f : \mathfrak{a}^n \rightarrow M$ restricts to zero on U . Its image N is finite, and $\widetilde{N}$ is supported on $V(\mathfrak{a})$. Thus $\mathfrak{a}^r N = 0$ for some r . It follows that the restriction of f to $\mathfrak{a}^{n+r}$ is zero, so the class of f vanishes in the colimit. Hence Φ is injective.

For surjectivity, first suppose that M is finite. Set

$$T_j = \{m \in M : \mathfrak{a}^j m = 0\}, \quad T = \bigcup_j T_j.$$

The ascending chain (T_j) stabilizes, so $T = T_t$ for some t . Artin–Rees, applied to $T \subseteq M$, gives an integer c such that

$$\mathfrak{a}^q M \cap T \subseteq \mathfrak{a}^{q-c} T \quad (q \geq c).$$

Choose $q \geq c + t$ and put $M' = \mathfrak{a}^q M$. Then $M' \cap T = 0$, so M' has no nonzero $\mathfrak{a}$ -power torsion, while $\widetilde{M'}|_U = \widetilde{M}|_U$. We may therefore replace M by M' and assume

$$\Gamma_{\mathfrak{a}}(M) = 0. \quad (2)$$

Write $\mathfrak{a} = (f_1, \dots, f_s)$, and take $z \in \Gamma(U, \widetilde{M})$. Condition (2) makes the natural map $M \rightarrow \Gamma(U, \widetilde{M})$ injective. On $D(f_i)$, write

$$z = \frac{m_i}{f_i^{e_i}} \quad (m_i \in M).$$

The section $m_i - f_i^{e_i} z$ on U vanishes on $D(f_i)$. The finite cover $U = \bigcup_j D(f_j)$ then supplies an exponent r_i for which

$$f_i^{r_i} (m_i - f_i^{e_i} z) = 0 \quad \text{on } U.$$

Choose one $L \geq 1$ at least as large as every $e_i + r_i$. Then $f_i^L z$ belongs to the embedded copy of M for every i . Every monomial of total degree sL in the f_i has some exponent at least L ; hence

$$\mathfrak{a}^{sL} z \subseteq M \quad \text{inside } \Gamma(U, \widetilde{M}). \quad (3)$$

Multiplication by z in (3) defines an A -linear map

$$\mathfrak{a}^{sL} \longrightarrow M, \quad b \longmapsto bz,$$

whose image under Φ is z . This proves surjectivity for finite modules.

For arbitrary M , write it as the filtered union of its finite submodules. The open set U is quasi-compact and quasi-separated, so sections of quasi-coherent sheaves over U commute with this filtered union. Thus every $z \in \Gamma(U, \widetilde{M})$ comes from $\Gamma(U, \widetilde{N})$ for some finite submodule $N \subseteq M$. The finite case produces a map $\mathfrak{a}^n \rightarrow N$; composing with $N \rightarrow M$ shows that z lies in the image of (1). This completes the proof of Deligne's formula.

(b) Let $W' \subseteq W \subseteq X$ be open. Since X is noetherian, there are radical ideals $\mathfrak{b} \subseteq \mathfrak{a}$ such that

$$W = X \setminus V(\mathfrak{a}), \quad W' = X \setminus V(\mathfrak{b}).$$

By (a), a section of $\widetilde{I}$ over W' is represented by a map $g : \mathfrak{b}^n \rightarrow I$. The inclusion $\mathfrak{b}^n \subseteq \mathfrak{a}^n$ and injectivity of I extend g to a map $\mathfrak{a}^n \rightarrow I$. The corresponding section over W restricts to the given section. Every restriction map is therefore surjective, so $\widetilde{I}$ is flasque. $\square$

Exercise III.3.8

Permitted context.

Let

$$A = k[x_0, x_1, x_2, \dots] / (x_0^n x_n : n \geq 1),$$

and let I be an injective A -module containing A as a submodule. Equality of two fractions in I_{x_0} means that a power of x_0 kills their cross-multiplied difference.

Problem. Prove that the localization map

$$I \longrightarrow I_{x_0}$$

is not surjective. This gives the stated failure, without a noetherian hypothesis, of the flasqueness result for sheaves associated to injective modules.

Solution. The element $1/x_0 \in I_{x_0}$ is well defined because $1 \in A \subseteq I$. Suppose it were the image of some $u \in I$. Equality

$$\frac{u}{1} = \frac{1}{x_0} \quad \text{in } I_{x_0}$$

would give an integer $r \geq 0$ such that

$$x_0^r(x_0 u - 1) = 0, \quad \text{or equivalently} \quad x_0^{r+1}u = x_0^r. \quad (1)$$

Multiply (1) by x_{r+1} . The defining relation $x_0^{r+1}x_{r+1} = 0$ yields

$$x_0^r x_{r+1} = x_0^{r+1} x_{r+1} u = 0 \quad \text{in } I. \quad (2)$$

But $x_0^r x_{r+1} \neq 0$ in A : the defining ideal is a monomial ideal, and this monomial is not divisible by any generator $x_0^n x_n$. Equation (2) contradicts the injectivity of $A \rightarrow I$. Hence $1/x_0$ is not in the image, and $I \rightarrow I_{x_0}$ is not surjective.

Indeed, this map is the restriction

$$\Gamma(\operatorname{Spec} A, \widetilde{I}) \longrightarrow \Gamma(D(x_0), \widetilde{I}),$$

so $\widetilde{I}$ is not flasque. $\square$

III.4 Čech Cohomology

Exercise III.4.1

Permitted context.

For a quasi-coherent sheaf on a noetherian separated scheme, an affine open cover computes sheaf cohomology by its Čech complex. The inverse image of an affine open under an affine morphism is affine, and the pushforward of a quasi-coherent sheaf under such a morphism is quasi-coherent.

Problem. Let $f : X \rightarrow Y$ be an affine morphism of noetherian separated schemes. Prove that for every quasi-coherent sheaf $\mathcal{F}$ on X there are natural isomorphisms

$$H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F}) \quad (i \geq 0).$$

Solution. Choose an affine open cover $\mathfrak{U} = (U_j)$ of Y . Because Y is separated, every finite intersection $U_{j_0 \dots j_p}$ is affine. Since f is affine, every $f^{-1}(U_{j_0 \dots j_p})$ is affine as well. Thus both covers are acyclic for the relevant quasi-coherent sheaves.

By the definition of direct image, for every finite intersection there is a natural identity

$$\Gamma(U_{j_0 \dots j_p}, f_*\mathcal{F}) = \Gamma(f^{-1}(U_{j_0 \dots j_p}), \mathcal{F}). \quad (1)$$

The identities (1) commute with all restriction maps and therefore give an isomorphism of augmented Čech complexes

$$C^\bullet(\mathfrak{U}, f_*\mathcal{F}) \cong C^\bullet(f^{-1}\mathfrak{U}, \mathcal{F}).$$

The Čech comparison theorem identifies the cohomology of the two complexes with the two sheaf-cohomology groups in the statement. Taking cohomology of this natural isomorphism proves the result in every degree. $\square$

Exercise III.4.2**Permitted context.**

Finite morphisms are affine, closed, and stable under base change. On a noetherian scheme, every quasi-coherent sheaf embeds in a flasque quasi-coherent sheaf. We may use generic freeness, the affine morphism equivalence for quasi-coherent modules, and Serre's cohomological criterion for affineness. In part (b), retain the integral hypothesis of (a) when referring to the generic point.

Problem. Let $f : X \rightarrow Y$ be a finite surjective morphism of noetherian separated schemes, with X affine.

(a) If X and Y are integral, construct a coherent sheaf $\mathcal{M}$ on X and a morphism

$$\alpha : \mathcal{O}_Y^r \longrightarrow f_*\mathcal{M}$$

which is an isomorphism at the generic point of Y .

(b) For every coherent $\mathcal{F}$ on Y , construct a coherent $\mathcal{G}$ on X and a morphism

$$\beta : f_*\mathcal{G} \longrightarrow \mathcal{F}^r$$

which is an isomorphism at the generic point.

(c) Prove Chevalley's theorem: Y is affine.

Solution. (a) Let $V = \operatorname{Spec} B \subseteq Y$ be a nonempty affine open and write $f^{-1}(V) = \operatorname{Spec} A$. The map $B \rightarrow A$ is finite and injective. After shrinking V , generic freeness permits us to assume that A is a free B -module with basis $a_1, \dots, a_r$.

Let $j : f^{-1}(V) \rightarrow X$ be the open immersion. The sheaf $j_*\mathcal{O}_{f^{-1}(V)}$ is quasi-coherent; embed it into a flasque quasi-coherent sheaf $\mathcal{I}$. Denote by $s \in \Gamma(f^{-1}(V), \mathcal{I})$ the image of 1. Flasqueness allows each $a_i s$ to extend to a section $\tilde{s}_i \in \Gamma(X, \mathcal{I})$. Let $\mathcal{M}$ be the $\mathcal{O}_X$ -submodule generated by these finitely many sections. It is coherent because X is noetherian.

On $f^{-1}(V)$, the a_i generate A , so $\mathcal{M}|_{f^{-1}(V)} = \mathcal{O}_{f^{-1}(V)} s$, a copy of the structure sheaf. The global sections $\tilde{s}_i$ define

$$\alpha : \mathcal{O}_Y^r \longrightarrow f_*\mathcal{M}.$$

Over V , this is the map of B -modules

$$B^r \longrightarrow A, \quad (b_1, \dots, b_r) \longmapsto \sum_i b_i a_i,$$

and hence is an isomorphism. In particular it is an isomorphism at the generic point.

(b) Apply the sheaf-Hom functor $\mathcal{H}om_Y(-, \mathcal{F})$ to α . Precomposition gives

$$\beta : \mathcal{H} := \mathcal{H}om_Y(f_*\mathcal{M}, \mathcal{F}) \longrightarrow \mathcal{F}^r. \quad (1)$$

The sheaf $\mathcal{H}$ is coherent, and (1) is an isomorphism at the generic point. It also has a natural $f_*\mathcal{O}_X$ -module structure:

$$(a\varphi)(m) = \varphi(am).$$

The affine-morphism equivalence therefore gives a quasi-coherent $\mathcal{O}_X$ -module $\mathcal{G}$ with $f_*\mathcal{G} \cong \mathcal{H}$. It is coherent: finitely many $\mathcal{O}_Y$ -generators of $\mathcal{H}$ also generate it as an $f_*\mathcal{O}_X$ -module, and finiteness can be checked on affine opens. With this identification, (1) is the required morphism.

(c) We first make two reductions. Passing to reductions preserves finiteness and surjectivity, X_{red} is affine, and a noetherian scheme is affine exactly when its reduction is affine. Thus we may suppose that X and Y are reduced. For each irreducible component Y_j , some irreducible component X_i maps onto Y_j : apply surjectivity to the generic point of Y_j , and use that a finite map is closed. The induced map $X_i \rightarrow Y_j$ is finite and surjective, and X_i is affine. If the theorem is known in the integral case, every Y_j is affine; the component criterion for a reduced noetherian scheme then makes Y affine. It remains to treat the integral case.

We use noetherian induction on the closed subsets of the integral scheme Y . Assume the theorem has been proved over every proper closed subset of Y . Let $\mathcal{J} \subseteq \mathcal{O}_Y$ be a coherent ideal sheaf. Part (b) supplies

$$\beta : f_*\mathcal{G} \longrightarrow \mathcal{J}^r$$

which is an isomorphism at the generic point. Put $\mathcal{K} = \ker \beta$, $\mathcal{Q} = \text{im } \beta$, and $\mathcal{C} = \text{coker } \beta$. The coherent sheaves $\mathcal{K}$ and $\mathcal{C}$ have proper closed support.

Give either support a closed scheme structure annihilating the corresponding sheaf. Its inverse image in X is affine and maps to it finitely and surjectively. The inductive hypothesis, together with the reduction criterion, says that each such support scheme is affine. Consequently

$$H^i(Y, \mathcal{K}) = H^i(Y, \mathcal{C}) = 0 \quad (i > 0). \quad (2)$$

Moreover, the result for affine morphisms and the affineness of X give

$$H^i(Y, f_*\mathcal{G}) = H^i(X, \mathcal{G}) = 0 \quad (i > 0). \quad (3)$$

Apply cohomology to

$$0 \rightarrow \mathcal{K} \rightarrow f_*\mathcal{G} \rightarrow \mathcal{Q} \rightarrow 0, \quad 0 \rightarrow \mathcal{Q} \rightarrow \mathcal{J}^r \rightarrow \mathcal{C} \rightarrow 0.$$

Equations (2) and (3) first give $H^1(Y, \mathcal{Q}) = 0$, and then $H^1(Y, \mathcal{J}^r) = 0$. Since $r > 0$, this implies $H^1(Y, \mathcal{J}) = 0$. Serre's criterion now shows that Y is affine, completing the induction and the proof. $\square$

Exercise III.4.3

Permitted context.

Let $X = \mathbb{A}_k^2 = \text{Spec } k[x, y]$ and $U = X \setminus \{(0, 0)\}$. The cover $U = D(x) \cup D(y)$ is an affine cover with affine intersection, so its Čech complex computes the cohomology of $\mathcal{O}_U$.

Problem. Compute $H^1(U, \mathcal{O}_U)$ and prove that it is the infinite-dimensional k -vector space with basis

$$\{x^i y^j : i < 0, j < 0\}.$$

Solution. The degree-zero and degree-one terms of the Čech complex are

$$k[x, x^{-1}, y] \oplus k[x, y, y^{-1}] \longrightarrow k[x, x^{-1}, y, y^{-1}], \quad (1)$$

where the map is the difference of the two restrictions. Therefore

$$H^1(U, \mathcal{O}_U) \cong \frac{k[x, x^{-1}, y, y^{-1}]}{k[x, x^{-1}, y] + k[x, y, y^{-1}]} \quad (2)$$

The Laurent monomials $x^i y^j$, $(i, j) \in \mathbb{Z}^2$, form a k -basis of the numerator. The first summand in the denominator is spanned by those with $j \geq 0$, and the second by those with $i \geq 0$. Thus precisely the monomials with both exponents negative survive in (2), and their residue classes remain linearly independent. Hence

$$H^1(U, \mathcal{O}_U) = \bigoplus_{i < 0, j < 0} k x^i y^j,$$

which is infinite-dimensional. $\square$

Exercise III.4.4

Permitted context.

Let $\mathfrak{U} = (U_i)_{i \in I}$ be an open cover of a topological space X , and let $\mathcal{F}$ be a sheaf of abelian groups. We use alternating Čech cochains; thus a cochain is extended to all ordered index tuples by the sign rule and is zero when two indices coincide.

Problem. (a) Construct the maps on Čech cohomology induced by a refinement and the direct limit over all open covers.

(b) Prove that the comparison maps from Čech cohomology to derived sheaf cohomology are compatible with refinements.

(c) Prove that the resulting natural map

$$\varinjlim_{\mathfrak{U}} \check{H}^1(\mathfrak{U}, \mathcal{F}) \longrightarrow H^1(X, \mathcal{F})$$

is an isomorphism.

Solution. (a) Let $\mathfrak{V} = (V_j)_{j \in J}$ refine $\mathfrak{U}$, and choose a refinement function $\lambda : J \rightarrow I$ with $V_j \subseteq U_{\lambda(j)}$. Define

$$(\lambda^* c)_{j_0 \dots j_p} = c_{\lambda(j_0) \dots \lambda(j_p)}|_{V_{j_0} \cap \dots \cap V_{j_p}}. \quad (1)$$

The alternating convention makes (1) meaningful even when the $\lambda(j_q)$ are repeated or out of order. Direct substitution in the alternating differential shows that $d\lambda^* = \lambda^* d$, so (1) induces maps

$$\lambda^p : \check{H}^p(\mathfrak{U}, \mathcal{F}) \longrightarrow \check{H}^p(\mathfrak{V}, \mathcal{F}).$$

These maps do not depend on the choice of refinement function. In fact, for two choices $\lambda, \mu : J \rightarrow I$, the standard prism operator

$$(hc)_{j_0 \dots j_{p-1}} = \sum_{q=0}^{p-1} (-1)^q c_{\lambda(j_0) \dots \lambda(j_q) \mu(j_q) \dots \mu(j_{p-1})} |_{V_{j_0} \cap \dots \cap V_{j_{p-1}}}$$

satisfies $dh + hd = \mu^* - \lambda^*$. Thus the induced maps on cohomology agree. Any two covers have the common refinement $(U_i \cap V_j)_{i,j}$; consequently these maps define the filtered direct limit in the statement.

(b) Regard the augmented Čech complex as a resolution of $\mathcal{F}$ by sheaves and compare it with an injective resolution $\mathcal{F} \rightarrow \mathcal{I}^\bullet$. A refinement gives a morphism of augmented Čech resolutions which is the identity on $\mathcal{F}$. By the comparison theorem for resolutions, its composite with a lift to $\mathcal{I}^\bullet$ is homotopic to the lift used for the original cover. After taking global sections and cohomology, this

says precisely that the comparison map for $\mathfrak{U}$, composed with λ^p , is the comparison map for $\mathfrak{U}$. The comparison maps therefore induce a map from the direct limit.

(c) Choose a short exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I} \longrightarrow \mathcal{Q} \longrightarrow 0 \quad (2)$$

with $\mathcal{I}$ injective, hence flasque. The associated long exact sequence identifies

$$H^1(X, \mathcal{F}) \cong \text{coker}(\Gamma(X, \mathcal{I}) \rightarrow \Gamma(X, \mathcal{Q})). \quad (3)$$

Let a class in (3) be represented by $q \in \Gamma(X, \mathcal{Q})$. Since $\mathcal{I} \rightarrow \mathcal{Q}$ is an epimorphism of sheaves, there is an open cover (U_i) and lifts $t_i \in \mathcal{I}(U_i)$. On an overlap,

$$c_{ij} = t_j - t_i \in \mathcal{F}(U_i \cap U_j)$$

is a Čech 1-cocycle. Its comparison image is the connecting class of q , so every element of $H^1(X, \mathcal{F})$ is reached after choosing a suitable cover. The direct-limit map is surjective.

It is also injective, in fact already on each cover. Suppose a Čech cocycle c for $\mathfrak{U}$ maps to zero in $H^1(X, \mathcal{F})$. Viewed as an $\mathcal{I}$ -valued cocycle, it is a coboundary because a flasque sheaf has zero positive Čech cohomology. Thus $c_{ij} = t_j - t_i$ for sections $t_i \in \mathcal{I}(U_i)$. Their images in $\mathcal{Q}$ glue to a global section q . The comparison image of c is the class of q in (3). Its vanishing supplies $t \in \Gamma(X, \mathcal{I})$ mapping to q . Then $t_i - t|_{U_i}$ lies in $\mathcal{F}(U_i)$, and these sections exhibit c as an $\mathcal{F}$ -valued coboundary. Hence the direct-limit map is injective and therefore an isomorphism. $\square$

Exercise III.4.5

Permitted context.

For a ringed space $(X, \mathcal{O}_X)$, let $\mathcal{O}_X^*$ be the sheaf of multiplicative units and let $\text{Pic } X$ be the group of isomorphism classes of invertible sheaves, with tensor product as its operation. First sheaf cohomology is the direct limit of first Čech cohomology over open covers.

Problem. Prove the natural group isomorphism

$$\text{Pic } X \cong H^1(X, \mathcal{O}_X^*).$$

Solution. Fix an open cover $\mathfrak{U} = (U_i)$. If an invertible sheaf $\mathcal{L}$ is trivial on every U_i , choose generators e_i of $\mathcal{L}|_{U_i}$. On overlaps there are unique units $g_{ij} \in \mathcal{O}_X^*(U_i \cap U_j)$ such that

$$e_j = g_{ij}e_i.$$

On triple intersections these satisfy $g_{ij}g_{jk} = g_{ik}$, so they form a Čech 1-cocycle. Replacing e_i by $u_i e_i$ multiplies (g_{ij}) by a Čech coboundary. Thus $\mathcal{L}$, together with its local triviality over this cover, determines a class in $\check{H}^1(\mathfrak{U}, \mathcal{O}_X^*)$.

Conversely, a cocycle (g_{ij}) glues the sheaves $\mathcal{O}_{U_i}$ by the isomorphisms given by multiplication by g_{ij} . The cocycle identity is exactly the compatibility needed for gluing, and the resulting sheaf is invertible. A coboundary changes the local generators and hence gives an isomorphic sheaf. These two constructions are inverse.

Every invertible sheaf is trivial on some open cover. Passing to the direct limit over refinements therefore gives a bijection

$$\text{Pic } X \longrightarrow \varinjlim_{\mathfrak{U}} \check{H}^1(\mathfrak{U}, \mathcal{O}_X^*) \cong H^1(X, \mathcal{O}_X^*).$$

Under this bijection, tensoring line bundles multiplies their transition functions. Hence the bijection is an isomorphism of groups. $\square$

Exercise III.4.6**Permitted context.**

Let $(X, \mathcal{O}_X)$ be a ringed space, let $\mathcal{I} \subseteq \mathcal{O}_X$ be an ideal sheaf with $\mathcal{I}^2 = 0$, and put $X_0 = (X, \mathcal{O}_X/\mathcal{I})$. We identify Picard groups with first cohomology of the corresponding sheaves of units.

Problem. Prove that $a \mapsto 1 + a$ fits into an exact sequence of sheaves of abelian groups

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_X^* \longrightarrow \mathcal{O}_{X_0}^* \longrightarrow 0.$$

Deduce the exact sequence

$$\cdots \longrightarrow H^1(X, \mathcal{I}) \longrightarrow \text{Pic } X \longrightarrow \text{Pic } X_0 \longrightarrow H^2(X, \mathcal{I}) \longrightarrow \cdots.$$

Solution. Because $\mathcal{I}^2 = 0$,

$$(1 + a)(1 + b) = 1 + a + b \quad (a, b \in \mathcal{I}),$$

so $a \mapsto 1 + a$ is a homomorphism from the additive sheaf $\mathcal{I}$ to the multiplicative sheaf $\mathcal{O}_X^*$.

Exactness can be checked on stalks. Let $R = \mathcal{O}_{X,x}$ and $I = \mathcal{I}_x$. The kernel of $R^* \rightarrow (R/I)^*$ consists exactly of the elements $1 + a$, $a \in I$, and this representation is unique. The map on units is surjective as well. Indeed, if the class of $r \in R$ is invertible modulo I , choose $s \in R$ with $rs = 1 + c$, $c \in I$. Since $(1 + c)^{-1} = 1 - c$, the element r is a unit in R . This proves the short exact sequence of sheaves.

Its long exact cohomology sequence contains

$$H^1(X, \mathcal{I}) \longrightarrow H^1(X, \mathcal{O}_X^*) \longrightarrow H^1(X, \mathcal{O}_{X_0}^*) \longrightarrow H^2(X, \mathcal{I}).$$

The identifications $H^1(X, \mathcal{O}_X^*) \cong \text{Pic } X$ and $H^1(X, \mathcal{O}_{X_0}^*) \cong \text{Pic } X_0$ turn this into the required exact sequence. $\square$

Exercise III.4.7**Permitted context.**

Let $X \subseteq \mathbb{P}_k^2$ be the hypersurface defined by a homogeneous polynomial $f(x_0, x_1, x_2)$ of degree $d > 0$, not necessarily irreducible. Assume $(1 : 0 : 0) \notin X$. Standard affine opens and all their intersections are acyclic for $\mathcal{O}_X$, so the associated Čech complex computes cohomology.

Problem. Show that

$$U = X \cap D_+(x_1), \quad V = X \cap D_+(x_2)$$

are affine and cover X . Compute their Čech complex explicitly and prove

$$\dim_k H^0(X, \mathcal{O}_X) = 1, \quad \dim_k H^1(X, \mathcal{O}_X) = \frac{(d-1)(d-2)}{2}.$$

Solution. The two sets cover X , because a point outside both would have $x_1 = x_2 = 0$ and hence would be $(1 : 0 : 0)$. Each is closed in the corresponding standard affine open of $\mathbb{P}_k^2$, and is therefore affine.

Put

$$S = k[x_0, x_1, x_2]/(f).$$

The rings in the Čech complex are

$$\Gamma(U, \mathcal{O}_X) = S_{(x_1)}, \quad \Gamma(V, \mathcal{O}_X) = S_{(x_2)}, \quad \Gamma(U \cap V, \mathcal{O}_X) = S_{(x_1 x_2)}, \quad (1)$$

where parentheses denote the degree-zero part of a homogeneous localization.

The hypothesis on $(1 : 0 : 0)$ says that the coefficient of x_0^d in f is nonzero. After scaling f , it is monic as a polynomial in x_0 . Polynomial division therefore gives a unique normal form in the last ring of (1) as a finite k -linear combination of

$$x_0^a x_1^b x_2^c, \quad 0 \leq a < d, \quad b, c \in \mathbb{Z}, \quad a + b + c = 0. \quad (2)$$

The image of $S_{(x_1)}$ is spanned by the normal monomials (2) with $c \geq 0$, while the image of $S_{(x_2)}$ is spanned by those with $b \geq 0$. Indeed, division by the monic polynomial f introduces no negative power of a variable that was not already inverted.

The Čech differential is

$$S_{(x_1)} \oplus S_{(x_2)} \longrightarrow S_{(x_1 x_2)}, \quad (g, h) \longmapsto h - g. \quad (3)$$

For H^0 , the intersection of the two image subspaces has $b, c \geq 0$. Together with $a \geq 0$ and $a + b + c = 0$, this forces $a = b = c = 0$. Hence

$$H^0(X, \mathcal{O}_X) = k.$$

For H^1 , quotienting the target of (3) by the two images leaves exactly the normal monomials for which $b < 0$ and $c < 0$. Write $p = -b$ and $q = -c$. Then $p, q \geq 1$ and $a = p + q$, with $a < d$. Thus

$$H^1(X, \mathcal{O}_X) \cong \bigoplus_{\substack{p, q \geq 1 \\ p+q < d}} k x_0^{p+q} x_1^{-p} x_2^{-q}.$$

For a fixed $a = 2, \dots, d-1$, there are $a-1$ pairs (p, q) with $p+q = a$. Consequently

$$\dim_k H^1(X, \mathcal{O}_X) = \sum_{a=2}^{d-1} (a-1) = \frac{(d-1)(d-2)}{2}.$$

□

Exercise III.4.8**Permitted context.**

For a noetherian separated scheme X , define $\text{cd}(X)$ to be the least n such that $H^i(X, \mathcal{F}) = 0$ for every quasi-coherent $\mathcal{F}$ and every $i > n$. Cohomology on a noetherian space commutes with filtered direct limits, and every quasi-coherent sheaf is the filtered union of its coherent subsheaves. For $r = 0$ in (e), an empty complement has zero cohomology in every degree.

Problem. (a) Show that coherent sheaves suffice in the definition of $\text{cd}(X)$.

(b) If X is quasi-projective over a field, show that locally free coherent sheaves suffice.

(c) If X is covered by $r + 1$ affine opens, prove $\text{cd}(X) \leq r$.

(d) Prove that a quasi-projective scheme of dimension r over a field can be covered by $r + 1$ affine opens. Deduce $\text{cd}(X) \leq \dim X$.

(e) If $Y \subseteq \mathbb{P}_k^n$ is a set-theoretic complete intersection of codimension r , prove $\text{cd}(\mathbb{P}_k^n \setminus Y) \leq r - 1$.

Solution. (a) Write a quasi-coherent sheaf as a filtered colimit $\mathcal{F} = \varinjlim_{\lambda} \mathcal{F}_{\lambda}$ of coherent subsheaves. Then

$$H^i(X, \mathcal{F}) \cong \varinjlim_{\lambda} H^i(X, \mathcal{F}_{\lambda}).$$

Thus vanishing above a fixed degree for all coherent sheaves implies the same vanishing for every quasi-coherent sheaf. The converse is immediate.

(b) Suppose $H^i(X, \mathcal{E}) = 0$ for $i > n$ and every locally free coherent $\mathcal{E}$. The ample-sheaf theorem for a quasi-projective scheme gives, for each coherent $\mathcal{F}_j$, a surjection from a finite direct sum of twists,

$$0 \longrightarrow \mathcal{F}_{j+1} \longrightarrow \mathcal{E}_j \longrightarrow \mathcal{F}_j \longrightarrow 0, \quad \mathcal{F}_0 = \mathcal{F}, \quad (1)$$

with $\mathcal{E}_j$ locally free and $\mathcal{F}_{j+1}$ coherent. For fixed $i > n$, the long exact sequences of (1) give dimension shifting

$$H^i(X, \mathcal{F}) \cong H^{i+t}(X, \mathcal{F}_t) \quad (2)$$

for every $t \geq 0$. Choose t so that $i + t > \dim X$. Grothendieck vanishing makes the right side of (2) zero. Hence all coherent sheaves have the required vanishing, and part (a) finishes the claim.

(c) Let $\mathfrak{U} = (U_0, \dots, U_r)$ be such a cover. Since X is separated, every finite intersection of the U_i is affine. For a quasi-coherent $\mathcal{F}$, this is an acyclic cover, so Čech cohomology computes $H^*(X, \mathcal{F})$. Its Čech complex has no terms in degrees greater than r . Therefore

$$H^i(X, \mathcal{F}) = 0 \quad (i > r),$$

and $\text{cd}(X) \leq r$.

(d) Realize X as a closed subscheme of an open $W \subseteq \mathbb{P}_k^N$, and let $Z = \mathbb{P}_k^N \setminus W$. We inductively choose homogeneous polynomials $f_0, \dots, f_r$ vanishing on Z . Start with $T_0 = X$. If $T_j \neq \emptyset$, none of its finitely many generic points lies in Z . Homogeneous prime avoidance, after raising candidates to a common degree, gives

$$f_j \in I(Z)$$

which vanishes at none of those generic points. Set

$$T_{j+1} = T_j \cap V_+(f_j).$$

Every irreducible component drops in dimension, so $\dim T_{j+1} < \dim T_j$. Since $\dim X = r$, after $r + 1$ choices we have $T_{r+1} = \emptyset$.

Now $D_+(f_j) \subseteq W$, because f_j vanishes on Z , and $D_+(f_j)$ is affine. Hence

$$X_j = X \cap D_+(f_j)$$

is closed in an affine scheme and is affine. The equality $T_{r+1} = \emptyset$ says that the X_j cover X . Part (c) then gives $\text{cd}(X) \leq r = \dim X$.

(e) Write Y set-theoretically as

$$Y = V_+(f_1) \cap \cdots \cap V_+(f_r)$$

for homogeneous polynomials f_i . Then

$$\mathbb{P}_k^n \setminus Y = D_+(f_1) \cup \cdots \cup D_+(f_r).$$

These are r affine opens. Applying (c) gives $\text{cd}(\mathbb{P}_k^n \setminus Y) \leq r - 1$. □

Exercise III.4.9

Permitted context.

Let

$$X = \text{Spec } k[x_1, x_2, x_3, x_4], \quad Y_1 = V(x_1, x_2), \quad Y_2 = V(x_3, x_4), \quad Y = Y_1 \cup Y_2.$$

An open cover by two affine subsets has cohomological dimension at most one.

Problem. Prove that Y is not a set-theoretic complete intersection in X . Deduce that its projective closure in $\mathbb{P}_k^4$ is not a set-theoretic complete intersection.

Solution. The closed set Y has codimension two. If it were a set-theoretic complete intersection, we could write

$$Y = H_1 \cap H_2$$

with affine hypersurfaces $H_i = V(f_i)$. Its complement would then be

$$X \setminus Y = D(f_1) \cup D(f_2),$$

a union of two affine opens. This would imply $\text{cd}(X \setminus Y) \leq 1$.

We show instead that $H^2(X \setminus Y, \mathcal{O}_X) \neq 0$. Put

$$U_1 = X \setminus Y_1, \quad U_2 = X \setminus Y_2.$$

Then

$$U_1 \cap U_2 = X \setminus Y, \quad U_1 \cup U_2 = X \setminus \{0\}. \tag{1}$$

Each U_i is covered by two principal affine opens, so $H^q(U_i, \mathcal{O}_X) = 0$ for $q \geq 2$. The Mayer–Vietoris sequence for the cover in (1) therefore gives an isomorphism

$$H^2(X \setminus Y, \mathcal{O}_X) \cong H^3(X \setminus \{0\}, \mathcal{O}_X). \tag{2}$$

Cover $X \setminus \{0\}$ by the four opens $D(x_i)$. The cover is acyclic, and the top Čech quotient is

$$\frac{k[x_1^{\pm 1}, x_2^{\pm 1}, x_3^{\pm 1}, x_4^{\pm 1}]}{\sum_{i=1}^4 k[x_1, \dots, x_4, x_j^{-1} : j \neq i]}. \quad (3)$$

The residue classes of Laurent monomials

$$x_1^{a_1} x_2^{a_2} x_3^{a_3} x_4^{a_4}, \quad a_i < 0 \text{ for every } i,$$

form a basis of (3). In particular, the class of $(x_1 x_2 x_3 x_4)^{-1}$ is nonzero. Thus the right side of (2) is nonzero, contradicting the cohomological-dimension consequence of a set-theoretic complete-intersection presentation.

Finally, if the projective closure $\bar{Y} \subseteq \mathbb{P}_k^4$ were the set-theoretic intersection of two projective hypersurfaces, restriction to the standard affine chart containing $X = \mathbb{A}_k^4$ would express Y as the intersection of two affine hypersurfaces. This is impossible by what we have just proved. $\square$

Exercise III.4.10

Permitted context.

Let X be a nonsingular variety over an algebraically closed field k , let $\mathcal{F}$ be coherent, and let $\mathcal{T}_X = \mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{O}_X)$. An infinitesimal extension of X by $\mathcal{F}$ is a square-zero extension of its structure sheaf whose ideal is identified with $\mathcal{F}$. Such an extension is split on a nonsingular affine open.

Problem. Prove that the isomorphism classes of infinitesimal extensions of X by $\mathcal{F}$ are in natural one-to-one correspondence with

$$H^1(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{T}_X).$$

Solution. Choose an affine open cover $\mathfrak{U} = (U_i)$ of X . On each U_i , local splitting gives an isomorphism of square-zero extensions with

$$\mathcal{O}_{U_i} \oplus \mathcal{F}|_{U_i}, \quad (a, m)(b, n) = (ab, an + bm). \quad (1)$$

Choose such a trivialization φ_i .

We first identify the possible transition automorphisms. An automorphism of (1) which induces the identity on both the quotient $\mathcal{O}_{U_i}$ and the ideal $\mathcal{F}|_{U_i}$ has the form

$$(a, m) \mapsto (a, m + D(a)). \quad (2)$$

Compatibility with multiplication is equivalent to

$$D(ab) = aD(b) + bD(a),$$

so D is a k -derivation $\mathcal{O}_{U_i} \rightarrow \mathcal{F}|_{U_i}$. Conversely every such derivation defines (2), and composition of these automorphisms corresponds to addition of derivations.

On $U_i \cap U_j$, the two chosen trivializations therefore differ by a derivation

$$D_{ij} \in \text{Der}_k(\mathcal{O}_X, \mathcal{F})(U_i \cap U_j).$$

The compatibility on triple intersections is

$$D_{ij} + D_{jk} = D_{ik}, \quad (3)$$

so (D_{ij}) is a Čech 1-cocycle. Replacing φ_i by a different local trivialization adds a local derivation D_i , and changes (D_{ij}) by its Čech coboundary. Isomorphic extensions give the same cohomology class.

Conversely, a cocycle satisfying (3) gives transition automorphisms of the split algebras (1). Gluing them produces a square-zero extension of X by $\mathcal{F}$, and cohomologous cocycles give isomorphic extensions. Thus the desired set of isomorphism classes is

$$\check{H}^1(\mathfrak{U}, \text{Der}_k(\mathcal{O}_X, \mathcal{F})). \quad (4)$$

The universal property of differentials gives

$$\text{Der}_k(\mathcal{O}_X, \mathcal{F}) \cong \mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{F}).$$

Since X is nonsingular, $\Omega_{X/k}$ is locally free of finite rank, and hence

$$\mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{F}) \cong \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{T}_X. \quad (5)$$

The intersections in the affine cover are affine because X is separated, so Čech cohomology of the quasi-coherent sheaf (5) equals derived sheaf cohomology. Combining (4) and (5) gives the asserted correspondence. $\square$

Exercise III.4.11

Permitted context.

Let X be a topological space, $\mathcal{F}$ a sheaf of abelian groups, and $\mathfrak{U} = (U_i)$ an open cover. Assume that for every nonempty finite intersection

$$V = U_{i_0} \cap \cdots \cap U_{i_p}$$

one has $H^q(V, \mathcal{F}|_V) = 0$ for all $q > 0$.

Problem. Prove that every natural comparison map

$$\check{H}^p(\mathfrak{U}, \mathcal{F}) \longrightarrow H^p(X, \mathcal{F})$$

is an isomorphism. Deduce the Čech comparison theorem for a quasi-coherent sheaf on a noetherian separated scheme and an affine open cover.

Solution. Choose an injective resolution $\mathcal{F} \rightarrow \mathcal{I}^\bullet$, and form the first-quadrant double complex

$$K^{p,q} = \prod_{i_0 < \cdots < i_p} \Gamma(U_{i_0} \cap \cdots \cap U_{i_p}, \mathcal{I}^q). \quad (1)$$

The horizontal differential is the Čech differential and the vertical differential comes from the resolution.

Compute the cohomology of the total complex in the horizontal direction first. Every injective sheaf is flasque, and a flasque sheaf has zero positive Čech cohomology for every cover. Hence the augmented horizontal row

$$0 \longrightarrow \Gamma(X, \mathcal{I}^q) \longrightarrow K^{0,q} \longrightarrow K^{1,q} \longrightarrow \dots$$

is exact. It follows that

$$H^n(\text{Tot } K^{\bullet, \bullet}) \cong H^n(\Gamma(X, \mathcal{I}^{\bullet})) = H^n(X, \mathcal{F}). \quad (2)$$

Now compute vertically first. Restricting $\mathcal{I}^{\bullet}$ to an open set gives a flasque resolution of the restricted sheaf, so

$$H^q(\Gamma(U_{i_0} \cap \dots \cap U_{i_p}, \mathcal{I}^{\bullet})) \cong H^q(U_{i_0} \cap \dots \cap U_{i_p}, \mathcal{F}). \quad (3)$$

By hypothesis, (3) is zero for $q > 0$, while for $q = 0$ it is $\Gamma(U_{i_0} \cap \dots \cap U_{i_p}, \mathcal{F})$. Thus the vertical cohomology of (1) is concentrated in its bottom row, and that row is exactly the Čech complex $C^{\bullet}(\mathfrak{U}, \mathcal{F})$. Therefore

$$H^n(\text{Tot } K^{\bullet, \bullet}) \cong \check{H}^n(\mathfrak{U}, \mathcal{F}). \quad (4)$$

The identifications (2) and (4) are induced by the augmentations and their composite is the natural comparison map. Hence that map is an isomorphism in every degree.

Finally, let X be noetherian and separated, let $\mathcal{F}$ be quasi-coherent, and let $\mathfrak{U}$ be an affine open cover. Every finite intersection of members of $\mathfrak{U}$ is affine, and the restriction of $\mathcal{F}$ to it has no higher cohomology. The hypothesis above is satisfied, so the usual Čech comparison theorem follows. $\square$

III.5 The Cohomology of Projective Space

Exercise III.5.1

Permitted context.

Let X be projective over a field k . For a coherent sheaf $\mathcal{F}$, all $H^i(X, \mathcal{F})$ are finite-dimensional and vanish for i sufficiently large, so

$$\chi(\mathcal{F}) = \sum_i (-1)^i \dim_k H^i(X, \mathcal{F})$$

is well defined.

Problem. If

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

is a short exact sequence of coherent sheaves, prove

$$\chi(\mathcal{F}) = \chi(\mathcal{F}') + \chi(\mathcal{F}'').$$

Solution. The short exact sequence gives a long exact sequence of finite-dimensional vector spaces

$$0 \rightarrow H^0(\mathcal{F}') \rightarrow H^0(\mathcal{F}) \rightarrow H^0(\mathcal{F}'') \rightarrow H^1(\mathcal{F}') \rightarrow \dots,$$

which terminates because the cohomology groups vanish in high degree. For every finite exact sequence of finite-dimensional vector spaces, the alternating sum of the dimensions is zero. Applying this fact to the displayed sequence and collecting the terms belonging to each sheaf gives

$$\sum_i (-1)^i \dim H^i(\mathcal{F}) = \sum_i (-1)^i \dim H^i(\mathcal{F}') + \sum_i (-1)^i \dim H^i(\mathcal{F}''),$$

which is the required identity. $\square$

Exercise III.5.2**Permitted context.**

Let X be projective over k , let $\mathcal{O}_X(1)$ be very ample, and let $\mathcal{F}$ be coherent. After embedding $i : X \rightarrow \mathbb{P}_k^N$, the sheaf $i_*\mathcal{F}$ admits a finite resolution by finite direct sums of twists of $\mathcal{O}_{\mathbb{P}^N}$. Also

$$\chi(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m)) = \binom{m+N}{N}$$

as a polynomial identity for every $m \in \mathbb{Z}$. The section module $\Gamma_*(\mathcal{F})$, indexed over all integers, need not be finitely generated. Its Hilbert polynomial means that of a sufficiently high finite truncation, equivalently the polynomial agreeing with its dimensions in large degrees.

Problem. (a) Prove that there is a polynomial $P_{\mathcal{F}}(z) \in \mathbb{Q}[z]$ such that

$$\chi(\mathcal{F}(n)) = P_{\mathcal{F}}(n) \quad (n \in \mathbb{Z}).$$

(b) For $X = \mathbb{P}_k^r$ and $M = \Gamma_*(\mathcal{F})$, prove that $P_{\mathcal{F}}$ is the usual Hilbert polynomial of the graded module M .

Solution. (a) Use the projective embedding defined by $\mathcal{O}_X(1)$. Direct image under the closed immersion preserves cohomology and satisfies

$$i_*(\mathcal{F}(n)) = (i_*\mathcal{F})(n).$$

Choose a finite resolution

$$0 \longrightarrow \mathcal{E}_m \longrightarrow \cdots \longrightarrow \mathcal{E}_0 \longrightarrow i_*\mathcal{F} \longrightarrow 0,$$

where

$$\mathcal{E}_j = \bigoplus_{\ell} \mathcal{O}_{\mathbb{P}^N}(-a_{j\ell}).$$

Twisting is exact. Repeated additivity of Euler characteristic gives

$$\chi(\mathcal{F}(n)) = \sum_{j,\ell} (-1)^j \chi(\mathcal{O}_{\mathbb{P}^N}(n - a_{j\ell})) = \sum_{j,\ell} (-1)^j \binom{n - a_{j\ell} + N}{N}. \quad (1)$$

The right side is the value at n of a polynomial in $\mathbb{Q}[z]$, and (1) holds for every integer n . This defines $P_{\mathcal{F}}$; it is unique because two polynomials agreeing at all integers are equal.

(b) By Serre vanishing, for n sufficiently large,

$$H^i(\mathbb{P}^r, \mathcal{F}(n)) = 0 \quad (i > 0).$$

Consequently

$$P_{\mathcal{F}}(n) = \chi(\mathcal{F}(n)) = \dim_k H^0(\mathbb{P}^r, \mathcal{F}(n)) = \dim_k M_n \quad (n \gg 0).$$

The graded-module Hilbert polynomial is characterized as the polynomial eventually equal to $\dim_k M_n$. Hence it equals $P_{\mathcal{F}}$. $\square$

Exercise III.5.3**Permitted context.**

For a projective scheme X of dimension r , define

$$p_a(X) = (-1)^r (\chi(\mathcal{O}_X) - 1).$$

For a projective embedding, the Hilbert polynomial satisfies $P_X(n) = \chi(\mathcal{O}_X(n))$ for every integer n .

Problem. (a) If X is integral and k is algebraically closed, prove $H^0(X, \mathcal{O}_X) = k$, and deduce

$$p_a(X) = \sum_{j=0}^{r-1} (-1)^j \dim_k H^{r-j}(X, \mathcal{O}_X).$$

In particular, for a curve, $p_a(X) = \dim_k H^1(X, \mathcal{O}_X)$.

(b) Prove that for a closed projective variety this definition agrees with the Hilbert-polynomial definition of arithmetic genus.

(c) Prove that arithmetic genus is a birational invariant of nonsingular projective curves over an algebraically closed field. Deduce that a nonsingular plane curve of degree $d \geq 3$ is not rational.

Solution. (a) A global regular function gives a morphism $X \rightarrow \mathbb{A}_k^1$. Since X is proper, its image is a proper closed integral subscheme of $\mathbb{A}_k^1$, hence a single closed point. The field is algebraically closed, so the function is a constant in k . Thus $H^0(X, \mathcal{O}_X) = k$.

Writing $h^i = \dim_k H^i(X, \mathcal{O}_X)$, we obtain

$$\chi(\mathcal{O}_X) - 1 = \sum_{i=1}^r (-1)^i h^i.$$

Multiplication by $(-1)^r$, followed by the substitution $i = r - j$, gives the formula in the statement. When $r = 1$, only h^1 remains.

(b) If $e = \dim X$, the earlier definition from the Hilbert polynomial is

$$(-1)^e (P_X(0) - 1).$$

But $P_X(0) = \chi(\mathcal{O}_X)$. Hence this expression is exactly the present definition, and in particular does not depend on the chosen projective embedding.

(c) A birational map between nonsingular projective curves extends to an isomorphism. Their structure-sheaf cohomology, and therefore their arithmetic genera, are consequently equal. A rational smooth projective curve is isomorphic to $\mathbb{P}^1$ and has arithmetic genus zero. On the other hand, the Čech computation for a plane curve of degree d gives

$$p_a(X) = \dim_k H^1(X, \mathcal{O}_X) = \frac{(d-1)(d-2)}{2}.$$

This is positive for $d \geq 3$, so such a nonsingular plane curve cannot be rational. $\square$

Exercise III.5.4**Permitted context.**

Let $K(X)$ be the Grothendieck group of coherent sheaves on a noetherian scheme. It is generated by symbols $\gamma(\mathcal{F})$, with the relation

$$\gamma(\mathcal{F}) = \gamma(\mathcal{F}') + \gamma(\mathcal{F}'')$$

for every short exact sequence $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$. For an arbitrary choice of linear subspaces, first prove the result for a flag. Injectivity of the Hilbert-polynomial map then identifies any other subspace of the same dimension with its flag representative.

Problem. (a) For a projective scheme X with a fixed very ample $\mathcal{O}_X(1)$, prove that there is a unique additive map

$$P : K(X) \longrightarrow \mathbb{Q}[z]$$

sending $\gamma(\mathcal{F})$ to the Hilbert polynomial of $\mathcal{F}$.

(b) Let $X = \mathbb{P}_k^r$, and for $0 \leq i \leq r$ let $L_i \subseteq X$ be a linear subspace of dimension i . Prove that $K(X)$ is freely generated by the classes $\gamma(\mathcal{O}_{L_i})$, and that P is injective.

Solution. (a) Hilbert polynomials are additive in short exact sequences because Euler characteristics are additive after every twist. Thus the assignment

$$\gamma(\mathcal{F}) \longmapsto P_{\mathcal{F}}$$

respects all defining relations of $K(X)$, and so induces an additive homomorphism P . The classes of coherent sheaves generate $K(X)$, which also proves uniqueness.

(b) We first prove generation by induction on r . Let $H = \mathbb{P}_k^{r-1}$ be a hyperplane and $U = X \setminus H \cong \mathbb{A}_k^r$. The localization sequence is

$$K(H) \longrightarrow K(X) \longrightarrow K(U) \longrightarrow 0. \quad (1)$$

Every finite module over $k[x_1, \dots, x_r]$ has a finite free resolution. Hence $K(U) \cong \mathbb{Z}$, generated by $\gamma(\mathcal{O}_U)$; the rank map shows that this generator has infinite order. The class $\gamma(\mathcal{O}_{L_r}) = \gamma(\mathcal{O}_X)$ restricts to this generator. By induction, $K(H)$ is generated by the classes of linear spaces $L_0, \dots, L_{r-1}$ lying in H . Exactness of (1) therefore shows that

$$\gamma(\mathcal{O}_{L_0}), \dots, \gamma(\mathcal{O}_{L_r})$$

generate $K(X)$.

Their Hilbert polynomials are

$$P(\gamma(\mathcal{O}_{L_i})) = \binom{z+i}{i} = \frac{(z+1) \cdots (z+i)}{i!}. \quad (2)$$

The polynomials in (2) have distinct degrees $0, 1, \dots, r$, with nonzero leading coefficients, and are therefore linearly independent over $\mathbb{Q}$. Any integral relation among the classes would, after applying P , give a relation among these polynomials, so all coefficients must vanish. The displayed classes are consequently a free basis. Finally, every element of $K(X)$ has a unique expression in that basis; (2) shows that an element with zero Hilbert polynomial has all coefficients zero. Thus P is injective. $\square$

Exercise III.5.5**Permitted context.**

Let $X = \mathbb{P}_k^r$, and let $Y \subseteq X$ be a complete intersection of dimension $q \geq 1$. Write $c = r - q$. Its homogeneous ideal is generated by a regular sequence $f_1, \dots, f_c$, with $d_i = \deg f_i$. Put

$$Y_i = V_+(f_1, \dots, f_i), \quad Y_0 = X, \quad Y_c = Y.$$

Problem. (a) Prove that for every $n \in \mathbb{Z}$, the restriction map

$$H^0(X, \mathcal{O}_X(n)) \longrightarrow H^0(Y, \mathcal{O}_Y(n))$$

is surjective.

(b) Prove that Y is connected.

(c) Prove

$$H^j(Y, \mathcal{O}_Y(n)) = 0 \quad (0 < j < q, \ n \in \mathbb{Z}).$$

(d) Prove

$$p_a(Y) = \dim_k H^q(Y, \mathcal{O}_Y).$$

Solution. (a) Regularity of the sequence gives, for $1 \leq i \leq c$, exact sequences

$$0 \longrightarrow \mathcal{O}_{Y_{i-1}}(n - d_i) \xrightarrow{f_i} \mathcal{O}_{Y_{i-1}}(n) \longrightarrow \mathcal{O}_{Y_i}(n) \longrightarrow 0. \quad (1)$$

We prove simultaneously, by induction on i , that

$$\begin{aligned} H^0(X, \mathcal{O}_X(n)) &\longrightarrow H^0(Y_i, \mathcal{O}_{Y_i}(n)) \quad \text{is surjective for every } n, \\ H^j(Y_i, \mathcal{O}_{Y_i}(n)) &= 0 \quad \text{for } 0 < j < \dim Y_i. \end{aligned} \quad (2)$$

For $i = 0$, these assertions are the standard cohomology calculation for projective space.

Assume (2) for Y_{i-1} , whose dimension is $r - i + 1 \geq q + 1 \geq 2$. The H^1 -vanishing in (2), applied to $\mathcal{O}_{Y_{i-1}}(n - d_i)$, shows from (1) that

$$H^0(Y_{i-1}, \mathcal{O}_{Y_{i-1}}(n)) \longrightarrow H^0(Y_i, \mathcal{O}_{Y_i}(n))$$

is surjective. Composing with the map from X proves the first line of (2) for Y_i .

Now let $0 < j < \dim Y_i = r - i$. The portion of the long exact sequence of (1) surrounding $H^j(Y_i, \mathcal{O}_{Y_i}(n))$ has outer terms

$$H^j(Y_{i-1}, \mathcal{O}_{Y_{i-1}}(n)) \quad \text{and} \quad H^{j+1}(Y_{i-1}, \mathcal{O}_{Y_{i-1}}(n - d_i)).$$

Both vanish by the inductive hypothesis, because $0 < j < j + 1 < \dim Y_{i-1}$. This proves the second line of (2). Taking $i = c$ proves (a).

(b) Part (a) with $n = 0$ gives a surjection

$$k = H^0(X, \mathcal{O}_X) \longrightarrow H^0(Y, \mathcal{O}_Y).$$

Constants inject, so $H^0(Y, \mathcal{O}_Y) = k$. A decomposition of Y into two nonempty open-and-closed subsets would give a nontrivial idempotent in this ring. Hence Y is connected.

(c) This is the second assertion of (2) for $i = c$.

(d) By (b), $H^0(Y, \mathcal{O}_Y) = k$, and by (c) all cohomology in degrees $1, \dots, q - 1$ vanishes. Therefore

$$\chi(\mathcal{O}_Y) = 1 + (-1)^q \dim_k H^q(Y, \mathcal{O}_Y).$$

The definition $p_a(Y) = (-1)^q (\chi(\mathcal{O}_Y) - 1)$ gives the claimed equality. $\square$

Exercise III.5.6**Permitted context.**

Let $Q = V(xy - zw) \subseteq \mathbb{P}_k^3$. The Segre map identifies Q with $\mathbb{P}_k^1 \times_k \mathbb{P}_k^1$, and under this identification

$$\mathcal{O}_Q(a, b) = p_1^* \mathcal{O}_{\mathbb{P}^1}(a) \otimes p_2^* \mathcal{O}_{\mathbb{P}^1}(b), \quad \mathcal{O}_Q(n) = \mathcal{O}_Q(n, n).$$

For $q > 0$, unions of q distinct lines in either ruling give exact sequences

$$\begin{aligned} 0 \rightarrow \mathcal{O}_Q(a, b - q) \rightarrow \mathcal{O}_Q(a, b) \rightarrow \mathcal{O}_{\mathbb{P}^1}(a)^{\oplus q} \rightarrow 0, \\ 0 \rightarrow \mathcal{O}_Q(a - q, b) \rightarrow \mathcal{O}_Q(a, b) \rightarrow \mathcal{O}_{\mathbb{P}^1}(b)^{\oplus q} \rightarrow 0. \end{aligned} \quad (1)$$

For the cohomological assertions one may first extend k to an algebraic closure: coherent cohomology commutes with field extension, and the extension is faithfully flat. We may use the cohomology of $\mathcal{O}_{\mathbb{P}^r}(m)$ and Bertini's theorem. A smooth curve of type $(1, 3)$ on a quadric has degree four and a degree-one projection to $\mathbb{P}^1$, hence is rational.

Problem. (a) Prove the following assertions.

(1) If $|a - b| \leq 1$, then $H^1(Q, \mathcal{O}_Q(a, b)) = 0$.

(2) If $a, b < 0$, then $H^1(Q, \mathcal{O}_Q(a, b)) = 0$.

(3) If $a \leq -2$, then $H^1(Q, \mathcal{O}_Q(a, 0)) \neq 0$.

(b) Let $Y \subseteq Q$ be a locally principal closed subscheme of type (a, b) .

(1) If $a, b > 0$, prove that Y is connected.

(2) If k is algebraically closed and $a, b > 0$, prove that there is an irreducible nonsingular curve of type (a, b) .

(3) If Y is irreducible and nonsingular and $a, b > 0$, prove that its given embedding in $\mathbb{P}^3$ is projectively normal if and only if $|a - b| \leq 1$. In particular, type $(1, 3)$ gives a nonsingular rational quartic that is not projectively normal.

(c) Prove

$$p_a(Y) = ab - a - b + 1.$$

Solution. (a) The quadric sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^3}(n - 2) \longrightarrow \mathcal{O}_{\mathbb{P}^3}(n) \longrightarrow \mathcal{O}_Q(n, n) \longrightarrow 0$$

and the vanishing of the intermediate cohomology of line bundles on $\mathbb{P}^3$ give

$$H^1(Q, \mathcal{O}_Q(n, n)) = 0 \quad (n \in \mathbb{Z}). \quad (2)$$

Let L be a ruling line of type $(0, 1)$. The case $q = 1$ of (1) gives

$$0 \rightarrow \mathcal{O}_Q(n, n - 1) \rightarrow \mathcal{O}_Q(n, n) \rightarrow \mathcal{O}_L(n) \rightarrow 0. \quad (3)$$

If $n < 0$, then $H^0(L, \mathcal{O}_L(n)) = 0$. If $n \geq 0$, the restriction map from $\mathbb{P}^3$, and hence the one from Q , to L is surjective on degree- n forms. The cohomology sequence of (3) and (2) therefore show that $H^1(Q, \mathcal{O}_Q(n, n-1)) = 0$ for every n . Interchanging the two rulings gives the same conclusion for $\mathcal{O}_Q(n-1, n)$. Together with (2), this proves (1).

For (2), after the harmless field extension described in the context, choose a reduced divisor D consisting of $-a$ lines of one ruling and $-b$ lines of the other. It has type $(-a, -b)$, and it is connected because every line of one ruling meets every line of the other. The divisor sequence is

$$0 \rightarrow \mathcal{O}_Q(a, b) \rightarrow \mathcal{O}_Q \rightarrow \mathcal{O}_D \rightarrow 0.$$

Both Q and D are proper, reduced, and connected, so their rings of global functions are k . Since $H^1(Q, \mathcal{O}_Q) = 0$ by (2), the cohomology sequence gives $H^1(Q, \mathcal{O}_Q(a, b)) = 0$.

For (3), let D be the disjoint union of $-a$ lines in one ruling. From

$$0 \rightarrow \mathcal{O}_Q(a, 0) \rightarrow \mathcal{O}_Q \rightarrow \mathcal{O}_D \rightarrow 0$$

we obtain

$$k \longrightarrow k^{-a} \longrightarrow H^1(Q, \mathcal{O}_Q(a, 0)) \longrightarrow 0.$$

The first map is diagonal, so its cokernel has dimension $-a - 1 > 0$. Faithful flatness then gives the same nonvanishing over the original field.

(b) For (1), the divisor sequence is

$$0 \longrightarrow \mathcal{O}_Q(-a, -b) \longrightarrow \mathcal{O}_Q \longrightarrow \mathcal{O}_Y \longrightarrow 0. \quad (4)$$

Since $-a, -b < 0$, part (a2) shows that the restriction

$$k = H^0(Q, \mathcal{O}_Q) \longrightarrow H^0(Y, \mathcal{O}_Y)$$

is surjective. It is injective because it preserves constants, so $H^0(Y, \mathcal{O}_Y) = k$. This ring has no nontrivial idempotent; hence Y is connected.

For (2), if $a, b > 0$, the bundle $\mathcal{O}_Q(a, b)$ is very ample. Bertini's theorem gives a nonsingular general member Y of its complete linear system. Every such member is connected by part (1). A nonsingular curve over an algebraically closed field is the disjoint union of its irreducible components, so this connected Y is irreducible.

For (3), since Y is nonsingular, it is normal. Thus it is projectively normal precisely when

$$H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(n)) \longrightarrow H^0(Y, \mathcal{O}_Y(n)) \quad (5)$$

is surjective for every $n \geq 0$. The quadric sequence

$$0 \rightarrow \mathcal{O}_{\mathbb{P}^3}(n-2) \rightarrow \mathcal{O}_{\mathbb{P}^3}(n) \rightarrow \mathcal{O}_Q(n, n) \rightarrow 0$$

and the vanishing of $H^1(\mathbb{P}^3, \mathcal{O}(m))$ show that the restriction from $\mathbb{P}^3$ to Q is always surjective. Meanwhile, twisting (4) gives

$$0 \rightarrow \mathcal{O}_Q(n-a, n-b) \rightarrow \mathcal{O}_Q(n, n) \rightarrow \mathcal{O}_Y(n) \rightarrow 0. \quad (6)$$

Because $H^1(Q, \mathcal{O}_Q(n, n)) = 0$, its cohomology sequence identifies the cokernel of the restriction in (6) with $H^1(Q, \mathcal{O}_Q(n-a, n-b))$. If $|a-b| \leq 1$, part (a1) makes this group zero for every n , so (5) is surjective.

Conversely, suppose for instance that $a \geq b + 2$. Taking $n = b$ in (6), part (a3) gives

$$H^1(Q, \mathcal{O}_Q(b - a, 0)) \neq 0.$$

Thus restriction fails to be surjective in degree $b \geq 0$, and Y is not projectively normal. The case $b \geq a + 2$ is symmetric.

For $(a, b) = (1, 3)$, the criterion shows failure of projective normality; this curve is the rational quartic from Exercise I.3.18.

(c) A ruling line and additivity of Euler characteristic give, for all integers u, v ,

$$\begin{aligned}\chi(\mathcal{O}_Q(u, v)) - \chi(\mathcal{O}_Q(u, v - 1)) &= u + 1, \\ \chi(\mathcal{O}_Q(u, 0)) - \chi(\mathcal{O}_Q(u - 1, 0)) &= 1.\end{aligned}$$

The quadric sequence gives $\chi(\mathcal{O}_Q) = 1$. The second recurrence, followed by the first, therefore yields

$$\chi(\mathcal{O}_Q(u, v)) = (u + 1)(v + 1) \quad (u, v \in \mathbb{Z}).$$

Taking Euler characteristics in (4),

$$\chi(\mathcal{O}_Y) = 1 - (1 - a)(1 - b) = a + b - ab.$$

Since Y has dimension one,

$$p_a(Y) = 1 - \chi(\mathcal{O}_Y) = ab - a - b + 1.$$

□

Exercise III.5.7

Permitted context.

Let X and Y be proper schemes over a noetherian ring A , and let $\mathcal{L}$ denote an invertible sheaf. We may use both standard criteria for ampleness: sufficiently high twists of every coherent sheaf are globally generated, and, on a proper scheme,

$$\mathcal{L} \text{ is ample} \iff H^i(X, \mathcal{F} \otimes \mathcal{L}^n) = 0 \quad (i > 0, n \gg 0)$$

for every coherent $\mathcal{F}$.

Problem. (a) If $\mathcal{L}$ is ample on X and $i : Y \hookrightarrow X$ is a closed immersion, prove that $i^*\mathcal{L}$ is ample on Y .

(b) Prove that $\mathcal{L}$ is ample on X if and only if its restriction $\mathcal{L}_{\text{red}}$ is ample on X_{red} .

(c) Suppose that X is reduced, with irreducible components $X_1, \dots, X_s$. Prove that $\mathcal{L}$ is ample on X if and only if every $\mathcal{L}|_{X_j}$ is ample.

(d) Let $f : X \rightarrow Y$ be finite and surjective. Prove that $\mathcal{L}$ is ample on Y if and only if $f^*\mathcal{L}$ is ample on X .

Solution. (a) Let $\mathcal{F}$ be coherent on Y . The sheaf $i_*\mathcal{F}$ is coherent, so for $n \gg 0$,

$$i_*\mathcal{F} \otimes \mathcal{L}^n \cong i_*(\mathcal{F} \otimes i^*\mathcal{L}^n)$$

is generated by its global sections. Its global sections are exactly those of $\mathcal{F} \otimes i^*\mathcal{L}^n$, and restricting the evaluation map to Y shows that the latter sheaf is globally generated. Hence $i^*\mathcal{L}$ is ample.

(b) The forward implication is (a). Conversely, let $\mathcal{N} \subseteq \mathcal{O}_X$ be the nilradical. Since X is noetherian, $\mathcal{N}^m = 0$ for some m . For any coherent $\mathcal{F}$, the finite filtration

$$\mathcal{F} \supseteq \mathcal{N}\mathcal{F} \supseteq \cdots \supseteq \mathcal{N}^m\mathcal{F} = 0 \quad (1)$$

has coherent successive quotients annihilated by $\mathcal{N}$, and therefore coherent on X_{red} . If $\mathcal{L}_{\text{red}}$ is ample, Serre vanishing applies to every quotient of (1). The long exact cohomology sequences for the filtration then give

$$H^i(X, \mathcal{F} \otimes \mathcal{L}^n) = 0 \quad (i > 0, n \gg 0).$$

The cohomological criterion proves that $\mathcal{L}$ is ample.

(c) Necessity follows from (a). For sufficiency, let $\mathcal{I}_j$ be the ideal of X_j . Reducedness gives $\bigcap_j \mathcal{I}_j = 0$, hence $\mathcal{I}_1 \cdots \mathcal{I}_s = 0$. Filter a coherent sheaf $\mathcal{F}$ by

$$\mathcal{F} \supseteq \mathcal{I}_1\mathcal{F} \supseteq \mathcal{I}_1\mathcal{I}_2\mathcal{F} \supseteq \cdots \supseteq 0. \quad (2)$$

The j -th successive quotient is annihilated by $\mathcal{I}_j$, so it is a coherent sheaf on X_j . If every $\mathcal{L}|_{X_j}$ is ample, its higher cohomology vanishes after a sufficiently high twist. There are only finitely many quotients in (2), so the long exact sequences and the cohomological criterion show that $\mathcal{L}$ is ample on X .

(d) First suppose that $\mathcal{L}$ is ample, and let $\mathcal{F}$ be coherent on X . The finite morphism f is affine and $f_*\mathcal{F}$ is coherent. For $n \gg 0$, the projection formula makes

$$f_*(\mathcal{F} \otimes f^*\mathcal{L}^n) \cong f_*\mathcal{F} \otimes \mathcal{L}^n$$

globally generated. Pull its evaluation map back to X , and compose with the counit

$$f^*f_*(\mathcal{F} \otimes f^*\mathcal{L}^n) \longrightarrow \mathcal{F} \otimes f^*\mathcal{L}^n. \quad (3)$$

The map (3) is surjective: over $\text{Spec } B \subseteq Y$, it is the multiplication map $C \otimes_B M \rightarrow M$, and $1 \otimes m \mapsto m$. Thus every high twist by $f^*\mathcal{L}$ is globally generated, proving that $f^*\mathcal{L}$ is ample.

Conversely, assume that $f^*\mathcal{L}$ is ample. We use noetherian induction on closed subsets of Y . Parts (b) and (c), together with base change of f , reduce the inductive step to the case in which Y is integral and $\mathcal{L}$ is already known to be ample on every proper closed subscheme of Y . If necessary, choose an irreducible component of X_{red} that dominates Y ; its map to Y is still finite and surjective, and the restriction of $f^*\mathcal{L}$ is ample by (a). We may therefore also take X integral.

Let $\mathcal{F}$ be coherent on Y . The finite-morphism lemma used in the proof of Chevalley's theorem supplies a coherent sheaf $\mathcal{G}$ on X , an integer $r > 0$, and a morphism

$$\beta : f_*\mathcal{G} \longrightarrow \mathcal{F}^{\oplus r}$$

which is an isomorphism at the generic point of Y . Put $\mathcal{K} = \ker \beta$, $\mathcal{Q} = \text{im } \beta$, and $\mathcal{C} = \text{coker } \beta$. Both $\mathcal{K}$ and $\mathcal{C}$ are supported on proper closed subsets of Y . The inductive hypothesis and Serre vanishing therefore give, for $i > 0$ and $n \gg 0$,

$$H^i(Y, \mathcal{K} \otimes \mathcal{L}^n) = H^i(Y, \mathcal{C} \otimes \mathcal{L}^n) = 0. \quad (4)$$

On the other hand, finiteness of f , the projection formula, and the assumed ampleness upstairs give

$$\begin{aligned} H^i(Y, f_* \mathcal{G} \otimes \mathcal{L}^n) &\cong H^i(Y, f_*(\mathcal{G} \otimes f^* \mathcal{L}^n)) \\ &\cong H^i(X, \mathcal{G} \otimes f^* \mathcal{L}^n) = 0 \end{aligned} \quad (5)$$

for $i > 0$ and $n \gg 0$. Apply cohomology first to

$$0 \rightarrow \mathcal{K} \rightarrow f_* \mathcal{G} \rightarrow \mathcal{Q} \rightarrow 0$$

and then to

$$0 \rightarrow \mathcal{Q} \rightarrow \mathcal{F}^{\oplus r} \rightarrow \mathcal{C} \rightarrow 0.$$

Equations (4) and (5) imply successively that $H^i(Y, \mathcal{Q} \otimes \mathcal{L}^n) = 0$ and $H^i(Y, \mathcal{F}^{\oplus r} \otimes \mathcal{L}^n) = 0$ for every $i > 0$ and all sufficiently large n . Hence the same vanishing holds for $\mathcal{F}$. The cohomological criterion proves that $\mathcal{L}$ is ample on Y . $\square$

Exercise III.5.8

Permitted context.

Let k be algebraically closed and let X be a one-dimensional proper k -scheme. We may use that a proper scheme over k with an ample invertible sheaf is projective, and the ampleness results of Exercise III.5.7. We use the projectivity theorem for nonsingular proper curves, equivalently ampleness of divisors of positive degree on such a curve.

Problem. Prove that X is projective by treating successively the following cases.

- (a) X is irreducible and nonsingular.
- (b) X is integral. Use its normalization and descend an ample invertible sheaf.
- (c) X is reduced. If $X_1, \dots, X_s$ are its irreducible components, prove that

$$\mathrm{Pic} X \longrightarrow \bigoplus_{j=1}^s \mathrm{Pic} X_j$$

is surjective, and use this to construct an ample invertible sheaf on X .

- (d) Treat an arbitrary X by lifting an ample invertible sheaf from X_{red} .

Solution. (a) An irreducible nonsingular proper curve has a positive effective divisor, and the associated invertible sheaf is ample. Equivalently, this is the projectivity theorem for nonsingular proper curves.

(b) Let $f: \tilde{X} \rightarrow X$ be the normalization. Normalization is finite for an integral scheme of finite type over a field, so $\tilde{X}$ is proper. A normal noetherian curve is regular: each local ring at a closed point is a one-dimensional noetherian normal local domain, hence a discrete valuation ring. Since k is perfect, $\tilde{X}$ is nonsingular and is projective by (a).

Choose a very ample invertible sheaf $\mathcal{M}$ on $\tilde{X}$. The finite birational map f is an isomorphism away from a finite set $S \subseteq \tilde{X}$. Embed $\tilde{X}$ by $\mathcal{M}$ and choose a hyperplane avoiding S , which is

possible because k is infinite. Its intersection with $\tilde{X}$ is an effective Cartier divisor D , supported where f is an isomorphism, and

$$\mathcal{O}_{\tilde{X}}(D) \cong \mathcal{M}. \quad (1)$$

The divisor D therefore descends to an effective Cartier divisor D_0 on X : near its support use the isomorphism furnished by f , and use the zero divisor elsewhere. Put $\mathcal{L}_0 = \mathcal{O}_X(D_0)$. Then

$$f^*\mathcal{L}_0 \cong \mathcal{O}_{\tilde{X}}(D) \cong \mathcal{M}.$$

Since f is finite and surjective, Exercise III.5.7(d) implies that $\mathcal{L}_0$ is ample. Thus X is projective.

(c) Let $i_j : X_j \hookrightarrow X$ be the inclusions. Reducedness gives an injection of sheaves of abelian groups

$$1 \longrightarrow \mathcal{O}_X^\times \longrightarrow \prod_{j=1}^s i_{j*} \mathcal{O}_{X_j}^\times \longrightarrow \mathcal{Q} \longrightarrow 1. \quad (2)$$

Away from the intersections of distinct components, the middle map is an isomorphism. Those intersections form a finite set, so $\mathcal{Q}$ is supported at finitely many closed points and $H^1(X, \mathcal{Q}) = 0$. The cohomology sequence of (2) consequently gives a surjection

$$\mathrm{Pic} X = H^1(X, \mathcal{O}_X^\times) \longrightarrow \prod_{j=1}^s H^1(X_j, \mathcal{O}_{X_j}^\times) = \bigoplus_{j=1}^s \mathrm{Pic} X_j. \quad (3)$$

Every one-dimensional component X_j is projective by (b), while a zero-dimensional component is already projective. Choose an ample invertible sheaf $\mathcal{L}_j$ on each component and lift their tuple through (3) to an invertible sheaf $\mathcal{L}$ on X . Exercise III.5.7(c) says that $\mathcal{L}$ is ample, so X is projective.

(d) Let $\mathcal{N}$ be the nilradical. It is nilpotent, say $\mathcal{N}^{m+1} = 0$, and the schemes

$$X_j = (X, \mathcal{O}_X/\mathcal{N}^j), \quad 1 \leq j \leq m+1,$$

form a chain from $X_1 = X_{\mathrm{red}}$ to $X_{m+1} = X$. The ideal $\mathcal{I}_j = \mathcal{N}^j/\mathcal{N}^{j+1}$ of $X_j \subseteq X_{j+1}$ has square zero. Hence there is an exact sequence

$$0 \longrightarrow \mathcal{I}_j \xrightarrow{a \mapsto 1+a} \mathcal{O}_{X_{j+1}}^\times \longrightarrow \mathcal{O}_{X_j}^\times \longrightarrow 1. \quad (4)$$

Grothendieck vanishing gives $H^2(X_j, \mathcal{I}_j) = 0$, because the underlying space has dimension one. The cohomology sequence of (4) therefore makes

$$\mathrm{Pic} X_{j+1} \longrightarrow \mathrm{Pic} X_j$$

surjective. Iterating, every invertible sheaf on X_{red} lifts to X . By (c), choose an ample one and lift it to $\mathcal{L}$ on X . Exercise III.5.7(b) shows that $\mathcal{L}$ is ample, and therefore X is projective. $\square$

Exercise III.5.9**Permitted context.**

Let k be algebraically closed of characteristic zero, let $X = \mathbb{P}_k^2$, and put $\omega = \Omega_X^2 \cong \mathcal{O}_X(-3)$. Write $U_i = D_+(x_i)$ for the standard affine cover. Under $\omega \otimes \mathcal{T}_X \cong \Omega_X^1$, the Čech cocycle

$$\xi_{ij} = \frac{x_j}{x_i} d\left(\frac{x_i}{x_j}\right) \in \Gamma(U_i \cap U_j, \Omega_X^1) \quad (1)$$

defines a square-zero extension X' of X by ω :

$$0 \longrightarrow \omega \longrightarrow \mathcal{O}_{X'} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

The associated sequence of units yields

$$H^1(X, \omega) \longrightarrow \text{Pic } X' \longrightarrow \text{Pic } X \xrightarrow{\delta} H^2(X, \omega). \quad (2)$$

Problem. Compute $\delta(\mathcal{O}_X(1))$ in Čech cohomology and prove that δ is injective. Deduce that $\text{Pic } X' = 0$, that X' has no ample invertible sheaf, and hence that X' is not projective.

Solution. Choose local splittings

$$\mathcal{O}_{X'}|_{U_i} \cong \mathcal{O}_{U_i} \oplus \omega|_{U_i},$$

where the second summand is square zero. With the natural identification used in (1), passage from the j -splitting to the i -splitting on U_{ij} is

$$(a, m) \longmapsto (a, m + D_{ij}(a)), \quad D_{ij}(a) = da \wedge \xi_{ij}. \quad (3)$$

The sign changes if one chooses the opposite convention for $\omega \otimes \mathcal{T}_X \cong \Omega_X^1$, but the nonvanishing below is unaffected.

Represent $\mathcal{O}_X(1)$ by

$$g_{ij} = \frac{x_j}{x_i}, \quad g_{01}g_{12}g_{20} = 1.$$

Lift g_{01}, g_{12}, g_{20} using respectively the splittings on U_0, U_1, U_2 . Expressing all three lifts in the U_0 -splitting and multiplying them with

$$(a, m)(b, n) = (ab, an + bm)$$

gives $(1, \eta)$, where

$$\begin{aligned} \eta &= g_{20}^{-1}D_{02}(g_{20}) + g_{12}^{-1}D_{01}(g_{12}) \\ &= d \log(g_{20}) \wedge \xi_{02} + d \log(g_{12}) \wedge \xi_{01}. \end{aligned} \quad (4)$$

Here $d \log(h)$ is shorthand for $h^{-1}dh$; no logarithm of a regular function is being chosen. The first term is zero because $\xi_{02} = d \log(x_0/x_2) = d \log(g_{20})$. On U_0 , set

$$u = \frac{x_1}{x_0}, \quad v = \frac{x_2}{x_0}.$$

Then the second term in (4) is

$$d \log(v/u) \wedge d \log(1/u) = \frac{du}{u} \wedge \frac{dv}{v}. \quad (5)$$

Thus $\delta(\mathcal{O}_X(1))$ is represented, up to the harmless overall sign just mentioned, by

$$\frac{1}{uv} du \wedge dv \in \Gamma(U_0 \cap U_1 \cap U_2, \omega). \quad (6)$$

We verify directly that this class is nonzero. On the triple intersection,

$$\Gamma(U_{012}, \omega) = k[u^{\pm 1}, v^{\pm 1}] du \wedge dv.$$

The restrictions from U_{01} span the Laurent monomials $u^a v^b du \wedge dv$ with $b \geq 0$, and those from U_{02} span the ones with $a \geq 0$. On U_1 , use $s = x_0/x_1 = u^{-1}$ and $t = x_2/x_1 = v/u$. Since

$$ds \wedge dt = -u^{-3} du \wedge dv,$$

the restrictions from U_{12} span precisely the monomials with $a + b \leq -3$. Modulo these three subspaces, the only remaining Laurent monomial is $u^{-1}v^{-1} du \wedge dv$. Consequently

$$\check{H}^2(\{U_i\}, \omega) = k \left[\frac{du \wedge dv}{uv} \right] \cong H^2(X, \omega),$$

and (6) is a generator. In particular, $\delta(\mathcal{O}_X(1)) \neq 0$.

Now $\text{Pic } X \cong \mathbb{Z}[\mathcal{O}_X(1)]$, while $H^2(X, \omega) \cong (k, +)$. Since δ is a group homomorphism,

$$\delta(\mathcal{O}_X(n)) = n \delta(\mathcal{O}_X(1)).$$

Characteristic zero makes this zero only when $n = 0$, so δ is injective. Also $H^1(X, \omega) = H^1(X, \mathcal{O}_X(-3)) = 0$. Exactness of (2) now gives $\text{Pic } X' = 0$.

If X' possessed an ample invertible sheaf, it would be trivial; its restriction to the closed subscheme X would then be $\mathcal{O}_X$. But the restriction of an ample sheaf is ample, whereas $\mathcal{O}_{\mathbb{P}^2}$ is not ample. Thus X' has no ample invertible sheaf and cannot be projective. $\square$

Exercise III.5.10**Permitted context.**

Let X be projective over a noetherian ring A , with a fixed very ample sheaf $\mathcal{O}_X(1)$. Kernels and images of morphisms of coherent sheaves on X are coherent, and Serre vanishing holds for every coherent sheaf.

Problem. Let

$$\mathcal{F}^1 \longrightarrow \mathcal{F}^2 \longrightarrow \cdots \longrightarrow \mathcal{F}^r$$

be an exact sequence of coherent sheaves. Prove that there is an integer n_0 such that, for every $n \geq n_0$, the sequence

$$\Gamma(X, \mathcal{F}^1(n)) \longrightarrow \Gamma(X, \mathcal{F}^2(n)) \longrightarrow \cdots \longrightarrow \Gamma(X, \mathcal{F}^r(n))$$

is exact.

Solution. Write $d_i : \mathcal{F}^i \rightarrow \mathcal{F}^{i+1}$, and put $\mathcal{K}_i = \ker d_i$. Exactness of the given sequence says that

$$\operatorname{im} d_{i-1} = \mathcal{K}_i \quad (2 \leq i \leq r-1).$$

Consequently, for each such i there is a short exact sequence of coherent sheaves

$$0 \longrightarrow \mathcal{K}_{i-1} \longrightarrow \mathcal{F}^{i-1} \longrightarrow \mathcal{K}_i \longrightarrow 0. \quad (1)$$

After twisting, the only obstruction to surjectivity of

$$\Gamma(X, \mathcal{F}^{i-1}(n)) \longrightarrow \Gamma(X, \mathcal{K}_i(n))$$

is $H^1(X, \mathcal{K}_{i-1}(n))$. By Serre vanishing this group is zero for all sufficiently large n . Since there are only finitely many indices, choose one n_0 that works for every sequence (1).

For $n \geq n_0$, left exactness of global sections gives

$$\ker \Gamma(d_i(n)) = \Gamma(X, \mathcal{K}_i(n)),$$

while the preceding surjectivity gives

$$\operatorname{im} \Gamma(d_{i-1}(n)) = \Gamma(X, \mathcal{K}_i(n)).$$

Thus image equals kernel at every interior term, as required. If exactness at either endpoint is part of the original sequence, the same argument gives surjectivity at the last term, while injectivity at the first is preserved automatically by the left exact functor $\Gamma(X, -)$. $\square$

III.6 Ext Groups and Sheaves

Exercise III.6.1

Permitted context.

Let $(X, \mathcal{O}_X)$ be a ringed space, and let $\mathcal{A}, \mathcal{C} \in \text{Mod}(X)$. An extension of $\mathcal{C}$ by $\mathcal{A}$ is a short exact sequence

$$0 \longrightarrow \mathcal{A} \longrightarrow \mathcal{E} \longrightarrow \mathcal{C} \longrightarrow 0.$$

Two extensions are equivalent if an isomorphism between their middle terms induces the identity on $\mathcal{A}$ and $\mathcal{C}$.

Problem. For an extension as above, apply $\text{Hom}(\mathcal{C}, -)$ and let

$$\xi_{\mathcal{E}} = \delta(1_{\mathcal{C}}) \in \text{Ext}^1(\mathcal{C}, \mathcal{A}).$$

Prove that $\mathcal{E} \mapsto \xi_{\mathcal{E}}$ is a bijection from equivalence classes of extensions of $\mathcal{C}$ by $\mathcal{A}$ to $\text{Ext}^1(\mathcal{C}, \mathcal{A})$.

Solution. Choose an injective sheaf $\mathcal{I}$ containing $\mathcal{A}$, and set $\mathcal{Q} = \mathcal{I}/\mathcal{A}$. From

$$0 \longrightarrow \mathcal{A} \longrightarrow \mathcal{I} \xrightarrow{p} \mathcal{Q} \longrightarrow 0 \tag{1}$$

and the injectivity of $\mathcal{I}$, we obtain an exact sequence

$$\text{Hom}(\mathcal{C}, \mathcal{I}) \longrightarrow \text{Hom}(\mathcal{C}, \mathcal{Q}) \xrightarrow{\partial} \text{Ext}^1(\mathcal{C}, \mathcal{A}) \longrightarrow 0. \tag{2}$$

Given $u : \mathcal{C} \rightarrow \mathcal{Q}$, form the pullback

$$\mathcal{E}_u = \mathcal{I} \times_{\mathcal{Q}} \mathcal{C} = \{(i, c) : p(i) = u(c)\}.$$

It fits into an exact sequence

$$0 \longrightarrow \mathcal{A} \longrightarrow \mathcal{E}_u \longrightarrow \mathcal{C} \longrightarrow 0. \tag{3}$$

The boundary map in (2) sends u to the pullback of (1) along u , namely (3). Equivalently, the connecting map belonging to (3) sends $1_{\mathcal{C}}$ to $\partial(u)$.

If $u' = u + p \circ v$ for some $v : \mathcal{C} \rightarrow \mathcal{I}$, then

$$(i, c) \longmapsto (i + v(c), c)$$

is an equivalence $\mathcal{E}_u \cong \mathcal{E}_{u'}$. Thus (2) shows that the construction depends only on the class in $\text{Ext}^1(\mathcal{C}, \mathcal{A})$, and every such class is represented by an extension.

It remains to show that no extensions have been identified incorrectly. Start with an arbitrary extension

$$0 \rightarrow \mathcal{A} \rightarrow \mathcal{E} \xrightarrow{\pi} \mathcal{C} \rightarrow 0. \tag{4}$$

Injectivity of $\mathcal{I}$ extends the inclusion $\mathcal{A} \hookrightarrow \mathcal{I}$ to a map $s : \mathcal{E} \rightarrow \mathcal{I}$. The composite $p \circ s$ vanishes on $\mathcal{A}$, so it descends uniquely through π to a map $u : \mathcal{C} \rightarrow \mathcal{Q}$. There is then a commutative diagram of short exact sequences

$$\begin{array}{ccccccc}
0 & \rightarrow & \mathcal{A} & \rightarrow & \mathcal{E} & \xrightarrow{\pi} & \mathcal{C} \rightarrow 0 \\
& & \parallel & & \downarrow & & \parallel \\
0 & \rightarrow & \mathcal{A} & \rightarrow & \mathcal{E}_u & \rightarrow & \mathcal{C} \rightarrow 0,
\end{array}$$

where the middle vertical map is $e \mapsto (s(e), \pi(e))$. Since the outer vertical maps are identities, the middle one is an isomorphism. Hence (4) is equivalent to $\mathcal{E}_u$.

A different choice of s changes u by an element in the image of $\text{Hom}(\mathcal{C}, \mathcal{I})$, so it has the same class in (2). The two constructions are therefore mutually inverse, which proves the bijection. $\square$

Exercise III.6.2

Permitted context.

Let $X = \mathbb{P}_k^1$, where k is infinite. Write $\text{Mod}(X)$, $\text{Qco}(X)$, and $\text{Coh}(X)$ for the categories of all $\mathcal{O}_X$ -modules, quasi-coherent modules, and coherent modules, respectively.

Problem. (a) Prove that there is no projective object $\mathcal{P} \in \text{Mod}(X)$ admitting an epimorphism $\mathcal{P} \twoheadrightarrow \mathcal{O}_X$.

(b) Prove the same assertion when projective means projective in either $\text{Qco}(X)$ or $\text{Coh}(X)$.

Solution. (a) Suppose that $\alpha : \mathcal{P} \twoheadrightarrow \mathcal{O}_X$ exists. Choose a closed k -rational point x . Surjectivity on the stalk at x gives a neighborhood U of x and $s \in \Gamma(U, \mathcal{P})$ whose image has residue 1 at x . Because k is infinite, choose another k -rational point $y \in U \setminus \{x\}$, put $V = X \setminus \{y\}$, and let $j : V \hookrightarrow X$.

There is a stalkwise surjection

$$j_! \mathcal{O}_V \twoheadrightarrow i_{x*} k(x). \quad (1)$$

The composite $\mathcal{P} \xrightarrow{\alpha} \mathcal{O}_X \twoheadrightarrow i_{x*} k(x)$ must lift through (1), since $\mathcal{P}$ is projective. Let $\varphi : \mathcal{P} \rightarrow j_! \mathcal{O}_V$ be such a lift. Then $\varphi(s) \in \Gamma(U, j_! \mathcal{O}_V)$ has nonzero germ at x .

On the other hand,

$$\Gamma(U, j_! \mathcal{O}_V) = 0. \quad (2)$$

Indeed, a section on the left is a regular function on $U \cap V = U \setminus \{y\}$ whose support is closed in U and avoids y . If it were nonzero, its support would contain the generic point of the integral open set U , and its closure in U would be all of U , including y . This contradicts the defining support condition for extension by zero. Equation (2) contradicts the nonzero germ of $\varphi(s)$.

(b) Suppose now that $\mathcal{P}$ is projective in either of the two stated subcategories and that $\alpha : \mathcal{P} \twoheadrightarrow \mathcal{O}_X$. Fix a rational point x . On some open neighborhood U there is $s \in \Gamma(U, \mathcal{P})$ whose image in $k(x)$ is nonzero. After shrinking U to a principal open, write $U = X \setminus D$, where D is an effective Cartier divisor. Let $\mathcal{L} = \mathcal{O}_X(D)$ and let $f \in \Gamma(X, \mathcal{L})$ be its canonical section, which is invertible on U .

The extension lemma for quasi-coherent sheaves says that for some $N \geq 0$, the section $f^N s$ extends to

$$t \in \Gamma(X, \mathcal{P} \otimes \mathcal{L}^N). \quad (3)$$

Its image in the fiber $\mathcal{L}_x^N \otimes k(x)$ is nonzero. Set

$$\mathcal{E} = \mathcal{L}^{-N} \otimes \mathcal{O}_X(-1).$$

After choosing an identification of its one-dimensional fiber at x with $k(x)$, there is an epimorphism in both $\text{Coh}(X)$ and $\text{Qco}(X)$,

$$\mathcal{E} \rightarrow i_{x*}k(x). \quad (4)$$

Projectivity of $\mathcal{P}$ lifts $\mathcal{P} \rightarrow \mathcal{O}_X \rightarrow i_{x*}k(x)$ through (4), producing $\varphi : \mathcal{P} \rightarrow \mathcal{E}$. Tensoring by $\mathcal{L}^N$, the section (3) maps to a global section of

$$\mathcal{E} \otimes \mathcal{L}^N \cong \mathcal{O}_X(-1).$$

That global section is zero. Its value in the fiber at x is therefore zero, whereas commutativity with (4) says that it is the nonzero image of t . This contradiction proves the assertion in both categories. $\square$

Exercise III.6.3

Permitted context.

Let X be a noetherian scheme. For sheaves $\mathcal{F}, \mathcal{G} \in \text{Mod}(X)$, write $\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})$ for the sheaf Ext . Sheaf Ext is local on X , and a locally free resolution of finite rank may be used to compute it.

Problem. (a) If $\mathcal{F}$ and $\mathcal{G}$ are coherent, prove that $\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})$ is coherent for every $i \geq 0$.

(b) If $\mathcal{F}$ is coherent and $\mathcal{G}$ is quasi-coherent, prove that $\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})$ is quasi-coherent for every $i \geq 0$.

Solution. Both assertions are local. On an affine open $U = \text{Spec } A$, the coherent sheaf $\mathcal{F}|_U$ corresponds to a finite A -module. Since A is noetherian, repeatedly taking kernels of surjections from finite free modules gives a resolution

$$\cdots \rightarrow \mathcal{E}_2 \rightarrow \mathcal{E}_1 \rightarrow \mathcal{E}_0 \rightarrow \mathcal{F}|_U \rightarrow 0 \quad (1)$$

in which every $\mathcal{E}_j$ is locally free of finite rank. Applying $\mathcal{H}om(-, \mathcal{G}|_U)$ gives the complex

$$0 \rightarrow \mathcal{E}_0^\vee \otimes \mathcal{G}|_U \rightarrow \mathcal{E}_1^\vee \otimes \mathcal{G}|_U \rightarrow \cdots, \quad (2)$$

whose i -th cohomology sheaf is $\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})|_U$.

(a) If $\mathcal{G}$ is coherent, every term of (2) is coherent. Kernels, images, and cokernels of maps of coherent sheaves are coherent on a noetherian scheme, so every cohomology sheaf of (2) is coherent.

(b) If $\mathcal{G}$ is quasi-coherent, every term of (2) is quasi-coherent. Kernels and cokernels of morphisms of quasi-coherent sheaves on a scheme are quasi-coherent. Hence the cohomology sheaves of (2), and therefore all the stated sheaf Ext s, are quasi-coherent. $\square$

Exercise III.6.4**Permitted context.**

Let X be noetherian, and suppose that every coherent sheaf is a quotient of a locally free coherent sheaf. In this situation we say that $\text{Coh}(X)$ has enough locally free sheaves. Fix $\mathcal{G} \in \text{Mod}(X)$.

Problem. Prove that

$$(\mathcal{E}xt_X^i(-, \mathcal{G}))_{i \geq 0}$$

is a contravariant universal delta functor from $\text{Coh}(X)$ to $\text{Mod}(X)$.

Solution. The long exact sheaf-Ext sequence makes the displayed family a contravariant delta functor. By the standard universality criterion, it is enough to prove that its positive-degree terms are coexact: for each coherent $\mathcal{F}$, there must be an epimorphism $\mathcal{E} \twoheadrightarrow \mathcal{F}$ such that the induced maps

$$\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}) \longrightarrow \mathcal{E}xt_X^i(\mathcal{E}, \mathcal{G})$$

vanish for all $i > 0$.

By hypothesis, choose such an epimorphism with $\mathcal{E}$ locally free of finite rank. Then

$$\mathcal{H}om_X(\mathcal{E}, -) \cong \mathcal{E}^\vee \otimes_{\mathcal{O}_X} -$$

is exact. Consequently

$$\mathcal{E}xt_X^i(\mathcal{E}, \mathcal{G}) = 0 \quad (i > 0),$$

so all the maps above have zero target. The delta functor is coexact and hence universal. $\square$

Exercise III.6.5**Permitted context.**

Let X be noetherian and assume that $\text{Coh}(X)$ has enough locally free sheaves. For a coherent sheaf $\mathcal{F}$, its homological dimension $\text{hd}(\mathcal{F})$ is the least length of a locally free resolution, or $+\infty$ if no finite resolution exists.

Problem. (a) Prove that $\mathcal{F}$ is locally free if and only if

$$\mathcal{E}xt_X^1(\mathcal{F}, \mathcal{G}) = 0$$

for every $\mathcal{G} \in \text{Mod}(X)$.

(b) Prove that $\text{hd}(\mathcal{F}) \leq n$ if and only if

$$\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}) = 0 \quad (i > n)$$

for every $\mathcal{G} \in \text{Mod}(X)$.

(c) Prove

$$\text{hd}(\mathcal{F}) = \sup_{x \in X} \text{pd}_{\mathcal{O}_{X,x}} \mathcal{F}_x.$$

Solution. (a) If $\mathcal{F}$ is locally free of finite rank, then $\mathcal{H}om(\mathcal{F}, -) \cong \mathcal{F}^\vee \otimes -$ is exact, so every positive sheaf Ext vanishes.

Conversely, choose an exact sequence

$$0 \longrightarrow \mathcal{K} \longrightarrow \mathcal{E} \xrightarrow{q} \mathcal{F} \longrightarrow 0 \quad (1)$$

with $\mathcal{E}$ locally free. Apply $\mathcal{H}om(\mathcal{F}, -)$. The relevant part of the long exact sequence is

$$\mathcal{H}om(\mathcal{F}, \mathcal{E}) \longrightarrow \mathcal{H}om(\mathcal{F}, \mathcal{F}) \longrightarrow \mathcal{E}xt_X^1(\mathcal{F}, \mathcal{K}).$$

The last sheaf is zero by hypothesis. Hence the identity of $\mathcal{F}$ lifts locally through q , so (1) is locally split. Thus $\mathcal{F}$ is locally a direct summand of a finite locally free sheaf. Such a coherent direct summand is locally free.

(b) If $\text{hd}(\mathcal{F}) \leq n$, apply $\mathcal{H}om(-, \mathcal{G})$ to a length- n locally free resolution. The resulting complex has no terms in degrees greater than n , so its cohomology gives

$$\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}) = 0 \quad (i > n).$$

For the converse, set $\mathcal{K}_0 = \mathcal{F}$. Using enough locally free sheaves, construct exact sequences

$$0 \longrightarrow \mathcal{K}_{j+1} \longrightarrow \mathcal{E}_j \longrightarrow \mathcal{K}_j \longrightarrow 0, \quad 0 \leq j < n, \quad (2)$$

with each $\mathcal{E}_j$ locally free and each $\mathcal{K}_j$ coherent. Dimension shifting in (2) gives, for every $\mathcal{G}$,

$$\mathcal{E}xt_X^1(\mathcal{K}_n, \mathcal{G}) \cong \mathcal{E}xt_X^{n+1}(\mathcal{F}, \mathcal{G}) = 0.$$

Part (a) shows that $\mathcal{K}_n$ is locally free. Splicing the sequences (2) therefore gives a locally free resolution of $\mathcal{F}$ of length at most n . The case $n = 0$ is exactly part (a).

(c) Any locally free resolution of $\mathcal{F}$ remains a free resolution after taking a stalk. Therefore

$$\sup_{x \in X} \text{pd}_{\mathcal{O}_{X,x}} \mathcal{F}_x \leq \text{hd}(\mathcal{F}). \quad (3)$$

Let the supremum on the left be $d < \infty$. For a coherent first argument, sheaf Ext has the stalk formula

$$(\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}))_x \cong \text{Ext}_{\mathcal{O}_{X,x}}^i(\mathcal{F}_x, \mathcal{G}_x). \quad (4)$$

If $i > d$, the right side of (4) vanishes for every x and every $\mathcal{G}$. Hence the sheaf on the left is zero. Part (b) gives $\text{hd}(\mathcal{F}) \leq d$, which together with (3) proves equality. If the supremum is infinite, (3) already proves the assertion. $\square$

Exercise III.6.6

Permitted context.

Let $(A, \mathfrak{m})$ be a regular local ring and let M be a finite A -module. We may use that A has finite global dimension

$$\text{gldim } A = \dim A.$$

In particular, every finite A -module has finite projective dimension.

Problem. (a) Prove that M is projective if and only if

$$\text{Ext}_A^i(M, A) = 0 \quad (i > 0).$$

(b) For every $n \geq 0$, prove that

$$\text{pd}_A M \leq n \iff \text{Ext}_A^i(M, A) = 0 \quad (i > n).$$

Solution. (a) The forward implication is immediate. Conversely, choose a finite free presentation

$$0 \longrightarrow N \longrightarrow A^r \longrightarrow M \longrightarrow 0. \quad (1)$$

It is enough to prove that (1) splits, or equivalently that $\text{Ext}_A^1(M, N) = 0$. Resolve N by finite free modules, writing

$$0 \longrightarrow N_{j+1} \longrightarrow A^{r_j} \longrightarrow N_j \longrightarrow 0, \quad N_0 = N. \quad (2)$$

The hypothesis gives $\text{Ext}_A^i(M, A^{r_j}) = 0$ for every $i > 0$. The long exact sequences from (2) therefore give isomorphisms

$$\text{Ext}_A^1(M, N) \cong \text{Ext}_A^2(M, N_1) \cong \cdots \cong \text{Ext}_A^{d+1}(M, N_d),$$

where $d = \dim A$. The last group is zero because $\text{pd}_A M \leq d$. Thus (1) splits and M is projective. Since A is local, it is in fact free.

(b) If $\text{pd}_A M \leq n$, the asserted Ext vanishing follows by applying $\text{Hom}_A(-, A)$ to a projective resolution of length n .

Conversely, take the first n steps of a finite free resolution:

$$0 \longrightarrow K_n \longrightarrow F_{n-1} \longrightarrow \cdots \longrightarrow F_0 \longrightarrow M \longrightarrow 0. \quad (3)$$

For $j > 0$, dimension shifting in (3) gives

$$\mathrm{Ext}_A^j(K_n, A) \cong \mathrm{Ext}_A^{j+n}(M, A) = 0.$$

Part (a) shows that K_n is projective. Thus (3) is a projective resolution of length n , so $\mathrm{pd}_A M \leq n$. For $n = 0$, this is precisely part (a). $\square$

Exercise III.6.7

Permitted context.

Let $X = \mathrm{Spec} A$ be noetherian. Let M, N be A -modules, with M finite, and let $\widetilde{M}, \widetilde{N}$ be their associated quasi-coherent sheaves.

Problem. Prove naturally, for every $i \geq 0$,

$$\mathrm{Ext}_X^i(\widetilde{M}, \widetilde{N}) \cong \mathrm{Ext}_A^i(M, N)$$

and

$$\mathcal{E}xt_X^i(\widetilde{M}, \widetilde{N}) \cong \widetilde{\mathrm{Ext}_A^i(M, N)}.$$

Solution. Because A is noetherian and M is finite, there is a free resolution

$$\cdots \longrightarrow P_2 \longrightarrow P_1 \longrightarrow P_0 \longrightarrow M \longrightarrow 0 \quad (1)$$

with every P_j finite free. Sheafifying (1) gives a locally free resolution of $\widetilde{M}$. Applying sheaf Hom into $\widetilde{N}$ produces

$$\mathcal{H}om_X(\widetilde{P}_j, \widetilde{N}) \cong \widetilde{\mathrm{Hom}_A(P_j, N)} \quad (2)$$

in every degree. Sheafification is exact, so it commutes with the cohomology of this complex. Therefore

$$\mathcal{E}xt_X^i(\widetilde{M}, \widetilde{N}) \cong H^i(\widetilde{\mathrm{Hom}_A(P_\bullet, N)}) = \widetilde{\mathrm{Ext}_A^i(M, N)}. \quad (3)$$

Now use the local-to-global Ext spectral sequence

$$E_2^{p,q} = H^p(X, \mathcal{E}xt_X^q(\widetilde{M}, \widetilde{N})) \implies \mathrm{Ext}_X^{p+q}(\widetilde{M}, \widetilde{N}). \quad (4)$$

The sheaves in (3) are quasi-coherent, and higher cohomology of a quasi-coherent sheaf on the affine scheme X vanishes. Thus (4) has only its $p = 0$ column and degenerates. Taking global sections in (3) gives

$$\mathrm{Ext}_X^i(\widetilde{M}, \widetilde{N}) \cong \Gamma\left(X, \widetilde{\mathrm{Ext}_A^i(M, N)}\right) \cong \mathrm{Ext}_A^i(M, N).$$

Every construction used above is functorial in M and N , so the isomorphisms are natural. $\square$

Exercise III.6.8**Permitted context.**

Let X be a noetherian, integral, separated, locally factorial scheme. If $\mathcal{L}$ is invertible and $s \in \Gamma(X, \mathcal{L})$, write

$$X_s = \{x \in X : s_x \text{ generates } \mathcal{L}_x\}.$$

This is the complement of the zero locus of s .

Problem. (a) Prove that the opens X_s , as $\mathcal{L}$ and s vary, form a basis for the topology of X .
 (b) Deduce that every coherent sheaf on X is a quotient of a locally free sheaf of finite rank.

Solution. (a) We first prove a lemma: if $W \subseteq X$ is a nonempty affine open, every irreducible component of $X \setminus W$ has codimension one.

Let Z be such a component, with generic point η , and choose an affine neighborhood V of η . Because X is separated, $V \cap W$ is affine. Put $R = \mathcal{O}_{X, \eta}$, with maximal ideal $\mathfrak{m}$. The inverse image of $V \cap W$ in $\text{Spec } R$ is

$$\text{Spec } R \setminus \{\mathfrak{m}\}. \quad (1)$$

Indeed, a point of the local spectrum other than its closed point corresponds to a proper generalization of η . If that generalization also lay in $X \setminus W$, its closure would be an irreducible subset of $X \setminus W$ strictly containing Z , contrary to the maximality of Z . The open in (1) is affine, because it is obtained by base change from the affine morphism $V \cap W \rightarrow V$.

The local ring R is a unique factorization domain, hence normal. If $\dim R \geq 2$, the Hartogs property for a noetherian normal local domain gives

$$\Gamma(\text{Spec } R \setminus \{\mathfrak{m}\}, \mathcal{O}) = R. \quad (2)$$

If the punctured spectrum were affine, (2) would identify it with $\text{Spec } R$; under this identification its open immersion into $\text{Spec } R$ would be induced by the identity of R , an impossibility because the closed point is missing. Hence $\dim R < 2$. The point η is not the generic point of X , since every nonempty open contains that point, so $\dim R = 1$. This proves the lemma.

Now let $x \in U$, with U open. Choose an affine open W such that $x \in W \subseteq U$. The closed set $X \setminus W$ has finitely many irreducible components $D_1, \dots, D_r$, and the lemma says that each D_j is a prime Weil divisor. Since X is locally factorial, every D_j is Cartier. Thus

$$D = D_1 + \dots + D_r$$

is an effective Cartier divisor. Its canonical section $s_D \in \Gamma(X, \mathcal{O}_X(D))$ has zero locus $|D| = X \setminus W$, and consequently

$$X_{s_D} = W.$$

This gives $x \in X_{s_D} \subseteq U$, proving the basis assertion.

(b) Let $\mathcal{F}$ be coherent. By part (a), noetherian quasi-compactness, and coherence, choose finitely many affine basis opens

$$U_i = X_{s_i}, \quad s_i \in \Gamma(X, \mathcal{L}_i),$$

together with finitely many sections $t_{ij} \in \Gamma(U_i, \mathcal{F})$ that generate $\mathcal{F}|_{U_i}$. The extension lemma for sections says that, after replacing an exponent by a sufficiently large integer n_{ij} , the section

$$s_i^{n_{ij}} t_{ij} \in \Gamma(U_i, \mathcal{F} \otimes \mathcal{L}_i^{n_{ij}})$$

extends to a global section. It defines a morphism

$$\mathcal{L}_i^{-n_{ij}} \longrightarrow \mathcal{F}. \quad (3)$$

On U_i , multiplication by $s_i^{n_{ij}}$ is a unit, so the images of the maps (3) generate exactly when the t_{ij} do. Therefore the combined map

$$\bigoplus_{i,j} \mathcal{L}_i^{-n_{ij}} \longrightarrow \mathcal{F}$$

is surjective. Its source is a finite direct sum of invertible sheaves, hence is locally free of finite rank. $\square$

Exercise III.6.9

Permitted context.

Let X be a noetherian, integral, separated, regular scheme. Let $K(X)$ be the Grothendieck group of coherent sheaves. Define $K_1(X)$ from finite-rank locally free sheaves, with the relations

$$[\mathcal{E}]_1 = [\mathcal{E}']_1 + [\mathcal{E}'']_1$$

for every short exact sequence $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$ of locally free sheaves. There is a natural homomorphism

$$\epsilon : K_1(X) \longrightarrow K(X).$$

The notation $K_1(X)$ here is the book's group of vector bundles with exact-sequence relations, usually denoted K_0 of vector bundles; it is not higher algebraic K_1 .

Problem. (a) Prove that every coherent sheaf on X has a finite locally free resolution.
(b) If

$$0 \rightarrow \mathcal{E}_n \rightarrow \cdots \rightarrow \mathcal{E}_0 \rightarrow \mathcal{F} \rightarrow 0$$

is such a resolution, set

$$\delta(\mathcal{F}) = \sum_{i=0}^n (-1)^i [\mathcal{E}_i]_1.$$

Prove that this class is independent of the resolution, is additive in short exact sequences of coherent sheaves, and induces an inverse $\delta : K(X) \rightarrow K_1(X)$ to ϵ .

Solution. (a) A regular local ring is factorial, so X is locally factorial. Exercise III.6.8 therefore shows that coherent sheaves on X have enough locally free sheaves. Starting with a coherent $\mathcal{F} = \mathcal{K}_0$, construct exact sequences

$$0 \longrightarrow \mathcal{K}_{i+1} \longrightarrow \mathcal{E}_i \longrightarrow \mathcal{K}_i \longrightarrow 0 \quad (1)$$

with each $\mathcal{E}_i$ locally free of finite rank. All $\mathcal{K}_i$ are coherent.

Let

$$C_i = \{x \in X : (\mathcal{K}_i)_x \text{ is not a free } \mathcal{O}_{X,x}\text{-module}\}.$$

The locally free locus of a coherent sheaf is open, so C_i is closed. If $(\mathcal{K}_i)_x$ is free, the stalk of (1) splits and $(\mathcal{K}_{i+1})_x$ is a finite projective module over a local ring, hence free. Thus

$$C_0 \supseteq C_1 \supseteq C_2 \supseteq \cdots \quad (2)$$

At every x , the regular local ring $\mathcal{O}_{X,x}$ has finite global dimension. The stalk resolution therefore reaches a free syzygy, so $x \notin C_i$ for some i . Hence $\bigcap_i C_i = \emptyset$. Since the topological space of X is noetherian, the descending chain (2) stabilizes; its stable value must be empty. Thus $\mathcal{K}_N$ is locally free for some N , and (1) splices to the required finite resolution.

(b) We first record that a bounded exact complex of locally free sheaves has alternating class zero in $K_1(X)$. Starting at one end, its cycle sheaves are locally free: each is the kernel of a surjection of locally free sheaves onto a locally free cycle, and that sequence splits locally. The short exact sequences involving successive cycles then make all terms cancel in the alternating sum.

Next let $\mathcal{E}_\bullet \rightarrow \mathcal{F}$ and $\mathcal{E}'_\bullet \rightarrow \mathcal{F}$ be two finite locally free resolutions. They admit a common locally free refinement $\mathcal{A}_\bullet \rightarrow \mathcal{F}$ with augmentation-preserving chain maps

$$\mathcal{A}_\bullet \longrightarrow \mathcal{E}_\bullet, \quad \mathcal{A}_\bullet \longrightarrow \mathcal{E}'_\bullet. \quad (3)$$

Here is the construction. In degree zero, choose a locally free sheaf mapping onto the coherent fiber product $\mathcal{E}_0 \times_{\mathcal{F}} \mathcal{E}'_0$. Inductively, after the lower degrees have been constructed, take the fiber product of the two lifting problems over the current syzygy and choose a locally free sheaf surjecting onto it. This simultaneously continues the resolution and both chain maps. Part (a), applied to the coherent syzygies through the same descending nonfree-locus argument, shows that the current syzygy is locally free after finitely many stages. Continue until both original finite resolutions have also ended, and only then use the syzygy as the final term. At this degree its required maps to the original resolutions are zero, and the chain identities already follow from exactness there. This gives a finite truncation without assuming global lifting from a locally free sheaf into an unfinished resolution.

Each map in (3) is a quasi-isomorphism, so its mapping cone is a bounded exact complex of locally free sheaves. The preceding observation applied to the two cones gives

$$\sum_i (-1)^i [\mathcal{E}_i]_1 = \sum_i (-1)^i [\mathcal{A}_i]_1 = \sum_i (-1)^i [\mathcal{E}'_i]_1. \quad (4)$$

Thus $\delta(\mathcal{F})$ is independent of all choices.

Now take a short exact sequence

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0. \quad (5)$$

A fiber-product version of the horseshoe construction gives bounded locally free resolutions in a termwise short exact sequence

$$0 \longrightarrow \mathcal{E}'_{\bullet} \longrightarrow \mathcal{E}_{\bullet} \longrightarrow \mathcal{E}''_{\bullet} \longrightarrow 0 \quad (6)$$

resolving (5). For completeness, in degree zero choose a locally free $\mathcal{E}''_0 \twoheadrightarrow \mathcal{F}''$, form $\mathcal{F} \times_{\mathcal{F}''} \mathcal{E}''_0$, and choose a locally free $\mathcal{E}_0$ surjecting onto that fiber product. Then $\mathcal{E}_0 \rightarrow \mathcal{E}''_0$ is surjective and its kernel $\mathcal{E}'_0$ is locally free; the induced map $\mathcal{E}'_0 \rightarrow \mathcal{F}'$ is surjective. The snake lemma gives a new short exact sequence of syzygies, and the construction repeats. The nonfree-locus argument from part (a), applied simultaneously to the three syzygies, makes the resulting resolutions finite.

Each degree of (6) gives

$$[\mathcal{E}_i]_1 = [\mathcal{E}'_i]_1 + [\mathcal{E}''_i]_1.$$

Taking alternating sums and using (4) yields

$$\delta(\mathcal{F}) = \delta(\mathcal{F}') + \delta(\mathcal{F}'').$$

Therefore δ descends to a homomorphism $K(X) \rightarrow K_1(X)$.

If $\mathcal{E}$ is locally free, its length-zero resolution gives $\delta(\epsilon[\mathcal{E}]_1) = [\mathcal{E}]_1$, so $\delta \circ \epsilon$ is the identity. Conversely, exactness of a finite resolution gives in $K(X)$

$$[\mathcal{F}] = \sum_i (-1)^i [\mathcal{E}_i].$$

Hence $\epsilon \circ \delta$ is also the identity, and ϵ is an isomorphism. □

Exercise III.6.10**Permitted context.**

Let $f : X \rightarrow Y$ be a finite morphism of noetherian schemes, and put

$$\mathcal{B} = f_* \mathcal{O}_X.$$

This is a coherent $\mathcal{O}_Y$ -algebra. Quasi-coherent $\mathcal{O}_X$ -modules are equivalent to quasi-coherent $\mathcal{B}$ -modules on Y . The derived comparison used here is the equivalence between the derived category of quasi-coherent modules and the subcategory of complexes with quasi-coherent cohomology on a noetherian scheme (Stacks Project, Proposition 36.8.3, tag 09T1). Thus one may resolve inside quasi-coherent modules and compare Ext afterwards.

Problem. (a) For a quasi-coherent $\mathcal{O}_Y$ -module $\mathcal{G}$, show that

$$\mathcal{H}om_Y(\mathcal{B}, \mathcal{G})$$

is a quasi-coherent $\mathcal{B}$ -module. Denote the corresponding quasi-coherent sheaf on X by $f^! \mathcal{G}$.

(b) If $\mathcal{F}$ is coherent on X , prove the natural isomorphism

$$f_* \mathcal{H}om_X(\mathcal{F}, f^! \mathcal{G}) \xrightarrow{\sim} \mathcal{H}om_Y(f_* \mathcal{F}, \mathcal{G}).$$

(c) Construct natural maps, for $i \geq 0$,

$$\varphi_i : \mathrm{Ext}_X^i(\mathcal{F}, f^! \mathcal{G}) \longrightarrow \mathrm{Ext}_Y^i(f_* \mathcal{F}, \mathcal{G}).$$

(d) Assume in addition that X and Y are separated, that $\mathrm{Coh}(X)$ has enough locally free sheaves, and that $\mathcal{B}$ is locally free on Y , equivalently that f is flat. Prove that every φ_i is an isomorphism.

Solution. (a) Work over an affine open $U = \mathrm{Spec} A \subseteq Y$, and write $f^{-1}U = \mathrm{Spec} B$ and $\mathcal{G}|_U = \tilde{N}$. The A -module B is finite and therefore finitely presented. Consequently localization commutes with $\mathrm{Hom}_A(B, N)$, and

$$\mathcal{H}om_Y(\mathcal{B}, \mathcal{G})|_U \cong \widetilde{\mathrm{Hom}_A(B, N)}.$$

This proves quasi-coherence. Give this Hom module the B -action

$$(b \cdot \lambda)(b') = \lambda(bb'). \quad (1)$$

These local actions glue to a $\mathcal{B}$ -module structure, which corresponds to the asserted sheaf $f^! \mathcal{G}$ on X .

(b) The assertion is the sheaf form of coinduction adjunction. For an A -algebra B , a B -module M , and an A -module N , there is a natural isomorphism

$$\mathrm{Hom}_B(M, \mathrm{Hom}_A(B, N)) \xrightarrow{\sim} \mathrm{Hom}_A(M, N). \quad (2)$$

It sends ψ to $m \mapsto \psi(m)(1)$; its inverse sends $h : M \rightarrow N$ to

$$m \longmapsto (b \longmapsto h(bm)).$$

Applying (2) over every affine open $U \subseteq Y$, with $M = \Gamma(f^{-1}U, \mathcal{F})$, gives compatible isomorphisms on sections. They sheafify to

$$f_* \mathcal{H}om_X(\mathcal{F}, f^! \mathcal{G}) \cong \mathcal{H}om_Y(f_* \mathcal{F}, \mathcal{G}). \quad (3)$$

(c) Resolve in the category of quasi-coherent modules and use the derived comparison from the context to identify its Ext groups with the sheaf Ext groups in the problem. Since f is affine,

$$f_* : \mathbf{Qco}(X) \longrightarrow \mathbf{Qco}(Y)$$

is exact, and (3) says that $f^!$ is its right adjoint. In particular, $f^!$ carries injective objects of $\mathbf{Qco}(Y)$ to injective objects of $\mathbf{Qco}(X)$. The Grothendieck spectral sequence for

$$\mathrm{Hom}_X(\mathcal{F}, -) \circ f^! \cong \mathrm{Hom}_Y(f_* \mathcal{F}, -)$$

is

$$E_2^{p,q} = \mathrm{Ext}_X^p(\mathcal{F}, R^q f^! \mathcal{G}) \implies \mathrm{Ext}_Y^{p+q}(f_* \mathcal{F}, \mathcal{G}). \quad (4)$$

The edge morphism from the bottom row of (4) is

$$E_2^{i,0} = \mathrm{Ext}_X^i(\mathcal{F}, f^! \mathcal{G}) \longrightarrow \mathrm{Ext}_Y^i(f_* \mathcal{F}, \mathcal{G}).$$

Define this to be φ_i . The spectral-sequence construction makes it natural, and for $i = 0$ it is exactly the adjunction (3) on global Hom groups.

(d) If $\mathcal{B}$ is locally free of finite rank, then

$$f^! \mathcal{G} \quad \leftrightarrow \quad \mathcal{H}om_Y(\mathcal{B}, \mathcal{G}) \cong \mathcal{B}^\vee \otimes \mathcal{G}. \quad (5)$$

Thus $f^!$ is exact and $R^q f^! = 0$ for $q > 0$. The spectral sequence (4) has only its bottom row, so it degenerates and every edge map φ_i is an isomorphism.

Equivalently, if $\mathcal{G} \rightarrow \mathcal{I}^\bullet$ is an injective resolution in $\mathbf{Qco}(Y)$, exactness of $f^!$ and the adjunction with the exact functor f_* make $f^! \mathcal{I}^\bullet$ an injective resolution of $f^! \mathcal{G}$. Termwise adjunction then gives

$$\mathrm{Hom}_X(\mathcal{F}, f^! \mathcal{I}^\bullet) \cong \mathrm{Hom}_Y(f_* \mathcal{F}, \mathcal{I}^\bullet),$$

whose cohomology is the desired isomorphism. In fact, this exact adjunction argument proves a slightly stronger statement; the extra separatedness and enough-locally-free hypotheses ensure the standard comparison framework used in the exercise but are not otherwise needed once (4) is available. $\square$

III.7 The Serre Duality Theorem

Exercise III.7.1

Permitted context.

Let X be an integral projective scheme of dimension at least one over a field k , and let $\mathcal{L}$ be an ample invertible sheaf on X .

Problem. Prove that

$$H^0(X, \mathcal{L}^{-1}) = 0.$$

Solution. Suppose that $0 \neq s \in H^0(X, \mathcal{L}^{-1})$. Because X is integral, the corresponding map

$$\mathcal{O}_X \xrightarrow{s} \mathcal{L}^{-1}$$

is injective: after trivializing $\mathcal{L}^{-1}$, it is multiplication by a nonzero element of a domain. Tensoring by $\mathcal{L}$ gives an injection $\mathcal{L} \hookrightarrow \mathcal{O}_X$, and taking tensor powers gives

$$\mathcal{L}^r \hookrightarrow \mathcal{O}_X \quad (r \geq 1). \tag{1}$$

Thus

$$\dim_k H^0(X, \mathcal{L}^r) \leq \dim_k H^0(X, \mathcal{O}_X) \tag{2}$$

for every r . The number on the right is finite because X is projective.

Choose $m > 0$ such that $\mathcal{L}^m$ is very ample. For $r \gg 0$, the dimension of $H^0(X, \mathcal{L}^{mr})$ is the value of the Hilbert polynomial of X in this embedding. That polynomial has degree $\dim X \geq 1$ and positive leading coefficient, so these dimensions are unbounded as $r \rightarrow \infty$. This contradicts the uniform bound (2). Hence no nonzero section of $\mathcal{L}^{-1}$ exists. $\square$

Exercise III.7.2**Permitted context.**

Let $f : X \rightarrow Y$ be a finite morphism of projective k -schemes, both of dimension n . Let ω_Y° be a dualizing sheaf on Y , with trace

$$t_Y : H^n(Y, \omega_Y^\circ) \longrightarrow k.$$

Use the functor $f^!$ from Exercise III.6.10.

Problem. (a) Prove that $f^!\omega_Y^\circ$ is a dualizing sheaf on X .

(b) If X and Y are nonsingular and k is algebraically closed, construct the natural trace morphism

$$\mathrm{Tr}_f : f_*\omega_X \longrightarrow \omega_Y.$$

Solution. (a) For every coherent $\mathcal{F}$ on X , finite duality adjunction and Serre duality on Y give natural isomorphisms

$$\begin{aligned} \mathrm{Hom}_X(\mathcal{F}, f^!\omega_Y^\circ) &\cong \mathrm{Hom}_Y(f_*\mathcal{F}, \omega_Y^\circ) \\ &\cong H^n(Y, f_*\mathcal{F})^\vee. \end{aligned} \tag{1}$$

Since f is finite, it is affine and has no higher direct images on quasi-coherent sheaves. The Leray spectral sequence therefore gives

$$H^n(Y, f_*\mathcal{F}) \cong H^n(X, \mathcal{F}). \tag{2}$$

Combining (1) and (2), we obtain functorially

$$\mathrm{Hom}_X(\mathcal{F}, f^!\omega_Y^\circ) \cong H^n(X, \mathcal{F})^\vee. \tag{3}$$

Let $t_X \in H^n(X, f^!\omega_Y^\circ)^\vee$ be the functional corresponding under (3), with $\mathcal{F} = f^!\omega_Y^\circ$, to its identity endomorphism. Naturality of (3) says that a map $\alpha : \mathcal{F} \rightarrow f^!\omega_Y^\circ$ corresponds exactly to

$$H^n(X, \mathcal{F}) \xrightarrow{H^n(\alpha)} H^n(X, f^!\omega_Y^\circ) \xrightarrow{t_X} k.$$

Thus the pair $(f^!\omega_Y^\circ, t_X)$ has the defining representing property of a dualizing sheaf on X .

(b) On a nonsingular projective n -fold, the canonical sheaf $\omega = \Omega^n$, with its trace, is dualizing. By the uniqueness of a representing object with trace, part (a) gives a canonical trace-preserving isomorphism

$$\omega_X \xrightarrow{\sim} f^!\omega_Y. \tag{4}$$

The counit of the adjunction $f_* \dashv f^!$ is

$$f_*f^!\omega_Y \longrightarrow \omega_Y. \tag{5}$$

Affine-locally, for a finite map $A \rightarrow B$, it is the evaluation map

$$\mathrm{Hom}_A(B, \omega_A) \longrightarrow \omega_A, \quad \lambda \longmapsto \lambda(1).$$

Composing f_* of (4) with (5) gives the natural morphism

$$\mathrm{Tr}_f : f_*\omega_X \longrightarrow \omega_Y.$$

□

Exercise III.7.3

Permitted context.

Let $X = \mathbb{P}_k^n$, and write $\Omega_X^p = \bigwedge^p \Omega_X^1$. The Euler sequence is

$$0 \longrightarrow \Omega_X^1 \longrightarrow \mathcal{O}_X(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Problem. For $0 \leq p, q \leq n$, prove

$$H^q(X, \Omega_X^p) \cong \begin{cases} k, & p = q, \\ 0, & p \neq q. \end{cases}$$

Solution. Because the quotient in the Euler sequence has rank one, its p -th exterior-power sequence is short exact:

$$0 \longrightarrow \Omega_X^p \longrightarrow \bigwedge^p \left(\mathcal{O}_X(-1)^{\oplus(n+1)} \right) \longrightarrow \Omega_X^{p-1} \longrightarrow 0.$$

Equivalently,

$$0 \longrightarrow \Omega_X^p \longrightarrow \mathcal{O}_X(-p)^{\oplus \binom{n+1}{p}} \longrightarrow \Omega_X^{p-1} \longrightarrow 0. \quad (1)$$

For $1 \leq p \leq n$, the line bundle $\mathcal{O}_X(-p)$ is acyclic. Indeed it has no global sections, all its intermediate cohomology vanishes, and by Serre duality

$$H^n(X, \mathcal{O}_X(-p)) \cong H^0(X, \mathcal{O}_X(p-n-1))^\vee = 0.$$

The long exact sequence of (1) consequently gives

$$H^q(X, \Omega_X^p) \cong H^{q-1}(X, \Omega_X^{p-1}) \quad (2)$$

with the convention that a negative cohomology group is zero. The initial case is

$$H^q(X, \Omega_X^0) = H^q(X, \mathcal{O}_X) \cong \begin{cases} k, & q = 0, \\ 0, & q > 0. \end{cases}$$

Iterating (2) proves that the only nonzero group occurs when $q = p$, and in that case it is k . □

Exercise III.7.4**Permitted context.**

Let X be a nonsingular projective variety of dimension n over an algebraically closed field k , and let $i : Y \hookrightarrow X$ be a nonsingular subvariety of codimension p . Restriction of differential forms and the trace on Y give a functional

$$H^{n-p}(X, \Omega_X^{n-p}) \xrightarrow{i^*} H^{n-p}(Y, \omega_Y) \xrightarrow{t_Y} k.$$

Under the Serre-duality pairing

$$H^p(X, \Omega_X^p) \times H^{n-p}(X, \Omega_X^{n-p}) \xrightarrow{\smile, \wedge} H^n(X, \omega_X) \xrightarrow{t_X} k,$$

this functional corresponds to a unique class $\eta(Y) \in H^p(X, \Omega_X^p)$. Thus

$$t_X(\eta(Y) \smile \alpha) = t_Y(i^* \alpha) \quad (1)$$

for every $\alpha \in H^{n-p}(X, \Omega_X^{n-p})$.

Problem. (a) If $P \in X$ is a closed point, prove

$$t_X(\eta(P)) = 1.$$

(b) If $X = \mathbb{P}_k^n$, use the identification $H^p(X, \Omega_X^p) \cong k$ from Exercise III.7.3, normalized so that $c(\mathcal{O}(1))^p$ maps to 1, to prove

$$\eta(Y) = (\deg Y) \cdot 1.$$

(c) For a finite-type k -scheme Z , show that

$$d\log : \mathcal{O}_Z^\times \longrightarrow \Omega_Z^1, \quad f \longmapsto f^{-1}df,$$

is a homomorphism of sheaves of abelian groups. Denote the induced homomorphism

$$c : \text{Pic } Z = H^1(Z, \mathcal{O}_Z^\times) \longrightarrow H^1(Z, \Omega_Z^1)$$

by c .

(d) If $p = 1$, prove

$$\eta(Y) = c(\mathcal{O}_X(Y)).$$

Solution. (a) Here $p = n$, so the functional defining $\eta(P)$ is

$$H^0(X, \mathcal{O}_X) \longrightarrow H^0(P, \mathcal{O}_P) \xrightarrow{t_P} k.$$

Because k is algebraically closed, $P = \text{Spec } k$ and t_P is the identity. Taking $\alpha = 1$ in (1) gives

$$t_X(\eta(P)) = t_P(1) = 1.$$

(c) The Leibniz rule gives

$$d\log(fg) = (fg)^{-1}(f dg + g df) = f^{-1}df + g^{-1}dg.$$

The construction commutes with restriction, so $d\log$ is a homomorphism of sheaves from the multiplicative group $\mathcal{O}_Z^\times$ to the additive group Ω_Z^1 . Applying H^1 gives c . If an invertible sheaf has transition functions g_{ij} , its class c is represented by the Čech cocycle $d\log(g_{ij})$. This description also shows that c is additive under tensor products and compatible with pullback.

(d) Let Y be a nonsingular effective Cartier divisor. We use the residue sequence

$$0 \longrightarrow \omega_X \longrightarrow \omega_X(Y) \xrightarrow{\text{res}} i_*\omega_Y \longrightarrow 0. \quad (2)$$

Write

$$\partial : H^{n-1}(Y, \omega_Y) \longrightarrow H^n(X, \omega_X) \quad (3)$$

for its connecting map. The residue normalization of the trace satisfies

$$t_X \circ \partial = t_Y. \quad (4)$$

There is also the standard divisor identity

$$\partial(i^*\alpha) = c(\mathcal{O}_X(Y)) \smile \alpha \quad (\alpha \in H^{n-1}(X, \Omega_X^{n-1})). \quad (5)$$

To see (5), choose local equations f_i for Y . The transition functions of $\mathcal{O}_X(Y)$ are, with compatible frame conventions, $g_{ij} = f_i/f_j$, so its c -class is represented by

$$d\log(g_{ij}) = \frac{df_i}{f_i} - \frac{df_j}{f_j}.$$

Local logarithmic lifts in (2) have exactly these differences. Computing the boundary in Čech cohomology therefore gives cup product with this cocycle, which is (5).

Combining (4) and (5), for every α we obtain

$$t_X(c(\mathcal{O}_X(Y)) \smile \alpha) = t_Y(i^*\alpha).$$

Comparison with (1), followed by perfection of the Serre-duality pairing, proves $\eta(Y) = c(\mathcal{O}_X(Y))$.

(b) Put

$$h = c(\mathcal{O}_{\mathbb{P}^n}(1)) \in H^1(\mathbb{P}^n, \Omega^1).$$

Under the recursive isomorphisms of Exercise III.7.3, h^r is the element $1 \in H^r(\mathbb{P}^n, \Omega^r) \cong k$. One way to fix the normalization is to choose a transverse flag of hyperplanes. Part (d), iterated through the residue sequences, gives

$$h^r = \eta(\mathbb{P}^{n-r}). \quad (6)$$

For $r = n$, part (a) consequently gives

$$t_{\mathbb{P}^n}(h^n) = 1. \quad (7)$$

We use the same residue calculation in the following general form: if smooth divisors $D_1, \dots, D_r$ meet transversely on a smooth projective variety, then

$$\eta(D_1 \cap \cdots \cap D_r) = \prod_{j=1}^r c(\mathcal{O}(D_j)). \quad (8)$$

Indeed, (8) follows by iterating (5), or equivalently by the projection formula for the successive residue maps.

Let $m = \dim Y = n - p$. Since $H^p(\mathbb{P}^n, \Omega^p) = kh^p$, write

$$\eta(Y) = ah^p.$$

Pairing with h^m and using (1) and (7) gives

$$a = t_{\mathbb{P}^n}(\eta(Y) \smile h^m) = t_Y((i^*h)^m). \quad (9)$$

Choose m general hyperplanes meeting Y transversely. Their intersection with Y is a reduced zero-dimensional scheme consisting of $\deg Y$ points. By (8) on Y , $(i^*h)^m$ is the sum of the cohomology classes of those points. Part (a) says that the trace of each point class is 1, so the right side of (9) is $\deg Y$. Hence

$$\eta(Y) = (\deg Y)h^p = (\deg Y) \cdot 1$$

under the identification of Exercise III.7.3. □

III.8 Higher Direct Images of Sheaves

Exercise III.8.1

Permitted context.

Let $f : X \rightarrow Y$ be a continuous map. Direct image is a left-exact functor on sheaves of abelian groups, and its right-derived functors are denoted by $R^q f_*$. Injective sheaves are flasque, direct image preserves flasque sheaves, and flasque resolutions compute sheaf cohomology.

Problem. Let $\mathcal{F}$ be a sheaf of abelian groups on X , and suppose

$$R^q f_* \mathcal{F} = 0 \quad (q > 0).$$

Prove that, naturally in $\mathcal{F}$,

$$H^i(X, \mathcal{F}) \cong H^i(Y, f_* \mathcal{F}) \quad (i \geq 0).$$

Solution. Choose an injective resolution

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I}^0 \longrightarrow \mathcal{I}^1 \longrightarrow \cdots. \quad (1)$$

Set $\mathcal{K}^0 = \mathcal{F}$, and for $q \geq 0$ let $\mathcal{K}^{q+1}$ be the cokernel of $\mathcal{K}^q \hookrightarrow \mathcal{I}^q$. Thus there are short exact sequences

$$0 \longrightarrow \mathcal{K}^q \longrightarrow \mathcal{I}^q \longrightarrow \mathcal{K}^{q+1} \longrightarrow 0. \quad (2)$$

Because $\mathcal{I}^q$ is f_* -acyclic, the long exact sequence of higher direct images associated to (2) gives

$$R^1 f_* \mathcal{K}^q \cong R^{q+1} f_* \mathcal{F}. \quad (3)$$

Indeed, this follows by dimension shifting, starting with $\mathcal{K}^0 = \mathcal{F}$. The hypothesis makes the right-hand side of (3) zero. The beginning of the same long exact sequence is therefore

$$0 \longrightarrow f_* \mathcal{K}^q \longrightarrow f_* \mathcal{I}^q \longrightarrow f_* \mathcal{K}^{q+1} \longrightarrow 0.$$

Consequently, applying f_* to (1) produces an exact resolution

$$0 \longrightarrow f_* \mathcal{F} \longrightarrow f_* \mathcal{I}^0 \longrightarrow f_* \mathcal{I}^1 \longrightarrow \cdots. \quad (4)$$

Every $f_* \mathcal{I}^q$ is flasque: if $V \subseteq U \subseteq Y$, its restriction map is the restriction map

$$\mathcal{I}^q(f^{-1}U) \longrightarrow \mathcal{I}^q(f^{-1}V),$$

which is surjective. Thus (4) is a flasque resolution and computes $H^i(Y, f_* \mathcal{F})$. On the other hand, (1) computes $H^i(X, \mathcal{F})$, and in every degree

$$\Gamma(Y, f_* \mathcal{I}^q) = \Gamma(X, \mathcal{I}^q).$$

The two cohomology groups are therefore the cohomology groups of the same complex. The comparison isomorphism is independent of the chosen resolution by the comparison theorem for resolutions, so it is natural in $\mathcal{F}$. $\square$

Exercise III.8.2

Permitted context.

Let $f : X \rightarrow Y$ be a morphism of schemes with X noetherian. For a quasi-coherent sheaf $\mathcal{F}$ on X , higher direct images may be computed locally on the target: on an open subset $U \subseteq Y$,

$$(R^i f_* \mathcal{F})|_U \cong R^i (f_U)_* (\mathcal{F}|_{f^{-1}U}),$$

where $f_U : f^{-1}U \rightarrow U$. Equivalently, on an affine-open basis they are obtained from the groups $H^i(f^{-1}U, \mathcal{F})$. Quasi-coherent sheaves have no positive cohomology on noetherian affine schemes.

Problem. Suppose that $f : X \rightarrow Y$ is affine, X is noetherian, and $\mathcal{F}$ is quasi-coherent. Prove

$$R^i f_* \mathcal{F} = 0 \quad (i > 0),$$

and deduce natural isomorphisms

$$H^i(X, \mathcal{F}) \cong H^i(Y, f_* \mathcal{F}) \quad (i \geq 0).$$

Solution. Let $U \subseteq Y$ be affine. Since f is affine, $f^{-1}U$ is affine; it is also noetherian because it is an open subscheme of the noetherian scheme X . The restriction of $\mathcal{F}$ to $f^{-1}U$ is quasi-coherent, so affine vanishing gives

$$H^i(f^{-1}U, \mathcal{F}) = 0 \quad (i > 0). \quad (1)$$

The same assertion holds after replacing U by any affine open subset of it. By the local description of higher direct images, (1) says that $(R^i f_* \mathcal{F})|_U = 0$ for every affine open $U \subseteq Y$. Such opens form a basis, hence

$$R^i f_* \mathcal{F} = 0 \quad (i > 0).$$

Exercise III.8.1 now applies and yields

$$H^i(X, \mathcal{F}) \cong H^i(Y, f_* \mathcal{F})$$

in every degree. Naturality follows from the naturality of both affine cohomology and the comparison in that exercise. $\square$

Exercise III.8.3

Permitted context.

Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. Higher direct images commute with restriction to an open subset of the target and with finite direct sums. If $\mathcal{E}$ is finite locally free, tensoring with $\mathcal{E}$ is exact, and the ordinary projection formula is

$$f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^* \mathcal{E}) \cong f_* \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{E}.$$

Problem. For an $\mathcal{O}_X$ -module $\mathcal{F}$ and a finite locally free $\mathcal{O}_Y$ -module $\mathcal{E}$, prove the higher projection formula

$$R^i f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^* \mathcal{E}) \cong R^i f_* \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{E} \quad (i \geq 0).$$

Solution. Choose an open subset $U \subseteq Y$ on which $\mathcal{E}|_U \cong \mathcal{O}_U^{\oplus r}$, and write $f_U : f^{-1}U \rightarrow U$. Then

$$(\mathcal{F} \otimes f^* \mathcal{E})|_{f^{-1}U} \cong (\mathcal{F}|_{f^{-1}U})^{\oplus r}.$$

Additivity of higher direct images now gives an isomorphism

$$R^i f_{U*}((\mathcal{F}|_{f^{-1}U})^{\oplus r}) \cong (R^i f_{U*}(\mathcal{F}|_{f^{-1}U}))^{\oplus r}. \quad (1)$$

Via the chosen frame, the left side of (1) is the restriction of $R^i f_*(\mathcal{F} \otimes f^* \mathcal{E})$, and the right side is the restriction of $R^i f_* \mathcal{F} \otimes \mathcal{E}$.

This local isomorphism does not depend on the frame. Indeed, a change of frame is represented by an invertible matrix with entries in $\mathcal{O}_U$. Functoriality and $\mathcal{O}_U$ -linearity of $R^i f_{U*}$ make that matrix act in exactly the same way on the two sides of (1). The local isomorphisms therefore agree on overlaps and glue to the required global isomorphism. For $i = 0$, the resulting map is precisely the ordinary projection formula. $\square$

Exercise III.8.4**Permitted context.**

Let Y be noetherian, let $\mathcal{E}$ be a locally free $\mathcal{O}_Y$ -module of rank $n+1$, where $n \geq 1$, and use the convention

$$X = \mathbb{P}(\mathcal{E}) = \operatorname{Proj}_Y(\operatorname{Sym} \mathcal{E}).$$

Write $\pi : X \rightarrow Y$ for the projection and $\mathcal{O}_X(1)$ for the tautological invertible sheaf. Thus the convention is normalized by $\pi_* \mathcal{O}_X(l) = \operatorname{Sym}^l \mathcal{E}$ for $l \geq 0$. Put $\mathcal{L} = \bigwedge^{n+1} \mathcal{E}$.

When arithmetic and geometric genera occur below, assume in addition that Y is a projective variety over a field k ; for the geometric-genus assertion assume that Y is nonsingular. Then X is also nonsingular and projective.

Problem. (a) Compute the direct images of the twists sufficiently to prove

$$\pi_* \mathcal{O}_X(l) = \begin{cases} \operatorname{Sym}^l \mathcal{E}, & l \geq 0, \\ 0, & l < 0, \end{cases}$$

$$R^i \pi_* \mathcal{O}_X(l) = 0 \quad (0 < i < n, l \in \mathbb{Z}), \quad R^n \pi_* \mathcal{O}_X(l) = 0 \quad (l > -n-1).$$

(b) Prove the relative Euler sequence

$$0 \longrightarrow \Omega_{X/Y} \longrightarrow \pi^* \mathcal{E} \otimes \mathcal{O}_X(-1) \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Deduce

$$\omega_{X/Y} := \bigwedge^n \Omega_{X/Y} \cong \pi^* \mathcal{L} \otimes \mathcal{O}_X(-n-1),$$

and construct a natural isomorphism $R^n \pi_* \omega_{X/Y} \cong \mathcal{O}_Y$.

(c) For every $l \in \mathbb{Z}$, prove

$$R^n \pi_* \mathcal{O}_X(l) \cong (\pi_* \mathcal{O}_X(-l-n-1))^\vee \otimes \mathcal{L}^\vee.$$

(d) Prove

$$p_a(X) = (-1)^n p_a(Y), \quad p_g(X) = 0.$$

(e) Let Y be a nonsingular projective curve of genus g , and let $\mathcal{E}$ have rank 2. Show that X is a projective surface with

$$p_a(X) = -g, \quad p_g(X) = 0, \quad q(X) = g.$$

Solution. (a) All assertions are local on Y . Take an affine open $U = \operatorname{Spec} A$ on which $\mathcal{E}|_U \cong \mathcal{O}_U^{\oplus(n+1)}$. Then

$$X_U := \pi^{-1}U \cong \mathbb{P}_A^n = \operatorname{Proj} A[x_0, \dots, x_n].$$

The standard affine cover $D_+(x_0), \dots, D_+(x_n)$ and its Čech complex give

$$H^q(\mathbb{P}_A^n, \mathcal{O}(l)) = 0 \quad (0 < q < n) \quad (1)$$

for every l , while

$$H^0(\mathbb{P}_A^n, \mathcal{O}(l)) = \begin{cases} A[x_0, \dots, x_n]_l, & l \geq 0, \\ 0, & l < 0. \end{cases} \quad (2)$$

The only possible remaining positive cohomology is in degree n . The same Čech calculation identifies it with the free A -module having basis

$$x_0^{-a_0} \cdots x_n^{-a_n}, \quad a_j \geq 1, \quad \sum_{j=0}^n a_j = -l. \quad (3)$$

In particular, this module is zero when $l \geq -n$, equivalently when $l > -n - 1$. These computations commute with localization in A , so they describe the restrictions of the higher direct-image sheaves to U . Equations (1)–(3) prove all the vanishing statements. Finally, (2) is compatible with change of frame in $\mathcal{E}$ and glues to

$$\pi_* \mathcal{O}_X(l) = \text{Sym}^l \mathcal{E}$$

for $l \geq 0$; for $l < 0$ it glues to zero.

(b) The tautological quotient

$$\pi^* \mathcal{E} \longrightarrow \mathcal{O}_X(1)$$

becomes, after tensoring by $\mathcal{O}_X(-1)$, a surjection

$$\pi^* \mathcal{E} \otimes \mathcal{O}_X(-1) \longrightarrow \mathcal{O}_X. \quad (4)$$

On every open set U used in part (a), (4) is the last map in the usual Euler sequence on $\mathbb{P}_A^n$; its kernel is $\Omega_{\mathbb{P}_A^n/A}$. Since (4) is intrinsic, these local kernel identifications agree on overlaps. This proves the asserted relative Euler sequence.

Taking determinants in that sequence gives

$$\begin{aligned} \omega_{X/Y} &\cong \det(\pi^* \mathcal{E} \otimes \mathcal{O}_X(-1)) \\ &\cong \pi^*(\det \mathcal{E}) \otimes \mathcal{O}_X(-n-1) \\ &= \pi^* \mathcal{L} \otimes \mathcal{O}_X(-n-1). \end{aligned} \quad (5)$$

There is a canonical local residue isomorphism

$$R^n \pi_* \mathcal{O}_X(-n-1) \cong \mathcal{L}^\vee. \quad (6)$$

To see both the isomorphism and its canonicity, choose a frame $x_0, \dots, x_n$ of $\mathcal{E}|_U$. Formula (3) says that the left side of (6) is freely generated by the Čech class

$$\left[\frac{1}{x_0 \cdots x_n} \right].$$

Pair it with $x_0 \wedge \cdots \wedge x_n$ by sending the displayed pair to 1. Under a change of frame the exterior product is multiplied by the determinant, while the top Čech class is multiplied by the inverse determinant modulo a coboundary. Hence the pairing is independent of the frame and glues to (6).

Using (5), the higher projection formula, and (6), we obtain

$$\begin{aligned} R^n \pi_* \omega_{X/Y} &\cong R^n \pi_* (\pi^* \mathcal{L} \otimes \mathcal{O}_X(-n-1)) \\ &\cong \mathcal{L} \otimes R^n \pi_* \mathcal{O}_X(-n-1) \\ &\cong \mathcal{L} \otimes \mathcal{L}^\vee \cong \mathcal{O}_Y. \end{aligned} \quad (7)$$

This is the natural relative trace isomorphism.

(c) Multiplication of twists, followed by the trace in (7), gives a pairing

$$\pi_* \mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_X(l), \omega_{X/Y}) \otimes R^n \pi_* \mathcal{O}_X(l) \longrightarrow \mathcal{O}_Y. \quad (8)$$

It is perfect. This may be checked on a trivializing affine open $U = \text{Spec } A$. Put $d = -l - n - 1$. If $d < 0$, both terms relevant to the asserted duality vanish by part (a). If $d \geq 0$, degree- d monomials

$$x_0^{b_0} \cdots x_n^{b_n}, \quad b_j \geq 0, \quad \sum b_j = d,$$

form a basis of $H^0(\mathbb{P}_A^n, \mathcal{O}(d))$. The basis (3) of $H^n(\mathbb{P}_A^n, \mathcal{O}(l))$ is indexed by $a_j = b_j + 1$. Multiplication followed by residue sends the corresponding two basis elements to 1, and sends distinct paired basis elements to 0. Thus (8) is a perfect pairing over A ; the residue construction shows that these local pairings glue.

Now (5) and the ordinary projection formula give

$$\begin{aligned} \pi_* \mathcal{H}om(\mathcal{O}_X(l), \omega_{X/Y}) &\cong \pi_* (\pi^* \mathcal{L} \otimes \mathcal{O}_X(-l-n-1)) \\ &\cong \mathcal{L} \otimes \pi_* \mathcal{O}_X(-l-n-1). \end{aligned}$$

Dualizing this identity through the perfect pairing (8) yields

$$R^n \pi_* \mathcal{O}_X(l) \cong (\pi_* \mathcal{O}_X(-l-n-1))^\vee \otimes \mathcal{L}^\vee,$$

as required.

(d) Taking $l = 0$ in part (a) gives

$$\pi_* \mathcal{O}_X = \mathcal{O}_Y, \quad R^i \pi_* \mathcal{O}_X = 0 \quad (i > 0). \quad (9)$$

For $i = n$, the second assertion follows because $0 > -n - 1$; for $i > n$, it follows from the same affine Čech cover, whose complex has length n . Exercise III.8.1 applied to (9) therefore gives

$$H^i(X, \mathcal{O}_X) \cong H^i(Y, \mathcal{O}_Y) \quad (i \geq 0). \quad (10)$$

In particular, $\chi(\mathcal{O}_X) = \chi(\mathcal{O}_Y)$. If $m = \dim Y$, then $\dim X = m + n$, and the definition

$$p_a(Z) = (-1)^{\dim Z} (\chi(\mathcal{O}_Z) - 1)$$

implies

$$p_a(X) = (-1)^{m+n} (\chi(\mathcal{O}_Y) - 1) = (-1)^n p_a(Y).$$

For the geometric genus, smoothness of the projective bundle and the exact sequence of relative differentials give

$$\omega_X \cong \pi^* \omega_Y \otimes \omega_{X/Y} \cong \pi^* (\omega_Y \otimes \mathcal{L}) \otimes \mathcal{O}_X(-n-1).$$

By part (a) and the projection formula,

$$\pi_*\omega_X \cong (\omega_Y \otimes \mathcal{L}) \otimes \pi_*\mathcal{O}_X(-n-1) = 0.$$

Hence $H^0(X, \omega_X) = 0$, so $p_g(X) = 0$.

(e) Here $n = 1$, so X has dimension 2. It is projective over the projective curve Y , and hence projective over k . Since $p_a(Y) = g$, part (d) gives

$$p_a(X) = -g, \quad p_g(X) = 0.$$

Finally, (10) gives

$$q(X) := h^1(X, \mathcal{O}_X) = h^1(Y, \mathcal{O}_Y) = g.$$

Thus X has all the stated invariants. □

III.9 Flat Morphisms

Exercise III.9.1

Permitted context.

A finite-type morphism of noetherian schemes has constructible image. In a noetherian topological space, a constructible subset is open if and only if it is stable under generization. Flatness is preserved by restriction and base change. A flat module over a domain is torsion-free.

Problem. Prove that a flat morphism $f : X \rightarrow Y$ of finite type between noetherian schemes is open.

Solution. It suffices to show that the image of X is open. Indeed, if $W \subseteq X$ is open, then W is noetherian and therefore quasi-compact. The restriction $W \rightarrow Y$ is again flat and of finite type, so the same assertion applied to W makes $f(W)$ open.

The question is local on Y , and a finite affine-open cover of X reduces it to

$$f : \operatorname{Spec} B \longrightarrow \operatorname{Spec} A,$$

where B is a finitely generated flat A -algebra. Its image is constructible. We prove that it is stable under generization.

Let $\mathfrak{q} \subseteq \mathfrak{p}$ be primes of A , and suppose $\mathfrak{p}$ is in the image. Make the base change

$$A' = A_{\mathfrak{p}}/\mathfrak{q}A_{\mathfrak{p}}, \quad B' = B \otimes_A A'.$$

The ring A' is a local domain. Its closed point is the point induced by $\mathfrak{p}$, and its generic point is induced by $\mathfrak{q}$. Since the fiber over $\mathfrak{p}$ was nonempty,

$$B' \otimes_{A'} \kappa(\mathfrak{p}) \neq 0.$$

In particular $B' \neq 0$. The A' -module B' is flat, so it is torsion-free. If $K = \operatorname{Frac}(A')$, the localization map

$$B' \longrightarrow B' \otimes_{A'} K$$

is therefore injective. Its source is nonzero, and hence the generic fiber $\text{Spec}(B' \otimes_{A'} K)$ is nonempty. This says that $\mathfrak{q}$ belongs to the image of $\text{Spec } B \rightarrow \text{Spec } A$.

Thus the constructible image is stable under generization, so it is open. Applying this argument to every open $W \subseteq X$ proves that f is an open map. $\square$

Exercise III.9.2

Permitted context.

Let a be the coordinate on $\mathbb{A}_k^1$, and use homogeneous coordinates x, y, z, w on $\mathbb{P}_k^3$. The twisted cubic has parametrization

$$[u : v] \mapsto [u^3 : u^2v : uv^2 : v^3]$$

and ideal

$$(xz - y^2, xw - yz, yw - z^2).$$

Apply the one-parameter coordinate change used in the flat-limit construction by replacing the third coordinate by a times that coordinate, and take the scheme-theoretic closure over $\mathbb{A}^1$. We use the monomial basis theorem and Buchberger criterion for a Groebner basis.

Problem. Compute the special fiber of this family and prove that it is the plane cuspidal cubic together with an embedded point at its cusp.

Solution. Over $a \neq 0$, the transformed twisted cubic is parametrized by

$$x = u^3, \quad y = u^2v, \quad z = auv^2, \quad w = v^3. \quad (1)$$

Its ideal in $k[a, a^{-1}, x, y, z, w]$ is

$$(xz - ay^2, axw - yz, a^2yw - z^2). \quad (2)$$

The ideal of the closure is the contraction of (2) to $k[a, x, y, z, w]$. It is

$$J = (xz - ay^2, axw - yz, a^2yw - z^2, x^2w - y^3). \quad (3)$$

Here is an explicit verification that (3) is the contraction. Use lexicographic order $z > y > x > w > a$ and write

$$g_1 = xz - ay^2, \quad g_2 = yz - axw, \quad g_3 = z^2 - a^2yw, \quad g_4 = y^3 - x^2w.$$

The leading monomials are xz, yz, z^2, y^3 . The six S-polynomials have the following reductions:

$$\begin{array}{l|l} (1, 2) & -ag_4 \\ (1, 3) & -ayg_2 \\ (1, 4) & x^2wg_1 - ay^2g_4 \\ (2, 3) & -awg_1 \\ (2, 4) & xwg_1 \\ (3, 4) & x^2wg_3 - a^2ywg_4. \end{array}$$

In each row successive leading-term division gives zero. Buchberger's criterion therefore makes these polynomials a Groebner basis. Their leading monomials do not involve a ; the standard monomials give a free $k[a]$ -basis of the quotient. Thus multiplication by a is injective modulo J , and J is saturated with respect to a . After inverting a , its fourth generator follows from the first two by

$$x(axw - yz) + y(xz - ay^2) = a(x^2w - y^3).$$

Hence $J[a^{-1}]$ is (2), and saturation proves the required contraction equality. All leading coefficients are one, so this verification works in every characteristic.

Setting $a = 0$ gives the special-fiber ideal

$$J_0 = (xz, yz, z^2, x^2w - y^3). \quad (4)$$

Its radical is

$$\sqrt{J_0} = (z, x^2w - y^3),$$

the plane cubic in the plane $z = 0$. This cubic is cuspidal at $P = [0 : 0 : 0 : 1]$.

To see the embedded structure, work on the affine chart $w = 1$. Then (4) becomes

$$I = (xz, yz, z^2, x^2 - y^3) \subseteq k[x, y, z].$$

There is a primary decomposition

$$I = (z, x^2 - y^3) \cap (x, y, z^2). \quad (5)$$

For completeness, the inclusion from left to right is immediate. Conversely, if $h = cz + d(x^2 - y^3)$ lies in (x, y, z^2) , then $cz \in (x, y, z^2)$, whence $c \in (x, y, z)$; it follows that $h \in (xz, yz, z^2, x^2 - y^3)$.

The first component in (5) is the cuspidal cubic. The second is primary to the maximal ideal (x, y, z) , and that maximal ideal strictly contains the minimal prime of the cubic. Hence it is an embedded associated point, supported at P , with its nilpotent direction transverse to the plane $z = 0$. $\square$

Exercise III.9.3**Permitted context.**

For a finite module over a noetherian local ring, flatness is equivalent to freeness. A regular local ring has finite global dimension, and the Auslander–Buchsbaum formula says

$$\mathrm{pd}_A M + \mathrm{depth}_A M = \mathrm{depth} A$$

for every nonzero finite A -module M . Regular local rings are Cohen–Macaulay. Flatness can also be detected by vanishing of Tor_1 .

Problem. (a) Let $f : X \rightarrow Y$ be a finite surjective morphism of nonsingular varieties over an algebraically closed field. Prove that f is flat.

(b) Put $A = k[x, y]$, and let

$$B = \frac{k[x, y, z, w]}{(z, w) \cap (x + z, y + w)}.$$

Geometrically, $\mathrm{Spec} B$ is the union of two planes meeting at one point, and each plane maps isomorphically to $\mathrm{Spec} A$. Prove that B is not flat over A .

(c) Put

$$C = \frac{A[z, w]}{(z^2, zw, w^2, xz - yw)}.$$

Show that $(\mathrm{Spec} C)_{\mathrm{red}} \cong \mathrm{Spec} A$, that $\mathrm{Spec} C$ has no embedded points, and that C is nevertheless not flat over A .

Solution. (a) Fix $y \in Y$, set $A = \mathcal{O}_{Y, y}$, and use the finite semilocal algebra

$$B = (f_* \mathcal{O}_X)_y.$$

Its maximal ideals correspond to the finitely many points x above y , and $B_{\mathfrak{n}_x} = \mathcal{O}_{X, x}$. The dimension formula for finite dominant morphisms of varieties gives $\dim B_{\mathfrak{n}_x} = \dim A = d$ for all these points. Each $B_{\mathfrak{n}_x}$ is regular, hence Cohen–Macaulay.

A regular system of parameters $t_1, \dots, t_d$ of A generates a primary ideal in each $B_{\mathfrak{n}_x}$, because the finite fiber is zero-dimensional. It is therefore a regular sequence in each such localization. Injectivity of multiplication on the successive quotients can be tested at all maximal ideals, so the same sequence is B -regular. Thus $\mathrm{depth}_A B = d$. Auslander–Buchsbaum applies to this finite A -module and gives $\mathrm{pd}_A B = 0$. Consequently B is free over A . Localizing B at $\mathfrak{n}_x$ preserves flatness over A , so all the local maps of f are flat.

(b) After identifying the coordinate ring of either plane with A , the coordinate ring of their union is the fiber product

$$B = \{(r, s) \in A \oplus A : r(0, 0) = s(0, 0)\}.$$

Thus there is an exact sequence of A -modules

$$0 \longrightarrow B \longrightarrow A \oplus A \xrightarrow{(r, s) \mapsto r(0, 0) - s(0, 0)} k \longrightarrow 0. \quad (2)$$

Let $\mathfrak{m} = (x, y)$, so $k = A/\mathfrak{m}$. The Koszul resolution of k is

$$0 \longrightarrow A \longrightarrow A^{\oplus 2} \longrightarrow A \longrightarrow k \longrightarrow 0.$$

In particular, $\mathrm{Tor}_2^A(k, k) \cong k$. Applying $-\otimes_A k$ to (2), and using that $A \oplus A$ is free, gives

$$\mathrm{Tor}_1^A(B, k) \cong \mathrm{Tor}_2^A(k, k) \cong k \neq 0.$$

Therefore B is not flat over A . Equivalently, the generic fiber has length 2, whereas the fiber over $\mathfrak{m}$ has length 3.

(c) Let $N = (z, w) \subseteq C$. Then $N^2 = 0$ and $C/N \cong A$. Since A is reduced, N is exactly the nilradical, and hence

$$(\mathrm{Spec} C)_{\mathrm{red}} \cong \mathrm{Spec} A.$$

As an A -module, C has the presentation

$$0 \longrightarrow A \xrightarrow{1 \mapsto (0, x, -y)} A^{\oplus 3} \xrightarrow{(a, b, c) \mapsto a + bz + cw} C \longrightarrow 0. \quad (3)$$

In particular,

$$C \cong A \oplus \frac{Az \oplus Aw}{A(xz - yw)}.$$

The second summand is isomorphic to the ideal $(x, y) \subseteq A$: send $z \mapsto y$ and $w \mapsto x$. Thus C , viewed as an A -module, is torsion-free.

Every prime of C contains the nilradical N , and primes of C correspond to primes of $A = C/N$. If $\mathfrak{p} \in \mathrm{Ass}_C(C)$, then $\mathfrak{p} \cap A = (0)$, because every nonzero element of the domain A acts injectively on the torsion-free A -module C . It follows that $\mathfrak{p} = N$. Hence the only associated prime of C is its minimal prime N , so $\mathrm{Spec} C$ has no embedded points.

Finally, tensor (3) with $k = A/(x, y)$. Its left-hand map becomes the zero map, so

$$\mathrm{Tor}_1^A(C, k) \cong k \neq 0.$$

Therefore C is not flat over A , despite being generically a double structure without embedded points. Its generic fiber has length 2, while its fiber over the origin has length 3. $\square$

Exercise III.9.4**Permitted context.**

We use the local criterion for flatness in the following form. Let $(R, \mathfrak{m}) \rightarrow (S, \mathfrak{n})$ be a local homomorphism of noetherian local rings, let $I \subseteq \mathfrak{m}$, and let M be a finite S -module. Then M is flat over R if and only if

$$M/IM \text{ is flat over } R/I \quad \text{and} \quad \mathrm{Tor}_1^R(R/I, M) = 0.$$

We also use that Tor commutes with localization. The topological generization criterion is applied to the noetherian sober space $\mathrm{Spec} B$; generic points exist there.

Problem. Let $f : X \rightarrow Y$ be a finite-type morphism of noetherian schemes. Prove that the flat locus

$$\{x \in X : f \text{ is flat at } x\}$$

is open in X .

Solution. We first prove the generic-freeness lemma needed below. If R is a noetherian domain, C is a finite-type R -algebra, and M is a finite C -module, then there is a nonzero $r \in R$ such that M_r is a free R_r -module.

Write C as a quotient of $R[x_1, \dots, x_n]$, and regard M as a finite module over that polynomial ring. We argue by induction on n . If $n = 0$, choose elements $m_1, \dots, m_s \in M$ whose images form a basis of $M \otimes_R K$, where $K = \mathrm{Frac} R$. After inverting one nonzero element of R , the m_i generate M , and their linear independence over K makes them a basis.

Suppose $n > 0$, and put $R' = R[x_1, \dots, x_{n-1}]$. Choose $R'[x_n]$ -generators of M , and let N be their R' -span. Define

$$F_j = N + x_n N + \cdots + x_n^{j-1} N \quad (j \geq 1), \quad F_0 = 0.$$

Then $M = \bigcup_j F_j$. Multiplication by x_n induces surjections

$$F_1/F_0 \twoheadrightarrow F_2/F_1 \twoheadrightarrow F_3/F_2 \twoheadrightarrow \cdots. \quad (1)$$

All these quotients are finite R' -modules. Their kernels, viewed as an increasing sequence of submodules of the noetherian module F_1 , eventually stabilize. Thus only finitely many quotients in (1) occur. By induction, after inverting one nonzero element $r \in R$, every quotient $(F_{j+1}/F_j)_r$ is free over R_r . Each exact sequence

$$0 \longrightarrow (F_j)_r \longrightarrow (F_{j+1})_r \longrightarrow (F_{j+1}/F_j)_r \longrightarrow 0$$

then splits as a sequence of R_r -modules. Taking the increasing union expresses M_r as a direct sum of free R_r -modules, so it is free. This proves generic freeness.

We now prove a slightly more general affine statement. Let $A \rightarrow B$ be a finite-type homomorphism of noetherian rings, and let M be a finite B -module. Set

$$U = \{\mathfrak{p} \in \mathrm{Spec} B : M_{\mathfrak{p}} \text{ is flat over } A_{\mathfrak{q}}, \mathfrak{q} = \mathfrak{p} \cap A\}. \quad (2)$$

First, U is stable under generization. Indeed, suppose $\mathfrak{p}_1 \subseteq \mathfrak{p}_2$, put $\mathfrak{q}_i = \mathfrak{p}_i \cap A$, and assume $\mathfrak{p}_2 \in U$. Base change from $A_{\mathfrak{q}_2}$ to $A_{\mathfrak{q}_1}$, followed by localization from $\mathfrak{p}_2$ to $\mathfrak{p}_1$, shows that $M_{\mathfrak{p}_1}$ is flat over $A_{\mathfrak{q}_1}$.

Fix $\mathfrak{p} \in U$ and put $\mathfrak{q} = \mathfrak{p} \cap A$. We claim that $U \cap V(\mathfrak{p})$ contains an open neighborhood of the generic point $\mathfrak{p}$ of $V(\mathfrak{p})$.

Consider the finite B -module

$$T = \mathrm{Tor}_1^A(A/\mathfrak{q}, M). \quad (3)$$

It is finite because $\mathfrak{q}$ is finitely generated and Tor in degree one can be computed from a finite presentation of $A/\mathfrak{q}$. Since $M_{\mathfrak{p}}$ is flat, $T_{\mathfrak{p}} = 0$. Hence there is $s \in B \setminus \mathfrak{p}$ such that $T_s = 0$.

Apply generic freeness to the domain $R = A/\mathfrak{q}$, the finite-type R -algebra $B/\mathfrak{q}B$, and the finite module

$$\overline{M} = M/\mathfrak{q}M.$$

There is $a \in A \setminus \mathfrak{q}$ such that $\overline{M}_a$ is free over $(A/\mathfrak{q})_a$. Because $\mathfrak{p} \cap A = \mathfrak{q}$, both s and a are outside $\mathfrak{p}$. Therefore

$$V(\mathfrak{p}) \cap D(s) \cap D(a) \quad (4)$$

is a nonempty open neighborhood of $\mathfrak{p}$ in $V(\mathfrak{p})$.

Let $\mathfrak{p}'$ belong to (4), and put $\mathfrak{q}' = \mathfrak{p}' \cap A$. Localization of (3) gives

$$\mathrm{Tor}_1^{A_{\mathfrak{q}'}}(A_{\mathfrak{q}'}/\mathfrak{q}A_{\mathfrak{q}'}, M_{\mathfrak{p}'}) = 0. \quad (5)$$

Moreover,

$$M_{\mathfrak{p}'}/\mathfrak{q}M_{\mathfrak{p}'} \cong \overline{M}_{\mathfrak{p}'}$$

is flat over $A_{\mathfrak{q}'}/\mathfrak{q}A_{\mathfrak{q}'}$: it is obtained from the free module $\overline{M}_a$ by module localization, which is a filtered colimit and therefore preserves flatness. The local criterion for flatness, applied with the ideal $\mathfrak{q}A_{\mathfrak{q}'}$, now shows that $M_{\mathfrak{p}'}$ is flat over $A_{\mathfrak{q}'}$. Thus (4) is contained in U .

We finish with a noetherian topological argument. In a noetherian space, a subset U is open if

$$U \text{ is stable under generization, and} \quad (6)$$

$$U \cap \overline{\{u\}} \text{ contains a nonempty open subset of } \overline{\{u\}} \text{ for every } u \in U.$$

Indeed, let Z be the complement of U . If an irreducible component of $\overline{Z}$ had generic point in U , the second condition in (6) would give a nonempty open part of that component disjoint from Z , contradicting the density of Z there. Every generic point of $\overline{Z}$ therefore lies in Z . The first condition says that Z is stable under specialization, so every such irreducible component is contained in Z . Hence $Z = \overline{Z}$, proving that U is open.

The two conditions in (6) have just been verified for (2), so its flat locus is open. Finally, cover Y by affine opens and their inverse images in X by affine opens. On each such pair take $M = B$. The set (2) is then precisely the locus where the local ring of X is flat over the corresponding local ring of Y . These affine flat loci glue, proving the assertion for f . $\square$

Exercise III.9.5**Permitted context.**

For a closed subscheme $Z \subseteq \mathbb{P}_F^n$, let $I_Z \subseteq F[x_0, \dots, x_n]$ be its saturated homogeneous ideal and put

$$h_Z(d) = \dim_F(F[x_0, \dots, x_n]/I_Z)_d.$$

The projective cone $C(Z) \subseteq \mathbb{P}_F^{n+1}$ is defined by the extension of I_Z to $F[x_0, \dots, x_n, x_{n+1}]$.

We use the Hilbert-polynomial flatness criterion for projective morphisms over an integral noetherian base. We also use cohomology and base change in the following form: for a projective morphism and a coherent sheaf flat over the base, constancy of the fiber dimensions of H^0 makes the direct image locally free and its formation compatible with base change.

In (d), the scheme-theoretic fibers themselves are assumed projectively normal. This implies $H^1(\mathbb{P}_{\kappa(t)}^n, \mathcal{I}_t(d)) = 0$ for every $d \geq 0$. Vanishing in this degree gives cohomology and base change for $p_*\mathcal{I}(d)$. Over a DVR its torsion-free coherent direct image is locally free.

Problem. (a) Give a flat family $X_\lambda \subseteq \mathbb{P}^n$ for which the fiberwise projective cones $C(X_\lambda)$ cannot be the fibers of a flat projective family.

(b) Let T be an integral noetherian scheme and $X \subseteq \mathbb{P}_T^n$ a closed subscheme. For $t \in T$, write

$$S_t = \kappa(t)[x_0, \dots, x_n]$$

and let I_t be the saturated homogeneous ideal of the fiber X_t . Call the family very flat if

$$\dim_{\kappa(t)}(S_t/I_t)_d$$

is independent of t for every $d \geq 0$.

(c) Prove that every very flat family is flat. Prove also that its family of projective cones is very flat, and therefore flat.

(d) Let T be an integral nonsingular curve over an algebraically closed field, and let $X \subseteq \mathbb{P}_T^n$ be flat over T . Assume that every scheme-theoretic fiber is projectively normal. Prove that this is a very flat family. Explain why normality of the reduced members alone would not suffice.

Solution. (a) Let λ be the coordinate on $\mathbb{A}_k^1$, and consider

$$\begin{aligned} \Phi : \mathbb{P}_k^1 \times \mathbb{A}_k^1 &\longrightarrow \mathbb{P}_k^4 \times \mathbb{A}_k^1, \\ ([u : v], \lambda) &\longmapsto ([u^4 : u^3v : \lambda u^2v^2 : uv^3 : v^4], \lambda). \end{aligned} \tag{1}$$

This is an immersion. On $u \neq 0$, the ratio of the second coordinate to the first recovers v/u , and on $v \neq 0$, the ratio of the fourth coordinate to the fifth recovers u/v ; these two calculations also identify the maps on local coordinate rings. Since the source is proper over $\mathbb{A}^1$, the immersion is closed. Its image X is isomorphic to $\mathbb{P}^1 \times \mathbb{A}^1$, so $X \rightarrow \mathbb{A}^1$ is flat.

For $\lambda \neq 0$, the fiber is projectively equivalent to the rational normal quartic. Its degree- d coordinate functions are all monomials of degree $4d$ in u, v , and hence

$$h_{X_\lambda}(d) = 4d + 1 \quad (d \geq 0, \lambda \neq 0). \tag{2}$$

For $\lambda = 0$, the degree-one coordinate functions have v -exponents 0, 1, 3, 4. Thus

$$h_{X_0}(0) = 1, \quad h_{X_0}(1) = 4.$$

Every integer from 0 through 8 is a sum of two elements of $\{0, 1, 3, 4\}$. Inductively, adding either 0 or 4 shows that every integer from 0 through $4d$ occurs for every $d \geq 2$. Consequently,

$$h_{X_0}(d) = 4d + 1 \quad (d \geq 2). \quad (3)$$

If R_Z is the homogeneous coordinate ring of Z , then the coordinate ring of its projective cone is $R_Z[x_{n+1}]$, so

$$h_{C(Z)}(m) = \sum_{d=0}^m h_Z(d). \quad (4)$$

Equations (2)–(4) give, for $m \gg 0$,

$$\begin{aligned} P_{C(X_\lambda)}(m) &= \sum_{d=0}^m (4d + 1) = (2m + 1)(m + 1) \quad (\lambda \neq 0), \\ P_{C(X_0)}(m) &= (2m + 1)(m + 1) - 1. \end{aligned}$$

The Hilbert polynomials differ, so no flat projective family can have exactly these fiberwise cones as its scheme fibers. The scheme-theoretic closure of the generic cones must have extra structure in its special fiber; it cannot be identified with the cone of X_0 .

(b) This is the stated definition. Notice that it asks for constancy of the entire Hilbert function, not merely the eventual Hilbert polynomial.

(c) In a very flat family, the fiber Hilbert functions, and hence their eventual Hilbert polynomials, are independent of t . The Hilbert-polynomial criterion therefore shows that $X \rightarrow T$ is flat.

We next construct the cone family with the correct scheme fibers. Let

$$p : \mathbb{P}_T^n \longrightarrow T, \quad \mathcal{I} = \mathcal{I}_X,$$

and write

$$\mathcal{S} = \text{Sym}_{\mathcal{O}_T}(\mathcal{O}_T^{\oplus(n+1)}) = \bigoplus_{d \geq 0} \mathcal{S}_d.$$

For every d , the sequence

$$0 \longrightarrow \mathcal{I}(d) \longrightarrow \mathcal{O}_{\mathbb{P}_T^n}(d) \longrightarrow \mathcal{O}_X(d) \longrightarrow 0 \quad (5)$$

has its middle and right terms flat over T , so $\mathcal{I}(d)$ is flat over T . On a fiber,

$$\dim_{\kappa(t)} H^0(\mathbb{P}_{\kappa(t)}^n, \mathcal{I}_t(d)) = \dim_{\kappa(t)} (I_t)_d = \dim_{\kappa(t)} (S_t)_d - h_{X_t}(d),$$

which is constant by very flatness. Cohomology and base change applied to (5) therefore shows that

$$\mathcal{J}_d := p_* \mathcal{I}(d) \subseteq \mathcal{S}_d$$

is locally free and satisfies

$$\mathcal{J}_d \otimes \kappa(t) \cong (I_t)_d \quad (6)$$

for every t . Multiplication of homogeneous polynomials makes

$$\mathcal{J} = \bigoplus_{d \geq 0} \mathcal{J}_d$$

a homogeneous ideal in $\mathcal{S}$.

Introduce a new degree-one variable z and define

$$\mathcal{C} = \text{Proj}_T (\mathcal{S}[z]/\mathcal{J}\mathcal{S}[z]) \subseteq \mathbb{P}_T^{n+1}. \quad (7)$$

By (6), formation of (7) commutes with every fiber and $\mathcal{C}_t = C(X_t)$ scheme-theoretically. Moreover,

$$\left(\frac{S_t[z]}{I_t S_t[z]} \right)_m \cong \bigoplus_{d=0}^m (S_t/I_t)_d z^{m-d}. \quad (8)$$

Its dimension is

$$h_{C(X_t)}(m) = \sum_{d=0}^m h_{X_t}(d),$$

which is independent of t for every m . Thus the cone family is very flat. Its Hilbert polynomial is consequently constant, so the Hilbert-polynomial criterion also makes $\mathcal{C} \rightarrow T$ flat.

(d) Let $p : \mathbb{P}_T^n \rightarrow T$, with ideal sheaf $\mathcal{I}$. Flatness of X makes $\mathcal{I}(d)$ flat over T . Projective normality of each scheme fiber gives

$$H^1(\mathbb{P}_{\kappa(t)}^n, \mathcal{I}_t(d)) = 0 \quad (d \geq 0).$$

Hence $\mathcal{J}_d = p_* \mathcal{I}(d)$ commutes with passage to the fibers. This coherent sheaf is torsion-free on the nonsingular curve, hence locally free. Its inclusion into the free polynomial piece $\mathcal{S}_d$ is injective on every fiber, so its cokernel $\mathcal{R}_d$ is locally free as well. Its fiber is $(S_t/I_t)_d$. Its rank is constant on the integral curve, which proves constancy of every Hilbert-function value.

To see the need for the precise fiber assumption, in characteristic different from two consider

$$x_0^2 - tx_1x_2 = 0 \quad \text{in } \mathbb{P}^2 \times \mathbb{A}^1.$$

This is a flat integral total family. Its reduced members are smooth conics for $t \neq 0$ and a line for $t = 0$, all projectively normal. Their Hilbert functions are $2d + 1$ and $d + 1$. The scheme-theoretic special fiber is a double line. Thus an assertion about reduced members alone cannot yield very flatness. $\square$

Exercise III.9.6**Permitted context.**

Let k be algebraically closed. A projective complete intersection is arithmetically Cohen–Macaulay; this follows from its Koszul resolution. In particular, if $W \subseteq \mathbb{P}^N$ is a complete intersection of positive dimension, then

$$H^1(\mathbb{P}^N, \mathcal{I}_W(m)) = 0 \quad (m \in \mathbb{Z}).$$

A complete intersection has a Cohen–Macaulay homogeneous coordinate ring and consequently has no embedded associated points.

Problem. Let $Y \subseteq \mathbb{P}_k^n$ be a nonsingular variety with $\dim Y \geq 2$. Let $H \cong \mathbb{P}_k^{n-1}$ be a hyperplane which does not contain Y , and suppose the scheme-theoretic intersection

$$Y' = Y \cap H$$

is nonsingular. Prove that Y is a complete intersection in $\mathbb{P}^n$ if and only if Y' is a complete intersection in H .

Solution. Write $H = V(h)$, where h is linear.

Suppose first that Y is a complete intersection of codimension r . Its ideal is generated by r homogeneous equations, and their restrictions to H define the scheme $Y \cap H = Y'$. Since h is a nonzerodivisor on the coordinate ring of the integral variety Y ,

$$\operatorname{codim}_H Y' = \operatorname{codim}_{\mathbb{P}^n} Y = r.$$

Thus the r restricted equations generate an ideal of height r in the Cohen–Macaulay coordinate ring of H , so they form a regular sequence. Hence Y' is a complete intersection.

Conversely, assume Y' is a complete intersection of codimension r in H . Let $\mathcal{I}_Y$ and $\mathcal{I}_{Y'/H}$ denote the corresponding ideal sheaves. Because h is a nonzerodivisor on $\mathcal{O}_Y$, restriction to H gives, for every m , an exact sequence

$$0 \longrightarrow \mathcal{I}_Y(m-1) \xrightarrow{h} \mathcal{I}_Y(m) \longrightarrow \mathcal{I}_{Y'/H}(m) \longrightarrow 0. \quad (1)$$

The complete intersection Y' has dimension $\dim Y - 1 \geq 1$, so its Koszul resolution gives

$$H^1(H, \mathcal{I}_{Y'/H}(m)) = 0 \quad (m \in \mathbb{Z}). \quad (2)$$

We claim that

$$H^1(\mathbb{P}^n, \mathcal{I}_Y(m)) = 0 \quad (m \in \mathbb{Z}). \quad (3)$$

For $m < 0$, both $H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(m))$ and $H^0(Y, \mathcal{O}_Y(m))$ vanish; the latter vanishing follows because the inverse of an ample line bundle has no nonzero section on a positive-dimensional integral projective variety. Also $H^1(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(m)) = 0$, since $n \geq 2$. The ideal-sheaf sequence for Y therefore proves (3) for $m < 0$.

Taking cohomology in (1) and using (2) shows that multiplication by h induces a surjection

$$H^1(\mathbb{P}^n, \mathcal{I}_Y(m-1)) \twoheadrightarrow H^1(\mathbb{P}^n, \mathcal{I}_Y(m))$$

for every m . Starting with the negative-degree vanishing and inducting upward proves (3) in all degrees.

Let

$$I_{Y'} = (f_1, \dots, f_r) \subseteq k[x_0, \dots, x_n]/(h)$$

be the saturated homogeneous ideal of Y' , with the f_i forming a regular sequence. The cohomology sequence of (1), together with (3), makes

$$H^0(\mathbb{P}^n, \mathcal{I}_Y(d)) \longrightarrow H^0(H, \mathcal{I}_{Y'/H}(d)) \quad (4)$$

surjective for every d . We may therefore lift each f_i to a homogeneous

$$g_i \in H^0(\mathbb{P}^n, \mathcal{I}_Y(\deg f_i)).$$

Put $S = k[x_0, \dots, x_n]$, $J = (g_1, \dots, g_r)$, and

$$Z = V(J) \subseteq \mathbb{P}^n.$$

Since the images of the g_i modulo h are the regular sequence $f_1, \dots, f_r$,

$$\text{ht}(J, h) = r + 1.$$

Krull's height theorem gives $\text{ht } J = r$. The polynomial ring S is Cohen–Macaulay, so $g_1, \dots, g_r$ is a regular sequence. Thus Z is a complete intersection. Moreover, h is a nonzerodivisor on S/J : otherwise it would lie in a minimal prime of the unmixed ideal J , contradicting $\text{ht}(J, h) = r + 1$.

The choices of the g_i give

$$Y \subseteq Z, \quad Z \cap H = Y' \quad (5)$$

scheme-theoretically. Indeed, modulo h , both hyperplane-section ideals are generated by $f_1, \dots, f_r$.

Because h is a nonzerodivisor on both Z and Y , (5) gives exact sequences

$$\begin{aligned} 0 \longrightarrow \mathcal{O}_Z(m-1) &\xrightarrow{h} \mathcal{O}_Z(m) \longrightarrow \mathcal{O}_{Y'}(m) \longrightarrow 0, \\ 0 \longrightarrow \mathcal{O}_Y(m-1) &\xrightarrow{h} \mathcal{O}_Y(m) \longrightarrow \mathcal{O}_{Y'}(m) \longrightarrow 0. \end{aligned} \quad (6)$$

Taking Hilbert polynomials in (6) shows

$$P_Z(m) - P_Z(m-1) = P_Y(m) - P_Y(m-1).$$

Hence $P_Z - P_Y$ is constant. From

$$0 \longrightarrow \mathcal{K} \longrightarrow \mathcal{O}_Z \longrightarrow \mathcal{O}_Y \longrightarrow 0, \quad \mathcal{K} = \mathcal{I}_{Y/Z}, \quad (7)$$

we conclude that either $\mathcal{K} = 0$, or $\mathcal{K}$ is supported on finitely many closed points.

The second possibility cannot occur. A nonzero finite-length submodule of a local ring contains a nonzero socle element, so a closed point in $\text{Supp } \mathcal{K}$ would be an associated point of Z . It would be embedded because $\dim Z = \dim Y \geq 2$. But the complete intersection Z is Cohen–Macaulay and has no embedded associated points. Therefore $\mathcal{K} = 0$, so $Y = Z$. Since Z is a complete intersection, so is Y . $\square$

Exercise III.9.7**Permitted context.**

Let $D = k[\epsilon]/(\epsilon^2)$. A D -module M is flat if and only if multiplication by ϵ induces an isomorphism

$$M/\epsilon M \xrightarrow{\sim} \epsilon M.$$

Let $Y \hookrightarrow X$ be a closed immersion with ideal sheaf $\mathcal{I}$. Its normal sheaf is

$$\mathcal{N}_{Y/X} = \mathcal{H}om_{\mathcal{O}_Y}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_Y).$$

Problem. Assume X is of finite type over k . An embedded infinitesimal deformation of Y in X is a closed subscheme

$$Y' \subseteq X \times_k \operatorname{Spec} D$$

which is flat over D and whose closed fiber is Y . Prove that such deformations are naturally classified by

$$H^0(Y, \mathcal{N}_{Y/X}).$$

Solution. We first work on an affine open $X = \operatorname{Spec} A$, with Y defined by an ideal $I \subseteq A$. Put $A_D = A[\epsilon]/(\epsilon^2)$. A deformation is determined by an ideal $I' \subseteq A_D$ such that

$$B = A_D/I'$$

is flat over D and the reduction of I' modulo ϵ is I .

Flatness of B implies the useful identity

$$I' \cap \epsilon A_D = \epsilon I. \quad (1)$$

Indeed, if $\epsilon c \in I'$, then the class of c in B is killed by ϵ . The flatness criterion says that this class lies in ϵB , so its reduction in $B/\epsilon B \cong A/I$ is zero; hence $c \in I$. Conversely, if $c \in I$, choose $c + \epsilon d \in I'$ and multiply by ϵ to obtain $\epsilon c \in I'$.

For $a \in I$, choose a lift

$$a + \epsilon b \in I'$$

and define

$$\varphi_{I'}(a) = b \bmod I \in A/I. \quad (2)$$

This is well defined: two choices differ by $\epsilon(b-b') \in I'$, and (1) then gives $b-b' \in I$. Multiplying a lift by an element of A shows that (2) is A -linear. Since I annihilates A/I , it kills I^2 and therefore factors uniquely as

$$\varphi_{I'} : I/I^2 \longrightarrow A/I. \quad (3)$$

Conversely, start with an A/I -linear homomorphism $\varphi : I/I^2 \rightarrow A/I$. Define

$$I_\varphi = \{a + \epsilon b \in A_D : a \in I, \quad b \bmod I = \varphi(a \bmod I^2)\}. \quad (4)$$

This is an ideal. In fact, for $c + \epsilon d \in A_D$,

$$(c + \epsilon d)(a + \epsilon b) = ca + \epsilon(cb + da),$$

and modulo I the second coefficient is $c\varphi(a) = \varphi(ca)$. The reduction of I_φ is plainly I , and (4) gives

$$I_\varphi \cap \epsilon A_D = \epsilon I. \quad (5)$$

Let $B_\varphi = A_D/I_\varphi$. Multiplication by ϵ therefore induces

$$B_\varphi/\epsilon B_\varphi \cong A/I \xrightarrow{\sim} \epsilon B_\varphi, \quad \bar{a} \longmapsto \epsilon a.$$

Thus B_φ is flat over D . Formulas (2) and (4) are inverse constructions, yielding a natural bijection

$$\left\{ \begin{array}{c} \text{embedded deformations of } Y \\ \text{inside } \operatorname{Spec} A \end{array} \right\} \longleftrightarrow \operatorname{Hom}_{A/I}(I/I^2, A/I). \quad (6)$$

The construction commutes with localization in A : both choosing the coefficient of ϵ in (2) and forming the ideal (4) do. Consequently, the local maps obtained from a global deformation agree on overlaps and define a morphism

$$\mathcal{I}/\mathcal{I}^2 \longrightarrow \mathcal{O}_Y.$$

Conversely, a global morphism of this kind produces through (4) compatible ideal sheaves on an affine cover, which glue to a unique closed subscheme $Y' \subseteq X \times \operatorname{Spec} D$. Flatness is local, so the glued deformation is flat. We have therefore obtained the required natural bijection

$$\{\text{embedded infinitesimal deformations of } Y \text{ in } X\} \cong \operatorname{Hom}_{\mathcal{O}_Y}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_Y) = H^0(Y, \mathcal{N}_{Y/X}).$$

□

Exercise III.9.8**Permitted context.**

Let A be a finitely generated k -algebra, choose a presentation

$$0 \longrightarrow J \longrightarrow P \longrightarrow A \longrightarrow 0, \quad P = k[x_1, \dots, x_n],$$

and let $D = k[\epsilon]/(\epsilon^2)$. Embedded first-order deformations of $\text{Spec } A$ in $\text{Spec } P$ are classified by

$$\text{Hom}_A(J/J^2, A).$$

The universal property of Kahler differentials identifies

$$\text{Hom}_A(\Omega_{P/k} \otimes_P A, A) \cong \text{Der}_k(P, A).$$

Problem. Apply $\text{Hom}_A(-, A)$ to the conormal sequence

$$J/J^2 \longrightarrow \Omega_{P/k} \otimes_P A \longrightarrow \Omega_{A/k} \longrightarrow 0$$

and define

$$T^1(A) = \text{coker} \left(\text{Hom}_A(\Omega_{P/k} \otimes_P A, A) \longrightarrow \text{Hom}_A(J/J^2, A) \right).$$

Prove that $T^1(A)$ classifies isomorphism classes of flat D -algebras A' equipped with an identification $A'/\epsilon A' \cong A$, where isomorphisms must induce the identity on A . Deduce that $T^1(A)$ is independent of the chosen polynomial presentation.

Solution. For $\varphi \in \text{Hom}_A(J/J^2, A)$, let $J_\varphi \subseteq P_D = P[\epsilon]/(\epsilon^2)$ be

$$J_\varphi = \{j + \epsilon q : j \in J, \quad q \bmod J = \varphi(j \bmod J^2)\}. \quad (1)$$

Then

$$A_\varphi = P_D/J_\varphi \quad (2)$$

is flat over D and has closed fiber A . Every deformation embedded in $\text{Spec } P_D$ arises uniquely in this way.

We determine when (2) and A_ψ are isomorphic as abstract deformations. An element

$$\delta \in \text{Hom}_A(\Omega_{P/k} \otimes_P A, A) = \text{Der}_k(P, A)$$

is determined by $a_i = \delta(x_i) \in A$. Choose lifts $\tilde{a}_i \in P$, and let

$$\tilde{\delta} = \sum_{i=1}^n \tilde{a}_i \frac{\partial}{\partial x_i} : P \longrightarrow P.$$

There is a D -algebra automorphism, reducing to the identity modulo ϵ ,

$$\begin{aligned} \Theta_\delta : P_D &\longrightarrow P_D, \\ p + \epsilon q &\longmapsto p + \epsilon(q + \tilde{\delta}(p)). \end{aligned} \quad (3)$$

Its inverse is obtained by replacing $\tilde{\delta}$ with $-\tilde{\delta}$. For $j + \epsilon q \in J_\varphi$, formula (3) gives

$$\Theta_\delta(j + \epsilon q) = j + \epsilon(q + \tilde{\delta}(j)).$$

Modulo J , the new coefficient of ϵ is

$$\varphi(j) + \delta(j).$$

Consequently,

$$\Theta_\delta(J_\varphi) = J_{\varphi + \delta|_J}, \quad (4)$$

where $\delta|_J$ is exactly the image of δ under the map appearing in the definition of $T^1(A)$. Thus elements of $\text{Hom}_A(J/J^2, A)$ differing by that image give isomorphic abstract deformations.

Conversely, suppose

$$\alpha : A_\varphi \xrightarrow{\sim} A_\psi \quad (5)$$

is a D -algebra isomorphism inducing the identity on A . For each polynomial generator, choose $a_i \in A$ such that

$$\alpha([x_i]) = [x_i + \epsilon a_i]. \quad (6)$$

Such a_i exist because the difference in (6) belongs to ϵA_ψ . Since the x_i generate P_D as a D -algebra, (6) says that α is induced by an automorphism of the form (3), with $\delta(x_i) = a_i$. The condition that this automorphism carry J_φ to J_ψ gives

$$\psi - \varphi = \delta|_J. \quad (7)$$

Hence the two embedded classes determine isomorphic abstract deformations if and only if their difference lies in the image of $\text{Der}_k(P, A)$.

It remains to see that every abstract deformation has an embedded representative. Let B be flat over D , together with $B/\epsilon B \cong A$. Lift the images of $x_1, \dots, x_n$ to elements $b_1, \dots, b_n \in B$. They define a D -algebra homomorphism

$$P_D \longrightarrow B. \quad (8)$$

This map is surjective. Indeed, for $b \in B$, first choose a polynomial p such that $b - p(b_1, \dots, b_n) \in \epsilon B$. Write this difference as ϵc . Choose a polynomial q whose value modulo ϵ is the reduction of c . Flatness is not needed for this last equality: the difference $c - q(b_1, \dots, b_n)$ lies in ϵB , and hence $\epsilon(c - q(b_1, \dots, b_n)) = 0$. Thus

$$\epsilon c = \epsilon q(b_1, \dots, b_n).$$

Thus b is the image of $p + \epsilon q$ under (8).

Let K be the kernel of (8). Tensoring $0 \rightarrow K \rightarrow P_D \rightarrow B \rightarrow 0$ with $k = D/(\epsilon)$ remains exact on the left because B is D -flat. It follows that $K/\epsilon K$ maps isomorphically to the kernel J of $P \rightarrow A$. Thus K is an embedded deformation ideal, so it is J_φ for a unique $\varphi \in \text{Hom}_A(J/J^2, A)$. Combining this fact with (4)–(7) produces a bijection

$$T^1(A) \longleftrightarrow \left\{ \begin{array}{l} \text{isomorphism classes of flat } D\text{-algebras } A' \\ \text{equipped with } A'/\epsilon A' \cong A \end{array} \right\}.$$

The right-hand side is intrinsic to A . Moreover, addition of first-order coefficients in (1) agrees with addition in the cokernel. Therefore any two polynomial presentations give canonically isomorphic k -vector spaces $T^1(A)$, proving the claimed independence. $\square$

Exercise III.9.9

Permitted context.

For a presentation $A = P/J$, where P is a polynomial k -algebra, first-order abstract deformations of A are classified by

$$T^1(A) = \operatorname{coker} (\operatorname{Der}_k(P, A) \longrightarrow \operatorname{Hom}_A(J/J^2, A)).$$

The algebra A is rigid if $T^1(A) = 0$.

Problem. Let

$$A = \frac{k[x, y, z, w]}{(x, y) \cap (z, w)}.$$

This is the union of two coordinate planes in $\mathbb{A}_k^4$ meeting at the origin. Prove that A is rigid.

Solution. Put $P = k[x, y, z, w]$. Since the variables in the two ideals are disjoint,

$$J = (x, y) \cap (z, w) = (xz, xw, yz, yw). \quad (1)$$

Write $e_{xz}, e_{xw}, e_{yz}, e_{yw}$ for these four generators. Their first syzygies are generated by

$$\begin{aligned} ye_{xz} - xe_{yz} &= 0, & ye_{xw} - xe_{yw} &= 0, \\ we_{xz} - ze_{xw} &= 0, & we_{yz} - ze_{yw} &= 0. \end{aligned} \quad (2)$$

For example, this presentation follows by tensoring the two Koszul presentations of the ideals (x, y) and (z, w) . Tensoring the resulting presentation of J with $A = P/J$ presents J/J^2 , so (2) gives all the conditions on a homomorphism $J/J^2 \rightarrow A$.

It is useful to write

$$A = \{(F, G) \in k[x, y] \times k[z, w] : F(0, 0) = G(0, 0)\}. \quad (3)$$

Let

$$\varphi \in \operatorname{Hom}_A(J/J^2, A),$$

and denote the images of the four generators by $a_{xz}, a_{xw}, a_{yz}, a_{yw}$. The first two relations in (2), applied to the $k[x, y]$ -components in (3), imply that there are $f_z, f_w \in k[x, y]$ such that

$$\begin{aligned} (a_{xz})_1 &= xf_z, & (a_{yz})_1 &= yf_z, \\ (a_{xw})_1 &= xf_w, & (a_{yw})_1 &= yf_w. \end{aligned} \quad (4)$$

Indeed, for example $y(a_{xz})_1 = x(a_{yz})_1$, and x, y are relatively prime. The last two relations in (2), applied to the $k[z, w]$ -components, similarly give $g_x, g_y \in k[z, w]$ such that

$$\begin{aligned}(a_{xz})_2 &= zg_x, & (a_{xw})_2 &= wg_x, \\ (a_{yz})_2 &= zg_y, & (a_{yw})_2 &= wg_y.\end{aligned}\tag{5}$$

Thus every φ has the form

$$\begin{aligned}a_{xz} &= (xf_z, zg_x), & a_{xw} &= (xf_w, wg_x), \\ a_{yz} &= (yf_z, zg_y), & a_{yw} &= (yf_w, wg_y).\end{aligned}\tag{6}$$

We now construct a derivation inducing it. Since P is polynomial, prescribe

$$\begin{aligned}\delta(x) &= (g_x(0,0), g_x), & \delta(y) &= (g_y(0,0), g_y), \\ \delta(z) &= (f_z, f_z(0,0)), & \delta(w) &= (f_w, f_w(0,0)).\end{aligned}\tag{7}$$

Each pair in (7) belongs to A by (3), and the assignments extend uniquely to a k -derivation $\delta : P \rightarrow A$. The Leibniz rule, together with the fact that every product of an element of (x, y) and an element of (z, w) is zero in A , gives

$$\begin{aligned}\delta(xz) &= (xf_z, zg_x), & \delta(xw) &= (xf_w, wg_x), \\ \delta(yz) &= (yf_z, zg_y), & \delta(yw) &= (yf_w, wg_y).\end{aligned}$$

By (6), the restriction of δ to J/J^2 is exactly φ . Therefore

$$\mathrm{Der}_k(P, A) \longrightarrow \mathrm{Hom}_A(J/J^2, A)$$

is surjective. Its cokernel $T^1(A)$ is zero, so A is rigid. $\square$

Exercise III.9.10**Permitted context.**

For a nonsingular k -scheme Z , first-order deformations are classified by $H^1(Z, \mathcal{T}_Z)$. Smoothness is open, and a flat morphism of finite presentation is smooth at a point when its geometric fiber is regular there. A smooth morphism has a section through a chosen rational point after an étale base change.

We also use cohomology and base change for a proper flat family of curves: if a line bundle has constant h^0 and vanishing h^1 on the fibers, its direct image is locally free, commutes with base change, and its evaluation map may be checked fiberwise.

Problem. A scheme over a field is called rigid if it has no nontrivial first-order deformations.

- (a) Prove that $\mathbb{P}_k^1$ is rigid.
- (b) Let k be algebraically closed. Construct a proper flat morphism

$$f : X \longrightarrow \mathbb{A}_k^2$$

whose fiber X_0 over the origin is isomorphic to $\mathbb{P}_k^1$, but for which there is no open neighborhood U of the origin satisfying $X_U \cong U \times \mathbb{P}^1$.

(c) Let $f : X \rightarrow T$ be flat and projective, where T is a nonsingular curve over an algebraically closed field. Suppose $X_t \cong \mathbb{P}_k^1$ for a closed point $t \in T$. Prove that there are a nonsingular curve T' and a flat morphism $g : T' \rightarrow T$, whose image contains t , such that

$$X \times_T T' \cong \mathbb{P}_{T'}^1.$$

Solution. (a) The tangent bundle of the projective line is

$$\mathcal{T}_{\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}(2).$$

Therefore

$$H^1(\mathbb{P}^1, \mathcal{T}_{\mathbb{P}^1}) = H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(2)) = 0.$$

The deformation classification in the context shows that $\mathbb{P}^1$ is rigid.

(b) Write a, b for the coordinates on $\mathbb{A}^2$, and $[x : y : z]$ for those on $\mathbb{P}^2$. If $\text{char } k \neq 2$, let

$$X = V(x^2 + (1+a)y^2 + (1+b)z^2) \subseteq \mathbb{P}_k^2 \times \mathbb{A}_k^2. \quad (1)$$

If $\text{char } k = 2$, use instead

$$X = V(x^2 + (1+a)xy + (1+a)y^2 + (1+b)z^2). \quad (2)$$

In either case the defining equation is monic in x^2 . Thus every graded piece of its homogeneous coordinate ring is a free $k[a, b]$ -module, with representatives having x -degree at most one. Hence $f : X \rightarrow \mathbb{A}^2$ is flat. It is projective, and therefore proper.

At $a = b = 0$, equation (1) is a nonsingular conic when $\text{char } k \neq 2$. In characteristic 2, the partial derivatives of the equation

$$x^2 + xy + y^2 + z^2$$

are $y, x, 0$; their only simultaneous affine zero has $x = y = 0$, and the equation would then force $z = 0$, so the projective conic is again nonsingular. A nonsingular conic over an algebraically closed field is isomorphic to $\mathbb{P}^1$.

Let $K = k(a, b)$. We show that the generic conic has no K -point. Suppose first that the characteristic is not 2. Clearing denominators in a hypothetical point gives relatively prime polynomials $X_1, Y_1, Z_1 \in k[a, b]$, not all zero, satisfying

$$X_1^2 + (1+a)Y_1^2 + (1+b)Z_1^2 = 0. \quad (3)$$

Set $a = -1$. Since $-(1+b)$ is not a square in $k(b)$, equation (3) implies

$$X_1(-1, b) = Z_1(-1, b) = 0.$$

Thus $1+a$ divides X_1 and Z_1 . Substitution back into (3) then shows that $1+a$ divides Y_1^2 , and hence Y_1 . This contradicts the choice of a relatively prime triple.

In characteristic 2, the same argument works with (2). After setting $a = -1$, the equation is

$$X_1^2 + (1+b)Z_1^2 = 0.$$

The element $1+b$ is not a square in $k(b)$, so $1+a$ divides X_1 and Z_1 . Substitution into

$$X_1^2 + (1+a)X_1Y_1 + (1+a)Y_1^2 + (1+b)Z_1^2 = 0$$

then forces $1+a \mid Y_1^2$, and hence $1+a \mid Y_1$, giving the same contradiction.

If X_U were isomorphic to $U \times \mathbb{P}^1$ on any open neighborhood U of the origin, then U would contain the generic point of $\mathbb{A}^2$, and the generic conic would have a K -rational point. The contradiction proves the required nontriviality.

(c) We may replace T by an open neighborhood of t . Because f is flat and the fiber X_t is smooth, f is smooth at every point of X_t . The nonsmooth locus is closed in X , and its image is closed because f is projective. After shrinking T , we may therefore assume that f is a smooth proper family of curves. The arithmetic genus and the number of geometric connected components are locally constant in such a family, so, after one more shrinking, its fibers are geometrically connected curves of genus 0.

Choose $x \in X_t(k)$. Smooth morphisms admit sections étale-locally, so there are an étale morphism

$$g_0 : T_0 \longrightarrow T,$$

a point $t_0 \in T_0$ above t , and a section

$$s : T_0 \longrightarrow X_0 := X \times_T T_0$$

through x . Replace T_0 by a sufficiently small affine open neighborhood of t_0 . It remains a nonsingular curve, and g_0 is flat.

The image $s(T_0)$ is an effective Cartier divisor on the smooth relative curve X_0/T_0 . Put

$$\mathcal{L} = \mathcal{O}_{X_0}(s(T_0)).$$

On every geometric fiber, $\mathcal{L}$ has degree 1. A geometrically connected smooth genus-zero curve with a rational point is a projective line, so

$$\mathcal{L}|_{(X_0)_u} \cong \mathcal{O}_{\mathbb{P}^1}(1), \quad h^0((X_0)_u, \mathcal{L}_u) = 2, \quad h^1((X_0)_u, \mathcal{L}_u) = 0.$$

Cohomology and base change now makes

$$\mathcal{E} = (f_0)_* \mathcal{L}$$

a locally free sheaf of rank 2, and the evaluation map

$$f_0^* \mathcal{E} \longrightarrow \mathcal{L}$$

is surjective. It defines a T_0 -morphism

$$\Phi : X_0 \longrightarrow \mathbb{P}_{T_0}(\mathcal{E}) \quad (4)$$

whose restriction to every geometric fiber is the isomorphism defined by $\mathcal{O}_{\mathbb{P}^1}(1)$. The morphism Φ is proper, since X_0 is proper over T_0 and the target is separated over T_0 . It is quasi-finite because this may be checked on the geometric fibers, so it is finite. Locally on the target, its finite algebra map becomes an isomorphism after tensoring with every residue field of T_0 . Surjectivity follows from Nakayama's lemma; because both source and target are flat over T_0 , formation of the kernel commutes with passage to a fiber, and a second application of Nakayama shows that the kernel is zero. Consequently (4) identifies

$$X_0 \cong \mathbb{P}_{T_0}(\mathcal{E}).$$

Finally, shrink T_0 once more around t_0 so that the rank-two bundle $\mathcal{E}$ is trivial. Calling this open curve T' , we obtain

$$X \times_T T' \cong \mathbb{P}_{T'}(\mathcal{O}_{T'}^{\oplus 2}) \cong \mathbb{P}_{T'}^1.$$

The composite $g : T' \rightarrow T$ is flat and its image contains t , as required. $\square$

Exercise III.9.11

Permitted context.

For an integral projective curve Z over an algebraically closed field,

$$p_a(Z) = 1 - \chi(\mathcal{O}_Z).$$

A finite birational morphism from a normal curve to an integral curve is its normalization. If $\nu : \tilde{C} \rightarrow C$ is the normalization of a projective curve, then

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \nu_* \mathcal{O}_{\tilde{C}} \longrightarrow \mathcal{Q} \longrightarrow 0,$$

where $\mathcal{Q}$ has finite length. We also use the usual dimension formula for Grassmannians and the fact that a proper quasi-finite morphism is finite.

Problem. Let Y be a nonsingular curve of degree d in $\mathbb{P}_k^n$, where k is algebraically closed. Prove that

$$0 \leq p_a(Y) \leq \frac{1}{2}(d-1)(d-2).$$

Solution. Because Y is integral and projective, $H^0(Y, \mathcal{O}_Y) = k$. Hence

$$p_a(Y) = 1 - \chi(\mathcal{O}_Y) = h^1(Y, \mathcal{O}_Y) \geq 0. \quad (1)$$

For the upper bound, replace the ambient projective space by the linear span of Y , say $\mathbb{P}^r$. This does not change the degree. If $r = 1$, then $Y = \mathbb{P}^1$, and the result is immediate. If $r = 2$, then Y is a plane curve of degree d , and the standard plane-curve computation already gives

$$p_a(Y) = \frac{(d-1)(d-2)}{2}.$$

Assume now that $r \geq 3$. We first construct a suitable linear projection. Let V be a vector space of dimension $r+1$, so that $\mathbb{P}^r = \mathbb{P}(V)$, and let

$$G = \text{Gr}(r-2, V).$$

A point $W \in G$ corresponds to a prospective center $\Lambda = \mathbb{P}(W) \cong \mathbb{P}^{r-3}$, and

$$\dim G = 3(r-2). \quad (2)$$

For a fixed line $L \subseteq \mathbb{P}(V)$, the locus of $W \in G$ for which $\mathbb{P}(W) \cap L \neq \emptyset$ has dimension at most

$$1 + \dim \text{Gr}(r-3, V/kq) = 1 + 3(r-3) = 3r-8 = \dim G - 2. \quad (3)$$

Here q varies on L , and the Grassmannian parametrizes the subspaces W containing the corresponding one-dimensional subspace kq .

Apply (3) first to the embedded tangent lines $T_y Y$, with $y \in Y$. The incidence variety of pairs (y, W) for which $\mathbb{P}(W)$ meets $T_y Y$ has dimension at most $1 + (\dim G - 2) = \dim G - 1$. Thus a general center meets no tangent line to Y . The analogous incidence for secant lines,

$$\{(y, z, W) : y \neq z, \mathbb{P}(W) \cap \overline{yz} \neq \emptyset\},$$

has dimension at most $2 + (\dim G - 2) = \dim G$. The fiber-dimension theorem therefore shows that, for general W , only finitely many ordered pairs (y, z) occur. Finally, the centers meeting Y form a proper closed subset of G : for fixed y , the locus of centers containing y has dimension $3(r-3) = \dim G - 3$.

Choose a center Λ satisfying all three conclusions. The linear projection from Λ restricts to a morphism

$$\pi : Y \longrightarrow C \subseteq \mathbb{P}^2. \quad (4)$$

The morphism is quasi-finite: distinct points in one fiber determine a secant meeting Λ , and there are only finitely many such pairs. Since it is also projective, it is finite. Moreover, its differential is nonzero at every point, because Λ meets no embedded tangent line. Thus the finite extension $k(C) \subseteq k(Y)$ is separable. If its degree were greater than one, then over a dense open subset of C every geometric fiber would contain at least two distinct points, contradicting the finiteness of the set of secant pairs above. Consequently π is birational.

The image C is therefore an integral plane curve. Since

$$\pi^* \mathcal{O}_C(1) = \mathcal{O}_Y(1),$$

the degree formula for a finite map of curves gives

$$\deg C = \deg \mathcal{O}_C(1) = \deg \mathcal{O}_Y(1) = d. \quad (5)$$

Because Y is nonsingular, it is normal, so the finite birational map (4) is the normalization of C . We consequently have an exact sequence

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \pi_* \mathcal{O}_Y \longrightarrow \mathcal{Q} \longrightarrow 0 \quad (6)$$

with $\mathcal{Q}$ of finite length. Taking Euler characteristics, and using that a finite morphism has no higher direct images, yields

$$p_a(Y) = p_a(C) - \text{length } \mathcal{Q} \leq p_a(C). \quad (7)$$

Finally, an integral plane curve of degree d has ideal sheaf $\mathcal{O}_{\mathbb{P}^2}(-d)$. The sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2}(-d) \longrightarrow \mathcal{O}_{\mathbb{P}^2} \longrightarrow \mathcal{O}_C \longrightarrow 0$$

gives

$$p_a(C) = \frac{(d-1)(d-2)}{2}. \quad (8)$$

Combining (1), (7), and (8) proves both inequalities. $\square$

III.10 Smooth Morphisms

Exercise III.10.1

Permitted context.

A localization of a regular ring is regular. A scheme of finite type over a field is smooth only if it remains regular after every field extension; equivalently, it is geometrically regular.

Problem. Let k_0 be a field of characteristic $p > 0$, put $k = k_0(t)$, and let

$$X = V(y^2 - x^p + t) \subseteq \mathbb{A}_k^2.$$

Show that every local ring of X is regular, but that X is not smooth over k .

Solution. Introduce an indeterminate T over k_0 , and map

$$k_0[T] \longrightarrow k_0[x, y], \quad T \longmapsto x^p - y^2.$$

This map is injective, and the coordinate ring of X can be written as

$$\begin{aligned} A &= k_0(T)[x, y]/(y^2 - x^p + T) \\ &\cong k_0[x, y] \otimes_{k_0[T]} k_0(T). \end{aligned} \quad (1)$$

Thus A is obtained from the polynomial ring $k_0[x, y]$ by inverting the nonzero elements of the subring $k_0[x^p - y^2]$. Since a polynomial ring over a field is regular and regularity is preserved by localization, A is a regular ring. Consequently every local ring of $X = \text{Spec } A$ is regular.

Now let

$$K = k(t^{1/p}), \quad \alpha = t^{1/p} \in K.$$

After this field extension the equation becomes

$$y^2 = (x - \alpha)^p. \quad (2)$$

At the K -rational point $(x, y) = (\alpha, 0)$, put $u = x - \alpha$. The local ring is the localization at (u, y) of

$$K[u, y]/(y^2 - u^p).$$

It has Krull dimension 1, whereas its maximal ideal has two independent generators modulo its square, because $y^2 - u^p \in (u, y)^2$. Its embedding dimension is therefore 2, so this local ring is not regular. Hence X_K is not regular. Smoothness is preserved by base change and a smooth scheme over a field is regular, so X cannot be smooth over k . $\square$

Exercise III.10.2

Permitted context.

For a flat morphism of finite presentation, a point of the source is a smooth point of the morphism if and only if the corresponding geometric fiber is regular there. The smooth locus is open. A proper morphism is closed.

Problem. Let $f : X \rightarrow Y$ be a proper flat morphism of varieties over k . Suppose that, for some $y \in Y$, the fiber X_y is smooth over $k(y)$. Show that there is an open neighborhood U of y such that

$$f^{-1}(U) \longrightarrow U$$

is smooth.

Solution. Let $X^{\text{sm}} \subseteq X$ be the smooth locus of f . It is open. Since f is flat and of finite presentation, the fiberwise criterion for smoothness applies. The hypothesis that X_y is smooth over $k(y)$ therefore gives

$$X_y \subseteq X^{\text{sm}}. \quad (1)$$

Set $Z = X \setminus X^{\text{sm}}$. This is closed, and its image $f(Z)$ is closed because f is proper. By (1), the point y does not belong to $f(Z)$. Hence

$$U = Y \setminus f(Z)$$

is an open neighborhood of y . Its inverse image is contained in the smooth locus, so the restricted morphism $f^{-1}(U) \rightarrow U$ is smooth. $\square$

Exercise III.10.3**Permitted context.**

For a morphism locally of finite type, formation of Kähler differentials commutes with base change. We use the following local criterion. If B is a local algebra essentially of finite type over a field K , with maximal ideal $\mathfrak{n}$ and residue field L , then

$$\Omega_{B/K} = 0 \iff \mathfrak{n} = 0 \text{ and } L/K \text{ is finite separable.} \quad (*)$$

For completeness, if the left side vanishes, then the quotient map to L gives $\Omega_{L/K} = 0$; the separating-basis theorem says that L/K is finite separable. Separability provides a K -algebra section

$$L \longrightarrow B/\mathfrak{n}^2.$$

Thus $B/\mathfrak{n}^2 = L \oplus \mathfrak{n}/\mathfrak{n}^2$, and projection onto the second summand is a K -derivation. But $\Omega_{(B/\mathfrak{n}^2)/K} = 0$, as follows from the quotient sequence for differentials, so this projection is zero. Hence $\mathfrak{n}/\mathfrak{n}^2 = 0$, and Nakayama's lemma gives $\mathfrak{n} = 0$. The reverse implication is the familiar vanishing of differentials for a finite separable field extension.

Problem. A morphism $f : X \rightarrow Y$ of schemes of finite type over k is called étale if it is smooth of relative dimension 0. It is called unramified if, for every $x \in X$, with $y = f(x)$,

$$\mathfrak{m}_y \mathcal{O}_{X,x} = \mathfrak{m}_x$$

and $k(x)/k(y)$ is a separable algebraic extension. Prove that the following are equivalent:

- (i) f is étale;
- (ii) f is flat and $\Omega_{X/Y} = 0$;
- (iii) f is flat and unramified.

Solution. Suppose first that f is étale. A smooth morphism is flat, and for a smooth morphism of relative dimension n , the sheaf $\Omega_{X/Y}$ is locally free of rank n . Here $n = 0$, so $\Omega_{X/Y} = 0$. Thus (i) implies (ii).

Assume (ii), and fix $x \in X$, with $y = f(x)$. The local ring of the fiber at x is

$$B = \mathcal{O}_{X,x}/\mathfrak{m}_y \mathcal{O}_{X,x},$$

and its residue field is $k(x)$. Base change for differentials gives

$$\Omega_{B/k(y)} \cong (\Omega_{X/Y})_x \otimes_{\mathcal{O}_{X,x}} B = 0.$$

Criterion (*) shows both that

$$\mathfrak{m}_x/\mathfrak{m}_y \mathcal{O}_{X,x} = 0$$

and that $k(x)/k(y)$ is finite separable. Hence f is unramified, proving (iii).

Finally, assume (iii). At the same point x , the equality of maximal ideals says that the local ring of the fiber is precisely

$$B = k(x),$$

and this is a finite separable extension of $k(y)$. Thus every fiber is zero-dimensional and is locally the spectrum of a finite separable residue-field extension. After any extension of $k(y)$ to an algebraic closure, the fiber becomes a disjoint union of reduced points. In other words, all geometric fibers are regular of dimension 0.

The morphism is flat by hypothesis and is of finite presentation because the schemes are of finite type over k . The fiberwise criterion for smoothness now shows that f is smooth of relative dimension 0, hence étale. This proves (i). $\square$

Exercise III.10.4

Permitted context.

Completion of a noetherian local ring is faithfully flat, and flatness descends through a faithfully flat algebra. We use the criterion that a finite-type morphism is étale exactly when it is flat and unramified.

If $(R, \mathfrak{m})$ is local and M is a finite flat R -module, then M is free. Its rank is the dimension of $M/\mathfrak{m}M$ over the residue field. Equicharacteristic complete noetherian local rings admit coefficient fields. If the residue extension of a local homomorphism is finite separable, a chosen coefficient field of the completed base has a compatible extension in the completed source: lift a primitive element by Hensel's lemma applied to its separable minimal polynomial. No compatibility is asserted for arbitrary inseparable residue extensions.

Problem. Let $f : X \rightarrow Y$ be a morphism of schemes of finite type over k . For $x \in X$, put $y = f(x)$, and let $\hat{\mathcal{O}}_{X,x}$ and $\hat{\mathcal{O}}_{Y,y}$ be the completed local rings. Prove that f is étale if and only if, for every x , the extension $k(x)/k(y)$ is finite separable and, after choosing a coefficient field in the completed base and its compatible finite separable lift in the completed source, the natural map

$$\hat{\mathcal{O}}_{Y,y} \otimes_{k(y)} k(x) \longrightarrow \hat{\mathcal{O}}_{X,x} \quad (1)$$

is an isomorphism.

Solution. Fix $x \in X$, set $y = f(x)$, and abbreviate

$$A = \mathcal{O}_{Y,y}, \quad B = \mathcal{O}_{X,x}, \quad \kappa = k(y), \quad \lambda = k(x).$$

Write $\mathfrak{m}_A$ and $\mathfrak{m}_B$ for the maximal ideals.

Suppose first that λ/κ is separable algebraic and that (1) is an isomorphism. Since f is of finite type, λ/κ is in fact finite. The maximal ideal of $\hat{A} \otimes_{\kappa} \lambda$ is $\hat{\mathfrak{m}}_A \otimes_{\kappa} \lambda$. Consequently (1) gives

$$\mathfrak{m}_A \hat{B} = \mathfrak{m}_B \hat{B}.$$

The map $B \rightarrow \hat{B}$ is faithfully flat, so contraction to B yields

$$\mathfrak{m}_A B = \mathfrak{m}_B. \quad (2)$$

Together with the separability of λ/κ , equation (2) says that $A \rightarrow B$ is unramified. Moreover,

$$\widehat{A} \longrightarrow \widehat{A} \otimes_{\kappa} \lambda \cong \widehat{B}$$

is flat. Since $A \rightarrow \widehat{A}$ is flat, the composite $A \rightarrow \widehat{B}$ is flat. Faithfully flat descent along $B \rightarrow \widehat{B}$ now shows that $A \rightarrow B$ is flat. This holds at every x , so f is flat and unramified, hence étale.

Conversely, suppose that f is étale. Then $A \rightarrow B$ is flat and unramified. In particular,

$$\mathfrak{m}_B = \mathfrak{m}_A B \quad \text{and} \quad \lambda/\kappa \text{ is finite separable.} \quad (3)$$

For each $n \geq 1$, put

$$A_n = A/\mathfrak{m}_A^n, \quad B_n = B/\mathfrak{m}_B^n.$$

By (3), $\mathfrak{m}_B^n = \mathfrak{m}_A^n B$, and hence

$$B_n = B \otimes_A A_n.$$

Thus B_n is flat over the local Artinian ring A_n . It is also finite over A_n : it is a zero-dimensional local algebra essentially of finite type whose residue field λ is finite over κ . Therefore B_n is finite free over A_n , of rank

$$\dim_{\kappa}(B_n/\mathfrak{m}_A B_n) = \dim_{\kappa} \lambda = [\lambda : \kappa]. \quad (4)$$

The compatible coefficient fields define an A_n -linear map

$$A_n \otimes_{\kappa} \lambda \longrightarrow B_n. \quad (5)$$

Modulo the maximal ideal of A_n , map (5) is the identity of λ , so Nakayama's lemma makes it surjective. By (4), its source and target are finite free A_n -modules of the same rank; therefore (5) is an isomorphism.

Taking inverse limits over n , and using the finiteness of λ/κ , gives

$$\widehat{A} \otimes_{\kappa} \lambda \cong \varprojlim_n (A_n \otimes_{\kappa} \lambda) \cong \varprojlim_n B_n = \widehat{B}.$$

This is precisely (1), and the proof is complete. $\square$

Exercise III.10.5

Permitted context.

A flat local homomorphism of local rings is faithfully flat. Flatness of modules descends along faithfully flat maps, and a finitely presented flat module is projective. A finite projective module over a local ring is free.

Problem. An étale neighborhood of a point x of a scheme X is an étale morphism $f : U \rightarrow X$, together with a point $x' \in U$ satisfying $f(x') = x$.

Let $\mathcal{F}$ be a coherent sheaf on X . Suppose every point of X has an étale neighborhood $f : U \rightarrow X$ for which $f^* \mathcal{F}$ is a free $\mathcal{O}_U$ -module. Prove that $\mathcal{F}$ is locally free on X .

Solution. Fix $x \in X$, and choose such a neighborhood (U, x') . Set

$$A = \mathcal{O}_{X,x}, \quad B = \mathcal{O}_{U,x'}, \quad M = \mathcal{F}_x.$$

Because f is étale, the local homomorphism $A \rightarrow B$ is flat. It is a local homomorphism, and is therefore faithfully flat. The hypothesis gives

$$M \otimes_A B \cong (f^* \mathcal{F})_{x'},$$

which is a finite free B -module. It is consequently flat over B , and hence over A . Faithfully flat descent now implies that M is flat over A .

The sheaf $\mathcal{F}$ is coherent, so M is a finitely presented A -module. A finitely presented flat module is projective, and over the local ring A it is therefore free, say of rank r .

Choose sections near x representing a basis of M . They give a homomorphism

$$\mathcal{O}_X^{\oplus r} \longrightarrow \mathcal{F}$$

which is an isomorphism on the stalk at x . Its finite-type cokernel vanishes after shrinking around x , giving a surjection. Since $\mathcal{F}$ is finitely presented, the kernel of this surjection from a finite free sheaf is of finite type. Its zero stalk at x therefore makes it vanish after a further shrinking. Thus $\mathcal{F}$ is free on that neighborhood. Since x was arbitrary, $\mathcal{F}$ is locally free. $\square$

Exercise III.10.6

Permitted context.

Because the cubic in the problem is nodal, we assume $\text{char } k \neq 2$. At an ordinary node the completed local ring is $k[[u, v]]/(uv)$. A finite-type morphism inducing an isomorphism on completed local rings, with identical residue fields, is étale at the points in question. In this record the nodal cubic is affine. For its projective closure, replace the two affine normalization lines by projective lines and perform the same crosswise gluing.

Problem. Let Y be the plane nodal cubic

$$y^2 = x^2(x + 1).$$

Construct a finite étale covering $p : X \rightarrow Y$ of degree 2 such that X is the union of two irreducible components, each isomorphic to the normalization of Y .

Solution. Put

$$A = k[x, y]/(y^2 - x^2(x + 1)).$$

The normalization $\nu : N \rightarrow Y$ is

$$N = \text{Spec } k[t], \quad x = t^2 - 1, \quad y = t(t^2 - 1). \quad (1)$$

Indeed, t is integral over A , since $t^2 = x + 1$, and away from $x = 0$ it is the rational function y/x . The inverse image of the node $P = (0, 0)$ consists of the two points

$$P_+ = (t - 1), \quad P_- = (t + 1).$$

Take two copies N_1, N_2 of N . Glue $P_+ \in N_1$ to $P_- \in N_2$, and glue $P_- \in N_1$ to $P_+ \in N_2$. The resulting affine curve can be written explicitly as

$$X = \operatorname{Spec} B,$$

where

$$B = \left\{ (f_1, f_2) \in k[t] \times k[t] : \begin{array}{l} f_1(1) = f_2(-1), \\ f_1(-1) = f_2(1) \end{array} \right\}. \quad (2)$$

The two projection maps $B \rightarrow k[t]$ are surjective: prescribed values at 1 and -1 can be attained by a linear polynomial. Their kernels are the two minimal primes of B . Thus X has exactly two irreducible components, and each component is isomorphic to N .

The two normalization maps in (1) agree at every pair of points identified in (2), so they induce a morphism

$$p : X \longrightarrow Y. \quad (3)$$

Algebraically, $A \rightarrow B$ is the diagonal map through $A \rightarrow k[t]$. The A -module $k[t]$ is finite, and B is an A -submodule of $k[t] \oplus k[t]$. Since A is noetherian, B is finite over A ; hence p is finite. Over $Y \setminus \{P\}$, the normalization is an isomorphism, so (3) is the disjoint union of two isomorphisms there.

It remains to check the two points of X over P . First,

$$\widehat{\mathcal{O}}_{Y,P} \cong k[[x, y]]/(y^2 - x^2(1 + x)) \cong k[[r, s]]/(rs). \quad (4)$$

For the second isomorphism, Hensel's lemma gives a power series $h(x)$ with $h(0) = 1$ and $h(x)^2 = 1 + x$. One may take

$$r = y - xh(x), \quad s = y + xh(x);$$

the change of variables is invertible because 2 is a unit and $xh(x)$ is a local parameter.

At either of the two crosswise identifications in X , the completed local ring is the fiber product of the completed rings of the two branches:

$$k[[u]] \times_k k[[v]] \cong k[[u, v]]/(uv). \quad (5)$$

Under p , these two branches map respectively to the two distinct branches of Y at P . The induced homomorphism from (4) to (5) is therefore an isomorphism. The residue fields are both k , so the completed-local-ring criterion shows that p is étale at both points over P . We already saw that it is an isomorphism locally away from those points. Hence p is finite étale.

The generic fiber consists of one point on each of the two components, so the finite étale cover has degree 2, as required. $\square$

Exercise III.10.7**Permitted context.**

The classification assertion in part (b), as printed, is incomplete: there are also members consisting of a smooth conic and its tangent line. The solution proves all the valid assertions, gives an explicit counterexample to the stated dichotomy, and proves the corrected classification.

We use the standard classification of singular irreducible plane cubics: a double point with two distinct tangent lines is a node, whereas a double point with one double tangent line is a cusp.

Problem. Let k be an algebraically closed field of characteristic 2, let

$$S = \mathbb{P}^2(\mathbb{F}_2) = \{P_1, \dots, P_7\} \subseteq X = \mathbb{P}_k^2,$$

and let $\mathfrak{d}$ be the linear system of cubic curves through the seven points.

(a) Show that $\mathfrak{d}$ has dimension 2, has base locus exactly S , and determines an inseparable morphism of degree 2

$$X \setminus S \longrightarrow \mathbb{P}^2.$$

(b) Show that every member of $\mathfrak{d}$ is singular and that sending a member to its singular point gives a bijection

$$\mathfrak{d} \longleftrightarrow \mathbb{P}^2.$$

Prove the following classification, writing $\mathcal{A}$ for the union of the seven $\mathbb{F}_2$ -lines and P for the unique singular point of a member C :

1. if $P \in S$, then C is the union of the three $\mathbb{F}_2$ -lines through P ;
2. if $P \in \mathcal{A} \setminus S$, then C is the union of an $\mathbb{F}_2$ -line and a smooth conic tangent to it at P ;
3. if $P \notin \mathcal{A}$, then C is an irreducible cuspidal cubic with cusp P .

Correction to the printed statement. The original dichotomy lists only three concurrent lines or an irreducible cuspidal cubic. It omits the line-plus-tangent-conic case above, as the explicit factorization in the solution demonstrates.

Solution. Use homogeneous coordinates $[x_0 : x_1 : x_2]$, and put

$$\begin{aligned} F_0 &= x_1x_2(x_1 + x_2), \\ F_1 &= x_2x_0(x_2 + x_0), \\ F_2 &= x_0x_1(x_0 + x_1). \end{aligned} \tag{1}$$

(a) Let us first determine the linear system. Let F be a homogeneous cubic vanishing on S . Vanishing at the three coordinate points kills the coefficients of x_0^3, x_1^3, x_2^3 . Vanishing at $[1 : 1 : 0]$, $[1 : 0 : 1]$, and $[0 : 1 : 1]$ identifies the two coefficients involving each pair of variables. Finally, vanishing at $[1 : 1 : 1]$ kills the coefficient of $x_0x_1x_2$. Therefore

$$H^0(\mathbb{P}^2, \mathcal{I}_S(3)) = kF_0 \oplus kF_1 \oplus kF_2. \tag{2}$$

Thus the projective linear system has dimension 2.

The equations $F_0 = F_1 = F_2 = 0$ say that, for each pair of coordinates, either one coordinate is zero or the two are equal. After projective scaling, every common zero consequently has all coordinates in $\mathbb{F}_2$. Conversely, the forms in (1) vanish at every such point. Hence the base locus is exactly S , and the system defines

$$\phi : \mathbb{P}^2 \setminus S \longrightarrow \mathbb{P}^2, \quad [x_0 : x_1 : x_2] \longmapsto [F_0 : F_1 : F_2]. \quad (3)$$

We compute its function-field degree. On the chart $x_1 \neq 0$, set

$$q = \frac{x_0}{x_1}, \quad r = \frac{x_2}{x_1},$$

and, on the target chart $F_2 \neq 0$, set

$$A = \frac{F_0}{F_2} = \frac{r(r+1)}{q(q+1)}, \quad B = \frac{F_1}{F_2} = \frac{r(r+q)}{q+1}. \quad (4)$$

The identities

$$\frac{F_1(F_0 + F_1)}{F_2(F_0 + F_2)} = r^2 \quad (5)$$

and

$$\frac{F_0 + F_1 + F_2}{F_2(1+r)} + 1 = \frac{r}{q} \quad (6)$$

show that

$$k(q, r) = k(A, B)(r), \quad r^2 \in k(A, B). \quad (7)$$

In particular, ϕ is dominant and its degree is at most 2. It is not generically separable. Indeed, if

$$D = q(q+1), \quad N = r(r+1),$$

then differentiation of (4) in characteristic 2 gives

$$dA = \frac{D dr + N dq}{D^2}, \quad dB = \frac{D dr + N dq}{(q+1)^2}. \quad (8)$$

These two differentials are nonzero and proportional, so the generic differential of ϕ has rank 1, not 2. The extension in (7) is therefore nontrivial and purely inseparable. Its degree is exactly 2.

(b) We begin with the singular-point correspondence. Every element of k has a unique square root. Thus a member of $\mathfrak{d}$ can be written uniquely, up to simultaneous scaling, as

$$C_{a,b,c} : \quad a^2 F_0 + b^2 F_1 + c^2 F_2 = 0 \quad (9)$$

for a point $[a : b : c] \in \mathbb{P}^2$. Its partial derivatives are

$$\begin{aligned} \frac{\partial F}{\partial x_0} &= (bx_2 + cx_1)^2, \\ \frac{\partial F}{\partial x_1} &= (ax_2 + cx_0)^2, \\ \frac{\partial F}{\partial x_2} &= (ax_1 + bx_0)^2. \end{aligned} \quad (10)$$

The alternating matrix of the three linear forms inside the squares has rank 2, and its kernel is spanned by (a, b, c) . Consequently the three derivatives have the unique common zero

$$P_{a,b,c} = [a : b : c]. \quad (11)$$

Euler's identity gives $F(P_{a,b,c}) = 0$, so this point lies on the curve and is singular. This proves both that every member is singular and that

$$C_{a,b,c} \longmapsto P_{a,b,c}$$

is a bijection from $\mathfrak{d}$ to $\mathbb{P}^2$.

We prove these claims. The group $\mathrm{PGL}_3(\mathbb{F}_2)$ preserves S , the linear system, and the singular-point correspondence. It is therefore enough, for a point on an $\mathbb{F}_2$ -line, to consider the line $x_0 = 0$. Write such a point as $P = [0 : b : c]$. Its member factors as

$$C_{0,b,c} = V(x_0Q), \quad Q = b^2x_2(x_2 + x_0) + c^2x_1(x_0 + x_1). \quad (12)$$

If $[0 : b : c]$ is an $\mathbb{F}_2$ -point, (12) splits into the three $\mathbb{F}_2$ -lines through that point. For example, at $[1 : 0 : 0]$ the corresponding equation is

$$F_0 = x_1x_2(x_1 + x_2).$$

Now suppose $b, c, b + c \neq 0$, which is precisely the case in which $[0 : b : c]$ lies on $x_0 = 0$ but is not an $\mathbb{F}_2$ -point. The derivatives of Q are

$$Q_{x_0} = b^2x_2 + c^2x_1, \quad Q_{x_1} = c^2x_0, \quad Q_{x_2} = b^2x_0.$$

Their only common projective zero is $[0 : b^2 : c^2]$, and

$$Q(0, b^2, c^2) = b^2c^2(b + c)^2 \neq 0.$$

Thus Q is a smooth conic. Its restriction to $x_0 = 0$ is

$$Q|_{x_0=0} = (bx_2 + cx_1)^2,$$

so the line is tangent to the conic at $[0 : b : c]$. This proves the first two cases.

Conversely, suppose a member is reducible. It has a line component L . If L were not defined over $\mathbb{F}_2$, it could contain at most one point of S , so the residual conic would contain at least six points of S . No nonzero conic does so. To see this, use an $\mathbb{F}_2$ -projective transformation to make the omitted point $[1 : 1 : 1]$. A quadratic through the other six points first has all three square coefficients zero by evaluating at the coordinate points, and then has all three mixed coefficients zero by evaluating at $[1 : 1 : 0]$, $[1 : 0 : 1]$, and $[0 : 1 : 1]$.

Hence L is an $\mathbb{F}_2$ -line. Every intersection of L with the residual conic is singular on the cubic. Since (11) is the unique singular point, $P_{a,b,c} \in L$. We have therefore proved that a member is reducible exactly when its singular point lies in $\mathcal{A}$.

It remains to identify the singularity outside $\mathcal{A}$. After permuting coordinates, assume $a \neq 0$ and scale so that $a = 1$. On $x_0 = 1$, translate the singular point by writing

$$x_1 = b + u, \quad x_2 = c + v.$$

A direct expansion of (9) gives the local equation

$$uv(u+v) + c(c+1)u^2 + b(b+1)v^2 = 0. \quad (13)$$

Outside $\mathcal{A}$, the quadratic part is nonzero, and over the algebraically closed field k it is the square

$$\left(\sqrt{c(c+1)}u + \sqrt{b(b+1)}v\right)^2. \quad (14)$$

Thus the singularity has multiplicity 2 and a single double tangent. The curve is irreducible by the criterion just proved, so the standard classification of singular plane cubics shows that (13) is a cusp. This completes the corrected classification.

For an explicit counterexample to the dichotomy printed in the exercise, choose $\lambda \in k \setminus \mathbb{F}_2$. The member singular at $[0 : 1 : \lambda]$ is

$$x_0(x_2(x_0 + x_2) + \lambda^2 x_1(x_0 + x_1)) = 0,$$

which is a line together with a smooth tangent conic, not either of the two printed alternatives. $\square$

Exercise III.10.8

Permitted context.

The assertion labelled any characteristic is false for the printed equation in characteristic 2: the alleged conic becomes a double line. We prove the statement when $\text{char } k \neq 2$, explain the failure in characteristic 2, and give a replacement that works in every characteristic. There is a second qualification: the cones are defined only for $t \neq 0$, then completed to a pencil by taking the limiting quadric. The literal cone from $P_0 \in C$ is the plane $z = 0$, not that limiting quadric.

Problem. In affine 3-space with coordinates x, y, z , let C be the conic

$$(x-1)^2 + y^2 = 1$$

in the xy -plane, and let $P_t = (0, 0, t)$ be a point of the z -axis. For $t \neq 0$, let Y_t be the closure in $\mathbb{P}^3$ of the cone over C with vertex P_t . Show that, as t varies, the surfaces Y_t extend to a linear system of dimension 1, with a moving singularity at P_t , and that its base locus is C together with the z -axis.

Solution. Use homogeneous coordinates $[x : y : z : w]$ on $\mathbb{P}^3$, and assume first that $\text{char } k \neq 2$. The projective closure of C lies in $z = 0$ and has equation

$$Q = x^2 + y^2 - 2xw = 0. \quad (1)$$

For $t \neq 0$, project a point $[x : y : z : w]$ from $P_t = [0 : 0 : t : 1]$ to the plane $z = 0$. The resulting point is

$$[tx : ty : 0 : tw - z].$$

Requiring this point to satisfy (1) gives

$$tQ + 2xz = 0. \quad (2)$$

Thus the cones belong to the pencil

$$Y_{[s:t]} = V(tQ + 2sxz), \quad [s : t] \in \mathbb{P}^1. \quad (3)$$

The two quadrics Q and $2xz$ are linearly independent, so (3) is a linear system of projective dimension 1. The completed affine family is the chart $s = 1$; at $t = 0$, equation (3) gives its flat limiting member $V(xz)$. This is a prescribed limit, not the literal cone from P_0 . At $t = 0$, the lines joining $P_0 \in C$ to the other points of C fill the plane $z = 0$. The complete pencil also includes the member at $[s : t] = [0 : 1]$.

For $s = 1$, let

$$G_t = t(x^2 + y^2 - 2xw) + 2xz.$$

Its partial derivatives are

$$\begin{aligned} (G_t)_x &= 2tx - 2tw + 2z, & (G_t)_y &= 2ty, \\ (G_t)_z &= 2x, & (G_t)_w &= -2tx. \end{aligned}$$

They all vanish at $P_t = [0 : 0 : t : 1]$, so P_t is singular on Y_t . As t varies, this singular point moves along the z -axis. For $t \neq 0$, it is the vertex of the cone. The limiting member $Y_0 = V(xz)$ is singular along the intersection of its two plane components, which still contains P_0 .

The set-theoretic base locus of the pencil is

$$V(Q, xz).$$

On $z = 0$, this is precisely the projective closure of C . On $x = 0$, equation $Q = 0$ becomes $y^2 = 0$, whose underlying set is the line $x = y = 0$, namely the projective z -axis. Hence

$$|V(Q, xz)| = C \cup V(x, y). \quad (4)$$

Scheme-theoretically, the base scheme has a nonreduced structure along the axis; (4) is the base locus asserted in the exercise.

In characteristic 2, however,

$$Q = x^2 + y^2 = (x + y)^2, \quad 2xz = 0.$$

The curve called C is a double line, and all the equations (2) with $t \neq 0$ define the same double plane. Thus the displayed family has dimension 0, not 1, and the statement as printed cannot hold in characteristic 2.

For comparison, the intended phenomenon does have a characteristic-free version. Replace C by the smooth conic

$$xw - y^2 = 0, \quad z = 0.$$

For $t \neq 0$, the cones with vertex P_t have equations

$$t(xw - y^2) - xz = 0,$$

and extend at $t = 0$ by the flat limit $V(xz)$. These equations span the pencil generated by $xw - y^2$ and xz in every characteristic. The same derivative and base-locus calculations give the moving vertex P_t and the base set consisting of this conic and the z -axis. $\square$

Exercise III.10.9**Permitted context.**

We use the local criterion for flatness in the following form. Let $(A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$ be a local homomorphism of noetherian local rings. If $a \in \mathfrak{m}$ is a nonzerodivisor on both A and B , and B/aB is flat over A/aA , then B is flat over A .

In a Cohen–Macaulay local ring, every system of parameters is a regular sequence.

Problem. Let $f : X \rightarrow Y$ be a morphism of varieties over k . Assume that Y is regular, X is Cohen–Macaulay, and every fiber of f has dimension

$$\dim X - \dim Y.$$

Prove that f is flat.

Solution. Put

$$d = \dim X - \dim Y.$$

The fiber-dimension theorem first shows that f is dominant: if the closure of its image had dimension smaller than $\dim Y$, the generic fiber over that image would have dimension greater than d . Moreover, every irreducible component of every fiber has dimension at least d . Since every fiber has dimension exactly d , all fibers are pure of dimension d .

Fix $x \in X$, put $y = f(x)$, and set

$$A = \mathcal{O}_{Y,y}, \quad B = \mathcal{O}_{X,x}.$$

The local ring of the fiber is

$$B/\mathfrak{m}_A B = \mathcal{O}_{X_y,x}. \tag{1}$$

Dimension theory for varieties, together with the purity just proved, gives

$$\dim B = \dim A + \dim(B/\mathfrak{m}_A B). \tag{2}$$

For clarity, this can be seen from transcendence degrees:

$$\begin{aligned} \dim B - \dim A &= (\dim X - \operatorname{trdeg}_k k(x)) - (\dim Y - \operatorname{trdeg}_k k(y)) \\ &= d - \operatorname{trdeg}_{k(y)} k(x) = \dim \mathcal{O}_{X_y,x}. \end{aligned}$$

The ring A is regular. Choose a regular system of parameters

$$a_1, \dots, a_n, \quad n = \dim A.$$

By (2),

$$\dim B/(a_1, \dots, a_n)B = \dim B - n. \tag{3}$$

Since B is Cohen–Macaulay, equation (3) says that the images of $a_1, \dots, a_n$ form part of a system of parameters of B . They are therefore a B -regular sequence.

We now prove that $A \rightarrow B$ is flat by induction on n . If $n = 0$, then A is a field and there is nothing to prove. If $n > 0$, the element a_1 is a nonzerodivisor on both A and B . The quotient A/a_1A is regular, the quotient B/a_1B is Cohen–Macaulay, and the remaining images $a_2, \dots, a_n$ satisfy the same dimension equality. By induction,

$$B/a_1B \text{ is flat over } A/a_1A.$$

The local criterion for flatness from the context then implies that B is flat over A . This holds for every $x \in X$, so f is flat. $\square$

III.11 The Theorem on Formal Functions

Exercise III.11.1

Permitted context.

For a quasi-compact separated morphism of noetherian schemes, higher direct images of quasi-coherent sheaves are quasi-coherent. On an affine base, the Leray spectral sequence therefore identifies their global sections with the cohomology of the source.

Problem. Show that the fiber-dimension vanishing theorem for higher direct images is false without the projective hypothesis. More precisely, let $n \geq 2$,

$$X = \mathbb{A}_k^n, \quad P = (0, \dots, 0), \quad U = X \setminus \{P\},$$

and let $j : U \hookrightarrow X$ be the inclusion. Show that every fiber of j has dimension at most 0, but

$$R^{n-1}j_*\mathcal{O}_U \neq 0.$$

Solution. An open immersion has either an empty fiber or a fiber consisting of one reduced point, so all fibers have dimension at most 0.

Write

$$A = k[x_1, \dots, x_n].$$

The principal opens $D(x_1), \dots, D(x_n)$ cover U . Their intersections are affine, so this cover computes the cohomology of $\mathcal{O}_U$. The last differential in its Čech complex has cokernel

$$H^{n-1}(U, \mathcal{O}_U) \cong \frac{A_{x_1 \cdots x_n}}{\sum_{i=1}^n A_{x_1 \cdots \widehat{x_i} \cdots x_n}}. \quad (1)$$

Every Laurent monomial in a summand of the denominator has nonnegative exponent in at least one variable. Hence the class of

$$\frac{1}{x_1 \cdots x_n}$$

is nonzero in (1). More precisely, the quotient has k -basis

$$\{x_1^{-a_1} \cdots x_n^{-a_n} : a_1, \dots, a_n \geq 1\}. \quad (2)$$

Since X is affine and the higher direct images are quasi-coherent, Leray gives

$$\Gamma(X, R^{n-1}j_*\mathcal{O}_U) \cong H^{n-1}(U, \mathcal{O}_U).$$

The right side is nonzero by (2), so $R^{n-1}j_*\mathcal{O}_U \neq 0$, despite the zero-dimensional fibers. $\square$

Exercise III.11.2

Permitted context.

If a projective morphism has fibers of dimension at most r , then $R^i f_*\mathcal{F} = 0$ for every coherent $\mathcal{F}$ and every $i > r$. Serre's cohomological criterion says that a noetherian scheme Z is affine if $H^i(Z, \mathcal{F}) = 0$ for every coherent $\mathcal{F}$ and every $i > 0$. An affine proper morphism is finite. The schemes in this exercise are noetherian.

Problem. Prove that a projective morphism with finite fibers, equivalently a projective quasi-finite morphism, is finite.

Solution. Let $f : X \rightarrow Y$ be projective and quasi-finite. It is enough to prove that $f^{-1}(V)$ is affine for every affine open $V \subseteq Y$. Replace Y by such a V , and let $\mathcal{F}$ be any coherent sheaf on X .

Every fiber of f has dimension 0. The projective fiber-dimension theorem therefore gives

$$R^i f_*\mathcal{F} = 0 \quad (i > 0).$$

These higher direct images are coherent, and the base is affine, so the Leray spectral sequence yields

$$H^i(X, \mathcal{F}) = \Gamma(Y, R^i f_*\mathcal{F}) = 0 \quad (i > 0).$$

Serre's criterion now shows that X is affine. Thus f is an affine morphism. It is also proper because it is projective, and an affine proper morphism is finite. Hence f is finite. $\square$

Exercise III.11.3**Permitted context.**

Stein factorization writes a projective morphism as a morphism with connected fibers followed by a finite morphism. The pullback of an ample line bundle by a finite morphism is ample. On a normal projective variety of dimension at least 2, the support of every effective ample Cartier divisor is connected.

Problem. Let X be a normal projective variety over an algebraically closed field k . Let $\mathfrak{d}$ be a base-point-free linear system of effective Cartier divisors. Suppose that $\mathfrak{d}$ is not composite with a pencil: if

$$f : X \longrightarrow \mathbb{P}_k^n$$

is the morphism defined by $\mathfrak{d}$, then $\dim f(X) \geq 2$. Prove that every divisor in $\mathfrak{d}$ is connected.

Solution. Take the Stein factorization of f :

$$X \xrightarrow{g} Z \xrightarrow{h} f(X) \subseteq \mathbb{P}^n, \quad (1)$$

where g has connected fibers, h is finite, and $g_*\mathcal{O}_X = \mathcal{O}_Z$.

The variety Z is normal. Indeed, for an affine open $V \subseteq Z$,

$$\Gamma(V, \mathcal{O}_Z) = \Gamma(g^{-1}(V), \mathcal{O}_X).$$

The latter ring is integrally closed in $k(X)$, since $g^{-1}(V)$ is an open subset of the normal integral variety X . Also, because h is finite,

$$\dim Z = \dim f(X) \geq 2. \quad (2)$$

The finite morphism h makes Z projective, and

$$\mathcal{L} = h^*\mathcal{O}_{\mathbb{P}^n}(1)$$

is ample on Z .

Let $D \in \mathfrak{d}$, and let $H \subseteq \mathbb{P}^n$ be the hyperplane corresponding to D . Put

$$E = h^{-1}(H).$$

The hyperplane does not contain $f(X)$, so E is an effective Cartier divisor associated to the ample line bundle $\mathcal{L}$. By the connectedness theorem and (2), E is connected. Moreover,

$$D = f^{-1}(H) = g^{-1}(E). \quad (3)$$

It remains only to pass connectedness through g . The restriction

$$g^{-1}(E) \longrightarrow E$$

is a proper surjection with connected fibers. If its source were a disjoint union of two nonempty open-and-closed subsets, their images would be closed, would cover E , and could not meet because each fiber is connected. This would disconnect E , a contradiction. Thus $g^{-1}(E)$, and hence D by (3), is connected. $\square$

Exercise III.11.4**Permitted context.**

We use the degree-zero theorem on formal functions. We also use the following standard facts. Idempotents, equivalently open-and-closed decompositions, lift uniquely across nilpotent thickenings. A finite torsion-free module over a discrete valuation ring is free. Finally, if the geometric generic fiber of a finite-type morphism over an integral curve is disconnected, then, after a finite extension of the function field and normalization of the base, that disconnection spreads out over a nonempty open subset.

Problem. Principle of connectedness. Let $\{X_t\}$ be a flat family of closed subschemes of $\mathbb{P}_k^n$, parametrized by an irreducible curve T of finite type over the algebraically closed field k . Suppose that there is a nonempty open subset $U \subseteq T$ such that X_t is connected for every closed point $t \in U$. Prove that X_t is connected for every $t \in T$.

Solution. The hypothesis also makes the generic fiber geometrically connected: otherwise its disconnection spreads out after a finite extension of the function field, producing a disconnected closed fiber over the given open set U . This is impossible. It remains to treat the closed fibers.

Let

$$p : X \longrightarrow T$$

be the projective flat morphism representing the family. First replace T by its normalization $\nu : T' \rightarrow T$ and X by $X \times_T T'$. This does not change the question: ν is finite and surjective, and, because k is algebraically closed, the fiber over a closed point t' above t is just X_t . The inverse image of U is again nonempty. We may therefore assume that T is regular.

Fix a closed point $t \in T$, and put

$$A = \mathcal{O}_{T,t}, \quad \mathfrak{m} = (\pi), \quad K = \text{Frac}(A).$$

Thus A is a discrete valuation ring. Set

$$B = (p_* \mathcal{O}_X)_t.$$

Properness makes B a finite A -module. It is torsion-free: multiplication by any nonzero element of A is injective on $\mathcal{O}_X$, because X is flat over T , and remains injective after applying the left-exact functor p_* . Hence B is a finite free A -module.

Suppose, toward a contradiction, that X_t is disconnected. For $r \geq 0$, let

$$X_r = X \times_T \text{Spec}(A/\mathfrak{m}^{r+1}).$$

Every X_r has the same underlying topological space as X_t . The nontrivial open-and-closed decomposition of X_t consequently lifts uniquely to each X_r , and these lifts give compatible nontrivial idempotents

$$e_r \in H^0(X_r, \mathcal{O}_{X_r}).$$

The theorem on formal functions gives

$$\widehat{B} \cong \varprojlim_r H^0(X_r, \mathcal{O}_{X_r}), \tag{1}$$

so the e_r determine an idempotent $e \in \widehat{B}$. Its reduction modulo $\mathfrak{m}$ is neither 0 nor 1.

Because B is free over A , both $\widehat{B}$ and its direct summands $e\widehat{B}$ and $(1-e)\widehat{B}$ are torsion-free. Thus e remains nontrivial after inverting π . Flat base change identifies

$$\widehat{B}[1/\pi] \cong H^0(X_{\text{Frac}(\widehat{A})}, \mathcal{O}),$$

and the surviving idempotent shows that $X_{\text{Frac}(\widehat{A})}$ is disconnected. Therefore the generic fiber X_K is not geometrically connected.

This contradicts geometric connectedness of the generic fiber, established at the start. Thus X_t is connected. Since t was arbitrary, all fibers are connected. $\square$

Exercise III.11.5

Permitted context.

For a square-zero thickening $Z \subseteq Z'$ with ideal sheaf $\mathcal{J}$, the sequence

$$1 \longrightarrow 1 + \mathcal{J} \longrightarrow \mathcal{O}_{Z'}^* \longrightarrow \mathcal{O}_Z^* \longrightarrow 1, \quad 1 + \mathcal{J} \cong \mathcal{J},$$

gives the exact segment

$$H^1(Z, \mathcal{J}) \longrightarrow \text{Pic } Z' \longrightarrow \text{Pic } Z \longrightarrow H^2(Z, \mathcal{J}).$$

We also use $\text{Pic } \widehat{X} \cong \varprojlim_n \text{Pic } Y_n$ for the infinitesimal neighborhoods of Y .

Problem. Let Y be a hypersurface in $X = \mathbb{P}_k^N$, where $N \geq 4$, and let $\widehat{X}$ be the formal completion of X along Y . Prove that the natural map

$$\text{Pic } \widehat{X} \longrightarrow \text{Pic } Y$$

is an isomorphism.

Solution. Let Y have degree d , let $\mathcal{I} = \mathcal{O}_X(-d)$ be its ideal sheaf, and write

$$Y_n = (|Y|, \mathcal{O}_X/\mathcal{I}^{n+1}) \quad (n \geq 0).$$

Thus $Y_0 = Y$. The ideal of Y_n in Y_{n+1} is square-zero and is naturally

$$\mathcal{I}^{n+1}/\mathcal{I}^{n+2} \cong \mathcal{O}_Y(-(n+1)d). \quad (1)$$

Consequently the square-zero Picard sequence gives

$$H^1(Y, \mathcal{O}_Y(-(n+1)d)) \longrightarrow \text{Pic } Y_{n+1} \longrightarrow \text{Pic } Y_n \longrightarrow H^2(Y, \mathcal{O}_Y(-(n+1)d)). \quad (2)$$

For every integer m , tensoring the hypersurface sequence by $\mathcal{O}_X(m)$ gives

$$0 \longrightarrow \mathcal{O}_X(m-d) \longrightarrow \mathcal{O}_X(m) \longrightarrow \mathcal{O}_Y(m) \longrightarrow 0. \quad (3)$$

On $\mathbb{P}^N$,

$$H^q(\mathbb{P}^N, \mathcal{O}(m)) = 0 \quad (0 < q < N)$$

for every m . Since $N \geq 4$, the long exact sequence of (3) therefore yields

$$H^1(Y, \mathcal{O}_Y(m)) = H^2(Y, \mathcal{O}_Y(m)) = 0 \quad \text{for every } m \in \mathbb{Z}. \quad (4)$$

By (2) and (4), every transition map

$$\text{Pic } Y_{n+1} \xrightarrow{\sim} \text{Pic } Y_n$$

is an isomorphism. Taking inverse limits gives

$$\text{Pic } \widehat{X} \cong \varprojlim_n \text{Pic } Y_n \cong \text{Pic } Y_0 = \text{Pic } Y,$$

as required. $\square$

Exercise III.11.6**Permitted context.**

Let $Y \subseteq X = \mathbb{P}_k^N$ be a hypersurface of degree d , with ideal $\mathcal{I} = \mathcal{O}_X(-d)$, and put $Y_n = (|Y|, \mathcal{O}_X/\mathcal{I}^{n+1})$. Sections on the formal completion are inverse limits of sections on the Y_n . Completion is exact on coherent sheaves in this noetherian setting.

We use Serre vanishing and Serre duality on projective space. We also use that completion of a noetherian local ring along an ideal contained in its maximal ideal is faithfully flat.

Problem. Again let Y be a hypersurface in $X = \mathbb{P}_k^N$, this time with $N \geq 2$. (a) If $\mathcal{F}$ is locally free on X , prove that the natural map

$$H^0(X, \mathcal{F}) \longrightarrow H^0(\widehat{X}, \widehat{\mathcal{F}})$$

is an isomorphism.

(b) Prove that the following conditions are equivalent:

- (i) every locally free sheaf $\mathfrak{F}$ on $\widehat{X}$ is algebraizable, meaning that $\mathfrak{F} \cong \widehat{\mathcal{F}}$ for some coherent sheaf $\mathcal{F}$ on X ;
- (ii) for every locally free sheaf $\mathfrak{F}$ on $\widehat{X}$, there is an integer n_0 such that $\mathfrak{F}(n)$ is generated by global sections for all $n \geq n_0$.

(c) Show that these equivalent conditions imply that

$$\mathrm{Pic} X \longrightarrow \mathrm{Pic} \widehat{X}$$

is an isomorphism.

Solution. (a) Tensoring

$$0 \longrightarrow \mathcal{O}_X(-(n+1)d) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_{Y_n} \longrightarrow 0$$

with the locally free sheaf $\mathcal{F}$ gives

$$0 \longrightarrow \mathcal{F}(-(n+1)d) \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}|_{Y_n} \longrightarrow 0. \quad (1)$$

For $m \gg 0$, one has

$$H^0(X, \mathcal{F}(-m)) = H^1(X, \mathcal{F}(-m)) = 0. \quad (2)$$

The first vanishing is the standard negative-twist vanishing for a vector bundle. For the second, Serre duality gives

$$H^1(X, \mathcal{F}(-m))^\vee \cong H^{N-1}(X, \mathcal{F}^\vee(m-N-1)),$$

which vanishes for $m \gg 0$ by Serre vanishing; here $N \geq 2$. It follows from (1) and (2) that, for all sufficiently large n , restriction is an isomorphism

$$H^0(X, \mathcal{F}) \xrightarrow{\sim} H^0(Y_n, \mathcal{F}|_{Y_n}).$$

The inverse system is therefore eventually constant, and hence

$$H^0(\widehat{X}, \widehat{\mathcal{F}}) = \varprojlim_n H^0(Y_n, \mathcal{F}|_{Y_n}) \cong H^0(X, \mathcal{F}).$$

(b) Assume (i), and write $\mathfrak{F} \cong \widehat{\mathcal{F}}$ with $\mathcal{F}$ coherent on X . By Serre's theorem, $\mathcal{F}(n)$ is generated by global sections for all $n \gg 0$. Completing a surjection from a finite free sheaf shows that $\mathfrak{F}(n)$ is generated by global sections for the same values of n . Thus (ii) holds.

Conversely, assume (ii), and let $\mathfrak{F}$ be locally free on $\widehat{X}$. Choose n so that $\mathfrak{F}(n)$ is globally generated. Finitely many sections give a surjection

$$\widehat{\mathcal{E}}_1 = \mathcal{O}_{\widehat{X}}(-n)^{\oplus r} \rightarrow \mathfrak{F}. \quad (3)$$

Its kernel $\mathfrak{K}$ is locally free: locally the target of (3) is free, and a surjection onto a finite free module splits. Apply (ii) to $\mathfrak{K}$. For a suitable m , finitely many global sections give a surjection

$$\widehat{\mathcal{E}}_2 = \mathcal{O}_{\widehat{X}}(-m)^{\oplus s} \rightarrow \mathfrak{K}.$$

After composing with the inclusion in (3), we obtain a presentation

$$\widehat{\mathcal{E}}_2 \xrightarrow{\widehat{\varphi}} \widehat{\mathcal{E}}_1 \rightarrow \mathfrak{F} \rightarrow 0, \quad (4)$$

where

$$\mathcal{E}_1 = \mathcal{O}_X(-n)^{\oplus r}, \quad \mathcal{E}_2 = \mathcal{O}_X(-m)^{\oplus s}.$$

The bundle $\mathcal{H}om(\mathcal{E}_2, \mathcal{E}_1)$ is locally free, so part (a) identifies

$$\begin{aligned} \mathrm{Hom}_X(\mathcal{E}_2, \mathcal{E}_1) &= H^0(X, \mathcal{H}om(\mathcal{E}_2, \mathcal{E}_1)) \\ &\xrightarrow{\sim} H^0(\widehat{X}, \mathcal{H}om(\widehat{\mathcal{E}}_2, \widehat{\mathcal{E}}_1)). \end{aligned} \quad (5)$$

Thus $\widehat{\varphi}$ lifts uniquely to a morphism $\varphi : \mathcal{E}_2 \rightarrow \mathcal{E}_1$. Put

$$\mathcal{F} = \mathrm{coker} \varphi.$$

Exactness of completion and (4) give

$$\widehat{\mathcal{F}} \cong \mathfrak{F}.$$

Hence $\mathfrak{F}$ is algebraizable, proving (i).

(c) First consider injectivity. Since $\mathrm{Pic}(\mathbb{P}^N) \cong \mathbb{Z}$, every line bundle on X is $\mathcal{O}_X(a)$. If its completion is trivial, then $\mathcal{O}_Y(a)$ is trivial. For $a \neq 0$, this would make a positive power of the ample line bundle $\mathcal{O}_Y(1)$ trivial. That is impossible on the positive-dimensional projective scheme Y . Thus $a = 0$, and the map on Picard groups is injective.

For surjectivity, let $\mathfrak{L}$ be an invertible sheaf on $\widehat{X}$. By (i), there is a coherent sheaf $\mathcal{F}$ on X with

$$\widehat{\mathcal{F}} \cong \mathfrak{L}. \quad (6)$$

At every $y \in Y$, the completion of the finite $\mathcal{O}_{X,y}$ -module $\mathcal{F}_y$ is free of rank one. Faithful flatness of completion therefore implies that $\mathcal{F}_y$ itself is free of rank one. The locally free locus is open, so $\mathcal{F}$ is an invertible sheaf on some open neighborhood V of Y .

The complement $X \setminus V$ is finite. Indeed, a positive-dimensional closed projective subvariety of X must meet the ample hypersurface Y ; otherwise the defining section of $\mathcal{O}_X(d)$ would trivialize an ample line bundle on that subvariety. Hence $X \setminus V$ has dimension zero. Since $X = \mathbb{P}^N$ is regular and $N \geq 2$, restriction across a closed subset of codimension at least two induces an isomorphism

$$\mathrm{Pic} X \xrightarrow{\sim} \mathrm{Pic} V.$$

Thus $\mathcal{F}|_V$ extends to a line bundle $\mathcal{L}$ on X . Completion only sees a neighborhood of Y , so (6) gives $\widehat{\mathcal{L}} \cong \mathfrak{L}$. This proves surjectivity and completes the proof. $\square$

Exercise III.11.7**Permitted context.**

There is a characteristic issue in part (a). The kernel is naturally an infinite-dimensional k -vector space when $\text{char } k = 0$, by the formal logarithm. In positive characteristic it is generally not a vector space under tensor product. The characteristic-free statement is that it has a complete filtration whose associated graded pieces are finite dimensional k -vector spaces, with infinitely many nonzero pieces. We prove this precise statement, the stated characteristic-zero form, and the conclusions of (b) and (c) in every characteristic.

Problem. Let Y be a curve in $X = \mathbb{P}_k^2$, and let $\widehat{X}$ be the formal completion of X along Y . (a) Use the method of Exercise III.11.5 to show that

$$\text{Pic } \widehat{X} \longrightarrow \text{Pic } Y$$

is surjective and that its kernel is an infinite-dimensional vector space over k .

(b) Conclude that there is an invertible sheaf on $\widehat{X}$ which is not algebraizable.

(c) Conclude that there is a locally free sheaf $\mathfrak{F}$ on $\widehat{X}$ such that no twist $\mathfrak{F}(n)$ is generated by global sections.

Solution. Let Y have degree d , let $\mathcal{I} = \mathcal{O}_X(-d)$, and set

$$Y_n = (|Y|, \mathcal{O}_X/\mathcal{I}^{n+1}) \quad (n \geq 0).$$

Thus $Y_0 = Y$ and $\text{Pic } \widehat{X} = \varprojlim_n \text{Pic } Y_n$.

(a) The ideal of Y_n in Y_{n+1} is square-zero and equals

$$\mathcal{J}_n = \mathcal{I}^{n+1}/\mathcal{I}^{n+2} \cong \mathcal{O}_Y(-(n+1)d). \quad (1)$$

The exact sequence of units for this thickening gives

$$H^1(Y, \mathcal{J}_n) \longrightarrow \text{Pic } Y_{n+1} \longrightarrow \text{Pic } Y_n \longrightarrow H^2(Y, \mathcal{J}_n). \quad (2)$$

The last group vanishes because Y is a curve.

We can also identify the kernel in (2). From

$$0 \longrightarrow \mathcal{O}_X(-(n+1)d) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_{Y_n} \longrightarrow 0$$

and the vanishing of $H^0(\mathbb{P}^2, \mathcal{O}(-a))$ for $a > 0$ and of $H^1(\mathbb{P}^2, \mathcal{O}(m))$ for every m , we obtain

$$H^0(Y_n, \mathcal{O}_{Y_n}) = k.$$

Thus the restriction map on global units is $k^* \rightarrow k^*$, hence is an isomorphism. The full units sequence therefore turns (2) into the short exact sequence

$$0 \longrightarrow V_n \longrightarrow \text{Pic } Y_{n+1} \longrightarrow \text{Pic } Y_n \longrightarrow 0, \quad V_n = H^1(Y, \mathcal{O}_Y(-(n+1)d)). \quad (3)$$

In particular all transition maps are surjective, so every line bundle on Y lifts successively to all Y_n . Hence

$$\text{Pic } \widehat{X} \longrightarrow \text{Pic } Y$$

is surjective.

The vector spaces V_n are nonzero for all sufficiently large n , and their dimensions tend to infinity. Indeed, the hypersurface sequence gives

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2}(-(n+2)d) \longrightarrow \mathcal{O}_{\mathbb{P}^2}(-(n+1)d) \longrightarrow \mathcal{O}_Y(-(n+1)d) \longrightarrow 0.$$

Using $H^1(\mathbb{P}^2, \mathcal{O}(m)) = 0$ and $H^2(Y, -) = 0$, one obtains, for $n \gg 0$,

$$\dim_k V_n = \binom{(n+2)d-1}{2} - \binom{(n+1)d-1}{2}, \quad (4)$$

which is positive and grows linearly with n .

Let

$$K = \ker(\mathrm{Pic} \hat{X} \longrightarrow \mathrm{Pic} Y)$$

and, for $r \geq 1$, define

$$F^r K = \ker(\mathrm{Pic} \hat{X} \longrightarrow \mathrm{Pic} Y_{r-1}).$$

Surjectivity in (3), including the ability to continue every finite-level lift, gives

$$F^r K / F^{r+1} K \cong H^1(Y, \mathcal{O}_Y(-rd)). \quad (5)$$

Thus, in every characteristic, K is a complete filtered abelian group with infinitely many nonzero k -vector-space graded pieces.

If $\mathrm{char} k = 0$, the convergent formal power series

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

identifies the sheaf $1 + \hat{\mathcal{I}}$ with the additive sheaf $\hat{\mathcal{I}}$. Since part (a) of Exercise III.11.6 gives $H^0(\hat{X}, \mathcal{O}_{\hat{X}}) = k$, the global units on $\hat{X}$ and on Y are both k^* . The units sequence therefore identifies

$$K \cong H^1(\hat{X}, \hat{\mathcal{I}})$$

as a k -vector space. Its filtration has the graded pieces (5), so (4) proves that it is infinite-dimensional. This proves the assertion in the problem under its necessary characteristic-zero hypothesis.

Here is why that qualification cannot simply be omitted. Suppose $\mathrm{char} k = p > 0$ and take Y to be a line. The class in

$$H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-2))$$

represented on the standard overlap by t^{-1} lifts, using (3), to an element $L \in F^2 K$. Its leading class is nonzero. Taking the p -th tensor power sends a transition function $1+a$ to $1+a^p$. Hence the leading class of $L^{\otimes p}$ is represented by t^{-p} in

$$H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-2p)),$$

where it is still nonzero. Thus $L^{\otimes p} \neq \mathcal{O}$. But the additive group of a k -vector space in characteristic p is killed by p . Consequently the literal vector-space claim is false in positive characteristic, while (3)–(5) give its correct replacement.

(b) Choose a nontrivial $\mathfrak{L} \in K$, which exists by (4) and (5). Suppose that $\mathfrak{L}$ were algebraizable by a coherent sheaf $\mathcal{F}$ on X . The argument in Exercise III.11.6(c) does not require algebraizability

of every formal bundle: it shows that $\mathcal{F}$ is a line bundle near Y , and then extends it across the finite complement to a line bundle $\mathcal{M}$ on $\mathbb{P}^2$. Thus

$$\mathcal{L} \cong \widehat{\mathcal{M}}.$$

Now $\mathcal{M} \cong \mathcal{O}_{\mathbb{P}^2}(a)$ for some $a \in \mathbb{Z}$. Since $\mathcal{L}|_Y \cong \mathcal{O}_Y$, we have $\mathcal{O}_Y(a) \cong \mathcal{O}_Y$. Ampleness of $\mathcal{O}_Y(1)$ and $\dim Y = 1$ force $a = 0$. This would make $\mathcal{L}$ trivial, a contradiction. Hence $\mathcal{L}$ is not algebraizable.

(c) Suppose, to the contrary, that every formal locally free sheaf has at least one globally generated twist. Since $\mathcal{O}_{\widehat{X}}(1)$ is globally generated, each such sheaf would then have all sufficiently large twists globally generated. Condition (ii) of Exercise III.11.6(b) would hold for every formal vector bundle, including the kernels needed in that exercise's construction. Its equivalence would make every such bundle algebraizable, contradicting the line bundle in (b). Therefore some formal locally free sheaf has no globally generated twist. This argument does not identify that sheaf with the particular line bundle chosen in (b). $\square$

Exercise III.11.8

Permitted context.

For a projective morphism of noetherian schemes, higher direct images of coherent sheaves are coherent. We use the theorem on formal functions and Nakayama's lemma.

Problem. Let $f : X \rightarrow Y$ be a projective morphism, let $\mathcal{F}$ be a coherent sheaf on X which is flat over Y , and suppose that

$$H^i(X_y, \mathcal{F}_y) = 0$$

for some i and some $y \in Y$. Prove that $R^i f_* \mathcal{F}$ is zero in a neighborhood of y .

Solution. Put

$$A = \mathcal{O}_{Y,y}, \quad \mathfrak{m} = \mathfrak{m}_y,$$

and, for $n \geq 0$, set

$$A_n = A/\mathfrak{m}^{n+1}, \quad X_n = X \times_Y \operatorname{Spec} A_n, \quad \mathcal{F}_n = \mathcal{F}|_{X_n}.$$

Thus $X_0 = X_y$ and $\mathcal{F}_0 = \mathcal{F}_y$.

We first prove by induction that

$$H^i(X_n, \mathcal{F}_n) = 0 \quad \text{for every } n \geq 0. \quad (1)$$

The case $n = 0$ is the hypothesis. For $n \geq 1$, the exact sequence

$$0 \longrightarrow \mathfrak{m}^n/\mathfrak{m}^{n+1} \longrightarrow A_n \longrightarrow A_{n-1} \longrightarrow 0$$

remains exact after tensoring with $\mathcal{F}$, because $\mathcal{F}$ is flat over Y . It gives

$$0 \longrightarrow \mathcal{F}_0 \otimes_{k(y)} (\mathfrak{m}^n/\mathfrak{m}^{n+1}) \longrightarrow \mathcal{F}_n \longrightarrow \mathcal{F}_{n-1} \longrightarrow 0. \quad (2)$$

The left term in (2) is a finite direct sum of copies of $\mathcal{F}_0$, so its i -th cohomology vanishes. If (1) holds for $n - 1$, the degree- i part of the long exact cohomology sequence of (2) reads

$$0 \longrightarrow H^i(X_n, \mathcal{F}_n) \longrightarrow 0.$$

This proves (1).

The theorem on formal functions now yields

$$(\widehat{R^i f_* \mathcal{F}})_y \cong \varprojlim_n H^i(X_n, \mathcal{F}_n) = 0. \quad (3)$$

The stalk $(R^i f_* \mathcal{F})_y$ is a finite A -module. Reducing its completion modulo $\mathfrak{m}$, or equivalently using faithful flatness of completion, (3) implies

$$(R^i f_* \mathcal{F})_y = 0.$$

Finally, $R^i f_* \mathcal{F}$ is coherent, so the complement of its support is open. It therefore vanishes on an open neighborhood of y , as required. $\square$

III.12 The Semicontinuity Theorem

Exercise III.12.1

Permitted context.

If $\mathcal{F}$ is a coherent sheaf on a noetherian scheme, then

$$y \mapsto \dim_{k(y)}(\mathcal{F} \otimes k(y))$$

is upper semicontinuous.

Problem. Let Y be a scheme of finite type over an algebraically closed field k . Prove that the function

$$\varphi(y) = \dim_k(\mathfrak{m}_y/\mathfrak{m}_y^2)$$

is upper semicontinuous on the set of closed points of Y .

Solution. The sheaf of differentials $\Omega_{Y/k}$ is coherent. If y is a closed point, then $k(y) = k$, and localization gives

$$(\Omega_{Y/k})_y \cong \Omega_{\mathcal{O}_{Y,y}/k}.$$

The residue homomorphism $\mathcal{O}_{Y,y} \rightarrow k$ is split by the structure map $k \rightarrow \mathcal{O}_{Y,y}$. The cotangent-space sequence therefore identifies

$$\mathfrak{m}_y/\mathfrak{m}_y^2 \xrightarrow{\sim} \Omega_{\mathcal{O}_{Y,y}/k} \otimes_{\mathcal{O}_{Y,y}} k \cong \Omega_{Y/k} \otimes k(y). \quad (1)$$

Equivalently, both sides of the first isomorphism represent k -derivations from $\mathcal{O}_{Y,y}$ to a k -vector space.

It follows from (1) that, on closed points,

$$\varphi(y) = \dim_{k(y)}(\Omega_{Y/k} \otimes k(y)).$$

This is the fiber-dimension function of the coherent sheaf $\Omega_{Y/k}$, so it is upper semicontinuous. $\square$

Exercise III.12.2**Permitted context.**

For every field K , the dimensions of $H^q(\mathbb{P}_K^n, \mathcal{O}_{\mathbb{P}^n}(m))$ depend only on n, m, q , not on K . Also

$$H^q(\mathbb{P}_K^n, \mathcal{O}_{\mathbb{P}^n}) = 0 \quad (q > 0).$$

Assume $n \geq 1$.

Problem. Let $\{X_t\}$ be a family of hypersurfaces of the same degree in $\mathbb{P}_k^n$. Prove that, for every i , the function

$$t \mapsto h^i(X_t, \mathcal{O}_{X_t})$$

is constant.

Solution. Let the common degree be $d > 0$. Over the residue field $k(t)$, the fiber X_t is cut out by a nonzero homogeneous polynomial f_t of degree d , so there is an exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}_{k(t)}^n}(-d) \xrightarrow{f_t} \mathcal{O}_{\mathbb{P}_{k(t)}^n} \longrightarrow \mathcal{O}_{X_t} \longrightarrow 0. \quad (1)$$

Since $H^0(\mathbb{P}^n, \mathcal{O}(-d)) = 0$ and the positive cohomology of $\mathcal{O}_{\mathbb{P}^n}$ vanishes, the long exact sequence of (1) gives

$$0 \longrightarrow k(t) \longrightarrow H^0(X_t, \mathcal{O}_{X_t}) \longrightarrow H^1(\mathbb{P}_{k(t)}^n, \mathcal{O}(-d)) \longrightarrow 0 \quad (2)$$

and, for every $i \geq 1$,

$$H^i(X_t, \mathcal{O}_{X_t}) \cong H^{i+1}(\mathbb{P}_{k(t)}^n, \mathcal{O}(-d)). \quad (3)$$

The dimensions of the groups on the right sides of (2) and (3) depend only on n, d, i . Therefore all the numbers $h^i(X_t, \mathcal{O}_{X_t})$ are independent of t . $\square$

Exercise III.12.3**Permitted context.**

The two numerical assertions in the problem are false for the untwisted ideal sheaf: its zeroth and first cohomology vanish on every fiber. The asserted values are exactly correct for $\mathcal{I}(1)$. We construct the requested family, verify the failure as printed, and prove this standard corrected form. The ground field is algebraically closed. We use that all rational normal quartics are projectively equivalent and that elementary matrices generate $\mathrm{SL}_5(k)$.

Problem. Let $X_1 \subseteq \mathbb{P}_k^4$ be the rational normal quartic, the 4-uple embedding of $\mathbb{P}^1$. Let $X_0 \subseteq \mathbb{P}_k^3 \subseteq \mathbb{P}_k^4$ be a nonsingular rational quartic. Construct a flat family $\{X_a\}$ in $\mathbb{P}^4$, parametrized by $T = \mathbb{A}^1$, with the given fibers at $a = 1$ and $a = 0$.

Let $\mathcal{I} \subseteq \mathcal{O}_{\mathbb{P}^4 \times T}$ be the ideal sheaf of the total family. Show that $\mathcal{I}$ is flat over T , and show that

$$h^0(a, \mathcal{I}) = \begin{cases} 0, & a \neq 0, \\ 1, & a = 0, \end{cases} \quad h^1(a, \mathcal{I}) = \begin{cases} 0, & a \neq 0, \\ 1, & a = 0. \end{cases}$$

Solution. Work over the algebraically closed ground field k . Choose an isomorphism $X_0 \cong \mathbb{P}^1$, and choose coordinates so its containing hyperplane is $x_4 = 0$. The prescribed embedding is defined by a basis s_0, s_1, s_2, s_3 of a four-dimensional subspace $V \subset H^0(\mathbb{P}^1, \mathcal{O}(4))$. Extend it to a basis by a section s_4 , and set

$$\Phi(p, a) = ([s_0(p) : s_1(p) : s_2(p) : s_3(p) : as_4(p)], a). \quad (1)$$

Projection away from $[0 : 0 : 0 : 0 : 1]$ is defined on the image. There Φ is the graph over the fixed closed embedding defined by V , hence is an immersion. Since its source is proper over $\mathbb{A}^1$, it is a closed immersion into all of $\mathbb{P}^4 \times \mathbb{A}^1$. Its image is therefore flat, isomorphic to $\mathbb{P}^1 \times \mathbb{A}^1$. It has the prescribed fiber X_0 , and every nonzero fiber is a rational normal quartic.

If the embedded endpoint X_1 is prescribed exactly, choose $g \in \mathrm{PGL}_5(k)$ taking the fiber of (1) at 1 to X_1 . Scale a matrix representative to determinant one. Write it as a product of elementary matrices $\prod_j (I + c_j E_{i_j k_j})$, where $i_j \neq k_j$. Then

$$g(a) = \prod_j (I + ac_j E_{i_j k_j})$$

is invertible over $k[a]$, with $g(0) = I$ and $g(1) = g$. Applying this automorphism of $\mathbb{P}^4 \times \mathbb{A}^1$ to the family fixes X_0 and gives precisely X_1 at 1. It preserves flatness and all the fiber cohomology computations.

In the exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbb{P}^4 \times T} \longrightarrow \mathcal{O}_X \longrightarrow 0, \quad (2)$$

the middle and right terms are flat over T . The kernel $\mathcal{I}$ is therefore flat over T , and restricting (2) to a fiber remains exact.

We now encounter the error in the printed assertion. On every fiber,

$$0 \longrightarrow \mathcal{I}_a \longrightarrow \mathcal{O}_{\mathbb{P}^4} \longrightarrow \mathcal{O}_{X_a} \longrightarrow 0. \quad (3)$$

Both $X_a \cong \mathbb{P}^1$ and $\mathbb{P}^4$ have only the constant global functions, and the map between those constants is an isomorphism. Since $H^1(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}) = 0$, sequence (3) gives

$$H^0(\mathbb{P}^4, \mathcal{I}_a) = H^1(\mathbb{P}^4, \mathcal{I}_a) = 0 \quad \text{for every } a. \quad (4)$$

Thus the requested jump does not occur for $\mathcal{I}$.

For the corrected statement, twist (3) by $\mathcal{O}_{\mathbb{P}^4}(1)$. Since $\mathcal{O}_{\mathbb{P}^4}(1)|_{X_a} \cong \mathcal{O}_{\mathbb{P}^1}(4)$, cohomology gives

$$0 \longrightarrow H^0(\mathcal{I}_a(1)) \longrightarrow k^5 \xrightarrow{\rho_a} H^0(\mathbb{P}^1, \mathcal{O}(4)) \longrightarrow H^1(\mathcal{I}_a(1)) \longrightarrow 0. \quad (5)$$

Before applying the harmless fiberwise coordinate change $g(a)$, the map ρ_a sends the coordinate sections to s_0, s_1, s_2, s_3, as_4 . It has rank five for $a \neq 0$, and rank four for $a = 0$; at zero its kernel is ks_4 and its cokernel is represented by s_4 . Thus

$$h^0(a, \mathcal{I}(1)) = h^1(a, \mathcal{I}(1)) = \begin{cases} 0, & a \neq 0, \\ 1, & a = 0. \end{cases}$$

These are the displayed values intended by the exercise. □

Exercise III.12.4**Permitted context.**

We use the following consequence of cohomology and base change. If $f : X \rightarrow Y$ is projective, $\mathcal{F}$ is coherent and flat over Y , with Y reduced and noetherian, and

$$\dim_{k(y)} H^0(X_y, \mathcal{F}_y)$$

is constant, then $f_*\mathcal{F}$ is locally free and its formation commutes with passage to the fibers.

Problem. Let Y be an integral scheme of finite type over an algebraically closed field k . Let $f : X \rightarrow Y$ be a flat projective morphism whose fibers are integral schemes. Let $\mathcal{L}, \mathcal{M}$ be invertible sheaves on X , and suppose that

$$\mathcal{L}_y \cong \mathcal{M}_y$$

on X_y for every $y \in Y$. Prove that there is an invertible sheaf $\mathcal{N}$ on Y such that

$$\mathcal{L} \cong \mathcal{M} \otimes f^*\mathcal{N}.$$

Solution. Set

$$\mathcal{P} = \mathcal{L} \otimes \mathcal{M}^{-1}.$$

Then $\mathcal{P}_y \cong \mathcal{O}_{X_y}$ for every y . In particular, for every closed point y , the fiber X_y is an integral projective scheme over the algebraically closed field k , and hence

$$H^0(X_y, \mathcal{P}_y) \cong H^0(X_y, \mathcal{O}_{X_y}) = k. \quad (1)$$

For an arbitrary point y , the triviality of $\mathcal{P}_y$ gives at least one section. If its space of sections had dimension at least 2, upper semicontinuity would put y in the closed locus where $h^0 \geq 2$. Any nonempty such locus contains a closed point, contradicting (1). Thus $h^0(X_y, \mathcal{P}_y) = 1$ for every $y \in Y$.

The sheaf $\mathcal{P}$ is flat over Y , because it is locally free on the flat Y -scheme X . Cohomology and base change, applied to this constant value, therefore shows that

$$\mathcal{N} = f_*\mathcal{P}$$

is invertible on Y , and that

$$\mathcal{N} \otimes k(y) \xrightarrow{\sim} H^0(X_y, \mathcal{P}_y) \quad (2)$$

for every closed point y .

Consider the adjunction evaluation map

$$\varepsilon : f^*\mathcal{N} \longrightarrow \mathcal{P}. \quad (3)$$

After restricting (3) to X_y and using (2), it becomes

$$H^0(X_y, \mathcal{O}_{X_y}) \otimes_k \mathcal{O}_{X_y} \longrightarrow \mathcal{O}_{X_y},$$

the evaluation of constant functions. This is an isomorphism. Consequently (3) is an isomorphism at every point lying over a closed point of Y . If its cokernel were nonzero, its support would contain

a closed point of X , whose image in Y would also be closed, a contradiction. Thus (3) is surjective; being a surjection between invertible sheaves, it is an isomorphism.

We have proved

$$\mathcal{L} \otimes \mathcal{M}^{-1} = \mathcal{P} \cong f^* \mathcal{N},$$

or equivalently

$$\mathcal{L} \cong \mathcal{M} \otimes f^* \mathcal{N}.$$

□

Exercise III.12.5

Permitted context.

The assertion requires $\text{rank } \mathcal{E} \geq 2$; for a rank-one bundle, $\mathbb{P}(\mathcal{E}) \cong Y$ and there is no additional independent $\mathbb{Z}$ -factor.

We use constancy of the Hilbert polynomial in a flat projective family and Exercise III.12.4.

Problem. Let Y be an integral scheme of finite type over an algebraically closed field k . Let $\mathcal{E}$ be a locally free sheaf of rank at least 2 on Y , and put

$$X = \mathbb{P}(\mathcal{E}).$$

Prove that

$$\text{Pic } X \cong (\text{Pic } Y) \times \mathbb{Z}.$$

Solution. Let

$$\pi : X = \mathbb{P}(\mathcal{E}) \longrightarrow Y$$

be the projection, and write $\text{rank } \mathcal{E} = r + 1$, where $r \geq 1$. Define

$$\Psi : \text{Pic } Y \times \mathbb{Z} \longrightarrow \text{Pic } X, \quad (\mathcal{M}, d) \longmapsto \pi^* \mathcal{M} \otimes \mathcal{O}_X(d). \quad (1)$$

We first prove surjectivity. Let $\mathcal{L}$ be an invertible sheaf on X . For every point $y \in Y$,

$$X_y \cong \mathbb{P}_{k(y)}^r, \quad \mathcal{L}_y \cong \mathcal{O}_{\mathbb{P}^r}(d_y)$$

for a unique integer d_y . The sheaf $\mathcal{L}$ is flat over Y , and $\mathcal{O}_X(1)$ is relatively ample. Therefore the fiber Hilbert polynomial

$$\chi(X_y, \mathcal{L}_y(m)) = \binom{m + d_y + r}{r} \quad (2)$$

is locally constant in y . Since Y is integral, it is connected, so (2) is independent of y . Equality of these polynomials forces all the integers d_y to have one common value d .

Put

$$\mathcal{L}' = \mathcal{L} \otimes \mathcal{O}_X(-d).$$

Its restriction to every fiber is trivial. The morphism π is flat and projective with integral fibers, so Exercise III.12.4 gives an invertible sheaf $\mathcal{M}$ on Y such that

$$\mathcal{L}' \cong \pi^* \mathcal{M}.$$

Hence

$$\mathcal{L} \cong \pi^* \mathcal{M} \otimes \mathcal{O}_X(d),$$

and Ψ is surjective.

For injectivity, suppose

$$\pi^* \mathcal{M} \otimes \mathcal{O}_X(d) \cong \mathcal{O}_X.$$

Restriction to any fiber gives $\mathcal{O}_{\mathbb{P}^r}(d) \cong \mathcal{O}_{\mathbb{P}^r}$, so $d = 0$. We are left with $\pi^* \mathcal{M} \cong \mathcal{O}_X$. Since $\pi_* \mathcal{O}_X = \mathcal{O}_Y$, the projection formula gives

$$\mathcal{M} \cong \mathcal{M} \otimes \pi_* \mathcal{O}_X \cong \pi_* \pi^* \mathcal{M} \cong \mathcal{O}_Y.$$

Thus the kernel of Ψ is zero, and (1) is the desired isomorphism. $\square$

Exercise III.12.6

Permitted context.

We use Exercise III.11.8 and the cohomology-and-base-change theorem. In particular, if $p : Z \rightarrow S$ is projective, $\mathcal{A}$ is flat over S , and

$$H^1(Z_s, \mathcal{A}_s) = 0,$$

then, after shrinking around s , the fiber section map

$$(p_* \mathcal{A}) \otimes k(s) \longrightarrow H^0(Z_s, \mathcal{A}_s)$$

is surjective.

An integral projective scheme over an algebraically closed field has only constant global functions. Assume T is nonempty when choosing the reference point t_0 .

Problem. Let X be an integral projective scheme over an algebraically closed field k , and suppose that

$$H^1(X, \mathcal{O}_X) = 0.$$

Let T be a connected scheme of finite type over k .

(a) If $\mathcal{L}$ is an invertible sheaf on $X \times T$, prove that the invertible sheaves

$$\mathcal{L}_t = \mathcal{L}|_{X \times \{t\}}$$

on X are isomorphic for all closed points $t \in T$.

(b) Prove, without assuming that T is reduced, that

$$\text{Pic}(X \times T) \cong \text{Pic } X \times \text{Pic } T.$$

Solution. Write

$$p : X \times T \longrightarrow T, \quad q : X \times T \longrightarrow X$$

for the two projections.

We first establish a local observation. Suppose that $t \in T$ is a closed point and that an invertible sheaf $\mathcal{A}$ on $X \times T$ satisfies

$$\mathcal{A}_t \cong \mathcal{O}_X. \tag{1}$$

The sheaf $\mathcal{A}$ is flat over T , and

$$H^1(X, \mathcal{A}_t) = H^1(X, \mathcal{O}_X) = 0.$$

Exercise III.11.8 makes $R^1p_*\mathcal{A}$ zero near t , and cohomology and base change then makes

$$(p_*\mathcal{A}) \otimes k(t) \longrightarrow H^0(X, \mathcal{A}_t) \quad (2)$$

surjective. After replacing T by an affine neighborhood U of t , lift the section $1 \in H^0(X, \mathcal{O}_X)$, using (1) and (2), to

$$\sigma \in H^0(X \times U, \mathcal{A}).$$

Its zero locus $Z(\sigma)$ is closed and misses $X \times \{t\}$. Since p is proper, $p(Z(\sigma))$ is closed. Replacing U by $U \setminus p(Z(\sigma))$, the section is nowhere zero and therefore gives an isomorphism

$$\mathcal{O}_{X \times U} \xrightarrow{\sim} \mathcal{A}|_{X \times U}. \quad (3)$$

This argument is valid even when T is nonreduced.

(a) Fix a closed point $t_0 \in T$, and replace $\mathcal{L}$ by

$$\mathcal{L} \otimes q^*(\mathcal{L}_{t_0}^{-1}).$$

Its fiber at t_0 is trivial, so (3) shows that its fibers are trivial on a neighborhood of t_0 . Applying the same argument at every closed point shows that

$$t \longmapsto [\mathcal{L}_t] \in \text{Pic } X$$

is locally constant on the closed points of T . A scheme of finite type over k is Jacobson. Neighborhoods belonging to two different isomorphism classes cannot meet, since a nonempty intersection would contain a closed point belonging to both classes. Their union is all of T , since a nonempty closed complement would also contain a closed point. They therefore give a disjoint open partition of T . Since T is connected, there is only one part, and all $\mathcal{L}_t$ are isomorphic.

(b) Consider the homomorphism

$$\Theta : \text{Pic } X \times \text{Pic } T \longrightarrow \text{Pic}(X \times T), \quad (\mathcal{A}, \mathcal{N}) \longmapsto q^*\mathcal{A} \otimes p^*\mathcal{N}. \quad (4)$$

To prove surjectivity, let $\mathcal{L}$ be a line bundle on $X \times T$, choose a closed point t_0 , and put

$$\mathcal{A} = \mathcal{L}_{t_0}, \quad \mathcal{P} = \mathcal{L} \otimes q^*(\mathcal{A}^{-1}).$$

Part (a) gives $\mathcal{P}_t \cong \mathcal{O}_X$ for every closed t . The local observation (3), applied at every closed point, gives an open cover $\{U_i\}$ of T such that

$$\mathcal{P}|_{X \times U_i} \cong \mathcal{O}_{X \times U_i}. \quad (5)$$

The opens cover all of T , because a nonempty closed subset of a finite-type k -scheme contains a closed point.

Now

$$p_*\mathcal{O}_{X \times T} = \mathcal{O}_T, \quad (6)$$

since $H^0(X, \mathcal{O}_X) = k$; this equality remains valid for a nonreduced T . On $X \times (U_i \cap U_j)$, the two trivializations in (5) consequently differ by the pullback of a unique unit on $U_i \cap U_j$. These units form a cocycle defining an invertible sheaf $\mathcal{N}$ on T , and (5) glues to

$$\mathcal{P} \cong p^*\mathcal{N}.$$

Thus

$$\mathcal{L} \cong q^* \mathcal{A} \otimes p^* \mathcal{N},$$

so Θ is surjective.

Finally, suppose

$$q^* \mathcal{A} \otimes p^* \mathcal{N} \cong \mathcal{O}_{X \times T}.$$

Restriction to $X \times \{t_0\}$ gives $\mathcal{A} \cong \mathcal{O}_X$. Hence $p^* \mathcal{N} \cong \mathcal{O}_{X \times T}$. Choose a closed k -point $x \in X$, and pull back along the section

$$T \longrightarrow X \times T, \quad t \longmapsto (x, t).$$

This gives $\mathcal{N} \cong \mathcal{O}_T$. Therefore Θ is injective, and (4) is an isomorphism. $\square$

Chapter IV

IV.1 Riemann–Roch Theorem

Exercise IV.1.1

Permitted context.

Throughout, a curve is a complete nonsingular integral curve over an algebraically closed field k . For a divisor D on a curve X , set

$$L(D) = H^0(X, \mathcal{O}_X(D)) = \{f \in K(X)^\times : \operatorname{div}(f) + D \geq 0\} \cup \{0\}$$

and write $\ell(D) = \dim_k L(D)$. The Riemann–Roch theorem says

$$\ell(D) - \ell(K - D) = \deg D + 1 - g,$$

where K is a canonical divisor and g is the genus of X . A divisor of negative degree has no nonzero global section.

Problem. Let X be a curve and let $P \in X$ be a point. Prove that there is a nonconstant rational function $f \in K(X)$ that is regular on $X \setminus \{P\}$.

Solution. Choose an integer $m \geq 2g + 1$. Since $\deg K = 2g - 2$,

$$\deg(K - mP) = 2g - 2 - m < 0.$$

Thus $\ell(K - mP) = 0$, and Riemann–Roch gives

$$\ell(mP) = m + 1 - g \geq g + 2 > 1.$$

The constants form a one-dimensional subspace of $L(mP)$, so there is an $f \in L(mP)$ that is not constant. The condition

$$\operatorname{div}(f) + mP \geq 0$$

says that f has no pole away from P . Hence f is regular on $X \setminus \{P\}$, as required. In fact, f must have a pole at P , because every global regular function on the complete integral curve X is constant. $\square$

Exercise IV.1.2**Permitted context.**

For a divisor D on a complete nonsingular curve X , write

$$L(D) = H^0(X, \mathcal{O}_X(D)), \quad \ell(D) = \dim_k L(D).$$

If $\deg D > 2g - 2$, Riemann–Roch and $\deg K = 2g - 2$ give

$$\ell(D) = \deg D + 1 - g.$$

The ground field k is algebraically closed and therefore infinite; a finite-dimensional vector space over an infinite field is not the union of finitely many proper linear subspaces. For an empty list of prescribed poles, take a constant function.

Problem. Let X be a curve, and let $P_1, \dots, P_r \in X$ be points. Prove that there is a rational function $f \in K(X)$ having a pole of positive order at every P_i and no poles elsewhere.

Solution. Discarding repetitions, we may suppose that $P_1, \dots, P_r$ are distinct. Choose $m > 0$ so large that

$$mr - 1 > 2g - 2, \quad D = m(P_1 + \dots + P_r).$$

For each i , both D and $D - P_i$ have degree greater than $2g - 2$. Riemann–Roch therefore gives

$$\ell(D) = mr + 1 - g, \quad \ell(D - P_i) = mr - g.$$

Consequently $L(D - P_i)$ is a proper, codimension-one subspace of $L(D)$. Since k is infinite, the finite union

$$\bigcup_{i=1}^r L(D - P_i)$$

does not exhaust $L(D)$. Choose

$$f \in L(D) \setminus \bigcup_{i=1}^r L(D - P_i).$$

Membership in $L(D)$ shows that f is regular away from the P_i and has pole order at most m at each P_i . If its pole order at P_i were at most $m - 1$, then $f \in L(D - P_i)$, contrary to the choice of f . Thus f has pole order exactly m at every P_i and has no other poles. $\square$

Exercise IV.1.3**Permitted context.**

We use the following standard facts about curves over an algebraically closed field k .

An integral separated regular scheme of dimension one and finite type over k is an open subscheme of a complete nonsingular curve. A rational function on a complete nonsingular curve determines a morphism to $\mathbb{P}^1$; the inverse image of ∞ is precisely the support of its pole divisor. Finally, a nonconstant morphism between complete integral curves is finite, and a finite morphism is affine.

We may also use Exercise IV.1.2: given finitely many points on a curve, there is a rational function having a pole at every one of those points and no poles elsewhere.

Problem. Let X be an integral, separated, regular, one-dimensional scheme of finite type over k . Assume that X is not proper over k . Prove that X is affine.

Solution. Embed X as a dense open subscheme of a complete nonsingular curve $\overline{X}$. Its complement is a proper closed subset of the one-dimensional integral scheme $\overline{X}$, hence is a finite set of closed points:

$$\overline{X} \setminus X = \{P_1, \dots, P_r\}.$$

This set is nonempty, because otherwise $X = \overline{X}$ would be proper.

By Exercise IV.1.2, there is $f \in K(\overline{X})$ having a pole at every P_i and no poles elsewhere. The associated morphism

$$\overline{f} : \overline{X} \longrightarrow \mathbb{P}^1$$

therefore satisfies

$$\overline{f}^{-1}(\infty) = \{P_1, \dots, P_r\}, \quad \overline{f}^{-1}(\mathbb{A}^1) = X.$$

Because f has a pole, it is nonconstant. Thus $\overline{f}$ is a finite morphism of complete integral curves. Its restriction

$$X = \overline{f}^{-1}(\mathbb{A}^1) \longrightarrow \mathbb{A}^1$$

is finite by base change. A scheme finite over an affine scheme is affine, so X is affine. $\square$

Exercise IV.1.4**Permitted context.**

We may use Exercise IV.1.3, which says that every nonproper integral regular curve of finite type over k is affine, together with the following consequences of the cohomological criterion for affineness.

First, a noetherian scheme is affine if and only if its reduction is affine. Second, a reduced noetherian scheme is affine if and only if all of its irreducible components, with their reduced structures, are affine. Third, if $Y \rightarrow Z$ is a finite surjective morphism of noetherian schemes and Y is affine, then Z is affine.

The normalization of an integral curve of finite type over a field is finite and is a regular curve. Properness descends along a finite surjective morphism: if the source is proper over k , then so is the target.

Problem. Let X be a separated, one-dimensional scheme of finite type over k . Suppose that none of the irreducible components of X is proper over k . Prove that X is affine.

Solution. Let $X_1, \dots, X_s$ be the irreducible components of X_{red} , each with its reduced induced structure. Every X_i is an integral separated curve of finite type over k , and by hypothesis it is not proper. Let

$$\nu_i : \tilde{X}_i \longrightarrow X_i$$

be its normalization. This morphism is finite and surjective, and $\tilde{X}_i$ is regular. Moreover, $\tilde{X}_i$ cannot be proper: otherwise properness would descend through ν_i , contrary to the hypothesis on X_i .

Exercise IV.1.3 now shows that $\tilde{X}_i$ is affine. Since ν_i is finite and surjective, its target X_i is affine as well. Thus every irreducible component of the reduced noetherian scheme X_{red} is affine. The component criterion makes X_{red} affine, and the reduction criterion then makes X affine. This last step also accounts for every nilpotent structure on X . $\square$

Exercise IV.1.5**Permitted context.**

For a divisor D on a curve X , write

$$L(D) = H^0(X, \mathcal{O}_X(D)), \quad \ell(D) = \dim_k L(D).$$

If D is effective, then its complete linear system has dimension $\dim |D| = \ell(D) - 1$. Riemann–Roch says

$$\ell(D) - \ell(K - D) = \deg D + 1 - g,$$

and $\ell(K) = g$. A nonconstant rational function on a complete nonsingular curve defines a finite morphism to $\mathbb{P}^1$, whose degree is the degree of its pole divisor.

Problem. Let D be an effective divisor on a curve X of genus g . Prove that

$$\dim |D| \leq \deg D.$$

Prove also that equality holds if and only if $D = 0$ or $g = 0$.

Solution. Because $D \geq 0$, there is an inclusion

$$L(K - D) \subseteq L(K).$$

Hence $\ell(K - D) \leq \ell(K) = g$. Riemann–Roch gives

$$\dim |D| = \ell(D) - 1 = \deg D - g + \ell(K - D) \leq \deg D. \quad (1)$$

If $D = 0$, both sides of (1) are zero. If $g = 0$, then $\ell(K - D) = 0$, so Riemann–Roch gives $\ell(D) = \deg D + 1$; equality again holds.

Conversely, suppose equality holds and $g > 0$. We first show that the canonical linear system has no base point. Fix $P \in X$. If $\ell(P) \geq 2$, then $L(P)$ contains a nonconstant rational function whose pole divisor has degree at most one. Its morphism to $\mathbb{P}^1$ would have degree one, hence would be an isomorphism. This would give $g = 0$, a contradiction. Therefore $\ell(P) = 1$. Applying Riemann–Roch to P gives

$$1 - \ell(K - P) = 2 - g, \quad \text{so} \quad \ell(K - P) = g - 1. \quad (2)$$

Thus $L(K - P)$ is a proper subspace of $L(K)$ for every point P .

Equality in (1) forces $\ell(K - D) = g$, and hence $L(K - D) = L(K)$. If $D \neq 0$, choose a point P in its support. Then $D \geq P$, so

$$L(K - D) \subseteq L(K - P) \subsetneq L(K),$$

contradicting $L(K - D) = L(K)$. Therefore $D = 0$. This proves the stated characterization of equality. $\square$

Exercise IV.1.6**Permitted context.**

For a divisor D on a curve X , Riemann–Roch gives

$$\ell(D) = \deg D + 1 - g + \ell(K - D).$$

A nonconstant rational function $f \in K(X)$ defines a finite morphism $f : X \rightarrow \mathbb{P}^1$. Its degree is the degree of the pole divisor $f^*(\infty)$.

Problem. Let X be a curve of genus g . Prove that there is a finite morphism

$$f : X \longrightarrow \mathbb{P}^1$$

of degree at most $g + 1$.

Solution. Choose a point $P \in X$ and put $D = (g + 1)P$. Riemann–Roch gives

$$\ell(D) = \deg D + 1 - g + \ell(K - D) = 2 + \ell(K - D) \geq 2.$$

Thus $L(D)$ contains a nonconstant rational function f . Since $f \in L(D)$, its only possible pole is at P , with order at most $g + 1$. It must have a pole, because a global regular function on the complete integral curve X is constant. Consequently the morphism

$$f : X \longrightarrow \mathbb{P}^1$$

is nonconstant and therefore finite, and

$$\deg f = \deg f^*(\infty) \leq g + 1.$$

□

Exercise IV.1.7**Permitted context.**

A curve X is hyperelliptic if $g(X) \geq 2$ and it admits a finite morphism of degree two to $\mathbb{P}^1$. On a curve of positive genus, the canonical linear system is base-point-free: for every $P \in X$,

$$\ell(K - P) = g - 1 < g = \ell(K).$$

A base-point-free linear system of dimension one defines a morphism to $\mathbb{P}^1$. If the system is complete and corresponds to a divisor D , then the resulting morphism f satisfies $f^*\mathcal{O}_{\mathbb{P}^1}(1) \cong \mathcal{O}_X(D)$.

Identify a nonsingular quadric surface with $Q = \mathbb{P}^1 \times \mathbb{P}^1$. A curve of type (a, b) is the zero divisor of a section of $\mathcal{O}_Q(a, b)$; its intersection with a fiber of the first projection is b . Work over an algebraically closed field. Adjunction on the quadric gives genus $(a - 1)(b - 1)$ for a smooth curve of type (a, b) . For $a, b > 0$, $\mathcal{O}_Q(a, b)$ is very ample and Bertini gives a smooth integral general member.

Problem. (a) Let X be a curve of genus 2. Prove that its canonical divisor defines a complete linear system $|K|$ of degree 2 and dimension 1, with no base points. Deduce that X is hyperelliptic.

(b) For every $g \geq 2$, let X be a nonsingular irreducible curve of type $(g + 1, 2)$ on $\mathbb{P}^1 \times \mathbb{P}^1$. Prove that X admits a morphism of degree 2 to $\mathbb{P}^1$. Thus hyperelliptic curves exist in every genus $g \geq 2$.

Solution. (a) Since $g = 2$,

$$\deg K = 2g - 2 = 2, \quad \ell(K) = g = 2.$$

Therefore $|K|$ has degree 2 and projective dimension $\ell(K) - 1 = 1$. The canonical system has no base points, so it defines a morphism

$$f : X \longrightarrow \mathbb{P}^1$$

with $f^*\mathcal{O}_{\mathbb{P}^1}(1) \cong \mathcal{O}_X(K)$. It follows that

$$\deg f = \deg f^*\mathcal{O}_{\mathbb{P}^1}(1) = \deg K = 2.$$

The morphism is nonconstant and hence finite. Thus X is hyperelliptic.

(b) Let $\pi_1 : Q \rightarrow \mathbb{P}^1$ be the first projection. For a general point $t \in \mathbb{P}^1$, its fiber $F_t = \{t\} \times \mathbb{P}^1$ meets X in a zero-dimensional scheme of length

$$X \cdot F_t = 2,$$

because X has type $(g + 1, 2)$. Hence $\pi_1|_X : X \rightarrow \mathbb{P}^1$ is a nonconstant morphism of degree 2. It is finite because X is complete. Adjunction gives $g(X) = ((g + 1) - 1)(2 - 1) = g$. The system $\mathcal{O}_Q(g + 1, 2)$ is very ample, so Bertini supplies a smooth integral member. Thus the construction yields a hyperelliptic curve in each genus $g \geq 2$. $\square$

Exercise IV.1.8**Permitted context.**

Let X be an integral projective scheme of dimension one over an algebraically closed field k , and let $\nu: \tilde{X} \rightarrow X$ be its normalization. This is a finite morphism. There is an exact sequence

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \nu_* \mathcal{O}_{\tilde{X}} \longrightarrow \bigoplus_{P \in X} \overline{\mathcal{O}}_P / \mathcal{O}_{X,P} \longrightarrow 0, \quad (1)$$

where the final terms are skyscraper sheaves and only finitely many are nonzero. Put

$$\delta_P = \text{length}_{\mathcal{O}_{X,P}}(\overline{\mathcal{O}}_P / \mathcal{O}_{X,P}).$$

For a projective curve Y , $p_a(Y) = 1 - \chi(\mathcal{O}_Y)$. A finite morphism has no higher direct images on quasi-coherent sheaves, so $\chi(\nu_* \mathcal{O}_{\tilde{X}}) = \chi(\mathcal{O}_{\tilde{X}})$.

For a reduced curve singularity A of finite type over k , completion preserves the length of the normalization quotient:

$$\text{length}_A(\overline{A}/A) = \dim_k(\widehat{\overline{A}}/\widehat{A}). \quad (2)$$

Thus this length depends only on the analytic isomorphism class of the singularity.

Problem. With the notation above:

(a) Prove that

$$p_a(X) = p_a(\tilde{X}) + \sum_{P \in X} \delta_P.$$

(b) Prove that if $p_a(X) = 0$, then X is nonsingular and $X \cong \mathbb{P}^1$.

(c) If P is a node or an ordinary cusp, prove that $\delta_P = 1$.

Solution. (a) Taking Euler characteristics in (1) gives

$$\chi(\mathcal{O}_X) = \chi(\nu_* \mathcal{O}_{\tilde{X}}) - \sum_{P \in X} \delta_P = \chi(\mathcal{O}_{\tilde{X}}) - \sum_{P \in X} \delta_P.$$

Here a skyscraper sheaf of length n has n -dimensional space of global sections and no higher cohomology. Subtracting the displayed identity from 1 on both sides yields

$$p_a(X) = p_a(\tilde{X}) + \sum_{P \in X} \delta_P.$$

(b) The normalization $\tilde{X}$ is a complete nonsingular curve, so $p_a(\tilde{X}) = g(\tilde{X}) \geq 0$; every δ_P is also nonnegative. If $p_a(X) = 0$, part (a) therefore forces

$$g(\tilde{X}) = 0, \quad \delta_P = 0 \quad \text{for every } P \in X.$$

The vanishing of every normalization quotient says that $\mathcal{O}_X \rightarrow \nu_* \mathcal{O}_{\tilde{X}}$ is an isomorphism. Hence ν is an isomorphism and X is nonsingular. A complete nonsingular curve of genus zero over an algebraically closed field is isomorphic to $\mathbb{P}^1$, so $X \cong \mathbb{P}^1$.

(c) By (2), we may compute after replacing the singularity by its standard analytic model.

For a node, the completed local ring and its normalization are

$$A = k[[x, y]]/(xy), \quad \bar{A} = k[[x]] \oplus k[[y]].$$

The map $A \rightarrow \bar{A}$ sends the class of $h(x, y)$ to $(h(x, 0), h(0, y))$. Its image consists exactly of the pairs $(a(x), b(y))$ satisfying $a(0) = b(0)$. Therefore

$$\bar{A}/A \cong k,$$

with the quotient measured by $a(0) - b(0)$, and the node has $\delta_P = 1$.

For an ordinary cusp, the standard analytic model is

$$A = k[[x, y]]/(y^2 - x^3) \cong k[[t^2, t^3]] \subseteq k[[t]] = \bar{A},$$

under $x \mapsto t^2$ and $y \mapsto t^3$. Since every integer at least two lies in the semigroup generated by 2 and 3,

$$k[[t^2, t^3]] = k + t^2k[[t]].$$

Thus $\bar{A}/A$ is the one-dimensional vector space generated by the class of t . Hence an ordinary cusp also has $\delta_P = 1$. $\square$

Exercise IV.1.9**Permitted context.**

Let X be an integral projective scheme of dimension one over an algebraically closed field k , and let X_{reg} be its regular locus. A divisor supported in X_{reg} is a finite sum $D = \sum n_i P_i$; define $\deg D = \sum n_i$. Every such divisor is Cartier and determines an invertible sheaf $\mathcal{O}_X(D)$.

We use $p_a(X) = 1 - \chi(\mathcal{O}_X)$, additivity of Euler characteristic, Serre's theorem on twists of coherent sheaves, and the standard properties of ample and very ample invertible sheaves. The singular locus of an integral curve is finite.

If X is a locally complete intersection in projective space, its dualizing sheaf ω_X is invertible, and Serre duality gives

$$H^1(X, \mathcal{L})^\vee \cong H^0(X, \omega_X \otimes \mathcal{L}^{-1})$$

for every invertible sheaf $\mathcal{L}$.

Problem. (a) Let $D = \sum n_i P_i$ be a divisor whose support lies in X_{reg} . Prove that

$$\chi(\mathcal{O}_X(D)) = \deg D + 1 - p_a(X).$$

(b) Prove that every Cartier divisor on X is the difference of two very ample Cartier divisors.

(c) Deduce that every invertible sheaf $\mathcal{L}$ on X is isomorphic to $\mathcal{O}_X(D)$ for a divisor D supported in X_{reg} .

(d) Suppose in addition that X is a locally complete intersection in some projective space. Choose a divisor K , supported in X_{reg} , such that $\mathcal{O}_X(K) \cong \omega_X$. With $\ell(D) = h^0(X, \mathcal{O}_X(D))$, prove

$$\ell(D) - \ell(K - D) = \deg D + 1 - p_a(X).$$

Solution. (a) Let $P \in X_{\text{reg}}$. Its ideal sheaf is the invertible sheaf $\mathcal{O}_X(-P)$, and tensoring its defining sequence by $\mathcal{O}_X(D + P)$ gives

$$0 \longrightarrow \mathcal{O}_X(D) \longrightarrow \mathcal{O}_X(D + P) \longrightarrow k(P) \longrightarrow 0. \quad (1)$$

Since k is algebraically closed, the skyscraper sheaf $k(P)$ has Euler characteristic one. Thus

$$\chi(\mathcal{O}_X(D + P)) = \chi(\mathcal{O}_X(D)) + 1. \quad (2)$$

Starting with $D = 0$, every divisor supported in the regular locus can be reached by repeatedly adding or subtracting a point. Equation (2) therefore gives

$$\chi(\mathcal{O}_X(D)) = \chi(\mathcal{O}_X) + \deg D = \deg D + 1 - p_a(X).$$

(b) Fix a very ample Cartier divisor H , and let C be any Cartier divisor. For all sufficiently large n , the invertible sheaf

$$\mathcal{O}_X(C + (n - 1)H)$$

is generated by global sections. Since $\mathcal{O}_X(H)$ is very ample, their tensor product $\mathcal{O}_X(C+nH)$ is very ample. Increasing n if necessary, $\mathcal{O}_X(nH)$ is also very ample (indeed, its embedding is obtained from that of H by a Veronese embedding). Hence

$$C = (C + nH) - nH$$

expresses C as the difference of two very ample Cartier divisors.

(c) Choose a nonzero rational section of $\mathcal{L}$. Its local expressions in trivializations of $\mathcal{L}$ define a Cartier divisor C with

$$\mathcal{L} \cong \mathcal{O}_X(C).$$

By part (b), write $C = C_1 - C_2$, where each $\mathcal{O}_X(C_i)$ is very ample. Use each of these very ample sheaves to embed X in a projective space. Since the singular locus is finite and k is infinite, there is a hyperplane containing none of its points. Its pullback is an effective Cartier divisor E_i , supported entirely in X_{reg} , with $E_i \sim C_i$. Therefore

$$\mathcal{L} \cong \mathcal{O}_X(C) \cong \mathcal{O}_X(E_1 - E_2),$$

and $E_1 - E_2$ has support in the regular locus.

(d) Apply part (a) to D :

$$\deg D + 1 - p_a(X) = \chi(\mathcal{O}_X(D)) = h^0(X, \mathcal{O}_X(D)) - h^1(X, \mathcal{O}_X(D)).$$

Serre duality and the choice of K identify the final term as

$$h^1(X, \mathcal{O}_X(D)) = h^0(X, \omega_X \otimes \mathcal{O}_X(-D)) = h^0(X, \mathcal{O}_X(K - D)) = \ell(K - D).$$

The desired formula follows. □

Exercise IV.1.10**Permitted context.**

Let X be an integral projective locally complete-intersection curve over an algebraically closed field k , and let X_{reg} denote its regular locus. Every invertible sheaf is represented by a divisor supported in X_{reg} , and its degree is the degree of such a divisor. Write $\text{Pic}^0 X$ for the subgroup of degree-zero invertible sheaves.

The dualizing sheaf ω_X is invertible. If K represents it and D is supported in X_{reg} , singular Riemann–Roch and Serre duality give

$$\ell(D) - \ell(K - D) = \deg D + 1 - p_a(X). \quad (1)$$

A nonzero section of an invertible sheaf on an integral curve cuts out an effective Cartier divisor E . Its length satisfies

$$\text{length}(E) = \chi(\mathcal{O}_X(E)) - \chi(\mathcal{O}_X),$$

by the defining exact sequence. An effective Cartier divisor of length one is a single regular point: locally, length one means that its local equation generates the maximal ideal, so the one-dimensional local ring is regular.

Problem. Assume $p_a(X) = 1$, and fix $P_0 \in X_{\text{reg}}$. Prove that

$$X_{\text{reg}} \longrightarrow \text{Pic}^0 X, \quad P \longmapsto \mathcal{O}_X(P - P_0),$$

is a bijection.

Solution. First determine the canonical divisor. Since X is integral and projective,

$$h^0(X, \mathcal{O}_X) = 1, \quad \chi(\mathcal{O}_X) = 1 - p_a(X) = 0,$$

so $h^1(X, \mathcal{O}_X) = 1$. Serre duality gives $\ell(K) = h^0(X, \omega_X) = 1$. Applying (1) with $D = K$ and using $\ell(0) = 1$ yields

$$\deg K = 0. \quad (2)$$

Let $\mathcal{L} \in \text{Pic}^0 X$. Represent it by a divisor D supported in X_{reg} , so that $\deg D = 0$. Apply (1) to $D + P_0$. By (2),

$$\deg(K - D - P_0) = -1.$$

An invertible sheaf of negative degree has no nonzero section, because such a section would cut out an effective Cartier divisor of negative degree. Hence

$$\ell(D + P_0) = \deg(D + P_0) + 1 - p_a(X) = 1.$$

Choose its nonzero section, and let E be the resulting effective Cartier divisor. Its defining exact sequence and the Euler-characteristic formula for degree give

$$\text{length}(E) = \chi(\mathcal{O}_X(D + P_0)) - \chi(\mathcal{O}_X) = 1.$$

Hence E is a single point $P \in X_{\text{reg}}$. Thus

$$\mathcal{L}(P_0) \cong \mathcal{O}_X(D + P_0) \cong \mathcal{O}_X(P),$$

or $\mathcal{L} \cong \mathcal{O}_X(P - P_0)$. This proves surjectivity.

For uniqueness, let $P \in X_{\text{reg}}$. Equation (2) gives $\deg(K - P) = -1$, so another application of (1) yields

$$\ell(P) = 1.$$

Consequently the complete linear system $|P|$ consists only of the divisor P . If

$$\mathcal{O}_X(P - P_0) \cong \mathcal{O}_X(Q - P_0),$$

then $P \sim Q$; since both are effective and $Q \in |P|$, it follows that $Q = P$. The displayed map is therefore injective as well as surjective. $\square$

IV.2 Hurwitz's Theorem

Exercise IV.2.1

Permitted context.

A scheme Y is simply connected if every finite étale cover of Y is a disjoint union of copies of Y . We use the fact that $\mathbb{P}^1$ is simply connected.

We also use the connectedness theorem: on an integral normal projective variety of dimension at least two, the support of an effective ample divisor is connected. Finite étale morphisms and their degrees are preserved by base change. All schemes are over an algebraically closed field k .

Problem. Prove that $\mathbb{P}^n$ is simply connected for every $n \geq 1$.

Solution. The assertion for $n = 1$ is part of the permitted context. Let

$$f : X \longrightarrow \mathbb{P}^n$$

be a finite étale cover, and suppose first that X is connected. Because X is étale over the smooth variety $\mathbb{P}^n$, it is regular. A connected regular scheme has only one irreducible component, so X is integral.

Choose a flag of linear subspaces

$$\mathbb{P}^1 = L_1 \subset L_2 \subset \cdots \subset L_n = \mathbb{P}^n$$

and put $X_i = f^{-1}(L_i)$. We claim, by descending induction, that every X_i is connected. This is true for $X_n = X$. If $i \geq 2$ and X_i is connected, then X_i is again regular and hence integral and normal. The subscheme

$$X_{i-1} = f|_{X_i}^{-1}(L_{i-1})$$

is the support of the pullback of a hyperplane, which is an effective ample divisor on X_i . The connectedness theorem therefore makes X_{i-1} connected. The claim follows.

In particular,

$$X_1 \longrightarrow L_1 \cong \mathbb{P}^1$$

is a connected finite étale cover. Since $\mathbb{P}^1$ is simply connected, this cover has degree one. Degree is unchanged by the base change to L_1 , so f itself has degree one and is an isomorphism.

For an arbitrary finite étale cover, apply this argument to each connected component. Every component maps isomorphically to $\mathbb{P}^n$, so the cover is a finite disjoint union of copies of $\mathbb{P}^n$. $\square$

Exercise IV.2.2**Permitted context.**

Let k be algebraically closed of characteristic different from 2. Every curve X of genus 2 has a base-point-free canonical pencil, which defines a finite morphism $f : X \rightarrow \mathbb{P}^1$ of degree two. Changing a basis of $H^0(X, \omega_X)$ changes f only by an automorphism of $\mathbb{P}^1$.

For a separable finite morphism $f : X \rightarrow Y$ of degree n , Hurwitz's formula is

$$2g(X) - 2 = n(2g(Y) - 2) + \deg R.$$

In the tame case, a point of ramification index e_P contributes $e_P - 1$ to R .

A separable quadratic extension of $k(x)$ has the form $k(x)(\sqrt{h})$. A place is ramified precisely when the valuation of h there is odd. Two such extensions are isomorphic over $k(x)$ if their defining elements differ by a square.

Problem. (a) Let X be a curve of genus 2, and let $f : X \rightarrow \mathbb{P}^1$ be its canonical morphism. Prove that f is ramified at exactly six points of X , each with ramification index 2. Explain why X determines an unordered configuration of six points of $\mathbb{P}^1$, up to an automorphism of $\mathbb{P}^1$.

(b) Given six distinct elements $a_1, \dots, a_6 \in k$, let

$$K = k(x, z), \quad z^2 = \prod_{i=1}^6 (x - a_i),$$

and let $f : X \rightarrow \mathbb{P}^1$ be the morphism of nonsingular projective curves corresponding to $k(x) \subset K$. Prove that $g(X) = 2$, that f is the canonical morphism, and that it is ramified precisely over $x = a_1, \dots, a_6$.

(c) Prove that for any three distinct points P_1, P_2, P_3 of $\mathbb{P}^1$, there is a unique $\varphi \in \text{Aut } \mathbb{P}^1$ satisfying

$$\varphi(P_1) = 0, \quad \varphi(P_2) = 1, \quad \varphi(P_3) = \infty.$$

Deduce that an ordered branch configuration can be normalized to

$$(0, 1, \infty, \beta_1, \beta_2, \beta_3),$$

where the β_i are pairwise distinct and avoid $0, 1$.

(d) Let S_6 act on such triples as follows: reorder the six entries

$$(0, 1, \infty, \beta_1, \beta_2, \beta_3)$$

and then apply the unique automorphism that sends the first three entries to $0, 1, \infty$; the last three entries form the new triple. Prove that this defines an action.

(e) Prove that isomorphism classes of genus-2 curves over k are in bijection with triples

$$(\beta_1, \beta_2, \beta_3) \in (\mathbb{A}^1 \setminus \{0, 1\})^3$$

having pairwise distinct entries, modulo the action of S_6 in part (d). In this sense, curves of genus 2 depend on three parameters.

Solution. (a) Because $\text{char } k \neq 2$, the degree-two map f is separable and all of its ramification is tame. Hurwitz's formula gives

$$2 = 2g(X) - 2 = 2(2g(\mathbb{P}^1) - 2) + \deg R = -4 + \deg R.$$

Thus $\deg R = 6$. At a ramification point of a degree-two map the ramification index must be 2, so each ramification point contributes one to R . There are therefore exactly six ramification points. Their six images are distinct: a fiber of degree two cannot contain two different points that both have ramification index two.

The map f is defined by the complete canonical pencil. A different basis of that two-dimensional space of sections composes f with an element of $\text{PGL}_2(k)$. Hence X determines the unordered set of six branch points up to $\text{Aut } \mathbb{P}^1$.

(b) The polynomial

$$h(x) = \prod_{i=1}^6 (x - a_i)$$

is square-free, so $K/k(x)$ is a separable quadratic extension. At the place $x = a_i$, one has $v_{a_i}(h) = 1$; hence there is one point above a_i , with ramification index two. Every other finite place has even valuation zero unless it is one of these six places.

At infinity, put $u = x^{-1}$ and $w = zu^3$. The equation becomes

$$w^2 = \prod_{i=1}^6 (1 - a_i u).$$

At $u = 0$ the right side equals 1, so, since $\text{char } k \neq 2$, there are two distinct unramified points above infinity. Thus the six a_i are exactly the branch points.

Hurwitz's formula now yields

$$2g(X) - 2 = 2(-2) + 6 = 2,$$

so $g(X) = 2$. Let $D = f^*(\infty)$. It has degree two, and the pullbacks of 1 and a coordinate on $\mathbb{P}^1$ show that $\ell(D) \geq 2$. Riemann–Roch on the genus-two curve gives

$$\ell(D) - \ell(K_X - D) = \deg D + 1 - g(X) = 1.$$

Hence $\ell(K_X - D) \geq 1$. This divisor has degree zero, so a nonzero section forces $K_X - D \sim 0$. Therefore

$$\mathcal{O}_X(D) \cong \omega_X, \quad \ell(D) = 2,$$

and the pencil defining f is the complete canonical pencil. Thus f is the canonical morphism, up to the choice of coordinate on its target.

(c) Choose an affine coordinate t in which the three points are finite; after a preliminary change of coordinate this is always possible. Then

$$\varphi(t) = \frac{(t - P_1)(P_2 - P_3)}{(t - P_3)(P_2 - P_1)}$$

has the required values. This formula has the evident homogeneous interpretation if one of the original points is infinity.

For uniqueness, an automorphism fixing 0, 1, and ∞ is represented by a diagonal fractional linear transformation fixing 1, and is therefore the identity. Applying the unique φ to an ordering

of the six branch points gives the displayed normal form. Distinctness of the branch points implies the asserted conditions on the β_i .

(d) Let

$$\mathbf{q} = (0, 1, \infty, \beta_1, \beta_2, \beta_3).$$

Use the convention that $\sigma \in S_6$ replaces this ordered sextuple by $(q_{\sigma^{-1}(1)}, \dots, q_{\sigma^{-1}(6)})$, and then normalize its first three entries as in part (c). The identity permutation clearly fixes the triple.

For $\sigma, \tau \in S_6$, performing the construction first with τ and then with σ has the same effect on the ordered sextuple as the permutation $\sigma\tau$, followed by a projective automorphism. In both constructions the final first three entries are $0, 1, \infty$. The uniqueness in part (c) therefore says that the two final normalizations are identical. Hence

$$\sigma \cdot (\tau \cdot (\beta_1, \beta_2, \beta_3)) = (\sigma\tau) \cdot (\beta_1, \beta_2, \beta_3),$$

which proves the action law.

(e) By part (a), a genus-two curve gives an unordered six-point branch configuration modulo $\mathrm{PGL}_2(k)$. Ordering the points and applying part (c) produces a triple $(\beta_1, \beta_2, \beta_3)$. Changing the ordering changes this triple precisely by the action in part (d), so the resulting S_6 -orbit is intrinsic.

Conversely, a six-point configuration on $\mathbb{P}^1$ determines a quadratic cover branched at those points. After moving infinity away from the configuration it has the form in part (b); alternatively, if the normalized configuration contains infinity, one may use

$$z^2 = x(x-1)(x-\beta_1)(x-\beta_2)(x-\beta_3),$$

whose odd degree supplies ramification at infinity. Part (b) shows that its nonsingular projective model has genus two.

It remains to check uniqueness of the cover. If $h, h' \in k(x)^\times$ give quadratic extensions with the same six branch places, then $\mathrm{div}(h/h')$ is even. Since every degree-zero divisor on $\mathbb{P}^1$ is principal, there are $r \in k(x)^\times$ and $c \in k^\times$ with

$$h/h' = cr^2.$$

The constant c is a square because k is algebraically closed, so the two quadratic extensions are isomorphic. Projectively equivalent branch configurations therefore give isomorphic curves, and an isomorphism of genus-two curves preserves their canonical maps and hence their branch configurations. This proves the claimed bijection. The triples form a nonempty open subset of $\mathbb{A}^3$, and the remaining quotient is by the finite group S_6 , which explains the three-parameter statement. $\square$

Exercise IV.2.3**Permitted context.**

Let $X \subset \mathbb{P}^2$ be a nonsingular plane curve of degree $d \geq 2$ over an algebraically closed field k of characteristic zero. Its genus is

$$g(X) = \frac{(d-1)(d-2)}{2}.$$

For $P \in X$, write $T_P X$ for its tangent line. The Gauss map

$$\gamma : X \longrightarrow (\mathbb{P}^2)^*, \quad P \longmapsto [T_P X],$$

has image the dual curve X^* .

A point P is an inflection point if $r_P = I_P(X, T_P X) \geq 3$, and it is counted with weight $r_P - 2$. A line tangent at more than one point is a multiple tangent; a strict bitangent is tangent at exactly two distinct points.

We use Hurwitz's formula, Bézout's theorem, and the formula

$$p_a(C) = \frac{(m-1)(m-2)}{2}$$

for a plane curve C of degree m . A node and an ordinary cusp both have normalization defect one. A finite birational map from a nonsingular curve is its normalization, and the points of a normalization index the branches of a curve germ. For formal power series of orders two and three in characteristic zero, the corresponding plane branch is formally equivalent to the ordinary cusp.

Problem. (a) Fix a line $L \subset \mathbb{P}^2$ that is not tangent to X , and define

$$\varphi : X \longrightarrow L, \quad \varphi(P) = T_P X \cap L.$$

Prove that φ is ramified at P exactly when either $P \in L$ or P is an inflection point. Deduce that X has only finitely many inflection points.

(b) Let M be tangent to X at distinct points $P_1, \dots, P_r$, which are all its tangency points, none of which is an inflection point. Prove that $[M] \in X^*$ is an ordinary r -fold point. Deduce that X has only finitely many multiple tangents.

(c) Let $O \in \mathbb{P}^2$ lie neither on X nor on an inflectional or multiple tangent, and project X from O to a line. Prove that the projection is ramified at P exactly when $OP = T_P X$, and then has ramification index 2. Deduce that exactly $d(d-1)$ tangents to X pass through O , and hence

$$\deg X^* = d(d-1).$$

(d) Prove that, outside a finite set of choices of $O \in X$, exactly

$$(d+1)(d-2)$$

tangent lines other than $T_O X$ pass through O .

(e) Prove that the morphism φ in part (a) has degree $d(d-1)$. Deduce that the weighted number of inflection points is

$$3d(d-2).$$

Prove that an ordinary inflection point corresponds to an ordinary cuspidal branch of X^* . It is an ordinary cusp of the whole dual curve if the tangent has no other tangency.

(f) Suppose that X^* has only nodes and ordinary cusps. Prove that the number of bitangents is

$$\frac{1}{2}d(d-2)(d-3)(d+3).$$

(g) Deduce that a plane cubic has exactly nine inflection points, all ordinary, and that the line through any two distinct inflection points meets the cubic in a third inflection point.

(h) Deduce that a smooth plane quartic has 28 generalized bitangents, where a tangent with fourfold contact is counted as a degenerate bitangent. Clarify what happens to X^* at such a hyperflex.

Solution. We begin with a local calculation used throughout. Choose affine coordinates at P such that $P = (0, 0)$, $T_P X$ is $y = 0$, and $t = x$ is a uniformizer on X . Then

$$y(t) = at^r + \text{terms of order greater than } r, \quad a \neq 0, \quad r = I_P(X, T_P X) \geq 2. \quad (1)$$

The tangent line at $(t, y(t))$ has equation

$$-y'(t)X + Y + ty'(t) - y(t) = 0. \quad (2)$$

In the affine chart of the dual plane where the coefficient of Y is one, the Gauss map is therefore

$$t \mapsto (-y'(t), ty'(t) - y(t)) = (-art^{r-1} + \cdots, a(r-1)t^r + \cdots). \quad (3)$$

Characteristic zero ensures that the displayed leading coefficients are nonzero.

We will also use that γ is the normalization of X^* . It is nonconstant and finite onto its image. At a general point, characteristic zero makes its differential nonzero. Differentiating (3) shows that the tangent to its image is P^* . Two distinct points over a general smooth point of X^* would therefore give two different tangents to that smooth point. This is impossible; thus the generic degree is one. The finite birational map from the nonsingular curve X is its normalization. In particular its fibers index all local branches.

(a) If $P \notin L$, choose the projective coordinates above so that L is the line at infinity. The point $T_{(t, y(t))} X \cap L$ records the tangent direction, and a local coordinate for it is $y'(t)$. By (1), its order is $r - 1$. Thus

$$e_P(\varphi) = r - 1 \quad (P \notin L). \quad (4)$$

If $P \in L$, then $L \neq T_P X$; take L to be $x = 0$. The intersection of (2) with L has coordinate

$$y(t) - ty'(t) = a(1 - r)t^r + \cdots,$$

so

$$e_P(\varphi) = r \quad (P \in L). \quad (5)$$

Equations (4) and (5) say exactly that φ ramifies at the points of $X \cap L$ and at the inflection points. The morphism is nonconstant, and in characteristic zero it is separable, so its ramification locus is finite. Hence there are only finitely many inflection points.

(b) At a noninflectionary point, $r = 2$, and (3) has the form

$$t \mapsto (c_1 t + \cdots, c_2 t^2 + \cdots), \quad c_1 c_2 \neq 0.$$

Thus the corresponding branch of X^* is nonsingular. Its tangent line in the dual plane is the line P^* parametrizing all lines in $\mathbb{P}^2$ through P . At $[M]$, the r points $P_1, \dots, P_r$ therefore give r smooth

branches whose tangent directions are the distinct lines $P_1^*, \dots, P_r^*$. Hence $[M]$ is an ordinary r -fold point.

The singular locus of the integral plane curve X^* is finite. Every multiple tangent not passing through an inflection point gives one of these singular points, while the tangent lines at inflection points form a finite set by part (a). Thus only finitely many multiple tangents exist.

(c) Let

$$\psi : X \longrightarrow \mathbb{P}^1$$

be projection from O . Its fiber through P is cut out by the line OP , and the local ramification index is

$$e_P(\psi) = I_P(X, OP). \quad (6)$$

Consequently ψ ramifies precisely when $OP = T_P X$. The assumptions on O exclude inflectional tangents, so the contact order is then two. They also exclude multiple tangents, so distinct ramification points yield distinct tangent lines through O .

The map ψ has degree d . Since

$$2g(X) - 2 = d(d - 3),$$

Hurwitz gives

$$\deg R_\psi = 2g(X) - 2 + 2d = d(d - 1).$$

Every ramification point contributes one, proving the tangent count. The lines through a general point O form a line O^* in the dual plane, and $O^* \cap X^*$ parametrizes these tangents. Therefore $\deg X^* = d(d - 1)$.

(d) Exclude from X the inflection points and the intersections of X with every inflectional or multiple tangent. Parts (a) and (b) show that this removes only finitely many points. For a remaining point O , projection from O is defined by the pencil of lines through O after its simple base point at O is removed. It has degree $d - 1$. The local calculation gives ramification index $r_O - 1 = 1$ at O , so $T_O X$ contributes no ramification. Every other ramification point has index two and corresponds to exactly one tangent through O . Hurwitz now gives

$$\deg R = 2g(X) - 2 + 2(d - 1) = d(d - 3) + 2d - 2 = (d + 1)(d - 2),$$

which is the required number.

(e) For a general $Q \in L$, the fiber $\varphi^{-1}(Q)$ consists of the points P for which $T_P X$ passes through Q . Part (c) shows that there are $d(d - 1)$ of them, so

$$\deg \varphi = d(d - 1).$$

Hurwitz and the genus formula give

$$\deg R_\varphi = 2g(X) - 2 + 2d(d - 1) = d(3d - 5). \quad (7)$$

Because L is not tangent to X , it meets X transversely in d points. Formulas (4) and (5) show that every point of $X \cap L$ contributes a baseline amount one to R_φ , while an inflection point of contact order r contributes an additional $r - 2$, whether or not it lies on L . Thus

$$\deg R_\varphi = d + \sum_{P \text{ an inflection}} (r_P - 2).$$

Together with (7), this gives

$$\sum_{P \text{ an inflection}} (r_P - 2) = 3d(d - 2). \quad (8)$$

At an ordinary inflection $r = 3$, formula (3) parametrizes the dual branch by

$$(u, v) = (c_1 t^2 + \cdots, c_2 t^3 + \cdots), \quad c_1 c_2 \neq 0.$$

After formal coordinate changes this branch is an ordinary cusp. Another point of tangency on the same line contributes another branch at its dual point. Thus without the additional hypothesis in (e), the entire singular point need not be an ordinary cusp.

(f) We have already identified γ with the normalization. Put $m = \deg X^* = d(d - 1)$. Under the stated hypothesis, the nodes of X^* correspond exactly to bitangents, and its ordinary cusps correspond exactly to ordinary inflection points. Equation (8) says that the number of cusps is

$$\kappa = 3d(d - 2).$$

If b is the number of nodes, the normalization-genus formula gives

$$\frac{(m - 1)(m - 2)}{2} = \frac{(d - 1)(d - 2)}{2} + b + \kappa.$$

Solving and substituting $m = d(d - 1)$ and the value of κ yields

$$b = \frac{1}{2}d(d - 2)(d - 3)(d + 3).$$

(g) For $d = 3$, equation (8) gives nine inflection points counted with weight. Bézout bounds every tangent contact by three, so every inflection is ordinary and the nine points are distinct.

Let H denote a line section. If P is an inflection point, its tangent line cuts out $3P$, so $3P \sim H$. If a line through two distinct inflection points P, Q cuts out $P + Q + R$, then

$$P + Q + R \sim H.$$

Multiplying by three and using $3P \sim 3Q \sim H$ gives

$$3R \sim H.$$

The line-section system is complete: the hypersurface sequence on $\mathbb{P}^2$ gives $h^0(X, \mathcal{O}_X(1)) = 3$. Thus some line cuts out $3R$, so R is also an inflection point. The third point is distinct from P, Q : otherwise a degree-zero divisor such as $P - Q$ would be principal. The corresponding rational function would have one simple pole and define a degree-one morphism from the genus-one curve X to $\mathbb{P}^1$, an impossibility.

(h) For $d = 4$, equation (8) says that the total inflectionary weight is 24. Let f be the number of ordinary inflections, h the number of hyperflexes with contact order four, and b the number of strict bitangents. Then

$$f + 2h = 24. \quad (9)$$

The dual of an ordinary inflection is an ordinary cusp of defect one, and a strict bitangent gives a node of defect one. At a hyperflex, (3) has local parametrization

$$(u, v) = (c_1 t^3 + \cdots, c_2 t^4 + \cdots).$$

Write its complete local subring as $A = k[[u(t), v(t)]]$. Every integer $m \geq 6$ is $3a + 4b$ for nonnegative a, b . Thus a monomial in u, v has each possible leading order $m \geq 6$. Successively canceling leading terms, and using completeness, gives $t^6 k[[t]] \subseteq A$. Modulo t^6 , the ring A is spanned by $1, u, v$, which are independent because their leading orders are $0, 3, 4$. Consequently

$$\dim_k(k[[t]]/A) = 6 - 3 = 3.$$

This computes the normalization defect without identifying the subring in the original parameter with a monomial subring. These exhaust the singularities of X^* : a singular branch of the Gauss image comes from contact order at least three, and two branches have the same image exactly when their two points share a tangent line. For a quartic, Bézout allows only the cases just listed.

Now $\deg X^* = 12$, so

$$p_a(X^*) = \frac{11 \cdot 10}{2} = 55, \quad g(X) = 3.$$

The normalization formula therefore gives

$$55 = 3 + b + f + 3h. \quad (10)$$

Combining (9) and (10) yields

$$b + h = 28.$$

Thus there are 28 generalized bitangent lines: the b lines tangent at two distinct points together with the h hyperflex lines, each regarded as a bitangent whose two contact points coincide. Under the strict definition from part (b), the number of two-point bitangents is $28 - h$. The dual singularity at a hyperflex is the unibranch $(3, 4)$ -cusp computed above, not a tacnode. $\square$

Exercise IV.2.4

Permitted context.

Let k be an algebraically closed field of characteristic 3, and let

$$X = V(F) \subset \mathbb{P}_k^2, \quad F(x, y, z) = x^3y + y^3z + z^3x.$$

For a smooth plane curve, the Gauss map sends a point P to its tangent line in the dual projective plane. Its image has closure the dual curve X^* .

A point P is an inflection point when $I_P(X, T_P X) \geq 3$. We use the Jacobian criterion and the fact that the relative Frobenius of a smooth curve in characteristic p is a finite purely inseparable morphism of degree p .

Problem. Prove that X is nonsingular, that every point of X is an inflection point, and that X^* is isomorphic to X , although the natural map $X \rightarrow X^*$ is purely inseparable.

Solution. In characteristic 3, the partial derivatives are

$$F_x = z^3, \quad F_y = x^3, \quad F_z = y^3.$$

They cannot vanish simultaneously at a projective point. The Jacobian criterion therefore shows that X is nonsingular. In particular, X is irreducible: two positive-degree components in the projective plane would meet and give a singular point.

Let $P = [a : b : c] \in X$. Its tangent line is

$$T_P X : c^3 x + a^3 y + b^3 z = 0. \quad (1)$$

Choose a direction $V = (u, v, w)$ on this line, so that

$$c^3 u + a^3 v + b^3 w = 0. \quad (2)$$

Using $(\alpha + t\beta)^3 = \alpha^3 + t^3\beta^3$, expansion along the line $P + tV$ gives

$$\begin{aligned} F(P + tV) &= F(P) + t(c^3 u + a^3 v + b^3 w) \\ &\quad + t^3(u^3 b + v^3 c + w^3 a) + t^4(u^3 v + v^3 w + w^3 u). \end{aligned} \quad (3)$$

The constant term vanishes because $P \in X$, and the coefficient of t vanishes by (2). There is no quadratic term. Since the tangent line is not a component of X , equation (3) proves

$$I_P(X, T_P X) \geq 3.$$

Thus every point of X is an inflection point.

Write $[U : V : W]$ for coordinates in the dual plane, representing the line $Ux + Vy + Wz = 0$. Formula (1) shows that the Gauss map is

$$\gamma : X \longrightarrow (\mathbb{P}^2)^*, \quad [a : b : c] \longmapsto [c^3 : a^3 : b^3]. \quad (4)$$

The image satisfies

$$\begin{aligned} U^3 V + V^3 W + W^3 U &= c^9 a^3 + a^9 b^3 + b^9 c^3 \\ &= (c^3 a + a^3 b + b^3 c)^3 \\ &= F(a, b, c)^3 = 0. \end{aligned} \quad (5)$$

Consequently X^* is contained in the plane quartic

$$C : U^3 V + V^3 W + W^3 U = 0.$$

This quartic is nonsingular by the same partial-derivative calculation as above, hence integral. The map (4) is nonconstant, so its image is dense in C . Therefore $X^* = C$. The equation of C is the same as that of X after merely renaming the coordinates, and hence

$$X^* \cong X.$$

Finally, (4) factors as

$$X \xrightarrow{F_{X/k}} X^{(3)} \xrightarrow{[A:B:C] \mapsto [C:A:B]} X^*, \quad (6)$$

where the first arrow is relative Frobenius; the equation has coefficients in the prime field, so $X^{(3)}$ is identified with the same plane quartic. The second arrow is an isomorphism, whereas $F_{X/k}$ is purely inseparable of degree 3. Thus the natural Gauss map $X \rightarrow X^*$ is purely inseparable, even though its source and target are isomorphic smooth curves. $\square$

Exercise IV.2.5**Permitted context.**

Let X be a nonsingular projective curve of genus $g \geq 2$ over an algebraically closed field k of characteristic zero. Assume the known finiteness of

$$G = \text{Aut}(X), \quad |G| = n.$$

The fixed-field inclusion

$$K(X)^G = K(Y) \subset K(X)$$

defines a finite quotient morphism $f : X \rightarrow Y$ of degree n .

For a tame finite morphism, Hurwitz's formula is

$$2g(X) - 2 = \deg(f)(2g(Y) - 2) + \sum_{P \in X} (e_P - 1).$$

Problem. (a) If $P \in X$ is ramified and $e_P = r$, prove that $f^{-1}(f(P))$ consists of exactly n/r points, all of ramification index r . Choose one point P_i over each branch point of Y , and write $r_i = e_{P_i}$. Prove that

$$\frac{2g - 2}{n} = 2g(Y) - 2 + \sum_{i=1}^s \left(1 - \frac{1}{r_i}\right).$$

(b) Suppose $h, s \geq 0$ and $r_i \geq 2$ are integers for which

$$2h - 2 + \sum_{i=1}^s \left(1 - \frac{1}{r_i}\right) > 0.$$

Prove that the least possible value of this expression is $1/42$. Deduce Hurwitz's bound

$$|\text{Aut}(X)| \leq 84(g - 1).$$

Solution. (a) The extension $K(X)/K(Y)$ is Galois with group G . The group G acts transitively on the points of X above any given point of Y . For $P \in X$, its stabilizer

$$G_P = \{\sigma \in G : \sigma(P) = P\}$$

is the decomposition group of the corresponding discrete valuation. All residue fields are k , so the decomposition group equals the inertia group. Characteristic zero makes the ramification tame, and hence

$$|G_P| = e_P.$$

If $e_P = r$, the orbit-stabilizer formula now gives

$$\#f^{-1}(f(P)) = \#(G \cdot P) = \frac{n}{r}.$$

The indices at all points in this orbit are equal because their local extensions are conjugate.

Let $Q_1, \dots, Q_s$ be the branch points and choose $P_i \in f^{-1}(Q_i)$, with $r_i = e_{P_i}$. The fiber over Q_i contains n/r_i points, each contributing $r_i - 1$ to the ramification divisor. Consequently

$$\deg R = \sum_{i=1}^s \frac{n}{r_i} (r_i - 1) = n \sum_{i=1}^s \left(1 - \frac{1}{r_i}\right).$$

Substitution in Hurwitz's formula and division by n gives

$$\frac{2g-2}{n} = 2g(Y) - 2 + \sum_{i=1}^s \left(1 - \frac{1}{r_i}\right). \quad (1)$$

(b) Put

$$E = 2h - 2 + \sum_{i=1}^s \left(1 - \frac{1}{r_i}\right) > 0.$$

We determine its smallest possible value. If $h \geq 2$, then $E \geq 2$. If $h = 1$, positivity requires $s \geq 1$, and every summand is at least $1/2$, so $E \geq 1/2$.

It remains to take $h = 0$, when

$$E = s - 2 - \sum_{i=1}^s \frac{1}{r_i}. \quad (2)$$

Positivity forces $s \geq 3$. If $s \geq 5$, then

$$E \geq s - 2 - \frac{s}{2} = \frac{s}{2} - 2 \geq \frac{1}{2}.$$

If $s = 4$, the tuple $(2, 2, 2, 2)$ gives zero; every other tuple has at least one entry at least 3, and therefore

$$E \geq 2 - \left(\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{3}\right) = \frac{1}{6}.$$

Finally suppose $s = 3$, and reorder the indices so that

$$2 \leq r_1 \leq r_2 \leq r_3.$$

If $r_1 \geq 3$, the zero case $(3, 3, 3)$ is excluded, and the largest reciprocal sum below 1 is attained at $(3, 3, 4)$. Hence

$$E \geq 1 - \frac{1}{3} - \frac{1}{3} - \frac{1}{4} = \frac{1}{12}.$$

If $r_1 = 2$, then $r_2 = 2$ is impossible because the reciprocal sum would exceed 1. For $r_2 = 3$, positivity forces $r_3 \geq 7$, so

$$E \geq 1 - \frac{1}{2} - \frac{1}{3} - \frac{1}{7} = \frac{1}{42}.$$

For $r_2 \geq 4$, the zero tuple $(2, 4, 4)$ is excluded, and the largest reciprocal sum compatible with positivity is obtained from $(2, 4, 5)$. Thus

$$E \geq 1 - \frac{1}{2} - \frac{1}{4} - \frac{1}{5} = \frac{1}{20}.$$

The global minimum is therefore $1/42$, attained for

$$(h; s; r_1, r_2, r_3) = (0; 3; 2, 3, 7).$$

The left side of (1) is positive because $g \geq 2$. Applying the minimum just proved to its right side yields

$$\frac{2g-2}{n} \geq \frac{1}{42}.$$

Therefore

$$n \leq 42(2g-2) = 84(g-1),$$

which is Hurwitz's bound. □

Exercise IV.2.6**Permitted context.**

Let $f : X \rightarrow Y$ be a finite morphism of degree n between nonsingular projective curves over an algebraically closed field. For

$$D = \sum_P m_P P \in \text{Div } X$$

define its divisor pushforward by

$$f_*^{\text{div}} D = \sum_P m_P f(P).$$

The morphism f is finite flat, so for every invertible sheaf $\mathcal{M}$ on X , its sheaf-theoretic direct image $f_*\mathcal{M}$ is locally free of rank n .

For a rank- r locally free sheaf $\mathcal{E}$, write $\det \mathcal{E} = \bigwedge^r \mathcal{E}$. We use finite duality in the form

$$f_*\omega_X \cong \mathcal{H}om_Y(f_*\mathcal{O}_X, \omega_Y).$$

In positive characteristic, R is the different divisor with its actual multiplicities, including wild ramification.

Problem. (a) Prove that for every divisor D on X ,

$$\det(f_*\mathcal{O}_X(D)) \cong \det(f_*\mathcal{O}_X) \otimes \mathcal{O}_Y(f_*^{\text{div}} D).$$

(b) Deduce that divisor pushforward respects linear equivalence and therefore induces a homomorphism

$$f_* : \text{Pic } X \longrightarrow \text{Pic } Y.$$

Prove that f_*f^* is multiplication by n on $\text{Pic } Y$.

(c) Prove that

$$\det(f_*\Omega_X^1) \cong \det(f_*\mathcal{O}_X)^{-1} \otimes \omega_Y^{\otimes n}.$$

(d) Suppose that f is separable, let R be its ramification divisor, and put $B = f_*^{\text{div}} R$. Prove that

$$\det(f_*\mathcal{O}_X)^{\otimes 2} \cong \mathcal{O}_Y(-B).$$

Solution. (a) We first record the effect of an elementary modification. If $\mathcal{E}' \subset \mathcal{E}$ are rank- n vector bundles on Y and

$$0 \longrightarrow \mathcal{E}' \longrightarrow \mathcal{E} \longrightarrow k(Q) \longrightarrow 0 \tag{1}$$

is exact, then

$$\det \mathcal{E} \cong \det \mathcal{E}' \otimes \mathcal{O}_Y(Q). \tag{2}$$

Indeed, over the discrete valuation ring $\mathcal{O}_{Y,Q}$, suitable bases put the inclusion in (1) in diagonal form $\text{diag}(t, 1, \dots, 1)$, where t is a uniformizer. Taking top exterior powers gives (2).

For any divisor D and closed point $P \in X$, there is an exact sequence

$$0 \longrightarrow \mathcal{O}_X(D) \longrightarrow \mathcal{O}_X(D+P) \longrightarrow k(P) \longrightarrow 0. \tag{3}$$

Finite direct image is exact. If $Q = f(P)$, then $f_*k(P) = k(Q)$, so (2) applied to the direct image of (3) gives

$$\det f_*\mathcal{O}_X(D+P) \cong \det f_*\mathcal{O}_X(D) \otimes \mathcal{O}_Y(Q). \tag{4}$$

Starting with $D = 0$, repeated use of (4), in both the positive and negative directions, proves

$$\det f_* \mathcal{O}_X(D) \cong \det(f_* \mathcal{O}_X) \otimes \mathcal{O}_Y(f_*^{\text{div}} D) \quad (5)$$

for every divisor D .

(b) If $D \sim D'$, then $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$. Their direct images and determinants are therefore isomorphic. Formula (5), after cancelling $\det(f_* \mathcal{O}_X)$, shows that

$$\mathcal{O}_Y(f_*^{\text{div}} D) \cong \mathcal{O}_Y(f_*^{\text{div}} D').$$

Thus $f_*^{\text{div}} D \sim f_*^{\text{div}} D'$, and divisor pushforward induces the asserted map on Picard groups.

For a closed point $Q \in Y$, one has

$$f^* Q = \sum_{P \mapsto Q} e_P P.$$

Because the residue fields are all k , the degree formula for the fiber says $\sum_{P \mapsto Q} e_P = n$. Hence

$$f_*^{\text{div}} f^* Q = \sum_{P \mapsto Q} e_P Q = nQ.$$

Linearity gives $f_* f^* [D] = n[D]$ for every $[D] \in \text{Pic } Y$.

(c) Put $\mathcal{E} = f_* \mathcal{O}_X$. Since X and Y are nonsingular curves, $\Omega_X^1 = \omega_X$ and finite duality gives

$$f_* \Omega_X^1 \cong \mathcal{H}om_Y(\mathcal{E}, \omega_Y) \cong \mathcal{E}^\vee \otimes \omega_Y.$$

For a rank- n vector bundle $\mathcal{E}$ and a line bundle $\mathcal{L}$,

$$\det(\mathcal{E}^\vee \otimes \mathcal{L}) \cong (\det \mathcal{E})^{-1} \otimes \mathcal{L}^{\otimes n}.$$

Taking $\mathcal{L} = \omega_Y$ proves

$$\det(f_* \Omega_X^1) \cong \det(f_* \mathcal{O}_X)^{-1} \otimes \omega_Y^{\otimes n}. \quad (6)$$

(d) Separability gives the canonical-sheaf form of the ramification formula,

$$\omega_X \cong f^* \omega_Y \otimes \mathcal{O}_X(R).$$

The projection formula and then part (a) yield

$$\begin{aligned} \det(f_* \omega_X) &\cong \det(\omega_Y \otimes f_* \mathcal{O}_X(R)) \\ &\cong \omega_Y^{\otimes n} \otimes \det(f_* \mathcal{O}_X) \otimes \mathcal{O}_Y(B). \end{aligned} \quad (7)$$

On the other hand, part (c) says

$$\det(f_* \omega_X) \cong \det(f_* \mathcal{O}_X)^{-1} \otimes \omega_Y^{\otimes n}. \quad (8)$$

Comparing (7) and (8), and cancelling $\omega_Y^{\otimes n}$, gives

$$\det(f_* \mathcal{O}_X)^{\otimes 2} \cong \mathcal{O}_Y(-B),$$

as required. $\square$

Exercise IV.2.7**Permitted context.**

Let Y be a nonsingular projective curve over an algebraically closed field k of characteristic different from 2. Covers are considered up to isomorphism over Y , and line bundles up to isomorphism.

For a finite locally free morphism $f : X \rightarrow Y$, write $\mathcal{A} = f_*\mathcal{O}_X$. Then

$$X \cong \operatorname{Spec}_Y \mathcal{A}.$$

If f is finite étale of degree 2, its branch divisor is zero, so Exercise IV.2.6 gives

$$\det(\mathcal{A})^{\otimes 2} \cong \mathcal{O}_Y.$$

Problem. (a) Let $f : X \rightarrow Y$ be a finite étale morphism of degree 2. For the natural inclusion

$$\mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X,$$

prove that the cokernel $\mathcal{L}$ is invertible, that $\mathcal{L} \cong \det(f_*\mathcal{O}_X)$, and hence that $\mathcal{L}^{\otimes 2} \cong \mathcal{O}_Y$.

(b) Conversely, given an invertible sheaf $\mathcal{L}$ and an isomorphism $\phi : \mathcal{L} \otimes \mathcal{L} \rightarrow \mathcal{O}_Y$, give $\mathcal{O}_Y \oplus \mathcal{L}$ the multiplication

$$(a, b)(a', b') = (aa' + \phi(b \otimes b'), ab' + a'b).$$

Prove that its relative spectrum is a finite étale cover of Y of degree 2.

(c) Prove that the constructions in (a) and (b) are inverse. Conclude that degree-two finite étale covers of Y correspond bijectively to the 2-torsion elements of $\operatorname{Pic} Y$.

Solution. (a) Put $\mathcal{A} = f_*\mathcal{O}_X$. This is a rank-two locally free $\mathcal{O}_Y$ -algebra. Its trace map satisfies

$$\operatorname{Tr}_{\mathcal{A}/\mathcal{O}_Y}(1) = 2.$$

Because 2 is invertible, the map $\frac{1}{2} \operatorname{Tr} : \mathcal{A} \rightarrow \mathcal{O}_Y$ splits the inclusion of $\mathcal{O}_Y$. Thus its cokernel $\mathcal{L}$ is locally free of rank one, and the exact sequence

$$0 \longrightarrow \mathcal{O}_Y \longrightarrow \mathcal{A} \longrightarrow \mathcal{L} \longrightarrow 0 \tag{1}$$

gives

$$\det \mathcal{A} \cong \det \mathcal{O}_Y \otimes \det \mathcal{L} \cong \mathcal{L}.$$

Since f is étale, its ramification and branch divisors vanish. Exercise IV.2.6(d) therefore gives

$$\mathcal{L}^{\otimes 2} \cong \det(\mathcal{A})^{\otimes 2} \cong \mathcal{O}_Y.$$

(b) Let

$$\mathcal{A}_\phi = \mathcal{O}_Y \oplus \mathcal{L}$$

with the displayed multiplication. It is a commutative $\mathcal{O}_Y$ -algebra, locally free of rank two, so

$$X_\phi = \operatorname{Spec}_Y \mathcal{A}_\phi \longrightarrow Y$$

is finite flat of degree two.

To check that it is étale, work on an open set on which $\mathcal{L}$ has a generator e . Write $\phi(e \otimes e) = u$, where u is a unit. Then

$$\mathcal{A}_\phi \cong \mathcal{O}_Y[T]/(T^2 - u).$$

In this algebra T is a unit, and 2 is a unit by hypothesis. Thus the derivative $2T$ is invertible, so

$$\Omega_{\mathcal{A}_\phi/\mathcal{O}_Y} = 0.$$

The finite flat morphism is therefore étale.

The cover depends only on the isomorphism class of the 2-torsion line bundle, not on the chosen ϕ . Indeed, two trivializations of $\mathcal{L}^{\otimes 2}$ differ by a scalar $c \in k^\times$. Choose $\lambda \in k^\times$ with $c\lambda^2 = 1$. Multiplication by λ on the $\mathcal{L}$ -summand gives an isomorphism between the resulting algebras.

(c) Starting from $(\mathcal{L}, \phi)$, the natural inclusion

$$\mathcal{O}_Y \longrightarrow \mathcal{O}_Y \oplus \mathcal{L}$$

has cokernel $\mathcal{L}$. Hence the construction in (a) immediately recovers the original line bundle.

Conversely, begin with a degree-two étale cover and put $\mathcal{A} = f_*\mathcal{O}_X$. The splitting in (a) identifies the cokernel in (1) with the trace-zero line bundle

$$\mathcal{M} = \ker(\text{Tr} : \mathcal{A} \rightarrow \mathcal{O}_Y), \quad \mathcal{A} = \mathcal{O}_Y \oplus \mathcal{M}. \quad (2)$$

There is a canonical involution

$$\tau : \mathcal{A} \longrightarrow \mathcal{A}, \quad \tau(a) = \text{Tr}(a) - a. \quad (3)$$

Étale locally on Y , the algebra is $\mathcal{O}_Y \times \mathcal{O}_Y$, and τ interchanges the two factors. It follows by descent that τ is an algebra automorphism, that its invariant subalgebra is $\mathcal{O}_Y$, and that it acts as -1 on $\mathcal{M}$.

Consequently multiplication in $\mathcal{A}$ sends $\mathcal{M} \otimes \mathcal{M}$ to the invariant summand $\mathcal{O}_Y$. This pairing is an isomorphism: after the same étale localization, $\mathcal{M}$ consists of the pairs $(a, -a)$, and

$$(a, -a)(b, -b) = (ab, ab).$$

Thus (2), including its algebra structure, is exactly the algebra constructed in part (b) from

$$\mathcal{M}^{\otimes 2} \xrightarrow{\sim} \mathcal{O}_Y.$$

Taking relative spectra recovers $X \rightarrow Y$. The two constructions are therefore inverse and give the asserted bijection.

The trivial class $\mathcal{L} = \mathcal{O}_Y$ corresponds to the split cover $Y \sqcup Y \rightarrow Y$; if only connected covers are admitted, it is the unique class to omit. $\square$

IV.3 Embeddings in Projective Space

Exercise IV.3.1

Permitted context.

Let X be a nonsingular projective integral curve of genus 2 over an algebraically closed field. We use Riemann–Roch and the facts that a divisor of degree at least $2g + 1$ is very ample, and that a nonsingular plane curve of degree m has genus

$$\frac{(m-1)(m-2)}{2}.$$

A very ample complete linear system on a positive-genus curve has at least three independent sections. For a line bundle of nonnegative degree d , a nonzero section identifies it with an effective divisor; removing its d points one at a time gives $h^0 \leq d + 1$. A line bundle of negative degree has no nonzero sections.

Problem. Prove that a divisor D on X is very ample if and only if

$$\deg D \geq 5.$$

Solution. If $\deg D \geq 5 = 2g + 1$, the standard very-ampleness theorem for curves shows that D is very ample.

Conversely, suppose that D is very ample, and put $d = \deg D$. Then $d > 0$, and the complete linear system embeds X in $\mathbb{P}^{h^0(D)-1}$. Since a positive-genus curve cannot embed in $\mathbb{P}^0$ or $\mathbb{P}^1$, one must have

$$h^0(D) \geq 3. \tag{1}$$

For $d \leq 1$, the elementary estimate $h^0(D) \leq d + 1$ contradicts (1). If $d = 2$, Riemann–Roch gives

$$h^0(D) = 1 + h^0(K_X - D) \leq 2,$$

because $K_X - D$ has degree zero and a degree-zero line bundle has at most one independent section. Thus $d \geq 3$.

If $d = 3$, then $\deg(K_X - D) = -1$, so Riemann–Roch gives

$$h^0(D) = d + 1 - g = 2,$$

again contradicting (1). If $d = 4$, the same calculation gives

$$h^0(D) = 3.$$

Very ampleness would therefore identify X with a nonsingular plane curve of degree 4. Such a curve has genus

$$\frac{(4-1)(4-2)}{2} = 3,$$

contrary to $g(X) = 2$. Hence $d \geq 5$, proving the converse. $\square$

Exercise IV.3.2**Permitted context.**

Let $X \subset \mathbb{P}^2$ be a nonsingular plane curve of degree 4, over an algebraically closed field. We use Riemann–Roch. Then $g(X) = 3$, and adjunction gives

$$\omega_X \cong \mathcal{O}_X(1).$$

We use $H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(m)) = 0$ for every integer m . A curve is hyperelliptic precisely when it has a degree-two divisor D with $\dim |D| = 1$.

Problem. (a) Prove that the effective canonical divisors on X are exactly the divisors $X \cdot L$, where $L \subset \mathbb{P}^2$ is a line.

(b) If D is any effective divisor of degree 2 on X , prove that

$$\dim |D| = 0.$$

(c) Deduce that X is not hyperelliptic.

Solution. (a) Restricting linear forms to X gives the exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2}(-3) \longrightarrow \mathcal{O}_{\mathbb{P}^2}(1) \longrightarrow \mathcal{O}_X(1) \longrightarrow 0.$$

Since

$$H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-3)) = 0 \quad \text{and} \quad H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-3)) = 0,$$

the restriction map

$$H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(1)) \xrightarrow{\sim} H^0(X, \mathcal{O}_X(1)) = H^0(X, \omega_X)$$

is an isomorphism. The zero divisor of a nonzero restricted linear form is the scheme-theoretic line section $X \cdot L$, and every nonzero canonical section arises this way.

(b) Riemann–Roch, with $g(X) = 3$ and $\deg D = 2$, gives

$$h^0(D) = h^0(K_X - D). \tag{1}$$

Suppose that $h^0(D) \geq 2$. By (1), choose an effective divisor

$$E \in |K_X - D|.$$

It has degree two. For every $D' \in |D|$, the divisor

$$D' + E$$

is effective and canonical, so part (a) writes it as a line section.

There is at most one line whose intersection divisor with X contains E . Indeed, if $E = P + Q$ with $P \neq Q$, that line must be the unique line through P and Q ; if $E = 2P$, it must be the unique tangent line $T_P X$. Thus every $D' \in |D|$ gives the same line section as D , and

$$D' + E = D + E.$$

Cancellation of effective divisors gives $D' = D$. This contradicts $h^0(D) \geq 2$, which would make $|D|$ positive-dimensional. Therefore $h^0(D) = 1$, or equivalently $\dim |D| = 0$.

(c) A hyperelliptic curve possesses a g_2^1 , that is, an effective degree-two divisor D with $\dim |D| = 1$. Part (b) rules out every such divisor. Hence the plane quartic X is not hyperelliptic. $\square$

Exercise IV.3.3**Permitted context.**

Work over an algebraically closed field. Let $X \subset \mathbb{P}^n$ be a nonsingular integral curve, cut out scheme-theoretically by hypersurfaces of degrees $d_1, \dots, d_{n-1}$. We use the canonical degree and dimension formulas

$$\deg \omega_X = 2g(X) - 2, \quad h^0(X, \omega_X) = g(X).$$

The adjunction formula gives

$$\omega_X \cong \mathcal{O}_X \left(\sum_{i=1}^{n-1} d_i - n - 1 \right).$$

The line bundle $\mathcal{O}_X(1)$ is very ample, and every positive tensor power of a very ample line bundle is very ample.

Problem. If $g(X) \geq 2$, prove that the canonical divisor K_X is very ample. Deduce that a curve of genus 2 cannot be a complete intersection in any projective space.

Solution. Set

$$m = \sum_{i=1}^{n-1} d_i - n - 1.$$

The adjunction formula says

$$\omega_X \cong \mathcal{O}_X(m). \tag{1}$$

Taking degrees in (1) gives

$$2g(X) - 2 = m \deg \mathcal{O}_X(1).$$

The left side is positive because $g(X) \geq 2$, and $\deg \mathcal{O}_X(1) > 0$. Hence the integer m is positive. Thus ω_X is a positive tensor power of the very ample line bundle $\mathcal{O}_X(1)$, and is itself very ample.

Now suppose that $g(X) = 2$. The canonical space has dimension

$$h^0(X, \omega_X) = g(X) = 2,$$

so the canonical map has target $\mathbb{P}^1$. A genus-two curve cannot be embedded in $\mathbb{P}^1$: a closed integral one-dimensional subscheme of $\mathbb{P}^1$ is $\mathbb{P}^1$ itself, which has genus zero. Hence K_X is not very ample. This contradicts the first paragraph if X is a complete intersection, so no genus-two curve is a complete intersection in any $\mathbb{P}^n$. $\square$

Exercise IV.3.4**Permitted context.**

Work over an algebraically closed field; curves below are nonsingular, projective, and integral.

Let

$$\nu_d : \mathbb{P}^1 \longrightarrow \mathbb{P}^d, \quad [s : t] \longmapsto [s^d : s^{d-1}t : \cdots : st^{d-1} : t^d]$$

be the d -uple embedding. Its image is the rational normal curve of degree d .

We use the criterion that a normal projective variety is projectively normal when all restriction maps

$$H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m)) \longrightarrow H^0(X, \mathcal{O}_X(m))$$

are surjective. For an effective divisor D of degree d on a curve, successively removing its points gives $h^0(D) \leq d + 1$.

Problem. (a) Prove that the rational normal curve is projectively normal and that its homogeneous ideal is generated by quadrics.

(b) Let $X \subset \mathbb{P}^n$ be a curve of degree $d \leq n$ that is not contained in a hyperplane. Prove that

$$d = n, \quad g(X) = 0,$$

and that, up to an automorphism of $\mathbb{P}^d$, X is the rational normal curve.

(c) Deduce that every curve of degree 2 in projective space is a conic in some $\mathbb{P}^2$.

(d) Prove that every curve of degree 3 in projective space is either a plane cubic or the twisted cubic in $\mathbb{P}^3$.

Solution. (a) Let $S = k[X_0, \dots, X_d]$, and consider the graded map

$$\theta : S \longrightarrow k[s, t]^{(d)} = \bigoplus_{m \geq 0} k[s, t]_{md}, \quad X_i \longmapsto s^{d-i}t^i. \quad (1)$$

Every monomial $s^{md-q}t^q$, $0 \leq q \leq md$, is a product of m monomials of degree d . Hence θ is surjective in every degree. Equivalently,

$$\mathrm{Sym}^m H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(d)) \longrightarrow H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(md))$$

is surjective for every m . Since $\mathbb{P}^1$ is normal, this proves projective normality. Alternatively, the target of (1) is the d -th Veronese subring of $k[s, t]$, which is integrally closed.

It remains to identify the kernel. Let J be generated by the quadrics

$$X_i X_j - X_{i+1} X_{j-1}, \quad 0 \leq i < j-1 \leq d-1. \quad (2)$$

These are the 2×2 minors of

$$\begin{pmatrix} X_0 & X_1 & \cdots & X_{d-1} \\ X_1 & X_2 & \cdots & X_d \end{pmatrix},$$

and their images under θ vanish, so $J \subseteq \ker \theta$.

Modulo (2), if two indices in a monomial differ by at least two, they may be moved one step toward each other without changing their sum. Each move strictly decreases the sum of the squares of the indices, so the process terminates with all indices differing by at most one. It follows that every degree- m monomial, for $m \geq 1$, is congruent to

$$X_a^{m-r} X_{a+1}^r, \quad q = ma + r, \quad 0 \leq r < m, \quad (3)$$

where q is the sum of its indices; for $q = md$, (3) means X_d^m . Thus $(S/J)_m$ is spanned by at most $md + 1$ elements, one for each $q = 0, \dots, md$. But the induced map

$$(S/J)_m \longrightarrow k[s, t]_{md}$$

is surjective, and its target has dimension $md + 1$. It is therefore an isomorphism for every $m \geq 1$; in degree zero it is the identity of k . Hence $J = \ker \theta$, proving that the homogeneous ideal is generated by quadrics.

(b) Let H be a hyperplane-section divisor on X . Since X is nondegenerate, the $n + 1$ coordinate linear forms restrict to linearly independent sections, and therefore

$$n + 1 \leq h^0(X, \mathcal{O}_X(H)). \quad (4)$$

Choose an effective divisor in $|H|$, of degree d . The point-removal estimate and the hypothesis $d \leq n$ give

$$n + 1 \leq h^0(H) \leq d + 1 \leq n + 1. \quad (5)$$

All inequalities are equalities. In particular,

$$d = n, \quad h^0(H) = d + 1. \quad (6)$$

Write an effective representative as

$$H = P_1 + \dots + P_d$$

with repetitions allowed. At each step, adding one point can increase h^0 by at most one. Since the total increase from $h^0(\mathcal{O}_X) = 1$ to $h^0(H) = d + 1$ is exactly d , equality holds at every step. In particular,

$$h^0(P_1) = 2.$$

The pencil $|P_1|$ has no base point: otherwise $h^0(P_1) = h^0(\mathcal{O}_X) = 1$. It therefore defines a degree-one morphism $X \rightarrow \mathbb{P}^1$, which is an isomorphism. Hence $g(X) = 0$.

Under $X \cong \mathbb{P}^1$, the line bundle $\mathcal{O}_X(H)$ has degree d , so it is $\mathcal{O}_{\mathbb{P}^1}(d)$. Equations (4)–(6) show that the given embedding uses its complete space of sections. Changing the basis of that space only applies an automorphism of $\mathbb{P}^d$. Thus the embedding is the rational normal curve up to projective automorphism.

(c) Let $\mathbb{P}^r$ be the linear span of a degree-two curve. Then the curve is nondegenerate in $\mathbb{P}^r$. It cannot have $r = 1$, since the only integral curve in $\mathbb{P}^1$ is $\mathbb{P}^1$ itself, of degree one. If $r \geq 2$, part (b), with $2 \leq r$, gives $r = 2$ and identifies the curve with the rational normal conic.

(d) Let $\mathbb{P}^r$ be the linear span of a degree-three curve. Again $r \neq 1$. If $r = 2$, the curve is a plane cubic. If $r \geq 3$, part (b) gives $r = 3$ and identifies it with the rational normal curve of degree three, namely the twisted cubic. $\square$

Exercise IV.3.5**Permitted context.**

Let $X \subset \mathbb{P}^3$ be a nonsingular curve not contained in a plane, over an algebraically closed field, and let $O \notin X$. The generic-projection theorem supplies a nonempty open set of centers outside X inducing birational morphisms to plane curves. We use the vanishing of $H^1(\mathbb{P}^2, \mathcal{O}(m))$ for all integers m , and the fact that finite birational morphisms of nonsingular projective curves are isomorphisms. Projection from O is defined by the three-dimensional space of hyperplanes through O , and its associated line bundle on X is $\mathcal{O}_X(1)$.

For an integral plane curve C of degree d , its arithmetic genus is

$$p_a(C) = \frac{(d-1)(d-2)}{2}.$$

If $\tilde{C} \rightarrow C$ is its normalization, then

$$p_a(C) = g(\tilde{C}) + \sum_{Q \in C} \dim_k((\nu_* \mathcal{O}_{\tilde{C}} / \mathcal{O}_C)_Q).$$

We also use constancy of the Hilbert polynomial in a flat projective family.

Problem. (a) Suppose projection from O induces a birational morphism $\varphi : X \rightarrow C = \varphi(X) \subset \mathbb{P}^2$. Prove that C is singular.

(b) If X has degree d and genus g , deduce that

$$g < \frac{(d-1)(d-2)}{2}.$$

(c) Let $\{X_t\}$ be the flat family induced by the projection, with $X_1 = X$ and with X_0 supported on C . Prove that X_0 has nilpotent elements.

Solution. (a) Suppose, to the contrary, that C is nonsingular. A birational morphism between nonsingular projective curves is an isomorphism, so

$$X \cong C \quad \text{and} \quad \mathcal{O}_X(1) \cong \varphi^* \mathcal{O}_C(1). \quad (1)$$

The restriction sequence for a plane curve shows that

$$h^0(C, \mathcal{O}_C(1)) = 3 \quad (2)$$

when $\deg C \geq 2$; the case that C is a line would give only two sections and leads to the same contradiction. From (1) and (2),

$$h^0(X, \mathcal{O}_X(1)) \leq 3.$$

On the other hand, because X is not contained in a plane, the four coordinate linear forms on $\mathbb{P}^3$ restrict to linearly independent sections of $\mathcal{O}_X(1)$. Thus

$$h^0(X, \mathcal{O}_X(1)) \geq 4,$$

a contradiction. Therefore C is singular.

(b) Choose a center giving a birational projection, as provided by the generic-projection theorem. The morphism $\varphi : X \rightarrow C$ is finite and birational, so it is the normalization of C . Moreover,

$$\varphi^* \mathcal{O}_C(1) \cong \mathcal{O}_X(1),$$

and hence $\deg C = d$. The normalization-genus formula gives

$$\frac{(d-1)(d-2)}{2} = g + \sum_{Q \in C} \delta_Q. \quad (3)$$

Part (a) says that at least one δ_Q is positive. Equation (3) therefore yields the strict inequality

$$g < \frac{(d-1)(d-2)}{2}.$$

(c) The Hilbert polynomial of a projective curve of degree d and arithmetic genus p_a is

$$P(m) = dm + 1 - p_a.$$

Flatness of the family gives

$$P_{X_0}(m) = P_X(m) = dm + 1 - g. \quad (4)$$

If X_0 were reduced, then, because its support is the integral curve C , it would equal C with its reduced scheme structure. Its Hilbert polynomial would then be

$$P_C(m) = dm + 1 - \frac{(d-1)(d-2)}{2}. \quad (5)$$

The constant terms in (4) and (5) differ by part (b), contradicting flatness. Hence X_0 is nonreduced, or equivalently its structure sheaf contains nonzero nilpotent elements. $\square$

Exercise IV.3.6

Permitted context.

Let X be a nonsingular projective curve of degree 4, and let $\mathbb{P}^r$ be its linear span, over an algebraically closed field. Thus X is nondegenerate in $\mathbb{P}^r$.

We use the preceding results that a nondegenerate curve of degree $d \leq r$ in $\mathbb{P}^r$ is the rational normal curve with $d = r$, and that a nonplanar curve of degree d in $\mathbb{P}^3$ has genus strictly less than $(d-1)(d-2)/2$. A genus-two curve has no very ample divisor of degree four. We also use Riemann–Roch, the plane-curve genus formula, and the following complete-intersection facts: two surfaces without a common component meet in degree equal to the product of their degrees, and their intersection is Cohen–Macaulay, with no embedded associated points. The existence examples are the rational normal quartic in $\mathbb{P}^4$, the smooth rational quartic in $\mathbb{P}^3$, a genus-one curve embedded by a degree-four line bundle, and a smooth plane quartic.

Problem. (a) Prove that exactly the following possibilities can occur:

$g(X)$	linear span	description
0	$\mathbb{P}^4$	rational normal quartic
0	$\mathbb{P}^3$	rational quartic
1	$\mathbb{P}^3$	elliptic quartic
3	$\mathbb{P}^2$	nonsingular plane quartic.

(b) In the genus-one case, prove that $X \subset \mathbb{P}^3$ is a complete intersection of two irreducible quadric surfaces.

Solution. (a) If $r \geq 4$, then $\deg X = 4 \leq r$. The rational-normal-curve result gives

$$r = 4, \quad g(X) = 0,$$

and X is the rational normal quartic in $\mathbb{P}^4$.

Suppose $r = 3$. The genus bound for nonplanar degree-four curves gives

$$g(X) < \frac{(4-1)(4-2)}{2} = 3,$$

so $g(X) \in \{0, 1, 2\}$. The line bundle $\mathcal{O}_X(1)$ is very ample of degree four, which is impossible when $g(X) = 2$. Hence $g(X) = 0$ or 1 . In genus zero, $X \cong \mathbb{P}^1$ and $\mathcal{O}_X(1) \cong \mathcal{O}_{\mathbb{P}^1}(4)$; the four coordinate sections give the rational quartic in $\mathbb{P}^3$. The remaining possibility is the elliptic quartic.

If $r = 2$, then X is a nonsingular plane quartic and therefore

$$g(X) = \frac{(4-1)(4-2)}{2} = 3.$$

Finally, $r = 1$ is impossible: the only integral one-dimensional closed subscheme of $\mathbb{P}^1$ with its reduced structure is $\mathbb{P}^1$, which has degree one. This exhausts the possibilities.

(b) Now assume $g(X) = 1$, so $X \subset \mathbb{P}^3$ is nondegenerate. Twist its ideal sequence by 2:

$$0 \longrightarrow \mathcal{I}_X(2) \longrightarrow \mathcal{O}_{\mathbb{P}^3}(2) \longrightarrow \mathcal{O}_X(2) \longrightarrow 0. \quad (1)$$

There are ten quadratic forms on $\mathbb{P}^3$. The line bundle $\mathcal{O}_X(2)$ has degree 8, and Riemann–Roch on the genus-one curve gives

$$h^0(X, \mathcal{O}_X(2)) = 8.$$

Taking global sections in (1), without needing surjectivity of the restriction map, therefore yields

$$h^0(\mathbb{P}^3, \mathcal{I}_X(2)) \geq 10 - 8 = 2. \quad (2)$$

Choose linearly independent quadrics Q_1, Q_2 containing X .

Each Q_i is irreducible. Indeed, over an algebraically closed field a reducible quadric is a union of two planes, possibly the same plane twice. Since X is integral, containment in such a union would put X in one plane, contrary to nondegeneracy.

The irreducible quadrics Q_1 and Q_2 are not associates, so they have no common surface component. Their scheme-theoretic intersection is consequently a complete-intersection curve of degree

$$\deg Q_1 \deg Q_2 = 4.$$

It contains the integral degree-four curve X . A complete intersection has no embedded components and is pure of dimension one; equality of the degrees leaves no residual component or multiplicity. More explicitly, if $Y = Q_1 \cap Q_2$, then the kernel of $\mathcal{O}_Y \rightarrow \mathcal{O}_X$ is zero at the generic point and is supported at finitely many closed points. A nonzero such submodule would have a closed associated point of $\mathcal{O}_Y$, contrary to the absence of embedded associated points. Hence

$$X = Q_1 \cap Q_2$$

scheme-theoretically, proving the assertion. □

Exercise IV.3.7**Permitted context.**

Assume that k is algebraically closed of characteristic different from 2. A plane quartic has arithmetic genus 3, and normalization at one ordinary node lowers the genus by one. We use the Jacobian criterion and Bézout's theorem. Projection here means projection from a point outside the source curve.

If a nonsingular curve $X \subset \mathbb{P}^3$ projects birationally onto a plane curve C , then X is the normalization of C and

$$\mathcal{O}_X(1) \cong \nu^* \mathcal{O}_C(1).$$

In particular, their degrees agree. A genus-two curve has no very ample divisor of degree four.

Problem. Show that the affine curve

$$xy + x^4 + y^4 = 0$$

provides a plane curve with only nodes that is not the birational projection of any nonsingular curve in $\mathbb{P}^3$.

Solution. Let $C \subset \mathbb{P}^2$ be the projective closure

$$F(x, y, z) = xyz^2 + x^4 + y^4 = 0.$$

Its partial derivatives are

$$F_x = yz^2 + 4x^3, \quad F_y = xz^2 + 4y^3, \quad F_z = 2xyz.$$

At a singular point, $F_z = 0$, so $xyz = 0$. If $z = 0$, the first two derivatives force $x = y = 0$, which is not a projective point. If $x = 0$ or $y = 0$, the equation $F = 0$ forces both x and y to vanish. Thus the unique singular point is

$$P = [0 : 0 : 1].$$

In the affine chart $z = 1$, the lowest-degree part of the local equation at P is xy . It has two distinct tangent directions, so P is an ordinary node.

The curve C is reduced: a repeated irreducible factor of F would divide all its partial derivatives and produce a whole curve of singular points, contradicting the computation above. It is also irreducible. Indeed, if it were reducible, distinct components could meet only at the unique singular point P . The node has just two transverse branches, whose intersection multiplicity is one. But a decomposition of a quartic into components of positive degrees a and b , with $a + b = 4$, has $ab \geq 3$, contradicting Bézout's theorem. More than two components are also impossible at a two-branch node.

Let $\nu : \tilde{C} \rightarrow C$ be the normalization. Since a plane quartic has arithmetic genus three and an ordinary node has δ -invariant one,

$$g(\tilde{C}) = 3 - 1 = 2. \tag{1}$$

Suppose that C were the birational projection of a nonsingular curve $X \subset \mathbb{P}^3$. Then $X \cong \tilde{C}$, and the projection linear system satisfies

$$\mathcal{O}_X(1) \cong \nu^* \mathcal{O}_C(1).$$

Consequently $\mathcal{O}_X(1)$ is a very ample line bundle of degree four on the genus-two curve X . The very-ampleness bound in the context says that every very ample divisor on a genus-two curve has degree at least five. This contradiction proves that C is not such a projection. $\square$

Exercise IV.3.8**Permitted context.**

Work over an algebraically closed field. An integral curve $C \subset \mathbb{P}^n$ is called strange if some point $A \in \mathbb{P}^n$ lies on every tangent line to C at its nonsingular points.

For projection π_A from A , the kernel of the differential at a point $P \neq A$ is the tangent direction of the line AP . In characteristic zero, a nonconstant morphism between integral curves is separable and has nonzero differential at the generic point.

Problem. (a) Over a field of characteristic $p > 0$, prove that the projective closure of the parametrized affine curve

$$x = t, \quad y = t^p, \quad z = t^{2p}$$

is a singular strange curve.

(b) Prove that in characteristic zero every strange integral curve is a line, hence isomorphic to $\mathbb{P}^1$. In particular, there are no singular strange curves.

Solution. (a) Use homogeneous coordinates $[W : X : Y : Z]$ on $\mathbb{P}^3$. The projective closure C is parametrized by

$$\nu : \mathbb{P}^1 \longrightarrow \mathbb{P}^3, \quad [s : u] \longmapsto [s^{2p} : us^{2p-1} : u^p s^p : u^{2p}]. \quad (1)$$

On the affine chart $s \neq 0$, this is

$$t \longmapsto [1 : t : t^p : t^{2p}].$$

Because $t = x$, the parametrization is birational onto its integral image, and this affine part is nonsingular. The morphism is nonconstant and proper with finite fibers, hence finite; since $\mathbb{P}^1$ is normal, it is the normalization of C .

Formal differentiation in characteristic p gives

$$\frac{d}{dt}(t, t^p, t^{2p}) = (1, 0, 0).$$

Thus the tangent line at every affine point has the x -direction. After projective closure, all these lines pass through

$$A = [0 : 1 : 0 : 0].$$

The affine points are all the nonsingular points except possibly the single point at infinity, which we now show is singular.

It remains to see what happens at infinity. Formula (1) sends $[0 : 1]$ to $Q = [0 : 0 : 0 : 1]$. On the chart $u = 1$, with local parameter $q = s/u$ on the normalization and $Z = 1$ on the target, the three local coordinate functions at Q are

$$W = q^{2p}, \quad X = q^{2p-1}, \quad Y = q^p.$$

Their least order is $p > 1$. Hence the branch has multiplicity p , so Q is singular. This proves that C is a singular strange curve.

(b) Let $C \subset \mathbb{P}^n$ be strange with strange point A , and let $\tilde{C} \rightarrow C$ be its normalization. Projection from A gives a rational map

$$C \dashrightarrow \mathbb{P}^{n-1},$$

which induces a morphism from $\tilde{C}$ because the latter is a nonsingular projective curve.

At every nonsingular $P \in C$ with $P \neq A$, the tangent line $T_P C$ is the line AP . It is precisely the direction killed by the differential of projection. The induced morphism on $\tilde{C}$ therefore has zero differential on a dense open set, hence at the generic point. In characteristic zero a nonconstant morphism of curves is separable and cannot have identically zero differential. The projected map is consequently constant.

Thus all points of C lie on one fixed line through A . Since C is an integral closed curve, it equals that line. Therefore $C \cong \mathbb{P}^1$, and in particular no singular strange curve exists in characteristic zero. $\square$

Exercise IV.3.9

Permitted context.

Let $X \subset \mathbb{P}^3$ be a nonsingular curve of degree d that is not contained in a plane, over an algebraically closed field of arbitrary characteristic. The family of secant lines is the image of the two-dimensional parameter space $X^{(2)}$ in the Grassmannian of lines.

We use the multisequant and strange-curve results of the generic-projection theorem (Hartshorne IV, Proposition 3.8 and Theorem 3.9), in the following form: if a general secant of a nonplanar curve is a multisequant, then the curve is strange. A nonsingular strange curve is a line, or in characteristic 2 a plane conic. Thus a nonplanar nonsingular curve is not strange.

Problem. Prove the following Bertini lemma: for a nonempty open subset of planes $H \subset \mathbb{P}^3$, the intersection $X \cap H$ consists of exactly d distinct points, no three of which are collinear.

Solution. First consider transversality. A plane H fails to meet X transversely at P precisely when it contains the tangent line $T_P X$. For a fixed P , the planes containing $T_P X$ form a projective line. Hence the incidence correspondence

$$\{(P, H) : T_P X \subset H\} \subset X \times (\mathbb{P}^3)^*$$

has dimension two. Its image is therefore a proper closed subset of the three-dimensional dual projective space. A plane outside this subset cuts X transversely, and Bézout's theorem then says that $X \cap H$ consists of d distinct points.

It remains to exclude three collinear points. Let $\mathcal{T}$ be the family of lines meeting X in at least three distinct points. We claim that

$$\dim \mathcal{T} \leq 1. \quad (1)$$

Indeed, the full secant family has dimension two. Its map from $X^{(2)}$ has finite general fiber, since a line not contained in X meets X in only finitely many points. If $\dim \mathcal{T} = 2$, a general secant would be a multisequant. The multisequant argument would then make X strange. This is impossible for a nonplanar nonsingular curve, proving (1).

Now form the incidence correspondence

$$\mathcal{I} = \{(L, H) : L \in \mathcal{T}, L \subset H\}.$$

For each line L , the planes containing it form a projective line. Thus (1) gives $\dim \mathcal{I} \leq 2$. The closure of its image in $(\mathbb{P}^3)^*$ is again a proper closed subset. A plane H outside it contains no trisecant of X .

Choose H outside both exceptional closed subsets. It meets X in d distinct points. If three of them were collinear, their line would be a trisecant contained in H , contrary to the choice of H . These good planes form the required nonempty open subset. $\square$

Exercise IV.3.10**Permitted context.**

Let $X \subset \mathbb{P}^n$ be an integral nondegenerate curve over an algebraically closed field of characteristic zero. Points are always chosen in a suitable dense open subset of the nonsingular locus.

We use the characteristic-zero trisecant lemma in the following precise form: for an integral projective curve $C \subset \mathbb{P}^m$ not contained in a plane, the closure in the Grassmannian of the lines meeting C in three distinct points has dimension at most one. The curve may be singular and m may exceed three. We also use normalization of projective curves, separability in characteristic zero, and the dimension theorem for fibers.

Problem. Assume $n \geq 2$. Prove that for general points

$$P_1, \dots, P_{n-1} \in X,$$

their linear span is an $(n-2)$ -plane and contains no other point of X .

Solution. We prove a slightly stronger statement by induction on n , allowing the curve to be singular but choosing all points in its nonsingular locus. This is useful because an intermediate projected image need not be nonsingular.

For $n = 2$, one general point spans a point and plainly contains no other point of the reduced integral curve. Now let $n \geq 3$, choose a general point $P_1 \in X$, and project from it:

$$\pi_{P_1} : X \dashrightarrow X' \subset \mathbb{P}^{n-1}.$$

Here X is not planar, since it is nondegenerate and $n \geq 3$. Let $\mathcal{T}$ denote its trisecant family. The incidence of pairs (P, ℓ) with $P \in X \cap \ell$ and $\ell \in \mathcal{T}$ has dimension at most one: its fibers over $\mathcal{T}$ are finite, since X is not a line. Consequently, only finitely many trisecants pass through a general P_1 . Projection from P_1 is therefore one-to-one away from finitely many points of X . On the normalization this rational map extends to a nonconstant morphism. It is finite onto X' , and in characteristic zero generic injectivity implies degree one, hence birationality. The image X' is nondegenerate: if it lay in a hyperplane of $\mathbb{P}^{n-1}$, then X would lie in the corresponding hyperplane of $\mathbb{P}^n$ through P_1 .

Write $\nu : \tilde{X} \rightarrow X$ for normalization and $f : \tilde{X} \rightarrow X'$ for the finite birational morphism just obtained. Remove from X' its singular points, the images under f of $\nu^{-1}(X_{\text{sing}})$, and $f(\nu^{-1}(P_1))$. This removes only finitely many points and leaves a dense open U' for which the entire inverse image $f^{-1}(U') \rightarrow U'$ is an isomorphism. Indeed, a finite birational map onto a normal curve is an isomorphism. Normalization identifies $f^{-1}(U')$ with an open $U \subset X$. Thus we have an isomorphism

$$U \xrightarrow{\sim} U'$$

and no additional points of $X \setminus \{P_1\}$ map into U' . Choose the remaining general points $P_2, \dots, P_{n-1}$ in U , and put

$$P'_i = \pi_{P_1}(P_i).$$

They are general points of X' . By the induction hypothesis,

$$L' = \langle P'_2, \dots, P'_{n-1} \rangle \subset \mathbb{P}^{n-1}$$

has dimension $n-3$ and contains no other point of X' .

The inverse linear span of L' under projection is

$$L = \langle P_1, P_2, \dots, P_{n-1} \rangle,$$

and

$$\dim L = \dim L' + 1 = n - 2.$$

Suppose $Q \in X \cap L$. If $Q = P_1$, it is already listed. Otherwise,

$$\pi_{P_1}(Q) \in X' \cap L',$$

so the induction hypothesis gives

$$\pi_{P_1}(Q) = P'_i$$

for some $i \geq 2$. The points P'_i were chosen in the open set over which the birational projection is an isomorphism, and hence $Q = P_i$. Thus L contains exactly the chosen $n - 1$ points of X , completing the induction. These conditions hold on a dense open subset of the parameter space of tuples: linear independence is open, while tuples admitting an extra distinct point in their span form a constructible set by the incidence correspondence. The argument excludes that set at the generic tuple, so its closure is proper. This gives the asserted meaning of general points. $\square$

Exercise IV.3.11

Permitted context.

Work over an algebraically closed field of arbitrary characteristic. A variety in projective space is taken to be closed and integral. For a smooth projective variety, a base-point-free space of sections defines a closed immersion if it separates distinct points and nonzero tangent vectors.

The secant variety $S(X)$ is the closure of the union of lines joining distinct points of X . The embedded tangent space $T_P X$ is a projective linear space of dimension $\dim X$. We use dimensions of incidence varieties and their fibers.

Problem. (a) Let $X \subset \mathbb{P}^n$ be a nonsingular projective variety of dimension r , with $n > 2r + 1$. Prove that projection from a suitable point $O \notin X$ induces a closed immersion $X \rightarrow \mathbb{P}^{n-1}$.

(b) Let $X = \nu_2(\mathbb{P}^2) \subset \mathbb{P}^5$ be the Veronese surface. Prove that every point on a secant line lies on infinitely many secant lines. Show that $\dim S(X) = 4$, and deduce that projection from a suitable point embeds X into $\mathbb{P}^4$.

Solution. (a) If $r = 0$, then X is a single reduced point, and any center outside X gives the required embedding. Assume $r \geq 1$. The incidence of triples

$$(P, Q, O), \quad P, Q \in X, \quad P \neq Q, \quad O \in \langle P, Q \rangle$$

has dimension $2r + 1$. Thus its image has closure of dimension at most $2r + 1$, and

$$\dim S(X) \leq 2r + 1.$$

The incidence of pairs (P, O) with $O \in T_P X$ has dimension $2r$. Its image is closed, since it is the image of a projective bundle over the projective variety X . As $n > 2r + 1$, we can choose O outside X , $S(X)$, and all embedded tangent spaces.

Projection from O is defined everywhere on X . Two distinct points could have the same image only if their joining line passed through O , which was excluded. Its differential could kill a nonzero tangent vector at P only if $O \in T_P X$, also excluded. The separation criterion therefore proves that the projection is a closed immersion.

(b) Write the Veronese coordinates as

$$\nu_2([x : y : z]) = [x^2 : y^2 : z^2 : xy : xz : yz].$$

For a line $L \subset \mathbb{P}^2$, its image is a nonsingular conic C_L spanning a plane $\Pi_L \subset \mathbb{P}^5$. If $P, Q \in X$ are distinct, let L join their preimages. Then $\langle P, Q \rangle \subset \Pi_L$.

Let R lie on this secant. If $R \in C_L$, the lines joining R to the other points of C_L already give infinitely many secants through R . If $R \notin C_L$, projection of the conic from R is a finite morphism of degree two to the pencil of lines through R . The given secant meets the conic in two distinct points, so this morphism cannot be purely inseparable. It is therefore separable; a nonempty open subset of the pencil consists of lines meeting the conic in two distinct points. These are infinitely many secants of X through R . This argument also works in characteristic two: the nucleus of a conic lies on no secant with two distinct endpoints and hence is not a possible R in this case.

We compute the dimension without assuming that every point of a conic plane is on an actual secant. The closure of the secants of a nonsingular plane conic is its entire plane. Indeed, for fixed P on the conic, the lines PQ , as Q varies, are dense in the pencil through P . Consequently,

$$S(X) = \bigcup_{L \in (\mathbb{P}^2)^*} \Pi_L. \quad (1)$$

The right side is closed: it is the image of the projective plane bundle whose fiber at L is Π_L . This bundle has dimension $2 + 2 = 4$, so $\dim S(X) \leq 4$.

Identify the coordinates of $\mathbb{P}^5$ with nonzero symmetric 3×3 matrices up to scalar, so that a point $[v]$ of $\mathbb{P}^2$ maps to $[vv^t]$. If $L = \mathbb{P}(W)$, where $W \subset k^3$ has dimension two, every matrix represented in Π_L has column space contained in W . The plane Π_L contains rank-two matrices, for example $uu^t + vv^t$ for a basis u, v of W . A rank-two matrix in Π_L determines W uniquely as its column space. Thus the map from the bundle in (1) has one-point fibers on its nonempty rank-two open subset. The dimension theorem gives $\dim S(X) = 4$.

Finally every tangent line to X at $\nu_2(p)$ is the tangent line to $\nu_2(L)$ for the line L through p in the specified tangent direction. It is contained in Π_L , hence in $S(X)$. Therefore $S(X)$ contains all embedded tangent planes as well as all secants. A point of $\mathbb{P}^5 \setminus S(X)$ separates points and tangent vectors under projection. The criterion in the context gives the required closed immersion into $\mathbb{P}^4$. $\square$

Exercise IV.3.12**Permitted context.**

Work over an algebraically closed field of arbitrary characteristic. A node is a double point whose quadratic tangent cone is a product of two distinct linear forms. An integral plane curve of degree d with exactly r nodes and no other singularities has normalization genus

$$g = \frac{(d-1)(d-2)}{2} - r.$$

We use the Jacobian criterion, the dimension theorem for incidence varieties, and Bézout's theorem. The first jet of a local equation at Q is its residue modulo $\mathfrak{m}_Q^2$; prescribing it imposes three independent linear conditions if the available sections surject onto this three-dimensional space.

We also use Riemann–Roch and $\deg K_Y = 2g - 2$. On a nonsingular projective curve of genus g , every divisor of degree at least $2g + 1$ is very ample. The generic-projection theorems for curves give embeddings into $\mathbb{P}^3$ by successive projections from larger projective spaces, and a birational projection of a smooth nonplanar space curve onto a plane curve having only nodes. All these projections preserve degree.

Problem. For every $d \in \{2, 3, 4, 5\}$ and every integer

$$0 \leq r \leq \frac{(d-1)(d-2)}{2},$$

construct an irreducible plane curve of degree d with exactly r nodes and no other singularities.

Solution. We first construct curves with a small number of prescribed nodes. The remaining cases will come from projecting smooth curves.

Let A be a selected subset of the three coordinate vertices of $\mathbb{P}^2$. Let $V_d(A)$ be the vector space of homogeneous degree- d forms vanishing to order at least two at every point of A . We claim that a general member has precisely the nodes in A , and is irreducible, in the cases

$$(d, |A|) = (3, 1), (4, 1), (5, 1), (5, 2), (5, 3). \quad (1)$$

First we show that $V_d(A)$ separates first jets at every $Q \notin A$. It suffices to find a degree- $(d-1)$ form B vanishing doubly on A with $B(Q) \neq 0$: the forms $B\ell$, as ℓ ranges over linear forms, then give all first jets at Q , since multiplication by the local unit B is invertible.

If $A = \{[1 : 0 : 0]\}$, one of y, z is nonzero at Q , so one of y^{d-1}, z^{d-1} works; here $d-1 \geq 2$. For $d = 5$ and two or three selected vertices, use the quartics

$$x^2y^2, \quad x^2z^2, \quad y^2z^2.$$

They vanish doubly at all three vertices and do not vanish simultaneously away from those vertices. If Q is an unselected vertex, its corresponding pure fourth power is nonzero there and vanishes doubly at every selected vertex. This proves the claim about first jets in all cases of (1).

Write $N = \dim \mathbb{P}(V_d(A))$. For fixed $Q \notin A$, being singular at Q imposes three independent linear conditions. The incidence of such pairs $(Q, [F])$ consequently has dimension at most

$$2 + (N - 3) = N - 1.$$

The closure of its image in $\mathbb{P}(V_d(A))$ is proper. A member outside this closure has no singularity away from A , including on the line at infinity.

At $P = [1 : 0 : 0] \in A$, the forms

$$x^{d-2}y^2, \quad x^{d-2}yz, \quad x^{d-2}z^2$$

belong to $V_d(A)$ in all cases of (1) and realize arbitrary quadratic terms in the local coordinates $y/x, z/x$. The same holds at each selected vertex by permuting coordinates. The condition that this quadratic term have two distinct linear factors is a nonempty open condition, also in characteristic two; the quadratic yz is one example. The parameter space is irreducible, so these finitely many nonempty open conditions can be imposed simultaneously with absence of other singularities. This gives a curve with exactly $|A|$ nodes.

Such a curve is reduced, since a repeated component would make every point of that component singular. If it were reducible, divide its components into two nonempty groups of total degrees a and $d - a$. Their intersections must all occur among the nodes, and at a node belonging to both groups the intersection multiplicity is one. Bézout would give

$$a(d - a) \leq |A|.$$

But $a(d - a) \geq d - 1 > |A|$ in every case of (1). This contradiction proves irreducibility.

The same first-jet incidence argument, now in the full space of degree- d forms, gives a nonsingular plane curve for every $d \geq 2$. It is irreducible: two positive-degree components would meet by Bézout and produce a singular point. Thus the case $r = 0$ is available for each required d .

We now produce the remaining curves using the following construction. Take a nonsingular projective curve Y of genus g and an effective divisor D of degree $d \geq 2g + 1$. Its complete linear system embeds Y with degree d . If the ambient dimension exceeds three, project successively to an embedding in $\mathbb{P}^3$. A birational generic projection to $\mathbb{P}^2$ then has integral image of degree d , with only nodes, and normalization Y . Its number of nodes is therefore exactly

$$r = \frac{(d-1)(d-2)}{2} - g. \quad (2)$$

In the applications below the complete embedding has dimension at least three and is nonplanar, as required for this projection theorem.

For $g = 0$, take $Y = \mathbb{P}^1$ and D of degree $d = 3, 4, 5$. The complete embedding is the rational normal curve in $\mathbb{P}^d$, and (2) gives $r = 1, 3, 6$, respectively.

For $g = 1$, take Y to be any nonsingular plane cubic, whose existence was just proved. Divisors of degrees 4 and 5 are very ample. Riemann–Roch gives $h^0(D) = d$, so the complete embeddings lie in $\mathbb{P}^3$ and $\mathbb{P}^4$, respectively. Formula (2) gives a quartic with $r = 2$ and a quintic with $r = 5$.

For $g = 2$, normalize the one-node plane quartic constructed in (1). The normalization has genus $3 - 1 = 2$. A divisor of degree $5 = 2g + 1$ is very ample and has $h^0(D) = 5 + 1 - 2 = 4$. Its embedding into $\mathbb{P}^3$, followed by generic projection, gives a quintic with $r = 6 - 2 = 4$.

Together the constructions cover exactly the following ranges:

d	direct plane construction	projection construction
2	$r = 0$	
3	$r = 0, 1$	$r = 1$
4	$r = 0, 1$	$r = 2, 3$
5	$r = 0, 1, 2, 3$	$r = 4, 5, 6$.

All the curves are integral and have no singularities other than the stated nodes. No restriction on the characteristic was used. $\square$

IV.4 Elliptic Curves

Exercise IV.4.1

Permitted context.

Let X be a nonsingular projective integral curve of genus one over an algebraically closed field k of characteristic different from two, and let $P \in X(k)$. Write

$$L(nP) = \{f \in k(X) : \operatorname{div}(f) + nP \geq 0\} \cup \{0\}.$$

Riemann–Roch gives $h^0(nP) = n$ for $n \geq 1$ and $h^0(0) = 1$. The Legendre normal form with P at infinity supplies functions u, v with their only poles at P , of orders two and three, and an equation

$$v^2 = u(u-1)(u-\lambda), \quad \lambda \in k \setminus \{0, 1\}.$$

Problem. Show that the section ring

$$R = \bigoplus_{n \geq 0} H^0(X, \mathcal{O}_X(nP))$$

admits a graded presentation

$$R \cong k[t, x, y] / (y^2 - x(x - t^2)(x - \lambda t^2)), \quad \deg t = 1, \quad \deg x = 2, \quad \deg y = 3.$$

Solution. Represent the direct sum as the subring $\bigoplus_{n \geq 0} L(nP)T^n \subset k(X)[T]$, where T is a formal variable. Define

$$t \mapsto T, \quad x \mapsto uT^2, \quad y \mapsto vT^3.$$

The Legendre equation makes the indicated weighted homogeneous polynomial vanish, so this defines a graded homomorphism from the claimed quotient to R .

Since the relation is monic of degree two in y , the quotient is a free $k[t, x]$ -module with basis $1, y$. Its degree- n part therefore has basis

$$t^{n-2a}x^a \quad (0 \leq 2a \leq n), \quad t^{n-3-2a}x^ay \quad (0 \leq 3+2a \leq n).$$

Their images, after removing T^n , are u^a and u^av . Their pole orders at P are respectively $2a$ and $2a+3$. These orders are all distinct. A linear combination cannot vanish: its term of largest pole order cannot be canceled by any other term. Thus the map is injective on each homogeneous part.

For $n \geq 1$, the displayed orders are precisely

$$0, 2, 3, \dots, n,$$

with only 0 present when $n = 1$. There are n of them, equal to $\dim L(nP)$. They therefore form a basis of $L(nP)$. In degree zero both sides are k . The homomorphism is consequently an isomorphism in every degree, proving the presentation. $\square$

Exercise IV.4.2**Permitted context.**

Let X be a nonsingular projective integral genus-one curve over an algebraically closed field. Its canonical bundle is trivial; for a positive-degree line bundle M , Riemann–Roch gives $h^0(M) = \deg M$ and $H^1(X, M) = 0$. A line bundle of degree at least three is very ample. A complete embedding of a normal variety by L is projectively normal if all multiplication maps

$$\mathrm{Sym}^m H^0(X, L) \longrightarrow H^0(X, L^m)$$

are surjective for $m \geq 0$. We use $H^1(\mathbb{P}^2, \mathcal{O}(q)) = 0$ for every integer q .

Problem. If D is a divisor of degree at least three on X , prove that the embedding defined by its complete linear system is projectively normal.

Solution. Put $L = \mathcal{O}_X(D)$ and $d = \deg L$. We first prove surjectivity of multiplication in degree two by induction on d . For $d = 3$, the complete embedding is a smooth plane cubic. Its restriction sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2}(m-3) \longrightarrow \mathcal{O}_{\mathbb{P}^2}(m) \longrightarrow L^m \longrightarrow 0$$

and the stated H^1 -vanishing give surjectivity of restriction in every degree, in particular degree two.

Let $d \geq 4$, choose $P \in X$, and put $L' = L(-P)$. By induction, products of sections of L' span

$$H^0((L')^2) = H^0(L^2(-2P)).$$

Choose $s \in H^0(L)$ not vanishing at P . Also choose $t \in H^0(L(-P))$ vanishing to order exactly one at P ; this is possible because

$$h^0(L(-P)) - h^0(L(-2P)) = (d-1) - (d-2) = 1.$$

The quotient $H^0(L^2)/H^0(L^2(-2P))$ has dimension two. In a local trivialization at P , the classes of s^2 and st have vanishing orders zero and one, so they are linearly independent in this quotient. Together with the products from L' , they span $H^0(L^2)$. This proves the induction step.

To pass to higher degrees, choose two sections $a, b \in H^0(L)$ with no common zero. One can first choose nonzero a , then choose b avoiding its finite zero set, since L is globally generated and k is infinite. Their evaluation sequence is

$$0 \longrightarrow L^{-1} \longrightarrow \mathcal{O}_X^2 \xrightarrow{(a,b)} L \longrightarrow 0.$$

Tensor by L^m , where $m \geq 2$. Since $H^1(L^{m-1}) = 0$, the resulting cohomology sequence gives a surjection

$$H^0(L^m)^2 \longrightarrow H^0(L^{m+1}), \quad (u, v) \longmapsto au + bv.$$

Starting from degree two, induction on m now shows that every section of L^m is a sum of products of m sections of L . Degrees zero and one require no argument. The normal-generation criterion proves projective normality. $\square$

Exercise IV.4.3**Permitted context.**

Let k be algebraically closed with $\text{char } k \neq 2$, and let E be the smooth projective completion of

$$y^2 = F(x) = x(x-1)(x-\lambda), \quad \lambda \neq 0, 1,$$

with origin $O = [0 : 1 : 0]$. The functions x, y have poles of orders two and three at O . Riemann–Roch gives

$$L(2O) = \langle 1, x \rangle, \quad L(3O) = \langle 1, x, y \rangle.$$

An automorphism fixing O is a group automorphism, so commutes with inversion $(x, y) \mapsto (x, -y)$. The invariant is

$$j = 256 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}.$$

The Legendre parameters for a fixed j are the orbit under $\lambda \mapsto 1 - \lambda$ and $\lambda \mapsto 1/\lambda$.

Problem. Prove that every automorphism of E fixing O is induced by a projective linear transformation whose affine expression is

$$x' = ax + b, \quad y' = cy.$$

Describe these transformations and determine the group $G = \text{Aut}(E, O)$ in the four cases

$$j \neq 0, 1728; \quad j = 1728, \text{ char } k \neq 3; \quad j = 0, \text{ char } k \neq 3; \quad j = 0, \text{ char } k = 3.$$

Solution. An automorphism σ preserving O preserves each pole-order space, hence

$$\sigma^*x = ax + b, \quad \sigma^*y = cy + dx + e, \quad a, c \neq 0.$$

The nonzero leading coefficients follow from the exact pole orders. Commuting with inversion forces $d = e = 0$, since the characteristic is not two. Thus σ extends to

$$[X : Y : Z] \mapsto [aX + bZ : cY : Z]$$

in the plane. Such a transformation preserves E precisely when

$$F(ax + b) = c^2 F(x). \tag{1}$$

Equivalently, the affine map $x \mapsto ax + b$ permutes $\{0, 1, \lambda\}$ and $c^2 = a^3$. Each allowed affine permutation has exactly two lifts, differing by inversion. These conditions describe the transformations for every original choice of λ .

For completeness, a nonidentity affine permutation of the three roots either exchanges two roots and fixes the third, or cyclically permutes them. In the first case the fixed root is the midpoint of the other two, giving $\lambda \in \{-1, 2, 1/2\}$. In the second case, after ordering the roots, the map is

$$x \mapsto (\lambda - 1)x + 1,$$

and its sending λ to zero is equivalent to $\lambda^2 - \lambda + 1 = 0$. Thus there are no other exceptional parameters.

If $j \neq 0, 1728$, neither exception occurs, so $a = 1, b = 0, c = \pm 1$. The group is cyclic of order two.

If $j = 1728$ and the characteristic is not three, an affine relabeling and rescaling puts the equation in the form $y^2 = x^3 - x$, corresponding to $\lambda = -1$. Its automorphisms are

$$(x, y) \mapsto (u^2x, u^3y), \quad u^4 = 1.$$

These four transformations form a cyclic group of order four. Conjugating by the change of coordinates gives the transformations for $\lambda = 2$ or $1/2$.

If $j = 0$ and the characteristic is not three, then $\lambda^2 - \lambda + 1 = 0$. The transformation

$$\rho(x, y) = ((\lambda - 1)x + 1, y)$$

cycles the three roots and has order three; indeed $(\lambda - 1)^3 = 1$. Together with $\iota(x, y) = (x, -y)$, it gives all six automorphisms $\rho^a \iota^b$, $0 \leq a < 3$, $0 \leq b < 2$. The generators commute, so G is cyclic of order six.

Finally, in characteristic three, $j = 0$ gives $\lambda = -1$ and the equation $y^2 = x^3 - x$. The affine permutations of its root set $\mathbb{F}_3$ are

$$x \mapsto ax + b, \quad a = \pm 1, \quad b \in \mathbb{F}_3.$$

Equation (1) says $c^2 = a$. Hence the twelve automorphisms are

$$(x, y) \mapsto (c^2x + b, cy), \quad c^4 = 1, \quad b \in \mathbb{F}_3.$$

Choose $i \in k$ with $i^2 = -1$. The maps

$$\tau(x, y) = (x + 1, y), \quad \sigma(x, y) = (-x, iy)$$

generate G and satisfy

$$\tau^3 = 1, \quad \sigma^4 = 1, \quad \sigma\tau\sigma^{-1} = \tau^{-1}.$$

Thus G is the semidirect product of a cyclic group of order three by a cyclic group of order four, with the latter acting by inversion. In particular this group is not cyclic. $\square$

Exercise IV.4.4**Permitted context.**

As in the elliptic-curve section, work over an algebraically closed field k of characteristic different from two. For a Legendre equation $y^2 = x(x-1)(x-\lambda)$, the invariant is

$$j = 256 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}.$$

A monic cubic with coefficients A, B, C has discriminant

$$\delta = A^2B^2 - 4B^3 - 4A^3C - 27C^2 + 18ABC.$$

This is the product of the squared pairwise differences of its roots; it is nonzero precisely when the roots are distinct. The projective Weierstrass cubic $y^2 = x^3 + Ax^2 + Bx + C$ is smooth if and only if $\delta \neq 0$.

Problem. Let E be the nonsingular projective cubic with affine equation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6.$$

Express $j(E)$ as a rational function of the coefficients with rational coefficients. Prove that $j(E) \in k_0$ if all a_i belong to a subfield $k_0 \subset k$. Conversely, for each $j_0 \in k_0$, construct an elliptic curve over k_0 with invariant j_0 .

Solution. Complete the square by setting $Y = y + (a_1x + a_3)/2$. The equation becomes $Y^2 = x^3 + Ax^2 + Bx + C$, where

$$A = a_2 + \frac{a_1^2}{4}, \quad B = a_4 + \frac{a_1a_3}{2}, \quad C = a_6 + \frac{a_3^2}{4}.$$

We claim that

$$j(E) = 256 \frac{(A^2 - 3B)^3}{A^2B^2 - 4B^3 - 4A^3C - 27C^2 + 18ABC}. \quad (1)$$

To see this, let the distinct roots of the cubic be α, β, γ , put $h = \beta - \alpha$, and set $\lambda = (\gamma - \alpha)/h$. After $x = \alpha + hu$ and rescaling Y , the equation is Legendre form. Directly from the symmetric expressions in the roots,

$$A^2 - 3B = h^2(1 - \lambda + \lambda^2), \quad \delta = h^6\lambda^2(1 - \lambda)^2.$$

Substitution into the Legendre formula gives (1). No division by three is used, so the formula also holds in characteristic three.

Formula (1), with the displayed expressions for A, B, C , is the required rational function over $\mathbb{Q}$. Its only constant denominators are powers of two, and its discriminant denominator is nonzero because E is smooth. If the original coefficients lie in k_0 , the formula therefore gives $j(E) \in k_0$.

We now construct the converse over k_0 itself. First suppose its characteristic is different from both two and three. For $j_0 = 0$, take $y^2 = x^3 + 1$; for $j_0 = 1728$, take $y^2 = x^3 + x$. For $j_0 \notin \{0, 1728\}$, put

$$t = \frac{j_0}{1728 - j_0}, \quad E : y^2 = x^3 + 3tx + 2t.$$

Here $t \neq 0, -1$, and the cubic discriminant is $-108t^2(t+1) \neq 0$. Formula (1) gives

$$j(E) = 1728 \frac{t}{t+1} = j_0.$$

In characteristic three, take $y^2 = x^3 - x$ when $j_0 = 0$. Its discriminant is $4 = 1$, and (1) gives $j = 0$. When $j_0 \neq 0$, take

$$E : y^2 = x^3 + x^2 - \frac{1}{j_0}.$$

Now $A = 1, B = 0, C = -1/j_0$, so $\delta = 1/j_0 \neq 0$ and $256(A^2 - 3B)^3 = 1$ in k_0 . Again (1) gives $j(E) = j_0$. Every constructed cubic is smooth and has the k_0 -rational point at infinity, hence is an elliptic curve over k_0 . $\square$

Exercise IV.4.5

Permitted context.

Let k be algebraically closed of characteristic different from two. A nonconstant morphism between elliptic curves is finite. An origin-preserving morphism is a group homomorphism, and a separable such morphism is unramified by Hurwitz. Every degree-two morphism from an elliptic curve to $\mathbb{P}^1$ is the quotient by an involution $Q \mapsto A - Q$. If it is ramified at the origin, its involution is inversion. A double cover of $\mathbb{P}^1$ is determined, up to isomorphism, by its branch set; four branch points give genus one. The Legendre invariant is

$$j(\lambda) = 256 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2},$$

and two elliptic curves over k are isomorphic if and only if their invariants agree.

Problem. Let $f : E \rightarrow E$ be an endomorphism of degree two, and let $\pi : E \rightarrow \mathbb{P}^1$ have degree two and be ramified at the origin.

- Find a degree-two map $\pi' : E \rightarrow \mathbb{P}^1$ and a degree-two map $g : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ such that $\pi f = g\pi'$.
- Choose coordinates so that $g(u) = u^2$.
- Relate the branch points of π, π' , and g , and derive an equation for a Legendre parameter of E .

(d) Determine the possible j -invariants and corresponding parameters. In characteristic zero there are three distinct invariants; in positive characteristic allow their reductions to coincide.

Solution. (a) Since f is a homomorphism, it commutes with inversion. Use the inversion quotient also on its source as π' . Then πf descends to a map g of the two quotient lines. The equality of degrees in $\pi f = g\pi'$ gives $\deg g = 2$. Since the characteristic is not two, all these double covers are separable, and f is unramified.

(b) Hurwitz shows that g has two ramification points, with distinct images. Send its ramification points to $0, \infty$ in the source and their images to $0, \infty$ in the target. The resulting rational function has zero divisor $2(0)$ and pole divisor $2(\infty)$, hence is cu^2 . Rescale a coordinate to obtain $g(u) = u^2$.

(c) For $P \in E$, compare ramification indices in the square:

$$e_\pi(f(P)) = e_g(\pi'(P))e_{\pi'}(P). \quad (1)$$

The left side is at most two. Above either ramification point of g , formula (1) forces $e_{\pi'} = 1$ and $e_\pi = 2$. Thus the two branch values of g belong to the branch set of π . Above each of the other

two branch values of π , the map g is unramified; (1) then forces π' to ramify at each of their four preimages. These are all four branch points of π' .

Scale coordinates so that the branch set of π is $\{0, \infty, 1, \lambda\}$, where $\lambda \neq 0, 1$, and that $g(u) = u^2$ still holds. Write $s^2 = \lambda$. The branch set of π' is

$$\{1, -1, s, -s\}.$$

Its cross-ratio can be taken to be $\mu = ((s-1)/(s+1))^2$. Substitution into the Legendre formula gives

$$j(\mu) = 16 \frac{(\lambda^2 + 14\lambda + 1)^3}{\lambda(\lambda - 1)^4}.$$

Both covers have source isomorphic to E , so $j(\lambda) = j(\mu)$. Clearing the nonzero denominators yields

$$16(\lambda^2 - \lambda + 1)^3(\lambda - 1)^2 - \lambda(\lambda^2 + 14\lambda + 1)^3 = 0. \quad (2)$$

(d) The polynomial in (2) factors as

$$(\lambda + 1)^2(\lambda^2 - 6\lambda + 1)(\lambda^2 - \lambda + 16)(16\lambda^2 - \lambda + 1). \quad (3)$$

The resulting possibilities, with this choice of branch ordering, are

condition on λ	j	
$\lambda = -1$	1728	
$\lambda^2 - 6\lambda + 1 = 0$	8000	
$\lambda^2 - \lambda + 16 = 0$ or $16\lambda^2 - \lambda + 1 = 0$	-3375.	(4)

The table follows by substituting each quadratic relation in the formula for $j(\lambda)$. In characteristic zero its parameters are

$$-1, \quad 3 \pm 2\sqrt{2}, \quad \frac{1 \pm 3\sqrt{-7}}{2}, \quad \frac{1 \pm 3\sqrt{-7}}{32}.$$

Other orderings of the four branch points give the usual six Legendre transforms of these parameters.

We verify sufficiency as well. For any root λ of (3), take the smooth completion D of

$$v^2 = (u^2 - 1)(u^2 - \lambda).$$

None of these roots is zero or one: the left side of (2) takes the nonzero values 16 and -4096 there. Thus D is a smooth genus-one double cover with branch set $\{\pm 1, \pm s\}$. The formulas

$$x = u^2, \quad y = uv$$

give a degree-two morphism $D \rightarrow E_\lambda$, where E_λ is the Legendre curve. Its function-field extension has degree at most two, since $u^2 = x$ and $v = y/u$. It has degree exactly two because the nontrivial involution $(u, v) \mapsto (-u, -v)$ fixes x, y . By (2), D and E_λ have equal invariant, hence are isomorphic. Choose as origin of D a point above the origin of E_λ ; composing an isomorphism with a translation makes it preserve these origins. The displayed map then gives a degree-two endomorphism of E_λ .

Consequently the possible invariants are precisely the reductions of 1728, 8000, -3375. They are distinct in characteristic zero. In odd positive characteristic collisions occur exactly in characteristics 3, 5, 7, 13, as is seen by factoring their pairwise differences. The statement about three distinct invariants must be understood with this qualification. $\square$

Exercise IV.4.6**Permitted context.**

Work over an algebraically closed field of characteristic zero. For a base-point-free linear series $V \subset H^0(X, L)$ of dimension $n + 1$ on a smooth projective curve, its vanishing sequence at P is

$$0 = a_0(P) < a_1(P) < \cdots < a_n(P),$$

the orders of vanishing of a basis adapted to P . Its ramification weight is

$$w(P) = \sum_{i=0}^n (a_i(P) - i).$$

We use $\deg \omega_X = 2g - 2$, Riemann–Roch, and the identification $P \mapsto \mathcal{O}_X(P - O)$ of an elliptic curve with its degree-zero Picard group. For an immersed plane branch, the vanishing sequence of line sections is $(0, 1, m)$, where m is its tangent-line contact.

Problem. (a) Let a smooth genus- g curve X be the normalization of an integral plane curve of degree $d \geq 2$ whose only singularities are nodes. Establish the inflection formula $6(g - 1) + 3d$, with ramification multiplicities, and explain the qualification needed if points lying over nodes are excluded.

(b) Let $X \subset \mathbb{P}^n$, $n \geq 3$, be a nondegenerate smooth curve of degree d . A hyperplane with contact at least n is osculating; contact at least $n + 1$ is hyperosculating. Show that every point has an osculating hyperplane and that the total hyperosculation weight is

$$n(n + 1)(g - 1) + (n + 1)d.$$

(c) For an elliptic curve E and $d \geq 3$, apply the formula to the embedding by $|dO|$ to prove $\#E[d](k) = d^2$. Here $E[d]$ means points killed by d , rather than points of exact order d .

Solution. We first prove the ramification formula for any such linear series. On an open set choose a local parameter z and a frame e for L , and write a basis of V as $s_j = f_j e$. Form the Wronskian

$$W = \det \left(\frac{d^i f_j}{dz^i} \right)_{0 \leq i, j \leq n}.$$

The expressions

$$W e^{n+1} (dz)^{n(n+1)/2}$$

glue to a global section of

$$L^{n+1} \otimes \omega_X^{n(n+1)/2}. \quad (1)$$

Indeed, replacing a frame or a parameter changes successive derivatives by triangular row operations; their diagonal factors are exactly canceled by those of the displayed tensor factors. Changing the basis of V multiplies the section by a nonzero constant, which leaves its zero divisor unchanged.

At P , take an adapted basis with $f_j = c_j z^{a_j} + \text{higher terms}$, where $c_j \neq 0$. The lowest coefficient of the determinant is

$$\left(\prod_j c_j \right) \prod_{i < j} (a_j - a_i),$$

up to the fixed nonzero sign determined by the basis ordering. It is nonzero in characteristic zero. Thus the section in (1) is nonzero, and its order at P is $\sum_j a_j - n(n+1)/2 = w(P)$. Taking degrees gives

$$\sum_{P \in X} w(P) = (n+1)d + n(n+1)(g-1). \quad (2)$$

(a) Pull back the plane line sections to the normalization. They form a base-point-free three-dimensional series of degree d . With $n = 2$, formula (2) gives $3d + 6(g-1)$. At a smooth point of the plane curve the contribution is $m - 2$, where m is tangent contact. In particular an ordinary flex has weight one and a higher flex has larger weight.

Each branch over a node is also immersed, with sequence $(0, 1, m)$. Therefore, if the nodes themselves are excluded, the precise weighted count of smooth inflection points is

$$3d + 6(g-1) - \sum_{\nu(P) \text{ a node}} (m_P - 2). \quad (3)$$

When neither branch of any node has tangent contact above two, the correction vanishes and the claimed formula holds. A node contributes no automatic weight, but an inflectionary branch above it does contribute to the Wronskian.

This qualification is necessary. For instance, the quartic

$$xyz^2 + x^4 + y^4 = 0$$

has a unique node at $[0 : 0 : 1]$, as its three partial derivatives show. It is reduced, since a repeated component would give infinitely many singular points. If it were reducible, its components could meet only at that node, with total intersection multiplicity one, contradicting Bézout for degree four. Its normalization thus has genus two. The two local branches have expansions $y = -x^3 + \text{higher terms}$ and $x = -y^3 + \text{higher terms}$. Each has tangent contact three and weight one. Consequently the weighted number of smooth flexes is $18 - 2 = 16$, although the total ramification weight on the normalization is 18.

(b) A basis adapted to P has $a_n(P) \geq n$. Its last section defines an osculating hyperplane. There is a hyperosculating hyperplane exactly when $a_n(P) > n$, equivalently $w(P) > 0$, since the increasing nonnegative sequence equals $0, 1, \dots, n$ precisely when its last term is n . Formula (2) proves the assertion, counting these points with weights $w(P)$; the unweighted count can be smaller.

(c) Put $L = \mathcal{O}_E(dO)$. It has $h^0(L) = d$ and is very ample. For $0 \leq m < d$, Riemann–Roch gives

$$h^0(L(-mP)) = d - m.$$

For $m = d$, the dimension is one if $L \cong \mathcal{O}_E(dP)$ and zero otherwise, since a degree-zero line bundle has a nonzero section exactly when it is trivial. Thus the vanishing sequence is $0, 1, \dots, d-1$, except that its last term is d when $d(P - O) = 0$ in $\text{Pic}^0(E)$. The exceptional points are precisely $E[d](k)$, and each has weight one. Formula (2), with $g = 1, n = d - 1$, gives

$$\#E[d](k) = d^2.$$

This is a count of distinct points because all their weights have just been shown to be one. $\square$

Exercise IV.4.7**Permitted context.**

Let E, E' be elliptic curves over an algebraically closed field of characteristic different from two, with origins O, O' . Use $Q \mapsto \mathcal{O}_E(Q - O)$ to identify $E \cong \text{Pic}^0(E)$, including the functorial identification for algebraic families. Write $[n]_E$ for multiplication by n . An origin-preserving morphism is a group homomorphism; if nonconstant, it is finite. Its separable part is unramified, and its purely inseparable part has one geometric point in each fiber with multiplicity equal to its degree.

We use the following form of the see-saw principle. If a line bundle N on a product of smooth projective curves restricts trivially to every fiber of the first projection, then N is pulled back from the first factor. Indeed, cohomology and base change give a rank-one direct image, and the evaluation from its pullback is an isomorphism on every fiber, hence an isomorphism. Restriction to a horizontal section identifies the line bundle on the first factor. Finally $[2]_E$ is separable of degree four: its differential is multiplication by two, and its kernel consists of the origin and the three finite branch points of the Legendre map.

Problem. (a) Show that pullback by a morphism $f : E \rightarrow E'$ induces a homomorphism of elliptic curves $\widehat{f} : E' \rightarrow E$, called its dual.

(b) Prove $\widehat{g \circ f} = \widehat{f} \circ \widehat{g}$.

(c) If f preserves the origins and has degree $n > 0$, prove

$$\widehat{f} \circ f = [n]_E, \quad f \circ \widehat{f} = [n]_{E'}.$$

Explain the separable and inseparable cases.

(d) For origin-preserving $f, g : E \rightarrow E'$, prove $\widehat{f + g} = \widehat{f} + \widehat{g}$.

(e) Prove $\widehat{[n]_E} = [n]_{E'}$ for every integer n , and $\deg[n]_E = n^2$ for $n \neq 0$.

(f) Prove $\deg \widehat{f} = \deg f$ for nonconstant f .

Solution. (a) Pullback takes degree-zero line bundles to degree-zero line bundles and respects tensor products. For a constant f , its pullback on Pic^0 is zero. Otherwise degrees of divisors are multiplied by $\deg f$, which also proves preservation of degree zero. The functorial Picard identification makes this pullback an algebraic morphism of the representing elliptic curves. It preserves the identity and is a homomorphism; explicitly,

$$\mathcal{O}_E(\widehat{f}(Q') - O) \cong f^* \mathcal{O}_{E'}(Q' - O'). \quad (1)$$

(b) The identity $(g \circ f)^* = f^* g^*$ on line bundles and (1) give the asserted reversal of composition.

(c) First suppose f is separable. Its kernel K has n points, and translation identifies the fiber over $f(Q)$ with $Q + K$. Consequently

$$f^*((f(Q)) - (O')) = \sum_{T \in K} ((Q + T) - (T)).$$

The degree-zero divisor on the right represents nQ in the elliptic group law. Equation (1) gives $\widehat{f}(f(Q)) = nQ$. If f is purely inseparable, its fibers over $f(Q)$ and O' are respectively $n(Q)$ and $n(O)$, giving the same result. In general let the inseparable degree be e and the separable degree be s . There are s kernel points, each fiber point has multiplicity e , and the same computation gives $\widehat{f}(f(Q)) = esQ = nQ$. Thus $\widehat{f}f = [n]_E$. Since f is surjective,

$$(f\widehat{f})f = f[n]_{E'} = [n]_{E'}f$$

implies $f\hat{f} = [n]_{E'}$. Equality on geometric points here is equality of morphisms, since the curves are reduced over an algebraically closed field.

(d) A degree-zero line bundle M on E' is invariant under translation. To verify this, represent it by a degree-zero divisor $\sum_i a_i(Q_i)$. Translating its pullback shifts all the Q_i by the same point, and their group sum changes by that point times $\sum_i a_i = 0$. The degree-zero Picard identification therefore leaves its class unchanged.

Let $m : E' \times E' \rightarrow E'$ be addition and p_1, p_2 the projections. Set

$$N = m^*M \otimes (p_1^*M)^{-1} \otimes (p_2^*M)^{-1}.$$

Translation invariance makes N trivial on every fiber of p_1 . Its restriction to $E' \times \{O'\}$ is also trivial. The see-saw principle gives $N \cong \mathcal{O}_{E' \times E'}$. Pulling back along $(f, g) : E \rightarrow E' \times E'$ yields

$$(f + g)^*M \cong f^*M \otimes g^*M.$$

Under (1), this is exactly the additivity of the dual.

(e) Duality sends the identity to itself and the zero homomorphism to zero. Part (d), including additive inverses, gives $\widehat{[n]} = [n]$ for every integer n . We check that no nonzero integer acts as zero, without assuming the degree formula to be proved. The separable map $[2^r]$ has degree 4^r , so $\#E[2^r](k) = 4^r$. If a nonzero integer $n = 2^v u$, u odd, killed all points of E , then Bezout's identity would imply $E[2^r](k) \subset E[2^v](k)$ for every $r > v$. This contradicts their respective sizes. Hence $[n] \neq 0$ for $n \neq 0$, and the homomorphism $\mathbb{Z} \rightarrow \text{End}(E)$ is injective.

Let $a = \deg[n]$. Apply part (c) and self-duality:

$$[a] = \widehat{[n]}[n] = [n]^2 = [n^2].$$

Injectivity of the integer action gives $a = n^2$.

(f) First suppose f preserves the origins, and put $n = \deg f > 0$. From $f\hat{f} = [n]$, both maps are nonconstant and

$$(\deg f)(\deg \hat{f}) = \deg[n] = n^2.$$

Thus $\deg \hat{f} = n$. An arbitrary nonconstant f is an origin-preserving homomorphism followed by translation. Translation has degree one and acts trivially on Pic^0 , so it does not affect the dual or the degree. The same conclusion follows. $\square$

Exercise IV.4.8**Permitted context.**

Let E be an elliptic curve over an algebraically closed field k of characteristic zero or $p > 2$, with origin O . Its algebraic fundamental group is the inverse limit of the deck groups of connected pointed finite étale Galois covers. Write $\mathbb{Z}_\ell = \varprojlim_r \mathbb{Z}/\ell^r \mathbb{Z}$.

We use Hurwitz, and the following isogeny results: $\deg[n] = n^2$, the differential of $[n]$ is multiplication by n , and an isogeny f has a dual satisfying $f\hat{f} = \hat{f}f = [\deg f]$. A finite map of smooth curves factors uniquely through a purely inseparable map followed by a separable map; for an isogeny the separable factor is étale. The tangent space of $\text{Pic}^0(E)$ at the identity is $H^1(E, \mathcal{O}_E)$, and the differential of a pullback map on Picard varieties is pullback on this cohomology. The Hasse invariant $h \in \{0, 1\}$ records whether Frobenius on this one-dimensional space is zero or nonzero.

Problem. Compute the algebraic fundamental group of E , and prove

$$\pi_1(E, O) \cong \begin{cases} \prod_{\ell \text{ prime}} \mathbb{Z}_\ell^2, & \text{char } k = 0, \\ \prod_{\ell \neq p} \mathbb{Z}_\ell^2, & \text{char } k = p, \ h = 0, \\ \mathbb{Z}_p \times \prod_{\ell \neq p} \mathbb{Z}_\ell^2, & \text{char } k = p, \ h = 1. \end{cases}$$

Solution. Let $f : Y \rightarrow E$ be a connected finite étale cover. Hurwitz gives $2g(Y) - 2 = (\deg f)(2g(E) - 2) = 0$. Choose a point of Y over O as origin. Then Y is an elliptic curve and f is an isogeny. Its fibers are translates of its kernel, so translation by the kernel gives all deck transformations. In particular every connected finite étale cover is Galois and its deck group is abelian.

For every positive integer n , factor multiplication as

$$E \xrightarrow{q_n} E_n \xrightarrow{h_n} E, \quad [n] = h_n q_n,$$

with q_n purely inseparable and h_n separable, hence étale. In characteristic zero q_n is the identity. These covers are cofinal: if $f : Y \rightarrow E$ has degree n , then $[n] = f\hat{f}$. The function field of Y is a separable intermediate extension of that defined by $[n]$, hence is contained in its maximal separable subextension $k(E_n)$. Thus h_n factors through f .

The map q_n is bijective on geometric points and induces a group isomorphism

$$E[n](k) \cong \ker(h_n)(k).$$

Indeed a point belongs to the kernel on the right exactly when its unique preimage is killed by $[n]$. If $n \mid m$, the identity $[m] = [n] \circ [m/n]$ induces the transition on these groups given by multiplication by m/n . It follows that

$$\pi_1(E, O) \cong \varprojlim_n E[n](k). \quad (1)$$

For a prime ℓ different from the characteristic, $[\ell^r]$ is separable of degree ℓ^{2r} , so its kernel has that many points. This finite abelian group is killed by ℓ^r , and its subgroup killed by ℓ has order ℓ^2 . The classification of finite abelian groups consequently gives

$$E[\ell^r](k) \cong (\mathbb{Z}/\ell^r \mathbb{Z})^2.$$

Multiplication by ℓ from the next level is surjective: lift a point along the surjective isogeny $[\ell]$, and note that the lift is killed by ℓ^{r+1} . One may therefore choose compatible bases noncanonically, giving inverse limit $\mathbb{Z}_\ell^2$.

In characteristic p , consider relative Frobenius $F : E \rightarrow E^{(p)}$ and its dual $V : E^{(p)} \rightarrow E$. They have degree p and $VF = [p]$. The differential of V , viewed as pullback on Picard varieties, is the Frobenius pullback on $H^1(\mathcal{O})$, with the scalar twist understood. Thus it is nonzero exactly when $h = 1$. If $h = 1$, V is separable of degree p ; if $h = 0$, V is purely inseparable. Since F is bijective on points, this proves

$$\#E[p](k) = p \quad (h = 1), \quad \#E[p](k) = 1 \quad (h = 0).$$

Separable degrees multiply, so in the first case $\#E[p^r](k) = p^r$. There is only one cyclic factor, since the subgroup killed by p has order p . Hence $E[p^r](k) \cong \mathbb{Z}/p^r\mathbb{Z}$, compatibly under multiplication by p , with inverse limit $\mathbb{Z}_p$. In the second case $[p]$ and all its iterates are purely inseparable, and every $E[p^r](k)$ is trivial.

Finally, primary decomposition of finite torsion groups and the Chinese remainder theorem decompose the inverse limit in (1) as the product of these prime-power inverse limits. This gives exactly the three groups stated. $\square$

Exercise IV.4.9

Permitted context.

Work over an algebraically closed field of characteristic different from two. Two elliptic curves are isogenous if there is a finite surjective morphism between them. After composing with a translation, such a morphism preserves origins and is a homomorphism. A nonconstant homomorphism f has a dual isogeny $\hat{f}$ with $f\hat{f} = [\deg f]$. Multiplication by $n \neq 0$ has degree n^2 , so $E[n](k)$ is finite. In characteristic p , any isogeny factors as an iterated relative Frobenius $E \rightarrow E^{(p^r)}$, followed by a separable isogeny. A separable isogeny is the quotient by translations by its finite kernel; its target function field is the invariant subfield.

Problem. (a) Prove that isogeny is an equivalence relation on elliptic curves.

(b) For a fixed elliptic curve E , prove that only countably many isomorphism classes of elliptic curves are isogenous to E .

Solution. (a) The identity proves reflexivity. The composite of two finite surjective morphisms is again finite and surjective, proving transitivity. For symmetry, translate a given finite map to an isogeny preserving origins and take its dual. This is a finite map in the opposite direction.

(b) First count the targets of separable isogenies from a fixed elliptic curve A . The torsion group

$$A(k)_{\text{tors}} = \bigcup_{n \geq 1} A[n](k)$$

is countable, being a countable union of finite sets. It has only countably many finite subsets, hence only countably many finite subgroups. Every separable isogeny $A \rightarrow B$ has one of these groups H as its kernel, and B is determined up to isomorphism by H : its function field is $k(A)^H$, and a one-variable function field determines its smooth projective curve. Thus there are only countably many such B .

In characteristic zero all isogenies are separable, so this already proves the assertion. In characteristic p , every isogeny from E first passes through one of the countably many curves $E^{(p^r)}$, $r \geq 0$,

and then is separable. Each such twist has only countably many possible separable quotients by the argument just given. A countable union of countable sets is countable. By symmetry from (a), considering maps out of E includes its whole isogeny class. $\square$

Exercise IV.4.10

Permitted context.

Let E be an elliptic curve over an algebraically closed field, with origin O , and let $p_1, p_2 : E \times E \rightarrow E$ be the projections. The identification $E \cong \text{Pic}^0(E)$ represents algebraic families of degree-zero line bundles normalized along the origin. Its normalized Poincare bundle is

$$\mathcal{P} = \mathcal{O}_{E \times E}(\Delta) \otimes p_1^* \mathcal{O}_E(-O) \otimes p_2^* \mathcal{O}_E(-O).$$

Its restriction to $\{Q\} \times E$ represents $\mathcal{O}_E(Q - O)$, and its restrictions to both axes through (O, O) are trivial. A morphism $E \rightarrow E$ fixing O is a group homomorphism. Degree is constant in a connected family of line bundles on a projective curve. The see-saw principle says that a line bundle trivial on every fiber of p_1 is pulled back from the first factor.

Problem. Writing $R = \text{End}(E, O)$ with its additive group structure, construct and prove exactness of

$$0 \longrightarrow \text{Pic}(E) \oplus \text{Pic}(E) \xrightarrow{(A,B) \mapsto p_1^* A \otimes p_2^* B} \text{Pic}(E \times E) \longrightarrow R \longrightarrow 0.$$

Solution. For $L \in \text{Pic}(E \times E)$, put

$$A = L|_{E \times \{O\}}, \quad B = L|_{\{O\} \times E}, \quad L_0 = L \otimes p_1^* A^{-1} \otimes p_2^* B^{-1}.$$

Both restrictions of L_0 to the axes are trivial as line bundles. Their possible constant one-dimensional fiber factors do not affect their Picard classes. The restriction of L_0 to every vertical fiber has degree zero, because its degree is constant and it is zero on the fiber over O .

The family of restrictions therefore defines a morphism

$$\Phi_L : E \longrightarrow \text{Pic}^0(E) \cong E, \quad Q \longmapsto [L_0|_{\{Q\} \times E}].$$

At $Q = O$ this is the trivial class, so $\Phi_L(O) = O$. Thus $\Phi_L \in R$. Normalization and restriction commute with tensor product, which shows that $L \mapsto \Phi_L$ is a group homomorphism.

If $L = p_1^* A \otimes p_2^* B$, its normalized bundle is trivial, so $\Phi_L = 0$. Conversely, if $\Phi_L = 0$, every vertical restriction of L_0 is trivial. By see-saw, $L_0 = p_1^* C$ for some $C \in \text{Pic}(E)$. Restriction to $E \times \{O\}$ gives $C \cong \mathcal{O}_E$. Hence $L = p_1^* A \otimes p_2^* B$, proving exactness at the middle term.

Given $f \in R$, define

$$L_f = (f \times \text{id}_E)^* \mathcal{P}.$$

It is normalized on both axes, and its restriction to $\{Q\} \times E$ represents $\mathcal{O}_E(f(Q) - O)$. Consequently $\Phi_{L_f} = f$, proving surjectivity.

Finally, if $p_1^* A \otimes p_2^* B$ is trivial, restricting successively to the two axes proves A and B trivial. This proves injectivity of the first nonzero map and completes exactness of the sequence. $\square$

Exercise IV.4.11**Permitted context.**

Let $E = \mathbb{C}/\Lambda$, where $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ and $\text{Im } \tau > 0$. Endomorphisms of the elliptic curve preserving the origin correspond to complex numbers α satisfying $\alpha\Lambda \subseteq \Lambda$. A nonzero endomorphism is an unramified covering; its degree is the number of points in its kernel. Its dual satisfies $\widehat{f}f = [\deg f]$. A quadratic algebraic integer satisfies a monic quadratic polynomial over $\mathbb{Z}$.

Problem. (a) If a nonzero endomorphism f corresponds to multiplication by α , prove $\deg f = |\alpha|^2$.
 (b) Show that its dual corresponds to $\bar{\alpha}$.
 (c) If τ is an algebraic integer in an imaginary quadratic field, prove $\text{End}(E) = \mathbb{Z}[\tau]$.

Solution. (a) The kernel of multiplication by α is $\alpha^{-1}\Lambda/\Lambda$. Multiplication by α identifies this finite group with $\Lambda/\alpha\Lambda$, so

$$\deg f = [\Lambda : \alpha\Lambda].$$

The index is the ratio of the areas of fundamental parallelograms for $\alpha\Lambda$ and Λ . As a real-linear transformation of $\mathbb{C}$, multiplication by α has determinant $|\alpha|^2$. Therefore the ratio, and hence $\deg f$, equals $|\alpha|^2$.

(b) Write β for the complex scalar corresponding to $\widehat{f}$. The identity $\widehat{f}f = [\deg f]$ gives

$$\beta\alpha = \deg f = \alpha\bar{\alpha}.$$

Since $\alpha \neq 0$, cancellation yields $\beta = \bar{\alpha}$. The zero endomorphism and its dual both correspond to zero as well.

(c) Any scalar α preserving Λ sends 1 into Λ , so $\alpha \in \mathbb{Z} + \mathbb{Z}\tau$. Conversely, integrality of the quadratic number τ gives

$$\tau^2 = a\tau + b, \quad a, b \in \mathbb{Z}.$$

Thus multiplication by τ preserves Λ , and so does multiplication by every element of $\mathbb{Z} + \mathbb{Z}\tau = \mathbb{Z}[\tau]$. These two inclusions prove the equality. $\square$

Exercise IV.4.12**Permitted context.**

Write $E_\tau = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$, with the unique representative satisfying $\text{Im } \tau > 0$, $-1/2 \leq \text{Re } \tau < 1/2$, $|\tau| \geq 1$, and $\text{Re } \tau \leq 0$ when $|\tau| = 1$. Lattices define isomorphic curves exactly when homothetic. Endomorphisms are lattice-preserving scalars α , of degree $|\alpha|^2$, with integral trace and determinant. We use the established facts that degree-two endomorphisms occur exactly at $j = 1728, 8000, -3375$, and order-four origin-preserving automorphisms exactly at $j = 1728$. The Weierstrass function's nonzero half-period values are the three cubic roots; for a conjugation-invariant lattice, conjugating its argument conjugates its value.

Problem. (a) Prove that E_τ has an origin-preserving automorphism other than 1, -1 only when

$$\tau = i \quad \text{or} \quad \tau = \frac{-1 + \sqrt{-3}}{2}.$$

(b) Prove that E_τ has an endomorphism of degree two exactly for

$$\tau = i, \quad \tau = \sqrt{-2}, \quad \tau = \frac{-1 + \sqrt{-7}}{2},$$

and match these with the invariants 1728, 8000, -3375 .

Solution. We first give an elementary lattice observation. Each of the rings

$$\mathbb{Z}[i], \quad \mathbb{Z}[\sqrt{-2}], \quad \mathbb{Z}\left[\frac{-1 + \sqrt{-3}}{2}\right], \quad \mathbb{Z}\left[\frac{-1 + \sqrt{-7}}{2}\right] \quad (1)$$

is Euclidean for the norm $|z|^2$. For the first two, a nearest rectangular-grid point leaves squared distance at most $1/2$ and $3/4$, respectively. For either of the last two, first approximate the imaginary coordinate by a multiple of $\sqrt{m}/2$, then approximate the real coordinate in that row by an integer, where $m = 3, 7$. The squared error is at most

$$\frac{1}{4} + \frac{m}{16} < 1.$$

Approximating a quotient of two ring elements and multiplying by the denominator gives Euclidean division with smaller norm. Thus all four rings are principal ideal domains.

If a lattice $\Lambda \subset \mathbb{C}$ is stable under one of these rings $A = \mathbb{Z}[\alpha]$, choose $0 \neq \omega \in \Lambda$. The two vectors $\omega, \alpha\omega$ span $\Lambda \otimes_{\mathbb{Z}} \mathbb{Q}$ over $\mathbb{Q}$, since α is nonreal. It follows that $\omega^{-1}\Lambda$ lies in $\mathbb{Q}(\alpha)$. After clearing the denominators of a lattice basis, it is a fractional ideal of A . Since A is principal, for some $c \in \mathbb{C}^\times$ we have

$$\Lambda = cA. \quad (2)$$

(a) An automorphism corresponds to a scalar α of absolute value one. If α is real, preservation of $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ forces $\alpha \in \mathbb{Z}$, so $\alpha = \pm 1$. Otherwise its integral trace satisfies

$$\alpha + \bar{\alpha} = t, \quad t \in \{-1, 0, 1\},$$

and its norm is one. Thus α is a primitive fourth, third, or sixth root of unity. Its ring $\mathbb{Z}[\alpha]$ is respectively $\mathbb{Z}[i]$ or $\mathbb{Z}[(-1 + \sqrt{-3})/2]$. By (2), these give the only two lattice classes. Their

representatives in the specified fundamental region are the two values stated; each indeed has the indicated extra root-of-unity automorphisms.

(b) A degree-two endomorphism has norm two. Its scalar is nonreal, since a real scalar preserving Λ is an integer and has square degree. Its trace is an integer with $|t| < 2\sqrt{2}$, so

$$t \in \{-2, -1, 0, 1, 2\}, \quad \alpha^2 - t\alpha + 2 = 0.$$

Up to replacing α by its negative or conjugate, the possibilities are

$$1 + i, \quad \sqrt{-2}, \quad \frac{1 + \sqrt{-7}}{2}.$$

Their rings are the first, second, and fourth rings of (1). By (2), the corresponding lattices have exactly the three claimed representatives. Conversely each displayed scalar preserves its ring lattice and has norm two, so each curve does have such an endomorphism.

The square lattice has an automorphism of order four and therefore $j = 1728$. For $\tau = \sqrt{-2}$, the rectangular lattice is invariant under conjugation. Each of its nonzero half-periods is fixed by conjugation modulo the lattice and sign, so the three Weierstrass branch values are real and distinct. Ordering them gives a Legendre parameter $0 < \lambda < 1$. In particular $j(\lambda) > 0$. This lattice class differs from the square one, so its invariant is not 1728; among the three possible invariants it must be 8000. The remaining, distinct lattice class $\tau = (-1 + \sqrt{-7})/2$ therefore has $j = -3375$. $\square$

Exercise IV.4.13

Permitted context.

Work over an algebraically closed field of characteristic thirteen. Every elliptic curve is isomorphic to a Legendre curve $y^2 = x(x-1)(x-\lambda)$, with $\lambda \neq 0, 1$. For an equation $y^2 = f(x)$ in odd characteristic p , the Hasse invariant vanishes exactly when the coefficient of x^{p-1} in $f(x)^{(p-1)/2}$ is zero. The invariant of a Legendre curve is

$$j = 256 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}.$$

Problem. Find all j -invariants of elliptic curves in characteristic thirteen whose Hasse invariant is zero.

Solution. The desired coefficient is that of x^{12} in $x^6(x-1)^6(x-\lambda)^6$. Expanding the last two factors, it equals

$$H(\lambda) = \sum_{i=0}^6 \binom{6}{i}^2 \lambda^i.$$

Modulo thirteen this is

$$H(\lambda) = \lambda^6 + 10\lambda^5 + 4\lambda^4 + 10\lambda^3 + 4\lambda^2 + 10\lambda + 1.$$

A direct expansion gives the useful identity

$$H(\lambda) = (\lambda^2 - \lambda + 1)^3 - 2\lambda^2(\lambda - 1)^2.$$

Since $256 = 9$ and $9 \cdot 2 = 5$ in characteristic thirteen, the Legendre formula implies

$$j - 5 = \frac{9H(\lambda)}{\lambda^2(\lambda - 1)^2}.$$

The denominator is nonzero, so the Hasse invariant vanishes if and only if $j = 5$.

There do exist such curves: the nonconstant polynomial H has a root in the algebraically closed field, and $H(0) = H(1) = 1$, so its roots are allowed Legendre parameters. Thus $j = 5$ is exactly the one required value. $\square$

Exercise IV.4.14

Permitted context.

For an elliptic curve in characteristic p , the Hasse invariant is zero precisely when absolute Frobenius acts as zero on $H^1(E, \mathcal{O}_E)$. For a smooth plane cubic $F = 0$, the restriction sequence identifies

$$H^1(E, \mathcal{O}_E) \cong H^2(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-3)).$$

In the standard three-chart Čech computation the latter space is one-dimensional, represented by $1/(xyz)$. A homogeneous Laurent monomial with a nonnegative exponent in any coordinate represents zero in this quotient. We use the prime-progression density theorem: each reduced residue class modulo three has Dirichlet density $1/2$.

Problem. For each prime $p \neq 3$, consider the Fermat elliptic curve

$$E_p : x^3 + y^3 = z^3$$

over an algebraic closure of $\mathbb{F}_p$. Determine the primes for which its Hasse invariant is zero, and determine their density.

Solution. The partial derivatives of $F = x^3 + y^3 - z^3$ have no common projective zero when $p \neq 3$, so E_p is smooth. It has the rational point $[1 : 0 : 1]$, hence is elliptic. This also applies when $p = 2$.

Under the stated cohomology identification, Frobenius sends the class of $1/(xyz)$ to the class of

$$\frac{F^{p-1}}{x^p y^p z^p}. \quad (1)$$

Indeed Frobenius first raises the Čech representative to the p -th power, putting it in degree $-3p$. Multiplication by F^{p-1} returns to degree -3 . This multiplication is the map on the ideal term of the restriction sequence that makes the Frobenius square commute, since $F \cdot F^{p-1} = F^p$.

Only the coefficient of $x^{p-1}y^{p-1}z^{p-1}$ in F^{p-1} survives in (1): to have three negative exponents after division, each numerator exponent must be at most $p-1$, and their sum is $3(p-1)$, forcing equality for all three. Every monomial of F^{p-1} has exponents divisible by three. Thus the surviving coefficient is zero unless $p \equiv 1 \pmod{3}$.

If $p = 3m + 1$, that coefficient is

$$(-1)^m \frac{(p-1)!}{(m!)^3},$$

which is nonzero modulo p , since every factorial involved has argument less than p . It follows that the Hasse invariant vanishes exactly for

$$p \equiv 2 \pmod{3},$$

including $p = 2$. The density theorem for the two reduced classes modulo three gives this set Dirichlet density $1/2$. Its natural density among primes is also $1/2$ if one uses the prime number theorem for arithmetic progressions. $\square$

Exercise IV.4.15**Permitted context.**

Let E be an elliptic curve over an algebraically closed field of characteristic $p > 2$, as in the section's characteristic convention. Let E_p denote the inverse Frobenius twist, and $F' : E_p \rightarrow E$ the k -linear Frobenius, of degree p . It is purely inseparable and bijective on geometric points. Write $V = \widehat{F'} : E \rightarrow E_p$. Duality gives $\deg V = p$ and $F'V = [p]_E$.

The tangent space of $\text{Pic}^0(E)$ is $H^1(E, \mathcal{O}_E)$; the differential of a map induced by pullback is the corresponding pullback on this cohomology. The Hasse invariant $h \in \{0, 1\}$ is zero or one according as the semilinear absolute Frobenius on this one-dimensional space is zero or nonzero. A homomorphism of elliptic curves is separable exactly when its differential at the origin is nonzero.

Problem. Prove that V is separable if and only if $h = 1$. Deduce that the group of geometric points killed by p is

$$E[p](k) \cong \begin{cases} \mathbb{Z}/p\mathbb{Z}, & h = 1, \\ 0, & h = 0. \end{cases}$$

Solution. By definition of the dual, V represents the pullback

$$(F')^* : \text{Pic}^0(E) \longrightarrow \text{Pic}^0(E_p).$$

Its tangent map is therefore

$$(F')^* : H^1(E, \mathcal{O}_E) \longrightarrow H^1(E_p, \mathcal{O}_{E_p}).$$

Undoing the scalar twist identifies its vanishing or nonvanishing with that of absolute Frobenius on $H^1(E, \mathcal{O}_E)$. Thus $dV \neq 0$ precisely when $h = 1$, proving the first assertion by the separability criterion.

If $h = 1$, the degree- p homomorphism V is étale: translation carries its nonzero differential at the origin to every point. Its kernel consists of p geometric points and is a group of prime order, hence cyclic. Since F' has only the origin in its geometric kernel,

$$[p](Q) = F'(V(Q)) = O \iff V(Q) = O_p.$$

Thus $E[p](k) = \ker V(k) \cong \mathbb{Z}/p\mathbb{Z}$.

If $h = 0$, V is inseparable. Its degree is the prime p , so it is purely inseparable and has a one-point geometric kernel. The same equivalence then gives $E[p](k) = 0$. This statement concerns geometric points; the finite group scheme $E[p]$ still has length p^2 . $\square$

Exercise IV.4.16**Permitted context.**

Let E be an elliptic curve defined over $\mathbb{F}_q$, where $q = p^r$, $p > 2$, and choose its origin in $E(\mathbb{F}_q)$. We work over an algebraic closure k . Let $F' : E_q \rightarrow E$ denote k -linear Frobenius from the inverse q -power twist. For an isogeny u , duality satisfies

$$u\widehat{u} = [\deg u], \quad \widehat{u+v} = \widehat{u} + \widehat{v}, \quad \widehat{[m]} = [m], \quad \deg[m] = m^2.$$

The action of $\mathbb{Z}$ on E is faithful. The degree of the zero homomorphism is defined to be zero. A separable isogeny has degree equal to the number of its geometric kernel points. The dual of degree- p Frobenius is separable exactly when the Hasse invariant is one.

Problem. (a) Identify E_q with E over k and describe F' on projective coordinates.

(b) Prove that $1 - F'$ is separable and that its kernel is $E(\mathbb{F}_q)$.

(c) Write $N = \#E(\mathbb{F}_q)$. Prove that

$$F' + \widehat{F'} = [a], \quad N = q + 1 - a$$

for an integer a .

(d) Use nonnegativity of $\deg([m] + [n]F')$ to prove $|a| \leq 2\sqrt{q}$.

(e) When $q = p$, show that the Hasse invariant is zero if and only if $a \equiv 0 \pmod{p}$. Deduce that for $p \geq 5$ this is equivalent to $N = p + 1$.

Solution. (a) Forming the inverse q -power twist replaces the coefficients of a defining equation by their q -th roots. Coefficients in $\mathbb{F}_q$ are unchanged. Thus $E_q \cong E$, and F' , henceforth denoted F , is

$$[X : Y : Z] \mapsto [X^q : Y^q : Z^q].$$

It fixes the chosen rational origin, has degree q , and has zero differential.

(b) The differential of $1 - F$ is the identity, so this homomorphism is nonconstant and separable. Its kernel consists of the points fixed by F . In any projective chart with one nonzero coordinate normalized to one, fixedness says that the other coordinates satisfy $u^q = u$, hence lie in $\mathbb{F}_q$. Therefore the kernel is precisely $E(\mathbb{F}_q)$, and $\deg(1 - F) = N$.

(c) Put $V = \widehat{F}$; then $FV = VF = [q]$. Duality and part (b) give

$$[N] = (1 - F)(1 - V) = [q + 1] - (F + V).$$

It follows that $F + V = [q + 1 - N]$. Taking $a = q + 1 - N \in \mathbb{Z}$ proves both assertions.

(d) For integers m, n , additivity of duality gives

$$\begin{aligned} ([m] + [n]F)([m] + [n]V) &= [m^2] + [mn](F + V) + [n^2]FV \\ &= [m^2 + amn + qn^2]. \end{aligned}$$

Comparison with $u\widehat{u} = [\deg u]$, including $u = 0$, and faithfulness of integer multiplication imply

$$\deg([m] + [n]F) = m^2 + amn + qn^2 \geq 0. \quad (1)$$

If $a^2 > 4q$, the real quadratic $t^2 + at + q$ is negative on a nonempty open interval between its roots. Choose a rational m/n in that interval. Multiplying by $n^2 > 0$ contradicts (1). Hence $a^2 \leq 4q$,

or $|a| \leq 2\sqrt{q}$. Nonnegativity is the necessary assertion; strict positivity cannot hold for the zero homomorphism.

(e) Suppose $q = p$. Differentiating $F + V = [a]$ gives $dV = a$, since $dF = 0$. The Hasse invariant is zero exactly when V is inseparable, equivalently when $dV = 0$, hence exactly when $p \mid a$. For $p \geq 5$, the strict inequality $2\sqrt{p} < p$, together with the bound in (d), shows that the only possible multiple of p is $a = 0$. By (c), this is equivalent to $N = p + 1$. $\square$

Exercise IV.4.17

Permitted context.

Let $E/\mathbb{Q}$ have equation $y^2 + y = x^3 - x$, with origin the point $O = [0 : 1 : 0]$ at infinity. Three intersections of a line with this cubic, counted with multiplicity, sum to O . Inversion is $(x, y) \mapsto (x, -1 - y)$. The projective equation over any field is

$$F(X, Y, Z) = Y^2Z + YZ^2 - X^3 + XZ^2 = 0.$$

We use the Jacobian criterion to check smoothness.

Problem. (a) Put $P = (0, 0)$. Compute $P + Q$ for $Q = (a, b)$ and list the coordinates of nP for $1 \leq n \leq 10$, including the exceptional cases in the addition formula.

(b) Prove that the displayed integral equation has nonsingular reduction modulo every prime other than 37.

Solution. (a) If $a \neq 0$, the line through P and Q has equation $y = mx$, where $m = b/a$. Substitution into the cubic gives

$$x(x^2 - m^2x - (m + 1)) = 0.$$

The third intersection has x -coordinate $m^2 - a$. Negating it gives

$$P + Q = (m^2 - a, -1 - m(m^2 - a)), \quad m = \frac{b}{a}. \quad (1)$$

If $Q = -P = (0, -1)$, then $P + Q = O$. If $Q = P$, the tangent slope is

$$m = \left. \frac{3x^2 - 1}{2y + 1} \right|_{(0,0)} = -1,$$

and the tangent computation gives $2P = (1, 0)$. Finally $P + O = P$. These exhaust the cases with $a = 0$ or Q at infinity.

Iterating (1), starting with the stated tangent value, yields

n	$x(nP)$	$y(nP)$
1	0	0
2	1	0
3	-1	-1
4	2	-3
5	1/4	-5/8
6	6	14
7	-5/9	8/27
8	21/25	-69/125
9	-20/49	-435/343
10	161/16	-2065/64.

In particular the sixth multiple is the claimed $(6, 14)$. Each row is obtained from the preceding one by the rational operations in (1), so the table also verifies membership in the subgroup generated by P .

(b) At infinity $Z = 0$ forces $X = 0$, so the only point is $O = [0 : 1 : 0]$. There $F_Z = 1$, and hence it is smooth in every characteristic. In the affine chart the partial derivatives of $y^2 + y - x^3 + x$ are

$$1 - 3x^2, \quad 2y + 1.$$

In characteristic two the second is one; in characteristic three the first is one. Thus both these reductions are smooth. For a prime other than two or three, a singular point must have

$$y = -\frac{1}{2}, \quad x^2 = \frac{1}{3}.$$

The original equation then becomes $-1/4 = -2x/3$, so $x = 3/8$. Combining with $x^2 = 1/3$ gives $27 = 64$, which holds only in characteristic 37. This proves smoothness at every other prime. Conversely in characteristic 37, the point $(x, y) = (5, 18)$ satisfies the equation and both derivatives, so that prime really is exceptional. $\square$

Exercise IV.4.18

Permitted context.

On the elliptic curve $E/\mathbb{Q}$ given by $y^2 = x^3 - 7x + 10$, the origin is the point at infinity and $-(x, y) = (x, -y)$. For finite points $A = (x_1, y_1)$, $B = (x_2, y_2)$ with $B \neq -A$, define

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad (x_1 \neq x_2), \quad m = \frac{3x_1^2 - 7}{2y_1} \quad (A = B).$$

Then the sum has coordinates

$$x_3 = m^2 - x_1 - x_2, \quad y_3 = -y_1 + m(x_1 - x_3).$$

Problem. Find at least twenty-six distinct integral points on E . For each, verify membership in the subgroup of $E(\mathbb{Q})$ generated by $P = (1, 2)$ and $Q = (2, 2)$.

Solution. The following table lists thirteen points with positive y -coordinate and expressions for them in terms of P, Q :

x	y	group expression
-3	2	$-P - Q$
-2	4	$4P + Q$
-1	4	$-2P$
1	2	P
2	2	Q
3	4	$-2P - Q$
5	10	$3P + Q$
9	26	$-3P$
13	46	$P - Q$
31	172	$2P + 2Q$
41	262	$-5P - 2Q$
67	548	$6P + Q$
302	5248	$-4P + Q$.

(1)

Negating these expressions supplies thirteen further points with the same x -coordinates and opposite y -coordinates. All twenty-six points are distinct, since the listed x -coordinates are distinct and none of the listed y 's is zero.

Here are intermediate additions verifying the larger entries without an integral-point search assumption. The addition formulas give

$$2P = (-1, -4), \quad 3P = (9, -26), \quad 4P = (9/4, 19/8), \quad P + Q = (-3, -2), \quad P - Q = (13, 46).$$

Continuing with the same formulas gives

$$2P + Q = (3, -4), \quad 3P + Q = (5, 10), \quad 4P + Q = (-2, 4).$$

Doubling $P + Q$ gives

$$2P + 2Q = (31, 172).$$

Adding P to $4P + Q = (-2, 4)$ first yields

$$5P + Q = \left(\frac{13}{9}, -\frac{46}{27} \right).$$

Adding Q to this point gives $(41, -262)$, so

$$5P + 2Q = (41, -262). \tag{2}$$

Adding P instead gives

$$6P + Q = (67, 548).$$

Finally, $4P = (9/4, 19/8)$ gives $-4P + Q = (302, 5248)$ by direct addition with Q .

Negating (2) gives the positive- y entry at $x = 41$:

$$(41, -262) = 5P + 2Q, \quad (41, 262) = -5P - 2Q.$$

The other expressions in (1) follow from the displayed intermediate additions or their negatives. Substitution of each pair into $y^2 = x^3 - 7x + 10$ is an independent integer check of its coordinates. This constructs the requested twenty-six points; it does not assert that no other integral or rational points exist. $\square$

Exercise IV.4.19**Permitted context.**

Let $E/\mathbb{Q}$ be an elliptic curve with a proper integral plane model, and let $T \subset \operatorname{Spec} \mathbb{Z}$ be its open set of good reduction with the prime two removed. Write $\mathcal{E} \rightarrow T$ for the resulting smooth proper genus-one model. Its rational origin extends by the valuative criterion of properness. A smooth proper genus-one family with a section has a unique elliptic-scheme group law compatible with that section; it is obtained from its relative degree-zero Picard functor.

We use these facts: multiplication by a nonzero integer on a geometric elliptic curve is finite of degree n^2 ; a proper quasi-finite morphism is finite; a finite dominant morphism between regular equidimensional schemes of the same dimension is flat (the finite case of miracle flatness). A finite flat morphism with étale geometric fibers is étale. Over the complete local ring $\mathbb{Z}_p$, sections of a finite étale scheme are uniquely determined by their reductions.

Problem. For $n \neq 0$, prove that $[n] : E \rightarrow E$ extends over T to a finite flat morphism $[n] : \mathcal{E} \rightarrow \mathcal{E}$, and that its kernel is flat over T . For a prime $p \in T$ with $(n, p) = 1$, prove that reduction injects $E(\mathbb{Q})[n]$ into $\mathcal{E}_p(\mathbb{F}_p)$. Apply this to prove that the rational torsion subgroups of

$$y^2 + y = x^3 - x, \quad y^2 = x^3 - 7x + 10$$

are trivial.

Solution. The group law of $\mathcal{E}/T$ defines $[n]$ by repeated addition and inversion, and it restricts to the original multiplication map on the generic fiber. For $n \neq 0$, every geometric fiber of this morphism is finite. Since the source is proper over T and the target separated over T , the morphism is proper. It is therefore finite.

The scheme $\mathcal{E}$ is regular of dimension two, being smooth over the regular one-dimensional base T . The multiplication map is surjective and has finite fibers. Miracle flatness applies to its source and target local rings, so $[n]$ is flat. It is consequently finite locally free; its rank is n^2 , as may be read on any geometric fiber. The kernel is its base change along the identity section:

$$\mathcal{E}[n] = \mathcal{E} \times_{\mathcal{E}, [n]} T.$$

It is finite locally free of rank n^2 , hence flat over T .

Fix $p \in T$ with $p \nmid n$. On the base change to $\mathbb{Z}_p$, the differential of $[n]$ on each fiber is multiplication by the invertible scalar n . Its geometric fibers are étale, so $\mathcal{E}[n]$ is finite étale over $\mathbb{Z}_p$. Every rational point of E extends to a $\mathbb{Z}_p$ -point by properness. Its reduction is compatible with the group law. If two rational n -torsion points have equal reduction, their extended sections of $\mathcal{E}[n]$ coincide by uniqueness of finite-étale lifting over $\mathbb{Z}_p$. Their generic points, and hence the original rational points, are equal. This proves injectivity.

For the first displayed curve, direct counting gives

$$\#E(\mathbb{F}_3) = 7, \quad \#E(\mathbb{F}_5) = 8.$$

For the second, the corresponding counts are 7 and 10. Both curves have good reduction at three and five, as the Jacobian criterion or their discriminants show. For a prime $\ell \neq 3, 5$, rational ℓ -torsion injects into both reduction groups; their relatively prime orders exclude such torsion. For $\ell = 3$, reduction at five excludes it, since neither eight nor ten is divisible by three. For $\ell = 5$, reduction at three excludes it, since seven is not divisible by five. Thus neither curve has a rational

point of any prime order. Any nonzero torsion point would have a multiple of prime order, so both rational torsion subgroups are trivial.

The restriction $n \neq 0$ is necessary for flatness of the multiplication morphism: $[0]$ maps each elliptic fiber to its origin and is not flat. Its kernel is the whole family, which is nevertheless flat over T . $\square$

Exercise IV.4.20

Permitted context.

Let E be an elliptic curve over an algebraically closed field of characteristic $p > 2$, and put $R = \text{End}(E, O)$. The inverse Frobenius twist E_p is obtained by replacing coefficients of an equation by their p -th roots. The k -linear Frobenius $F' : E_p \rightarrow E$ is purely inseparable of degree p . Dual isogenies have the same degree, and $f\hat{f} = \hat{f}f = [\deg f]$. The dual of Frobenius is separable if and only if the Hasse invariant is one. In particular, $[p]$ is purely inseparable if the invariant is zero and has separable degree p otherwise.

For a smooth curve over a perfect field, its purely inseparable degree- p^r quotient is uniquely its r -fold Frobenius twist, up to isomorphism. This follows from $[k(E) : k(E)^{p^r}] = p^r$. The invariant j is a rational function of Weierstrass coefficients over the prime field and classifies elliptic curves over k . The tangent space of $\text{Pic}^0(E)$ is $H^1(E, \mathcal{O}_E)$.

Problem. (a) Prove $j(E_p) = j(E)^{1/p}$, and conclude $E \cong E_p$ if and only if $j(E) \in \mathbb{F}_p$.

(b) Show that $[p]$ factors as $\pi\hat{\pi}$ for an endomorphism π of degree p if and only if $E \cong E_p$. In this case show that the Hasse invariant is zero if and only if $\pi, \hat{\pi}$ differ by an automorphism.

(c) If the Hasse invariant is zero, prove $j(E) \in \mathbb{F}_{p^2}$.

(d) Let λ_f be the scalar by which f^* acts on $H^1(E, \mathcal{O}_E)$. Show that $f \mapsto \lambda_f$ is a ring homomorphism $R \rightarrow k$, and that its image is $\mathbb{F}_p$ when the Hasse invariant is nonzero. Deduce that R has a prime ideal with quotient $\mathbb{F}_p$.

Solution. (a) Taking p -th roots of the coefficients commutes with the rational formula for j , because its constants lie in the prime field. Thus $j(E_p) = j(E)^{1/p}$. The two curves are isomorphic exactly when these invariants are equal, equivalently $j(E)^p = j(E)$, which means $j(E) \in \mathbb{F}_p$.

(b) If $E \cong E_p$, compose such an origin-preserving isomorphism with F' to obtain a degree- p endomorphism π of E . Duality gives $\pi\hat{\pi} = [p]$.

Conversely, in a factorization into two degree- p maps, at least one factor is inseparable, since $[p]$ is inseparable. An inseparable degree- p map is purely inseparable. The uniqueness of the purely inseparable quotient therefore implies $E \cong E^{(p)}$, equivalently $E \cong E_p$.

If the Hasse invariant is zero, $[p]$ is purely inseparable, so both factors are purely inseparable of degree p . As maps with the same source, they have the same function-field image $k(E)^p$, and differ by an isomorphism of their targets. Thus π and $\hat{\pi}$ differ by a unit of R . If the invariant is one, the separable degree of $[p]$ is p , so one factor is separable and the other is purely inseparable. Composing with an automorphism cannot change separability, so they cannot be associates.

(c) When the invariant is zero, the degree- p^2 map $[p]$ is purely inseparable. Uniqueness of this quotient gives $E \cong E^{(p^2)}$. Their invariants satisfy $j(E) = j(E)^{p^2}$. The roots of $T^{p^2} - T$ in k are exactly $\mathbb{F}_{p^2}$, proving the claim.

(d) Pullback reverses composition on H^1 , but the resulting endomorphisms of its one-dimensional vector space are scalars and commute. Thus $\lambda_{fg} = \lambda_f \lambda_g$. Additivity follows from the Picard interpretation: $\widehat{f+g} = \widehat{f} + \widehat{g}$, and the differential of addition of group homomorphisms is the sum

of their differentials. The differential of $\widehat{f}$ is exactly f^* on H^1 . Also $\lambda_1 = 1$, so this is a unital ring homomorphism.

Absolute Frobenius $F : E \rightarrow E$, although not k -linear, commutes with every morphism of schemes by functoriality. On cohomology this gives $f^*F^* = F^*f^*$. Choose a nonzero vector v and write $F^*v = cv$. If the Hasse invariant is nonzero, then $c \neq 0$, and semilinearity gives

$$c\lambda_f v = f^*F^*v = F^*f^*v = \lambda_f^p cv.$$

Hence $\lambda_f^p = \lambda_f$, so every λ_f lies in $\mathbb{F}_p$. The image contains $\mathbb{F}_p$ because the homomorphism is unital. Its kernel is consequently a maximal, hence prime, two-sided ideal with quotient $\mathbb{F}_p$. $\square$

Exercise IV.4.21

Permitted context.

Let K be a quadratic number field and $\mathcal{O}_K$ its ring of integers. It has an integral basis $1, \omega$, so $\mathcal{O}_K = \mathbb{Z} \oplus \mathbb{Z}\omega$, with $\omega^2 = a\omega + b$ for integers a, b . Subrings are required to contain the identity, hence contain $\mathbb{Z}$. Every nonzero additive subgroup of $\mathbb{Z}$ is $f\mathbb{Z}$ for a unique positive integer f .

Problem. Prove that every subring $R \subseteq \mathcal{O}_K$, $R \neq \mathbb{Z}$, has the form

$$R = \mathbb{Z} + f\mathcal{O}_K$$

for a unique integer $f \geq 1$, called its conductor.

Solution. Project the additive group of $\mathcal{O}_K$ onto $\mathcal{O}_K/\mathbb{Z} \cong \mathbb{Z}$, using the coefficient of ω . The image of R is nonzero because $R \neq \mathbb{Z}$. It is therefore $f\mathbb{Z}$ for a unique $f \geq 1$. Choose $c + f\omega \in R$ lifting its generator. Since $c \in \mathbb{Z} \subset R$, subtraction shows $f\omega \in R$. Thus

$$\mathbb{Z} + f\mathbb{Z}\omega \subseteq R.$$

The reverse inclusion follows from the definition of the image $f\mathbb{Z}$: every element of R has its ω -coefficient divisible by f . Hence

$$R = \mathbb{Z} + f\mathbb{Z}\omega = \mathbb{Z} + f\mathcal{O}_K.$$

The subgroup image determines f uniquely.

Conversely this additive subgroup is a ring, since $(f\omega)^2 = f^2b + fa(f\omega)$ belongs to it. Thus all positive f do produce subrings. The usual conductor ideal also equals $f\mathcal{O}_K$: if $r = c + df\omega \in R$ satisfies $r\mathcal{O}_K \subset R$, then the coefficient of ω in $r\omega = dfb + (c + dfa)\omega$ must be divisible by f . This forces $f \mid c$, hence $r \in f\mathcal{O}_K$. The reverse inclusion is immediate. $\square$

Exercise IV.4.22**Permitted context.**

A family of elliptic curves with section over $S = \mathbb{A}_{\mathbb{C}}^1$ means a smooth proper morphism $\mathcal{E} \rightarrow S$ with geometrically connected genus-one fibers and a chosen section O . It carries an elliptic-scheme group law. The kernel $\mathcal{E}[2]$ is finite etale of rank four. The quotient by inversion is a projective-line bundle $Y \rightarrow S$; the quotient map $\mathcal{E} \rightarrow Y$ is finite flat of degree two and branched at the images of the four two-torsion points. It can be constructed by the relative linear system $|2O|$.

We use Hurwitz for finite maps of smooth projective complex curves and $\text{Pic}(\mathbb{P}^1 \times S) \cong \mathbb{Z}$, generated by the degree-one bundle on $\mathbb{P}^1$. For a finite flat double cover in characteristic different from two, the trace decomposition of its algebra is $\mathcal{O}_Y \oplus M^{-1}$; multiplication on the second summand is a section of M^2 , whose zero divisor is the branch divisor.

Problem. Prove that every family of elliptic curves with a section over $\mathbb{A}_{\mathbb{C}}^1$ is trivial: it is isomorphic, with section, to $E_0 \times \mathbb{A}_{\mathbb{C}}^1$ for a fixed complex elliptic curve E_0 .

Solution. First, every connected finite etale cover $Z \rightarrow \mathbb{A}_{\mathbb{C}}^1$ has degree one. To prove this, compactify it to a finite map $\bar{Z} \rightarrow \mathbb{P}^1$ of degree d , with $\bar{Z}$ smooth and projective. All ramification is over infinity. If there are $r \geq 1$ points over infinity, their total ramification contribution is $d - r$, since the characteristic is zero. Hurwitz gives

$$2g(\bar{Z}) - 2 = -2d + (d - r) = -d - r.$$

As $g(\bar{Z}) \geq 0$, this forces $d + r \leq 2$, hence $d = r = 1$.

It follows that $\mathcal{E}[2]$ is the disjoint union of four sections. Label them O, T_1, T_2, T_3 . Their images in the quotient line bundle Y are disjoint sections. The first three determine a unique trivialization

$$Y \cong \mathbb{P}^1 \times S$$

carrying their images to $\infty, 0, 1$, respectively. Indeed the unique projective linear transformation carrying three distinct points to these values varies regularly in a local trivialization, and uniqueness makes these transformations glue. The fourth image is consequently a morphism

$$\lambda : S \longrightarrow \mathbb{A}^1 \setminus \{0, 1\}.$$

It is a polynomial with no zero on $\mathbb{C}$, so it is constant. Write its value as $\lambda_0 \neq 0, 1$. The branch divisor is therefore the fixed divisor

$$B = (\infty) + (0) + (1) + (\lambda_0)$$

pulled back from $\mathbb{P}^1$.

It remains to rule out a varying double-cover twist, rather than merely conclude that the fibers are isomorphic. By the trace decomposition, the double cover is described by a line bundle M on $\mathbb{P}^1 \times S$ and a section $b \in H^0(M^2)$ with divisor $B \times S$. In particular $M^2 \cong \mathcal{O}(4)$. Since the Picard group of this product is $\mathbb{Z}$, we have $M \cong \mathcal{O}(2)$, pulled back from $\mathbb{P}^1$. Under this identification, b differs from the fixed binary quartic defining B by a global unit. The global units on $\mathbb{P}^1 \times \mathbb{A}_{\mathbb{C}}^1$ are $\mathbb{C}^\times$, so that unit is a constant c . Choose a square root of c and rescale the trace-zero generator. This identifies the double-cover algebra with the pullback of the fixed double-cover algebra on $\mathbb{P}^1$.

Thus $\mathcal{E} \cong E_0 \times S$, where E_0 is the double cover branched at $0, 1, \lambda_0, \infty$, equivalently the Legendre elliptic curve with parameter λ_0 . The chosen section O is the unique ramification section above infinity, so this isomorphism preserves the section as well as the family. $\square$

IV.5 The Canonical Embedding

Exercise IV.5.1

Permitted context.

Work over an algebraically closed field. A hyperelliptic curve is a nonsingular projective integral curve of genus $g \geq 2$ with a degree-two morphism to $\mathbb{P}^1$; its canonical map factors through that morphism and hence is not an embedding. For a nonsingular complete intersection of hypersurfaces of degrees $d_1, \dots, d_{n-1}$ in $\mathbb{P}^n$, adjunction gives $\omega_X = \mathcal{O}_X(\sum d_i - n - 1)$. The canonical degree is $2g - 2$, and a positive tensor power of a very ample line bundle is very ample.

Problem. Prove that a hyperelliptic curve is not a complete intersection in any projective space.

Solution. Suppose an embedding made X such a complete intersection. Set $m = \sum d_i - n - 1$, so adjunction gives $\omega_X = \mathcal{O}_X(m)$. Taking degrees yields

$$2g - 2 = m \deg \mathcal{O}_X(1).$$

Both the left side and $\deg \mathcal{O}_X(1)$ are positive, so $m > 0$. Consequently ω_X is very ample. Its complete linear system would embed X , contradicting the degree-two factorization of the canonical map of a hyperelliptic curve. Since the argument applies to every embedding and every ambient projective dimension, no complete-intersection embedding exists. $\square$

Exercise IV.5.2

Permitted context.

Let X be a nonsingular projective integral curve over an algebraically closed field, of genus $g \geq 2$. A line bundle of degree at least $2g + 1$ is very ample. The tangent sheaf is $T_X = \omega_X^{-1}$, and a negative-degree line bundle has no nonzero global section. The subgroup scheme of PGL_{N+1} preserving a fixed projective subscheme is closed and of finite type: preservation of finitely many homogeneous ideal generators gives its defining equations. Its tangent vectors at the identity induce derivations on that subscheme. We use that local dimension is at most tangent-space dimension and that a zero-dimensional finite-type scheme has finitely many geometric points.

Problem. In characteristic zero, prove that $\mathrm{Aut}(X)$ is finite. In fact, prove this assertion in arbitrary characteristic.

Solution. The line bundle $L = \omega_X^{\otimes 3}$ has degree $6g - 6$, which is at least $2g + 1$ for every $g \geq 2$. Embed X by the complete system $|L|$ into $\mathbb{P}^N$. Every automorphism acts functorially on $H^0(X, L)$, hence is induced by a projective transformation. Conversely, a projective transformation preserving X restricts to an automorphism. These restrictions are faithful. Indeed, if a matrix fixes every point of X , each such point is represented by an eigenvector. The finitely many eigenspaces cover the integral curve, so one contains it. Since the complete embedding is nondegenerate, that eigenspace is the whole vector space and the matrix is scalar. Thus $\mathrm{Aut}(X)$ is the group of geometric points of the closed stabilizer $G \subset \mathrm{PGL}_{N+1}$.

The tangent action at the identity gives a linear map

$$T_1 G \longrightarrow H^0(X, T_X).$$

It is injective by the same eigenvector argument: a tangent vector is represented by a matrix A modulo scalar matrices, and its projective vector field is zero at $[v]$ exactly when $Av \in kv$. If its restriction to X is zero, the preceding argument forces A scalar. But $\deg T_X = 2 - 2g < 0$, so $H^0(X, T_X) = 0$. Hence $T_1G = 0$, and the local dimension of G at the identity is zero. Translation gives the same conclusion at every geometric point. Therefore G is zero-dimensional and of finite type, and has finitely many geometric points. This proves the assertion without a characteristic restriction. $\square$

Exercise IV.5.3

Permitted context.

Work over an algebraically closed field, in any characteristic. We measure a family's dimension in the moduli space of smooth curves. We use finite automorphism groups in genus at least two, the dimension-of-fibers theorem, and the uniqueness of the hyperelliptic pencil. A nonhyperelliptic genus-four canonical curve is the complete intersection of a unique quadric $Q \subset \mathbb{P}^3$ and a cubic. The quadric is smooth or a cone; its rulings give exactly two or one degree-three pencils, respectively. Finite morphisms between smooth curves are flat. On $\mathbb{P}^1$, line bundles are classified by degree and $H^1(\mathcal{O}(m)) = 0$ for $m \geq -1$. We use smoothness as an open condition and Bertini for a very ample system.

Problem. Show that the hyperelliptic genus-four curves form an irreducible family of dimension seven, the nonhyperelliptic ones an irreducible family of dimension nine, and those with just one g_3^1 an irreducible family of dimension eight.

Solution. First parametrize double covers without excluding characteristic two. For a smooth hyperelliptic curve with covering map $\pi : X \rightarrow \mathbb{P}^1$, the quotient of $\pi_*\mathcal{O}_X$ by the unit subbundle is a line bundle: locally the nonzero unit is part of a basis of this rank-two bundle. Euler characteristics show that the quotient is $\mathcal{O}(-5)$. The extension splits since $H^1(\mathcal{O}(5)) = 0$. Consequently the algebra has the form

$$\mathcal{O} \oplus \mathcal{O}(-5), \quad w^2 + aw = b, \quad a \in H^0(\mathcal{O}(5)), \quad b \in H^0(\mathcal{O}(10)). \quad (1)$$

Conversely, (1) defines a finite flat double cover with $\chi(\mathcal{O}_X) = 1 - 4 = -3$, so a smooth connected such cover has genus four. Smooth covers cannot be disconnected: then their two degree-one components would both be $\mathbb{P}^1$, contradicting this Euler characteristic. The coefficient space is affine of dimension $6 + 11 = 17$. Its smooth locus is open and nonempty. In odd characteristic, take $a = 0$ and b a binary form with ten distinct roots. In characteristic two, take homogeneous base coordinates $[s : t]$, $a = s^5$, and $b = st^9$. On $s \neq 0$ the derivative with respect to w is one; on $t \neq 0$, the equation is $v^2 + z^5v = z$, whose derivative with respect to z is nonzero at the only point over $z = 0$. Thus this example also is smooth.

Changing the algebra generator over a fixed base has the form $w \mapsto uw + c$, where $u \in k^\times$ and $c \in H^0(\mathcal{O}(5))$; this group has dimension seven. Allowing a base automorphism adds three dimensions. Equivalently, the group of these changes fits over PGL_2 with the preceding seven-dimensional kernel; lifts of a base transformation to $\mathcal{O}(5)$ exist and differ by a scalar. Uniqueness of the hyperelliptic pencil implies that two parameters define isomorphic curves exactly when related by these changes. The stabilizers are finite, since they act faithfully as curve automorphisms. Thus every moduli fiber has dimension ten, and the image is irreducible of dimension

$$17 - 10 = 7.$$

For nonhyperelliptic curves, the quadrics that are smooth or cones form an irreducible open subset of the projective space $\mathbb{P}^9$ of quadratic forms. For each such Q , the restriction sequence gives

$$h^0(Q, \mathcal{O}_Q(3)) = h^0(\mathbb{P}^3, \mathcal{O}(3)) - h^0(\mathbb{P}^3, \mathcal{O}(1)) = 20 - 4 = 16.$$

The cubic sections therefore form a projective bundle with fiber $\mathbb{P}^{15}$. Its open subset of smooth complete intersections is nonempty over both types of quadric: on a cone require the cubic to avoid its vertex, and apply Bertini on the smooth locus. These curves are connected complete intersections, and adjunction gives $\omega_X = \mathcal{O}_X(1)$ and genus four. Thus they are canonical curves, and they parametrize exactly the nonhyperelliptic locus. The parameter space has dimension $9 + 15 = 24$. Uniqueness of the quadric and completeness of the canonical embedding identify each moduli fiber with a PGL_4 -orbit. Its dimension is fifteen, because its stabilizer is finite. The image therefore is irreducible of dimension $24 - 15 = 9$.

Finally, the cone quadrics form a single projective orbit of dimension eight. One can also count them by choosing their vertex (three parameters) and a smooth conic in the quotient plane (five parameters). This description works in characteristic two as well. Restricting the cubic-section bundle to this irreducible locus gives dimension $8 + 15 = 23$, hence moduli dimension $23 - 15 = 8$. The canonical classification in the context identifies its image exactly with the curves having one g_3^1 . $\square$

Exercise IV.5.4

Permitted context.

Work over an algebraically closed field. A nonhyperelliptic genus-four canonical curve is a degree-six complete intersection of a cubic and a quadric; a smooth quadric gives two g_3^1 's and a cone gives one. A smooth quadric is $\mathbb{P}^1 \times \mathbb{P}^1$, and its curve has type $(3, 3)$. Resolving a quadric cone gives the ruled surface $\mathbb{F}_2$, with section E , fiber F , $E^2 = -2$, $EF = 1$, and hyperplane class $E + 2F$. We use blowup and blowdown charts, the plane genus formula, and the generic-projection theorem giving a nodal plane model of the same degree from a smooth space curve. A purely inseparable degree-three cover of $\mathbb{P}^1$ over this field is rational. A node has delta invariant one. A tacnode here means two smooth branches with intersection multiplicity two; it has delta invariant two, including in characteristic two.

Problem. Let X be nonhyperelliptic of genus four. (a) Prove that it has two g_3^1 's exactly when it has a plane quintic model with two nodes and no other singularities.

(b) If it has one g_3^1 , construct a plane quintic model with one tacnode and no other singularities, and prove that the least degree of a nodal plane model is six.

Solution. (a) Put the canonical curve on its smooth quadric Q . Choose $P \in X$ so that each of the two ruling fibers through P meets X in three distinct points. Such choices form a nonempty open subset: both degree-three maps to $\mathbb{P}^1$ are separable. Indeed, an inseparable degree-three map could occur only in characteristic three and would be purely inseparable, forcing X to be rational.

Project Q from P to $\mathbb{P}^2$. On the blowup of Q at P , this is a birational morphism contracting exactly the strict transforms of the two ruling fibers through P ; they are disjoint curves of self-intersection minus one. This description follows as well by writing the quadric in Segre coordinates with $P = [1 : 0 : 0 : 0]$ and projecting onto the last three coordinates. The strict transform of X meets each contracted curve in two distinct points, transversely. Blowing down such a curve therefore produces a node: its two intersection points specify two distinct tangent directions at the

image point. Away from these curves the surface morphism is an isomorphism. The restricted plane system is $K_X - P$, of degree five. Consequently the image is a plane quintic with exactly the two stated nodes.

Conversely, projection from either node of a plane quintic gives, on its normalization, a base-point-free degree-three pencil. These pencils are distinct. In fact, the pair of projections from two distinct plane points is birational on the plane: outside their joining line, a point is recovered as the intersection of its two projection lines. Thus their restrictions jointly generate the function field of the normalization. If the pencils were equal, their pair would factor through a degree-three map to $\mathbb{P}^1$, which is impossible for a birational map. Hence X has two g_3^1 's.

(b) Now the canonical quadric is a cone. Its vertex does not lie on X : if it did, the quadric gradient there would vanish and the complete intersection would be singular. On $\mathbb{F}_2$, the curve therefore has class $3E + 6F$ and is disjoint from E . Choose a fiber meeting X in three distinct points P, A, B , and blow up P . The strict transform F' of that fiber has self-intersection minus one. Contract F' ; this changes $\mathbb{F}_2$ to $\mathbb{F}_1$. The image E_1 of E now has self-intersection minus one, so contracting E_1 gives $\mathbb{P}^2$. This composite is the resolution of projection of the cone from P , and restricts to the system $K_X - P$.

The first contraction identifies A, B to a node R on E_1 . Both branches are transverse to E_1 , since their points on F' differ from $E \cap F'$. The second contraction makes these branches tangent. For precision, take coordinates (u, v) at R with $E_1 = (u = 0)$. The branches have expansions $v = a_i u + O(u^2)$, with $a_1 \neq a_2$. The blowdown has coordinates $x = u$, $y = uv$, so the images are $y = a_i x^2 + O(x^3)$. Each branch is smooth and their intersection multiplicity is exactly two. Thus the image has a tacnode. The surface map is an isomorphism elsewhere along the curve, so there are no other singularities. Its degree is five.

A plane curve of degree at most four has arithmetic genus at most three, so cannot normalize to X . A nodal quintic with normalization genus four would have exactly two nodes, and by (a) would give two g_3^1 's. Thus no nodal model of degree at most five exists in the one-pencil case. Finally, generic projection of the smooth degree-six canonical space curve gives a nodal sextic. The least possible degree is therefore six. $\square$

Exercise IV.5.5**Permitted context.**

Work over an algebraically closed field. We use Riemann–Roch, Clifford’s theorem with its equality cases, adjunction, Bertini, and finite automorphism groups for curves of genus at least two. A plane curve of degree five has arithmetic genus six; a reduced plane singularity with delta invariant one is a node or a cusp. Here cusp means the unibranch delta-one singularity, also in positive characteristic. On $S = \mathrm{Bl}_p \mathbb{P}^2$, write H for the pullback of a line and E for the exceptional curve. Then $H^2 = 1$, $HE = 0$, $E^2 = -1$, and $K_S = -3H + E$. The conics through p , of class $2H - E$, embed S in $\mathbb{P}^4$. Curves of type (a, b) on $\mathbb{P}^1 \times \mathbb{P}^1$ have arithmetic genus $(a - 1)(b - 1)$. We use the dimension-of-fibers theorem.

Problem. Let X be a nonhyperelliptic curve of genus five. (a) Show that the canonical curves which are complete intersections of three quadrics form a family of dimension twelve.

(b) Prove the corrected statement: X has a g_3^1 if and only if it has a plane quintic model with one node or one cusp and no other singularities. These curves form an irreducible family of dimension eleven. The source’s assertion with only a node omits the cuspidal case.

(c) Show that the conics through the singular point induce the canonical embedding and place X on a smooth cubic scroll $V \cong \mathrm{Bl}_p \mathbb{P}^2$. Prove that its ruling lines are the trisecants from the degree-three pencil, that every quadric containing X contains V , and that both V and the pencil are unique.

Solution. (a) Quadratic forms in five variables form a vector space of dimension fifteen. The three-dimensional subspaces form $\mathrm{Gr}(3, 15)$, irreducible of dimension $3(15 - 3) = 36$. The locus whose common zero scheme is a smooth curve complete intersection is nonempty and open: successive general quadrics give examples by Bertini. Adjunction gives $K_X = \mathcal{O}_X(1)$; the degree is eight, so the genus is five. These embeddings are complete canonical embeddings: restriction of linear forms is injective, and both spaces of sections have dimension five. The degree-two part of the ideal has dimension three, as follows from its Koszul resolution, so it recovers the original point of the Grassmannian. Consequently moduli fibers are precisely the PGL_5 -orbits. Stabilizers are finite, so these fibers have dimension $25 - 1 = 24$. The resulting family is irreducible of dimension $36 - 24 = 12$.

(b) Let D be the divisor class of a g_3^1 . It has no base points, since otherwise X would have a degree-one or degree-two map to $\mathbb{P}^1$. Clifford’s theorem gives $h^0(D) = 2$, and Riemann–Roch then gives

$$\deg(K_X - D) = 5, \quad h^0(K_X - D) = 3. \quad (1)$$

This residual system is base-point-free. A base point P would give a degree-four divisor $K_X - D - P$ with at least three sections. It is special by Riemann–Roch; equality in Clifford’s theorem would force hyperellipticity, since this divisor is neither zero nor canonical.

Thus (1) maps X to a nondegenerate plane curve. The product of the degree of the map and the degree of its image is five. Since the image is not a line, this prime degree forces the map to be birational and the image to be a quintic. The difference between arithmetic genus six and normalization genus five is one. There is therefore exactly one singularity, a node or a cusp, and no others. Conversely, projection from a multiplicity-two singularity of either type gives a degree-three pencil on the normalization. Its degree is $5 - 2 = 3$, because the two fixed intersections at the center are removed on normalization.

We will need uniqueness. Suppose there are two distinct degree-three pencils. The pair of maps to $\mathbb{P}^1 \times \mathbb{P}^1$ has degree dividing three onto its image. If that degree were three, both projections of the

image would have degree one; the pencils would then differ by an automorphism of $\mathbb{P}^1$. Otherwise the pair is birational and its image has type $(3, 3)$, of arithmetic genus four, which cannot have normalization genus five. Thus the pencil is unique.

For the parameter count, fix a marked point p of the plane. Vanishing to order at least two at p imposes three independent linear conditions on the twenty-one quintic coefficients. Allowing p to vary gives an irreducible projective bundle of dimension

$$2 + (20 - 3) = 19.$$

The locus where the quintic is integral, its strict transform on the blowup at p is smooth, and the multiplicity at p is exactly two, is nonempty and open. The strict transform has class $5H - 2E$ and genus five by adjunction; its intersection of length two with E gives a node if reduced and a cusp if supported at one point. There are no other singularities. Nonemptiness follows from Bertini for the very ample system $5H - 2E$, whose general restriction to E has two distinct zeros.

Each resulting normalization is nonhyperelliptic: a degree-two map together with its degree-three map would be birational onto a curve of type $(2, 3)$, with arithmetic genus two. On the blowup, adjunction and projection from p give, on the normalization,

$$K_X = (2H - E)|_X, \quad D = (H - E)|_X. \quad (2)$$

Thus its plane system is $H|_X = K_X - D$, uniquely determined by the unique pencil. Two of our plane models of the same curve are therefore projectively equivalent. The moduli fibers are PGL_3 -orbits of dimension eight, so the image is irreducible of dimension $19 - 8 = 11$.

The omitted cuspidal case really occurs. On the blowup, impose a double intersection at a specified point of E and take a general member of $|5H - 2E|$ with that condition. Its local equation can be chosen to have the form $v^2 - u + \text{higher terms} = 0$, where $E = (u = 0)$. It is smooth there and tangent to E ; Bertini away from that point makes it smooth elsewhere. For example these conditions come from plane quintics with quadratic term y^2 and nonzero x^3 term at p ; the freely varying terms of degree at least three remove base points away from p . The smooth strict transform is connected (its class is ample), hence integral, and blowing down gives the cusp. By the uniqueness of D and $K_X - D$, this normalization cannot also have a nodal plane quintic model.

(c) Formula (2) shows that the conics through p restrict to canonical divisors. Their vector space has dimension five, and restriction is injective because a conic cannot contain an integral quintic. Hence it is the full canonical space $H^0(X, K_X)$. The conic embedding of S has degree $(2H - E)^2 = 3$; call its image V . The fibers $H - E$ of the pencil of lines through p map to lines, since

$$(2H - E)(H - E) = 1,$$

and each meets X in degree $(5H - 2E)(H - E) = 3$. Thus V is the union of the ruling trisecants of the degree-three pencil.

Every quadric containing X contains each of these lines: its restriction to a line has degree two but vanishes on the length-three intersection with X . It consequently contains their union V . This already shows that these trigonal canonical curves are not complete intersections of three quadrics. For uniqueness of the surface, the unique pencil recovers each ruling line as the span of its fiber in the canonical embedding, so recovers their union. Indeed, a length-three fiber has canonical span a line: Riemann–Roch gives $h^0(K_X - D) = 3$, hence the span has dimension $h^0(K_X) - h^0(K_X - D) - 1 = 1$. This proves uniqueness of both the pencil and the scroll. $\square$

Exercise IV.5.6**Permitted context.**

Work over an algebraically closed field. A smooth plane quintic has genus six and canonical bundle $\mathcal{O}_X(2)$; its canonical sections are restrictions of plane conics. We use Riemann–Roch and that a curve with a purely inseparable map of prime degree to $\mathbb{P}^1$ is rational. A curve of bidegree (a, b) on $\mathbb{P}^1 \times \mathbb{P}^1$ has arithmetic genus $(a-1)(b-1)$, and its normalization has no larger genus. Bertini supplies smooth integral members of the very ample system $\mathcal{O}(a, b)$ for positive a, b .

Problem. Prove that a nonsingular plane quintic has no g_3^1 . Construct a nonhyperelliptic genus-six curve that has no nonsingular plane quintic model.

Solution. Let X be a smooth plane quintic. Its canonical bundle $\mathcal{O}_X(2)$ is very ample, so X is nonhyperelliptic. A degree-three pencil on it could have no fixed part: removing that part would give a map of degree at most two to $\mathbb{P}^1$, contradicting either genus six or nonhyperellipticity. Suppose such a pencil exists. Its map is separable, since an inseparable degree-three map would force rationality. A general fiber is therefore $D = P + Q + R$ with three distinct points. These points impose three independent conditions on plane conics. To see independence at P , choose a line through Q avoiding P and a line through R avoiding P ; their product vanishes at Q, R but not at P , and similarly for the other two points. Since the six-dimensional conic space maps isomorphically to $H^0(X, K_X)$, it follows that

$$h^0(X, K_X - D) = 6 - 3 = 3.$$

But Riemann–Roch and $h^0(X, D) \geq 2$ give

$$h^0(X, K_X - D) = h^0(X, D) - 3 - 1 + 6 \geq 4,$$

a contradiction.

For the example, choose a smooth integral curve C of bidegree $(3, 4)$ on $\mathbb{P}^1 \times \mathbb{P}^1$. It has genus $(3-1)(4-1) = 6$, and one projection gives a degree-three map. It is not hyperelliptic. Indeed, if it also had a degree-two map, the pair of maps would have degree dividing both two and three onto its image, and therefore would be birational. The image would have bidegree $(2, 3)$, with arithmetic genus two, contradicting the genus of its normalization C . If C admitted a smooth plane quintic model, its degree-three pencil would contradict the first part. Thus C is the required example. $\square$

Exercise IV.5.7**Permitted context.**

Work over an algebraically closed field k . Nonhyperelliptic genus-three curves embed canonically as smooth plane quartics, forming an open dense locus in genus-three moduli. Automorphism groups of curves of genus at least two are finite. We use the tame Hurwitz bound $84(g-1)$. For a cyclic group of prime order ℓ , Hurwitz reads $2g-2 = \ell(2h-2) + \sum d_P$; in the tame case each ramified point contributes $\ell-1$, and in characteristic ℓ it contributes $(\ell-1)(m_P+1)$, with $m_P \geq 1$.

We explicitly permit one finite-group representation fact: over $\mathbb{C}$, the matrices S, T, U displayed in part (b) below generate the faithful Klein representation of the simple group $\mathrm{PSL}_2(\mathbb{F}_7)$, of order 168. This is the matrix representation lemma, not an assumption about the full automorphism group of a curve; see Elkies, *The Klein Quartic in Number Theory*, Section 1.1, equations (1.2)–(1.3). We also use Jordan normal form, Sylow's and Cauchy's theorems, and dimension of fibers.

Problem. (a) Prove that every automorphism of a nonhyperelliptic genus-three curve is induced by a projective transformation of its canonical plane. Explain the missing hypothesis in the source statement.

(b) For

$$X : \quad x^3y + y^3z + z^3x = 0,$$

prove that, in characteristic different from three and seven, $\mathrm{Aut}(X) \cong \mathrm{PSL}_2(\mathbb{F}_7)$ has order 168. Explain the necessary exclusion of characteristic seven and the scope of the Hurwitz bound.

(c) Show that a general genus-three curve has no nonidentity automorphism, in any characteristic.

Solution. (a) An automorphism pulls back canonical differentials, giving an invertible linear map on the three-dimensional space $H^0(X, \omega_X)$. The canonical map is equivariant for its projectivization, so an automorphism is induced by the corresponding projective transformation. The canonical embedding makes this action faithful. For a hyperelliptic genus-three curve the canonical map has degree two onto a conic and is not an embedding. Its hyperelliptic involution acts trivially on the canonical image, so the assertion as phrased in the source needs the nonhyperelliptic hypothesis.

(b) Write $F = x^3y + y^3z + z^3x$. First check smoothness. At a coordinate point on X , one partial derivative is nonzero. If all coordinates are nonzero and all partial derivatives vanish, put $A = x^3y$, $B = y^3z$, $C = z^3x$. Multiplying the three derivative equations by x, y, z gives $3A + C = A + 3B = B + 3C = 0$. Hence $A = 9C$, $B = -3C$, and $F = 7C$. Thus a singular point on the curve forces characteristic seven. In that characteristic $[1 : 2 : 4]$ is indeed on the curve and all three derivatives vanish. For every other characteristic the curve is smooth; adjunction makes it a canonical genus-three curve.

Choose a primitive seventh root ζ , and set

$$a = \zeta - \zeta^6, \quad b = \zeta^2 - \zeta^5, \quad c = \zeta^4 - \zeta^3, \quad \delta = a + b + c.$$

Then $\delta^2 = -7$. The promised matrices are

$$S = \begin{pmatrix} \zeta^4 & 0 & 0 \\ 0 & \zeta^2 & 0 \\ 0 & 0 & \zeta \end{pmatrix}, \quad T = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad U = -\delta^{-1} \begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}. \quad (1)$$

They preserve F . For S, T this follows term by term; for U , writing $M = -\delta U$, expansion using $1 + \zeta + \cdots + \zeta^6 = 0$ gives

$$M^2 = -7I, \quad F(M(x, y, z)) = 49F(x, y, z). \quad (2)$$

Since $\delta^4 = 49$, (2) proves the assertion.

By the permitted representation lemma, in characteristic zero they generate a copy of the simple group of order 168. All matrices and their inverses lie in $\mathbb{Z}[\zeta, 1/7]$, since $\delta^{-1} = -\delta/7$. Thus all multiplication relations reduce to characteristic $p \neq 7$, yielding a homomorphism from that simple group to $\mathrm{PGL}_3(k)$. Its image is nontrivial, since S still has projective order seven. Simplicity makes this homomorphism injective. Consequently X has a subgroup G of 168 automorphisms in every characteristic different from seven.

Characteristic zero and tame characteristic. In characteristic zero, Hurwitz immediately gives $\mathrm{Aut}(X) = G$. The same holds when $p \geq 5$, $p \neq 7$, because no genus-three curve has an automorphism of order p in those characteristics. Indeed, for a cyclic action of order p , the quotient genus h is at most one. If $h = 1$, the different would have degree four, whereas each ramified point contributes at least $2(p-1) > 4$. If $h = 0$, Hurwitz requires $p-1$ to divide $4+2p$, hence to divide six, forcing $p = 7$ among primes $p \geq 5$. Cauchy's theorem therefore shows that the entire finite automorphism group has order prime to p . It is tame, so its order is at most 168.

Characteristic-two bitangents. It remains to give an upper bound in characteristic two, where a tame-bound argument would be invalid. Consider lines whose intersection divisor with X is twice a degree-two divisor. Such a line has equation $z = \alpha x + \beta y$: no line with zero z -coefficient has this property, as direct substitution into F shows. The coefficients of x^3y and xy^3 in the restriction are respectively

$$1 + \alpha^2\beta, \quad \alpha + \beta^3.$$

The restriction is a square precisely when both vanish, so these seven lines have

$$\alpha = \beta^3, \quad \beta^7 = 1. \quad (3)$$

They are ordinary bitangents: the coefficient of x^2y^2 is $\alpha\beta^2 = \beta^5 \neq 0$, so the quadratic square root has two distinct zeros.

In the dual plane, (3) gives the points $(\beta^3 : \beta : 1) = (\beta^4 : \beta^2 : \beta)$. As β runs over $\mathbb{F}_8^\times$, these form a projective copy of the seven points of $\mathbb{P}^2(\mathbb{F}_2)$. To verify this, the map $t \mapsto (t^4, t^2, t)$ is $\mathbb{F}_2$ -linear, and the images of an $\mathbb{F}_2$ -basis of $\mathbb{F}_8$ are independent over k : a dependence among the rows would give a nonzero polynomial of degree at most four vanishing on all eight elements of $\mathbb{F}_8$. Their projective incidence is therefore the Fano plane. Its collineation group is $\mathrm{GL}_3(\mathbb{F}_2)$, of order $(8-1)(8-2)(8-4) = 168$. For example a collineation is determined by the images of a projective frame, and the Fano incidence forces the remaining points to be the corresponding sums. Every projective automorphism of X permutes these seven bitangents, faithfully because they contain a dual projective frame. Hence $|\mathrm{Aut}(X)| \leq 168$. Together with G , this proves equality also in characteristic two. The value agrees with $84(g-1)$, but here that equality is a computation, not an application of the tame Hurwitz bound.

(c) *General quartics.* Smooth plane quartics form an open subset of $\mathbb{P}^{14}$. We bound the locus with a nonidentity automorphism by dimension twelve. Every such finite group contains an element of prime order ℓ . Hurwitz implies $\ell \leq 7$. For completeness, if $\ell \geq 11$ the quotient genus is at most one. Quotient genus one would require a different of degree four, impossible since every nonzero contribution is at least $\ell-1$. For quotient genus zero, all different exponents are multiples of $\ell-1$, so $\ell-1 \mid 4+2\ell$, hence $\ell-1 \mid 6$, again impossible. This argument includes wild cyclic actions. Only the primes 2, 3, 5, 7 need be considered.

If $\ell \neq \text{char } k$, the projective transformation is diagonalizable. Up to projective conjugacy there are finitely many possibilities, with eigenvalues normalized to ℓ -th roots of unity. With one repeated eigenvalue its conjugacy class has dimension four; with three distinct eigenvalues it has dimension six. A preserved quartic form is in an eigenspace of the substitution action on degree-four forms. The largest possible eigenspace dimensions, and hence bounds for the corresponding families, are

ℓ	eigenvalue type	eigenspace dimension	family dimension
2	repeated	9	$4 + 8 = 12$
3	repeated	7	$4 + 6 = 10$
3	distinct	5	$6 + 4 = 10$
5, 7	repeated	5	$4 + 4 = 8$
5, 7	distinct	≤ 5	$\leq 6 + 4 = 10$

(4)

These counts come from the fifteen monomials $x^i y^j z^k$ with $i + j + k = 4$. For repeated eigenvalues, group by the exponent of the exceptional variable modulo ℓ . For three distinct cube roots each weight has five monomials. For $\ell \geq 5$ and distinct eigenvalues, fixing one exponent leaves at most one monomial of each weight among the at most five possibilities for a second exponent, giving the stated bound.

In characteristic ℓ , an element of projective order ℓ is unipotent. Its nontrivial Jordan type is either $(2, 1)$, with conjugacy-class dimension four, or (3) , with dimension six. The latter has order four in characteristic two and so does not enter the prime-order count there. For type $(2, 1)$, use $x \mapsto x + y$ with y, z fixed. The invariant polynomial ring is

$$k[y, z, x^\ell - xy^{\ell-1}].$$

Its degree-four part has dimension nine, seven, or five as $\ell = 2, 3$, or $5, 7$, respectively, giving the same bounds as the repeated-eigenvalue rows of (4). The ring formula also follows by division by the monic invariant $x^\ell - xy^{\ell-1}$: an invariant remainder of x -degree less than ℓ must be independent of x .

For type (3) , use $(x, y, z) \mapsto (x + y, y + z, z)$. On degree-four forms, the kernel of substitution minus identity has dimension five in characteristic three and three in characteristics five and seven. Here is an explicit coefficient computation of these small ranks. Index rows and columns by triples of nonnegative integers summing to four. In column (a, b, c) , expand

$$(x + y)^a (y + z)^b z^c - x^a y^b z^c = \sum_{i=0}^a \sum_{j=0}^b \binom{a}{i} \binom{b}{j} x^i y^{a-i+j} z^{b-j+c} - x^a y^b z^c.$$

The resulting fifteen-by-fifteen matrices have ranks ten, twelve, twelve modulo three, five, seven. Thus the projective fixed-form loci, with their conjugacy classes, have dimensions at most $6 + 4 = 10$ or $6 + 2 = 8$.

There are only finitely many conjugacy types and eigenspaces in this argument. For each, the family of preserved quartics is the image of the conjugacy class with the projective eigenspace over it; its dimension has just been bounded by twelve. The closures of their images therefore form a proper closed subset of $\mathbb{P}^{14}$. A smooth quartic outside that subset has no prime-order automorphism, and hence no nonidentity automorphism. This property is invariant under projective equivalence. Since the nonhyperelliptic locus is open dense in genus-three moduli, it proves the assertion for a general genus-three curve. $\square$

IV.6 Classification of Curves in Projective 3-Space

Exercise IV.6.1

Permitted context.

Work over an algebraically closed field. Curves are nonsingular, projective, and integral. A plane curve of degree d has genus $(d-1)(d-2)/2$. On $\mathbb{P}^1$, $h^0(\mathcal{O}(m)) = m+1$ for $m \geq 0$. We use Bezout and the complete-intersection genus formula $g = 1 + ab(a+b-4)/2$. A smooth curve on a quadric cone of even degree $2a$ has genus $(a-1)^2$; one of odd degree $2a+1$ has genus $a(a-1)$. These cone formulas are the classification recalled in Hartshorne IV, Remark 6.4.1(d).

Problem. Show that a rational quartic curve in $\mathbb{P}^3$ lies on a unique quadric, which is nonsingular.

Solution. Let $L = \mathcal{O}_X(1)$. Since $X \cong \mathbb{P}^1$ and $\deg L = 4$, $h^0(X, L^2) = 9$, whereas the space of ambient quadratic forms has dimension ten. The restriction map therefore has a nonzero kernel, giving a quadric Q through X . The curve cannot be planar: a nonsingular plane quartic has genus three. Thus no quadric containing it can be a union or double of planes, since integrality would put X in one of those planes.

If two distinct quadrics contained X , their irreducibility would make their intersection a complete-intersection curve of degree four. Since it contains the degree-four integral curve X , it equals X scheme-theoretically. To justify the last assertion, a complete intersection is Cohen–Macaulay with no embedded points, and equality of degrees forces generic multiplicity one and leaves no other component. Its ideal is consequently the radical ideal of X . The complete-intersection genus would be one, a contradiction. Hence Q is unique.

Finally, if Q were a cone, the even-degree cone formula with $2a = 4$ would again give genus one. Thus Q is nonsingular. $\square$

Exercise IV.6.2

Permitted context.

Work over an algebraically closed field, in any characteristic. A rational nonsingular projective curve is $\mathbb{P}^1$, and $h^0(\mathbb{P}^1, \mathcal{O}(m)) = m+1$ for $m \geq 0$. An injective morphism of a smooth projective curve that separates tangent vectors is a closed immersion. There are twenty cubic forms and ten quadratic forms in four variables.

Problem. Show that every rational quintic in $\mathbb{P}^3$ is contained in a cubic surface, and construct one contained in no quadric surface.

Solution. For a rational quintic X , the restriction of cubic forms has target

$$H^0(X, \mathcal{O}_X(3)) \cong H^0(\mathbb{P}^1, \mathcal{O}(15)),$$

of dimension sixteen. Its kernel in the twenty-dimensional space of cubic forms is therefore nonzero. Any nonzero element defines the required cubic surface.

For the second assertion, consider

$$[s : t] \longmapsto [s^5 : s^4t : s^3t^2 + st^4 : t^5]. \quad (1)$$

The coordinates have no common zero. On $s \neq 0$, put $u = t/s$; the map is $[1 : u : u^2 + u^4 : u^5]$, so it separates points and tangent vectors there. The point $s = 0$ has image $[0 : 0 : 0 : 1]$, which is not an image of a finite u . Near it put $q = s/t$. In the chart where the last coordinate is one, the third coordinate is $q^3 + q$, with derivative one at $q = 0$. Thus (1) is a closed immersion, and its image is a smooth rational curve of degree five.

The ten quadratic coordinate products, restricted to $s = 1$, are

$$1, u, u^2 + u^4, u^5, u^2, u^3 + u^5, u^6, u^4 + 2u^6 + u^8, u^7 + u^9, u^{10}. \quad (2)$$

They are linearly independent over every field. In a relation, successively compare coefficients of u^{10}, u^9, u^8, u^6 to remove the last, next-to-last, eighth, and seventh terms. Then the coefficient of u^4 removes the third term, that of u^3 the sixth, and that of u^5 the fourth. The remaining $1, u, u^2$ are independent. This reasoning does not divide by two. Therefore no nonzero quadratic form vanishes on the image of (1), as required. $\square$

Exercise IV.6.3

Permitted context.

Work over an algebraically closed field. On a genus-two curve X , the canonical pencil $|K_X|$ is the unique degree-two pencil, and $\deg K_X = 2$. Every line bundle of degree at least five is very ample, and Riemann–Roch applies. The variety $\text{Pic}^5(X)$ is irreducible of dimension two; the image of a morphism from X has dimension at most one. We use Bezout and the quadric-cone resolution as well as the plane genus formula. $\mathbb{F}_2 \rightarrow Q$ denotes the resolution, with $E^2 = -2$, $EF = 1$, $F^2 = 0$ and hyperplane class $E + 2F$. A smooth degree-five curve on the cone has strict transform $2E + 5F$, meeting E transversely once; the restriction of F is its degree-two pencil.

Problem. Prove that a degree-five genus-two curve in $\mathbb{P}^3$ lies on a unique quadric. For every abstract genus-two curve, construct degree-five embeddings whose quadric is smooth, and others whose quadric is singular.

Solution. For an embedded curve, put $L = \mathcal{O}_X(1)$. The curve is not planar, since a smooth plane quintic has genus six. Riemann–Roch gives $h^0(L^2) = 10 + 1 - 2 = 9$; hence some quadric contains X . It is irreducible, since otherwise X would lie in a plane. Two distinct such quadrics would intersect in a degree-four curve and could not contain the degree-five curve X , by Bezout. This proves uniqueness.

Now start with an arbitrary genus-two curve. Every degree-five line bundle L is very ample, and $h^0(L) = 5 + 1 - 2 = 4$. Thus its complete system embeds X with the required degree into $\mathbb{P}^3$. We claim that its unique quadric is a cone exactly when

$$L \cong \omega_X^{\otimes 2}(P) \quad \text{for some } P \in X. \quad (1)$$

For this choice of L , let u, v be a basis of $H^0(X, \omega_X)$. The three independent sections u^2, uv, v^2 form a basis of $H^0(X, \omega_X^{\otimes 2})$; their independence follows by pulling back independent quadratics on the canonical $\mathbb{P}^1$. Include them in $H^0(X, L)$ using the canonical section of $\mathcal{O}_X(P)$, and extend to a basis with one further section. The resulting image satisfies $x_0x_2 - x_1^2 = 0$, an irreducible quadric cone (including in characteristic two). This proves one direction.

Conversely, for a curve on a cone, use its strict transform on $\mathbb{F}_2$. The section E cuts out one point P , and the ruling cuts out the canonical degree-two pencil. Restricting the hyperplane class $E + 2F$ therefore gives $L \cong \mathcal{O}_X(P + 2K_X)$, proving (1).

The bundles in (1) form the image of the proper curve $X \rightarrow \text{Pic}^5(X)$, $P \mapsto 2K_X + P$. This is a closed subset of dimension at most one, so it does not fill the two-dimensional Picard variety. Choose L outside it: its unique irreducible quadric cannot be a cone, and hence is smooth. Choosing any P in (1) gives the singular-quadric embedding. Both kinds therefore exist on every X . $\square$

Exercise IV.6.4

Permitted context.

Work over an algebraically closed field. We use Riemann–Roch, the plane genus formula, and Clifford’s theorem. A degree-two divisor with two sections on a curve of positive genus is its hyperelliptic pencil A ; then $K_X \sim (g-1)A$ and $h^0(nA) = n+1$ for $0 \leq n \leq g$. A curve of type (a, b) on a smooth quadric has degree $a+b$ and genus $(a-1)(b-1)$. A smooth curve of degree $2a+1$ on a quadric cone has genus $a(a-1)$.

Problem. Prove that there is no nonsingular integral curve of degree nine and genus eleven in $\mathbb{P}^3$.

Solution. Suppose X were such a curve and let D be its hyperplane divisor. It is not planar, since a smooth plane curve of degree nine has genus twenty-eight. Riemann–Roch gives

$$h^0(2D) = 18 + 1 - 11 + h^0(K_X - 2D) = 8 + h^0(K_X - 2D). \quad (1)$$

The residual divisor has degree two. We claim it has at most one section. Otherwise it is a hyperelliptic pencil A , and $K_X \sim 10A$. Then $2D \sim 9A$, whose complete linear system is pulled back from $|\mathcal{O}_{\mathbb{P}^1}(9)|$: pullback supplies ten sections, and $h^0(9A) = 10$. It cannot separate the degree-two fibers. But D is very ample, so $2D$ is very ample as well, a contradiction. This proves the claim.

By (1), $h^0(2D) \leq 9$, whereas ambient quadratic forms have dimension ten. Thus X lies on a quadric Q . The quadric is irreducible, because X is nonplanar. If Q is smooth, writing its curve class as (a, b) gives

$$a + b = 9, \quad ab - a - b + 1 = 11,$$

so $ab = 19$. The roots of $t^2 - 9t + 19$ are not integers, since its discriminant is five. If Q is a cone, degree nine means $2a+1 = 9$, so the cone genus formula gives genus $4 \cdot 3 = 12$, again a contradiction. These exhaust the possibilities. $\square$

Exercise IV.6.5**Permitted context.**

Work over an algebraically closed field. For a scheme-theoretic complete intersection of two hypersurfaces of degrees a, b in $\mathbb{P}^3$, the Koszul resolution of its ideal sheaf is

$$0 \rightarrow \mathcal{O}(-a-b) \rightarrow \mathcal{O}(-a) \oplus \mathcal{O}(-b) \rightarrow \mathcal{I}_X \rightarrow 0.$$

On $\mathbb{P}^3$, $H^1(\mathcal{O}(t)) = 0$ for all integers t , and $H^0(\mathcal{O}(t)) = 0$ when $t < 0$.

Problem. If $X \subset \mathbb{P}^3$ is a complete intersection of surfaces of degrees a, b , prove that it lies on no surface of degree less than $\min(a, b)$.

Solution. A surface of degree m containing X scheme-theoretically corresponds to a nonzero element of $H^0(\mathbb{P}^3, \mathcal{I}_X(m))$. Twist the Koszul resolution by $\mathcal{O}(m)$. Its cohomology gives

$$H^0(\mathcal{O}(m-a)) \oplus H^0(\mathcal{O}(m-b)) \rightarrow H^0(\mathcal{I}_X(m)) \rightarrow H^1(\mathcal{O}(m-a-b)).$$

If $m < \min(a, b)$, the first group is zero by negativity and the last is zero by projective-space cohomology. Exactness forces $H^0(\mathcal{I}_X(m)) = 0$, which proves the assertion. For the nonsingular curves of the section, set-theoretic containment also gives scheme-theoretic containment, since their defining ideals are radical. $\square$

Exercise IV.6.6**Permitted context.**

Work over an algebraically closed field. For a smooth projectively normal curve in projective space, the restriction maps $H^0(\mathbb{P}^3, \mathcal{O}(n)) \rightarrow H^0(X, \mathcal{O}_X(n))$ are surjective for all $n \geq 0$. A nonplanar degree- d space curve satisfies Castelnuovo's bound $g \leq \lfloor (d-2)^2/4 \rfloor$. We use Riemann–Roch and the dimensions four and ten of the spaces of linear and quadratic forms on $\mathbb{P}^3$.

Problem. Let $X \subset \mathbb{P}^3$ be a nonsingular projectively normal curve not contained in a plane. Prove that degree six implies $g = 3$ or 4 , and degree seven implies $g = 5$ or 6 .

Solution. Write D for the hyperplane divisor. Nonplanarity makes restriction of linear forms injective, and projective normality makes it surjective. Hence $h^0(D) = 4$. Riemann–Roch yields

$$4 = d + 1 - g + h^1(D), \quad g = d - 3 + h^1(D) \geq d - 3.$$

For $d = 6$, this gives $g \geq 3$, while Castelnuovo's bound gives $g \leq 4$, proving the first assertion.

For $d = 7$, the same arguments give $4 \leq g \leq 6$. If $g = 4$, then Riemann–Roch gives

$$h^0(2D) = 14 + 1 - 4 + h^1(2D) \geq 11.$$

But projective normality requires the ten-dimensional space of quadrics to surject onto this space, which is impossible. Thus $g = 5$ or 6 . $\square$

Exercise IV.6.7**Permitted context.**

Work over an algebraically closed field. A multisection of a nonsingular integral curve $X \subset \mathbb{P}^3$ means a line not contained in X whose scheme-theoretic intersection with X has length at least three. Thus tangential intersections count with multiplicity. A morphism from a smooth projective curve is an embedding if its defining system separates every length-two subscheme. We use the plane genus formula and Castelnuovo's bound $g \leq \lfloor (d-2)^2/4 \rfloor$ for nonplanar curves. A nonplanar smooth degree-three curve is a twisted cubic, whose ideal is generated by quadrics. A smooth genus-one degree-four space curve is a complete intersection of two quadrics.

Problem. Prove that a line, conic, twisted cubic, and elliptic quartic have no multisections, whereas every other nonsingular integral curve in $\mathbb{P}^3$ has infinitely many multisections.

Solution. For a line the assertion follows from intersection of two distinct lines. For a conic, a line outside its plane meets it in length at most one, and a line in the plane meets it in length at most two. For the twisted cubic and elliptic quartic, any line meeting the curve in length at least three would be contained in every defining quadric: a polynomial of degree two on the line cannot have three zeros counted with multiplicity unless it is zero. Since their ideals are generated by quadrics, such a line would be contained in the curve, which is impossible.

A plane curve of degree at least three has infinitely many multisections, namely its general plane lines. It remains to treat nonplanar curves. Suppose there were only finitely many multisections. Choose $P \in X$ on none of them, which is possible because each has finite intersection with X . Projection from P , after removing its fixed point, is defined everywhere on X by the three-dimensional space of hyperplanes through P , a subsystem of $|D - P|$, where D is the hyperplane divisor. It is base-point-free since the original embedding separates length-two subschemes.

This projection is an embedding into $\mathbb{P}^2$. Indeed, failure to separate a length-two subscheme $Z \subset X$ would mean that the length-three divisor $P + Z$ fails to impose three independent conditions on ambient linear forms. Its projective span would then be a line, which is a multisection through P . This also covers $P \in Z$ and nonreduced Z : the divisors $2P + Q$ and $3P$ record tangent and osculating failures. Such a multisection was excluded by the choice of P .

The image is thus a smooth plane curve of degree $d - 1$, so

$$g = \frac{(d-2)(d-3)}{2} \leq \left\lfloor \frac{(d-2)^2}{4} \right\rfloor.$$

For $d \geq 5$, even the inequality without the floor would give $2(d-3) \leq d-2$, or $d \leq 4$, a contradiction. Thus $d = 3$ or 4 , since a nonplanar curve has degree at least three. The displayed genus formula gives respectively $g = 0$ or 1 . These are precisely the twisted cubic and elliptic quartic. Therefore any curve outside the four listed classes has infinitely many multisections. $\square$

Exercise IV.6.8**Permitted context.**

Work over an algebraically closed field. A divisor D is nonspecial if $H^1(X, \mathcal{O}_X(D)) = 0$. Base-point-free means that its global sections generate the line bundle, so an empty linear system does not qualify. We use Riemann–Roch, Serre duality, and the irreducibility and dimension g of $\text{Pic}^d(X)$. Effective divisors of degree $e \geq 0$ are parametrized by the proper e -dimensional variety $X^{(e)}$; there are none of negative degree. On $\mathbb{P}^1$, $\mathcal{O}(d)$ is generated by global sections exactly when $d \geq 0$.

Problem. For $g \geq 1$, prove that a genus- g curve admits a nonspecial base-point-free divisor of degree d exactly when $d \geq g + 1$. For $g = 0$, identify the additional case omitted by the literal statement in the source.

Solution. Suppose $L = \mathcal{O}_X(D)$ is nonspecial and generated by its sections. Then Riemann–Roch gives $h^0(L) = d + 1 - g \geq 1$. If $h^0(L) = 1$, its sole section must vanish nowhere, so $L \cong \mathcal{O}_X$. Nonspeciality would then say $g = h^1(\mathcal{O}_X) = 0$, and necessarily $d = 0$. Apart from this case, $h^0(L) \geq 2$, so $d \geq g + 1$. This proves necessity and identifies the possible exception.

Now let $g \geq 1$ and $d \geq g + 1$. The special bundles in $\text{Pic}^d(X)$ have the form

$$L \cong \omega_X(-E), \quad E \text{ effective of degree } 2g - 2 - d.$$

When this degree is nonnegative their locus is a proper closed image of dimension at most $2g - 2 - d \leq g - 3 < g$; otherwise it is empty.

For a nonspecial L , the exact sequence at a point P shows that P is a base point exactly when $H^1(L(-P)) \neq 0$. In detail,

$$H^0(L) \rightarrow L|_P \rightarrow H^1(L(-P)) \rightarrow H^1(L) = 0;$$

the first map is zero or surjective because its target has dimension one. By Serre duality, this exceptional condition gives

$$L \cong \omega_X(P - E), \quad E \text{ effective of degree } 2g - 1 - d.$$

Allowing P and E to vary gives a closed image of dimension at most $1 + (2g - 1 - d) \leq g - 1 < g$, when the degree is nonnegative, and an empty locus otherwise. Irreducibility of the g -dimensional Picard variety leaves a bundle outside both proper closed loci. It is nonspecial and has no base points. A line bundle on a smooth curve is represented by a divisor, so it supplies the requested D .

Finally, when $g = 0$, the curve is $\mathbb{P}^1$. The bundles $\mathcal{O}(d)$ for all $d \geq 0$ are generated and have $H^1 = 0$, whereas no negative degree bundle is generated. Thus the correct condition is $d \geq 0$. The source's inequality $d \geq g + 1$ omits precisely the valid case $g = 0, d = 0$. $\square$

Exercise IV.6.9**Permitted context.**

Let $X \subset \mathbb{P}_k^3$ be a nonsingular integral curve over an algebraically closed field. Its ideal sheaf $\mathcal{I}_X(t)$ is globally generated for all sufficiently large t , by Serre's theorem. The conormal bundle $\mathcal{I}_X/\mathcal{I}_X^2$ has rank two. A hypersurface through a point of a smooth ambient variety is singular there exactly when its equation has zero first-order jet. We use the dimension-of-fibers theorem and the facts $H^1(\mathbb{P}^3, \mathcal{O}(t)) = 0$ for all t and $H^0(\mathbb{P}^3, \mathcal{O}) = k$.

Problem. Prove that for every sufficiently large integer m there is a nonsingular surface of degree m containing X .

Solution. Choose m so large that both $\mathcal{I}_X(m-1)$ and $\mathcal{I}_X(m)$ are globally generated, and put $W = H^0(\mathbb{P}^3, \mathcal{I}_X(m))$. We exhibit two proper bad loci in the affine space W .

For $P \in X$, taking the first normal derivative gives a surjection

$$W \longrightarrow (\mathcal{I}_X/\mathcal{I}_X^2)(m)|_P,$$

whose target has dimension two. Surjectivity follows from global generation and Nakayama's lemma: the target is $\mathcal{I}_{X,P}/\mathfrak{m}_P\mathcal{I}_{X,P}$, with its twist. A section vanishes on X , so its tangential derivative along X already vanishes. Its hypersurface is singular at P exactly when its normal derivative is also zero. The incidence of such pairs $(s, P) \in W \times X$ therefore has dimension

$$(\dim W - 2) + 1 = \dim W - 1.$$

Since X is proper, its image in W is closed and proper.

For $P \notin X$, global generation of $\mathcal{I}_X(m-1)$ supplies a section h with $h(P) \neq 0$. The products $h\ell$, for ambient linear forms ℓ , belong to W . Linear forms give all first-order jets at P , and multiplication by the unit h is invertible on the local first-jet space. Hence

$$W \longrightarrow \mathcal{O}_{\mathbb{P}^3}(m) \otimes (\mathcal{O}_{\mathbb{P}^3,P}/\mathfrak{m}_P^2)$$

is surjective; its target has dimension four. The incidence of a section with zero first-order jet at some $P \in \mathbb{P}^3 \setminus X$ has dimension

$$(\dim W - 4) + 3 = \dim W - 1.$$

The closure of its image in W is again proper.

Choose s outside the union of these two proper closed subsets. Its hypersurface F is smooth at every point, contains X , and has degree m . The choice is nonzero because the zero section is in both bad loci. For completeness F is integral: the hypersurface sequence and $H^1(\mathcal{O}(-m)) = 0$ give $H^0(F, \mathcal{O}_F) = k$, so it is connected. Distinct components of a smooth scheme cannot intersect, and smoothness also implies reducedness. Thus this connected smooth hypersurface has just one integral component. The construction works for every m beyond the fixed global-generation threshold. $\square$

Chapter V

V.1 Geometry on a Surface

Exercise V.1.1

Permitted context.

Let X be a nonsingular projective integral surface over an algebraically closed field, with canonical divisor K . For a divisor A , surface Riemann–Roch says

$$\chi(\mathcal{O}_X(A)) = \chi(\mathcal{O}_X) + \frac{A(A - K)}{2}.$$

Intersection is symmetric, bilinear, and invariant under linear equivalence. The Euler characteristic is the alternating sum of coherent-cohomology dimensions.

Problem. For divisors C, D , put $L = \mathcal{O}_X(C)$ and $M = \mathcal{O}_X(D)$. Prove

$$CD = \chi(\mathcal{O}_X) - \chi(L^{-1}) - \chi(M^{-1}) + \chi(L^{-1} \otimes M^{-1}).$$

Solution. Apply Riemann–Roch to $-C, -D, -C - D$. The four copies of $\chi(\mathcal{O}_X)$ on the right side cancel, leaving

$$\frac{(C + D)(C + D + K) - C(C + K) - D(D + K)}{2}.$$

By bilinearity the numerator is $2CD$, proving the formula. The calculation concerns integers and rational numbers in the Riemann–Roch formula, so it is valid also when the ground field has characteristic two. $\square$

Exercise V.1.2

Permitted context.

Let X be a nonsingular projective integral surface over an algebraically closed field, and H a very ample divisor defining an embedding into $\mathbb{P}^N$. We use $\chi(\mathcal{O}_X(A)) = \chi(\mathcal{O}_X) + A(A - K_X)/2$, adjunction $2g(H) - 2 = H(H + K_X)$ for a smooth member, and Bertini's existence of such a member. The Hilbert polynomial equals $\chi(\mathcal{O}_X(nH))$. For a dimension- r projective scheme its leading coefficient is its degree divided by $r!$. The arithmetic genus of a surface is $p_a(X) = \chi(\mathcal{O}_X) - 1$. Here a curve means an integral one-dimensional closed subscheme; on a nonsingular surface it is an effective Cartier divisor.

Problem. Write the Hilbert polynomial as $F(z) = az^2/2 + bz + c$, and let π be the genus of a smooth curve in $|H|$. Show $a = H^2$, $b = H^2/2 + 1 - \pi$, and $c = 1 + p_a(X)$. Deduce $\deg X = H^2$ and $\deg C = CH$ for every curve $C \subset X$.

Solution. Riemann–Roch gives the polynomial identity

$$F(z) = \frac{H^2}{2}z^2 - \frac{HK_X}{2}z + \chi(\mathcal{O}_X).$$

Thus $a = H^2$ and $c = 1 + p_a(X)$. Adjunction gives $HK_X = 2\pi - 2 - H^2$, so $b = -HK_X/2 = H^2/2 + 1 - \pi$. By the leading-coefficient definition of degree, $\deg X = 2!(H^2/2) = H^2$.

Every curve on a nonsingular surface is a Cartier divisor. Its exact sequence, twisted by nH , gives

$$\chi(\mathcal{O}_C(nH)) = \chi(\mathcal{O}_X(nH)) - \chi(\mathcal{O}_X(nH - C)).$$

Substitution in Riemann–Roch makes the right side

$$nHC - \frac{C^2 + CK_X}{2}.$$

This is the Hilbert polynomial of C , whose linear coefficient is its projective degree. Hence $\deg C = HC$, without any smoothness assumption on C . $\square$

Exercise V.1.3

Permitted context.

Let X be a nonsingular projective integral surface over an algebraically closed field, with canonical divisor K . For a nonzero effective divisor D , $p_a(D) = 1 - \chi(\mathcal{O}_D)$, whether or not D is reduced or connected. We use the exact sequence $0 \rightarrow \mathcal{O}_X(-D) \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_D \rightarrow 0$ and surface Riemann–Roch. Intersection depends only on linear equivalence. For the empty divisor, the convention $1 - \chi(\mathcal{O}_\emptyset) = 1$ will agree with the virtual genus below.

Problem. (a) Prove $2p_a(D) - 2 = D(D + K)$ for an effective divisor. (b) Deduce invariance under linear equivalence. (c) Define the virtual arithmetic genus of any divisor by this same formula and prove

$$p_a(-D) = D^2 - p_a(D) + 2, \quad p_a(C + D) = p_a(C) + p_a(D) + CD - 1.$$

Solution. (a) Additivity of Euler characteristic gives

$$\chi(\mathcal{O}_D) = \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-D)) = -\frac{D(D + K)}{2}.$$

Since $p_a(D) = 1 - \chi(\mathcal{O}_D)$, this is exactly the desired formula. It also yields $p_a(0) = 1$ under the stated empty-divisor convention.

(b) Linearly equivalent divisors have the same self-intersection and the same intersection with K . The formula from (a) therefore gives the same arithmetic genus for linearly equivalent effective divisors.

(c) Set $p_a(A) = 1 + (A^2 + AK)/2$ for every divisor A . This is an integer: Riemann–Roch applied to $-A$ shows that $A(A + K)$ is even. Then

$$p_a(-D) + p_a(D) = 2 + D^2,$$

which is the first identity. Similarly,

$$\begin{aligned} p_a(C + D) &= 1 + \frac{C^2 + D^2 + 2CD + CK + DK}{2} \\ &= p_a(C) + p_a(D) + CD - 1. \end{aligned}$$

Thus the virtual extension agrees with the ordinary genus on effective divisors and has both stated identities. $\square$

Exercise V.1.4**Permitted context.**

Work over an algebraically closed field. For a nonsingular degree- d surface $X \subset \mathbb{P}^3$, $K_X = (d-4)H$, where H is the hyperplane class. Adjunction on a nonsingular curve $C \subset X$ gives $2g(C) - 2 = C(C + K_X)$. A line has genus zero and intersects H in degree one. We use the projective hypersurface Jacobian criterion.

Problem. (a) If a nonsingular degree- d surface contains a line C , prove $C^2 = 2 - d$. (b) In characteristic zero, construct such a surface for every $d \geq 1$, containing the line $x = y = 0$.

Solution. (a) Adjunction gives

$$-2 = C^2 + (d-4)CH = C^2 + d - 4,$$

so $C^2 = 2 - d$.

(b) For $d = 1$, take the plane $x = 0$. For $d \geq 2$, use homogeneous coordinates $[x : y : z : w]$ and take

$$F = xz^{d-1} + yw^{d-1} + x^d + y^d = 0. \quad (1)$$

It contains $x = y = 0$. When $d = 2$, the four partial derivatives are $z+2x$, $w+2y$, x , y ; they cannot vanish at a projective point. When $d \geq 3$, vanishing of $F_z = (d-1)xz^{d-2}$ forces $x = 0$ or $z = 0$. In either case vanishing of $F_x = z^{d-1} + dx^{d-1}$ forces both $x = z = 0$. The equations for y, w likewise force $y = w = 0$. Characteristic zero ensures that d and $d-1$ are nonzero. Thus the gradient never vanishes at a projective point, and (1) defines a nonsingular surface. A reducible hypersurface in $\mathbb{P}^3$ would have intersecting components and would be singular there. Thus this surface is integral as well. $\square$

Exercise V.1.5**Permitted context.**

Work over an algebraically closed field. For a nonsingular degree- d surface in $\mathbb{P}^3$, adjunction gives $K_X = (d-4)H$, and $H^2 = d$. For smooth curves C, C' , the cotangent bundle of their product is $p_1^*\Omega_C \oplus p_2^*\Omega_{C'}$. A curve's canonical degree is $2g-2$. Fibers $l = C \times \{P'\}$, $m = \{P\} \times C'$ satisfy $l^2 = m^2 = 0$, $lm = 1$. Pullbacks of divisors from either curve have intersections determined by their degrees.

Problem. (a) For a nonsingular surface of degree d in $\mathbb{P}^3$, show $K_X^2 = d(d-4)^2$. (b) For $X = C \times C'$, with genera g, g' , show $K_X^2 = 8(g-1)(g'-1)$.

Solution. (a) Bilinearity and adjunction give

$$K_X^2 = ((d-4)H)^2 = (d-4)^2 H^2 = d(d-4)^2.$$

(b) Taking determinants of the product cotangent bundle gives

$$\omega_X = p_1^*\omega_C \otimes p_2^*\omega_{C'}.$$

Thus the canonical divisor is the sum of the two canonical pullbacks. Each pullback has self-intersection zero, since it is numerically a multiple of a fiber. Their mutual intersection is the product of their degrees, $(2g-2)(2g'-2)$. Therefore

$$K_X^2 = 2(2g-2)(2g'-2) = 8(g-1)(g'-1).$$

The same calculation includes genus zero and genus one. $\square$

Exercise V.1.6

Permitted context.

Let C be a nonsingular projective integral curve of genus g over an algebraically closed field. For the diagonal $\Delta \subset C \times C$, its conormal sheaf is $\mathcal{I}_\Delta/\mathcal{I}_\Delta^2 \cong \Omega_C$. The self-intersection of a smooth divisor is the degree of its normal bundle. Numerical equivalence means equality of intersections with all curves; $\text{Num}(X)$ is the group of divisor classes modulo this relation. For the fibers $l = C \times \{P\}$, $m = \{P\} \times C$, $l^2 = m^2 = 0$ and $lm = 1$. A pullback of a degree- d line bundle from one factor is numerically d times the corresponding fiber.

Problem. (a) Prove $\Delta^2 = 2 - 2g$. (b) For $g \geq 1$, prove that l, m, Δ are linearly independent in $\text{Num}(C \times C)$. Deduce that the numerical group has rank at least three and that pullbacks from the two factors do not generate $\text{Pic}(C \times C)$.

Solution. (a) Dualizing the conormal identification gives $N_{\Delta/(C \times C)} \cong T_C$. Its degree is $-\deg \Omega_C = 2 - 2g$, so this is Δ^2 .

(b) Each fiber meets the diagonal once, transversely. The intersection matrix on l, m, Δ is therefore

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 2-2g \end{pmatrix},$$

with determinant $2g$. For $g \geq 1$, this integer is nonzero. If an integral linear combination of these classes were numerically zero, pairing it with each of the three classes would give a homogeneous linear system with this invertible matrix over $\mathbb{Q}$. All coefficients must vanish. This proves independence, regardless of the characteristic of the ground field: the intersection matrix has integer entries, not entries reduced modulo that characteristic.

If every line bundle on $C \times C$ were a tensor product of pullbacks, $\mathcal{O}(\Delta)$ would have the form $p_1^*L \otimes p_2^*M$. Its numerical class would then lie in the span of m and l , contradicting the independence just proved. Hence these pullbacks do not generate the Picard group. $\square$

Exercise V.1.7**Permitted context.**

Let X be a nonsingular projective integral surface over an algebraically closed field. Two effective divisors are prealgebraically equivalent if they occur as fibers of an effective Cartier divisor on $X \times T$, flat over a nonsingular connected curve T . For arbitrary divisors, extend this by differences of such pairs; algebraic equivalence is the equivalence generated by finite chains of these pairs. A relative effective Cartier divisor on the flat family $X \times T$ is flat when each fiber is Cartier. Hilbert polynomials are constant in a projective flat family over connected T . For very ample H , the linear coefficient of a divisor's Hilbert polynomial is its intersection with H . Every divisor becomes very ample after adding a sufficiently large multiple of a fixed very ample divisor.

Problem. (a) Prove that the divisors algebraically equivalent to zero form a subgroup of $\text{Div}(X)$. (b) Show that linear equivalence implies algebraic equivalence. (c) Show that algebraic equivalence implies numerical equivalence.

Solution. (a) Constant families show that every effective divisor is prealgebraically equivalent to itself. Adding a fixed effective divisor to a family gives a new flat family: its fiber equation is the product of the old Cartier equation and that of the fixed divisor, again a nonzerodivisor. Write any fixed divisor as $A - B$ with A, B effective. By adding the corresponding constant families to the positive and negative pairs, translation by $A - B$ preserves prealgebraic equivalence. Interchanging positive and negative pairs shows compatibility with negation. These assertions also hold for finite chains.

In particular, zero is algebraically equivalent to itself. If D and E are each equivalent to zero, translate a chain from E to zero by D , then append a chain from D to zero. This connects $D + E$ to zero. Negating a chain connects $-D$ to zero. Hence the zero class is a subgroup.

(b) Suppose $D \sim D'$. Add a sufficiently positive fixed divisor A so that $D + A, D' + A$ are effective. They are zero divisors of sections s_0, s_1 of the same line bundle L . If the sections are proportional, the effective divisors coincide. Otherwise the pencil

$$us_0 + vs_1, \quad [u : v] \in \mathbb{P}^1,$$

defines an effective Cartier divisor on $X \times \mathbb{P}^1$. No linear combination at a parameter point is the zero section, so each fiber is an effective Cartier divisor on X ; the total divisor is flat over $\mathbb{P}^1$. Its fibers at $[1 : 0]$ and $[0 : 1]$ are $D + A$ and $D' + A$. Thus these are prealgebraically equivalent, and subtracting the constant divisor A proves D algebraically equivalent to D' .

(c) Fix a very ample divisor H . In a flat family of effective divisors, constancy of the Hilbert polynomial gives constancy of its linear coefficient, hence of intersection with H . Taking differences and chains shows

$$DH = D'H$$

whenever D, D' are algebraically equivalent. This holds for every very ample H .

Put $B = D - D'$ and let C be any curve on X . Choose n so large that both $C + nH$ and nH are very ample. Then

$$BC = B(C + nH) - B(nH) = 0 - 0 = 0.$$

Thus B intersects every curve in zero, which is exactly numerical equivalence of D and D' . □

Exercise V.1.8**Permitted context.**

Let X be a nonsingular projective integral surface over an algebraically closed field k . Applying H^1 to $d\log : \mathcal{O}_X^\times \rightarrow \Omega_X^1$ defines $c : \text{Pic}(X) \rightarrow H^1(X, \Omega_X^1)$. It is additive and compatible with pullback. The pairing on $H^1(X, \Omega_X^1)$ is cup product, followed by wedge of forms and the Serre trace on $H^2(X, \omega_X)$.

We use the divisor-class identity from Hartshorne III.7.4: if $i : Y \hookrightarrow X$ is a smooth divisor, then

$$\text{tr}_X(c(Y) \smile \alpha) = \text{tr}_Y(i^* \alpha).$$

On a smooth projective curve, $\text{tr}(c(\mathcal{O}(P))) = 1$. Every divisor is linearly equivalent to a difference of smooth curves, by twisting to very ample systems and applying Bertini. Intersection with a smooth curve is the degree of the restricted line bundle.

Problem. (a) Prove $(c(D), c(E)) = (DE) \cdot 1$ in k . (b) In characteristic zero, deduce that $\text{Num}(X)$ is a finitely generated free abelian group from finite dimensionality of $H^1(X, \Omega_X^1)$.

Solution. (a) First suppose D is a smooth curve. The trace identity in the context and functoriality of $d\log$ give

$$(c(D), c(E)) = \text{tr}_D(c(\mathcal{O}_X(E)|_D)).$$

Write the restricted line bundle as the line bundle of a divisor $\sum n_i P_i$ on D . Additivity of c , and the trace-one normalization for a closed point, make the right side

$$\sum n_i \cdot 1 = \deg_D(\mathcal{O}_X(E)|_D) \cdot 1 = (DE) \cdot 1.$$

For arbitrary D , express it up to linear equivalence as $D_1 - D_2$ with both D_i smooth and apply bilinearity. No smoothness condition on E is needed. This proves the formula in every characteristic, where the integer on the right is mapped into k .

(b) Let $N = \text{Num}(X)$ and $V = N \otimes_{\mathbb{Z}} \mathbb{Q}$. The group N is torsion-free: if nD is numerically zero with $n \neq 0$, its integer intersections show that D is numerically zero. The intersection pairing on V has zero radical by the definition of numerical equivalence.

Put $r = \dim_k H^1(X, \Omega_X^1)$. We first prove $\dim_{\mathbb{Q}} V \leq r$, without assuming any finite generation. Take n independent divisor classes $D_1, \dots, D_n$ in V . The functions

$$E \mapsto D_i E$$

are independent rational linear functionals: a dependence would put a nonzero combination of the D_i in the radical. Hence there exist divisor classes $E_1, \dots, E_n$ such that the matrix $(D_i E_j)$ is invertible over $\mathbb{Q}$. Denominators can be cleared so all chosen classes are represented by integral divisors.

In characteristic zero, its nonzero integer determinant remains nonzero in k . By (a) it is the pairing matrix of $c(D_1), \dots, c(D_n)$ with $c(E_1), \dots, c(E_n)$. Thus the $c(D_i)$ are independent over k , and $n \leq r$. This proves the finite-rank assertion.

Now choose integral divisors $A_1, \dots, A_s$ whose numerical classes form a rational basis of V . The homomorphism

$$N \longrightarrow \mathbb{Z}^s, \quad [D] \mapsto (DA_1, \dots, DA_s)$$

is injective: vanishing of these pairings forces vanishing against all of V , hence against every curve. A subgroup of the finitely generated free abelian group $\mathbb{Z}^s$ is itself finitely generated and free. This proves the assertion. The integral-pairing step is essential; finite rational rank alone would not imply finite generation. $\square$

Exercise V.1.9**Permitted context.**

Let X be a nonsingular projective integral surface over an algebraically closed field. An ample divisor H has $H^2 > 0$. The Hodge index theorem says that $AH = 0$ implies $A^2 \leq 0$ for a rational divisor class A . Numerical equivalence means equality of intersections with all divisors. On $C \times C'$, the fibers $l = C \times \{P'\}$, $m = \{P\} \times C'$ satisfy $l^2 = m^2 = 0$, $lm = 1$. Positive-degree line bundles on curves are ample, and the exterior tensor product of ample line bundles is ample, as follows by taking very ample powers and using the Segre embedding.

Problem. (a) Prove $D^2H^2 \leq (DH)^2$ for ample H . (b) On $X = C \times C'$, say D has type (a, b) if $Dl = a$, $Dm = b$. Show $D^2 \leq 2ab$, with equality exactly when D is numerically equivalent to $bl + am$.

Solution. (a) Put $A = D - (DH/H^2)H$. Then $AH = 0$, so the Hodge index theorem gives

$$0 \geq A^2 = D^2 - \frac{(DH)^2}{H^2}.$$

Multiplying by $H^2 > 0$ proves the inequality.

We also record its equality consequence. If $AH = 0$ and $A^2 = 0$, take any rational divisor class B orthogonal to H . For every rational t , Hodge index gives

$$(A + tB)^2 = 2tAB + t^2B^2 \leq 0.$$

For both signs of sufficiently small nonzero t , this is possible only if $AB = 0$. An arbitrary divisor is the sum of a rational multiple of H and a class orthogonal to H . Thus A intersects every divisor in zero; it is numerically trivial. Conversely, a numerically trivial A has $A^2 = 0$.

(b) The divisor $H = l + m$ is ample by the exterior product criterion in the context. Set

$$R = D - bl - am.$$

Using the definitions of a, b gives $Rl = Rm = 0$, hence $RH = 0$. Its square is

$$R^2 = D^2 - 2ab.$$

Hodge index therefore gives $D^2 \leq 2ab$. The equality consequence proved in (a) says that $R^2 = 0$ exactly when R is numerically trivial, which is exactly $D \equiv bl + am$. $\square$

Exercise V.1.10**Permitted context.**

Let C_0 be a smooth projective geometrically integral curve of genus g over $\mathbb{F}_q$, and put $C = C_0 \times_{\mathbb{F}_q} \overline{\mathbb{F}}_q$. Base change of the q -power Frobenius of C_0 defines an $\overline{\mathbb{F}}_q$ -morphism $f : C \rightarrow C$ of degree q , with zero differential and fixed points precisely $C_0(\mathbb{F}_q)$. For a smooth divisor its square is the degree of its normal bundle. A transverse intersection contributes one. The diagonal in $C \times C$ has square $2 - 2g$. For $l = C \times \{P\}$, $m = \{P\} \times C$, the product form of Hodge index says

$$D^2 \leq 2(Dl)(Dm)$$

for every divisor D . We use $\deg T_C = 2 - 2g$.

Problem. Let $N = \#C_0(\mathbb{F}_q)$, Γ be the graph of f , and Δ the diagonal. Show $\Gamma^2 = q(2 - 2g)$, $\Gamma\Delta = N$, and derive

$$N = 1 - a + q, \quad |a| \leq 2g\sqrt{q}.$$

Solution. Identify Γ with C by its first projection. The tangent sequence of its graph identifies its normal bundle as the quotient of $T_C \oplus f^*T_C$ by vectors $(v, df(v))$. The map $(v, w) \mapsto w - df(v)$ identifies that quotient with f^*T_C . Therefore

$$\Gamma^2 = \deg f^*T_C = q(2 - 2g).$$

The graph and diagonal intersect at exactly the fixed points of f , namely the $\mathbb{F}_q$ -points. At such a point their tangent lines are $\{(v, 0)\}$ and $\{(v, v)\}$, since $df = 0$. They are distinct and span the surface tangent space, so every intersection is transverse. Hence $\Gamma\Delta = N$.

The intersections with the fibers are

$$\Gamma l = q, \quad \Gamma m = 1, \quad \Delta l = \Delta m = 1.$$

For arbitrary integers r, s , apply the product inequality to $D = r\Gamma + s\Delta$. It gives

$$r^2q(2 - 2g) + 2rsN + s^2(2 - 2g) \leq 2(rq + s)(r + s).$$

Set $a = q + 1 - N$ and rearrange:

$$gqr^2 + ars + gs^2 \geq 0 \quad \text{for all integers } r, s. \quad (1)$$

By homogeneity it holds for rational r, s , and by continuity for real r, s . If $g > 0$, the discriminant of this nonnegative quadratic form is nonpositive, so $a^2 - 4g^2q \leq 0$. If $g = 0$, (1) says $ars \geq 0$ for both signs of rs , forcing $a = 0$. Thus in every genus $N = q + 1 - a$ with $|a| \leq 2g\sqrt{q}$. $\square$

Exercise V.1.11**Permitted context.**

Let X be a nonsingular projective integral surface over an algebraically closed field. Assume its numerical divisor group N is finitely generated. It is torsion-free, so its image in $N_{\mathbb{R}}$ is a lattice. For ample H , intersection is negative definite on $H^{\perp}$. An integral curve C has $C^2 + CK_X = 2p_a(C) - 2 \geq -2$. Distinct integral curves have nonnegative intersection. On a product $C \times C$, the fibers l, m have $l + m$ ample and the diagonal has square $2 - 2g(C)$. We use the degree and intersection rules for graphs of morphisms of curves.

Problem. (a) For ample H and fixed integer d , prove that effective divisors D with $DH = d$ have only finitely many numerical classes. (b) Assuming this finite-generation hypothesis for $C \times C$, deduce that a curve C of genus at least two has finite automorphism group.

Solution. (a) First bound integral curves of fixed degree $e = CH > 0$. Set $h = H^2 > 0$, $k = K_X H$, and write in $N_{\mathbb{R}}$

$$C = \frac{e}{h}H + v, \quad K_X = \frac{k}{h}H + w, \quad v, w \in H^{\perp}.$$

On $H^{\perp}$ use the positive-definite inner product $\langle v, w \rangle = -vw$. Adjunction yields

$$-2 \leq C^2 + CK_X = \frac{e^2 + ek}{h} - \|v\|^2 - \langle v, w \rangle.$$

Completing the square gives

$$\left\| v + \frac{w}{2} \right\|^2 \leq 2 + \frac{e^2 + ek}{h} + \frac{\|w\|^2}{4}. \quad (1)$$

For fixed e , this is a bounded closed ball (or an empty set if the right side is negative). Since the numerical classes of the corresponding C 's lie in the affine hyperplane $CH = e$ inside the lattice N , there are only finitely many. This remains true when $H^{\perp}$ has dimension zero.

Now let $D = \sum n_i C_i$ be an effective divisor of degree $d > 0$. Each $C_i H$ is a positive integer at most d , and $\sum n_i \leq d$. By the first step there are finitely many possible classes for all C_i , and the multiplicity bound leaves finitely many sums. For $d < 0$ there are no effective divisors; for $d = 0$ only the zero divisor occurs, by ampleness. This proves (a).

(b) For an automorphism α of C , let $\Gamma_{\alpha} \subset C \times C$ be its graph. Both projections have degree one, so $\Gamma_{\alpha}(l + m) = 2$. Part (a) gives finitely many numerical classes of such graphs.

These classes determine the graphs themselves. First, if an integral graph Γ is numerically equivalent to Δ , then

$$\Gamma\Delta = \Delta^2 = 2 - 2g < 0.$$

Distinct integral curves have nonnegative intersection, so $\Gamma = \Delta$. More generally, if $\Gamma_{\alpha} \equiv \Gamma_{\beta}$, applying the surface automorphism $\text{id} \times \alpha^{-1}$ preserves numerical equivalence and takes the graphs to Δ and $\Gamma_{\alpha^{-1}\beta}$. The preceding argument gives $\alpha^{-1}\beta = \text{id}$. Thus distinct automorphisms have distinct numerical graph classes. Finiteness of those classes proves that $\text{Aut}(C)$ is finite. $\square$

Exercise V.1.12**Permitted context.**

Work over an algebraically closed field. For a nonsingular projective surface, Nakai–Moishezon says that a divisor D is ample exactly when $D^2 > 0$ and $DE > 0$ for every integral curve E . The exterior tensor product of very ample line bundles is very ample. Restricting a very ample line bundle to a closed subvariety gives a very ample line bundle. A smooth plane quartic has genus three and canonical bundle $\mathcal{O}_C(1)$. Bertini gives smooth plane quartics in every characteristic. We use curve Riemann–Roch and that a nontrivial degree-zero line bundle has no nonzero section.

Problem. Prove that numerical equivalence preserves ampleness of divisors on a nonsingular projective surface. Give an example showing that it does not preserve very ampleness.

Solution. If $D' \equiv D$, their intersections with every curve agree. Bilinearity also gives

$$(D')^2 = D^2,$$

since $(D' - D)(D' + D) = 0$. Thus both conditions of Nakai–Moishezon hold for D' whenever they hold for D , proving the ampleness assertion.

For the example, take a smooth plane quartic C and distinct points $P, Q \in C$. The degree-zero bundle $\mathcal{O}_C(Q - P)$ is nontrivial: a rational function with divisor $Q - P$ would give a degree-one map to $\mathbb{P}^1$, contradicting genus three. Let

$$L = \omega_C, \quad L' = \omega_C(P - Q).$$

Both have degree four, but Riemann–Roch gives

$$h^0(C, L') = 4 + 1 - 3 + h^0(C, \mathcal{O}_C(Q - P)) = 2.$$

Thus L' cannot be very ample: its complete system could map only to $\mathbb{P}^1$, where a genus-three curve cannot embed. In contrast, $L = \mathcal{O}_C(1)$ is very ample.

On the smooth surface $X = C \times \mathbb{P}^1$, with projections p_1, p_2 , set

$$A = p_1^*L \otimes p_2^*\mathcal{O}_{\mathbb{P}^1}(1), \quad A' = p_1^*L' \otimes p_2^*\mathcal{O}_{\mathbb{P}^1}(1).$$

The bundle A is very ample by the Segre embedding. Their quotient is $p_1^*\mathcal{O}_C(P - Q)$, which is numerically trivial on X . Indeed, on the normalization of any integral curve in X , its degree is zero: projection to C is either constant, giving a trivial pullback, or finite, multiplying the degree-zero divisor degree. Hence the divisors of A, A' are numerically equivalent.

Finally, A' restricts on $C \times \{t\}$ to L' , which is not very ample. Therefore A' is not very ample, giving the required counterexample. $\square$

V.2 Ruled Surfaces

Exercise V.2.1

Permitted context.

Work over an algebraically closed field. A birationally ruled surface is birational to $C \times \mathbb{P}^1$, with C a smooth projective integral curve. We use Luroth's theorem: a curve dominated by $\mathbb{P}^1$ has rational function field, also for an inseparable map. Rational maps from a smooth projective curve to a projective curve extend to morphisms, and birational smooth projective curves are isomorphic. A rational map from $C \times \mathbb{P}^1$ constant on the generic ruling fiber factors rationally through C : constancy on the generic fiber makes the pulled-back functions algebraic over $k(C)$, and $k(C)$ is algebraically closed in the purely transcendental extension $k(C)(t)$. Thus those functions lie in $k(C)$.

Problem. Prove that the base curve of a birationally ruled surface is unique up to isomorphism.

Solution. Suppose $C \times \mathbb{P}^1$ is birational to $C' \times \mathbb{P}^1$. If C' has positive genus, compose the birational map with projection to C' . On the generic ruling fiber this is a rational map from $\mathbb{P}^1_{k(C)}$ to the base change of C' . It must be constant by Luroth's theorem, since the latter curve has positive genus. Thus the composite factors through a dominant rational map $u : C \dashrightarrow C'$.

If both curves have positive genus, the inverse birational map similarly gives $v : C' \dashrightarrow C$. Composing with the original birational maps shows $vu = \text{id}_C$ and $uv = \text{id}_{C'}$ as rational maps: both are the maps induced on the respective bases by the identity map of the product surface. Therefore u, v are inverse birational maps of smooth projective curves, hence isomorphisms.

It remains to handle a rational base. If $C \cong \mathbb{P}^1$ and C' had positive genus, the same composite $\mathbb{P}^1 \times \mathbb{P}^1 \dashrightarrow C'$ would be constant on each ruling. In function fields, its pullback would lie in both $k(s)$ and $k(t)$ inside $k(s, t)$, whose intersection is k . It would be constant, contradicting dominance. Thus both bases have genus zero, and over the algebraically closed field both are $\mathbb{P}^1$. This completes the uniqueness proof. $\square$

Exercise V.2.2

Permitted context.

Let $\mathcal{E}$ be a rank-two vector bundle on a smooth projective integral curve C over an algebraically closed field. Use the quotient convention for $X = \mathbb{P}(\mathcal{E})$: a section corresponds to a line-bundle quotient $\mathcal{E} \twoheadrightarrow L$. Two such quotient maps represent the same point of a fiber precisely when their one-dimensional kernels agree. A morphism of vector bundles that is an isomorphism on every fiber is an isomorphism, by Nakayama's lemma.

Problem. Prove that $\mathcal{E}$ is decomposable exactly when $X \rightarrow C$ has two disjoint sections.

Solution. If $\mathcal{E} = L_1 \oplus L_2$, its two projection quotients give sections. In every fiber their kernels are the two different direct summands, so the sections have no common point.

Conversely, let two disjoint sections correspond to quotients $q_i : \mathcal{E} \twoheadrightarrow L_i$. The map

$$(q_1, q_2) : \mathcal{E} \longrightarrow L_1 \oplus L_2$$

is an isomorphism on every fiber: its two quotient functionals have distinct kernels and hence are independent. By the fiberwise criterion it is an isomorphism of bundles. Thus $\mathcal{E}$ decomposes into the two line bundles. $\square$

Exercise V.2.3**Permitted context.**

Work over an algebraically closed field. On a nonsingular curve, a coherent torsion-free sheaf is locally free, since its local modules are torsion-free over discrete valuation rings. Saturating a subsheaf means enlarging it until the quotient is torsion-free. On $\mathbb{P}^2$, every line bundle is $\mathcal{O}(a)$, $H^1(\mathcal{O}(a)) = 0$ for every integer a , and

$$0 \rightarrow \Omega_{\mathbb{P}^2}^1(1) \rightarrow \mathcal{O}^{\oplus 3} \rightarrow \mathcal{O}(1) \rightarrow 0$$

is the Euler sequence. Extensions of line bundles are classified by $\text{Ext}^1(\mathcal{O}(b), \mathcal{O}(a)) = H^1(\mathcal{O}(a-b))$.

Problem. (a) Prove that every rank- r vector bundle on a nonsingular curve has a filtration $0 = \mathcal{E}_0 \subset \mathcal{E}_1 \subset \cdots \subset \mathcal{E}_r = \mathcal{E}$ whose successive quotients are line bundles. (b) Show that this fails in dimension at least two; in particular, $\Omega_{\mathbb{P}^2}^1$ is not an extension of line bundles.

Solution. (a) Choose a one-dimensional subspace of the generic fiber of $\mathcal{E}$. Extending a generic generator after allowing poles, then saturating its image in $\mathcal{E}$, gives a coherent rank-one subsheaf L with torsion-free quotient. Locally this saturation is the intersection of the generic line with the free module $\mathcal{E}$; over a discrete valuation ring it is a rank-one direct summand. Hence L is a line bundle and $\mathcal{E}/L$ is a vector bundle of rank $r-1$. Induct on the rank, and pull the quotient's filtration back to $\mathcal{E}$. For rank zero the empty filtration suffices.

(b) An extension $0 \rightarrow \mathcal{O}(a) \rightarrow \Omega_{\mathbb{P}^2}^1 \rightarrow \mathcal{O}(b) \rightarrow 0$ would split, by the stated H^1 -vanishing. Taking determinants in the Euler sequence gives $\det \Omega_{\mathbb{P}^2}^1 = \mathcal{O}(-3)$, so $a+b = -3$. But the map on global sections $H^0(\mathcal{O}^{\oplus 3}) \rightarrow H^0(\mathcal{O}(1))$ is an isomorphism, and therefore $H^0(\Omega_{\mathbb{P}^2}^1(1)) = 0$. A splitting would force $a+1 < 0$ and $b+1 < 0$, hence $a, b \leq -2$, contradicting $a+b = -3$. Thus no such extension exists.

For each dimension $n > 2$, pull this rank-two bundle back to $\mathbb{P}^2 \times \mathbb{P}^{n-2}$. A filtration with line-bundle quotients would restrict, still exactly because those quotients are locally free, to such a filtration on a fiber $\mathbb{P}^2$. This is impossible by the preceding argument. Hence the failure occurs in every dimension at least two. $\square$

Exercise V.2.4**Permitted context.**

Let C be a smooth projective integral genus- g curve over an algebraically closed field. A section of $C \times \mathbb{P}^1 \rightarrow C$ is the graph of a morphism $h : C \rightarrow \mathbb{P}^1$. The normal bundle of a graph is $h^*T_{\mathbb{P}^1}$; its degree is the self-intersection. We use $T_{\mathbb{P}^1} = \mathcal{O}(2)$ and the existence, for $r \geq g+1$, of a nonspecial base-point-free degree- r line bundle. A nonhyperelliptic genus-three curve is a canonical plane quartic. A curve of type (a, b) on $\mathbb{P}^1 \times \mathbb{P}^1$ has arithmetic genus $(a-1)(b-1)$.

Problem. For a section D of $C \times \mathbb{P}^1$, show $D^2 = 2r$ with $r \geq 0$. (a) Prove that $r = 0$ and every $r \geq g+1$ are possible. (b) If $g = 3$, show that $r = 1$ is impossible, and exactly one of $r = 2, 3$ is possible, according as C is hyperelliptic or nonhyperelliptic.

Solution. For the graph of h , the normal-bundle formula gives $D^2 = \deg h^*\mathcal{O}(2) = 2r$, where $r = \deg h^*\mathcal{O}(1) \geq 0$. For nonconstant h this is its degree; for constant h it is zero.

(a) Constant maps provide $r = 0$. For $r \geq g + 1$, choose a generated nonspecial degree- r line bundle L . Riemann–Roch gives $h^0(L) = r + 1 - g \geq 2$. Choose a nonzero section s ; its zeros are finite. Because L is generated, there is a second section t nonzero at all of these zeros: each excluded vanishing condition is a proper linear subspace, and the field is infinite. Thus s, t have no common zero and give a map $h : C \rightarrow \mathbb{P}^1$ with $h^*\mathcal{O}(1) = L$. It has degree r , as desired.

(b) A degree-one map would make $C \cong \mathbb{P}^1$, so $r = 1$ is impossible. If C is hyperelliptic its degree-two pencil gives $r = 2$. It cannot also have a degree-three map: the pair of maps would be birational onto a curve of type $(2, 3)$, since its degree onto the image divides both two and three. That image has arithmetic genus two, less than the genus three of its normalization.

If C is nonhyperelliptic, $r = 2$ is impossible by definition. Its canonical plane quartic admits a degree-three map by projection from any point on the curve: the lines through that point give a pencil of hyperplane sections with that one fixed point removed. Smoothness makes the residual pencil base-point-free, of degree $4 - 1 = 3$. Thus $r = 3$ is possible. $\square$

Exercise V.2.5

Permitted context.

Let C be a smooth projective integral genus- g curve over an algebraically closed field, with $g \geq 1$. A rank-two bundle is normalized if it has a section and no positive-degree line subbundle; its ruled surface invariant is $e = -\deg \mathcal{E}$. A nonzero section of a normalized bundle is nowhere zero. A normalized decomposable bundle is $\mathcal{O}_C \oplus L$ with $\deg L \leq 0$. We use Riemann–Roch, Serre duality, and $\text{Ext}^1(L, \mathcal{O}_C) = H^1(L^{-1})$. The curve symmetric power $C^{(n)}$ has dimension n . Degree at least $2g + 1$ line bundles are very ample. Projection of an embedded smooth curve from a point is an embedding if that point lies on none of its secant or tangent lines, interpreted by length-two subschemes. A smooth plane quintic has genus six.

Problem. (a) For every $0 \leq e \leq 2g - 2$, construct an indecomposable normalized bundle of invariant e . (b) Let $d = -e > 0$, $\deg D = d$, and let $0 \neq \xi \in H^1(\mathcal{O}_C(-D))$ define $0 \rightarrow \mathcal{O}_C \rightarrow \mathcal{E} \rightarrow \mathcal{O}_C(D) \rightarrow 0$. Regard ξ as a functional on $H^0(K_C + D)$ and write $H = \mathbb{P}(\ker \xi)$. For each effective E of degree $d - 1$, put $L_E = |K_C + D - E| + E$. Show that $\mathcal{E}$ is normalized exactly when no L_E is contained in H . (c) Deduce existence of ruled surfaces with $-g \leq e < 0$. (d) If $g = 2$, prove the necessary bound $e \geq -2$.

Solution. (a) *Construction.* Choose an effective divisor A of degree e contained in a canonical divisor. Such a choice exists when $g \geq 2$; for $g = 1$ the only case is $e = 0$ and take $A = 0$. Serre duality gives

$$H^1(\mathcal{O}_C(A))^* = H^0(\mathcal{O}_C(K_C - A)) \neq 0.$$

Choose a nonzero extension class to obtain

$$0 \rightarrow \mathcal{O}_C \rightarrow \mathcal{E} \rightarrow \mathcal{O}_C(-A) \rightarrow 0. \quad (1)$$

A positive-degree line bundle can map nontrivially to neither the first nor last term, so none can be a subbundle of $\mathcal{E}$. Thus (1) is normalized and $-\deg \mathcal{E} = e$. If it decomposed, normalization would give $\mathcal{E} \cong \mathcal{O}_C \oplus \mathcal{O}_C(-A)$. For $e > 0$ its only global sections come from the first summand, so the displayed extension would split. For $e = 0, A = 0$, a nowhere-zero constant section of $\mathcal{O}_C^{\oplus 2}$ also splits off a summand. Both contradict the nonzero class. Hence $\mathcal{E}$ is indecomposable.

(b) *The normalization criterion.* The bundle already has a nonzero section. It fails normalization precisely when some positive-degree line bundle maps nontrivially into it. By replacing such a bundle by its twist down by an effective divisor, it is enough to test degree-one line bundles M . A nonzero map $M \rightarrow \mathcal{E}$ has nonzero composite with $\mathcal{O}_C(D)$, since $\text{Hom}(M, \mathcal{O}_C) = 0$. Such a composite identifies

$$M = \mathcal{O}_C(D - E)$$

for an effective divisor E of degree $d - 1$. It lifts to $\mathcal{E}$ exactly when the pulled-back extension class is zero in $H^1(\mathcal{O}_C(E - D))$.

By naturality of Serre duality, the dual of this map on extension classes is the inclusion

$$H^0(K_C + D - E) \longrightarrow H^0(K_C + D),$$

multiplication by the section of E . Thus its vanishing on ξ says precisely that ξ annihilates this subspace, or $L_E \subset H$. Excluding every such E is exactly normalization.

(c) *The incidence dimension bound.* Let $1 \leq d \leq g$ and fix any degree- d divisor D . The extension space $W = H^1(\mathcal{O}_C(-D))$ has dimension $g + d - 1$. For any effective E of degree $d - 1$, the divisor $K_C + D - E$ has degree $2g - 1$, so its section space has dimension g . Consequently its annihilator in W has dimension $d - 1$. The incidence of all such bad pairs (ξ, E) has dimension

$$(d - 1) + (d - 1) = 2d - 2 < g + d - 1 = \dim W.$$

The section spaces have constant dimension and vary algebraically with E , so these annihilators form a vector bundle over $C^{(d-1)}$; in particular the stated dimension bound applies. Their image cannot fill W . Choose ξ outside its closure. It is nonzero, since zero belongs to every annihilator. By (b) its extension is normalized, giving invariant $e = -d$. This proves all the claimed negative values.

(d) *The genus-two bound.* Suppose $g = 2$ and a normalized bundle has positive degree $d = -e$. It has exactly one independent global section. Indeed, every nonzero section is nowhere zero. Two independent sections cannot generically span a line, since that saturated line would have degree at most zero and at most one independent section. If they generically span rank two, their determinant is a nonzero section of a positive-degree line bundle; at one of its zeros a nontrivial constant combination of the two sections vanishes. This contradicts normalization. Riemann–Roch now gives $1 = h^0(\mathcal{E}) \geq \chi(\mathcal{E}) = d - 2$, so $d \leq 3$.

It remains to exclude $d = 3$. A nonzero section then gives an extension as in (b) with $\deg D = 3$ and nonzero class ξ . The very ample system $|K_C + D|$ embeds C as a degree-five curve in $\mathbb{P}(H^0(K_C + D)^*) = \mathbb{P}^3$. For a length-two effective divisor E , the projective annihilator of $H^0(K_C + D - E)$ is exactly its secant or tangent line. Part (b) would put the point $[\xi]$ on none of these lines. Projection from $[\xi]$ would therefore embed C as a smooth plane quintic, without changing its degree. Such a quintic has genus six, contradicting $g = 2$. Thus $d \leq 2$, or $e \geq -2$. $\square$

Exercise V.2.6**Permitted context.**

Work over an algebraically closed field. Every line bundle on $\mathbb{P}^1$ is $\mathcal{O}(a)$, and $H^1(\mathcal{O}(a)) = 0$ for $a \geq -1$. Vector bundles on a nonsingular curve admit line subbundles with locally free quotient, by saturation. For a vector bundle $\mathcal{E}$, Serre generation of $\mathcal{E}^*(M)$ for large M gives an injection $\mathcal{E} \hookrightarrow \mathcal{O}(M)^{\oplus N}$. Extensions of line bundles are classified by their H^1 groups.

Problem. Prove that every finite-rank vector bundle on $\mathbb{P}^1$ is a direct sum of line bundles.

Solution. Induct on the rank; ranks zero and one are immediate. Choose a line subbundle $\mathcal{O}(a) \subset \mathcal{E}$ of maximal degree. Such a maximum exists: there is a line subbundle by saturation, and the injection into $\mathcal{O}(M)^{\oplus N}$ bounds its degree above by M . Its quotient $\mathcal{F}$ is locally free. By induction, $\mathcal{F} = \bigoplus_i \mathcal{O}(b_i)$.

We claim $b_i \leq a$ for every i . Pull back the extension along the summand $\mathcal{O}(b_i) \subset \mathcal{F}$ and twist by $\mathcal{O}(-a-1)$, obtaining

$$0 \rightarrow \mathcal{O}(-1) \rightarrow \mathcal{E}_i(-a-1) \rightarrow \mathcal{O}(b_i - a - 1) \rightarrow 0.$$

Since $H^1(\mathcal{O}(-1)) = 0$, a nonzero section of the last term lifts to the middle. If $b_i \geq a+1$, this gives a nonzero map $\mathcal{O}(a+1) \rightarrow \mathcal{E}$. Saturating its image yields a line subbundle of degree at least $a+1$, contradicting maximality of a . Hence the claim holds.

It follows that

$$\mathrm{Ext}^1(\mathcal{F}, \mathcal{O}(a)) = \bigoplus_i H^1(\mathcal{O}(a - b_i)) = 0.$$

The original extension therefore splits: $\mathcal{E} \cong \mathcal{O}(a) \oplus \bigoplus_i \mathcal{O}(b_i)$. This completes the induction. $\square$

Exercise V.2.7**Permitted context.**

Let C be an elliptic curve over an algebraically closed field and $\mathcal{E}$ the normalized nonsplit extension $0 \rightarrow \mathcal{O}_C \rightarrow \mathcal{E} \rightarrow \mathcal{O}_C(P_0) \rightarrow 0$. Thus $\deg \mathcal{E} = 1$, it has no positive-degree line subbundle, and $X = \mathbb{P}(\mathcal{E})$ is the elliptic ruled surface of invariant -1 . A section with kernel $M \subset \mathcal{E}$ has divisor bundle $\mathcal{O}_X(1) \otimes \pi^* M^{-1}$ and square $\deg \mathcal{E} - 2 \deg M$. We use Riemann–Roch, Serre duality with $\omega_C = \mathcal{O}_C$, and $\mathrm{Pic}^0(C) \cong C$. A normalized Poincaré bundle represents its degree-zero line bundles; constant cohomology dimensions give locally free pushforward and base change by Grauert’s theorem.

Problem. Show that the sections of $X \rightarrow C$ with self-intersection one form a one-dimensional algebraic family parametrized by C , and that no two of them are linearly equivalent.

Solution. Fix $M \in \mathrm{Pic}^0(C)$. By Serre duality,

$$H^1(C, \mathcal{E} \otimes M^{-1}) \cong \mathrm{Hom}(\mathcal{E}, M)^*.$$

The Hom space is zero. A nonzero map $\mathcal{E} \rightarrow M$ would have image a line bundle of degree at most zero; its kernel would be a line subbundle of degree at least one, contrary to normalization. Riemann–Roch therefore gives

$$h^0(C, \mathcal{E} \otimes M^{-1}) = 1.$$

Its unique nonzero map $M \rightarrow \mathcal{E}$, up to scalar, is nowhere zero: a zero would make its saturation a positive-degree line subbundle. Its quotient is a line bundle, giving a unique section D_M of X . The self-intersection formula gives $D_M^2 = 1$. Conversely, a section of square one has $\deg M = 0$, so every such section is obtained.

These sections vary algebraically. Let $B = \text{Pic}^0(C)$, and let $\mathcal{P}$ be the Poincaré bundle on $C \times B$. The pushforward to B of $\text{pr}_C^* \mathcal{E} \otimes \mathcal{P}^{-1}$ is a line bundle, by the cohomology calculation and Grauert's theorem. Its evaluation, tensored with $\mathcal{P}$, gives a line subbundle of $\text{pr}_C^* \mathcal{E}$. The quotient defines an algebraic family of the sections D_M . Its parameter space $B \cong C$ has dimension one, and the preceding uniqueness shows it parametrizes exactly the claimed sections.

Finally, if $D_M \sim D_{M'}$, their divisor bundles give $\pi^*(M^{-1} \otimes M') \cong \mathcal{O}_X$. Pullback from C is injective on Picard groups, since restricting to any fixed section of π is a left inverse. Hence $M = M'$, and the uniqueness already proved gives $D_M = D_{M'}$. Distinct sections in the family are therefore not linearly equivalent. $\square$

Exercise V.2.8

Permitted context.

Let C be a smooth projective integral curve of genus g over an algebraically closed field. For a vector bundle E , put $\mu(E) = \deg E / \text{rank } E$. Stability requires every nonzero proper vector-bundle quotient to have slope greater than $\mu(E)$; semistability replaces greater than by at least. In rank two this is equivalent to every line subbundle having degree less than, or at most, $\deg E/2$, respectively. A normalized rank-two bundle has a nowhere-zero section and maximal line-subbundle degree zero. We use $\text{Ext}^1(L, \mathcal{O}_C) = H^1(L^{-1})$ and Serre duality. Normalization is the convention of the section's Notation 2.8.1; twisting by a line bundle preserves stability, semistability, and indecomposability.

Problem. (a) Show that a decomposable bundle is not stable. (b) For normalized rank-two E , prove that it is stable exactly when $\deg E > 0$, and semistable exactly when $\deg E \geq 0$. (c) Classify normalized indecomposable rank-two bundles that are not semistable by $0 < e \leq 2g - 2$, a line bundle L of degree $-e$, and a nonzero class in $H^1(L^{-1})$ up to scalar. Explain the additional twist needed if normalization is not imposed.

Solution. (a) If $E = E_1 \oplus E_2$ with both summands nonzero, then $\mu(E)$ is a weighted average of $\mu(E_1), \mu(E_2)$. At least one summand, viewed as a proper quotient, has slope at most $\mu(E)$. This violates stability.

(b) All line subbundles have degree at most zero, and the degree zero is attained by the nowhere-zero section $\mathcal{O}_C \subset E$. Thus all these degrees are strictly below $\deg E/2$ exactly when $\deg E > 0$, and are at most $\deg E/2$ exactly when $\deg E \geq 0$. These are precisely the two slope criteria.

(c) Let E be normalized, indecomposable, and not semistable. By (b), $\deg E = -e < 0$. Its section gives

$$0 \rightarrow \mathcal{O}_C \rightarrow E \rightarrow L \rightarrow 0, \quad L = \det E, \quad \deg L = -e. \quad (1)$$

Indecomposability makes its class $\xi \in H^1(L^{-1})$ nonzero. Serre duality gives $H^1(L^{-1})^* = H^0(\omega_C \otimes L) \neq 0$, so $2g - 2 - e \geq 0$, proving $e \leq 2g - 2$.

Conversely, any nonzero extension (1) with $\deg L = -e < 0$ is normalized: a positive-degree line bundle maps to neither $\mathcal{O}_C$ nor L , and hence cannot be a subbundle of E . Also $h^0(E) = 1$, since $h^0(L) = 0$. If E decomposed, normalization would force $E = \mathcal{O}_C \oplus L$; its unique section would then split (1), contrary to $\xi \neq 0$. Thus it is indecomposable and, by (b), not semistable.

The unique section up to scalar makes the subbundle $\mathcal{O}_C$ in (1) intrinsic. An isomorphism of middle bundles must preserve it, and induces automorphisms of that line and of L . These are nonzero scalars. The extension classes therefore give isomorphic bundles exactly when they differ by a nonzero scalar, proving the stated classification.

For an arbitrary unnormalized indecomposable bundle that is not semistable, let M be its maximal-degree line subbundle. Its degree is greater than half the bundle degree, and it is unique: a different such line could not map nontrivially into the quotient by M , whose degree is smaller. After twisting by M^{-1} one obtains the normalized bundle classified above. Thus in the unnormalized formulation one must also specify M , and the bundle is $M \otimes E$ with E as in (1). $\square$

Exercise V.2.9

Permitted context.

Work over an algebraically closed field. The resolution of a quadric cone $Q \subset \mathbb{P}^3$ is $\pi: \mathbb{F}_2 \rightarrow Q$, contracting a section E . With F a ruling fiber, $E^2 = -2$, $EF = 1$, $F^2 = 0$, the hyperplane pullback is $H = E + 2F$, and $K_{\mathbb{F}_2} = -2E - 4F$. Also $\pi^{-1}\mathcal{I}_{\text{vertex}} \cdot \mathcal{O}_{\mathbb{F}_2} = \mathcal{O}(-E)$ and $\pi_*\mathcal{O}_{\mathbb{F}_2} = \mathcal{O}_Q$. We use adjunction, the universal property of blowing up, and the projection formula. Restriction of degree- a forms from $\mathbb{P}^3$ to Q is surjective: the quadric ideal is $\mathcal{O}(-2)$, and $H^1(\mathbb{P}^3, \mathcal{O}(t)) = 0$.

Problem. Let Y be a nonsingular integral curve on Q . Show that either it is a complete intersection of Q with a degree- a surface, with $\deg Y = 2a$ and $g(Y) = (a-1)^2$, or its degree is odd, $2a+1$, and $g(Y) = a(a-1)$.

Solution. Let $\tilde{Y}$ be the strict transform. It is isomorphic to Y . This is immediate if Y avoids the vertex. If it passes through the vertex P , the vertex ideal restricts to the invertible ideal $\mathfrak{m}_{Y,P}$ on the smooth curve Y . The universal property of blowing up therefore lifts Y to the resolution, giving its strict transform isomorphically. Since the exceptional ideal restricts to $\mathfrak{m}_{Y,P}$, the intersection with E has length exactly one. Consequently

$$\tilde{Y}E = \delta, \quad \delta = 0 \text{ or } 1.$$

Write $\tilde{Y} \sim aE + bF$. Then

$$\deg Y = H\tilde{Y} = b, \quad b - 2a = \delta. \quad (1)$$

Here $a = \tilde{Y}F \geq 0$; when $\delta = 0$, positivity of the degree forces $a \geq 1$. Adjunction gives

$$g(Y) = 1 + \frac{\tilde{Y}^2 + K_{\mathbb{F}_2}\tilde{Y}}{2} = 1 - a^2 + ab - b.$$

If $\delta = 0$, then $b = 2a$ and this becomes $(a-1)^2$. If $\delta = 1$, then $b = 2a+1$ and it becomes $a(a-1)$. The latter includes a ruling line, where $a = 0$.

In the even case, $\tilde{Y} \sim aH$ and it avoids E . The section cutting it out comes from a section of $\mathcal{O}_Q(a)$, by $\pi_*\mathcal{O}_{\mathbb{F}_2} = \mathcal{O}_Q$ and the projection formula. Its value at the vertex is nonzero, since its pullback does not vanish on E . Its divisor on Q is therefore exactly Y . Lift this section to a degree- a form on $\mathbb{P}^3$. The corresponding surface cuts out Y scheme-theoretically on Q , proving the complete-intersection assertion. $\square$

Exercise V.2.10**Permitted context.**

Work over an algebraically closed field. On $\mathbb{F}_e$, $e \geq 0$, use $C_0^2 = -e$, $C_0f = 1$, $f^2 = 0$ and $K = -2C_0 - (e + 2)f$. For $n > e$, $H = C_0 + nf$ embeds the surface as a degree- $d = 2n - e$ scroll in $\mathbb{P}^{d+1}$, with $h^0(H) = d + 2$. A divisor $aC_0 + bf$, $a > 0$, is very ample if $b > ae$, and is generated by sections if $b \geq ae$. For $e > 0$, $|C_0 + ef|$ contracts only C_0 and is an isomorphism off it onto the complement of the vertex of the rational cone. We use Bertini and adjunction for smooth curves.

Problem. If $n > e \geq 0$ and $n \geq 2e - 2$, prove that the scroll contains a smooth curve Y of genus $d + 2$ canonically embedded by its ambient embedding. Deduce that every $g \geq 4$ occurs for a nonhyperelliptic curve with a g_3^1 .

Solution. Take a general member of

$$|H - K| = |3C_0 + (n + e + 2)f|. \quad (1)$$

The assumption $n \geq 2e - 2$ says $n + e + 2 \geq 3e$. If this is strict, the system is very ample, so Bertini gives a smooth integral member. In the equality case, (1) is $|3(C_0 + ef)|$. Here $n > e$ forces $e > 2$. It is the pullback of the cubic hyperplane system on the rational cone. A general member avoids the vertex and is smooth and integral by Bertini on that cone away from its vertex. Its pullback is likewise smooth and integral, and avoids C_0 . Thus in all cases we obtain the required smooth Y .

Adjunction gives

$$K_Y = (K + Y)|_Y = H|_Y,$$

and its degree is

$$HY = (C_0 + nf)(3C_0 + (n + e + 2)f) = 4n - 2e + 2 = 2d + 2.$$

Hence $2g(Y) - 2 = 2d + 2$, or $g(Y) = d + 2$. The restriction of sections of H to Y is injective, because $H - Y = K$ has degree minus two on a ruling fiber and has no global section. Both its source and $H^0(Y, K_Y)$ have dimension $d + 2$, so restriction is an isomorphism. Thus the scroll embedding restricts to the complete canonical embedding of Y . Also $Yf = 3$, so the ruling restricts to a degree-three pencil. In particular Y is nonhyperelliptic.

Finally, for any $g \geq 4$, put $d = g - 2$. If d is even, choose $e = 0$, $n = d/2$. If d is odd, choose $e = 1$, $n = (d + 1)/2$. In both cases $n > e$ and $n \geq 2e - 2$. The preceding construction produces the desired canonical trigonal genus- g curve. $\square$

Exercise V.2.11**Permitted context.**

Let $\pi : X = \mathbb{P}(\mathcal{E}) \rightarrow C$ be a ruled surface over a smooth projective curve. Use the normalized extension $0 \rightarrow \mathcal{O}_C \rightarrow \mathcal{E} \rightarrow \mathcal{O}_C(\epsilon) \rightarrow 0$ and $\mathcal{O}_X(C_0) = \mathcal{O}_X(1)$. For a divisor b on C , write $bf = \pi^*b$. The sections of $\mathcal{O}_X(C_0 + bf)$ are $H^0(C, \mathcal{E}(b))$. We use curve Riemann–Roch and the criterion that a linear system is very ample if it separates all length-two subschemes. A general section of a globally generated rank-two bundle on a curve is nowhere zero: vanishing at a point imposes two linear conditions over a one-dimensional parameter space.

Problem. (a) If $|b|$ and $|b + \epsilon|$ have no base points and b is nonspecial, show that $|C_0 + bf|$ has no base points and contains a section. (b) If $b, b + \epsilon$ are very ample and $b - P, b + \epsilon - P$ are nonspecial for every $P \in C$, prove $C_0 + bf$ very ample.

Solution. (a) Tensoring the normalized extension by $\mathcal{O}_C(b)$ gives

$$0 \rightarrow \mathcal{O}_C(b) \rightarrow \mathcal{E}(b) \rightarrow \mathcal{O}_C(b + \epsilon) \rightarrow 0. \quad (1)$$

Since $H^1(\mathcal{O}_C(b)) = 0$, global sections surject onto the quotient's sections. At any point, sections of $\mathcal{O}_C(b)$ span the subline and lifts of sections of $\mathcal{O}_C(b + \epsilon)$ span the quotient. Thus $\mathcal{E}(b)$ is globally generated. Its tautological quotient then shows that $\mathcal{O}_X(C_0 + bf)$ is globally generated.

Choose a nowhere-zero section of $\mathcal{E}(b)$. Composing its pullback with the tautological quotient defines a divisor in $|C_0 + bf|$. In each fiber $\mathbb{P}^1$ its equation is a nonzero linear form, so its zero scheme has one reduced point. Locally trivializing the nowhere-zero subline shows that this divisor is a section of π .

(b) Let $Z = P + Q$ be any length-two divisor on C , allowing $P = Q$. For $L = \mathcal{O}_C(b)$ and $L = \mathcal{O}_C(b + \epsilon)$, very ampleness gives $h^0(L(-P)) = h^0(L) - 1$ and $h^0(L(-Z)) = h^0(L) - 2$. Together with $H^1(L(-P)) = 0$, Riemann–Roch then gives $H^1(L(-Z)) = 0$. Twisting (1) by $-Z$ consequently gives $H^1(\mathcal{E}(b)(-Z)) = 0$. Hence

$$H^0(C, \mathcal{E}(b)) \longrightarrow H^0(Z, \mathcal{E}(b)|_Z)$$

is surjective for every such Z .

This separates every length-two subscheme of X . For two points over distinct base points, it prescribes the required quotient values independently. For a vertical length-two subscheme in one fiber, global generation supplies every linear form on that fiber, and $\mathcal{O}_{\mathbb{P}^1}(1)$ separates it. For a nonvertical tangent vector, use $Z = 2P$: local trivialization of $\mathcal{E}(b)$ identifies the two freely prescribed base jets with all first-order linear forms on the local projective bundle. They separate that tangent vector. These cases exhaust length-two subschemes on the smooth surface. Thus $C_0 + bf$ is very ample. $\square$

Exercise V.2.12**Permitted context.**

Let $X = \mathbb{P}(\mathcal{E}) \rightarrow C$ be a ruled surface over an elliptic curve, with $\mathcal{E}$ normalized and $e = -\deg \mathcal{E} \geq -1$. Write $\det \mathcal{E} = \mathcal{O}_C(\epsilon)$, $\mathcal{O}_X(C_0) = \mathcal{O}_X(1)$, and $n = \deg b$, with $C_0^2 = -e$. Line bundles on an elliptic curve are nonspecial in positive degree, generated in degree at least two, and very ample exactly in degree at least three. We use the two generation and very-ampleness criteria proved in V.2.11. For $e = -1$, $\mathcal{E}$ is stable of degree one; twists remain stable. On an elliptic curve a stable positive-degree bundle has $H^1 = 0$, by Serre duality and the quotient slope criterion. A stable negative-degree bundle has no section.

Problem. (a) If $n \geq e + 2$, prove that $|C_0 + bf|$ is base-point-free and contains a section. (b) Prove that $C_0 + bf$ is very ample exactly when $n \geq e + 3$, including $e = -1$.

Solution. (a) First suppose $e \geq 0$. If $n \geq e + 2$, the degrees of $b, b + \epsilon$ are $n \geq 2$ and $n - e \geq 2$, so both are generated and b is nonspecial. The first criterion from V.2.11 proves (a). Now suppose $e = -1$. The bundle $V = \mathcal{E}(b)$ is stable of degree $2n + 1$. If $n \geq 1$, then for every point P , $V(-P)$ is stable of positive degree $2n - 1$, and has $H^1 = 0$. The evaluation $H^0(V) \rightarrow V|_P$ is therefore surjective. Thus V is globally generated. As in V.2.11(a), its tautological system is generated and a nowhere-zero general section gives a section divisor. This proves (a) at $n \geq 1 = e + 2$.

(b) For $e \geq 0$ and $n \geq e + 3$, the degrees of $b, b + \epsilon$ are at least three, and subtracting any point leaves positive degree. Thus both are very ample and the required point-subtractions are nonspecial. The second criterion proves sufficiency. Conversely, restriction of $C_0 + bf$ to C_0 has degree $n - e$. Very ampleness on the surface forces very ampleness on this elliptic section, hence $n - e \geq 3$. This proves necessity for $e \geq 0$.

For $e = -1$, use $V = \mathcal{E}(b)$ as above. If $n \geq 2$, then for every length-two divisor Z , $V(-Z)$ is stable of positive degree $2n - 3$; again $H^1 = 0$. Evaluation to $V|_Z$ is surjective, so the length-two argument of V.2.11(b) makes $C_0 + bf$ very ample.

Finally, if $n < 0$, stability of the negative-degree V gives no sections. If $n = 0$ or 1 , positive-degree stability and Riemann–Roch give $h^0(V) = 2n + 1$, respectively one or three. Such a complete system cannot embed this surface: for $n = 0$ it maps to a point, and for $n = 1$ a surface embedding into $\mathbb{P}^2$ would identify X with $\mathbb{P}^2$. But X contains an elliptic section of square one; its image in $\mathbb{P}^2$ would have degree one, since a degree- d plane curve has square d^2 . A line is not elliptic. Thus $n \geq 2 = e + 3$ is necessary, completing (b). $\square$

Exercise V.2.13**Permitted context.**

Over an algebraically closed field, elliptic ruled surfaces exist with every invariant $e \geq -1$: use $\mathcal{O}_C \oplus L$ with $\deg L = -e$ for $e \geq 0$, and the normalized nonsplit degree-one bundle for $e = -1$. For a normalized bundle $\mathcal{E}$, $C_0^2 = -e$, $C_0 f = 1$, $f^2 = 0$, and $H^0(X, \mathcal{O}_X(C_0 + bf)) = H^0(C, \mathcal{E}(b))$. A divisor $C_0 + bf$ is very ample if $\deg b \geq e + 3$, by V.2.12. We use elliptic Riemann–Roch and the fact that a stable positive-degree elliptic bundle has $H^1 = 0$.

Problem. For $e \geq -1$ and $n \geq e + 3$, construct an elliptic scroll of degree $d = 2n - e$ in $\mathbb{P}^{d-1}$. In particular, construct one of degree five in $\mathbb{P}^4$.

Solution. Choose the indicated elliptic ruled surface, and a divisor b of degree n on its base. The divisor $H = C_0 + bf$ is very ample, and

$$H^2 = -e + 2n = d.$$

It has degree one on each ruling fiber, so the embedding maps those fibers to lines.

We compute the dimension of its complete system. For $e \geq 0$, tensoring $0 \rightarrow \mathcal{O}_C \rightarrow \mathcal{E} \rightarrow \mathcal{O}_C(e) \rightarrow 0$ by $\mathcal{O}_C(b)$ gives line terms of degrees n and $n - e$, both positive. Their H^1 's vanish, so $H^1(\mathcal{E}(b)) = 0$. For $e = -1$, the bundle $\mathcal{E}(b)$ is stable of positive degree $2n + 1$, and has the same vanishing. In either case elliptic Riemann–Roch gives

$$h^0(\mathcal{E}(b)) = \deg \mathcal{E}(b) = 2n - e = d.$$

Thus the complete embedding is into $\mathbb{P}^{d-1}$, with surface degree $H^2 = d$. Taking $e = -1$, $n = 2$ gives $d = 5$ and an elliptic scroll in $\mathbb{P}^4$. $\square$

Exercise V.2.14

Permitted context.

Let $X = \mathbb{P}(\mathcal{E}) \rightarrow C$ be a ruled surface over an algebraically closed field of characteristic $p > 0$. The base has genus $g \geq 2$, the bundle is normalized, and $e = -\deg \mathcal{E} < 0$. Numerically, $C_0^2 = -e$, $C_0 f = 1$, $f^2 = 0$ and $K_X \equiv -2C_0 + (2g - 2 - e)f$. Normalization means that $\mathcal{E}$ has a section and no line subbundle of positive degree. We use adjunction, the Hurwitz inequality for a separable map of smooth curves, and the Nakai–Moishezon criterion. A purely inseparable map of smooth curves over a perfect field preserves genus: it is an iterate of relative Frobenius followed by an isomorphism.

Problem. (a) For an integral curve $Y \equiv aC_0 + bf$, other than C_0 or a fiber, prove the following alternatives:

$$\begin{aligned} a = 1 : & \quad b \geq 0, \\ 2 \leq a < p : & \quad b \geq ae/2, \\ a \geq p : & \quad b \geq ae/2 + 1 - g. \end{aligned}$$

(b) Show that $a > 0$ and $b > a(e/2 + (g - 1)/p)$ imply ampleness of any divisor $D \equiv aC_0 + bf$. Show conversely that ampleness implies $a > 0$, $b > ae/2$.

Solution. (a) The projection of Y is finite and surjective; its degree is $a = Yf > 0$. Let $\tilde{Y}$ be its normalization, of genus h . Adjunction and normalization give

$$2h - 2 \leq 2p_a(Y) - 2 = (a - 1)(2b - ae) + a(2g - 2). \quad (1)$$

If $a = 1$, the finite degree-one map $Y \rightarrow C$ is an isomorphism: on affine opens its integral coordinate ring lies in $K(C)$, and the base ring is integrally closed. Thus Y is a section. Writing its line bundle as $\mathcal{O}_X(C_0) \otimes \pi^*B$, with $\deg B = b$, its defining section gives a nonzero map $B^{-1} \rightarrow \mathcal{E}$. Saturation and normalization imply $\deg B^{-1} \leq 0$, hence $b \geq 0$.

If $2 \leq a < p$, the map $\tilde{Y} \rightarrow C$ is separable, since a nontrivial inseparable degree is divisible by p . Hurwitz gives $2h - 2 \geq a(2g - 2)$. Together with (1), this yields $2b - ae \geq 0$.

For arbitrary a , factor the function-field extension into a purely inseparable part followed by a separable part. The former preserves genus, and Hurwitz for the latter gives $h \geq g$, since $g \geq 2$. For $a > 1$, (1) therefore implies

$$0 \leq (a - 1)(2b - ae + 2g - 2),$$

which is the last bound.

(b) Assume the sufficient inequalities. Then $Df = a > 0$ and $D^2 = a(2b - ae) > 0$. Also $DC_0 = b - ae > 0$, because $e < 0$. Consider any other integral curve $Y \equiv rC_0 + sf$. If $r = 1$, part (a) gives

$$DY = as + b - ae \geq b - ae > 0.$$

If $2 \leq r < p$, it gives

$$DY = as + rb - are \geq r(b - ae/2) > 0.$$

If $r \geq p$, it gives

$$DY \geq r(b - ae/2) - a(g - 1) > a(g - 1)(r/p - 1) \geq 0.$$

Thus D has positive square and positive intersection with every integral curve, so Nakai–Moishezon makes it ample.

Conversely, an ample divisor has $Df = a > 0$ and $D^2 = a(2b - ae) > 0$, whence $b > ae/2$. $\square$

Exercise V.2.15

Permitted context.

Work over an algebraically closed field k of characteristic three, and put

$$C = V(F) \subset \mathbb{P}^2, \quad F = x^3y + y^3z + z^3x.$$

Its partial derivatives are z^3, x^3, y^3 , so C is smooth and has genus three. We use the projective-space Čech calculation

$$H^1(C, \mathcal{O}_C) \simeq H^2(\mathbb{P}^2, \mathcal{O}(-4)),$$

where a Laurent monomial represents zero if any of its three exponents is nonnegative. Extensions of a line bundle L by $\mathcal{O}_C$ are classified by $H^1(C, L^{-1})$. For a normalized ruled surface, $C_0^2 = -e$, $C_0f = 1$, $f^2 = 0$, and $K_X \equiv -2C_0 + (2g - 2 - e)f$. We use adjunction and the fact that a degree- p purely inseparable map of smooth projective curves over a perfect field is relative Frobenius, up to isomorphism, and preserves genus.

Write F_C for absolute Frobenius. It fixes points and satisfies $F_C^*L = L^{\otimes 3}$; its cohomology action is semilinear. The coordinate-cubing endomorphism of this curve, which is defined over $\mathbb{F}_3$, is k -linear. Its cohomology matrix is the same, but it need not fix an arbitrary k -point. We use absolute Frobenius for the expressions involving P and $3P$.

Problem. (a) Prove that Frobenius acts as zero on $H^1(C, \mathcal{O}_C)$. (b) For any $P \in C$, find a nonzero $\xi \in H^1(C, \mathcal{O}_C(-P))$ killed by pullback to $H^1(C, \mathcal{O}_C(-3P))$. (c) Let

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(P) \longrightarrow 0$$

be its extension, and put $X = \mathbb{P}(\mathcal{E})$. Construct an integral curve $Y \equiv 3C_0 - 3f$ whose projection to C is purely inseparable, and use it to show that $2C_0$ is not ample despite satisfying the numerical inequalities in Hartshorne's Proposition V.2.21(b), a textbook result rather than an exercise in this collection. Check the textbook's additional assertion that Y is nonsingular.

Solution. (a) A basis for the indicated H^2 consists of

$$\frac{1}{x^2yz}, \quad \frac{1}{xy^2z}, \quad \frac{1}{xyz^2}.$$

Under this identification Frobenius sends a class u to F^2u^3 . Indeed, cubing the exact sequence defined by multiplication by F first gives the sequence defined by F^3 ; multiplication by F^2 then compares its degree-12 term to the original degree-4 term. In characteristic three,

$$F^2 = x^6y^2 + y^6z^2 + z^6x^2 + 2x^3y^4z + 2xy^3z^4 + 2x^4yz^3.$$

After division by $x^6y^3z^3$, every summand has at least one nonnegative exponent, and hence vanishes in Čech cohomology. Cyclic permutation proves the same for the other two basis vectors. Thus the semilinear action, and also the k -linear coordinate-Frobenius action, is zero.

(b) From $0 \rightarrow \mathcal{O}_C(-P) \rightarrow \mathcal{O}_C \rightarrow k(P) \rightarrow 0$ and the isomorphism on constants we get

$$H^1(\mathcal{O}_C(-P)) \simeq H^1(\mathcal{O}_C), \quad h^1(\mathcal{O}_C(-P)) = 3.$$

The composite of Frobenius pullback with

$$H^1(\mathcal{O}_C(-3P)) \rightarrow H^1(\mathcal{O}_C)$$

is zero by (a). The kernel of this last arrow has dimension two: it is the cokernel of the constants in $H^0(\mathcal{O}_{3P})$, a three-dimensional space. Thus a semilinear map from a three-dimensional space has image in a two-dimensional space. Since k is perfect, semilinear rank-nullity applies, and there is a nonzero ξ in its kernel.

(c) Set $L = \mathcal{O}_C(P)$. The nonzero extension class makes $\mathcal{E}$ nonsplit. It is normalized. To see this, a positive-degree line subbundle M cannot map nontrivially to $\mathcal{O}_C$, so it maps nontrivially to L . Its degree is at most one. If it is positive, that map has degree-zero divisor and is an isomorphism; it would split the extension. This is impossible. The given inclusion of $\mathcal{O}_C$ supplies the required section. Consequently $e = -1$.

Since $F_C^*\xi = 0$,

$$F_C^*\mathcal{E} \simeq \mathcal{O}_C \oplus L^3.$$

The local cube map is an $\mathcal{O}_C$ -linear inclusion $F_C^*\mathcal{E} \rightarrow \text{Sym}^3\mathcal{E}$, sending $a \otimes v$ to av^3 . Compose it with the splitting inclusion of L^3 . This gives a section of $\text{Sym}^3\mathcal{E} \otimes L^{-3}$, or equivalently a divisor

$$Y \in |\mathcal{O}_X(3) \otimes \pi^*L^{-3}|, \quad Y \equiv 3C_0 - 3f.$$

With a local frame e_0, e_1 adapted to the original extension, its equation is $e_1^3 + ce_0^3 = 0$. The coefficient of e_1^3 is a unit. Therefore Y is finite flat of degree three over C , has no vertical component, and is disjoint from the section $e_0 = 0$.

Over $K(C)$, this cubic is either irreducible or the cube of a linear polynomial. In the latter case its reduced support extends to a section S of the proper ruled surface, and $Y = 3S$, since there are no vertical components. Then $S \equiv C_0 - f$. But a section of this numerical class gives a nonzero map from a line bundle of degree one into $\mathcal{E}$, contradicting normalization as above. Thus the generic polynomial is irreducible. The finite flat algebra of Y embeds into its generic fiber, a field, so Y is integral. Its function-field extension over C is purely inseparable of degree three.

Now $D = 2C_0$ has $a = 2 > 0$ and $b = 0 > ae/2 = -1$, the numerical inequalities in question. However,

$$DY = 2C_0(3C_0 - 3f) = 0,$$

so D is not ample. The characteristic-zero or small-genus hypothesis of that textbook proposition is absent here.

Finally, the smoothness assertion is false. Here $K_X \equiv -2C_0 + 5f$, whence

$$Y^2 = -9, \quad K_X Y = 15, \quad p_a(Y) = 1 + \frac{-9 + 15}{2} = 4.$$

The normalization of Y maps purely inseparably of degree three to the smooth genus-three curve C ; it therefore has genus three. The difference between arithmetic and geometric genus is one, so Y is singular, with total δ -invariant one. This correction leaves the non-amenability example intact. $\square$

Exercise V.2.16

Permitted context.

Let C be a nonsingular affine curve over an algebraically closed field. Coherent sheaves on an affine scheme are generated by finitely many global sections and have no higher cohomology. On a nonsingular curve, torsion-free coherent sheaves are locally free. The Grothendieck group $K(C)$ has the relations $[\mathcal{F}] = [\mathcal{F}'] + [\mathcal{F}']$ for short exact sequences of coherent sheaves. Rank and determinant are additive and multiplicative, respectively, in such sequences, so equality of classes of vector bundles implies equality of their ranks and determinants. One can equivalently compute this group using vector bundles, since a coherent sheaf on a regular affine curve has a finite locally free resolution.

Problem. Prove that vector bundles of the same rank on C are isomorphic if and only if their classes in $K(C)$ coincide. Give a counterexample for a projective curve.

Solution. We prove that a rank- r bundle E , for $r \geq 1$, satisfies

$$E \simeq \mathcal{O}_C^{r-1} \oplus \det E. \quad (1)$$

Suppose $r \geq 2$, and choose a finite-dimensional space W of global sections generating E . For each point P , the subspace of sections vanishing at P has codimension r in W . Consequently the incidence variety of pairs (P, s) with $s(P) = 0$ has dimension $1 + \dim W - r < \dim W$. Its image has closure of dimension smaller than $\dim W$. A section outside that closure vanishes nowhere, and gives an exact sequence

$$0 \longrightarrow \mathcal{O}_C \longrightarrow E \longrightarrow Q \longrightarrow 0$$

with Q locally free of rank $r-1$. The sequence splits, because its extension group is $H^1(C, Q^\vee) = 0$. Induction, starting with rank one, proves (1); the determinant of the remaining line summand must be $\det E$.

If $[E] = [E']$, their determinants and ranks agree, and (1) gives an isomorphism. The converse follows immediately from the definition of $K(C)$. The zero-rank case is immediate as well.

On $\mathbb{P}^1$, the Euler sequence

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow \mathcal{O}^2 \longrightarrow \mathcal{O}(1) \longrightarrow 0$$

shows that $\mathcal{O}^2$ and $\mathcal{O}(-1) \oplus \mathcal{O}(1)$ have equal classes in $K(\mathbb{P}^1)$. They are not isomorphic: after twisting by $\mathcal{O}(-1)$, their spaces of sections have dimensions zero and one, respectively. $\square$

Exercise V.2.17**Permitted context.**

Work over an algebraically closed field. For a smooth curve in a smooth threefold, the conormal sheaf I/I^2 is locally free of rank two. We use the splitting theorem for bundles on $\mathbb{P}^1$, and the calculations

$$h^0(\mathcal{O}(d)) = \max(d+1, 0), \quad H^1(\mathcal{O}(-2)) \simeq k.$$

On the two standard affine charts with coordinate ζ and ζ^{-1} , the cocycle ζ^{-1} generates the last group.

Problem. (a) Determine the splitting of the conormal bundle of the twisted cubic, pulled back to $\mathbb{P}^1$. (b) Do the same for the rational quartic parametrized by $[s : t] \mapsto [s^4 : s^3t : st^3 : t^4]$, keeping track of the characteristic.

Solution. Treat both curves with

$$\phi_n([s : t]) = [s^{n+1} : s^n t : s t^n : t^{n+1}], \quad n = 2, 3.$$

The two charts $X_0 \neq 0$ and $X_3 \neq 0$ cover the curve. On the first, write $x = X_1/X_0$, $y = X_2/X_0$, $z = X_3/X_0$. The parameter is $x = t/s$, and the ideal is generated by

$$u = y - x^n, \quad v = z - xy.$$

On the second put $A = X_0/X_3$, $B = X_1/X_3$, $C = X_2/X_3$. The ideal is generated by

$$u' = B - C^n, \quad v' = A - BC.$$

These equations also show directly that the parametrizations are closed immersions.

On the overlap let $\zeta = x$. Substitute $y = \zeta^n + u$ and $z = \zeta^{n+1} + \zeta u + v$, and discard terms in $(u, v)^2$. The transition is

$$\begin{aligned} v' &= \zeta^{-2n-2}v, \\ u' &= -\zeta^{-2n}u + (n-1)\zeta^{-2n-1}v. \end{aligned} \tag{1}$$

For the first identity use $A - BC = (z - xy)/z^2$. For the second expand $x/z - y^n/z^n$ to first order; the coefficient of u is $-\zeta^{-2n}$, and that of v is $(n-1)\zeta^{-2n-1}$. This calculation is valid in every characteristic.

If $N^* = \phi_n^*(I/I^2)$, equation (1) gives

$$0 \longrightarrow \mathcal{O}(-2n-2) \longrightarrow N^* \longrightarrow \mathcal{O}(-2n) \longrightarrow 0. \tag{2}$$

After twisting by $\mathcal{O}(2n)$, the difference between the two local lifts of the quotient's constant section is, up to sign, $(n-1)\zeta^{-1}$. Thus (2) has extension class $n-1$, up to a nonzero scalar, in $H^1(\mathcal{O}(-2))$.

If $n-1 = 0$ in k , (2) splits. Otherwise put $G = N^*(2n)$. The boundary map $H^0(\mathcal{O}) \rightarrow H^1(\mathcal{O}(-2))$ is an isomorphism, so $H^0(G) = 0$. Its degree is -2 . Writing $G = \mathcal{O}(a) \oplus \mathcal{O}(b)$, the vanishing gives $a, b \leq -1$; their sum is -2 , so both equal -1 .

(a) For $n = 2$, the extension class is nonzero in every characteristic. Therefore

$$\phi_2^*(I/I^2) = \mathcal{O}(-5) \oplus \mathcal{O}(-5).$$

(b) For $n = 3$, the class is nonzero exactly when the characteristic is different from two. The answer is consequently

$$\phi_3^*(I/I^2) \simeq \begin{cases} \mathcal{O}(-7) \oplus \mathcal{O}(-7), & \text{char } k \neq 2, \\ \mathcal{O}(-6) \oplus \mathcal{O}(-8), & \text{char } k = 2. \end{cases}$$

□

V.3 Monoidal Transformations

Exercise V.3.1

Permitted context.

Work over an algebraically closed field. If X is smooth and $Y \subset X$ is a smooth closed subvariety of positive codimension r , then locally in the étale topology the pair is $(\mathbb{A}^n, \mathbb{A}^{n-r})$, with the second space a coordinate subspace. Blowing up commutes with flat base change, as do higher direct images of quasi-coherent sheaves under quasi-compact separated morphisms. The blow-up of $\mathbb{A}^r$ at the origin is the total space of $\mathcal{O}_{\mathbb{P}^{r-1}}(-1)$. We use the cohomology of $\mathcal{O}(m)$ on projective space, the Leray spectral sequence, and $p_a(X) = (-1)^{\dim X}(\chi(\mathcal{O}_X) - 1)$.

Problem. For a smooth projective variety X and a smooth proper closed center Y , prove that its blow-up $\pi : \tilde{X} \rightarrow X$ has the same arithmetic genus.

Solution. We show

$$\pi_* \mathcal{O}_{\tilde{X}} = \mathcal{O}_X, \quad R^i \pi_* \mathcal{O}_{\tilde{X}} = 0 \quad (i > 0). \quad (1)$$

This can be checked on the stated étale local models. For the model at the origin of $\mathbb{A}^r$, write $Z = \text{Tot}(\mathcal{O}_{\mathbb{P}^{r-1}}(-1))$. Its projection q to $\mathbb{P}^{r-1}$ is affine, with

$$q_* \mathcal{O}_Z = \bigoplus_{m \geq 0} \mathcal{O}(m).$$

A finite standard affine cover computes cohomology and commutes with this direct sum. For $i > 0$ all $H^i(\mathbb{P}^{r-1}, \mathcal{O}(m))$ with $m \geq 0$ vanish. Hence $H^i(Z, \mathcal{O}_Z) = 0$. Also

$$H^0(Z, \mathcal{O}_Z) = \bigoplus_{m \geq 0} H^0(\mathbb{P}^{r-1}, \mathcal{O}(m)) = k[x_1, \dots, x_r]$$

as a graded algebra, compatibly with the blow-up map. Since its target is affine, these calculations give (1) for that model. The codimension-one case is simply the identity blow-up and satisfies (1) as well.

Taking a product with $\mathbb{A}^{n-r}$ and then the étale base changes defining the pair (X, Y) proves (1) in general. The assertion is already the identity away from the center, so these local checks cover X . Leray now yields

$$H^i(\tilde{X}, \mathcal{O}_{\tilde{X}}) \simeq H^i(X, \mathcal{O}_X) \quad \text{for every } i.$$

Thus the Euler characteristics agree. The dimensions agree because the blow-up is birational, and the formula for p_a proves the claim. An empty center also gives the identity. $\square$

Exercise V.3.2**Permitted context.**

Let X be a smooth projective surface over an algebraically closed field, and let $\pi : \tilde{X} \rightarrow X$ blow up a point P , with exceptional curve E . For a curve C of multiplicity $m_P(C)$,

$$\tilde{C} = \pi^*C - m_P(C)E, \quad E^2 = -1, \quad \pi^*C E = 0.$$

Pullback preserves intersections of divisors. Curves with no common component have nonnegative intersection, equal to the sum of their positive local intersection lengths at intersection points. An infinitely near point is a point on a surface obtained by successive point blow-ups.

Problem. For curves C, D with no common component, prove

$$\tilde{C} \tilde{D} = CD - m_P(C)m_P(D).$$

Deduce that CD is the sum of the products of the multiplicities at all common ordinary and infinitely near points of the successive strict transforms.

Solution. Put $r = m_P(C)$ and $s = m_P(D)$. Bilinearity gives

$$(\pi^*C - rE)(\pi^*D - sE) = CD + rsE^2 = CD - rs.$$

The mixed terms vanish by the projection formula.

Whenever the current strict transforms meet, blow up one of their intersection points. Their intersection number decreases by the positive integer equal to the product of their multiplicities there. The strict transforms still have no common component, so the new intersection number remains nonnegative. This process must terminate after at most CD blow-ups. At termination the strict transforms are disjoint: otherwise another positive decrease would be possible. Their final intersection is therefore zero.

Telescope the displayed formula through this sequence. It gives

$$CD = \sum_Q m_Q(C_Q)m_Q(D_Q),$$

where C_Q, D_Q denote the strict transforms at the stage containing the center Q . The centers include each original intersection and every subsequent common infinitely near point until the branches separate. Thus this is exactly the asserted sum, and it is finite. $\square$

Exercise V.3.3**Permitted context.**

Let $\pi : \tilde{X} \rightarrow X$ blow up a point P on a smooth projective surface over an algebraically closed field, with exceptional curve E . We use $E^2 = -1$, $\pi^*D \cdot E = 0$, and $\tilde{C} = \pi^*C - m_P(C)E$. For an integral projective curve, a general hyperplane through a point cuts there with intersection multiplicity equal to the curve's multiplicity; the total hyperplane intersection is its degree. The Nakai–Moishezon criterion on a surface says that positive square and positive intersection with every integral curve imply ampleness.

Problem. If D is very ample on X , prove that $2\pi^*D - E$ is ample on $\tilde{X}$.

Solution. Use D to embed X into projective space. For every integral curve C through P , a general hyperplane through P , not containing C , gives

$$m_P(C) \leq DC. \quad (1)$$

For curves avoiding P , put $m_P(C) = 0$; then (1) holds as well.

Set $A = 2\pi^*D - E$. Since D^2 is the positive integer degree of the embedded surface,

$$A^2 = 4D^2 - 1 > 0, \quad AE = 1 > 0.$$

Every other integral curve of $\tilde{X}$ is the strict transform of an integral curve C on X . Its intersection is

$$A\tilde{C} = 2DC - m_P(C) \geq DC > 0$$

by (1) and very ampleness. Thus all conditions of the surface Nakai–Moishezon criterion hold, and A is ample. $\square$

Exercise V.3.4**Permitted context.**

Let $(A, \mathfrak{m}, \kappa)$ be a nonzero noetherian local ring and write ℓ for module length. The associated graded ring

$$G = \text{gr}_{\mathfrak{m}} A = \bigoplus_{j \geq 0} \mathfrak{m}^j / \mathfrak{m}^{j+1}$$

is a finitely generated standard graded κ -algebra. For such an algebra of dimension d , the sum of its graded dimensions below l is eventually a polynomial of degree d , with positive leading coefficient; this is the cumulative form of the Hilbert-polynomial theorem. We use homogeneous systems of parameters for graded algebras, systems of parameters in local rings, Krull's height theorem, and Nakayama's lemma. For a smooth surface local ring R at a closed point, $\text{gr}_{\mathfrak{m}} R = \kappa[u, v]$. An r -dimensional projective variety of degree d has Hilbert polynomial with leading term $dt^r/r!$.

Problem. (a) Show that $\ell(A/\mathfrak{m}^l)$ is eventually a polynomial $P_A(l)$ with rational coefficients. (b) Prove $\deg P_A = \dim A$. (c) Define the multiplicity as $(\dim A)!$ times the leading coefficient. (d) At a closed point of a curve on a smooth surface, show that this equals the order of a local equation. (e) Show that the vertex of the affine cone over a degree- d projective variety has multiplicity d .

Solution. (a) The filtration by powers of $\mathfrak{m}$ gives

$$\ell(A/\mathfrak{m}^l) = \sum_{j=0}^{l-1} \dim_{\kappa} G_j.$$

The cumulative Hilbert-polynomial theorem therefore gives the required polynomial, of degree $d = \dim G$.

(b) We verify $d = \dim A$, rather than assuming it. Put $n = \dim A$, and choose a parameter ideal $\mathfrak{q} = (a_1, \dots, a_n)$. Since $\mathfrak{q} \subseteq \mathfrak{m}$, and $\mathfrak{q}^j/\mathfrak{q}^{j+1}$ is generated over $A/\mathfrak{q}$ by the degree- j monomials in its n generators,

$$\ell(A/\mathfrak{m}^l) \leq \ell(A/\mathfrak{q}^l) \leq \ell(A/\mathfrak{q}) \binom{l+n-1}{n}.$$

Thus $d \leq n$. If $n = 0$, the ring is artinian and the eventual polynomial is the positive constant $\ell(A)$, which already proves the assertion in that case.

For the reverse inequality choose homogeneous parameters $g_1, \dots, g_d$ in G . Lift them to elements $b_i \in \mathfrak{m}$ of their respective orders, and put $J = (b_1, \dots, b_d)$. The quotient $G/(g_1, \dots, g_d)$ is finite dimensional. In sufficiently high degree N , every leading form lies in the ideal of the g_i ; lifting its expression gives

$$\mathfrak{m}^N \subseteq J + \mathfrak{m}^{N+1}.$$

In A/J , this says $\overline{\mathfrak{m}}^N = \overline{\mathfrak{m}} \overline{\mathfrak{m}}^N$. Nakayama gives $\overline{\mathfrak{m}}^N = 0$. Hence J is $\mathfrak{m}$ -primary. The height theorem, applied to its d generators, gives $n \leq d$. Together with the first bound, this proves $\deg P_A = n$.

(c) If $P_A(l) = cl^n + \text{lower terms}$, define

$$\mu(A) = n!c.$$

For a scheme point P , put $\mu_P(X) = \mu(\mathcal{O}_{X,P})$. The leading coefficient is positive by (a). It gives a positive integer: the n -th finite difference of an integer-valued eventual polynomial is the constant $n!c$. In dimension zero this is the length of the ring.

(d) Let R be the regular local ring of the surface, and $f \in R$ a local equation of the curve, of order r . Its initial form f_r is homogeneous of degree r in $\kappa[u, v]$. Since this graded ring is a domain, orders add under multiplication, so the initial ideal of (f) is exactly (f_r) . Consequently

$$\text{gr}(R/(f)) = \kappa[u, v]/(f_r).$$

Multiplication by f_r is injective in the polynomial ring. The degree- j component of the quotient has dimension $(j+1) - \max(j-r+1, 0) = \min(j+1, r)$. For $l \geq r$, summation yields

$$\ell((R/(f))/\mathfrak{m}^l(R/(f))) = rl - \frac{r(r-1)}{2}.$$

Its one-dimensional multiplicity is r , which is exactly the multiplicity used for curves in the surface.

(e) Let S be the homogeneous coordinate ring of the projective variety, of dimension $r+1$, and let $\mathfrak{n} = S_+$ be the vertex ideal. Because S is standard graded,

$$\text{gr}_{\mathfrak{n}S_{\mathfrak{n}}}(S_{\mathfrak{n}}) \simeq S.$$

Indeed, $\mathfrak{n}^j/\mathfrak{n}^{j+1} = S_j$, and localization does not change these vector spaces, which are annihilated by $\mathfrak{n}$. Thus the vertex Hilbert–Samuel function is $\sum_{j < l} \dim_k S_j$. The summands have leading term $dj^r/r!$, so the sum has leading term $dl^{r+1}/(r+1)!$. Its multiplicity is therefore d . $\square$

Exercise V.3.5**Permitted context.**

Work first over an algebraically closed field of characteristic different from two. For a reduced plane curve germ, δ drops by $\binom{m}{2}$ when a point of multiplicity m is blown up; the remaining contributions are those of the strict transform over that point. A germ $u^2 = v^n$ has $\delta = \lfloor n/2 \rfloor$: successive blow-ups replace n by $n - 2$, ending in a smooth branch or a node. A unit with constant term one has a square root in $k[[v]]$ in this characteristic. For a degree- r integral plane curve,

$$g(\tilde{C}) = \frac{(r-1)(r-2)}{2} - \sum_P \delta_P.$$

Problem. Let $a_1, \dots, a_r$ be distinct, with $r \geq 5$, and let C be the projective closure of

$$y^2 = \prod_{i=1}^r (x - a_i).$$

Describe its singular point at infinity, calculate its δ -invariant and normalization genus, and obtain hyperelliptic curves of every genus at least two. Explain why characteristic two must be excluded from this construction.

Solution. The homogeneous equation is

$$Y^2 Z^{r-2} = \prod_{i=1}^r (X - a_i Z).$$

Its only point at infinity is $P = [0 : 1 : 0]$. On the chart $Y = 1$, write $u = X/Y$, $v = Z/Y$. The equation is

$$v^{r-2} = \prod_i (u - a_i v).$$

Its lowest degree is $r - 2$, so P has multiplicity $r - 2 \geq 3$ and is singular.

Blow up using $v = uw$. After removing the exceptional factor u^{r-2} , the strict transform has equation

$$w^{r-2} = u^2 h(w), \quad h(w) = \prod_i (1 - a_i w).$$

It meets the exceptional curve $u = 0$ only at $Q = (0, 0)$. The formal change $u' = u\sqrt{h(w)}$ identifies its germ with $(u')^2 = w^{r-2}$. Therefore

$$\delta_P = \binom{r-2}{2} + \left\lfloor \frac{r-2}{2} \right\rfloor. \quad (1)$$

The affine curve is smooth: at a zero of the right side, squarefreeness makes its derivative nonzero, and elsewhere the derivative $2y$ is nonzero. Thus (1) accounts for every singularity. Subtracting it from the plane arithmetic genus gives

$$g(\tilde{C}) = \left\lfloor \frac{r-1}{2} \right\rfloor.$$

Equivalently, $\delta_P = 2(m-1)^2$ for $r = 2m$, and $\delta_P = 2m(m-1)$ for $r = 2m+1$.

The function x defines a separable degree-two map of the normalization to $\mathbb{P}^1$, because the squarefree polynomial is not a square in $k(x)$. Hence the normalization is hyperelliptic. For any $g \geq 2$, choosing $r = 2g + 1$, or $2g + 2$, gives the required genus.

In characteristic two the same assertion is false. If $f(x) = \sum c_i x^i$, choose square roots of the coefficients and put $h(t) = \sum \sqrt{c_i} t^i$. Then $x = t^2$, $y = h(t)$ parametrizes $y^2 = f(x)$. Since f is squarefree, $h(t) \notin k(t^2)$. The extension $k(t)/k(t^2)$ has degree two, so

$$k(t^2, h(t)) = k(t).$$

Thus the normalization is rational, rather than of the stated positive genus. The numerical δ -formula need not survive either: for $f = x^6 + x + 1$, the derivative is one, so the affine curve is smooth and its unique point at infinity has $\delta = 10$, the full plane arithmetic genus. Formula (1) would give eight. The characteristic restriction is therefore essential. $\square$

Exercise V.3.6

Permitted context.

For plane curve germs over an algebraically closed field, analytic isomorphism means isomorphism of their completed local k -algebras. The discrete resolution equivalence of V.3.9.4 records the local branches, exceptional components, and their multiplicities and incidences at each point blow-up, without recording cross-ratios of intersection points on an exceptional line. Blow-up charts are obtained by substitutions $y = xv$ or $x = yu$, followed by removing the exceptional factor. A formal coordinate substitution with invertible linear part is an automorphism of $k[[x, y]]$. Four distinct points on $\mathbb{P}^1$ are projectively equivalent as unordered sets only if their cross-ratios differ by one of the six transformations generated by $\lambda \mapsto 1 - \lambda$ and $\lambda \mapsto 1/\lambda$.

Problem. Show that analytically isomorphic plane curve singularities have equivalent embedded resolution data. Show that the converse fails.

Solution. Let the singular completed local rings be $k[[x, y]]/(f)$ and $k[[u, v]]/(g)$. An isomorphism between them is an isomorphism on their two-dimensional cotangent spaces. Lift the images of x, y to $k[[u, v]]$. Their linear terms are independent, so the resulting formal substitution is an automorphism of the two-variable power series rings. Its induced quotient map is the given isomorphism, and consequently it maps (f) onto (g) . In particular it preserves orders, tangent cones, and the branches of the completed curve.

Blowing up the maximal ideal is functorial under this formal isomorphism. In the explicit charts, the corresponding tangent directions give isomorphic completed local rings of the strict transforms, together with their exceptional ideals. Thus multiplicities, branch intersections, and incidences with exceptional components agree. Repeat at corresponding centers of an embedded resolution. The same argument at each step identifies the entire discrete resolution data, which proves the asserted implication. Smooth germs cause no exception; their local resolution data are already trivial.

For the converse, over $\mathbb{C}$ consider

$$f_\lambda = xy(x - y)(x - \lambda y), \quad \lambda \notin \{0, 1\}.$$

Each germ is an ordinary quadruple point. One blow-up separates its four smooth branches: they meet the exceptional line transversely at four distinct points. The multiplicities and incidence data are identical for every λ . Thus all these germs are equivalent in the discrete sense.

If two were analytically isomorphic, the linear part of the ambient formal isomorphism constructed above would identify their tangent cones. It would therefore carry the four tangent directions of one to those of the other by a projective linear transformation of $\mathbb{P}^1$. For $\lambda = 2$, the unordered cross-ratio orbit is $\{2, 1/2, -1\}$, whereas 3 is outside that orbit. Hence the germs for $\lambda = 2$ and $\lambda = 3$ are not analytically isomorphic, although their embedded resolutions are equivalent. $\square$

Exercise V.3.7

Permitted context.

Work over an algebraically closed field. The strict transform under a point blow-up is obtained in a chart by substitution and division by the exceptional factor of order equal to the curve's multiplicity. An embedded resolution requires the reduced total transform to have smooth components and only transverse double intersections. In particular, an ordinary triple intersection requires one more blow-up. For a reduced plane curve germ,

$$\delta = \sum_Q \binom{m_Q}{2},$$

where m_Q is the multiplicity of the strict transform at each ordinary or infinitely near singular point. Resolution equivalence preserves the branches, multiplicities, and incidence with exceptional components at every step.

Problem. Give embedded resolutions, calculate δ , and determine the equivalence classes of:

- (a) $x^3 + y^5 = 0$.
- (b) $x^3 + x^4 + y^5 = 0$.
- (c) $x^3 + y^4 + y^5 = 0$.
- (d) $x^3 + y^5 + y^6 = 0$.
- (e) $x^3 + xy^3 + y^5 = 0$.

Solution. Let E_i denote the exceptional component of the i -th blow-up in each calculation. All five curves have multiplicity three at the origin and unique tangent line $x = 0$. The first chart is $x = yu$, with exceptional line $E_1 : y = 0$. The strict-transform equations are

case	equation after the first blow-up	(1)
(a)	$u^3 + y^2$	
(b)	$u^3 + yu^4 + y^2$	
(c)	$u^3 + y + y^2$	
(d)	$u^3 + y^2 + y^3$	
(e)	$u^3 + yu + y^2$	

Each meets E_1 only at its origin.

(a) At that point the strict transform has multiplicity two and tangent $y = 0$. Blow up with $y = uv$. Its new equation is $u + v^2 = 0$, with $E_2 : u = 0$ and the strict transform of E_1 given by $v = 0$. The curve is smooth but tangent to E_2 at the intersection of these two exceptional curves. Blow up with $u = vw$. The curve becomes $w + v = 0$, while the exceptional components through the point are $E_3 : v = 0$ and $E_2 : w = 0$. These form three distinct tangent lines together with the curve. One final blow-up separates that ordinary triple point and gives normal crossings. The strict-curve multiplicities at the four centers are 3, 2, 1, 1. Hence

$$\delta_a = \binom{3}{2} + \binom{2}{2} = 4.$$

(b) Use the same successive charts. After the second and third blow-ups, the equations are

$$u + u^3v + v^2 = 0, \quad w + v + v^3w^3 = 0.$$

Their relevant linear terms and tangencies are the same as in (a): the second transform is tangent to E_2 , and the third forms an ordinary triple point with E_2, E_3 . A fourth blow-up resolves it. Thus the same multiplicities and incidences occur, $\delta_b = 4$, and (b) is equivalent to (a).

(c) The first equation in (1) for this case is already smooth, but it has contact order three with $E_1 : y = 0$. Blow up with $y = uv$; the strict transform is

$$u^2 + v + uv^2 = 0.$$

It is smooth and tangent to $E_1 : v = 0$ at the intersection with $E_2 : u = 0$. Next set $v = uw$; the equation becomes

$$u + w + u^2w^2 = 0.$$

This is an ordinary triple intersection with $E_3 : u = 0$ and $E_1 : w = 0$. A fourth blow-up gives normal crossings. The strict-curve multiplicities at the centers are 3, 1, 1, 1, so

$$\delta_c = \binom{3}{2} = 3.$$

In particular (c) is not equivalent to (a) or (b).

(d) The same charts as in (a) give, after the second and third blow-ups,

$$u + v^2 + uv^3 = 0, \quad w + v + v^3w = 0.$$

Again the second transform is smooth and tangent to E_2 , and the third gives the same ordinary triple point. The fourth blow-up resolves it. Thus $\delta_d = 4$, and (d) is equivalent to (a) and (b).

(e) The first transform has an ordinary node: its quadratic tangent cone is $y(u + y)$, with two distinct tangent directions in every characteristic. The line $E_1 : y = 0$ is tangent to one branch. Blow up with $y = uv$. After division by u^2 the equation is

$$u + v + v^2 = 0.$$

On $E_2 : u = 0$, it has two points, $v = 0$ and $v = -1$. At $v = -1$ it is smooth and transverse to E_2 ; no other exceptional curve passes there. At $v = 0$ it forms an ordinary triple point with $E_2 : u = 0$ and $E_1 : v = 0$, since its tangent is $u + v = 0$. One further blow-up at that point gives normal crossings. The multiplicities at the centers are 3, 2, 1, giving

$$\delta_e = \binom{3}{2} + \binom{2}{2} = 4.$$

This germ has two branches, whereas each of (a)–(d) has one, as their resolutions show. It is therefore inequivalent to all four.

The equivalence classes are consequently

$$\{(a), (b), (d)\}, \quad \{(c)\}, \quad \{(e)\},$$

with δ -values 4, 3, 4, respectively. All the chart calculations use unit linear coefficients and distinct coordinate directions; they remain valid in every characteristic. $\square$

Exercise V.3.8**Permitted context.**

Work over an algebraically closed field. The multiplicity of a reduced plane germ is the least degree in its local equation. After a point blow-up, remove that power of the exceptional equation to get the strict transform. The delta invariant is the sum of $\binom{m_Q}{2}$ over ordinary and infinitely near singular points of the strict transforms. Embedded resolution equivalence additionally preserves individual branch multiplicities and their incidences with exceptional components. The germ $x^a - y^b$, with coprime positive integers a, b , has one branch, parametrized by $x = t^b, y = t^a$; this holds in every characteristic.

Problem. Check the textbook's claim that the following two germs have the same multiplicities and configuration of infinitely near singular points, but are not equivalent:

- (a) $x^4 - xy^4 = 0$.
- (b) $x^4 - x^2y^3 - x^2y^5 + y^8 = 0$.

If the claimed numerical agreement fails, identify it explicitly and give a corrected pair that illustrates the intended distinction.

Solution. (a) The germ is $x(x^3 - y^4) = 0$, of multiplicity four, with a smooth branch and a branch of multiplicity three. Blow up with $x = yu$. Removing y^4 gives

$$u(u^3 - y) = 0.$$

Its only point on the exceptional line $y = 0$ is the origin. The strict transform has multiplicity two there, with two smooth branches having distinct tangents $u = 0$ and $y = 0$. Blowing up that node separates the branches and makes the strict curve nonsingular. Consequently its singular-point multiplicities are 4, 2, and

$$\delta_a = \binom{4}{2} + \binom{2}{2} = 7. \quad (1)$$

Extra blow-ups needed for normal crossings with the exceptional components do not change this delta calculation.

(b) Factor the second polynomial:

$$(x^2 - y^3)(x^2 - y^5).$$

It has two distinct branches, both of multiplicity two, and total multiplicity four. Under $x = yu$, the strict transform is

$$(u^2 - y)(u^2 - y^3) = 0.$$

At its only exceptional point, the lowest term is $-u^2y$, of degree three. Thus its infinitely near multiplicity is three, already different from (a). One branch, $u^2 - y = 0$, is smooth with tangent $y = 0$; the other, $u^2 - y^3 = 0$, is a cusp with tangent $u = 0$. Blow up this point. In the chart $y = uv$, the first branch becomes $u - v = 0$, and the second has no point at that chart's origin. In the other chart $u = yw$, the second becomes $w^2 - y = 0$, while the first has no point at that origin. Both strict branches are now smooth and disjoint. The singular-point multiplicities are therefore 4, 3, and

$$\delta_b = \binom{4}{2} + \binom{3}{2} = 9. \quad (2)$$

Equations (1)–(2) disprove the printed agreement. The printed germs are indeed inequivalent, but even their delta invariants differ.

The intended example is recovered by replacing the exponent four in the term xy^4 of (a) by five:

$$C'_a: \quad x^4 - xy^5 = x(x^3 - y^5) = 0.$$

After $x = yu$, its strict transform is $u(u^3 - y^2) = 0$. It has multiplicity three at the origin, with a smooth branch tangent to $u = 0$ and a cusp tangent to $y = 0$. Blowing up separates the two branches; the cusp becomes smooth, since in the chart $y = uv$ its equation is $u - v^2 = 0$. The other branch is smooth in the complementary chart. Thus C'_a and the germ in (b) each have exactly one infinitely near singular point: their multiplicities are 4, 3, giving $\delta = 9$ in both cases.

Nevertheless they are inequivalent. The original branches of C'_a have multiplicities 1, 3, while those of (b) have multiplicities 2, 2. No correspondence preserving individual branch multiplicities can identify their embedded resolution data. The corrected pair therefore demonstrates exactly why the total multiplicities at infinitely near singular points are weaker than the full embedded resolution invariant. $\square$

V.4 The Cubic Surface in Projective 3-Space

Exercise V.4.1

Permitted context.

Work over an algebraically closed field. The closure of the graph of a rational map defined by generating sections of an ideal sheaf is the blow-up of that base ideal. Multiplying the ideal by an invertible sheaf does not change its blow-up. The quadric $AD = BC$ in $\mathbb{P}^3$ is smooth, and is the Segre image of $\mathbb{P}^1 \times \mathbb{P}^1$. Inverse birational maps have the same graph closure, with its two projections interchanged.

Problem. Blow up two distinct points of $\mathbb{P}^2$. Show that the conics through those points define a morphism to a smooth quadric, and that this morphism identifies the blown-up plane with the quadric blown up at one point.

Solution. Choose the points to be $P_1 = [1 : 0 : 0]$ and $P_2 = [0 : 1 : 0]$. The complete vector space of conics through them has basis xy, xz, yz, z^2 , so its rational map is

$$[x : y : z] \dashrightarrow [xy : xz : yz : z^2]. \quad (1)$$

Near P_1 , where $x = 1$, its base ideal is (y, z) ; near P_2 it is (x, z) . There are no other base points. Consequently its graph closure is exactly $X' = \text{Bl}_{P_1, P_2} \mathbb{P}^2$, and the second projection is a morphism on X' .

The coordinates in (1) satisfy $AD = BC$. On $z \neq 0$, (1) is

$$(u, v) \mapsto [uv : u : v : 1],$$

an isomorphism onto the open subset $D \neq 0$ of that quadric. Thus the morphism is birational onto the entire smooth quadric Q . The inverse rational map is

$$Q \dashrightarrow \mathbb{P}^2, \quad [A : B : C : D] \mapsto [B : C : D]. \quad (2)$$

Its base locus is the single point $q = [1 : 0 : 0 : 0]$. In the chart $A = 1$, the equation is $D = BC$, so the base ideal (B, C, D) equals the maximal ideal (B, C) of that smooth point. Elsewhere the sections in (2) have no common zero. Hence the graph closure of (2) is $\text{Bl}_q Q$.

The two maps are birational inverses, so these graph closures are the same closed subscheme of $\mathbb{P}^2 \times Q$. It follows that

$$X' \simeq \text{Bl}_q Q,$$

compatibly with the conic-system morphism to Q . Its exceptional curve is the strict transform of the line $z = 0$ joining P_1, P_2 , which (1) sends to q . $\square$

Exercise V.4.2

Permitted context.

Let P_1, P_2, P_3 be noncollinear points of $\mathbb{P}^2$ over an algebraically closed field. Their quadratic transformation is resolved on $S = \text{Bl}_{P_1, P_2, P_3} \mathbb{P}^2$. Write l for the pullback of a line and e_i for the exceptional classes. Then $l^2 = 1$, $e_i^2 = -1$, $le_i = e_i e_j = 0$ for $i \neq j$. The other projection to $\mathbb{P}^2$ has line pullback $l' = 2l - e_1 - e_2 - e_3$ and exceptional classes $e'_i = l - e_j - e_k$, where $\{i, j, k\} = \{1, 2, 3\}$. The curve e'_i contracts to the inverse base point Q_i . A strict transform meets the exceptional curve of a point blow-up with intersection equal to its original multiplicity at that point.

Problem. For an integral degree- d plane curve C of multiplicities r_i at P_i , not one of the three contracted joining lines, prove that its transformed curve has

$$d' = 2d - r_1 - r_2 - r_3, \quad r'_i = d - r_j - r_k$$

at the inverse base points. Explain the excluded joining-line cases.

Solution. The strict transform of C on S has class

$$\tilde{C} = dl - r_1 e_1 - r_2 e_2 - r_3 e_3.$$

By assumption it is not exceptional for the second projection. Its image C' is therefore an integral curve, with the same strict transform $\tilde{C}$ on S . Intersecting with the pullback of a target line gives

$$d' = \tilde{C} l' = 2d - r_1 - r_2 - r_3.$$

Intersecting instead with the target exceptional curve gives

$$r'_i = \tilde{C} e'_i = \tilde{C}(l - e_j - e_k) = d - r_j - r_k.$$

No ordinary-singularity assumption was used: the strict-transform and intersection formulas apply to arbitrary singularities. The nonnegativity of each r'_i also follows from Bezout with the line $P_j P_k$, which is not a component of C .

If $C = P_j P_k$, its strict transform is e'_i and is contracted to Q_i . It has no transformed curve whose point multiplicities could satisfy the displayed formulas. Indeed, those expressions give $d' = 0$ and $r'_i = -1$ in this case. Thus the curve formulation implicitly requires excluding the three contracted lines; their actual images are the corresponding points. $\square$

Exercise V.4.3**Permitted context.**

Work over an algebraically closed field. A plane curve point is ordinary if its branches are smooth with distinct tangent directions. A quadratic transformation at three noncollinear points blows up those points and blows down their three joining lines. Blowing down an exceptional curve meeting the strict curve transversely at distinct smooth points creates an ordinary multiple point. At a multiplicity- m point, the sum of the delta invariants of the strict transform above it is $\delta - \binom{m}{2}$. An integral plane curve has only finitely many singular points.

Call a point P a strange center for a curve if every tangent at a general smooth point passes through P . Projection from P , on the normalization, is separable exactly when P is not a strange center: the differential of the pencil coordinate vanishes exactly when its line is tangent. For a separable projection, a general line through P meets the curve away from P transversely at smooth points. General lines without a prescribed center also have this transversality property.

Problem. Show that any integral plane curve can be transformed, by finitely many quadratic transformations, to a curve with only ordinary singularities.

Solution. Preparatory lemma. We first explain how to avoid a possible inseparability obstruction to using a singular point as a center.

Given finitely many points P on an integral curve C , a suitably general quadratic transformation with its three centers outside C has the following properties: it is a local isomorphism at every singular point of C , creates only ordinary new singularities, and makes the images of all the prescribed points non-strange centers. The first two assertions follow by requiring the three contracted joining lines to avoid the singular points and meet C transversely. These are nonempty open conditions on the ordered triples of centers.

Here is why the last assertion is also a nonempty open condition for any fixed P . Choose a smooth point $A \neq P$ and let T be its tangent line. If $P \in T$, choose the three centers off T . The strict image of T is a smooth conic: the quadratic transformation restricts to its degree-two embedding. The images of A, P are distinct points on that conic, and the tangent at the first does not pass through the second. The image curve has that same tangent at the image of A , so its image of P is not a strange center. If $P \notin T$, choose one center on T , outside C , and the other two generically. Then the strict image of T is a line. Arrange that A, P avoid the contracted lines. Their local images show that the image tangent at A does not contain the image of P . In both cases this proves the desired nonempty open condition, since nonincidence of that tangent and point is open. The parameter space of noncollinear triples is irreducible, so the finitely many conditions can be satisfied simultaneously.

The local improvement. Now let P be a nonordinary singular point. Apply such a preparatory transformation if necessary. It preserves all existing nonordinary germs and adds only ordinary ones, and we may suppose P is not a strange center. Choose two further centers Q, R outside the curve so that the lines PQ, PR have directions different from all branch tangents at P , and meet the curve elsewhere transversely at smooth points. Also require QR to meet the curve transversely at smooth points and all three lines to avoid the other singularities. Separability of projection from P and generic transversality make these simultaneous open conditions possible.

Resolve the quadratic transformation at P, Q, R . Since Q, R are off the curve, the only change to its singular germs on the blown-up surface is the point blow-up at P . The strict transforms of PQ, PR do not meet the strict curve above P , because their directions there were excluded. Each of the three lines to be contracted therefore meets the strict curve transversely at distinct smooth

points. Blowing them down creates only ordinary singularities. All remaining germs of the strict curve, including its points above P , are unchanged by these contractions.

Termination. Use the nonnegative integer

$$M(C) = \sum_{\substack{P \text{ singular} \\ P \text{ not ordinary}}} \delta_P.$$

A preparatory transformation leaves M unchanged. The subsequent transformation at a nonordinary point of multiplicity m decreases M by at least $\binom{m}{2} > 0$: the sum of all new delta invariants above that point has this decrease, and any of them that become ordinary are no longer counted. All other new points are ordinary. After finitely many repetitions $M = 0$, which means that every remaining singularity is ordinary. Each step was a quadratic transformation at three suitable ordinary points of the ambient plane, as required. $\square$

Exercise V.4.4

Permitted context.

Work over an algebraically closed field. The nine-point theorem (Theorem V.4.5 in Hartshorne, not Exercise V.4.5 in this collection) says that eight plane points, no four collinear and no seven on a conic, determine a ninth base point for all cubics through them, allowing the stated infinitely near versions. On an integral plane cubic, write $A \circ B$ for the third intersection with the chord AB , using the tangent if $A = B$. On the regular locus this construction extends algebraically to the tangency cases, and $A \circ (A \circ B) = B$, counting intersection multiplicities. Collinearity identities established for distinct intersections therefore extend to all cases where the chord construction is defined.

Problem. (a) Suppose lines cut an integral cubic in P, Q, R and P', Q', R' , respectively. Prove that $P \circ P', Q \circ Q', R \circ R'$ are collinear. (b) If O is a regular inflection point, use (a) to prove associativity of the operation

$$A + B = O \circ (A \circ B)$$

on the regular points.

Solution. (a) First take the intersections distinct and in general position for these constructions. The union of the three chords PP', QQ', RR' meets the cubic in the six original points and in

$$P'' = P \circ P', \quad Q'' = Q \circ Q', \quad R'' = R \circ R'.$$

Let M be the line through P'', Q'' . The cubic which is the union of the two original lines and M passes through eight of these nine intersection points. No four of those eight are collinear and no seven lie on a conic: either event would contradict Bezout with the integral cubic. The nine-point theorem forces this reducible cubic through R'' as well. In the general situation R'' is on neither original line, so it lies on M . The tangent and coincident-point versions follow by the algebraic extension of the chord construction, or by its infinitely near form.

(b) Part (a), applied to the triples $(A, B, A \circ B)$ and $(C, D, C \circ D)$, gives the medial identity

$$(A \circ B) \circ (C \circ D) = (A \circ C) \circ (B \circ D). \quad (1)$$

This uses the fact that the third point on the line through $A \circ C, B \circ D$ is $(A \circ B) \circ (C \circ D)$.

Put $j(A) = O \circ A$. The chord rule gives $j^2 = \text{id}$; the inflection hypothesis gives $O \circ O = O$, so $j(O) = O$. Taking $C = D = O$ in (1) yields

$$j(A \circ B) = j(A) \circ j(B). \quad (2)$$

Thus j preserves the chord operation. Since $A + B = j(A \circ B)$, applying (2) twice and then (1) shows

$$\begin{aligned} (A + B) + (C + D) &= (A \circ B) \circ (C \circ D) \\ &= (A \circ C) \circ (B \circ D) \\ &= (A + C) + (B + D). \end{aligned} \quad (3)$$

Also $A + O = j(A \circ O) = j^2(A) = A$. Set $C = O$ in (3) and replace D by C . The resulting equality is

$$(A + B) + C = A + (B + C),$$

which proves associativity directly from (a). The same construction is commutative, and $A + j(A) = O$, so it gives the asserted group law. $\square$

Exercise V.4.5

Permitted context.

Work over an algebraically closed field. The nine-point theorem for cubics says that, in a transverse complete intersection of two cubics with the usual general-position conditions, every cubic through eight of the nine points passes through the ninth. Collinearity is a closed algebraic condition. For six points on a fixed smooth conic, the parameter space is irreducible, and the intersection of two distinct joining lines varies algebraically wherever it is defined.

Problem. For six points A, B, C, A', B', C' on a smooth conic, prove Pascal's theorem:

$$P = AB' \cap A'B, \quad Q = AC' \cap A'C, \quad R = BC' \cap B'C$$

are collinear, whenever these joining lines and their intersections are defined.

Solution. First choose the six points generically. Consider the two reducible cubics

$$T_1 = AB' \cup BC' \cup CA', \quad T_2 = A'B \cup B'C \cup C'A.$$

Their nine intersection points are exactly the six conic points and P, Q, R . For example, the three intersections on AB' are P, B', A ; those on BC' are B, R, C' ; those on CA' are A', C, Q .

For a generic choice, these intersections are transverse and no four of the eight points consisting of the six conic points and P, Q are collinear. The unwanted incidences are excluded by the generic choice. No seven are on a conic: seven of these eight include at least five original conic points, and a conic through those five must be the original one by Bezout; neither P nor Q lies on it. These are the general-position hypotheses of the nine-point theorem.

Let $M = PQ$. The union of the original conic and M is another cubic, through eight of the nine points. The nine-point theorem makes it pass through R . In the generic configuration R does not lie on the conic, so it lies on M . This proves collinearity for a dense open set of configurations.

The sixfold product of the conic is irreducible, and the displayed line intersections are algebraic maps on their domain of definition. The determinant expressing collinearity vanishes on that dense open set, and therefore vanishes throughout its domain. Thus the conclusion holds for every configuration for which P, Q, R are defined; tangent versions are obtained by the same limiting construction. $\square$

Exercise V.4.6**Permitted context.**

Work over an algebraically closed field. The vector space of plane quartic forms has dimension fifteen. Two quartics without a common component form a complete intersection of length sixteen. Its ideal is generated by the two equations; in degree four its space of equations is exactly their two-dimensional span. The base scheme of a linear system is the common zero scheme of all its sections.

Problem. Explain under what hypotheses thirteen prescribed plane points force three further points through which every quartic containing the thirteen must pass. State what can go wrong without those hypotheses.

Solution. Let $Z = \{P_1, \dots, P_{13}\}$ be thirteen distinct points and put $V = H^0(\mathcal{I}_Z(4))$. The precise hypotheses for a pencil with three distinct additional ordinary base points are:

$$\dim V = 2, \quad \text{the pencil has no fixed curve component,} \quad \text{its base scheme is reduced.} \quad (1)$$

The first condition says that the thirteen point conditions are independent. Choose a basis F, G of V . The second says that they have no common component. Bezout then makes $B = V(F, G)$ a scheme of length sixteen. By the third condition it consists of sixteen distinct points. Thirteen are the assigned points, so the remaining three are uniquely determined:

$$B = Z \sqcup \{P_{14}, P_{15}, P_{16}\}.$$

Every quartic through Z has equation $\lambda F + \mu G$, so it contains all three residual points. The base scheme is independent of the basis of V , proving their intrinsic determination.

These conditions are also necessary if the common base scheme is to consist of precisely sixteen distinct ordinary points. Finiteness rules out a fixed curve. Choose two members with no common component; their intersection has length sixteen and contains the sixteen base points, so it is exactly that reduced scheme. Any other quartic in V belongs to its degree-four ideal, which is the span of the two chosen equations. Hence $\dim V = 2$.

If $\dim V > 2$ but there is no fixed component, a third independent member cannot contain the entire intersection of the first two: otherwise it would belong to their two-dimensional span. Thus the asserted three additional common ordinary points need not exist. If there is a fixed curve, the base locus is positive dimensional; for example, thirteen points on a conic force every such quartic to contain that conic by Bezout. Finally, without reducedness, the length-sixteen intersection can have coincident or infinitely near base points. One still has an intersection scheme, but cannot assert three distinct new ordinary points. This is the appropriate scheme-theoretic qualification of the statement. $\square$

Exercise V.4.7**Permitted context.**

Let X be a smooth cubic surface over an algebraically closed field, presented as the blow-up of six plane points in general position. Write l for the pullback of a line and e_i for the exceptional classes. Then

$$l^2 = 1, \quad e_i^2 = -1, \quad le_i = e_ie_j = 0 \quad (i \neq j), \quad H = -K_X = 3l - \sum_{i=1}^6 e_i.$$

The 27 lines have classes e_i , $l - e_i - e_j$, and $2l - \sum_{j \neq i} e_j$. They are the only integral curves of negative self-intersection. The arithmetic genus of a divisor is $p_a(D) = 1 + (D^2 - DH)/2$.

The Hodge index theorem gives $D^2H^2 \leq (DH)^2$. We use the smooth-member criterion proved in Exercise V.4.8: a class nonnegative on all 27 lines and of positive square contains a smooth integral curve. The class $H - L$ for a line L is a conic pencil with a smooth general member.

Problem. For every divisor D of degree $d = DH$, prove

$$p_a(D) \leq \begin{cases} (d-1)(d-2)/6, & d \equiv 1, 2 \pmod{3}, \\ (d-1)(d-2)/6 + 1/3, & d \equiv 0 \pmod{3}. \end{cases}$$

For each positive integer d , exhibit a smooth integral curve attaining this bound.

Solution. Since $H^2 = 3$, Hodge index and adjunction give

$$p_a(D) \leq 1 + \frac{d^2/3 - d}{2} = \frac{(d-1)(d-2)}{6} + \frac{1}{3}.$$

The arithmetic genus is an integer. If $d \equiv 1, 2 \pmod{3}$, the first term on the right is an integer, so rounding down removes $1/3$. If $3 \mid d$, the whole right side is an integer. This proves the asserted bounds, including for divisors of nonpositive degree.

Fix a line L , and put $F = H - L$. For positive degrees take

d	class	square
$3n$	nH	$3n^2$
$3n+1$	$nH + L$	$3n^2 + 2n - 1$
$3n+2$	$nH + F$	$3n^2 + 4n$

Here $n \geq 1$ in the first row and $n \geq 0$ in the others. For $n = 0$, use the line or a smooth conic, both of genus zero. For $n \geq 1$, all three squares are positive. The first class is ample. The second meets L in $n - 1$ and every other line nonnegatively. For a line $M \neq L$, $FM = 1 - LM \geq 0$, while $FL = 2$; thus the third class is nonnegative on all lines as well. The smooth-member criterion provides a smooth integral curve in each class. Substituting the displayed squares into adjunction gives equality in the corresponding bound. $\square$

Exercise V.4.8**Permitted context.**

Let X be a smooth cubic surface over an algebraically closed field, presented as the blow-up of six plane points in general position. Write l for the pullback of a line and e_i for the exceptional classes. Then

$$l^2 = 1, \quad e_i^2 = -1, \quad le_i = e_i e_j = 0 \quad (i \neq j), \quad H = -K_X = 3l - \sum_{i=1}^6 e_i.$$

The 27 lines have classes e_i , $l - e_i - e_j$, and $2l - \sum_{j \neq i} e_j$. They are the only integral curves of negative self-intersection. The arithmetic genus of a divisor is $p_a(D) = 1 + (D^2 - DH)/2$.

A conic class means a ruling class $H - L$, where L is one of the 27 lines; its general member is a smooth integral curve of degree two and square zero.

Quadratic transformations centered at three marked points give new blow-up presentations of the same cubic surface. For $D = al - \sum b_i e_i$, the new coefficients are

$$a' = 2a - b_1 - b_2 - b_3, \quad b'_i = a - b_j - b_k \quad (\{i, j, k\} = \{1, 2, 3\}),$$

with the other coefficients unchanged. Blowing up three, four, or five points in general position has very ample anticanonical class. The conics through two points give the morphism to a quadric in Exercise V.4.1. We use Bertini irreducibility for a birational linear system with two-dimensional image, and Bertini smoothness where its associated morphism is an immersion. These forms apply in every characteristic.

Problem. Prove that a divisor class D on X contains an integral curve if and only if it contains a smooth integral curve, if and only if it is one of the following: (a) a line class; (b) a conic class of square zero; (c) a class of positive square with $DL \geq 0$ for all 27 lines.

Solution. We first analyze any class nonnegative on the 27 lines. Write $D = al - \sum b_i e_i$, and reorder so $b_1 \geq \dots \geq b_6$. The line inequalities imply

$$b_i \geq 0, \quad a \geq b_i + b_j, \quad 2a \geq \sum_{j \neq i} b_j. \quad (1)$$

In particular $a \geq 0$. If $a < b_1 + b_2 + b_3$, perform the quadratic transformation at the first three points. Its new b'_i are nonnegative by (1), and $0 \leq a' < a$. Reordering and repeating must terminate. Thus after a new marking we have

$$a \geq b_1 + b_2 + b_3, \quad b_1 \geq \dots \geq b_6 \geq 0. \quad (2)$$

Define

$$G_0 = l, \quad G_1 = l - e_1, \quad G_2 = 2l - e_1 - e_2, \quad G_j = 3l - e_1 - \dots - e_j \quad (3 \leq j \leq 6).$$

The first is a pullback from the plane, the second a pencil, and the third the conic system giving a quadric. For $3 \leq j \leq 5$, G_j is the pullback of the very ample anticanonical system on the blow-up of j points; $G_6 = H$. Consequently all these classes are base point free and nef. There is the explicit decomposition

$$\begin{aligned} D = & (a - b_1 - b_2 - b_3)G_0 + (b_1 - b_2)G_1 \\ & + (b_2 - b_3)G_2 + (b_3 - b_4)G_3 \\ & + (b_4 - b_5)G_4 + (b_5 - b_6)G_5 + b_6 G_6. \end{aligned} \quad (3)$$

All coefficients are nonnegative integers. Thus D itself is base point free and nef.

Only G_1 has square zero; all the other G_i have positive square. Since pairwise intersections of nef classes are nonnegative, a nonzero D of square zero must be mG_1 for some $m > 0$. The morphism defined by G_1 is the connected-fiber ruling to $\mathbb{P}^1$. Hence its full system $|mG_1|$ is pulled back from $|\mathcal{O}_{\mathbb{P}^1}(m)|$. It has an integral member exactly when $m = 1$, and its general fiber is a smooth conic on the cubic surface.

If $D^2 > 0$, some generator $B \neq G_1$ occurs in (3). Write $D = B + N$, with N base point free. The morphism for B is birational and is an isomorphism onto its image off finitely many exceptional lines. They are the e_i for $B = G_0$, the lines $l - e_1 - e_2, e_3, \dots, e_6$ for $B = G_2$, and the e_i with $i > j$ for $B = G_j$, $3 \leq j \leq 6$. On their complement, multiply sections of B by a section of N nonvanishing at a given point. This shows that the system D is immersive there and is birational onto its image. Bertini makes its general member integral and smooth on this open.

Smoothness along the omitted lines requires a separate check. Every generator in (3) restricts to each such line E with degree zero or one. A base point free degree-one system on $E \cong \mathbb{P}^1$ is complete. Multiplication of their sections therefore shows that the restriction of $|D|$ to E contains the complete system $|\mathcal{O}_E(DE)|$. If $DE = 0$, a general member avoids E . If $DE > 0$, it meets E in distinct transverse points and is smooth at them. There are only finitely many such lines, so a general member is smooth everywhere.

Now let C be any integral member of a class D . If $C^2 < 0$, it is a line, giving (a). Otherwise C is distinct from every line, so its intersections with all lines are nonnegative. The preceding analysis gives (b) when $C^2 = 0$, and (c) when $C^2 > 0$. Conversely:

(a) Every line is smooth and integral.

(b) A square-zero conic class is the ruling class G_1 after a marking, and has a smooth integral general member.

(c) The argument following (3) gives a smooth integral member.

Thus each numerical alternative implies a smooth integral member, and such a member is in particular integral. All three conditions are equivalent. $\square$

Exercise V.4.9**Permitted context.**

Let X be a smooth cubic surface over an algebraically closed field, presented as the blow-up of six plane points in general position. Write l for the pullback of a line and e_i for the exceptional classes. Then

$$l^2 = 1, \quad e_i^2 = -1, \quad le_i = e_ie_j = 0 \quad (i \neq j), \quad H = -K_X = 3l - \sum_{i=1}^6 e_i.$$

The 27 lines have classes e_i , $l - e_i - e_j$, and $2l - \sum_{j \neq i} e_j$. They are the only integral curves of negative self-intersection. The arithmetic genus of a divisor is $p_a(D) = 1 + (D^2 - DH)/2$.

Exercise V.4.8 proves that every class nonnegative on all lines can be re-marked to satisfy $a \geq b_1 + b_2 + b_3$, $b_1 \geq \cdots \geq b_6 \geq 0$; such a class is nef. A positive-square class with these line inequalities contains a smooth integral member.

Problem. Let C be a smooth integral curve of degree d and positive genus g on X . Show

$$g \geq \begin{cases} (d-6)/2, & d \text{ even, } d \geq 8, \\ (d-5)/2, & d \text{ odd, } d \geq 13. \end{cases}$$

Prove that every indicated lower bound is attained.

Solution. The curve is not a line, so it is nonnegative on all lines. Choose a standard marking

$$C = al - \sum b_i e_i, \quad a \geq b_1 + b_2 + b_3, \quad b_1 \geq \cdots \geq b_6 \geq 0.$$

Put $D = C - H$. Adjunction gives

$$D^2 = 2g - d + 1. \tag{1}$$

We show that $D^2 \geq -5$, with the sole possible exception $C = 3l$, which has degree nine.

Let r be the number of positive b_i , and write

$$D = D_0 + \sum_{i>r} e_i, \quad D_0 = (a-3)l - \sum_{i \leq r} (b_i - 1)e_i.$$

The terms in this decomposition are orthogonal, so $D^2 = D_0^2 - (6-r)$. If $r \geq 3$, D_0 is standard and nef; hence $D^2 \geq -(6-r) \geq -3$. If $r = 2$ and $a \geq b_1 + b_2 + 1$, it is again standard, giving $D^2 \geq -4$. In the remaining $r = 2$ case, $a = b_1 + b_2$. The genus formula becomes $g = (b_1 - 1)(b_2 - 1) > 0$, so both $b_i \geq 2$. Then

$$D_0 = (l - e_1 - e_2) + (b_1 - 2)(l - e_1) + (b_2 - 2)(l - e_2).$$

The last two terms form a nef class orthogonal to $l - e_1 - e_2$. Thus $D_0^2 \geq -1$, giving $D^2 \geq -5$.

If $r = 1$, positive genus forces $a \geq b_1 + 2$, and

$$D_0 = (a - b_1 - 2)l + (b_1 - 1)(l - e_1)$$

is nef. Thus $D^2 \geq -5$. If $r = 0$, positive genus forces $a \geq 3$, and

$$D^2 = (a-3)^2 - 6.$$

This is at least -5 except when $a = 3$, the asserted degree-nine exception.

For even $d \geq 8$, (1) now gives $g \geq (d-6)/2$. For odd $d \geq 13$, the exception is absent and (1) is even. Thus $D^2 \geq -4$, giving $g \geq (d-5)/2$.

For equality, take the following classes:

C	d	g
$al - (a-2)e_1$	$2a+2$	$a-2 \quad (a \geq 3)$
$al - (a-2)e_1 - e_2$	$2a+1$	$a-2 \quad (a \geq 6).$

They have squares $4a-4$ and $4a-5$, respectively, and meet every one of the 27 lines nonnegatively. Exercise V.4.8 supplies smooth integral members. The table covers every degree in the requested ranges and attains exactly the corresponding lower bound. $\square$

Exercise V.4.10

Permitted context.

A convex function on a compact polytope achieves a maximum at a vertex: on any line segment its value does not exceed the larger endpoint value, and iterating on faces reduces to vertices.

A vertex in six-dimensional space has six linearly independent active defining constraints.

Problem. Let $a, b_1, \dots, b_6$ be real numbers such that

$$b_i > 0, \quad a > b_i + b_j \quad (i \neq j), \quad 2a > \sum_{i \neq j} b_i \quad \text{for each } j.$$

Prove directly that $a^2 > \sum_i b_i^2$.

Solution. Since $a > 0$, divide all b_i by a and reorder them. It suffices to consider $a = 1$ and $b_1 \geq \dots \geq b_6 > 0$. The strongest pair and five-term inequalities are $b_1 + b_2 < 1$ and $b_1 + \dots + b_5 < 2$. Consider their closed relaxation

$$P = \{b_1 \geq \dots \geq b_6 \geq 0, \quad b_1 + b_2 \leq 1, \quad b_1 + \dots + b_5 \leq 2\}.$$

It is compact. The convex function $q(b) = \sum b_i^2$ is bounded by its values at the vertices.

Here are all its vertices and their values:

$(b_1, b_2, b_3, b_4, b_5, b_6)$	q
$(0, 0, 0, 0, 0, 0)$	0
$(1, 0, 0, 0, 0, 0)$	1
$(1/2, 1/2, 0, 0, 0, 0)$	1/2
$(1/2, 1/2, 1/2, 0, 0, 0)$	3/4
$(1/2, 1/2, 1/2, 1/2, 0, 0)$	1
$(2/5, 2/5, 2/5, 2/5, 2/5, 0)$	4/5
$(2/5, 2/5, 2/5, 2/5, 2/5, 2/5)$	24/25
$(2/3, 1/3, 1/3, 1/3, 1/3, 0)$	8/9
$(2/3, 1/3, 1/3, 1/3, 1/3, 1/3)$	1
$(1/2, 1/2, 1/3, 1/3, 1/3, 0)$	5/6
$(1/2, 1/2, 1/3, 1/3, 1/3, 1/3)$	17/18
$(1/2, 1/2, 1/2, 1/4, 1/4, 0)$	7/8
$(1/2, 1/2, 1/2, 1/4, 1/4, 1/4)$	15/16.

Here is a direct two-block verification of completeness. Set $b_7 = 0$ and $x_i = b_i - b_{i+1}$ for $1 \leq i \leq 6$. This invertible linear change writes the constraints as

$$x_i \geq 0, \quad A \cdot x \leq 1, \quad B \cdot x \leq 2, \quad A = (1, 2, 2, 2, 2, 2), \quad B = (1, 2, 3, 4, 5, 5).$$

At a vertex at most two x_i can be positive: with three positive coordinates there are at most three active coordinate constraints and two active sum constraints, fewer than six. The zero vector is one vertex. With one positive coordinate, its value is $\min(1/A_i, 2/B_i)$, giving the next six rows. With two positive coordinates x_i, x_j , both sum constraints must be equalities. Their weighted average ratio is two, so $B_i/A_i < 2 < B_j/A_j$, after relabeling. The ratios are $1, 1, 3/2, 2, 5/2, 5/2$; hence exactly

$$i \in \{1, 2, 3\}, \quad j \in \{5, 6\}$$

are possible. Solving the two equations gives the final six rows. Thus the list is complete. Every displayed value is at most one, so $q \leq 1$ on P .

For the original strict inequalities, choose $\epsilon > 0$ small enough that $(1 + \epsilon)b$ still satisfies both weak sum inequalities. It lies in P , so

$$(1 + \epsilon)^2 \sum b_i^2 \leq 1.$$

Hence $\sum b_i^2 < 1$. Undoing normalization proves $a^2 > \sum b_i^2$. □

Exercise V.4.11**Permitted context.**

Let X be a smooth cubic surface over an algebraically closed field, presented as the blow-up of six plane points in general position. Write l for the pullback of a line and e_i for the exceptional classes. Then

$$l^2 = 1, \quad e_i^2 = -1, \quad le_i = e_ie_j = 0 \quad (i \neq j), \quad H = -K_X = 3l - \sum_{i=1}^6 e_i.$$

The 27 lines have classes e_i , $l - e_i - e_j$, and $2l - \sum_{j \neq i} e_j$. They are the only integral curves of negative self-intersection. The arithmetic genus of a divisor is $p_a(D) = 1 + (D^2 - DH)/2$.

The automorphism group G of the line configuration is generated by permutations of the six exceptional classes and the quadratic transformation at the first three marked points.

We use the elementary chamber theorem for a finite real reflection group. Reflections in the walls of a chamber have the presentation $s_i^2 = 1$, $(s_i s_j)^{m_{ij}} = 1$, where the angle between the walls is π/m_{ij} . The stabilizer of a vector is generated by the reflections whose walls contain that vector. One proves the presentation by following paths through adjacent chambers: a general deformation of a path changes its wall word only by backtracking or by passing a codimension-two wall intersection, giving precisely these relations. The same argument in a small neighborhood of a vector gives the stabilizer assertion.

Problem. For a graph, associate one involution to each vertex, imposing $(st)^3 = 1$ for joined vertices and $(st)^2 = 1$ for unjoined distinct vertices.

(a) For a chain of $n - 1$ vertices, prove that this group is S_n , under the map to adjacent transpositions.

(b) For a chain $x_1, \dots, x_5$ and a sixth vertex y joined to x_3 , construct a surjection of the associated group E_6 onto G , taking x_i to the adjacent permutation of exceptional curves and y to the quadratic transformation.

(c) Determine the order of E_6 , and prove this surjection is an isomorphism.

Solution. (a) Let the chain generators be $s_1, \dots, s_{n-1}$. The adjacent transpositions satisfy the relations and generate S_n , giving a surjection. For the reverse order bound, let H be generated by $s_1, \dots, s_{n-2}$, and put

$$r_k = s_{n-1} s_{n-2} \cdots s_k \quad (1 \leq k < n), \quad r_n = 1.$$

The union of the n right cosets Hr_k is stable under right multiplication by every generator. Indeed, $r_k s_i = s_i r_k$ for $i < k - 1$; $r_k s_{k-1} = r_{k-1}$, $r_k s_k = r_{k+1}$; and the braid relation gives $r_k s_i = s_{i-1} r_k$ for $i > k$. For r_n the assertion is immediate. The union contains the identity, hence the whole group. Induction gives $|H| \leq (n - 1)!$, so the group has at most $n!$ elements. Its surjection to S_n is therefore an isomorphism.

(b) On the real space $H^\perp$, use the positive definite form $\langle u, v \rangle = -uv$, and set

$$\alpha_i = e_i - e_{i+1} \quad (1 \leq i \leq 5), \quad \beta = l - e_1 - e_2 - e_3.$$

Each has square -2 . Its reflection on the Picard lattice is

$$s_\rho(D) = D + (D\rho)\rho. \tag{1}$$

The first five reflections permute neighboring e_i . The last sends l to $2l - e_1 - e_2 - e_3$ and e_i to $l - e_j - e_k$ for the first three indices, so it is the quadratic transformation. Direct pairings give

the chain relations and the single extra edge from β to α_3 . All these operations preserve the line configuration and generate G , proving the requested surjection.

(c) We compute the finite reflection group without assuming a classification of root systems. The integral classes ρ with $\rho H = 0$, $\rho^2 = -2$ are exactly

$$e_i - e_j \ (i \neq j), \quad \pm(l - e_i - e_j - e_k) \ (i < j < k), \quad \pm\theta, \quad \theta = 2l - \sum e_i. \quad (2)$$

There are $30 + 40 + 2 = 72$ of them. To verify the list, write $\rho = al - \sum b_i e_i$. Then $\sum b_i = 3a$, $\sum b_i^2 = a^2 + 2$. Cauchy gives $|a| \leq 2$. For $a = 0$ there are two nonzero coefficients, $1, -1$. For $a = 1$, $\sum(b_i^2 - b_i) = 0$ forces three ones. For $a = 2$, equality in Cauchy forces all six coefficients to be one. Negation handles negative a .

The six roots in (b) form a basis of $H^\perp$. Every root in (2) has coefficients all nonnegative or all nonpositive in this basis. For $e_i - e_j$, $i < j$, its expression is $\alpha_i + \cdots + \alpha_{j-1}$. For $l - e_i - e_j - e_k$, the coefficient of β is one and that of α_m is $\min(m, 3) - |\{i, j, k\} \cap \{1, \dots, m\}| \geq 0$. Finally

$$\theta = \alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4 + \alpha_5 + 2\beta.$$

Consequently the inequalities $\langle v, \alpha_i \rangle > 0$, $\langle v, \beta \rangle > 0$ define a chamber of the root hyperplanes.

Let W be generated by its wall reflections. They preserve the finite spanning root set, so W is finite. The chamber theorem gives exactly the E_6 presentation: $E_6 \cong W$. The roots form a single W -orbit. Indeed, for a positive root $\rho = \sum c_i \rho_i$,

$$2 = \langle \rho, \rho \rangle = \sum c_i \langle \rho, \rho_i \rangle$$

forces a positive pairing with a simple root. Unless ρ is that simple root, the pairing is one, and reflection subtracts that simple root, reducing height. This reaches a simple root. Adjacent simple roots are conjugate under their two reflections, the diagram is connected, and negation is achieved by reflection in the root itself.

The roots perpendicular to θ are exactly $e_i - e_j$: for a root as above, $\theta\rho = -a$. The stabilizer theorem identifies W_θ with the group generated by their reflections, which is S_6 of order 720. Orbit-stabilizer gives

$$|E_6| = |W| = 72 \cdot 720 = 51840.$$

Finally the action of W on the 27 line classes is faithful: these classes span the Picard space, since they include all e_i and $l - e_1 - e_2$. Thus the already surjective map $W \rightarrow G$ is injective as well. Hence $E_6 \cong G$. $\square$

Exercise V.4.12**Permitted context.**

Let X be a smooth cubic surface over an algebraically closed field, presented as the blow-up of six plane points in general position. Write l for the pullback of a line and e_i for the exceptional classes. Then

$$l^2 = 1, \quad e_i^2 = -1, \quad le_i = e_i e_j = 0 \quad (i \neq j), \quad H = -K_X = 3l - \sum_{i=1}^6 e_i.$$

The 27 lines have classes e_i , $l - e_i - e_j$, and $2l - \sum_{j \neq i} e_j$. They are the only integral curves of negative self-intersection. The arithmetic genus of a divisor is $p_a(D) = 1 + (D^2 - DH)/2$.

Every ample divisor on a smooth cubic surface is very ample (Theorem V.4.11). A general member is smooth and integral. The rational surface X satisfies $H^1(X, \mathcal{O}_X) = 0$; both X and an integral projective curve on it have only constant global regular functions.

Problem. Prove that $H^1(X, \mathcal{O}_X(-D)) = 0$ for every ample divisor D on a smooth cubic surface.

Solution. Choose a smooth integral $C \in |D|$. Its ideal sequence is

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_C \longrightarrow 0.$$

The resulting cohomology sequence contains

$$H^0(X, \mathcal{O}_X) \longrightarrow H^0(C, \mathcal{O}_C) \longrightarrow H^1(X, \mathcal{O}_X(-D)) \longrightarrow H^1(X, \mathcal{O}_X).$$

The first map is the identity on k , and the last group is zero. Exactness therefore forces the middle first-cohomology group to vanish. No restriction on the characteristic is used. $\square$

Exercise V.4.13**Permitted context.**

Let X be the blow-up of five plane points in general position over an algebraically closed field, embedded in $\mathbb{P}^4$ by the complete anticanonical system $H = 3l - e_1 - \cdots - e_5$. Thus $H^2 = 4$, $K_X = -H$, $l^2 = 1$, $e_i^2 = -1$, and all the mixed basis intersections are zero. The surface is rational, so $H^1(\mathcal{O}_X) = H^2(\mathcal{O}_X) = 0$. We use adjunction, Bertini, Riemann–Roch on a genus-one curve, and the fact that a complete intersection of two hypersurfaces is Cohen–Macaulay and has the product of their degrees.

Problem. (a) Prove that X contains exactly sixteen lines. (b) Prove that it is the scheme-theoretic complete intersection of two quadrics in $\mathbb{P}^4$.

Solution. (a) The exceptional curves give five lines. Any other line has class $C = al - \sum b_i e_i$ with $a \geq 1$, $b_i \geq 0$. Since $C \cong \mathbb{P}^1$ and $HC = 1$, adjunction gives $C^2 = -1$. Thus

$$\sum b_i = 3a - 1, \quad \sum b_i^2 = a^2 + 1.$$

Cauchy gives $(3a - 1)^2 \leq 5(a^2 + 1)$, hence $a \leq 2$. For $a = 1$, the equations force two coefficients one and the others zero: these are the ten transforms of joining lines. For $a = 2$, all five coefficients are

one: this is the unique conic through the five points. General position makes these curves irreducible and smooth; each has anticanonical degree one, so each is a line in the embedding. There are exactly $5 + 10 + 1 = 16$.

(b) Choose a smooth hyperplane section $E \in |H|$. Adjunction gives genus one. The restrictions of H and $2H$ to E have degrees four and eight, hence have respectively four and eight sections and zero first cohomology. The sequences for adding E , together with $H^1(\mathcal{O}_X) = H^2(\mathcal{O}_X) = 0$, give

$$h^0(\mathcal{O}_X(H)) = 5, \quad H^1(\mathcal{O}_X(H)) = 0, \quad h^0(\mathcal{O}_X(2H)) = 5 + 8 = 13.$$

There are fifteen quadrics in $\mathbb{P}^4$, so the kernel of restriction to X has dimension at least two. Choose independent quadrics Q_1, Q_2 in that kernel. Each is irreducible: otherwise the irreducible, nondegenerate surface X would lie in one of its hyperplane factors. Independence therefore ensures that the two quadrics have no common hypersurface component.

Their intersection Y is a complete intersection of pure dimension two and degree four, containing X , also of degree four. Hence X is its sole component, with generic multiplicity one. Since Y is Cohen–Macaulay, it has no embedded associated components; generic reducedness then makes it reduced everywhere. Thus $Y = X$ scheme-theoretically. $\square$

Exercise V.4.14

Permitted context.

Let X be a smooth cubic surface over an algebraically closed field, presented as the blow-up of six plane points in general position. Write l for the pullback of a line and e_i for the exceptional classes. Then

$$l^2 = 1, \quad e_i^2 = -1, \quad le_i = e_ie_j = 0 \ (i \neq j), \quad H = -K_X = 3l - \sum_{i=1}^6 e_i.$$

The 27 lines have classes e_i , $l - e_i - e_j$, and $2l - \sum_{j \neq i} e_j$. They are the only integral curves of negative self-intersection. The arithmetic genus of a divisor is $p_a(D) = 1 + (D^2 - DH)/2$.

A class $D = al - \sum b_ie_i$ is ample if

$$b_i > 0, \quad a > b_i + b_j, \quad 2a > \sum_{j \neq i} b_j$$

for all indicated distinct indices. An ample class on X is very ample and has a smooth integral general member.

Problem. Construct smooth integral curves in $\mathbb{P}^3$ with

$$(d, g) = (8, 6), (8, 7), (9, 7), (9, 8), (9, 9), (10, 8), (10, 9), (10, 10), (10, 11).$$

Solution. Use the following divisor classes on a fixed smooth cubic surface:

d	g	a	$(b_1, b_2, b_3, b_4, b_5, b_6)$
8	6	6	(2, 2, 2, 2, 1, 1)
8	7	7	(3, 2, 2, 2, 2, 2)
9	7	6	(2, 2, 2, 1, 1, 1)
9	8	7	(3, 2, 2, 2, 2, 1)
9	9	7	(2, 2, 2, 2, 2, 2)
10	8	6	(2, 2, 1, 1, 1, 1)
10	9	7	(3, 2, 2, 2, 1, 1)
10	10	7	(2, 2, 2, 2, 2, 1)
10	11	8	(3, 3, 2, 2, 2, 2).

The coefficients are sorted. Thus to check ampleness it suffices to check positivity of b_6 , $a - b_1 - b_2$, and $2a - \sum b_i + b_6$, which holds in each row. Each class consequently contains a smooth integral curve. Its degree and genus in the cubic embedding are

$$d = 3a - \sum b_i, \quad g = 1 + \frac{a^2 - \sum b_i^2 - d}{2}.$$

Substituting the entries gives the first two columns exactly. These are therefore the requested curves in $\mathbb{P}^3$. $\square$

Exercise V.4.15

Permitted context.

Work over an algebraically closed field k . A configuration consists of distinct ordinary points of $\mathbb{P}^2$. An admissible quadratic transformation has three noncollinear configuration points as centers; replace those centers by the three inverse centers and carry the other points along. Call a configuration Cremona general if every finite sequence of these operations leaves distinct ordinary points with no three collinear.

On the blow-up write $H, E_1, \dots, E_r$ for the line and exceptional classes. Their intersection matrix is $\text{diag}(1, -1, \dots, -1)$, and $K = -3H + \sum E_i$. Adjunction gives $2p_a(C) - 2 = C^2 + KC$. Distinct integral curves have nonnegative intersection. An integral projective curve of arithmetic genus zero over k is a smooth rational curve.

An admissible quadratic transformation lifts to an isomorphism between the surfaces obtained by blowing up the two full configurations. For the first three centers its action on divisor coordinates is

$$d' = 2d - m_1 - m_2 - m_3, \quad m'_1 = d - m_2 - m_3, \quad m'_2 = d - m_1 - m_3, \quad m'_3 = d - m_1 - m_2,$$

with the other multiplicities unchanged. In particular this is the reflection $s_\gamma(D) = D + (D\gamma)\gamma$, where $\gamma = H - E_1 - E_2 - E_3$. Permuting exceptional labels is also allowed. Polynomial equations in point coordinates describe incidence and nonvanishing conditions for every fixed finite sequence. The product $(\mathbb{P}^2)^m$ is irreducible, and a birational map restricts to an isomorphism between dense open subsets. Noether normalization gives a finite surjective morphism from an affine integral variety of dimension n to $\mathbb{A}^n$. We may use Bezout's theorem for plane curves. Vanishing and prescribed first-order vanishing at fixed points are linear conditions on homogeneous forms; degree- d forms form a vector space of dimension $\binom{d+2}{2}$.

Problem. (a) For six points, prove that Cremona general position is equivalent to having no three collinear and not having all six on a conic.

(b) Prove that an admissible sequence preserves Cremona general position.

(c) If k is uncountable and $P_1, \dots, P_r$ are fixed in Cremona general position, prove that a dense subset of choices of P_{r+1} extends this configuration to one in Cremona general position. Prove the countable-union lemma suggested by this assertion.

(d) For $r = 7, 8$, count the smooth rational curves of square -1 on the blow-up, obtaining respectively 56, 240, and show that there are no other negative integral curves.

(e) For $r = 9$, prove that there are infinitely many such curves.

Solution. (a) Normalize the centers to the coordinate vertices. The quadratic map is $[x : y : z] \mapsto [yz : xz : xy]$. A conic through the three vertices has equation $axy + bxz + cyz = 0$; its transform is a line. Thus six points on a conic would give three collinear transformed points.

Conversely, assume the two stated conditions. The three points not used as centers have all coordinates nonzero. A line through two inverse centers therefore cannot contain any of their images. A line through one inverse center and two of their images pulls back, after removing a coordinate-line factor, to a line through one original center and the two corresponding original points. A line through the three images pulls back to a conic through all six original points. Each possibility is excluded. Finally a conic through the three inverse centers and the three images pulls back to xyz times a linear equation through the three original noncenters, also impossible. The transformed configuration satisfies the same two conditions. Induction proves the assertion for every finite sequence.

(b) Concatenate any further sequence with the sequence already performed. The defining condition for the original configuration then gives the defining condition for the resulting one.

(c) *The countable-union lemma.* First prove the lemma, including density of the complement. A countable family of proper closed subsets of $\mathbb{A}^1(k)$ is countable and cannot cover an uncountable field. Inductively, for proper closed sets $Z_j \subset \mathbb{A}^n$, choose a nonzero polynomial f_j vanishing on each Z_j . View f_j as a polynomial in the last coordinate. Its coefficients do not vanish simultaneously outside a proper closed subset of $\mathbb{A}^{n-1}$. Choose a point outside all these exceptional sets by induction. On the resulting vertical line each f_j has only finitely many zeros, so a point avoiding all of them exists. Adding one further proper closed set to the family proves density. For an arbitrary integral variety, restrict to a nonempty affine open and use a finite surjective Noether-normalization map. The image of each proper closed subset is closed of smaller dimension. The affine-space case applies to these images. The same argument on every nonempty affine open proves density.

Reduction with the old configuration fixed. Here is the precise remaining reduction for the extension assertion. Hold $P_1, \dots, P_r$ fixed and let q vary in $\mathbb{P}^2$. For each finite word w of center triples, let U_w be the set of q for which all successive configurations are defined, ordinary, distinct, and have no collinear triple. Coordinate formulas and nonvanishing determinants make U_w open. There are only countably many words, and the desired set is

$$V = \bigcap_w U_w.$$

The lemma proves density if every U_w is nonempty. For words whose centers never involve the additional label, this nonemptiness follows by pulling back the finitely many forbidden lines under fixed birational maps. For arbitrary words involving the additional label, nonemptiness on the fixed-configuration fiber is an additional condition that this argument does not prove. Nonemptiness of the corresponding open subset of $(\mathbb{P}^2)^{r+1}$ does not suffice: an open subset of a product can miss an entire specified fiber.

Remaining gap in (c). For every fixed Cremona general configuration $P_1, \dots, P_r$, one must prove $U_w \neq \emptyset$ for every word w , including words whose centers involve the new point. This fixed-fiber nonemptiness has not been established here, so the original extension assertion remains unproved.

A proved weaker statement: a very general full configuration. If all $m = r + 1$ points are allowed to vary, the Cremona general configurations form a dense subset of $S = (\mathbb{P}^2)^m$. For $m < 3$ this follows just by excluding coincident points. For $m \geq 3$, let S° denote the nonempty open set of distinct configurations with no collinear triple.

For each triple of labels, we construct a birational involution of S representing the corresponding quadratic operation. On a dense coordinate chart, choose normalized column vectors for its three noncollinear centers and let A be their invertible matrix. Conjugate $\sigma([x : y : z]) = [yz : xz : xy]$ by A . Keep the three center labels at their original points: they are precisely the inverse centers

for $A\sigma A^{-1}$. Apply this map to all other points. On the dense open set where those points avoid the three joining lines, the construction is defined. Its second application uses the same matrix A , since the centers were retained, and σ^2 is the identity as a rational map. Thus this operation is a birational involution of the full parameter space.

Fix a finite word w . Every prefix gives a birational map of S . Requiring each operation to be defined and each prefix image to belong to S° gives a dense open subset G_w : it is a finite intersection of domains of definition and inverse images of dense opens under birational maps. There are countably many words. The countable-union lemma therefore makes $\bigcup_w G_w$ dense in S , and every configuration in this intersection is Cremona general. This proves the weaker whole-configuration statement. It supplies no nonemptiness assertion on an arbitrarily specified fiber with the first r points fixed.

(d) Every subset of a Cremona general configuration is Cremona general: a word on its labels can be performed on the full configuration. In particular (a) excludes six points on a conic. For eight points, a cubic through all of them singular at one would have an effective divisor of class

$$3H - 2E_1 - E_2 - \cdots - E_8.$$

Apply a quadratic transformation at labels 1, 2, 3. The new class is that of a conic through six points. A transformation at three of those six gives a line through three points, contradicting general position. This argument uses effectivity of divisor classes under the lifted isomorphisms, and also covers multiplicities greater than the specified minima.

Put $A = -K$. We first exhibit an integral member of $|A|$. For eight points, cubic equations through them form a nonzero vector space. A reducible cubic is a line plus a conic, or three lines; the restrictions on collinearity and conics allow it to contain at most seven of the points. Such a cubic is therefore integral. The preceding paragraph excludes singularities at the eight points, so its strict transform has class A . For seven points, the projective space of cubics through them has dimension at least two. A reducible member must be the union of a line through two of the points and the conic through the other five. There are only finitely many such members. Five points with no three collinear determine a unique smooth conic, by the elementary conic equations and Bezout. For each specified point, one of these line-plus-conic members is smooth there: put that point on the line, since the conic cannot contain a sixth point. Thus singularity at any one of the seven points is a proper linear condition. An integral member smooth at all seven exists, again giving an integral divisor of class A .

Since $A^2 = 9 - r > 0$, intersection with this member shows that $AC \geq 0$ for every integral curve C , including the member itself. If $C^2 < 0$, adjunction gives

$$0 \leq p_a(C) = 1 + \frac{C^2 - AC}{2}.$$

Consequently either $C^2 = -1$, $AC = 1$, $p_a(C) = 0$, or $C^2 = -2$, $AC = 0$, $p_a(C) = 0$. The latter case cannot be exceptional. Write its plane class as $dH - \sum m_i E_i$, with $d > 0$, $m_i \geq 0$. Then

$$\sum m_i = 3d, \quad \sum m_i^2 = d^2 + 2, \quad (9 - r)d^2 \leq 2r.$$

For seven points $d \leq 2$, yielding only a line through three points or a conic through six. For eight points $d \leq 4$; equality $d = 4$ in Cauchy-Schwarz would require all $m_i = 3/2$. The only additional case is $d = 3$, with multiplicities $(2, 1, 1, 1, 1, 1, 1)$. These are precisely the excluded cases.

For a nonexceptional (-1) -curve the equations instead are

$$\sum m_i = 3d - 1, \quad \sum m_i^2 = d^2 + 1.$$

Cauchy–Schwarz bounds $d \leq 3$ for seven points and $d \leq 7$ for eight. In the second case $d = 7$ would require every $m_i = 5/2$, so actually $d \leq 6$. Enumerating nonnegative integer partitions of $3d - 1$ subject to $\sum m_i(m_i - 1) = (d - 1)(d - 2)$ gives the following table. Exponents indicate repeated entries; omitted entries are zero.

class or degree	multiplicities	$r = 7$	$r = 8$
E_i		7	8
1	1^2	21	28
2	1^5	21	56
3	$2, 1^6$	7	56
4	$2^3, 1^5$	0	56
5	$2^6, 1^2$	0	28
6	$3, 2^7$	0	8

All listed classes occur. For each positive-degree row, perform a quadratic transformation at the three largest multiplicities, using a zero entry as the third center in degree one. The successive degree reductions are 1 to an exceptional class, 2 to 1, 3 to 2, 4 to 2, 5 to 4, and 6 to 5, with the lower-degree patterns in the table. General position makes every transformation admissible. Pulling back the final exceptional curve gives a smooth rational curve in the original class. Two different integral curves in one such class would have intersection -1 , impossible. The column sums are 56 and 240, as required.

(e) On the nine-point blow-up set

$$\delta = 3H - \sum_{i=1}^9 E_i, \quad \alpha = E_1 - E_2.$$

Thus $\delta^2 = \delta\alpha = 0$ and $\alpha^2 = -2$. The reflection s_α exchanges labels 1, 2. The class $\beta = \delta + \alpha$ has degree three and multiplicities $(0, 2, 1, 1, 1, 1, 1, 1, 1)$. Reflecting at its double point and two simple points reduces it to a conic-through-six class, then to a line-through-three class. Hence s_β is a conjugate of a quadratic reflection by a word of permutations and quadratic reflections.

Set $T = s_\beta s_\alpha$. Direct substitution gives

$$T(D) = D + (D\delta)\alpha + (D\delta - D\alpha)\delta.$$

Since $E_9\delta = 1$, $E_9\alpha = 0$, induction yields

$$T^n(E_9) = E_9 + n\alpha + n^2\delta \quad (n \geq 1).$$

Every word T^n can be realized by admissible transformations, by general position. The lifted isomorphisms carry an exceptional curve to a smooth rational (-1) -curve in this class. Its plane degree is $3n^2$, and its multiplicities are

$$(n^2 - n, n^2 + n, n^2, n^2, n^2, n^2, n^2, n^2, n^2 - 1).$$

In particular these curves are distinct and their degrees are unbounded. The line through two original points is also in the reflection orbit of an exceptional class, as the degree-one reduction in (d) shows, so this supplies the orbit required by the source's hint. $\square$

Exercise V.4.16**Permitted context.**

Work over an algebraically closed field k with $\text{char } k \neq 3$. A smooth cubic surface has 27 lines, and its anticanonical embedding is its given embedding in $\mathbb{P}^3$. Thus every automorphism extends to a projective linear map. In characteristic two, we use Hermitian linear algebra over $\mathbb{F}_4$, with conjugation $u \mapsto u^2$. A nondegenerate Hermitian space has an orthonormal basis.

Problem. For $X = V(x_0^3 + x_1^3 + x_2^3 + x_3^3)$, find all its lines and their incidences. Determine $\text{Aut}_k(X)$, including the exceptional characteristic-two case. Explain the exclusion of characteristic three.

Solution. Let every parameter below run through the three roots of -1 . The following three sets give 27 distinct lines:

$$\begin{aligned} A(\zeta, \eta) : \quad & x_0 = \zeta x_1, \quad x_2 = \eta x_3, \\ B(\alpha, \beta) : \quad & x_0 = \alpha x_2, \quad x_1 = \beta x_3, \\ C(\gamma, \delta) : \quad & x_0 = \gamma x_3, \quad x_1 = \delta x_2. \end{aligned} \tag{1}$$

Their equations make the cubic vanish identically. They are distinct within each family; lines from different families have different two-dimensional coordinate solution spaces. Since X is smooth and has 27 lines, the list is complete.

Within a family, two distinct lines meet exactly when one parameter agrees. Between families, substitution in (1) gives

pair	condition for intersection
$A(\zeta, \eta), B(\alpha, \beta)$	$\zeta\beta = \alpha\eta$
$A(\zeta, \eta), C(\gamma, \delta)$	$\gamma = \zeta\eta\delta$
$B(\alpha, \beta), C(\gamma, \delta)$	$\alpha\beta = \gamma\delta$.

(2)

When the indicated equality holds the common solution space is one-dimensional; otherwise it is zero. Thus every line meets four lines of its own family and three from each other family: ten in total. Equations (1)–(2) specify every pairwise incidence.

Suppose first that $\text{char } k \neq 2, 3$. The Hessian determinant of $F = \sum x_i^3$ is

$$6^4 x_0 x_1 x_2 x_3.$$

The chain rule for a linear change of coordinates shows that an automorphism preserving F up to scalar preserves this zero set. It must therefore permute the four coordinate hyperplanes, and its matrix is monomial. Conversely a permutation of coordinates preserves F . A diagonal matrix preserves F up to scalar exactly when the cubes of its diagonal entries agree. Modulo scalar matrices this gives

$$\text{Aut}(X) \cong \mu_3^3 \rtimes S_4, \quad |\text{Aut}(X)| = 27 \cdot 24 = 648. \tag{3}$$

Now let $\text{char } k = 2$. For a matrix A , expansion of $F(Ax)$ shows that preservation up to scalar is exactly

$$A^{(2)T} A = \lambda I, \tag{4}$$

where (2) squares every entry. Rescale A to make $\lambda = 1$. Taking squared transpose of (4) gives $A^{(2)T} A^{(4)} = I$, so comparison with (4) yields $A^{(4)} = A$. Every entry is therefore in $\mathbb{F}_4$. Conversely each unitary matrix over $\mathbb{F}_4$ satisfies (4) and preserves X . Two normalized matrices give the same projective map precisely when they differ by a scalar in μ_3 . Consequently

$$\text{Aut}(X) = U_4(2)/\mu_3 = \text{PGU}_4(2). \tag{5}$$

For completeness, a vector in the standard Hermitian $\mathbb{F}_4^n$ has norm one exactly when it has an odd number of nonzero coordinates. There are therefore

$$N_n = \sum_{j \text{ odd}} \binom{n}{j} 3^j = \frac{4^n - (-2)^n}{2}$$

such vectors, where the counting takes place in the integers. Choosing orthonormal basis vectors successively gives

$$|U_4(2)| = N_4 N_3 N_2 N_1 = 120 \cdot 36 \cdot 6 \cdot 3 = 77760.$$

Dividing by the three scalar matrices gives $|\text{Aut}(X)| = 25920$. These are automorphisms over k ; no field-semilinear maps are included.

Finally, in characteristic three, $F = (x_0 + x_1 + x_2 + x_3)^3$. It defines a triple plane, so the smooth cubic-surface question and its 27-line description do not apply. $\square$

V.5 Birational Transformations

Exercise V.5.1

Permitted context.

Work over an algebraically closed field k . A surface is smooth, integral, and projective over k . For the blow-up $\pi: \tilde{X} \rightarrow X$ at a closed point, with exceptional curve E ,

$$E \cong \mathbb{P}^1, \quad E^2 = -1, \quad (\pi^*D)E = 0, \quad (\pi^*D)(\pi^*D') = DD'.$$

If an effective divisor D has multiplicity m at the center, its strict transform has class $\pi^*D - mE$. Intersection multiplicities of effective divisors with no common component are nonnegative.

Regular local rings of a smooth surface are factorial. Thus a rational function with no polar component through a point is regular there.

Problem. For $f \in k(X)$, construct a finite sequence of point blow-ups $g: X' \rightarrow X$ such that the rational map defined by f extends to a morphism $X' \rightarrow \mathbb{P}^1$.

Solution. The zero function and all constant functions already define morphisms. Otherwise write

$$\text{div}(f) = Z - P$$

with effective divisors Z, P having no common component. Put $N = ZP$, a nonnegative integer. If their supports meet at x , let $m, n > 0$ be their respective multiplicities at x , and blow up x . Write $\tilde{Z}, \tilde{P}$ for their strict transforms. The new divisor of f is

$$\tilde{Z} - \tilde{P} + (m - n)E.$$

When $m \geq n$, the new zero and pole divisors are $\tilde{Z} + (m - n)E$ and $\tilde{P}$. Their intersection is

$$(ZP - mn) + (m - n)n = ZP - n^2.$$

When $n \geq m$, the analogous calculation gives $ZP - m^2$. In either case it decreases by $\min(m, n)^2 > 0$. The new zero and pole divisors again have no common component, so their intersection remains nonnegative. Consequently this operation can occur only finitely often.

At termination the supports of the zeros and poles are disjoint. Off the poles, f is regular; off the zeros, f^{-1} is regular. These two opens cover X' . On them define the map to $\mathbb{P}^1$ by $[f : 1]$ and $[1 : f^{-1}]$, respectively. On the overlap f is a unit and the expressions agree. They therefore glue to the required morphism. $\square$

Exercise V.5.2**Permitted context.**

Work over an algebraically closed field k . A surface is smooth, integral, and projective over k . For the blow-up $\pi: \tilde{X} \rightarrow X$ at a closed point, with exceptional curve E ,

$$E \cong \mathbb{P}^1, \quad E^2 = -1, \quad (\pi^*D)E = 0, \quad (\pi^*D)(\pi^*D') = DD'.$$

If an effective divisor D has multiplicity m at the center, its strict transform has class $\pi^*D - mE$. Intersection multiplicities of effective divisors with no common component are nonnegative.

We use Serre vanishing, the exact sequence for adding an effective Cartier divisor, and $H^1(\mathbb{P}^1, \mathcal{O}(d)) = 0$ for $d \geq -1$. A globally generated line bundle defines a morphism to projective space. The normalization of a projective variety is finite and projective; a dominant morphism from a normal variety factors through it.

Problem. Let $Y \subset X$ be a curve isomorphic to $\mathbb{P}^1$, with $Y^2 < 0$. Construct a birational morphism to a normal projective surface $f: X \rightarrow X_0$ which contracts Y to one point and is an isomorphism away from Y .

Solution. Put $n = -Y^2 > 0$. Choose a sufficiently large multiple H of a very ample divisor so that

$$H^1(X, \mathcal{O}_X(H)) = 0, \quad HY = kn$$

for a positive integer k . Divisibility is obtained by taking the multiple divisible by n , and vanishing by Serre's theorem. For $1 \leq j \leq k$, the restriction of $H + jY$ to Y has degree $(k - j)n \geq 0$. The sequences

$$0 \rightarrow \mathcal{O}_X(H + (j - 1)Y) \rightarrow \mathcal{O}_X(H + jY) \rightarrow \mathcal{O}_Y((H + jY)|_Y) \rightarrow 0$$

therefore show inductively that $H^1(X, \mathcal{O}_X(H + jY)) = 0$.

Set $A = H + kY$. Multiplication by the canonical section of $\mathcal{O}_X(kY)$ supplies enough sections of A to generate it away from Y . At Y , the last displayed sequence gives a surjection

$$H^0(X, \mathcal{O}_X(A)) \rightarrow H^0(Y, \mathcal{O}_Y(A)) = H^0(Y, \mathcal{O}_Y) = k,$$

since the preceding H^1 vanishes. A section lifting 1 generates along Y . Thus A is globally generated.

Let $h: X \rightarrow X_1 \subset \mathbb{P}^N$ be its morphism onto its image. Every section restricts to a constant on Y , so $h(Y)$ is a point p . The subsystem obtained from H vanishes on Y , while at each point outside Y some member is nonzero. Thus $h^{-1}(p) = Y$ set-theoretically. On a chart where such a member is nonzero, ratios of members of this subsystem are exactly the ratios giving the very ample embedding by H . They recover both points and local coordinates. Hence $h: X \setminus Y \rightarrow X_1 \setminus \{p\}$ is an isomorphism, and h is birational.

Normalize X_1 , obtaining X_0 , and factor h as $X \xrightarrow{f} X_0 \rightarrow X_1$. The factor f is proper, since X is projective and X_0 is separated, and its image is dense by birationality. Its closed image is therefore all of X_0 . The finite set over p has only one point: the surjective map f sends the connected curve Y onto that set. Off this point the normalization is an isomorphism because $X_1 \setminus \{p\}$ is smooth. Thus f has exactly the required properties. The contracted point is allowed to be singular. $\square$

Exercise V.5.3**Permitted context.**

Work over an algebraically closed field k . A surface is smooth, integral, and projective over k . For the blow-up $\pi : \tilde{X} \rightarrow X$ at a closed point, with exceptional curve E ,

$$E \cong \mathbb{P}^1, \quad E^2 = -1, \quad (\pi^*D)E = 0, \quad (\pi^*D)(\pi^*D') = DD'.$$

If an effective divisor D has multiplicity m at the center, its strict transform has class $\pi^*D - mE$. Intersection multiplicities of effective divisors with no common component are nonnegative.

Write $\Omega_X = \Omega_{X/k}^1$. For a point blow-up, $\pi_*\mathcal{O}_{\tilde{X}} = \mathcal{O}_X$ and $R^q\pi_*\mathcal{O}_{\tilde{X}} = 0$ for $q > 0$. Projection formula and Leray then give $H^q(\tilde{X}, \pi^*F) = H^q(X, F)$ for locally free F . Regular sections of a locally free sheaf on a smooth surface extend across a missing closed point. Serre duality gives $H^2(X, \Omega_X)^\vee = H^0(X, \Omega_X^\vee \otimes \omega_X)$; the wedge pairing identifies $\Omega_X^\vee \otimes \omega_X$ with Ω_X . We use $H^0(\mathbb{P}^1, \Omega_{\mathbb{P}^1}) = 0$ and $H^1(\mathbb{P}^1, \Omega_{\mathbb{P}^1}) = k$. The class $c_1(L) \in H^1(\Omega_X)$ is represented by logarithmic differentials of transition functions; on a smooth projective curve its trace is $\deg L$, viewed in k .

Problem. For the blow-up of a surface at one point, prove

$$H^1(\tilde{X}, \Omega_{\tilde{X}}) \cong H^1(X, \Omega_X) \oplus k.$$

Deduce that every sequence of contractions of exceptional curves of the first kind has bounded length.

Solution. Let $i : E \hookrightarrow \tilde{X}$. There is an exact sequence

$$0 \rightarrow \pi^*\Omega_X \rightarrow \Omega_{\tilde{X}} \rightarrow i_*\Omega_E \rightarrow 0. \quad (1)$$

To check it, take local smooth coordinates x, y at the center. On the chart $y = xt$, the pullbacks of dx, dy are $dx, t dx + x dt$. The map is injective, and its cokernel is $(\mathcal{O}/(x))dt$, the differentials on that chart of E . The other chart gives the same quotient, with the usual transition for $\Omega_{\mathbb{P}^1}$. These calculations can be made after an étale coordinate map, which suffices to verify (1).

The cohomology sequence and projection formula give

$$\begin{aligned} 0 \rightarrow H^1(X, \Omega_X) \rightarrow H^1(\tilde{X}, \Omega_{\tilde{X}}) \rightarrow k \\ \rightarrow H^2(X, \Omega_X) \rightarrow H^2(\tilde{X}, \Omega_{\tilde{X}}) \rightarrow 0. \end{aligned} \quad (2)$$

The initial zero follows from $H^0(E, \Omega_E) = 0$, and the final zero from the absence of second cohomology on E . Moreover pullback identifies the spaces of global one-forms on X and $\tilde{X}$: a form upstairs restricts to $X \setminus \{p\}$, extends across p , and pulls back to the original form because they agree on a dense open. Serre duality now shows that the last two nonzero terms of (2) have the same finite dimension. The displayed surjection between them is therefore an isomorphism. Hence the preceding boundary map from k is zero, and (2) reduces to a short exact sequence.

It splits explicitly: $c_1(\mathcal{O}_{\tilde{X}}(E))$ restricts to the class of $\mathcal{O}_E(-1)$, whose trace is $-1 \neq 0$, in every characteristic. Its span supplies a complement to the pullback of $H^1(X, \Omega_X)$. This proves the asserted decomposition.

Each contraction to a smooth surface reverses one point blow-up and lowers $h^1(\Omega)$ by one. Thus a sequence starting from X has length at most $h^1(X, \Omega_X)$, since the final dimension cannot be negative. $\square$

Exercise V.5.4**Permitted context.**

Work over an algebraically closed field k . A surface is smooth, integral, and projective over k . For the blow-up $\pi : \tilde{X} \rightarrow X$ at a closed point, with exceptional curve E ,

$$E \cong \mathbb{P}^1, \quad E^2 = -1, \quad (\pi^*D)E = 0, \quad (\pi^*D)(\pi^*D') = DD'.$$

If an effective divisor D has multiplicity m at the center, its strict transform has class $\pi^*D - mE$. Intersection multiplicities of effective divisors with no common component are nonnegative.

Every birational morphism between smooth projective surfaces factors as finitely many point blow-ups. Blowing up a smooth point of a smooth curve leaves its strict transform isomorphic to that curve, and decreases its self-intersection by one.

Problem. Let $f : X \rightarrow X'$ be a birational morphism of smooth projective surfaces.

(a) If an integral curve $Y \subset X$ is contracted by f , prove that $Y \cong \mathbb{P}^1$ and $Y^2 < 0$.

(b) For a point $p \in X'$, let $Y_1, \dots, Y_r$ be all the curve components of its fiber. Prove that their intersection matrix is negative definite when $r > 0$.

Solution. (a) Factor f into point blow-ups, starting from X' . Each new exceptional curve is a copy of $\mathbb{P}^1$ of square -1 . Subsequent blow-ups preserve the isomorphism type of its strict transform, and decrease its square only when their centers lie on it. Every contracted curve is one of these strict transforms: away from their union the composite is an isomorphism. This proves (a).

(b) Let $E_1, \dots, E_s$ be the total transforms on X of the successive new exceptional curves, ordered by their creation. The blow-up intersection formulas imply

$$E_i E_j = 0 \quad (i \neq j), \quad E_i^2 = -1.$$

Thus their real span has negative definite form, with $(\sum a_i E_i)^2 = -\sum a_i^2$. The strict transform of the i -th exceptional curve equals E_i minus an integer combination of E_j with $j > i$. Indeed each later blow-up subtracts its new exceptional class exactly when its center lies on the then-current strict transform. Consequently the change from total to strict transforms is triangular with diagonal entries one. All strict exceptional classes are therefore linearly independent and span the same negative definite space.

The curves $Y_1, \dots, Y_r$ form a subset of these strict transforms. For any nonzero real vector $(a_1, \dots, a_r)$, their linear independence makes $D = \sum a_i Y_i$ a nonzero vector in that space, so

$$\sum_{i,j} a_i a_j (Y_i Y_j) = D^2 < 0.$$

This is precisely negative definiteness of the requested matrix. □

Exercise V.5.5**Permitted context.**

Let C be a smooth projective integral curve over an algebraically closed field. A geometrically ruled surface over C is $\mathbb{P}_C(F)$ for a rank-two vector bundle F , using the convention that points are one-dimensional quotients. An elementary transformation at a point of a fiber blows up that point and contracts the strict transform of the fiber. A smooth rational curve of square -1 can be contracted to a smooth point. Local rings of C at closed points are discrete valuation rings. Every torsion-free coherent sheaf on C is locally free, and a torsion sheaf has finite length and a filtration with quotients $k(p)$.

Problem. Show that any two geometrically ruled surfaces over the same curve C are related, over C , by a finite sequence of elementary transformations.

Solution. First consider a surjection $F \rightarrow k(p)$ and its kernel F_1 . The kernel is a rank-two vector bundle. Locally at p , choose a parameter t and a basis e_1, e_2 so that the quotient sends e_1 to zero and e_2 to one. Then

$$F_1 = \mathcal{O}e_1 \oplus \mathcal{O}(te_2).$$

Restriction of a quotient of F to F_1 gives the rational map

$$(t, [u : v]) \longmapsto (t, [u : tv])$$

from $\mathbb{P}(F)$ to $\mathbb{P}(F_1)$. Its unique base point is $t = u = 0$, the chosen quotient point. On the chart $v = 1$, blowing up the ideal (u, t) resolves the pair $[u : t]$. On the other chart the pair was already regular. The strict transform of $t = 0$ is mapped to $[1 : 0]$, while the exceptional curve parametrizes the new fiber. The inverse rational map is $[a : b] \mapsto [ta : b]$, with its single base point at $t = b = 0$; the same two blow-up charts resolve it. Thus the common resolution is a point blow-up on either side, and $\mathbb{P}(F_1)$ is exactly the elementary transformation of $\mathbb{P}(F)$. This also shows that the inverse operation is an elementary transformation.

Now write the given surfaces as $\mathbb{P}(F)$ and $\mathbb{P}(G)$. Identify their generic fibers as two-dimensional vector spaces over $k(C)$, and view F, G as lattices in that common rational vector space. They agree away from finitely many points after this identification. Their intersection $L = F \cap G$ is coherent: locally it is the intersection of two finite lattices over a discrete valuation ring, hence again a finite free lattice. It is therefore a rank-two vector bundle, and both F/L and G/L are torsion sheaves of finite length.

Choose composition series of these two quotients and take preimages. They give chains from F to L and from G to L , each step the kernel of a surjection onto some $k(p)$. The local calculation realizes every step by an elementary transformation. Follow the chain from $\mathbb{P}(F)$ to $\mathbb{P}(L)$, then reverse the chain from $\mathbb{P}(G)$. This is the required finite sequence over C . $\square$

Exercise V.5.6**Permitted context.**

Work over an algebraically closed field k . A surface is smooth, integral, and projective over k . For the blow-up $\pi : \tilde{X} \rightarrow X$ at a closed point, with exceptional curve E ,

$$E \cong \mathbb{P}^1, \quad E^2 = -1, \quad (\pi^*D)E = 0, \quad (\pi^*D)(\pi^*D') = DD'.$$

If an effective divisor D has multiplicity m at the center, its strict transform has class $\pi^*D - mE$. Intersection multiplicities of effective divisors with no common component are nonnegative.

A valuation ring R of K/k contains k , has fraction field K , and is trivial on k . It has a unique center on any proper model, and dominates the local ring at that center. A valuation overring of a discrete valuation ring in its fraction field is either that ring or the field. Every rational function on a smooth projective surface becomes a morphism to $\mathbb{P}^1$ after finitely many point blow-ups. The center of a valuation lifts uniquely under every proper blow-up.

Problem. Let $K = k(X)$. Classify the nontrivial valuation rings $R \neq K$ of K/k as follows:

- (1) the local ring at the generic point of a curve on X ;
- (2) the local ring at the generic point of an exceptional curve on a surface obtained from X by finitely many point blow-ups;
- (3) the increasing union of local rings along an infinite sequence of closed valuation centers, each obtained by blowing up the preceding center.

In case (3), prove that the union equals R .

Solution. Properness supplies a center on X . It is not the generic point, since domination there would force $R = K$. If it is the generic point of a curve, its local ring is a discrete valuation ring. Domination and the overring assertion force equality, giving (1).

Otherwise the center x_0 is closed. Blow it up and take the new center. If this is a curve's generic point, that curve is exceptional, and the same argument gives (2). Repeat whenever the center remains closed. If this process never stops, write

$$X_0 = X, \quad X_{i+1} = \text{Bl}_{x_i} X_i, \quad A_i = \mathcal{O}_{X_i, x_i}.$$

The maps give local inclusions $A_i \subset A_{i+1} \subset R$, so their union R_0 is contained in R .

We verify the opposite inclusion. Fix $f \in R$, with $f \neq 0$. Choose a finite sequence of point blow-ups $Z \rightarrow X$ on which f defines a morphism to $\mathbb{P}^1$. The infinite sequence of centers eventually maps, in a neighborhood of its center, to Z . Here is the elementary lifting argument. Process the finite sequence defining Z one blow-up at a time. If its next center differs from the current valuation center, that blow-up is an isomorphism near the latter and needs no new step in our sequence. If they agree, perform the next blow-up in our sequence: it is the same local blow-up and its valuation center corresponds to the center on the chosen model. Induction through the finite list gives the asserted local map for some X_N .

Let z be the center on Z . Because $f \in R$, the map $\text{Spec } R \rightarrow \mathbb{P}^1$ defined by $[f : 1]$ has closed-point image in the affine chart with coordinate f . It agrees generically with the map induced from Z , hence everywhere by separatedness of $\mathbb{P}^1$. Thus z maps to this affine chart and $f \in \mathcal{O}_{Z, z}$. The local map just constructed gives $f \in A_N \subset R_0$. The zero element is already in every A_i , so $R \subset R_0$. Therefore $R = R_0$, establishing (3).

These alternatives exhaust the construction. If trivial valuation rings are included in the terminology, K itself must be added as a separate case. $\square$

Exercise V.5.7**Permitted context.**

Work over an algebraically closed field k . A surface is smooth, integral, and projective over k . For the blow-up $\pi : \tilde{X} \rightarrow X$ at a closed point, with exceptional curve E ,

$$E \cong \mathbb{P}^1, \quad E^2 = -1, \quad (\pi^*D)E = 0, \quad (\pi^*D)(\pi^*D') = DD'.$$

If an effective divisor D has multiplicity m at the center, its strict transform has class $\pi^*D - mE$. Intersection multiplicities of effective divisors with no common component are nonnegative.

For a surjective generically finite morphism $f : X \rightarrow X_0$ of projective integral surfaces and Cartier divisors H, H' on X_0 , $(f^*H)(f^*H') = \deg(f)(HH')$. A pulled-back divisor has intersection zero with every curve contracted by f . The Hodge index theorem says that the real numerical divisor space has signature $(1, \rho - 1)$; in particular, the orthogonal complement of any class of positive square is negative definite. An effective nonzero curve has positive intersection with an ample divisor, and therefore is not numerically zero.

Problem. Let $f : X \rightarrow X_0$ be a dominant morphism to a projective integral surface, and suppose that an integral curve $Y \subset X$ is contracted to a point p . Prove that $Y^2 < 0$. In particular this holds when Y is the whole set-theoretic fiber.

Solution. Since X is projective, its image is closed. Dominance therefore makes f surjective, and equality of dimensions makes it generically finite. For a very ample Cartier divisor H on X_0 , put $D = f^*H$. The projection formulas give

$$D^2 = \deg(f)H^2 > 0, \quad DY = 0.$$

The class of Y is nonzero numerically since an ample divisor on X has positive intersection with it. The Hodge index theorem, applied to the orthogonal complement of D , yields $Y^2 < 0$. $\square$

Exercise V.5.8**Permitted context.**

Let k be algebraically closed, of arbitrary characteristic. The blow-up of the origin in affine three-space has, for example, the z -chart $x = zu$, $y = zv$. For a hypersurface of multiplicity m , substitute these expressions and divide the equation by z^m to obtain its strict transform. Restriction gives the blow-up of the integral hypersurface itself.

We use the Jacobian criterion, and Nagata's factoriality criterion: if B is a noetherian domain, p is a prime element, and $B[p^{-1}]$ is a unique factorization domain, then B is one too. The binomial $x^2 + y^3$ is irreducible in $k[x, y]$: the homomorphism $x \mapsto s^3$, $y \mapsto -s^2$ has image $k[s^2, s^3]$ and kernel $(x^2 + y^3)$. Polynomial division by the monic quadratic in x verifies the kernel assertion, since the remaining even powers of s and the powers s^{3+2j} are disjoint.

A hypersurface double point with smooth projective tangent conic is resolved by one point blow-up. Indeed, on the exceptional divisor the strict-transform equation is that conic; its nonzero derivatives in projective tangent directions give smoothness of the transform there. The exceptional curve is the smooth conic, hence isomorphic to $\mathbb{P}^1$. In particular this applies to tangent cone $u^2 + st = 0$, also in characteristic two.

Problem. Let

$$X = V(x^2 + y^3 + z^5) \subset \mathbb{A}_k^3, \quad A = k[x, y, z]/(x^2 + y^3 + z^5).$$

(a) Prove that A is a unique factorization domain.

(b) Resolve its singularity at the origin by eight successive point blow-ups. Show that the reduced exceptional fiber consists of eight copies of $\mathbb{P}^1$, with intersection graph E_8 : a chain of seven vertices with an additional vertex attached at the third vertex from one end.

Solution. (a) The element z is a nonzerodivisor in A : in the polynomial ring, z is relatively prime to $x^2 + y^3 + z^5$, so $zg \in (x^2 + y^3 + z^5)$ implies g is in that ideal. Thus A injects into $A[z^{-1}]$. Set

$$t = z^{-1}, \quad u = t^3x, \quad v = t^2y.$$

Multiplying the defining equation by t^6 gives $u^2 + v^3 + t = 0$. Conversely $x = u/t^3$, $y = v/t^2$, $z = 1/t$. These substitutions yield inverse isomorphisms

$$A[z^{-1}] \cong k[u, v, t, t^{-1}]/(u^2 + v^3 + t) \cong k[u, v, (u^2 + v^3)^{-1}].$$

This is a localization of a polynomial ring, hence a domain and a unique factorization domain. The injection already proves that A is a domain. Furthermore

$$A/(z) \cong k[x, y]/(x^2 + y^3)$$

is a domain by the kernel calculation in the context, so z is prime in A . Nagata's criterion proves that A is a unique factorization domain.

(b) The Jacobian criterion shows that the only singular point is the origin. In characteristics two, three, or five, the two nonzero partial derivatives force the corresponding two coordinates to vanish, and the equation forces the remaining coordinate to vanish as well. We give all charts that can contain singular points above the origin; away from the exceptional locus every blow-up is an isomorphism. Denote the new exceptional curves, with their reduced structures, by $E_1, \dots, E_8$, in order of creation.

Blow up the origin first. In the z -chart, put $x = zu$, $y = zv$; the equation becomes

$$u^2 + zv^3 + z^3 = 0. \quad (1)$$

Its exceptional curve is $z = u = 0$. Along this curve the only possible singular point is $v = 0$, because the derivative with respect to z restricts to v^3 . In the y -chart the equation is $u^2 + y + y^3v^5 = 0$, which is smooth along $y = 0$. The x -chart has no exceptional points, since its equation restricts to $1 = 0$ there. Thus (1) has exactly one singular point above the origin.

Blow up that point. In its v -chart, put $u = va$, $z = vb$ and write $s = v$, $t = b$. The equation is

$$a^2 + s^2t + st^3 = 0. \quad (2)$$

Here E_2 is $s = a = 0$, and the strict transform of E_1 is $t = a = 0$. The derivative with respect to s along E_2 is t^3 , so the only singular point in this chart is its origin. The other relevant chart, $u = za$, $v = zb$, has equation $a^2 + z + z^2b^3 = 0$, smooth on $z = 0$. Again the chart of the squared variable has no exceptional points.

The third blow-up is at the origin of (2). Its two relevant charts are

$$\begin{aligned} a = su, \quad t = sv : \quad u^2 + sv + s^2v^3 &= 0, \\ a = tu, \quad s = tv : \quad u^2 + tv^2 + t^2v &= 0. \end{aligned} \quad (3)$$

On E_3 , the first chart has one singular point, $u = s = v = 0$; its quadratic tangent cone is $u^2 + sv = 0$. The second chart has one singular point, $u = t = v = 0$, with exact equation

$$u^2 + tv(v + t) = 0. \quad (4)$$

These are different points of E_3 : they correspond to the directions $[s : t] = [1 : 0]$ and $[0 : 1]$. They are all the singular points, since the relevant derivatives on E_3 are respectively v and v^2 .

Blow up the first point in (3) as the fourth blow-up. Its tangent conic is smooth in every characteristic, so this resolves that point and introduces the smooth conic E_4 . The strict transforms of E_1 and E_3 meet E_4 at two distinct points, corresponding to their two coordinate-axis directions in that conic. Their intersections are transverse: in a chart at either axis direction, the transformed equation solves for one coordinate, and the old curve and exceptional curve are the two remaining coordinate axes.

The fifth blow-up is at the point (4). In the t -chart, write $u = tw$, $v = tb$; the equation is

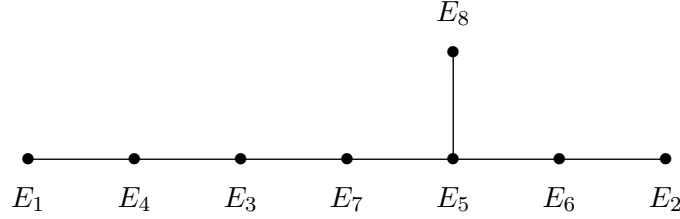
$$w^2 + tb(b + 1) = 0. \quad (5)$$

Its exceptional curve is $t = w = 0$, and its singular points on that curve are $b = 0, -1$. The v -chart gives $w^2 + vb(b + 1) = 0$ and supplies one more point, the direction missing from the t -chart. Thus E_5 contains exactly three singular points, corresponding to the directions $v = 0$, $t = 0$, $v + t = 0$ in (4). These directions are distinct even in characteristic two.

At each of these points the tangent cone has form $w^2 + crq = 0$ for $c = 1$ or -1 . For example, at $b = 0$ in (5) it is $w^2 + tb$; at $b = -1$, put $b = -1 + q$ and obtain $w^2 - tq$. Each projective tangent conic is smooth. Blow up these three points separately as blow-ups six, seven, and eight. They are disjoint centers, and each is resolved by its one blow-up. Name the resulting curves E_6, E_7, E_8 according to the directions $v = 0$, $t = 0$, $v + t = 0$, respectively. The surface is now smooth.

It remains to record the graph, including the older curves. After the second blow-up, E_1, E_2 pass through the point (2). The third blow-up separates their directions: E_1 meets E_3 at the first point in (3), and E_2 meets E_3 at (4). The fourth inserts E_4 between E_1 and E_3 . At (4), E_2 has

$v = u = 0$, whereas E_3 has $t = u = 0$. After the fifth blow-up they meet E_5 in the first two singular directions just listed. The sixth and seventh blow-ups insert E_6 between E_2, E_5 and E_7 between E_3, E_5 . The third singular direction contains neither older curve, so the eighth adds E_8 meeting only E_5 . Consequently the edges are exactly



Thus the chain is $E_1 - E_4 - E_3 - E_7 - E_5 - E_6 - E_2$, with E_8 attached only to E_5 . Every indicated intersection is transverse, by the same coordinate calculation at the two axes of a resolved quadratic cone; the three points on E_5 are distinct. All other pairs are disjoint. The exceptional curves of blow-ups one, two, three, and five are reduced projective lines; those of the other four blow-ups are smooth conics. Their later strict transforms remain isomorphic to $\mathbb{P}^1$. Thus there are exactly eight exceptional curves isomorphic to $\mathbb{P}^1$, with the stated E_8 graph; this does not assert that they are lines in a chosen projective embedding. $\square$

V.6 Classification of Surfaces

Exercise V.6.1

Permitted context.

Work over an algebraically closed field. A surface here is smooth, integral, and projective. Write H for its hyperplane divisor. Adjunction for a smooth complete intersection of degrees $d_1, \dots, d_c$ in $\mathbb{P}^n$ gives

$$K_X = \left(\sum_i d_i - n - 1 \right) H.$$

Its structure sheaf has the Koszul resolution with j -th term

$$Q_j = \bigoplus_{|J|=j} \mathcal{O}_{\mathbb{P}^n} \left(- \sum_{i \in J} d_i \right).$$

We use $H^i(\mathbb{P}^n, \mathcal{O}(t)) = 0$ for $0 < i < n$, Serre duality, and Castelnuovo's criterion: a smooth projective surface with $p_a = 0$ and $h^0(2K) = 0$ is rational. A surface with ample canonical divisor has Kodaira dimension two, hence is of general type. A K3 surface is a smooth projective surface with trivial canonical bundle and $h^1(\mathcal{O}) = 0$.

Problem. Let $X \subset \mathbb{P}^n$, $n \geq 3$, be a smooth complete intersection surface of degrees $d_1, \dots, d_{n-2}$, each at least two. Determine exactly which degree lists fail to give a surface of general type, and classify the exceptional surfaces. Regard permutations of a degree list as equivalent.

Solution. First $q(X) = h^1(X, \mathcal{O}_X) = 0$. For completeness, let $c = n - 2$, and let B_j be the image of $Q_j \rightarrow Q_{j-1}$, with $Q_0 = \mathcal{O}_{\mathbb{P}^n}$. Thus $B_1 = \mathcal{I}_X$, $B_c \cong Q_c$, and

$$0 \rightarrow B_{j+1} \rightarrow Q_j \rightarrow B_j \rightarrow 0$$

is exact for $j < c$. The sequence for $\mathcal{I}_X$ gives $H^1(\mathcal{O}_X) \cong H^2(B_1)$. Successive dimension shifts give

$$H^2(B_1) \cong H^3(B_2) \cong \dots \cong H^{c+1}(Q_c) = H^{n-1}(Q_c) = 0.$$

Each shift uses only the stated intermediate cohomology vanishing; when $c = 1$ the first group is already $H^2(Q_1) = 0$.

Put

$$m = \sum_i d_i - n - 1 = \sum_i (d_i - 1) - 3.$$

If $m > 0$, then $K_X = mH$ is ample and X is of general type. If $m = 0$, then K_X is trivial; together with $q = 0$ this makes X a K3 surface, with $p_g = 1$ and Kodaira dimension zero. If $m < 0$, every positive multiple of K_X has negative intersection with H , so it has no effective representative. In particular $p_g = h^0(K_X) = 0$ and $h^0(2K_X) = 0$. Since $p_a = p_g - q = 0$, Castelnuovo's criterion makes X rational, with Kodaira dimension -1 in the convention of this chapter.

Every $d_i - 1$ is a positive integer. Thus $m \leq 0$ is equivalent to a partition of one, two, or three into $c = n - 2$ positive parts. The complete list is

n	(d_i)	m	classification
3	(2)	-2	rational quadric
3	(3)	-1	rational cubic surface
4	(2, 2)	-1	rational degree-four del Pezzo surface
3	(4)	0	K3 surface
4	(2, 3)	0	K3 surface
5	(2, 2, 2)	0	K3 surface.

There are no further exceptional lists, since a sum of c positive integers at most three has $c \leq 3$. The labels del Pezzo in the negative cases agree with the ampleness of $-K_X$. All other smooth complete intersections in the problem have ample canonical bundle. $\square$

Exercise V.6.2**Permitted context.**

Work over $\mathbb{C}$. Let $X \subset \mathbb{P}^{n+1}$ be a smooth integral projective surface, and let H be its hyperplane divisor. A general $C \in |H|$ is smooth and integral, with $K_C = (K_X + H)|_C$. Clifford's theorem says that a special line bundle L with sections on a smooth curve satisfies $h^0(L) \leq \deg(L)/2 + 1$. For positive degree, equality occurs only for the canonical bundle or for a multiple of the degree-two pencil of a hyperelliptic curve; in the latter noncanonical case the complete system factors through the degree-two map and is not very ample.

We use Serre duality, surface Riemann–Roch

$$\chi(\mathcal{O}_X(D)) = \chi(\mathcal{O}_X) + \frac{D(D - K_X)}{2},$$

and Kodaira vanishing $H^i(X, \mathcal{O}_X(K_X + H)) = 0$ for $i > 0$. Write $p_g = h^0(K_X)$, $q = h^1(\mathcal{O}_X)$; then $\chi(\mathcal{O}_X) = 1 - q + p_g$. A K3 surface has trivial canonical bundle and $q = 0$.

Problem. Let $X \subset \mathbb{P}^{n+1}$ be a smooth integral surface not contained in a hyperplane, and let $d = \deg X$. Prove that $d < 2n$ implies $p_g = 0$. If $d = 2n$, prove that either $p_g = 0$, or $p_g = 1$ and X is a K3 surface.

Solution. For $n = 1$, a closed integral surface in $\mathbb{P}^2$ is $\mathbb{P}^2$ itself; it has degree one and $p_g = 0$. Assume $n \geq 2$, and suppose $p_g > 0$. Choose a nonzero canonical section, with effective zero divisor D , and a general smooth hyperplane section C not contained in D . Its restriction to C is a nonzero section of $K_X|_C$. Set $L = \mathcal{O}_C(H)$. Adjunction and duality on C give

$$h^1(C, L) = h^0(C, K_C \otimes L^{-1}) = h^0(C, K_X|_C) > 0,$$

so L is special.

The linear forms on $\mathbb{P}^{n+1}$ inject into $H^0(X, \mathcal{O}_X(H))$, by nondegeneracy. Their restrictions to C have a one-dimensional kernel: the exact sequence

$$0 \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X(H) \rightarrow \mathcal{O}_C(H) \rightarrow 0$$

identifies the kernel of restriction on all sections with the one-dimensional space generated by the equation of C . Consequently $h^0(C, L) \geq n + 1$. Clifford's theorem yields

$$n + 1 \leq h^0(C, L) \leq \frac{d}{2} + 1.$$

Hence $d \geq 2n$, proving the first assertion.

Now suppose $d = 2n$ and $p_g > 0$. Equality holds in Clifford. The line bundle L has positive degree and is very ample, since it gives the hyperplane embedding of C . The noncanonical hyperelliptic equality cases are therefore impossible, and $L \cong K_C$. Adjunction implies $K_X|_C \cong \mathcal{O}_C$, so $K_X H = 0$. The effective divisor D of the chosen canonical section satisfies $DH = 0$. Ampleness of H forces $D = 0$. Thus that section never vanishes,

$$\mathcal{O}_X(K_X) \cong \mathcal{O}_X, \quad p_g = 1.$$

It remains to prove $q = 0$. Kodaira vanishing now gives $H^i(X, \mathcal{O}_X(H)) = 0$ for $i > 0$. Since $H^2 = d = 2n$, Riemann–Roch becomes

$$h^0(X, \mathcal{O}_X(H)) = \chi(\mathcal{O}_X(H)) = 2 - q + n.$$

Nondegeneracy supplies at least $n + 2$ independent linear-form sections. Hence $2 - q + n \geq n + 2$, giving $q \leq 0$. As q is a dimension, it equals zero. Together with the trivial canonical bundle this says exactly that X is a K3 surface. $\square$